Quantum Chaos and Arithmetic Maps:

Exact Results and the Open BGS Frontier

A continuing research monograph on return laws,
resonance, spectral reductions, and related mathematics

Maksim Churilov*
BGS research programme

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Preface

This monograph collects the results of a single research programme on quantum chaos and arithmetic maps, centred on one open question: whether the microscopic spectra of a fixed deterministic quantum map follow random-matrix universality at the Heisenberg scale. The question is open; this volume does not claim it solved. The chapters are not a linear proof. Each states its own hypotheses, symmetry class, time scale and spectral object, and the claim index (Appendix A) records the status of every registered claim separately. Later chapters do not silently upgrade earlier ones, and chapter adjacency does not imply a dependency. Proofs, finite certificates, conditional reductions and open obligations are kept distinct throughout. Where a statement depends on an external theorem or a finite computation, the text says so at the statement, not only in the record.

Abstract

The Bohigas–Giannoni–Schmit (BGS) conjecture relates the microscopic spectra of appropriately desymmetrized quantum chaotic systems to random-matrix universality. The original chaotic-billiard evidence concerned the orthogonal symmetry class [1]; the specified unitary-class benchmark in this programme is the Sine_2 process. This *continuing research monograph* collects distinct, rigorously qualified project results and their proof boundaries. It does not establish BGS for the fixed deterministic map, let alone for all chaotic systems.

The arithmetic programme develops finite-field pair-key and Cayley-energy reductions, cyclotomic rigidity, exact quantum-map multiplicities, arithmetic-subsequence return laws, and smooth/Gevrey cohomological and resonance results. Some statements concern compressed observables or vanishing classical kicks; others concern fixed-strength but specially conjugate quantum systems. Related chapters address locality, harmonic analysis, spectral obstructions and quantum-information estimation. Their hypotheses, symmetry classes, time scales and spectral objects remain separate, rather than being combined into a universality claim.

The central deterministic problem uses the odd-dimensional family $U_N = \exp(i\tau N A_N)M_N$ with $\tau = 10^{-80}$ *fixed as* $N \to \infty$. The manuscript establishes exact trace/resolvent identities, conditional correlation bridges, quantitative intensity and selected interference estimates, and auxiliary noisy comparisons. An internally reconstructed continuation gives phase-memory bounds, an absorbed scalar real-flow comparison, and the sharp rational matrix-inner second-jet constant 4/3. The 11 October continuation adds maximal microscopic observable-flow paths, physical-source comparison in L^2, all fixed normal-source products, and separate frozen circular phase laws with an all-order critical CSE limit. The latest fixed-map results give a corrected inserted trace, a uniform power-trace bound with one dimension threshold

*GPT 6 and Muse Spark 1.3 were used for repository work, verification, and code implementation.

for all iterates, and global spectral matching of order $1/\log N$ with a matching lower order on Pell dimensions. These conclusions do not control microscopic statistics. The actual all-order, signed Heisenberg-scale trace comparison, including the pair plateau correction and deterministic removal of auxiliary noise, remains **open**. The reported proofs have not received a blanket independent external referee or novelty certification.

Keywords. BGS conjecture; quantum cat maps; Sine_2; signed joint traces; Weyl quantization; arithmetic return laws; cohomological resonance; exact certificates.
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Introduction and guide

The Bohigas–Giannoni–Schmit conjecture, in the unitary-class form studied here, concerns the local statistics of desymmetrized quantum chaotic spectra and their comparison with the Sine_2 process. This programme studies that question for fixed arithmetic quantum maps, where exact finite-dimensional computations are available but Heisenberg-scale control is not. The terminal open obligation is the signed Heisenberg-scale desymmetrized joint-trace estimate for the fixed map. Everything else in this volume is either an exact reduction toward it, a conditional bridge, or a result in a separate regime.

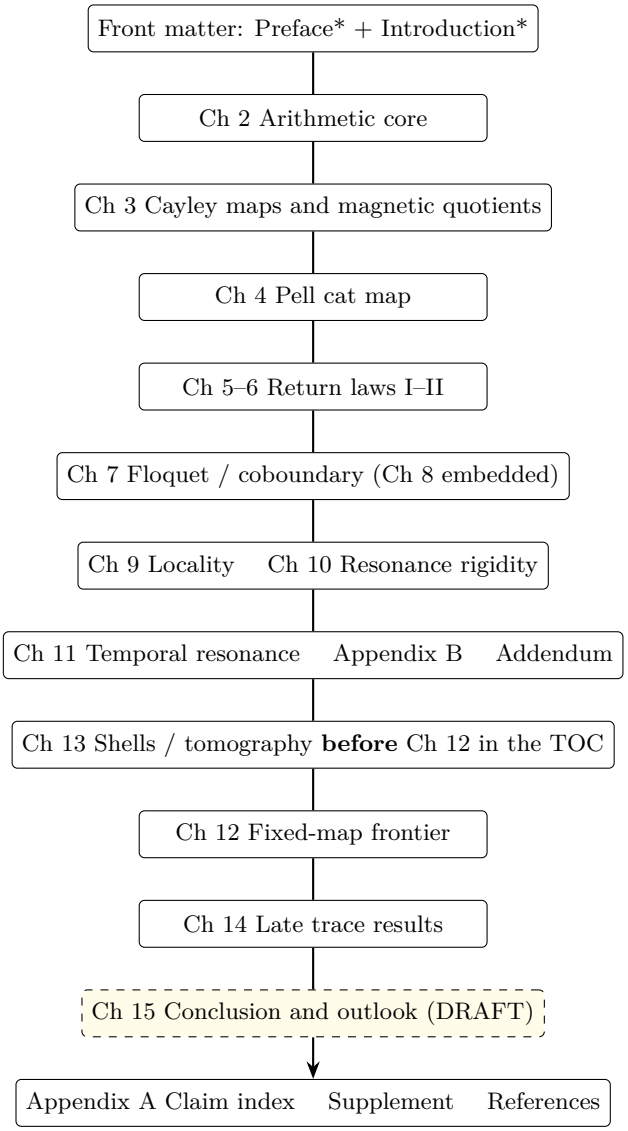


Figure 1: Reading-order schematic (structural, not data). The order follows the root file's input order; the Ch13-before-Ch12 order in the contents is structural (shell files precede fixed-map files in input order) and is kept by the no-reorder rule. Front matter and Chapter 15 are unnumbered/DRAFT additions.

1 Scope, normalizations, and the dependency structure

This is one consolidated research manuscript, not a concatenation of claims of equal evidentiary strength. Historical versions remain archived. Repeated proofs of the same mechanism have been

combined or cross-referenced; distinct asymptotic regimes have not been identified. In particular, a global Gaussian law of an averaged observable and a sine-kernel law for local propagator levels are different assertions. The former supplies neither the latter nor Heisenberg-scale joint trace control.

1.1 The deterministic microscopic target

In this manuscript the positive unitary-class benchmark is an explicitly prescribed quantum torus family, rather than the original GOE-class Sinai-billiard calculation [1]. The maintained frozen propagator is

$$S = \begin{pmatrix} 3 & 4 \\ 2 & 3 \end{pmatrix}, \qquad U_N = \exp(i\tau N A_N) M_N, \qquad \tau = 10^{-80}, \quad N \geq 3 \text{ odd}, \tag{1.1}$$

where M_N is the specified torus quantization and A_N is the specified Weyl-quantized kick. The coupling τ is *independent* of N; letting it tend to zero, averaging independent diagonal phases, or adding stochastic noise studies a different family.

A necessary CUE pair-reference target in the maintained normalization is

$$L_N(s) = N^{-2} \sum_{m \geq 1} e^{-2sm/N} |\operatorname{Tr} U_N^m|^2 \longrightarrow \frac{1 - e^{-2s}}{4s^2}, \qquad s > 0. \tag{1.2}$$

It has *not* been proved for this deterministic map. There is also an exact way to identify the missing *signed* contribution. Let $P_N = (|U_{N,ab}|^2)_{ab}$ and let D_m be the distinct-rotation cyclic-path sum defined in the position-mixing theorem. Let R_m^{rev} be its precisely defined reverse-cyclic off-orbit family, and set

$$R_m^{\mathrm{other}} := |\operatorname{Tr} U_N^m|^2 - D_m - R_m^{\mathrm{rev}}.$$

The established cyclic-ramp and reverse-path results give, as a *necessary* condition for the pair CUE limit, for each $w \in C_c^\infty((0,\infty))$,

$$\frac{1}{N^2} \sum_{m \geq 1} w(m/N) R_m^{\mathrm{other}} \longrightarrow \int_0^\infty (\min(x,1) - x) w(x)\, dx. \tag{1.3}$$

For $x > 1$ the correction is negative. Classical intensity mixing cannot supply this cancellation by itself; it is not legitimate to replace the complete signed remainder by a sum of termwise absolute estimates. Even proving the pair identity would not suffice for convergence of every fixed factorial order: the missing assertion is the fully *signed, symmetry-resolved Heisenberg-scale joint-trace comparison*, with the modulus taken outside each complete Fourier sum, together with the necessary local-process tightness conditions. For the ordinary unitary bulk, characterization through complete loop-equation hierarchies is another possible conditional route [2]; the required model-specific approximate equations and source bounds have not been established here. No conclusion about the unchanged U_N is imported from that external characterization.

1.2 The three spectral objects

We distinguish a quantum propagator U_N, a compressed observable $P_j B_N P_j|_{\operatorname{ran} P_j}$, and the matrix elements of B_N in an eigenbasis. Universality for one of these is not universality for another. An empirical law of U_N^K folds its phases K times; it is not the original nearest-neighbour law. A two-scale measure of one eigenvalue is not a two-eigenvalue correlation. These distinctions are used throughout.

In the finite-field sections, p is an odd prime, H is a multiplicative subgroup of odd prime order K. In the Pell sections, N is the Hilbert-space parameter and can be composite, while K

is an arithmetic return time with $N \asymp (3+2\sqrt{2})^{K/2}$. No unexplained identification of p and N is made. The notation D denotes the dimension of the active parity sector except where an operator is explicitly introduced under another local convention. Normalized Hilbert–Schmidt norms are $\|X\|_{2,D}^2 = D^{-1}\operatorname{Tr} X^*X$; operator norms carry the subscript op. Replacing one by the other can change a valid statement into a false one.

1.3 Weyl, classical, and perturbation conventions

For the fixed Fourier-covector matrix $T = \left(\begin{smallmatrix} 3 & 2 \\ 4 & 3 \end{smallmatrix}\right)$, the classical pullback matrix is $S = T^{\mathsf{T}}$. Define

$$W_N(a,b)\psi(x) = e_N(a(x-b/2))\psi(x-b), \qquad Q_N(a,b) = (-1)^{ab}W_N(a,b), \qquad \hbar_N = (2\pi N)^{-1},$$

where $e_N(t) = e^{2\pi i t/N}$. The half in W is modular; a,b in Q are integer Fourier frequencies before reduction. Thus, with $\omega((a,b),(a',b')) = ab' - ba'$, $Q_N(v)Q_N(w) = e^{\pi i\omega(v,w)/N}Q_N(v+w)$. For general hyperbolic matrices an explicitly stated invariant characteristic $Q_N^*(a,b) = (-1)^{ab+a+b}W_N(a,b)$ is used instead. These characteristics are not silently interchanged.

A quantum kick $e^{i\varepsilon \operatorname{Op}_N(f)}$ has classical action strength $\kappa = \hbar_N\varepsilon$. The fixed-strength convention in the new return-word section is $e^{i\kappa \operatorname{Op}_N(f)/\hbar_N}$. The older locality section writes $e^{i\kappa_0 N \operatorname{Op}_N(f)}$; there $\kappa = \kappa_0/(2\pi)$. Its formulas retain the explicitly stated κ_0 convention. This factor does not disappear on consolidation. We use $\{f,h\} = f_q h_p - f_p h_q$ and $X_f = (f_p, -f_q)$; sources using $(-f_p, f_q)$ have their flow-time sign stated alongside the map.

1.4 What is proved, imported, checked, or unresolved

The internal dependency chains are as follows. Finite Weyl covariance gives fixed-point trace identities and the exact cyclic multiplicities. Quadratic norm lifting, bounded-cluster counting and summable Pell phases give the observable moment laws. A finite Duhamel estimate then transfers selected laws to recurrence times of the actual propagator. The much stronger fixed-classical-strength results use support locality or noncommuting return words; they do not follow from the moment laws.

For the second resonance, the orbit expression of Green–Kubo variance gives its exact zero criterion. A local invariant differential operator gives the Hessian obstruction. The finite-box theorem additionally depends on an exact proof object, reconstructed and replayed by two implementations; it is not inferred from random sampling. The return-word example proves that a nonzero bracket can still lie in that zero set on the smooth infinite-frequency class. No implication from analytic nonzero bracket to positive variance is inserted.

The finite-field reductions form a parallel chain: exact pair keys, Cayley energy, fixed-degree row polynomials, scalar supports, and conditional arithmetic estimates. They are not a substitute for spectral trace estimates. The subresultant, cyclotomic-unit and subgroup-curve inputs used in those sections retain their named external hypotheses. Their sources are recorded in the bibliography; entries that are not cited at a point of use are collected there under "Further reading". Their historical derivations are preserved in this consolidated record; the present audit has not supplied new proofs of all external results. Exact-source checks and unretrieved leaves are listed separately in the source ledger, so no blanket referee verdict is attached to the whole historical corpus.

The scalar Fejer–Riesz theorem is used only after strict positivity of the proposed Laurent polynomial has been proved. The classical orbit/coboundary criterion is consistent with Cohen–Conze [17]; it is not imported as a quantum central limit theorem. Residual perturbations and their second-order commutator have direct precedent in Prosen–Znidaric [18]. Nonlinear cat-map pseudo-symmetries also predate this project [19]. The finite-return identity is group-valued coboundary algebra; no priority claim for that algebra is made. All new conclusions below are derived directly rather than assigned to a source merely from its title or abstract.

1.5 A precise universality obligation

For a D-dimensional spectrum $e^{2\pi i x_a}$, let U_D be a $D \times D$ unitary matrix with eigenvalues $e^{2\pi i x_a}$, and let $\phi \in \mathcal{S}(\mathbf{R})$ with Fourier transform $\widehat{\phi}(\xi) = \int_{\mathbf{R}} \phi(x) e^{-2\pi i x\xi}\, dx$. The global-density-unfolded factorial pair statistic is

$$R_D(\phi) = D^{-1} \sum_{a \neq b} \sum_{m \in \mathbb{Z}} \phi(D(x_a - x_b + m)).$$

Poisson summation yields

$$R_D(\phi) = D^{-2} \sum_{k \in \mathbb{Z}} \widehat{\phi}(k/D)(|\operatorname{Tr} U_D^k|^2 - D).$$

Thus a positive local-universality result requires the appropriate weighted joint trace comparison on the Heisenberg scale, with symmetry reduction and unfolding fixed beforehand. Positivity can sometimes exclude a proposed limit using only a few shorter-time traces; it cannot prove the full comparison. The two principal unresolved obligations remain a sharp arithmetic closure in the split finite-field branch and positive spectral universality for a natural fixed map not retaining the arithmetic model by smooth conjugacy. Neither is declared solved by the present consolidation.

1.6 Reading map for the continuing programme

The order of chapters records related research developments, not a single chain of implications to microscopic universality. For orientation the mathematical layers should be read with the following distinctions:

- **Finite-field and Cayley arithmetic:** exact combinatorial reductions and conditional or range-restricted energy bounds; the sharp centered pointwise arithmetic target remains open.
- **Return and Pell/Floquet theory:** asymptotic laws on explicitly declared arithmetic subsequences, regularity scales and observables; these cannot be silently reinterpreted as microscopic level statistics.
- **Resonance and cohomology:** sharp coercivity, rigidity and counterexamples in distinct smooth, analytic, linear-transport and nonlinear regimes; the all-members analytic extension remains unresolved.
- **Frozen-map microscopic programme:** symmetry, exact trace identities, selected interference sectors and conditional local-law criteria; the signed Heisenberg-scale comparison at every fixed factorial order, including its pair-scale plateau correction, is still missing.
- **Separate information-theory and noisy-law branches:** theorems with their own dynamics and source/measurement assumptions; these are not results about the unchanged deterministic fixed-strength spectrum.

The authoritative registered status and dependencies are in `record/claims.json`; the dated claim ledger explains changes, and the mathematical source files supply the proofs. A section's historical title, independent finite replay or successful typesetting run does not revise those statuses. This long-form manuscript is a living research exposition; focused possible articles have separate claim manifests and publication-review requirements.

Notation

This registry records the symbols whose meaning is not uniform across the compendium. A symbol listed here with a section scope is local to that section; the canonical (unqualified) form is the fixed-map propagator defined in the front matter. Renames applied in Pass 1 are mechanical token replacements, listed at the end of this registry; no statement was reworded by a rename.

Symbols not listed here are used with their standard meaning or are defined where they first appear.

Symbol	Meaning	Scope / location
U_N	Fixed kicked propagator $U_N = \exp(i\tau N A_N) M_N$, $\tau = 10^{-80}$ fixed	Canonical; front matter, fixed-map sections
$U_{N,\mathrm{loc}}$	A cat quantization with the exact covariance $UQ_N(v)U^* = Q_N(Tv)$	Locality section (Ch. 9), renamed in Pass 1
$U_{N,\mathrm{rig}}$	A cat quantization taken with the exact covariance, phase immaterial	Resonance-rigidity section (Ch. 10), renamed in Pass 1
$U_{N,\mathrm{Fl}}$	The Floquet operator $U_N = VU_0V^*$ of the finite Floquet construction	Floquet section (Ch. 7), renamed in Pass 1
$U_{N,\mathrm{test}}$	A generic test unitary (any unitary quantization satisfying the stated Egorov estimate)	Spectral-resolution section (Ch. 11), renamed in Pass 1
$U_N(T)$	The T-parametrized family $U_N(T) = VS_h^{-1}V^*$	Pell-blocks section; the argument distinguishes it from U_N
$\mathcal{M}_r$	$\sum_v \lvert\widehat{a}(v)\rvert(1+\lvert v\rvert_1)^r$	Locality section
M_r	$\sum_v \lvert\widehat{a}(v)\rvert\lvert v\rvert_2^r$ (Euclidean frequency weight)	Return-words section
M_j	The same object as M_r, with index letter j	Analytic section (regular-gradient rigidity)
M_r^{mom}, M_2^{mom}, M_4^{mom}	Moments $\mathbb{E}\lvert Z\rvert^r$ of $Z = s_N(w)$	Fixed-map frontier, early moment block, renamed in Pass 1
M_s	Occupancy moments $N^{-1}\sum_h n_h^s$	Fixed-map frontier, resonant-cubic section (defined in place)
M_3, M_4	Occupancy moments M_s at $s = 3, 4$	Fixed-map frontier, resonant-cubic and folded-square sections (verified Pass 2: same n_h occupancies and E_N as the M_s definition)
Φ^t	Flow of X_g (locality convention)	Locality section
Φ_g^t	Flow of X_g (same convention, explicit generator)	Locality section
Ψ_h^t	Flow of $X_h = (h_p, -h_q)$; inverse of the locality flow	Return-words section, renamed from Φ_h^t in Pass 1
κ_0	Rescaled time; locality writes $e^{i\kappa_0 N \operatorname{Op}_N(f)}$, return words write $\kappa = \kappa_0/(2\pi)$	Locality section
κ	Classical time itself, through $e = \kappa/\hbar_N$	Return-words section
κ_0^{thr}	The threshold constant $\frac{4}{3}(\sqrt{101}\sin(\pi/7) - 1)/100$	Fixed-map frontier, renamed in Pass 1
κ	A local exponent or contraction parameter (e.g. $m = \lceil \ell^{1+\kappa} \rceil$; $\kappa = 1 - \rho$)	Section-local; defined where used
κ_4	Fourth cumulant	Standard notation, wherever used

Pass 1 renames (mechanical, token-for-token). Fixed-map frontier, moment block (lines 378–414 only): $M_r, M_2, M_4 \to M_r^{\mathrm{mom}}, M_2^{\mathrm{mom}}, M_4^{\mathrm{mom}}$ (11 replacements). Fixed-map frontier, threshold block (lines 1679–1681): $\kappa_0 \to \kappa_0^{\mathrm{thr}}$ (3 replacements). Return words: $\Phi_h^t \to \Psi_h^t$ (5 replacements). Locality: $U_N \to U_{N,\mathrm{loc}}$ (15 replacements). Resonance rigidity: $U_N \to U_{N,\mathrm{rig}}$ (9 replacements). Floquet: $U_N \to U_{N,\mathrm{Fl}}$ (3 replacements), plus the coboundary cross-reference to the Floquet operator (1 replacement). Spectral resolution: $U_N \to U_{N,\mathrm{test}}$ (10 replacements) and $\widetilde{U}_N \to \widetilde{U_{N,\mathrm{test}}}$ (3 replacements). The tags were placed in the subscript ($U_{N,\mathrm{loc}}$ etc.) so that

existing superscripts ($U_N^*, U_N^j, U_N^{-j}, U_N^K$) keep working; a superscript tag was tried first and broke the build with double superscripts.

Open items and Pass 2 resolutions. (1) RESOLVED in Pass 2: Fixed-map frontier, late M_3, M_4 (lines 1917, 1944, 1964 in the resonant-cubic section itself, and line 2007 in the folded-square section, which reuses TRACE-011's E_N and counterterm budget and contrasts the M_4 bound with the new G_d bound) are the occupancy moments M_s defined at line 1910; see the M_3, M_4 row above. No token rename was applied. (2) κ as a section-local parameter is documented above but not unified; unification would require per-section renaming with proof reading, which is Pass 2 work. (2) DECIDED in Pass 2 (no rename): the microscopic-paths $\kappa > 0$ (free exponent parameter, e.g. $m = \lceil \ell^{1+\kappa} \rceil$) and the unmasked $\kappa = 1 - \rho > 0$ are section-local parameters with distinct definitions; renaming either would churn many formulas for no mathematical benefit, and the registry entries above already disambiguate them. Both stay as written. (3) The analytic M_j and return-words M_r are the same object under different index letters; no rename was applied, since the registry entry already disambiguates them.

Chapter 2 — Arithmetic core

2 Exact pair-key geometry

Throughout the following finite-field arithmetic sections, p is an odd prime, $H \leq \mathbb{F}_p^\times$ has odd prime order K, and $e_p(t) = e^{2\pi i t/p}$. Fixed-degree row presentations impose $K \geq 7$ where stated. These hypotheses are distinct from the composite Pell moduli in the later quantum sections.

For two admissible position pairs $(x_i, y_i) \in (\mathbb{F}_p \setminus \{0,1\})^2$, write their keys as (u_i, v_i, d_i). Let $B(t)$ count the ordered quadruples satisfying $u_1 u_2 \in H$, $v_1 v_2 \in H$, and $d_1 + d_2 = t$; let $O(t)$ count those for which both position pairs are off diagonal. Write V_4 for the set of all such quadruples without fixing t. These definitions include multiplicities in the collapsed diagonal fibre.

For $x, y \in \mathbf{F}_p \setminus \{0,1\}$ define

$$u = \frac{x}{y}, \qquad v = \frac{x-1}{y-1}, \qquad d = x - y. \tag{2.1}$$

Proposition 2.1 (Pair-key bijection)**.** *If $x \neq y$, then $u, v \in \mathbf{F}_p^\times \setminus \{1\}$, $u \neq v$, and*

$$y = \frac{1-v}{u-v}, \qquad x = \frac{u(1-v)}{u-v}, \qquad d = \frac{(u-1)(v-1)}{v-u}. \tag{2.2}$$

Conversely every such (u, v) gives a unique admissible off-diagonal pair. The diagonal collapses to $(u, v, d) = (1, 1, 0)$ with multiplicity $p - 2$.

Proof. The equations $x = uy$ and $x - 1 = v(y - 1)$ give $(u - v)y = 1 - v$, hence (2.2). The diagonal statement is immediate. Conversely, given $u, v \in \mathbf{F}_p^\times \setminus \{1\}$ with $u \neq v$, define x, y by (2.2). Then $y \neq 0$ because $v \neq 1$, and $y \neq 1$ because $u \neq 1$; $x \neq 0$ because $u \neq 0$ and $v \neq 1$; and $x \neq 1$, since $x = 1$ would give $u(1 - v) = u - v$, hence $uv = v$ and $u = 1$. Also $x = uy$ with $u \neq 1$ and $y \neq 0$ gives $x \neq y$, and substituting into (2.1) returns (u, v). □

Under the subgroup constraints $u, v \in H$, there are exactly $(K - 1)(K - 2)$ admissible off-diagonal ordered pairs. If $O(t)$ denotes the off/off contribution and

$$C_H(t) = \#\left\{(u, v) \in (H \setminus \{1\})^2 : u \neq v,\ \frac{(u-1)(v-1)}{v-u} = t\right\},$$

then exact bookkeeping gives

$$B(t) = O(t) + 2(p-2)C_H(t) + (p-2)^2 \mathbf{1}_{t=0}. \tag{2.3}$$

The factor two is essential: diagonal/off and off/diagonal are distinct ordered classes. The total count is

$$|V_4| = (K^2+1)p^2 - (6K^2+3K)p + 13K^2 - 3K. \tag{2.4}$$

If $\Sigma_4(q) = \sum_t B(t)e_p(qt)$, $V = \sum_t B(t)$ and $Z = B(0)$, finite Fourier orthogonality gives

$$\mathbb{E}_{q\neq 0}\Sigma_4(q) = \frac{pZ - V}{p-1}, \qquad \mathrm{Var}_{q\neq 0}\,\Sigma_4(q) = p\,\mathrm{Var}_{t\neq 0}\,B(t). \tag{2.5}$$

Here, and throughout, Var of a complex-valued quantity denotes $\mathbb{E}|\,\cdot\, - \mathbb{E}(\cdot)|^2$ over the indicated finite index set, with $\mathbb{E}$ the uniform average; for real-valued quantities this is the ordinary variance. No probabilistic independence assumption enters this reduction.

3 Cayley energy and a uniform bound

Put

$$c(h) = \frac{h+1}{h-1}, \qquad A = c(H \setminus \{1\}).$$

For $u \neq v$,

$$\frac{(u-1)(v-1)}{v-u} = \frac{2}{c(u)-c(v)}, \qquad c(h^{-1}) = -c(h). \tag{3.1}$$

Thus $A = -A$ and the nonzero multiplicities of C_H are additive-difference multiplicities of A up to the permutation $x \mapsto 2/x$. Hence

$$\sum_t C_H(t)^2 = E_+(A) - |A|^2, \qquad E_+(A) = \sum_x r_{A-A}(x)^2. \tag{3.2}$$

Identity and central inversion force the nonzero baseline

$$(K-1)(2K-5). \tag{3.3}$$

For the unnormalized Fourier transform,

$$E_+(A) = \frac{(K-1)^4}{p} + \frac{1}{p}\sum_{q\neq 0} |\widehat{1_A}(q)|^4. \tag{3.4}$$

The centered identity (3.4) separates the unavoidable zero-mode term $(K-1)^4/p$. In the regime where that term is $O(K^2)$ – in particular in the intended range $K = o(p^{1/4})$ – the forced baseline (3.3) sets the quadratic scale, and the sharp pointwise target (open, not a theorem) is

$$E_+(A) = \frac{(K-1)^4}{p} + O(K^{2+o(1)}). \tag{3.5}$$

For $r_A(t) = \#\{(a,b) \in A^2 : a+b = t\}$, clearing Cayley denominators gives

$$(2-t)xy + t(x+y) - (2+t) = 0,$$

an absolutely irreducible curve in odd characteristic: when $t \neq 2$ the bilinear coefficient determinant is $-4 \neq 0$, and when $t = 2$ it is a nonconstant linear polynomial. Its bidegree is $(1,1)$ in either case. For $t \neq 0, -2$ the constant term is nonzero and $\deg P(x,0) = 1$, exactly the hypotheses needed for the subgroup-curve input.

Theorem 3.1 (Direct pointwise Cayley bound)**.** *If*

$$100 < K < \frac{1}{3}p^{3/4},$$

then

$$E_+(A) < 34K^{8/3}. \tag{3.6}$$

Proof. The difference energy $E_+(A)$ of (3.2) equals the sum-representation energy: $\sum_x r_{A-A}(x)^2$ and $\sum_t r_A(t)^2$ both count $\{(a,b,c,d) \in A^4 : a+b=c+d\}$, after relabelling the second and fourth variables. Theorem 2 of Vyugin–Makarychev [9, 10], in the source form for bidegree $(m,n)=(1,1)$, applies to an absolutely irreducible P with $P(0,0)\neq 0$ and $\deg P(x,0)\geq 1$ under the size condition $100(mn)^{3/2} < K < \frac{1}{3}p^{3/4}$ – for $m=n=1$ exactly the hypothesis above – and gives $r_A(t) \leq 16mn^2(m+n)K^{2/3} = 32K^{2/3}$. On the displayed polynomial, $P_t(0,0) = -(2+t)$ and $P_t(x,0) = tx-(2+t)$, so the two degree hypotheses fail only at $t=-2$ and at $t=0$ respectively. The fibre $t=2$ is not exceptional: $P_2(x,y) = 2(x+y-2)$ is still absolutely irreducible of bidegree $(1,1)$, with $P_2(0,0)=-4$ and $\deg P_2(x,0)=1$, so the same bound $r_A(2)\leq 32K^{2/3}$ holds there. The exceptional fibres satisfy $r_A(0)=K-1$ and $r_A(-2)\leq K-1$. Since $\sum_t r_A(t) = (K-1)^2$,

$$E_+(A) \leq 2(K-1)^2 + 32K^{2/3} \sum_{t\neq 0,-2} r_A(t) < 34K^{8/3}.$$

□

4 Prime-order rigidity, square resultants and discriminants

Let $\zeta = e^{2\pi i/K}$. Then

$$c(\zeta^a) = -i\cot\frac{\pi a}{K}.$$

Chowla's prime-modulus cotangent independence [4, 5] identifies the characteristic-zero Cayley set with a rational cross-polytope.

Theorem 4.1 (Prime-order additive rigidity)**.** *Every characteristic-zero equal-difference collision among $\{c(\zeta^a) : a \in \mathbf{F}_K^*\}$ is identity or central inversion-swap.*

Proof. Write $b_a = \cot(\pi a/K)$ for $1\leq a\leq m=(K-1)/2$. By Chowla's theorem [4, 5] the numbers b_a are linearly independent over $\mathbf{Q}$. Since $c(\zeta^a) = -i\cot(\pi a/K)$ and $\cot(\pi(K-a)/K) = -\cot(\pi a/K)$, every value $c(\zeta^a)$ is $-i$ times a signed vertex $\pm b_a$; under $b_a \leftrightarrow e_a$ the Cayley values are the vertices of the rational cross-polytope $\{\pm e_a\}$. An equal-difference collision $c(\zeta^i)-c(\zeta^j) = c(\zeta^r)-c(\zeta^s)$, $i\neq j$, $r\neq s$, therefore gives an identity $x-y=u-v$ among signed vertices with $x\neq y$ and $u\neq v$, i.e. $x+v=u+y$. If $x+v=0$, then $v=-x$ and $u=-y$: the central inversion-swap $(r,s)=(-j,-i)$. Otherwise $x+v\neq 0$ is a sum of two signed basis vectors, and such a sum determines its unordered pair of summands uniquely – including the diagonal case $v=x$, where the sum is $\pm 2e_a$ and both summands are forced. Hence $\{x,v\}=\{u,y\}$; the alternative $u=y$, $v=x$ would give $x-y=y-x$, forcing $x=y$, a contradiction. Thus $u=x$ and $v=y$: the identity collision. □

For every admissible cleared collision polynomial $P(X)$, direct algebra gives $P(X) = (X-1)^3Q(X)$ and a reciprocal relation for Q. Passing to the maximal real subfield yields:

Theorem 4.2 (Square-resultant theorem)**.**

$$\mathrm{Res}(\Phi_K, P) = K^3M^2, \qquad M\in\mathbf{Z}. \tag{4.1}$$

For a non-forced characteristic-zero collision, $M\neq 0$.

Proof. Here the cleared equal-difference polynomial can be defined explicitly. For $1\leq a_i<K$ and $(\epsilon_1,\epsilon_2,\epsilon_3,\epsilon_4)=(1,-1,-1,1)$, put

$$P(X) = \sum_{i=1}^{4}\epsilon_i(X^{a_i}+1)\prod_{j\neq i}(X^{a_j}-1), \qquad s=\sum_i a_i.$$

Every term contains three factors vanishing at one, so $P = (X-1)^3 Q$ with $Q \in \mathbb{Z}[X]$. Direct inversion gives $P(X^{-1}) = -X^{-s}P(X)$ and hence $Q(X^{-1}) = X^{3-s}Q(X)$. Choose an integer k with $2k \equiv s-3 \pmod{K}$. Then $\xi = \zeta^{-k}Q(\zeta)$ is real and is an algebraic integer. The full cyclotomic norm of Q is the square of the real-subfield norm of ξ, because $\mathrm{Norm}(\zeta) = 1$. Also $\mathrm{Norm}(\zeta - 1) = K$. Therefore

$$\mathrm{Res}(\Phi_K, P) = K^3 \,\mathrm{Norm}_{\mathbb{Q}(\zeta+\zeta^{-1})/\mathbb{Q}}(\xi)^2.$$

The inner norm is an integer. In the non-forced case the additive rigidity input implies $P(\zeta) \neq 0$, so this integer is nonzero. This argument uses reciprocity as a Laurent identity and does not assume that a leading coefficient survives a finite-field specialization. □

The Fibonacci slice

$$\mathrm{Res}(\Phi_K(X), X^2 + 3X + 1) = F_K^2$$

shows that individual collision resultants can be exponentially large. At $K = 131$, $p = F_{131}$, a Lucas primality certificate and independent Python/C++ arbitrary-precision computations give exact nonzero Cayley excess 32 (finite certificate: `computations/verification/cayley_resultant_fibonacci.py` and `computations/cpp/cayley_fibonacci_verifier.cpp`); this is a finite certificate, not an asymptotic theorem. The primitive-divisor input for the associated rational model is from [8].

Let S_K be the primitive squarefree nonzero-difference polynomial obtained from the resultant of the characteristic-zero Cayley polynomial with its translate. For a split prime $p \equiv 1 \pmod{K}$, with H and $A = c(H \setminus \{1\})$ as in the preceding sections, define the *Cayley excess*

$$\Delta_K(p) = \sum_{d \neq 0} r_{A-A}(d)^2 - (K-1)(2K-5) = \sum_t C_H(t)^2 - (K-1)(2K-5) \in \mathbf{Z}_{\geq 0}. \tag{4.2}$$

The two expressions agree by (3.2) and $r_{A-A}(0) = |A| = K-1$. Nonnegativity holds because the identity and central inversion-swap collisions are distinct ordered collision quadruples already contributing exactly $(K-1)(2K-5)$ to the first sum: $\Delta_K(p)$ counts the remaining, non-forced, ordered collision quadruples. Then for split primes

$$\Delta_K(p) > 0 \iff p \mid \mathrm{Disc}(S_K), \qquad \Delta_K(p) \leq 4v_p(\mathrm{Disc}(S_K)). \tag{4.3}$$

The exact global form of $\mathrm{Disc}(S_K)$ comes from the ratio-orbit factorization. Write $n = K-1$ and $m = (K-1)/2$. Then $S_K = \prod_{[r]} F_r$, the product over the m ratio classes $[r]$, $r \sim r^{-1}$, where F_r is the degree-n Galois-orbit factor of the differences $c(\zeta^a) - c(\zeta^{ra})$. For one orbit, the index formula in the maximal real subfield, whose discriminant is K^{m-1}, gives $|\mathrm{Disc}(F_r)| = 2^{n^2} K^{K-2} I_r^4$ for an integer $I_r > 0$. For two distinct classes, $|\mathrm{Res}(F_r, F_s)| = 2^{n^2} K^{K+1} M_{r,s}^2$; the square comes from writing the even half-degree factors as $H_r(Y) = 1 + KP_r(Y)$ and using $\deg(P_s - P_r) = m-1$. That degree drop is the one place where Dedekind sums enter: the coefficient of Y^{m-1} in P_r is $(K-1)(K-2)/12 - Ks(r,K)$, and by the prime-modulus equality criterion [11, Corollary 1.2] (pp. 2197–2202), $s(r,K) = s(s,K)$ if and only if $r \equiv s$ or $rs \equiv 1 \pmod{K}$, so distinct ratio classes have distinct coefficients (for $K \geq 5$; at $K = 3$ there is a single ratio class and the formula is a direct check). Since $\mathrm{Disc}(\prod_r F_r) = \prod_r \mathrm{Disc}(F_r) \prod_{r<s} \mathrm{Res}(F_r, F_s)^2$, collecting the powers of 2 and K – the exponents are $mn^2 + m(m-1)n^2 = (K-1)^4/4$ and $m(K-2) + m(m-1)(K+1) = (K-1)(K^2-7)/4$ – gives

$$|\mathrm{Disc}(S_K)| = 2^{(K-1)^4/4} K^{(K-1)(K^2-7)/4} J_K^4 \tag{4.4}$$

for the integer $J_K = \prod_{[r]} I_r \prod_{[r]<[s]} |M_{r,s}| > 0$. Exact resultant computations through $K = 47$ (the normalized prime-to-$2K$ mass $4 \log J_K^{(2K)}/K^4$ increases from 0.0837 at $K = 23$ to 0.1182 at $K = 47$) show that the residual logarithmic mass genuinely remains K^4-scale on the computed range, so (4.4) is structural, not a hidden proof of (3.5).

5 Fixed-product rows and exact split support

For one valid subgroup pair put $u = xy$, $s = x + y$. Then

$$c(x) + c(y) = \frac{2(u-1)}{u-s+1}. \tag{5.1}$$

At fixed $u \neq 1$, regard the parameter as the finite value $\sigma = c(x) + c(y)$ in (5.1); the case $u - s + 1 = 0$ is the excluded infinite parameter. For finite σ – necessarily $\sigma \neq 0$, since $\sigma = 0$ would force $u = 1$ – (5.1) gives $s = u + 1 - 2(u-1)/\sigma$, and $\{x, y\}$ is then the root pair of $T^2 - sT + u$; hence there is at most one unordered pair. Let $I_K(p)$ be the maximum number of common parameters of two distinct product rows. The exact cross-row decomposition gives

$$\Delta_K(p) \leq 4I_K(p)(K-1)(K-2). \tag{5.2}$$

Normalize a row pair to $(1, r)$ and set

$$\tau = \zeta + \zeta^{-1}, \qquad \alpha_r = U_{r-1}(\tau/2), \qquad \beta_r = -2U_{r-2}(\tau/2).$$

The row equation is the affine trace relation

$$Y = \alpha_r X + \beta_r. \tag{5.3}$$

Let $C_K(T)$ be the minimal polynomial of τ and set

$$H_K(T) = (T-2)C_K(T).$$

The two full row polynomials have one unique characteristic-zero common trace, the invalid row trace. Removing it produces fixed-degree source/target cofactors over $\mathbf{Z}[\tau]$. For every split embedding prime $\mathfrak{q}$,

$$\deg \gcd(P_{K,r}, Q_{K,r} \bmod \mathfrak{q}) = I_{1,r}(\mathfrak{q}). \tag{5.4}$$

Let the coefficient ideal of the first three fixed-degree subresultants contract to $\Lambda^{(3)}_{K,r}\mathbf{Z}$.

Theorem 5.1 (Exact real-trace order-three support)**.** *For every split rational prime $p \equiv 1 \pmod K$,*

$$p \mid \Lambda^{(3)}_{K,r}$$

if and only if some row pair in ratio class $[r]$ has at least three common Cayley parameters.

Proof. Let S_0, S_1, S_2 be the first three fixed-degree subresultants of the two cofactors, and let $J^{\mathrm{tr}}_{K,r} \subset \mathbf{Z}[\tau]$ be the ideal generated by all their coefficients, so $\Lambda^{(3)}_{K,r}\mathbf{Z} = J^{\mathrm{tr}}_{K,r} \cap \mathbf{Z}$. This ideal is nonzero: the cofactors have no common characteristic-zero trace (the unique common trace was divided out), so $S_0 \neq 0$.

We use the contraction criterion: for a nonzero ideal J of $\mathbf{Z}[\tau]$ with $J \cap \mathbf{Z} = \Lambda\mathbf{Z}$, a rational prime p divides Λ if and only if $J \subseteq \mathfrak{q}$ for some prime ideal $\mathfrak{q}$ of $\mathbf{Z}[\tau]$ above p. Indeed, $J \subseteq \mathfrak{q}$ gives $\Lambda \in \mathfrak{q} \cap \mathbf{Z} = p\mathbf{Z}$. Conversely, if $p \mid \Lambda$ and $J + p\mathbf{Z}[\tau]$ were the whole ring, localizing at p would put $1 \in J_{(p)}$; clearing denominators in the domain $\mathbf{Z}[\tau]$ then gives an element of $J \cap \mathbf{Z}$ prime to p, contradicting $p \mid \Lambda$. Hence $J + p\mathbf{Z}[\tau]$ is proper, lies in a maximal ideal $\mathfrak{q}$, and $\mathfrak{q} \cap \mathbf{Z} = p\mathbf{Z}$.

The valid cofactors remain fixed-degree and squarefree after split reduction – their leading coefficients are 1 and the unit α_r^{m+1}, and $p \neq K$ – so (5.4) is exact and there is no specialization degree drop. By the criterion, $p \mid \Lambda^{(3)}_{K,r}$ if and only if some $\mathfrak{q} \mid p$ contains every coefficient of S_0, S_1, S_2, i.e. if and only if $S_0 \equiv S_1 \equiv S_2 \equiv 0 \pmod{\mathfrak{q}}$. Brown–Traub's fixed-degree subresultant criterion [12] identifies this with $\deg \gcd(P_{K,r}, Q_{K,r} \bmod \mathfrak{q}) \geq 3$, and (5.4) identifies that degree with $I_{1,r}(\mathfrak{q})$. As $\mathfrak{q}$ ranges over the primes above the split prime p, the associated real embeddings cover all row pairs in ratio class $[r]$, which gives the rational-prime statement. □

After removing factors $2, K$ and multiplying over ratio classes, write the result as ${\Lambda'_K}^{(3)}$. Then

$$p \nmid {\Lambda'_K}^{(3)} \Longrightarrow I_K(p) \le 2, \quad \Delta_K(p) \le 8(K-1)(K-2), \quad E_+(A) \le 11(K-1)(K-2). \tag{5.5}$$

The older dehomogenized D'/G' invariants remain valid only as one-way obstructions because specialization can reduce leading degree at projective infinity.

6 Characteristic-zero affine-intercept rigidity

Write

$$t_a = \zeta^a + \zeta^{-a}, \qquad s_a = \zeta^a - \zeta^{-a}, \qquad \gamma_r(a,b) = t_b - \alpha_r t_a.$$

Let $[a]$ denote the trace class $a \sim -a$, and put $E_j = |1-\zeta^j|$, $1 \le j \le m = (K-1)/2$.

Lemma 6.1 (Prime-conductor multiplicative independence)**.** *The positive numbers $E_1, \dots, E_m$ are multiplicatively independent.*

Proof. Suppose $\prod_{j=1}^m E_j^{n_j} = 1$ with $n_j \in \mathbf{Z}$. Since $E_j^2 = (1-\zeta^j)(1-\zeta^{-j})$ and $(1-\zeta^{-j}) = -\zeta^{-j}(1-\zeta^j)$, squaring the relation gives $\prod_j (1-\zeta^j)^{2n_j} = \omega$ for a root of unity ω (a sign times a power of ζ). Taking norms from $\mathbf{Q}(\zeta)$: every $1-\zeta^j$ is a Galois conjugate of $1-\zeta$, so $N(1-\zeta^j) = N(1-\zeta) = \Phi_K(1) = K$, while $N(\omega) = 1$. Hence $K^{2\sum_j n_j} = 1$, forcing

$$\sum_{j=1}^m n_j = 0.$$

The relation therefore rewrites as one among the ratios $(1-\zeta^j)/(1-\zeta)$, $2 \le j \le m$ (the $j=1$ ratio is 1, and the vanishing exponent sum absorbs the powers of $1-\zeta$). By Bass's theorem for prime conductor [6] (Theorem 3), these ratios are multiplicatively independent modulo torsion: their classes form a free basis of the cyclotomic units modulo torsion. Hence $2n_j = 0$ for $j \ge 2$, and the vanishing sum gives $n_1 = 0$. □

Use the elementary identities

$$|s_n| = E_{[2n]}, \qquad |t_x - t_y| = E_{[x-y]} E_{[x+y]}. \tag{6.1}$$

Theorem 6.2 (Trace-intercept fibre rigidity)**.** *For every odd prime $K \ge 7$ and every normalized ratio $r \ne 1$, every fibre of $([a],[b]) \mapsto \gamma_r(a,b)$ has cardinality at most two.*

If $r = -1$, a double fibre is only a swap. If $r \ne \pm 1$, every double fibre is, up to signs and exchange,

$$([a],[a+1-r]), \qquad ([a+2],[a+1+r]). \tag{6.2}$$

Proof. Suppose two distinct pairs have the same intercept. For $r \ne -1$,

$$t_b - t_d = \alpha_r (t_a - t_c).$$

Taking absolute values and applying (6.1) yields

$$E_{[b-d]} E_{[b+d]} E_{[2]} = E_{[a-c]} E_{[a+c]} E_{[2r]}.$$

Lemma 6.1 makes the multisets of folded indices equal. Since $[2r] \ne [2]$, one source difference must be $[2]$; after changing signs and ordering take $c = a+2$. The remaining two indices force the target classes in (6.2). The equality itself follows from

$$t_{x+r} - t_{x-r} = s_r s_x, \qquad t_{a+2} - t_a = s_1 s_{a+1}.$$

A third point would form a triangle in the step-two quotient of the odd K-cycle by $a \sim -a$; that quotient is a path. For $r = -1$ the intercept is $t_a + t_b$. If $t_a + t_b = t_c + t_d$, subtracting gives a rational relation among the powers $1, \zeta, \dots, \zeta^{K-1}$ whose coefficient sum is $4 - 4 = 0$. The space of such relations is spanned by $(1, \dots, 1)$, the coefficient vector of Φ_K; the zero coefficient sum forces the scalar to vanish, so every coefficient vanishes separately, the multisets $\{\pm a, \pm b\}$ and $\{\pm c, \pm d\}$ in $\mathbf{F}_K$ agree, and $\{[a], [b]\} = \{[c], [d]\}$: only identity or swap. This covers in particular $K = 7$, where the relation has length $8 > \deg \Phi_K = 6$. □

The distinguished characteristic-zero row solution $([1], [r])$ is therefore isolated. This mechanism is related to cyclotomic-unit cross-ratio methods in projective equivalence of roots of unity [7], but the affine theorem above is proved directly in the prime-conductor setting.

7 Cyclotomic-unit matching and a scalar shift resultant

Put $u_n = s_n/s_1$. For valid trace classes define

$$W_r(b) = u_{(b+r)/2} u_{(b-r)/2}, \qquad V_r(a) = u_r u_{(a+1)/2} u_{(a-1)/2}. \tag{7.1}$$

A direct difference factorization gives

$$\gamma_r(a, b) - \beta_r = s_1^2 (W_r(b) - V_r(a)). \tag{7.2}$$

Since s_1 is a unit away from K, every split row intersection is exactly a congruence $W_r(b) \equiv V_r(a) \pmod{\mathfrak{q}}$ between explicit products of cyclotomic units.

Let

$$P_1(T) = \frac{H_K(T)}{T - t_1}, \qquad R_r(U) = \frac{H_K(U)}{U - t_r}$$

and define the scalar shift resultant

$$\mathcal{R}_{K,r}(Z) = \mathrm{Res}_T(P_1(T), R_r(\alpha_r T + \beta_r + Z)) = c_0 + c_1 Z + c_2 Z^2 + \cdots. \tag{7.3}$$

Theorem 7.1 (Scalar support identity)**.** *For every split embedding prime* $\mathfrak{q}$*,*

$$\mathrm{ord}_{Z=0}(\mathcal{R}_{K,r}(Z) \bmod \mathfrak{q}) = I_{1,r}(\mathfrak{q}). \tag{7.4}$$

Consequently

$$I_{1,r}(\mathfrak{q}) \geq 3 \iff c_0 \equiv c_1 \equiv c_2 \equiv 0 \pmod{\mathfrak{q}}.$$

Proof. Over the split residue field the two valid trace polynomials factor linearly, so the product formula gives

$$\mathcal{R}_{K,r}(Z) = \prod_{[a] \neq [1]} \prod_{[b] \neq [r]} (Z + \alpha_r t_a + \beta_r - t_b).$$

Each factor is linear in Z and vanishes at zero exactly for one row intersection. □

The ideal (c_0, c_1, c_2) contracts to an integer $\Omega^{(3)}_{K,r}$ with the same split rational-prime order-three support as $\Lambda^{(3)}_{K,r}$. Equality of the full integers is not asserted. Moreover, the nonzero characteristic-zero constant coefficient gives the per-ratio deterministic budget, where $I_r(p) := \max_{\mathfrak{q} \mid p} I_{1,r}(\mathfrak{q})$ over the embedding primes $\mathfrak{q}$ of the split prime p:

$$\sum_{\substack{p \equiv 1 \,(K) \\ I_r(p) \geq 3}} \log p = O(K^3). \tag{7.5}$$

Indeed, $c_0 \neq 0$ in characteristic zero: by Theorem 6.2 the distinguished intercept is isolated, so no factor in the product formula for $\mathcal{R}_{K,r}(0)$ in Theorem 7.1 vanishes. Every prime counted in (7.5) divides the nonzero integer $|N_{\mathbf{Q}(\tau)/\mathbf{Q}}(c_0)|$: for such a prime, some $\mathfrak{q} \mid p$ has $c_0 \equiv 0 \pmod{\mathfrak{q}}$ by Theorem 7.1, and $\mathfrak{q}$ has residue degree one. It remains to bound this norm. Under the j-th real embedding, $|\sigma_j(u_n)| = |\sin(2\pi jn/K)|/|\sin(2\pi j/K)| \leq K/(4d_j)$ with $d_j = \min(j, K/2 - j)$, using $|\sin \pi x| \geq 2\,\mathrm{dist}(x, \mathbf{Z})$. Since $\alpha_r = u_r$ and $\beta_r = -2u_{r-1}$, each of the m^2 factors of c_0 has j-th embedding at most $2 + K/d_j$ in absolute value, so

$$\log |N(c_0)| \leq m^2 \sum_{j=1}^{m} \log(2 + K/d_j) = O(K^3) :$$

the last sum is $O(K)$ by the elementary Stirling estimate applied to the integers and half-integers d_j. Summing $\log p$ over the distinct counted primes, each dividing $|N(c_0)|$, gives (7.5). Summing over $O(K)$ ratio classes recovers an $O(K^4)$ scale, so (7.5) is not a polynomial largest-prime theorem.

8 Inverse reflection and the pair-sum square

The ratio $r = -1$ has additional symmetry. Here

$$\alpha = -1, \qquad \beta = 2\tau, \qquad Q(T) = -P(2\tau - T).$$

Set

$$F(X) = P(\tau + X) = E(X^2) + XO(X^2). \tag{8.1}$$

The reflection reduction gives

$$I_{-1}(\mathfrak{q}) = 2 \deg \gcd(E, O \bmod \mathfrak{q}). \tag{8.2}$$

Thus inverse-row intersections are automatically even. A fixed-degree half-size subresultant construction defines the exact split support integer Ξ_K.

The scalar shift resultant admits a stronger universal factorization.

Theorem 8.1 (Inverse pair-sum factorization)**.** *Let $x_1, \ldots, x_m$ be the roots of the monic polynomial F. Define*

$$S(Z) = \prod_{i<j} (Z - x_i - x_j).$$

Then $S \in \mathbf{Z}[\tau][Z]$ and

$$\mathrm{Res}_X(F(X), F(Z - X)) = 2^m F(Z/2) S(Z)^2. \tag{8.3}$$

For every split embedding prime,

$$I_{-1}(\mathfrak{q}) = 2\,\mathrm{ord}_{Z=0}(S(Z) \bmod \mathfrak{q}). \tag{8.4}$$

Hence

$$I_{-1}(\mathfrak{q}) \geq 3 \iff S(0) \equiv S'(0) \equiv 0 \pmod{\mathfrak{q}}. \tag{8.5}$$

Proof. The root-product formula gives

$$\mathrm{Res}_X(F(X), F(Z - X)) = \prod_{i,j} (Z - x_i - x_j).$$

The diagonal ordered pairs give $2^m F(Z/2)$, and every unordered pair $i < j$ occurs twice. The coefficients of S are symmetric integral polynomials in the roots. At a split prime, $p \neq K$ is unramified and $H_K \bmod \mathfrak{q}$ is separable – its roots, the trace values t_a, are pairwise distinct modulo $\mathfrak{q}$ – so τ is a simple root modulo $\mathfrak{q}$, i.e. $F(0) = P(\tau) = H_K'(\tau) \not\equiv 0 \pmod{\mathfrak{q}}$ (and $\tau \not\equiv 2 \pmod{\mathfrak{q}}$). Since p is odd, the diagonal factor $2^m F(0)$ is a unit at $Z = 0$. Combining with the general shift-resultant identity gives (8.4); (8.5) follows because the left side is even. □

Let $S(Z) = s_0 + s_1 Z + \cdots$ and let the ideal (s_0, s_1) contract to $\Theta_K \mathbf{Z}$. Then Θ_K, Ξ_K, and the inverse factor of $\Lambda^{(3)}_{K,-1}$ have the same split rational-prime support; their valuations need not agree.

9 The pair-sum polynomial as a translated half-degree resultant

The square root S itself can be obtained without constructing the degree-m^2 shift resultant. Translate

$$F_Z(X) = F(X + Z/2) = E_Z(X^2) + XO_Z(X^2). \tag{9.1}$$

Theorem 9.1 (Translated parity-resultant identity)**.** *There exists a convention-dependent sign $\varepsilon_m \in \{\pm 1\}$ such that*

$$S(Z) = \varepsilon_m \operatorname{Res}_Y(E_Z(Y), O_Z(Y)). \tag{9.2}$$

Consequently

$$S(0) = \varepsilon_m \operatorname{Res}_Y(E, O), \tag{9.3}$$

$$S'(0) = \varepsilon_m \left.\frac{d}{dZ} \operatorname{Res}_Y(E_Z, O_Z)\right|_{Z=0}. \tag{9.4}$$

The first-order deformation is

$$\partial_Z E_Z|_0 = \frac{O + 2YO'}{2}, \qquad \partial_Z O_Z|_0 = E'. \tag{9.5}$$

Proof. The roots of F_Z are $y_i = x_i - Z/2$. Apply the constant pair-sum identity from Theorem 8.1 to the monic polynomial F_Z. Its unordered pair-sum product at zero is

$$\prod_{i<j}(-y_i - y_j) = \prod_{i<j}(Z - x_i - x_j) = S(Z).$$

It remains to identify this product with the parity resultant. For a monic polynomial G with parity decomposition $G(X) = \mathcal{E}(X^2) + X\mathcal{O}(X^2)$ and roots y_i, the constant pair-sum coefficient satisfies $S_G(0)^2 = \operatorname{Res}_Y(\mathcal{E}, \mathcal{O})^2$. Indeed, (8.3) at $Z = 0$ gives $\operatorname{Res}_X(G(X), G(-X)) = 2^m G(0) S_G(0)^2$, while the root-product formula and $G(-y_i) - G(y_i) = -2y_i\mathcal{O}(y_i^2)$ give $\operatorname{Res}_X(G(X), G(-X)) = \prod_i G(-y_i) = 2^m \mathcal{E}(0) \operatorname{Res}_Y(\mathcal{E}, \mathcal{O})^2$: here $\prod_i y_i = (-1)^m \mathcal{E}(0)$, and $\prod_i \mathcal{O}(y_i^2) = \operatorname{Res}_X(G, \mathcal{O}(X^2)) = \operatorname{Res}_Y(\mathcal{E}, \mathcal{O})^2$, evaluating at the roots $\pm\sqrt{y}$ of $\mathcal{O}(X^2)$ and using that exactly one of $\mathcal{E}, \mathcal{O}$ is monic, in either parity case. Since $G(0) = \mathcal{E}(0) \neq 0$ in characteristic zero, $S_G(0)^2 = \operatorname{Res}_Y(\mathcal{E}, \mathcal{O})^2$. Apply this to $G = F_Z$ over $\mathbf{Q}(\tau)(Z)$: its pair-sum product at zero is $S(Z)$ by the display above, and its parity components are E_Z, O_Z by (9.1). Hence $S(Z)^2 = \operatorname{Res}_Y(E_Z, O_Z)^2$ as polynomials in Z; both are nonzero in the domain $\mathbf{Z}[\tau][Z]$, so they differ by a constant sign $\varepsilon_m \in \{\pm 1\}$, fixed by the monicity of S once the resultant convention is fixed. Differentiating $F(X + Z/2)$ at $Z = 0$ gives (9.5). □

Thus inverse order-three support is the simultaneous divisibility problem for a half-degree resultant and its translation derivative. In exact computation, both scalars can be obtained in one Sylvester determinant over dual numbers $\varepsilon^2 = 0$, reducing the determinant dimension to $O(K)$ rather than materializing a degree-m^2 polynomial.

Exact Smith contractions give, after removing factors $2K$,

$$\Theta'_7 = 1, \quad \Theta'_{11} = 1, \quad \Theta'_{13} = 5, \quad \Theta'_{17} = 1, \quad \Theta'_{19} = 37, \quad \Theta'_{23} = 47^2 \cdot 137.$$

For comparison $\Xi'_{23} = 47 \cdot 137$: split prime support agrees, valuations differ. The reference SymPy implementation reduces the $K = 17$ computation from roughly 51 seconds for the full

shift resultant to roughly 0.6 seconds for the translated half-degree route; at $K = 23$ the latter remains on the order of seconds, after which Smith normal form becomes the principal finite algebra bottleneck. These timings are reproducibility data, not asymptotic complexity theorems.

The norm of the inverse constant shift coefficient also has the exact shape

$$|\mathrm{N}(R_-(0))| = 2^{m^2} K^m \, |\mathrm{N}(\mathrm{Res}(E, O))|^2 , \tag{9.6}$$

using $|\mathrm{N}(F(0))| = K^m$. Hence every prime away from $2K$ occurs with even valuation in this particular norm. This explains part of the observed square pattern but does not imply equality of the support integers.

10 Global split-prime budget and minimal-extension Frobenius

Simultaneous multiplication of collision exponent indices by $a \in \mathbf{F}_K^\times$ acts freely on non-forced collision quadruples. The number of normalized scaling orbits is

$$N_K = K^3 - 5K^2 + 6K + 1.$$

Combining the square-resultant theorem with $|\operatorname{Res}(\Phi_K, P)| \le 12^{K-1}$ gives:

Theorem 10.1 (Weighted split-prime budget)**.**

$$\sum_{p \equiv 1 \,(K)} \Delta_K(p) \log p \le N_K\big((K-1)\log 12 - 3\log K\big). \tag{10.1}$$

Proof. Index the elements of A by $\mathbf{F}_K^*$. An admissible collision quadruple is $q = (i, j, r, s)$ with $i \ne j$, $r \ne s$; simultaneous multiplication by $a \in \mathbf{F}_K^*$ acts freely, every orbit has a unique representative with $i = 1$, and the number of non-forced orbits is

$$N_K = \frac{(K-1)^2(K-2)^2 - (K-1)(2K-5)}{K-1} = (K-1)(K-2)^2 - (2K-5) = K^3 - 5K^2 + 6K + 1.$$

For a normalized representative let P_q be the cleared Cayley collision polynomial and $R_q = \operatorname{Res}(\Phi_K, P_q)$. By the square-resultant theorem $R_q = K^3 M_q^2$ with $M_q \ne 0$, and $|R_q| \le 12^{K-1}$. For a split prime p and a generator h of H, the equation $P_q(h^a) = 0$ is exactly the collision attached to $a \cdot q$; writing $s_q(p)$ for the number of such a, one has $\Delta_K(p) = \sum_q s_q(p)$. Since Φ_K is separable modulo p and splits completely, every root counted by $s_q(p)$ contributes a factor p to R_q, hence $s_q(p) \le v_p(R_q) = 2v_p(M_q)$, the last equality because $p \ne K$. Summing over split primes and using $\sum_p v_p(M_q) \log p \le \log |M_q|$ gives

$$\sum_p \Delta_K(p) \log p \le \sum_q 2 \log |M_q| = \sum_q \log(|R_q|/K^3) \le N_K\big((K-1)\log 12 - 3\log K\big).$$

□

For prime $K \asymp X^\theta$ and split prime $p \asymp X$, Bombieri–Vinogradov and PNT [13, 14] imply

$$\text{average } \frac{\Delta_K(p)}{K^2} \ll_\theta X^{3\theta - 1}$$

for fixed $\theta < 1/3$. Integrality of the excess strengthens this to a density-one exact baseline for fixed $\theta < 1/5$:

$$\Delta_K(p) = 0, \qquad E_+(A) = 3(K-1)(K-2)$$

for asymptotically almost every split pair. Indeed, the average is over the dyadic family $Q/2 < K \le Q$, $X/2 < p \le X$, $p \equiv 1 \pmod K$, whose cardinality has order $X/(\log X)^2$ by

Bombieri–Vinogradov and the prime number theorem. Since $\Delta_K(p)$ is a nonnegative integer and $K^2 \le Q^2 = X^{2\theta}$ on this family, Markov's inequality gives

$$\frac{\#\{(K,p) : \Delta_K(p) \ge 1\}}{|\mathcal{D}_\theta(X)|} \le \text{average}\, \Delta_K(p) \le Q^2 \cdot \text{average}\, \frac{\Delta_K(p)}{K^2} \ll_\theta X^{5\theta-1} \longrightarrow 0 \qquad (\theta < 1/5),$$

and on the complementary pairs $\Delta_K(p) = 0$, whence $E_+(A) = (K-1)^2 + (K-1)(2K-5) = 3(K-1)(K-2)$.

Now let $p \ne 2, K$ be arbitrary and put

$$f = \mathrm{ord}_K(p), \qquad q = p^f.$$

In $\mathbf{F}_q$ the order-K subgroup exists uniquely. Put $A_{K,p} = c(H \setminus \{1\}) \subset \mathbf{F}_q$ and

$$\Delta_K^{\mathrm{ext}}(p) = \sum_{d \ne 0} r_{A_{K,p}}(d)^2 - (K-1)(2K-5),$$

the number of ordered non-forced collision quadruples in the minimal extension. It is a nonnegative integer, and $\Delta_K^{\mathrm{ext}}(p) = 0$ gives $E_+(A_{K,p}) = 3(K-1)(K-2)$.

Theorem 10.2 (Frobenius amplification)**.** *Every non-forced collision orbit under the p-power Frobenius has size exactly f. Hence*

$$f \mid \Delta_K^{\mathrm{ext}}(p).$$

The weighted resultant budget and the discriminant classification extend to all such rational primes in the minimal extension.

Proof. Frobenius multiplies every exponent by p modulo K. If a positive power $d < f$ fixed a collision tuple, any nonzero coordinate would imply $p^d \equiv 1 \pmod K$, contradicting the definition of f. Hence every non-forced Frobenius orbit has size exactly f, proving $f \mid \Delta_K^{\mathrm{ext}}(p)$.

It remains to extend the budget and the classification. Retain the N_K normalized scaling orbits and their resultants $R_Q = \mathrm{Res}(\Phi_K, P_Q) = K^3 M_Q^2$ from the split budget. For $p \ne K$ choose a prime ideal $\mathfrak{P}$ of $\mathbf{Z}[\zeta_K]$ above p; its residue degree is f, and reduction sends ζ_K to a primitive Kth root $h \in \mathbf{F}_q$. With $s_Q(p) = \#\{a \in \mathbf{F}_K^* : P_Q(h^a) = 0\}$ one has $\Delta_K^{\mathrm{ext}}(p) = \sum_Q s_Q(p)$, and the product formula $R_Q = \prod_a P_Q(\zeta_K^a)$ puts $R_Q \in \mathfrak{P}^{s_Q(p)}$. Since $p \ne K$ is unramified, $v_{\mathfrak{P}}$ of a rational integer is the ordinary p-valuation, so $s_Q(p) \le v_p(R_Q) = 2v_p(M_Q)$. Summing over p exactly as in the split case gives $\sum_{p \ne 2,K} \Delta_K^{\mathrm{ext}}(p) \log p \le N_K\big((K-1)\log 12 - 3\log K\big)$. For the classification, apply the maintained cluster argument to the squarefree difference polynomial S_K at $\mathfrak{P}$: its roots are $\mathfrak{P}$-integral, distinct roots coalesce modulo $\mathfrak{P}$ exactly when collisions occur, and $v_{\mathfrak{P}}(\mathrm{Disc}\, S_K) = v_p(\mathrm{Disc}\, S_K)$; hence $\Delta_K^{\mathrm{ext}}(p) > 0$ if and only if $p \mid \mathrm{Disc}(S_K)$, with $\Delta_K^{\mathrm{ext}}(p) \le 4v_p(\mathrm{Disc}(S_K))$. □

PNT, Brun–Titchmarsh and the scarcity of small multiplicative orders then give a density-one exact baseline for unrestricted prime pairs $K \asymp X^\theta$, $p \asymp X$ for every fixed $\theta < 1/3$. This mechanism is absent on the split slice because there $f = 1$. In detail, split the pairs according as $f = \mathrm{ord}_K(p) \le K^\beta$ or $f > K^\beta$, where $\max(0, 4 - 1/\theta) < \beta < 1$; such a β exists exactly when $\theta < 1/3$. The residue classes modulo K of order at most K^β number at most $\sum_{d \mid K-1,\, d \le K^\beta} \varphi(d) \le K^\beta \tau(K-1) = K^{\beta+o(1)}$, and Brun–Titchmarsh bounds the primes in each such class in $(X/2, X]$ by $O_\theta(X/(K \log X))$; summing over prime $K \asymp Q$ makes the small-order pairs a fraction $O(Q^{\beta-1+o(1)})$ of all pairs. If $f > K^\beta$ and the pair is bad, the orbit theorem gives $\Delta_K^{\mathrm{ext}}(p) \ge f > K^\beta$, so the extended budget bounds the bad primes for each K by $O(K^{4-\beta}/\log X)$, a fraction $O(Q^{4-\beta}/X)$ of all pairs. Both fractions tend to zero.

11 Full Cayley–Jacobi trace and corrected off/off geometry

Let $S = H^\perp$ be the multiplicative-character annihilator and define the normalized twisted-Jacobi matrix

$$M_S(q) = \frac{e_p(-q/2)}{p-1} J_S(q).$$

Character orthogonality gives the exact trace identity

$$\widehat{1_A}(r) = K \operatorname{Tr} M_S(2r), \tag{11.1}$$

and therefore

$$\sum_q |\operatorname{Tr} M_S(q)|^2 = \frac{p(K-1)}{K^2}, \tag{11.2}$$

$$\sum_q |\operatorname{Tr} M_S(q)|^4 = \frac{pE_+(A)}{K^4}. \tag{11.3}$$

The normalized fourth/second moment ratio tends to 3 in the maintained split double average below $\theta < 1/3$. This is a fourth-moment theorem, not a CLT.

The pure off/off family contains a coherent reducible fibre. For generic $\lambda = (h_1, h_2)$ it occurs at

$$t_\lambda = \frac{2}{c(h_1) - c(h_2)}$$

and has exactly $2p - 8$ valid affine points. After extracting the spike, Igusa's arithmetic open-image theorem [16] gives geometric irreducibility: the constants quotient is procyclic, so the geometric image contains the commutator of an open arithmetic image, hence an open subgroup of $\mathrm{SL}_2(\mathbf{Z}_\ell)$ for $\ell \neq p$. The normalized elliptic local systems are pure of weight zero on their smooth lisse loci; their multiplicative middle-extension stalks instead have normalized eigenvalues $\pm p^{-1/2}$. Apply the Fouvry–Kowalski–Michel trace-correlation argument [15] on the common smooth open and restore the boundedly many omitted points using their bounded normalized traces. This gives square-root quasi-orthogonality between non-isomorphic geometric parameter classes.

Theorem 11.1 (Corrected off/off variance)**.** *For every prime $p > 3$ and every subgroup $H \leq \mathbf{F}_p^*$ of odd order K,*

$$\operatorname{Var}_{t \neq 0} O(t) \ll pE_+(A) + K^2 p + K^4 p^{1/2} + K^4, \tag{11.4}$$

with an absolute implied constant.

Proof. Write $O(t) = \sum_{\lambda \in H^2} O_\lambda(t)$ and use Cayley coordinates $a = c(u)$, $b = c(v)$, $\alpha = c(h_1)$, $\beta = c(h_2)$, $s = \alpha + \beta$, $d = \alpha - \beta$. On the generic set $\Lambda_g = \{h_1, h_2 \neq 1,\ h_1 \neq h_2,\ h_1 h_2 \neq 1\}$, of size $(K-1)(K-3)$, clearing denominators in $2/(a-b) + 2/(T_\alpha(a) - T_\beta(b)) = t$ gives a branch quartic whose discriminant factors with residual quartic $R_{(s,d)}(t)$; one has $\operatorname{Disc}_t R_{(s,d)} = 2^{16} d^2 s^2 (s-d-2)^2 (s-d+2)^2 (s+d-2)^2 (s+d+2)^2 \neq 0$ on Λ_g and $R_{(s,d)}(2/d) = 16/d^2$. Hence the generic pencil has five distinct finite bad fibres: $t_\lambda = 2/d$ of multiplicative type I_4 and four of type I_1. At t_λ the cleared cubic factors into the two distinct lines $(\beta-1)(a-1) - (\alpha-1)(b-1)$ and $(\beta+1)(a+1) - (\alpha+1)(b+1)$; removing the seven distinct invalid points from their $2p-1$ affine points gives the exact spike count $O_\lambda(t_\lambda) = 2p - 8$. With $a_\lambda(t)$ the middle-extension Frobenius trace of the Jacobian pencil,

$$O_\lambda(t) = p + p\,\mathbf{1}_{t=t_\lambda} - a_\lambda(t) + O(1)$$

uniformly in λ and t. The pencil is non-isotrivial, so by Igusa's open-image theorem [16] the geometric image contains the commutator of an open arithmetic image, hence an open subgroup

of $\mathrm{SL}_2(\mathbf{Z}_\ell)$: the rank-two local system is geometrically irreducible, of bounded conductor (at most six tame singular places). Equal singular support forces $(s', d') = (s, d)$ or $(-s, d)$, so each geometric isomorphism class contains at most the two parameter classes λ and $\lambda^* = (h_2^{-1}, h_1^{-1})$. For non-isomorphic classes, the trace-correlation argument of [15] — run on the common smooth open via Schur orthogonality, the Grothendieck–Lefschetz formula, Deligne's Riemann hypothesis and the Grothendieck–Ogg–Shafarevich bound, with the boundedly many omitted points restored at cost $O(1)$ in the normalized sum — gives $\sum_t a_\lambda(t)\overline{a_\mu(t)} = O(p^{3/2})$; the $O(K^2)$ same/involution ordered pairs contribute $O(p^2)$ each by the Weil bound, so $\sum_t \bigl|\sum_{\lambda\in\Lambda_g} a_\lambda(t)\bigr|^2 = O(K^2p^2 + K^4p^{3/2})$. The spike multiplicity $D(t) = \#\{\lambda \in \Lambda_g : t_\lambda = t\}$ satisfies $\sum_{t\neq 0} D(t)^2 \le E_+(A)$, since $t \mapsto 2/t$ identifies it with Cayley difference counts. The complement $H^2 \setminus \Lambda_g$ has $4K - 3$ classes, each of variance $O(p)$ (a bilinear count for $h_1 = h_2$, Hasse bounds for $h_1h_2 = 1$ and for exactly one $h_i = 1$, and the zero class at $(1, 1)$), hence total variance $O(K^2p)$ by Cauchy–Schwarz. Combining the generic and special variances gives (11.4). □

Thus the off/off layer has collapsed onto the same pointwise Cayley-energy bottleneck as the mixed layer.

12 Exact trace-to-local-statistics bridge

Let U be an $N \times N$ unitary matrix with eigenvalues $e^{2\pi i\theta_1}, \dots, e^{2\pi i\theta_N}$. For a zero-sum integer tuple $m_1 + \cdots + m_k = 0$, define

$$D_U(m_1, \dots, m_k) = \sum_{j_1,\dots,j_k \text{ distinct}} e^{2\pi i \sum_{a=1}^k m_a\theta_{j_a}}.$$

For a set partition π of $\{1, \dots, k\}$, put

$$\mu(\pi) = \prod_{B\in\pi} (-1)^{|B|-1}(|B| - 1)!.$$

Theorem 12.1 (Factorial trace identity)**.**

$$D_U(m_1, \dots, m_k) = \sum_\pi \mu(\pi) \prod_{B\in\pi} \operatorname{Tr} U^{\sum_{a\in B} m_a}. \tag{12.1}$$

This is Möbius inversion on the partition lattice. Let $\phi \in \mathcal{S}(\mathbf{R}^{k-1})$ and periodize it at the mean-spacing scale. Poisson summation gives the exact microscopic statistic

$$C_{k,N}(U; \phi) = N^{-k} \sum_{m\in\mathbf{Z}^{k-1}} \widehat{\phi}(m/N) D_U(m_1, \dots, m_k), \tag{12.2}$$

where $m_k = -(m_1 + \cdots + m_{k-1})$. In particular,

$$C_{2,N}(U; \phi) = N^{-2} \sum_{m\in\mathbf{Z}} \widehat{\phi}(m/N)(|\operatorname{Tr} U^m|^2 - N). \tag{12.3}$$

If $\widehat{\phi}$ has compact support, the relevant powers satisfy $|m| = O(N)$: microscopic local statistics probe the Heisenberg scale.

A weighted comparison of the quantities $D_U(m)$ with the matching random matrix ensemble at every compact Fourier support is sufficient for the corresponding band-limited microscopic correlations. Under positive normalized averaging, the next theorem removes the band-limit. The required joint-trace estimates for the correct desymmetrized cat-map block remain open.

12.1 Positive correlation measures remove the band-limit

Write $d = k - 1$, unfold the circle to length N, and let $x_j = N\theta_j$. The lifted anchored factorial measure is

$$\mu_N = \frac{1}{N} \sum_{j_1,\dots,j_k} \sum_{\text{distinct}} \sum_{\ell \in \mathbb{Z}^d} \delta_{(x_{j_1} - x_{j_k} + N\ell_1, \dots, x_{j_d} - x_{j_k} + N\ell_d)}.$$

It is positive, even when eigenvalues coincide, and $\mu_N(\phi) = C_{k,N}(U;\phi)$.

Lemma 12.2 (Uniform translated-cube bound)**.** *For $N > 2$ and every $a \in \mathbb{R}^d$,*

$$\mu_N(a + [0,1)^d) \le 2k^k(1 + \mu_N([-2,2]^d)).$$

The same bound holds under positive normalized averaging.

Proof. Let $n(t)$ count the periodically lifted points in $[t, t+2)$, with multiplicity, and put $a_k = 0$. For each tuple in the cube, integrate starting points t in the unit interval immediately preceding its anchor. Each of its other points lies in $[t + a_i, t + a_i + 2)$ apart from irrelevant endpoints. Since $2 < N$, every window contains at most one lift of each labelled point. Integration over a period and then Hölder's inequality give

$$\mu_N(a + [0,1)^d) \le \frac{1}{N} \int_0^N \prod_{i=1}^{k} n(t + a_i)\, dt \le \frac{1}{N} \int_0^N n(t)^k\, dt.$$

For the falling factorial $(r)_k$ one has $r^k \le k^k((r)_k + r)$ for nonnegative integers r. Indeed, for $r < k$ use $r^k \le k^{k-1} r$; for $r \ge k$ each factor in $(r)_k$ is at least r/k. Moreover,

$$\frac{1}{N} \int_0^N n(t)\, dt = 2,$$

$$\frac{1}{N} \int_0^N (n(t))_k\, dt = \int \big(2 - \operatorname{span}(0, z_1, \dots, z_d)\big)_+ \, d\mu_N(z) \le 2\mu_N([-2,2]^d).$$

The identity counts the possible starts of a common length-two window for each distinct-label tuple; their total length is the displayed positive part. This proves the bound. Positivity and normalization preserve it on averaging. □

Theorem 12.3 (Extension from all compact Fourier supports)**.** *Fix $k \ge 2$. Let μ_N, ν_N be positive normalized averages of lifted k-point factorial measures of N-point spectra. Suppose that ν_N has a limit on all Schwartz tests and*

$$\mu_N(f) - \nu_N(f) \longrightarrow 0$$

for every Schwartz f with compact Fourier support, with no common upper bound on that support. Then the same comparison holds for every Schwartz f.

Proof. Choose a nonzero nonnegative smooth compactly supported Fourier bump η, and put $h = \mathcal{F}^{-1}\eta$. Since $h(0) > 0$, for sufficiently small $\epsilon > 0$ the nonnegative band-limited Schwartz function

$$B(z) = 4|h(\epsilon z)|^2 / h(0)^2$$

majorizes one on $[-2,2]^d$. The hypotheses bound $\nu_N(B)$ and $\mu_N(B)$. Lemma 12.2 therefore bounds both measures on every translated unit cube by one constant M. For $p > d$ and every Schwartz g this implies

$$|\mu_N(g)| + |\nu_N(g)| \le C_{d,p} M \sup_z (1 + |z|)^p |g(z)|.$$

For a smooth compactly supported χ equal to one near zero, set $f_R = \mathcal{F}^{-1}(\chi(\cdot/R)\widehat{f})$. Then $f_R \to f$ in Schwartz topology. The last inequality makes the two approximation errors uniform in N. First send $N \to \infty$ at fixed R, then $R \to \infty$. □

Thus a separate absolute high-frequency trace-tail estimate is unnecessary for this positive-average, all-bandwidth criterion. This does not establish such an absolute estimate, or extend a comparison confined to one fixed cap. For example, for even N, the unit clock and a period-two configuration with basis $\{0, a\}$, $0 < a < 2$, $a \neq 1$, have pair measures

$$\sum_{r\in\mathbb{Z}} \delta_r - \sum_{r\in\mathbb{Z}} \delta_{Nr}, \qquad \sum_{r\in\mathbb{Z}} \delta_{2r} + \frac{1}{2}\sum_{r\in\mathbb{Z}} (\delta_{2r+a} + \delta_{2r-a}) - \sum_{r\in\mathbb{Z}} \delta_{Nr}.$$

They agree on tests supported in Fourier space inside $(-1/2, 1/2)$ but have different full limits. The theorem also requires positivity of the averages. It addresses correlation tests, not a spacing-distribution theorem without the additional point-process hypotheses needed for that conclusion.

Chapter 3 — Cayley maps and magnetic quotients

13 Gaussian limits for the full Cayley–Jacobi trace

The fourth-moment results admit an all-order extension in sparse regimes. The external arithmetic inputs are Chowla's cotangent independence and, for the density statement only, Bombieri–Vinogradov and the prime number theorem. Priority for this specialization has not been established.

Let K be an odd prime, $p \equiv 1 \pmod{K}$ a prime, $H \leq \mathbf{F}_p^*$ of order K, and $A = c(H \setminus \{1\})$, where $c(h) = (h+1)/(h-1)$. Put $n = K - 1$, $d = n/2$, and

$$X_{K,p}(q) = n^{-1/2} \sum_{a\in A} e^{2\pi i q a/p}, \qquad q \in \mathbf{F}_p.$$

This is real since $A = -A$, and $X_{K,p}(0) = \sqrt{n}$. Probability within a design means uniform sampling of $q \in \mathbf{F}_p^*$, not sampling matrix eigenvalues.

Theorem 13.1 (Pointwise sparse trace limit)**.** *For every sequence of split prime designs with $K \to \infty$ and $K = o(\log p)$, the law of $X_{K,p}(q)$ tends to $N(0,1)$. For fixed distinct positive integers $m_1, \ldots, m_L$, more generally,*

$$(X_{K,p}(m_1 q), \ldots, X_{K,p}(m_L q)) \Longrightarrow N(0, I_L).$$

Theorem 13.2 (Density trace limit)**.** *Let $Q = Q(X) \to \infty$, $\log Q = o(\log X)$, and*

$$\mathcal{D}(X) = \{(K,p) : K, p \text{ prime},\ Q/2 < K \leq Q,\ X/2 < p \leq X,\ p \equiv 1 \pmod{K}\}.$$

Let $\mu_{K,p}^{\boldsymbol{m}}$ be the empirical joint law in the preceding theorem, for fixed distinct positive multipliers. For each $\varepsilon > 0$,

$$\frac{\#\{(K,p) \in \mathcal{D}(X) : d_{\mathrm{BL}}(\mu_{K,p}^{\boldsymbol{m}}, N(0, I_L)) > \varepsilon\}}{|\mathcal{D}(X)|} \longrightarrow 0,$$

where d_{BL} denotes the bounded-Lipschitz metric on probability measures.

These statements transfer to the full annihilator twisted-Jacobi trace through the exact identity from the arithmetic core,

$$X_{K,p}(mq) = \frac{K}{\sqrt{K-1}} \operatorname{Tr} M_{H^\perp}(2mq).$$

They concern parameter multiples, not $\operatorname{Tr}(M_{H^\perp}(q)^m)$. They imply no microscopic BGS or desymmetrized spectral universality theorem.

13.1 Weighted relations and their resultant budget

Fix a nonempty word $\boldsymbol{w} = (w_1, \dots, w_r)$ of positive integers, and set $W = \sum_i w_i$. A tuple $\boldsymbol{a} \in (\mathbf{F}_K^*)^r$ is balanced when

$$\sum_{i:a_i=b} w_i = \sum_{i:a_i=-b} w_i \qquad (1 \le b \le d).$$

Write $B_{K,\boldsymbol{w}}$ for its balanced count. Since $c(\zeta^a) = -i\cot(\pi a/K)$, Chowla's independence theorem says that $\sum_i w_i c(\zeta^{a_i}) = 0$ holds exactly for balanced tuples. Wang's *On a theorem of S. Chowla*, J. Number Theory 15 (1982), 1–4, Theorem A, states precisely this prime-order independence input.

Let $Z_{K,\boldsymbol{w}}(p)$ count all ordered tuples satisfying $\sum_i w_i c(h^{a_i}) = 0$ in $\mathbf{F}_p$, where h generates H, and put $\Delta_{K,\boldsymbol{w}}(p) = Z_{K,\boldsymbol{w}}(p) - B_{K,\boldsymbol{w}} \ge 0$. The inequality follows because every balanced relation survives reduction. For representatives $1 \le a_i < K$, use the integer polynomial

$$F_{\boldsymbol{a},\boldsymbol{w}}(T) = \sum_{i=1}^{r} w_i (T^{a_i} + 1) \prod_{j \ne i} (T^{a_j} - 1).$$

For a nonbalanced tuple, its resultant $R = \mathrm{Res}(\Phi_K, F)$ satisfies

$$R \ne 0, \qquad |R| \le (W2^r)^n, \qquad K^{r-1} \mid R. \tag{13.1}$$

Indeed $\|F\|_1 \le W2^r$, and all n conjugates of $F(\zeta)$ have this modulus bound. Each summand of F is divisible by $(T-1)^{r-1}$, so multiplicativity of the resultant and $\mathrm{Res}(\Phi_K, T-1) = K$ prove divisibility.

Simultaneous scaling of the exponents by $\mathbf{F}_K^*$ acts freely, preserves balance, and gives $(n^r - B_{K,\boldsymbol{w}})/n$ nonbalanced orbits. Choose one representative with $a_1 = 1$ from each orbit. At a split prime p, the n roots of Φ_K lift to $\mathbf{Z}_p$. Every colliding member of the orbit contributes a factor divisible by p to the product formula for its representative's resultant. Hence its number of colliding members is at most $v_p(R)$. Since $p \ne K$, removing the factor K^{r-1} and summing yields

Proposition 13.3 (All-order weighted budget)**.** *For $r \ge 1$,*

$$\sum_{\substack{p \text{ prime} \\ p \equiv 1 \pmod K}} \Delta_{K,\boldsymbol{w}}(p) \log p \le \frac{n^r - B_{K,\boldsymbol{w}}}{n} \left[n \log(W2^r) - (r-1)\log K \right] = O_{\boldsymbol{w}}(K^r). \tag{13.2}$$

Also $p > (W2^r)^n$ implies $\Delta_{K,\boldsymbol{w}}(p) = 0$.

The zeroth moment equals one and is not included in this resultant argument. For the scalar fourth moment, the earlier specialized coefficient 12 is sharper; the present gain is access to every fixed weighted order.

13.2 Balanced moments and the Gaussian comparison

For the scalar word of length $2s$, write $B_{K,2s}$. Counting equal positive and negative multiplicities gives

$$B_{K,2s} = (2s)![t^s] \left(\sum_{j \ge 0} \frac{t^j}{(j!)^2} \right)^d, \qquad B_{K,2s+1} = 0.$$

Thus $B_{K,2} = n$, $B_{K,4} = 3n(n-1)$, and $B_{K,6} = 15n^3 - 45n^2 + 40n$. The terms with s different antipodal coordinates give $(2s)!\binom{d}{s}$; the others are $O_s(d^{s-1})$. Consequently $B_{K,2s}/n^s = (2s-1)!! + O_s(n^{-1})$.

Additive orthogonality gives the exact mixed-moment identity

$$\frac{1}{p-1}\sum_{q\neq 0}\prod_{i=1}^{r}X_{K,p}(w_i q)=\frac{p(B_{K,\boldsymbol{w}}+\Delta_{K,\boldsymbol{w}}(p))-n^r}{(p-1)n^{r/2}}. \tag{13.3}$$

The subtraction removes $q=0$. The two lowest cases have no excess at any split prime, by direct $\mathbf{F}_p$ computations. For the word (1), a relation $c(h^a)=0$ forces $h^a=-1$, since $c(x)=(x+1)/(x-1)$ vanishes only at $x=-1$; but $-1\notin H$, as H has odd order K while -1 has order two. Hence $Z_{K,(1)}(p)=B_{K,(1)}=0$. For the word $(1,1)$, $c(x)+c(y)=0$ with $x,y\in H\setminus\{1\}$ becomes, after clearing denominators (legitimate since $x,y\neq 1$ and p is odd), the equation $2xy-2=0$, i.e. $xy=1$; these are exactly the balanced tuples. Hence $\Delta_{K,(1)}(p)=\Delta_{K,(1,1)}(p)=0$ for every split p, and (13.3) gives in particular the exact mean $-\sqrt{n}/(p-1)$ and the second moment $(p-n)/(p-1)$.

For independent uniform phases $\Theta_1,\dots,\Theta_d$ on $[0,2\pi]$, put

$$Y_{K,j}=n^{-1/2}\sum_{b=1}^{d}2\cos(m_j\Theta_b).$$

Expansion and phase integration show that their mixed moment for a word $\boldsymbol{w}$ is exactly $B_{K,\boldsymbol{w}}/n^{r/2}$. The vectors $(\sqrt{2}\cos(m_j\Theta_b))_{j=1}^{L}$ are bounded, independent over b, have mean zero and covariance I_L. In a fixed mixed-moment expansion, singleton blocks vanish. Pair partitions supply Wick's formula; all other partitions contribute $O_{\boldsymbol{w}}(d^{-1/2})$ or less. Their moments therefore tend to those of $N(0,I_L)$. Relations among the multipliers only affect lower-order terms: $B_{K,(1,1,2)}=n$ gives normalized mixed third moment $n^{-1/2}$.

For a fixed word, the difference between (13.3) and its independent-phase moment is at most

$$\frac{p}{p-1}\frac{\Delta_{K,\boldsymbol{w}}(p)}{n^{r/2}}+\frac{2n^{r/2}}{p-1}.$$

Under $K=o(\log p)$, the excess eventually vanishes for each fixed word, by (13.1). The remaining error tends to zero. Gaussian moment determinacy, or the scalar moment argument for every linear combination, proves the pointwise theorem.

13.3 The density denominator and the law-level limit

Bombieri–Vinogradov at X and $X/2$ gives

$$|\mathcal{D}(X)|=(\pi(X)-\pi(X/2))\sum_{\substack{Q/2<K\le Q\\ K\text{ prime}}}\frac{1}{K-1}+O_A(X/(\log X)^A).$$

Its modulus range contains $Q=X^{o(1)}$ eventually. PNT and partial summation give the reciprocal sum $(\log 2+o(1))/\log Q$. Therefore

$$|\mathcal{D}(X)|\sim\frac{\log 2}{2}\frac{X}{\log X\log Q}.$$

A fixed $A=3$ already makes the relative error tend to zero, even for arbitrarily slowly growing Q. No individual varying-modulus PNT is used.

By (13.2) and $\log p\ge\log(X/2)$,

$$\sum_{(K,p)\in\mathcal{D}(X)}\frac{\Delta_{K,\boldsymbol{w}}(p)}{(K-1)^{r/2}}\ll_{\boldsymbol{w}}\frac{1}{\log X}\sum_{\substack{Q/2<K\le Q\\ K\text{ prime}}}K^{r/2}\ll_{\boldsymbol{w}}\frac{Q^{r/2+1}}{\log X\log Q}.$$

After division by the design count, the average normalized excess is $O_{\boldsymbol{w}}(Q^{r/2+1}/X) = o(1)$. Markov's inequality proves convergence in probability of every mixed moment to its Gaussian value.

For completeness, this implies convergence of the random empirical laws. From any sequence, extract a subsequence on which the first j moments are simultaneously accurate with failure probability at most 2^{-j}. On a product coupling, Borel–Cantelli gives almost sure convergence of all mixed moments along this subsequence. Second moments give tightness, higher even moments give uniform integrability of lower polynomials, and Gaussian moment determinacy identifies every limit. Thus the bounded-Lipschitz distance tends to zero almost surely along a further subsequence, proving convergence in probability as claimed.

At a fixed scale $Q = X^{\theta}$, the same calculation proves only the order-r moment conclusion when $0 < \theta < \min(1/2, 2/(r+2))$. The allowed thresholds tend to zero with r; this argument does not give a fixed-polynomial-scale CLT.

13.4 A finite obstruction to the fourth-moment shortcut

Exact integer convolution for $(K, p) = (11, 199)$ gives

$$(Z_3, B_3) = (6, 0), \qquad (Z_4, B_4) = (270, 270), \qquad (Z_6, B_6) = (12250, 10900).$$

Thus the exact fourth-order baseline does not force the higher-order classification. The all-order budgets are needed. The maintained verifier checks 32 finite word/order designs, representing 6440 ordered tuples and 1242 scaling orbits, with 13293 orbit-valuation checks (run log `evidence/executions/20261006/corpus-revision/final/checks/cayley_higher_moments.log`). These are implementation regressions, not proofs of the external arithmetic inputs. The full statements, proof details, source locators, and limits of the BGS interpretation are in `mathematics/cayley_trace_gaussian_limit.md`.

13.5 A quantitative scalar refinement

The scalar limit admits a rate by keeping all constants uniform in the number of moments. Write $\mu_{K,p}$ for the empirical scalar law with $q \neq 0$, $\gamma = N(0,1)$, and $D_{K,p} = \sup_x |\mu_{K,p}((-\infty, x]) - \gamma((-\infty, x])|$.

Theorem 13.4 (Pointwise Kolmogorov bound)**.** *There is an absolute constant C such that, for every odd prime K and split prime p,*

$$D_{K,p} \leq C\left((K-1)^{-1/2} + \sqrt{\frac{K-1}{\log p}}\right).$$

Theorem 13.5 (Density Kolmogorov bound)**.** *For sufficiently large X, Q with $Q \leq X^{1/4}$, all but a proportion at most $C(\log X)X^{-1/4}$ of the designs in $\mathcal{D}(X)$ satisfy*

$$D_{K,p} \leq C\left(Q^{-1/2} + \sqrt{\frac{\log Q}{\log X}}\right).$$

The average distance is bounded by this right side plus $C(\log X)X^{-1/4}$. Constants are absolute and not optimized.

For fixed $Q = X^{\theta}$ this only gives an $O(\sqrt{\theta})$ discrepancy; it does not establish a fixed-power CLT.

Finite moments and smoothing. Let $\mu_{K,p}^{\mathrm{all}}$ include $q = 0$, and let ν_K be the scalar independent-phase law above. If $\Delta_{K,r}(p) = 0$ for $1 \le r \le 2s$, then these laws have exactly the same first $2s$ moments. The coefficientwise inequality

$$\sum_{j\ge 0} \frac{z^{2j}}{n^j (j!)^2} \preceq \exp(z^2/n)$$

is preserved by taking the d-th power. Thus $\int y^{2s}\, d\nu_K(y) \le (2s)!/(2^s s!)$. Taylor expansion of e^{ity} through degree $2s-1$ gives

$$|\widehat{\mu}_{K,p}^{\mathrm{all}}(t) - \widehat{\nu}_K(t)| \le \frac{2|t|^{2s}}{2^s s!}.$$

For $T^2 \le s/(2e)$ this implies

$$\int_0^T \frac{|\widehat{\mu}_{K,p}^{\mathrm{all}}(t) - \widehat{\nu}_K(t)|}{t}\, dt \le \frac{T^{2s}}{s\, 2^s s!} \le \frac{4^{-s}}{s}.$$

For $|t| \le \sqrt{n}/4$, put $v = t^2/n \le 1/16$. The cosine expansion gives $1 - v \le J_0(2t/\sqrt{n}) \le 1 - v + v^2/4$, so this quantity is positive and

$$|\log J_0(2t/\sqrt{n}) + v| \le \tfrac{8}{15} v^2.$$

Both $d \log J_0(2t/\sqrt{n})$ and $-t^2/2$ are at most $-t^2/3$. Since $\widehat{\nu}_K(t) = J_0(2t/\sqrt{n})^d$, the mean-value theorem gives

$$|\widehat{\nu}_K(t) - e^{-t^2/2}| \le \frac{4}{15n} t^4 e^{-t^2/3}, \qquad \int_0^T \frac{|\widehat{\nu}_K(t) - e^{-t^2/2}|}{t}\, dt \le \frac{6}{5n}.$$

The logarithmic bound follows from $\log(1-v) \ge -v - v^2/[2(1-v)]$ and $\log(1+u) \le u$.

Apply the classical Esseen smoothing inequality against the normal target with $T = \min\{\sqrt{s/(2e)}, \sqrt{n}/4\}$. This inequality permits an atomic first law; only the target needs a Lipschitz CDF. An exact statement is B. Borda, arXiv:2005.04925v2, Theorem A, p. 2. The smoothing remainder $O(1/T)$ and the preceding integrals imply

$$d_{\mathrm{K}}(\mu_{K,p}^{\mathrm{all}}, \gamma) \le C(s^{-1/2} + n^{-1/2}).$$

Removing zero changes the law by at most $1/p$ in Kolmogorov distance, because $\mu_{K,p}^{\mathrm{all}} = p^{-1}\delta_{\sqrt{n}} + (1 - p^{-1})\mu_{K,p}$. Hence

$$[\Delta_{K,r}(p) = 0 \ (1 \le r \le 2s)] \implies D_{K,p} \le C(s^{-1/2} + n^{-1/2}) + 1/p. \tag{13.4}$$

Uniform arithmetic cutoffs. For $r \le 2s$, $r2^r \le 2s\, 2^{2s} \le 2^{4s}$. Thus $p > 2^{4ns}$ excludes every nonbalanced scalar relation through order $2s$. Taking $s = \lfloor \log p/(8n \log 2) \rfloor$ when it is positive ensures the strict cutoff and gives the pointwise theorem by (13.4). If the floor is zero, the right side of the asserted bound is bounded below by an absolute positive constant, and $D_{K,p} \le 1$ suffices.

For density, the all-order budget and $\log(r2^r) \le 2r$ imply uniformly for every integer $s \ge 1$ that

$$\sum_{r=1}^{2s} \sum_{p \equiv 1 \pmod K} \Delta_{K,r}(p) \log p \le \sum_{r=1}^{2s} 2r n^r \le 8s n^{2s}.$$

A design with any excess through order $2s$ consumes at least $\log(X/2)$, by integrality. Bound $n^{2s} \le Q^{2s}$ before using the prime count $O(Q/\log Q)$. Dividing by the same uniform Bombieri–Vinogradov denominator shows that the exceptional fraction is at most CsQ^{2s+1}/X, with a constant independent of s. For $s = \lfloor \log X/(4\log Q) \rfloor$ and $Q \le X^{1/4}$,

$$s \ge 1, \qquad Q^{2s+1} \le X^{3/4}, \qquad s^{-1/2} \le C\sqrt{\log Q/\log X}.$$

The exceptional fraction is therefore at most $C(\log X)X^{-1/4}$. On its complement, use (13.4), $n \ge Q/3$ and $p > X/2$. This proves the density rate and, by bounding distance by one on the exceptional set, its averaged version. No dimension-uniform multivariate rate or microscopic spectral law is inferred.

Chapter 2 — Arithmetic core (continued)

14 Fixed-degree row formulation

Let $K = 2m + 1 \ge 7$ be an odd prime, let $\zeta = \exp(2\pi i/K)$, and put $\tau = \zeta + \zeta^{-1}$. The full subgroup trace polynomial is $H_K(T) = (T-2)C_K(T)$, where C_K is the minimal polynomial of τ. For a normalized row ratio $r \ne 1$ the affine trace map is

$$Y = \alpha_r X + \beta_r, \qquad \alpha_r = U_{r-1}(\tau/2), \qquad \beta_r = -2U_{r-2}(\tau/2).$$

Removing the shared invalid characteristic-zero row trace gives fixed-degree cofactors $P_{K,r}, Q_{K,r} \in \mathbf{Z}[\tau][T]$ of degree m. At every split prime $p \equiv 1 \pmod K$ and embedding prime $\mathfrak{q} \mid p$, the leading coefficients remain nonzero and

$$I_{1,r}(\mathfrak{q}) = \deg \gcd(P_{K,r}, Q_{K,r} \bmod \mathfrak{q}). \tag{14.1}$$

This fixed-degree statement is the specialization safeguard. Older projective dehomogenizations may lose degree at infinity and remain only one-way obstructions.

15 Principal subresultants are sufficient

For fixed-degree polynomials P, Q let $\mathrm{psc}_j(P,Q) = [T^j]S_j(P,Q;T)$. The standard subresultant theorem [12] states that, for $g = \deg\gcd(P,Q)$, the first nonzero subresultant is a nonzero scalar multiple of the gcd, of degree g.

Lemma 15.1 (Principal coefficients detect gcd degree)**.** *For $1 \le d \le \min(\deg P, \deg Q)$,*

$$\deg\gcd(P,Q) \ge d \iff \mathrm{psc}_0 = \cdots = \mathrm{psc}_{d-1} = 0.$$

Proof. If the gcd degree is at least d, these subresultants vanish. Otherwise its degree $g < d$ gives a first nonzero subresultant of degree exactly g, so its principal coefficient is nonzero. □

Set $\pi_j = \mathrm{psc}_j(P_{K,r}, Q_{K,r})$ for $j = 0, 1, 2$, and define

$$J^{\mathrm{psc}}_{K,r} = (\pi_0, \pi_1, \pi_2), \qquad J^{\mathrm{psc}}_{K,r} \cap \mathbf{Z} = \Pi^{(3)}_{K,r}\mathbf{Z}.$$

Theorem 15.2 (Three-scalar exact split support)**.** *For every split prime $p \equiv 1 \pmod K$,*

$$p \mid \Pi^{(3)}_{K,r} \iff \textit{some } \mathfrak{q} \mid p \textit{ has } I_{1,r}(\mathfrak{q}) \ge 3.$$

Proof. Use (14.1) and Lemma 15.1, applied over the residue field at $\mathfrak{q}$, where the leading coefficients remain nonzero by the fixed-degree setup. If some $\mathfrak{q} \mid p$ has $I_{1,r}(\mathfrak{q}) \geq 3$, the lemma gives $\pi_0, \pi_1, \pi_2 \in \mathfrak{q}$, hence $\Pi^{(3)}_{K,r} \in J^{\mathrm{psc}}_{K,r} \cap \mathbf{Z} \subseteq \mathfrak{q} \cap \mathbf{Z} = p\mathbf{Z}$.

Conversely, suppose $p \mid \Pi^{(3)}_{K,r}$. If $J^{\mathrm{psc}}_{K,r} = (0)$ the π_j vanish at every $\mathfrak{q}$ and there is nothing to prove. Otherwise, write $\mathcal{O}^+ = \mathbf{Z}[\tau]$ and localize at the multiplicative set $S = \mathbf{Z} \setminus p\mathbf{Z}$. The localized ideal is proper in $\mathcal{O}^+_S$: an element of $J^{\mathrm{psc}}_{K,r} \cap S$ would lie in $J^{\mathrm{psc}}_{K,r} \cap \mathbf{Z} = \Pi^{(3)}_{K,r}\mathbf{Z}$ yet be prime to p, while $p \mid \Pi^{(3)}_{K,r}$. By Krull's theorem the proper localized ideal is contained in a maximal ideal of $\mathcal{O}^+_S$; its contraction $\mathfrak{q}$ to $\mathcal{O}^+$ is a prime ideal with $\mathfrak{q} \cap S = \varnothing$ and $J^{\mathrm{psc}}_{K,r} \subseteq \mathfrak{q}$. The contraction $\mathfrak{q} \cap \mathbf{Z}$ is a prime of $\mathbf{Z}$ contained in $p\mathbf{Z}$; it is not (0): a prime of $\mathcal{O}^+$ contracting to (0) is itself (0), since $\mathcal{O}^+ \otimes \mathbf{Q}$ is a field, and (0) is not maximal in $\mathcal{O}^+_S$, as primes above the split prime p localize to strictly larger proper ideals. Hence $\mathfrak{q} \cap \mathbf{Z} = p\mathbf{Z}$, i.e. $\mathfrak{q} \mid p$. Then π_0, π_1, π_2 vanish mod $\mathfrak{q}$, and the lemma gives $I_{1,r}(\mathfrak{q}) \geq 3$. □

This is equality of split prime support, not of integral valuations. Exact finite Smith-normal-form controls are

(K, r)	$\Pi^{(3)}_{K,r}$	prime-to-$2K$ part
$(7, 2)$	7	1
$(11, 7)$	11^2	1
$(13, 3)$	13^3	1
$(13, 12)$	$2^4\, 5^3\, 13^3$	5^3
$(17, 10)$	$2^2\, 17^5\, 103$	103.

At $(13, 12)$ the full/shift support instead has prime-to-$2K$ valuation 5^2.

16 The intrinsic Sylvester/Fitting locus

Let

$$\mathcal{S}_{P,Q} : k[T]_{<m} \oplus k[T]_{<m} \longrightarrow k[T]_{<2m}, \qquad (U, V) \longmapsto UP + VQ.$$

Theorem 16.1 (Sylvester nullity)**.** *For nonzero degree-m polynomials,* $\dim \ker \mathcal{S}_{P,Q} = \deg \gcd(P, Q)$.

Proof. Write $P = GP_1, Q = GQ_1$, with $(P_1, Q_1) = 1$ and $g = \deg G$. Kernel vectors are $(WQ_1, -WP_1)$ with $\deg W < g$. □

Thus gcd degree at least three is equivalent to rank at most $2m - 3$, or to vanishing of all $(2m - 2)$-minors. At gcd degree two the corresponding compound matrix has rank one. These are intrinsic determinantal descriptions of the same field-valued support.

17 Shift-resultant transversality

The squarefreeness hypothesis must be stated on the correct input. A squarefree gcd alone does not suffice: $f = T^2$, $g = T(T - 1)$ have gcd degree one, but $\mathrm{Res}(f, g + Z) = Z^2$.

Theorem 17.1 (Universal initial resultant coefficient)**.** *Let $f = GA, g = GB$ have degrees n, m, with G monic of degree d. For degree-preserving perturbations $f_Z = f + Z\phi$, $g_Z = g + Z\psi$, put $D = A\psi - B\phi$. With the root-product resultant convention,*

$$\mathrm{Res}(f_Z, g_Z) = (-1)^{d(n-d)} \mathrm{Res}(A, B) \mathrm{Res}(G, D) Z^d + O(Z^{d+1}).$$

This coefficient identity does not require G to be squarefree. When $G = \gcd(f, g)$ and $\mathrm{Res}(G, D) \neq 0$, the order is exactly d.

Proof. For generic characteristic-zero coefficients, assume G squarefree and coprime to A. A root y of G contributes $ZD(y)/A(y) + O(Z^2)$ to the resultant root product. All other roots contribute $\operatorname{Res}(A, G) \operatorname{Res}(A, B)$, and $\operatorname{Res}(A, G) = (-1)^{d(n-d)} \prod_{G(y)=0} A(y)$. Multiplication proves the displayed identity and vanishing of every lower coefficient. These are polynomial identities over the integers in the coefficients of monic G and of A, B, ϕ, ψ. Therefore specialization proves them in arbitrary characteristic and on multiple-root strata. □

In particular, for monic same-degree f, g, squarefree $G = \gcd(f, g)$, $\gcd(A, G) = 1$, and ψ nonzero at every root of G, a fixed-f shift gives

$$\operatorname{ord}_{Z=0} \operatorname{Res}(f, g + Z\psi) = d, \qquad [Z^d] \operatorname{Res}(f, g + Z\psi) = \operatorname{Res}(A, B) \operatorname{Res}(A, G) \operatorname{Res}(G, \psi).$$

Up to the fixed subresultant sign, $\operatorname{Res}(A, B)$ is the first nonzero PSC; this is the content of the following lemma, in the monic case used below.

Lemma 17.2 (First nonzero principal subresultant)**.** *Let $f = GA$, $g = GB$ have the same degree m, with G of degree d and leading coefficient γ, and $\gcd(A, B) = 1$. Then, with the determinant normalization of psc_j,*

$$\operatorname{psc}_d(f, g) = \pm\gamma^{2(m-d)} \operatorname{Res}(A, B),$$

the sign depending only on m, d and the subresultant convention. In particular the factor is ± 1 when G is monic.

Proof. Up to transposition and reordering of rows and columns, the principal minor defining psc_d is the determinant of the linear map sending (U, V), with $\deg U, \deg V < m - d$, to the coefficients of $Uf + Vg$ at the powers $d, \dots, 2m - d - 1$. Since $Uf + Vg = G(UA + VB)$, this map factors as the Sylvester map $(U, V) \mapsto UA + VB$ of the coprime pair (A, B), whose determinant is $\pm \operatorname{Res}(A, B)$, followed by multiplication by G read in the same coefficient window. The latter is triangular in the window basis: the coefficient of T^{d+s} in GW is γW_s plus a combination of $W_{s+1}, \dots, W_{s+d}$. Its determinant is therefore $\gamma^{2(m-d)}$. □

The condition $\gcd(A, G) = 1$ says that the common roots are simple in f; squarefreeness of f is a stronger sufficient condition. Without it the coefficient formula still holds but can vanish.

For the row shift put $q_Z(T) = \alpha_r^{-m} R_r(\alpha_r T + \beta_r + Z)$, so $\psi(T) = \alpha_r^{-m} R_r'(\alpha_r T + \beta_r)$. The *maintained split locus* for (K, r) is the set of split primes p such that, for one (equivalently every, the condition being Galois-invariant) embedding prime $\mathfrak{q} \mid p$, the reductions of $P_{K,r}$ and $Q_{K,r}$ mod $\mathfrak{q}$ are squarefree and $\alpha_r \not\equiv 0 \pmod{\mathfrak{q}}$; equivalently $\mathfrak{q} \nmid \operatorname{Disc}_T(P_{K,r}) \operatorname{Disc}_T(Q_{K,r})$ and $\mathfrak{q} \nmid \alpha_r$, since a polynomial over a field is squarefree exactly when its discriminant is nonzero. On this locus $q_0'(y) = \alpha_r \psi(y) \neq 0$ at every common root y of P and q_0 mod $\mathfrak{q}$: with $P = GA$, $q_0 = GB$ and G monic, $q_0'(y) = G'(y)B(y)$; $G \mid q_0$ and q_0 squarefree give G squarefree, so $G'(y) \neq 0$, and $B(y) = 0$ would make y a multiple root of q_0. The same argument gives $\operatorname{Disc}(G) \neq 0$ and $\operatorname{Res}(G, B) \neq 0$, and with the roles of the two cofactors exchanged $\operatorname{Res}(A, G) \neq 0$ (a common root of A and G is a multiple root of P); $\operatorname{Res}(A, B) \neq 0$ since $\gcd(A, B) = 1$ by the choice of G as the gcd. Writing $P = GA, q_0 = GB$ with G monic gives

$$[Z^d] \operatorname{Res}(P, q_Z) = \pm\alpha_r^{-d} \operatorname{psc}_d(P, q_0) \operatorname{Disc}(G) \operatorname{Res}(A, G) \operatorname{Res}(G, B). \tag{17.1}$$

Every multiplier is nonzero. The first three shift coefficients therefore vanish precisely on the order-three PSC locus, with no conclusion about equality of integral valuations.

18 Inverse resultant transversality

Theorem 18.1 (Simple-gcd derivative)**.** *Let* $\gcd(f,g) = Y-y$*, write* $f = (Y-y)A, g = (Y-y)B$*, and preserve the displayed degrees in* $f_Z = f + Z\phi, g_Z = g + Z\psi$*. Then*

$$\frac{d}{dZ}\operatorname{Res}(f_Z, g_Z)\Big|_0 = (-1)^{\deg f - 1}\operatorname{Res}(A,B)\big(f'(y)\psi(y) - g'(y)\phi(y)\big).$$

A root may be multiple in one input; simplicity in both is not required.

Proof. Apply Theorem 17.1 with $G = Y - y$. □

For $F(X) = E(X^2) + XO(X^2)$, translation by $Z/2$ has first variations

$$\phi(Y) = \tfrac{1}{2}(O(Y) + 2YO'(Y)), \qquad \psi(Y) = E'(Y).$$

Suppose the characteristic is not two, F is nonconstant and squarefree, and $F(0) \neq 0$. Let $G = \gcd(E,O)$ be monic, with $E = GA, O = GB$. Then G is squarefree and its roots y are nonzero. At a common root,

$$D(y) = G'(y)(A(y)^2 - yB(y)^2),$$

and for $z^2 = y$,

$$F'(z)F'(-z) = -4yG'(y)^2(A(y)^2 - yB(y)^2) \neq 0.$$

Thus D is invertible modulo G.

To include all degree-drop cases, use the fixed ambient parity degrees in the Sylvester matrix. At least one of E, O has its full ambient degree, because F has its displayed degree. Swap the inputs if needed. For a full-degree first input f and a second input g of degree at most its ambient degree, the Sylvester kernel is still $(BW, -AW)$, $\deg W < d$, and its cokernel is identified with $k[Y]/(G)$. The derivative on kernel to cokernel is multiplication by $B\phi - A\psi = -D$ (up to the input swap). Its determinant is nonzero. Constant invertible row and column operations reduce the undeformed Sylvester matrix to an identity block and a zero d-block; the Schur complement then shows its determinant has order exactly d. This also covers $O = 0$, where the same kernel description uses $B = 0$. Consequently, with the fixed-degree convention,

$$\operatorname{ord}_{Z=0}\rho(Z) = \deg\gcd(E,O), \qquad \rho(Z) = \operatorname{Res}(E_Z, O_Z),$$

and

$$\deg\gcd(E,O) \geq 2 \iff \rho(0) = \rho'(0) = 0.$$

For the unique-common-root formula when the displayed input degrees are preserved, the local factor is

$$E'(y)^2 - yO'(y)^2 = -\frac{F'(z)F'(-z)}{4y}.$$

The example $E = (Y-1)^2, O = Y - 1$ over $\mathbf{Q}$ has $E'(1) = 0$ and squarefree $F = (X^2-1)(X^2 + X - 1)$ with $F(0) = 1$; its translated resultant has linear coefficient one. It is a finite control for the repaired multiple-input-root argument. These local identities do not bound the largest split prime in the integral support and do not establish microscopic BGS universality.

Chapter 3 — Cayley maps and magnetic quotients (continued)

19 Scaling rigidity of the Cayley collision family

Consider the off-diagonal collision polynomial

$$G_{\rho,\sigma}(U,V) = \sigma(\rho-1)U^2V^2 + \rho(1-\sigma)U^2 + 2(\sigma-\rho)UV + (1-\sigma)V^2 + \rho - 1.$$

Assume characteristic different from two, $\rho, \sigma \notin \{0, 1, -1\}$, and $\rho \neq \sigma$. These assumptions hold for distinct nonunit parameters in an odd-order multiplicative subgroup.

Proposition 19.1. *Among this family a scalar and diagonal-coordinate scaling class has exactly the two members (ρ, σ) and (ρ^{-1}, σ^{-1}) over an algebraic closure. Allowing invertible Laurent monomial coordinate changes and Laurent monomial multipliers enlarges it to at most four members, adding (σ, ρ) and (σ^{-1}, ρ^{-1}). For parameters in an odd-order subgroup all these equivalences can be realized by transformations preserving its Cartesian square.*

Proof. For coefficients (a, b, c, d, e) of $(U^2V^2, U^2, UV, V^2, 1)$, the quantities $I = ae/(bd)$ and $J = c^2/(bd)$ are scaling invariants. Here

$$I = \frac{\sigma(\rho-1)^2}{\rho(\sigma-1)^2}, \qquad J = \frac{4(\sigma-\rho)^2}{\rho(\sigma-1)^2}.$$

Writing $z = (\sigma-\rho)/(\sigma-1)$ and $c_0 = 1 - I + J/4$, direct algebra gives

$$z^2 = J\rho/4, \quad c_0\rho = (\rho+1)z, \quad J(\rho+1)^2 = 4c_0^2\rho.$$

The last equation is a reciprocal quadratic with $J \neq 0$, so it allows at most two values of ρ. Each determines z and then $\sigma = (z-\rho)/(z-1)$. Simultaneous inversion preserves the invariants and gives the second distinct pair. If $\alpha^2 = (\rho\sigma)^{-1}$ and $\beta = \rho\alpha$, then

$$G_{\rho^{-1},\sigma^{-1}}(U,V) = -\rho^{-1}G_{\rho,\sigma}(\alpha U, \beta V).$$

Squaring is bijective in an odd-order subgroup, proving that restricted statement as well.

The Newton polygon is the square $[0,2]^2$ with its centre as the only other support point. Its affine unimodular symmetries have signed-permutation linear parts. Interchanging the coordinates preserves (I, J); inverting one coordinate changes it to $(I^{-1}, J/I)$, the invariant pair obtained by interchanging ρ, σ. The first part therefore bounds the class by four, and the identity $G_{\sigma,\rho}(U,V) = -V^2G_{\rho,\sigma}(U,V^{-1})$ realizes the swap. There are only two distinct members when $\rho\sigma = 1$, and four otherwise. □

This rules out the proposed direct conversion of an unbounded two-parameter collision family into rescalings of one fixed polynomial by these coordinate changes. It does not rule out other curve-averaging methods, and it supplies no improvement of the active $K^{8/3}$ pointwise energy exponent.

20 A global Cayley subgroup-curve energy bound

The scaling-rigidity result above explains why the two-parameter collision family cannot be reduced to one polynomial by elementary coordinate changes. There is nevertheless a uniform bound for every subgroup size, obtained by applying the positive-characteristic curve theorem of

Corvaja–Zannier [3], *J. Eur. Math. Soc.* **15** (2013), Theorem 2/Corollary 2, to each off-diagonal ratio curve. The exponent remains $8/3$, but the earlier parity and size restrictions disappear.

Let p be an odd prime, let $H \le \mathbf{F}_p^*$ have order $K \ge 2$, and put

$$c(h) = \frac{h+1}{h-1}, \qquad A = c(H \setminus \{1\}),$$

with ratio-fibre map

$$d_\rho(x) = \frac{(\rho x - 1)(x-1)}{(1-\rho)x}, \qquad \rho \in H \setminus \{1\}.$$

For $\rho, \sigma \in H \setminus \{1\}$ define the extended collision count

$$N(\rho, \sigma) = \#\{(x, y) \in H^2 : d_\rho(x) = d_\sigma(y)\}.$$

The original pair-key energy satisfies

$$E_C := \sum_t C_H(t)^2 = E_+(A) - (K-1)^2. \tag{20.1}$$

Theorem 20.1 (Uniform Cayley curve bound)**.** *For every odd prime p, every subgroup $H \le \mathbf{F}_p^*$ of order $K \ge 2$, and every distinct $\rho, \sigma \in H \setminus \{1\}$,*

$$N(\rho, \sigma) \le \min\left\{2K, \max\left(3\sqrt[3]{32}\, K^{2/3}, \frac{48K^2}{p}\right)\right\}. \tag{20.2}$$

If

$$B(K, p) = \min\left\{2K, \max\left(3\sqrt[3]{32}\, K^{2/3}, \frac{48K^2}{p}\right)\right\}$$

and

$$D_K = 2K^2 - 7K + 4 + \gcd(2, K),$$

then

$$E_+(A) - (K-1)^2 \le D_K + (K-1)(K-2)(B(K,p) - 4). \tag{20.3}$$

Consequently, uniformly for every subgroup size,

$$E_+(A) \ll K^{8/3} + \frac{K^4}{p}, \tag{20.4}$$

with an absolute implied constant.

Proof. Clearing denominators in $d_\rho(x) = d_\sigma(y)$ gives

$$F_{\rho,\sigma}(x, y) = (1-\sigma)y(\rho x - 1)(x-1) - (1-\rho)x(\sigma y - 1)(y-1). \tag{20.5}$$

For $\rho \ne \sigma$ this has bidegree $(2, 2)$. Viewed as a quadratic in y, its discriminant is a quartic in x, and direct elimination gives

$$\operatorname{disc}_x(\operatorname{disc}_y F_{\rho,\sigma}) = 4096\rho^2\sigma^2(\rho-1)^6(\rho-\sigma)^2(\sigma-1)^6. \tag{20.6}$$

The right side is nonzero in odd characteristic for distinct nonzero nonunit parameters. Hence the quadratic discriminant is squarefree and is not a square in $\overline{\mathbf{F}}_p(x)$; therefore the curve is geometrically irreducible.

Its closure in $\mathbf{P}^1 \times \mathbf{P}^1$ has exactly the four boundary points

$$(0, 0), \quad (0, \infty), \quad (\infty, 0), \quad (\infty, \infty).$$

In the corresponding affine charts the pairs of first derivatives are

$$(\rho-1,1-\sigma),\quad (\sigma(\rho-1),1-\sigma),\quad (\rho-1,\rho(1-\sigma)),\quad (\sigma(\rho-1),\rho(1-\sigma)),$$

so none is singular. Equivalently, (20.6) exhibits the normalization as a double cover of $\mathbf{P}^1$ branched at four points, thus of genus one. Since a bidegree $(2,2)$ divisor has arithmetic genus one, there are no hidden projective singularities.

The coordinate functions x and y both have degree two. Their valuation pairs at the four boundary points are

$$(1,1),\quad (1,-1),\quad (-1,1),\quad (-1,-1),$$

which proves multiplicative independence modulo constants. With S the boundary divisor, the affine Euler characteristic entering Corvaja–Zannier is therefore

$$\chi = |S| + 2g - 2 = 4.$$

The degree-two coordinate maps are separable because p is odd, while $p \nmid K$ because $K \mid p-1$. Corollary 2 of that paper [3], applied with $m_1 = m_2 = K$, $d_1 = d_2 = 2$, and $\chi = 4$, now yields

$$N(\rho,\sigma) \le \max\left(3\sqrt[3]{32}\,K^{2/3}, \frac{48K^2}{p}\right).$$

For each fixed $x \in H$, equation (20.5) is quadratic in y, which also gives the trivial bound $N(\rho,\sigma) \le 2K$. This proves (20.2).

It remains to restore the exact pair-key domain. Extending one ratio fibre to all $x \in H$ adds precisely $x = 1$ and $x = \rho^{-1}$, and both have value zero. Hence every ordered parameter pair contributes exactly four artificial zero–zero collisions, giving the exact identity

$$E_C = \sum_{\rho,\sigma\neq 1} N(\rho,\sigma) - 4(K-1)^2. \tag{20.7}$$

On the diagonal,

$$F_{\rho,\rho}(x,y) = -(\rho-1)(x-y)(\rho xy - 1).$$

Summing the intersections of these two components over $\rho \neq 1$ gives $K - \gcd(2,K)$ and therefore

$$\sum_{\rho\neq 1} N(\rho,\rho) = 2K(K-1) - K + \gcd(2,K).$$

After subtracting the four artificial collisions for every diagonal ratio, the total diagonal contribution to E_C is exactly D_K. There are $(K-1)(K-2)$ ordered off-diagonal parameter pairs; inserting (20.2) into (20.7) proves (20.3). Finally $B(K,p) \ll K^{2/3} + K^2/p$, and the diagonal term is $O(K^2)$, which gives (20.4). □

Remark 20.2 (Range and sharp frontier)**.** *The characteristic-free term dominates whenever $128K^4 \le p^3$, equivalently $K \le 2^{-7/4}p^{3/4}$. Its coefficient is $3\sqrt[3]{32} = 9.5244063118\ldots$. This removes the old odd-order and subgroup-size restrictions from the project curve route, but it does not improve the exponent. The centered pointwise target*

$$E_+(A) = \frac{(K-1)^4}{p} + O\Big(K^{2+o(1)}\Big)$$

remains open and requires cancellation across the two-parameter family, not merely a worst-curve estimate.

Remark 20.3 (Exact regression boundary)**.** *The maintained exact verifier checks the discriminant identity, all four boundary gradients, diagonal factorization, restoration correction and the integer form of the energy estimate for every subgroup order $K \ge 3$ occurring for odd primes $p \le 257$. This is a finite algebra and implementation regression; it does not re-prove the external Corvaja–Zannier theorem.*

20.1 Dense-regime Fourier complement

The genus-one curve estimate is strongest in the sparse and intermediate regimes. A different argument gives a centered estimate for dense subgroups. Let $m = (p-1)/K$, $n = K-1$, and use the unnormalised transform

$$\widehat{1_A}(q) = \sum_{a \in A} e_p(qa).$$

If $S = H^\perp$ is the group of multiplicative characters trivial on H, then $|S| = m$ and for $q \neq 0$

$$\widehat{1_A}(q) = \frac{1}{m} \sum_{\chi \in S} T_\chi(q), \qquad T_\chi(q) = \sum_{h \in \mathbf{F}_p^*,\ h \neq 1} \chi(h) e_p\Big(q \frac{h+1}{h-1}\Big).$$

The trivial character contributes exactly $-e_p(q) - e_p(-q)$. For $\chi \neq 1$, the substitution $z = (h+1)/(h-1)$ gives the mixed multiplicative–additive sum

$$T_\chi(q) = \sum_{z \in \mathbf{F}_p \setminus \{-1,1\}} \chi\Big(\frac{z+1}{z-1}\Big)\, e_p(qz).$$

The classical mixed Weil bound gives $|T_\chi(q)| \leq 2\sqrt{p}$; the Kummer factor has one simple zero and one simple pole and the additive factor has its sole pole at infinity. Consequently

$$|\widehat{1_A}(q)| \leq M_m(p) := \frac{2((m-1)\sqrt{p}+1)}{m} \leq 2\sqrt{p}. \tag{20.8}$$

Theorem 20.4 (Dense centered Cayley energy bound)**.** *For every odd prime p and every subgroup $H \leq \mathbf{F}_p^*$ of order $K \geq 2$, with $A = c(H \setminus \{1\})$, $n = K-1$ and $m = (p-1)/K$,*

$$0 \leq E_+(A) - \frac{n^4}{p} \leq \frac{M_m(p)^2}{p}\, n(p-n) \tag{20.9}$$

$$= 4\frac{((m-1)\sqrt{p}+1)^2}{m^2 p}\, n(p-n) \tag{20.10}$$

$$\leq 4n(p-n). \tag{20.11}$$

Combining this with Theorem 20.1,

$$E_+(A) - \frac{(K-1)^4}{p} \ll \min\Big\{pK,\ K^{8/3} + \frac{K^4}{p}\Big\}.$$

Thus if $K = p^{\alpha+o(1)}$ with $\alpha > 3/5$, the centered error has strictly smaller exponent than $K^{8/3}$; in K-notation the Fourier branch is $K^{1+1/\alpha+o(1)}$. For example $\alpha = 2/3$ gives a $K^{5/2+o(1)}$ centered bound.

Proof. Parseval gives

$$\sum_q |\widehat{1_A}(q)|^2 = pn, \qquad \sum_{q \neq 0} |\widehat{1_A}(q)|^2 = n(p-n).$$

Also

$$E_+(A) = \frac{1}{p} \sum_q |\widehat{1_A}(q)|^4.$$

Apply (20.8) to one factor of the nonzero-frequency fourth moment and Parseval to the other. This proves (20.11); the hybrid estimate follows from Theorem 20.1. □

Remark 20.5. *This is a genuine dense-regime improvement, not the missing sparse theorem. The required inversive subgroup sums below the square-root scale are not controlled by this complete mixed-Weil argument. In particular the centered $K^{2+o(1)}$ target remains open in the arithmetic regime relevant to the present BGS route. The maintained Python and independent C++ exact regressions check 807 subgroup designs through $p \le 509$, including 364 strict dense-side cases $K^5 > p^3$ (certificate `cayley_fourier_hybrid_20260907.json`, indexed in `evidence/provenance.md`).*

21 Magnetic quotients: the restricted global semicircle regime

This is a complete project derivation, not a claim of priority or an externally refereed theorem. It concerns the **global empirical spectral measure of a compressed observable**, not microscopic BGS statistics of a quantum propagator. The arithmetic restriction is strong: prime subgroup orders $K = o(\log p)$. Subsection 21.9 proves that this regime is incompatible with the orders of any one fixed hyperbolic integral cat map. The exact operator identities and computational representation do not require that restriction.

The argument uses an explicit finite Heisenberg-group product, not the historical unverified word-rewrite claims. Independent implementations check its finite algebra; they are not substitutes for the limiting proof.

21.1 The exact matrix being studied

Let p be an odd prime, let $H \le \mathbb{F}_p^*$ have odd order K, and let $q \in \mathbb{F}_p^*$. All exponents and fractions inside $e_p(t) = \exp(2\pi i t/p)$ are interpreted in $\mathbb{F}_p$. On $\mathbb{C}^{\mathbb{F}_p}$, with the counting-measure inner product, define

$$W(a,b)f(x) = e_p(a(x-b/2))f(x-b), \qquad W_q = W(q,1),$$

$$S_h f(x) = f(h^{-1}x), \quad Q = \frac{1}{K}\sum_{h\in H} S_h, \quad R = S_{-1}, \quad P_- = (I-R)/2.$$

The projections Q, P_- commute. Set

$$\mathcal{E} = \operatorname{ran}(QP_-), \quad n = \dim \mathcal{E} = \frac{p-1}{2K}, \qquad B_q = W_q + W_q^*,$$

$$A_q = (QB_qQ)|_{\mathcal{E}}, \qquad X_q = \sqrt{K/2}\, A_q, \qquad \mu_m(q) = \frac{1}{n}\operatorname{Tr}_{\mathcal{E}} X_q^m. \tag{MQ.1}$$

The dimension follows by using signed indicators of the nonzero orbits of $H \cup (-H)$; odd functions vanish at zero. The identity $RW_qR = W_q^*$ shows that B_q preserves parity. Thus A_q is Hermitian and $\|A_q\| \le 2$. The notation A_q here is an operator, not the Cayley set in other notes.

For each odd multiplicative character χ trivial on H, extend $f_\chi(x) = \chi(x)/\sqrt{p-1}$ by zero at $x = 0$. These form an orthonormal basis of $\mathcal{E}$. Direct substitution gives

$$M_{\chi,\psi} = \frac{e_p(-q/2)}{p-1}\sum_{x\ne 0,1} \overline{\chi(x)}\psi(x-1)e_p(qx), \qquad [A_q] = M + M^*. \tag{MQ.2}$$

This is precisely the centered full-complex matrix in the maintained `spectral_diagnostics.py` (function `hermitian_matrix` in `computations/analysis/spectral_diagnostics.py`), not a surrogate ensemble. Compression of $RW_qR = W_q^*$ to one parity sector also gives $M = M^*$; it does not make M entrywise real.

21.2 An explicit antiunitary and an isospectral real gauge

Define

$$(F_q f)(x) = p^{-1/2} \sum_y e_p(qxy) f(y), \qquad (Cf)(x) = \overline{f(x)}, \qquad \Theta_q = F_q C.$$

Proposition 21.1. *The map Θ_q is an antiunitary involution preserving $\mathcal{E}$ and commuting with A_q. For a nontrivial character,*

$$\Theta_q f_\chi = a_\chi(q) f_\chi, \qquad a_\chi(q) = \frac{\chi(q)\tau(\overline{\chi})}{\sqrt{p}}, \qquad \tau(\overline{\chi}) = \sum_{y \neq 0} \overline{\chi(y)} e_p(y), \quad |a_\chi(q)| = 1. \tag{MQ.3}$$

Choose any square roots $d_\chi^2 = a_\chi(q)$ and put $D = \mathrm{diag}(d_\chi)$. Then $D^*[A_q]D$ is real symmetric and has exactly the spectrum of A_q.

Proof. Fourier orthogonality gives $F_q \overline{F_q} = I$. Substitution $y = z + 1$ in $\Theta_q W_q$ gives

$$e_p(qxy - q(y - 1/2)) = e_p(q(x - 1/2)) e_p(q(x - 1)z),$$

which proves $\Theta_q W_q = W_q \Theta_q$. Changing variables in the Fourier sum yields $\Theta_q S_h = S_{h^{-1}} \Theta_q$, hence invariance of Q, R. For $x \neq 0$, substitution $u = qxy$ proves (MQ.3); at zero the character sum vanishes. Expanding the square of the Gauss sum and using orthogonality gives $|\tau(\chi)|^2 = p$. Finally $\Theta_q(d_\chi f_\chi) = d_\chi f_\chi$. A commuting operator has real entries in this fixed basis, and Hermiticity makes them symmetric. □

This is a geometric conjugation, not conjugation chosen after diagonalizing an arbitrary Hermitian matrix. Nonetheless an antiunitary symmetry alone is not a GOE universality theorem. In particular, the old operation of taking $[A_q]$.real is not reinstated: it is generally not isospectral.

21.3 Orbit assembly with only p minus two contributions

Choose representatives r_i of $\mathbb{F}_p^*/(H \cup -H)$. Let $i(x)$ be the orbit index, and let $s(x)$ be $+1$ on Hr_i and -1 on $-Hr_i$. The orbit basis is $v_i(x) = s(x)/\sqrt{2K}$ on its orbit. Consequently

$$[M]_{ij}^{\mathrm{orb}} = \frac{1}{2K} \sum_{\substack{x \neq 0,1 \\ i(x)=i,\ i(x-1)=j}} s(x)s(x-1)e_p(q(x-1/2)), \qquad [A_q]^{\mathrm{orb}} = M^{\mathrm{orb}} + (M^{\mathrm{orb}})^*. \tag{MQ.4}$$

There are $p - 2$ contributions, so sparse arithmetic assembly costs $O(p)$ and there are at most $O(p)$ nonzero locations before coalescing. This avoids a naive $O(n^2 p)$ character-sum construction. The reference implementation returns a **dense** $n \times n$ matrix, so its storage is $O(n^2)$; sparse assembly does not remove eigensolution costs. The overlap of the two orthonormal bases proves exact unitary equivalence.

21.4 All-order trace identity with every word case included

The Weyl and dilation rules are

$$W(a,b)W(c,d) = e_p((ad - bc)/2) W(a + c, b + d), \qquad S_g W(a,b) S_g^{-1} = W(a/g, gb). \tag{MQ.5}$$

Expand $(B_q Q)^m$ into words with $\epsilon_j \in \{-1, 1\}$ and $h_j \in H$. Starting with $(g, a, b, c) = (1, 0, 0, 0)$, update at each letter by

$$a' = a + \epsilon/g, \quad b' = b + \epsilon g, \quad c' = c + \epsilon(ag - b/g)/2, \quad g' = gh. \tag{MQ.6}$$

The product is $e_p(qc) W(qa, b) S_g$. Define

$$T_g(a,b) = \begin{cases} p\, \mathbf{1}_{a=b=0}, & g = 1, \\ e_p\Big(\dfrac{qab(1+g)}{2(1-g)}\Big), & g \neq 1. \end{cases} \tag{MQ.7}$$

Proposition 21.2. *For every integer $m \geq 1$,*

$$\operatorname{Tr}_{\mathcal{E}} A_q^m = \frac{1}{2K^m} \sum_{\epsilon_1,\dots,\epsilon_m;\ h_1,\dots,h_m} e_p(qc)\big(T_g(a,b) - T_{-g}(a,b)\big). \tag{MQ.8}$$

Proof. *A diagonal entry of $W(qa,b)S_g$ requires $x = g^{-1}(x-b)$. For $g \neq 1$, its unique solution is $x = b/(1-g)$, giving (MQ.7). For $g = 1$, one needs $b = 0$; orthogonality of the remaining additive character gives the first line, since $q \neq 0$. Inserting R changes g to $-g$ and a, b to their negatives, preserving ab. The trace of the compressed power equals $\operatorname{Tr}(P_-(B_qQ)^m)$ by cyclicity and idempotence of Q. Expansion proves (MQ.8). Since H has odd order, $-g$ is never 1. No other cases are omitted.* □

In particular there is an integer, possibly signed, histogram $C_m(t)$ such that

$$\operatorname{Tr} A_q^m = \frac{1}{2K^m} \sum_t C_m(t) e_p(qt), \qquad \mathbb{E}_{q\neq 0} \operatorname{Tr} A_q^m = \frac{pC_m(0) - \sum_t C_m(t)}{2K^m(p-1)}. \tag{MQ.9}$$

Histograms differing by a constant vector are equal at every nontrivial pth root. Negative coefficients cannot be discarded.

21.5 Reciprocal short-relation rigidity

Lemma 21.3. *Suppose K is prime, $1 \leq m < K$, and*

$$p^2 > m^{K-1}. \tag{MQ.10}$$

If $t_j \in H$ and $\epsilon_j \in \{-1, 1\}$ satisfy

$$\sum_j \epsilon_j t_j = \sum_j \epsilon_j / t_j = 0 \quad \text{in } \mathbb{F}_p, \tag{MQ.11}$$

each label t occurs equally often with positive and negative sign. In particular m is even.

Proof. Write $t_j = h^{e_j}$, $0 \leq e_j < K$, and collect coefficients in $P(Z) = \sum_j \epsilon_j Z^{e_j}$. Its coefficient ℓ^1 norm is at most m. If $P(\zeta_K) \neq 0$, its norm N is a nonzero integer with

$$|N| = \prod_{a=1}^{K-1} |P(\zeta_K^a)| \leq m^{K-1}. \tag{MQ.12}$$

Multiplication by P in $\mathbb{Z}[Z]/\Phi_K$ has determinant N. Modulo p, Φ_K splits into distinct roots h^a, and (MQ.11) makes two distinct eigenvalues zero, at h, h^{-1}. Its nullity is at least two; Smith normal form then gives $p^2 \mid N$, contradicting (MQ.10)–(MQ.12). Thus $P(\zeta_K) = 0$. Irreducibility of Φ_K follows from Eisenstein applied after $Z \mapsto Z + 1$. Since $\deg P \leq K - 1$, we have $P = c\Phi_K$ for an integer c. A nonzero c would give coefficient ℓ^1 norm at least $K > m$. Hence $P = 0$ coefficientwise, proving balance. □

Both reciprocal equations and the strict cutoff are used. This sufficient cutoff is not an exact characterization of exceptional primes.

21.6 Main term and an explicit remainder

For words ending with $g = 1$, put $t_1 = 1$ and $t_j = h_1 \cdots h_{j-1}$ for $j \geq 2$. The labels $t_2, \dots, t_m$ are arbitrary in H, and the final h_m is determined. The state becomes

$$a = \sum_j \epsilon_j / t_j, \quad b = \sum_j \epsilon_j t_j, \quad c = \frac{1}{2} \sum_{i<j} \epsilon_i \epsilon_j (t_j/t_i - t_i/t_j).$$

Equations (MQ.1) and (MQ.8) give

$$\mu_m(q) = \frac{p}{p-1}\frac{1}{2^{m/2}K^{m/2-1}} \sum_{\substack{t_1=1,\ t_2,\dots,t_m\in H;\ \epsilon \\ a=b=0}} e_p(qc) + \mathcal{R}_m(q), \tag{MQ.13}$$

$$|\mathcal{R}_m(q)| \le \frac{2^{m/2+1}K^{m/2+1}}{p-1}. \tag{MQ.14}$$

Indeed every trace outside the $g = 1, a = b = 0$ contribution has modulus at most one; bounding the two traces of each of $(2K)^m$ words by two gives (MQ.14). Under Lemma 21.3 odd moments have no main term.

21.7 Pairings, overcounting, and crossing cancellation

Let $m = 2r$ and assume $2r < K$ throughout this subsection; this is the hypothesis under which the grid zero bound at (MQ.17) is applied, and it is also the hypothesis of Theorem 21.4. Every balanced tuple admits a matching of the $2r$ positions, with equal labels and opposite signs on matched positions. Write

$$P_r = (2r-1)!!, \qquad C_r = \frac{1}{r+1}\binom{2r}{r}.$$

These are the numbers of all and of noncrossing matchings. Order pairs by their first endpoints, so pair 1 contains position 1. Assign labels $x_1 = 1, x_2, \dots, x_r \in H$ and orientations $\sigma_i \in \{-1, 1\}$. With distinct labels the matching is unique. Overcounting is possible only when two labels coincide. A union bound bounds the number of bad matching assignments by

$$P_r 2^r \binom{r}{2} K^{r-2}, \tag{MQ.15}$$

interpreted as zero for $r = 1$. Each balanced tuple has at least one matching; therefore (MQ.15) also bounds the excess multiplicity of the replacement sum.

Summing the four interactions between two matched pairs gives the exact Laurent polynomial

$$F_{\pi,\sigma}(x) = \sum_{\substack{i<j \\ \text{pairs } i,j \text{ cross}}} \sigma_i\sigma_j(x_j/x_i - x_i/x_j). \tag{MQ.16}$$

Disjoint or nested pairs give zero; interlaced pairs give the indicated term. Noncrossing matchings have $F = 0$. Any crossing matching has nonconstant F. Indeed, a crossing of the unordered pair $\{i, j\}$ contributes the two monomials x_j/x_i and x_i/x_j, with coefficients $\sigma_i\sigma_j$ and $-\sigma_i\sigma_j$; their exponent vectors $\pm(e_j - e_i)$ determine the unordered pair $\{i, j\}$ from their support and the orientation from the sign pattern, so monomials contributed by distinct crossings are distinct. This persists after the normalization $x_1 = 1$: a crossing $\{1, j\}$ contributes $x_j^{\pm 1}$, whose exponent vectors $\pm e_j$ are supported on one coordinate, whereas a crossing $\{i, j\}$ with $i, j \ge 2$ contributes vectors supported on two coordinates, and different pairs give different supports. Hence every monomial of $F_{\pi,\sigma}$ occurs exactly once, with coefficient ± 1, nonzero in odd characteristic; no cancellation is possible, and F is nonconstant as soon as one crossing occurs.

For a crossing let $d = r - 1$ and

$$S_F(q) = K^{-d} \sum_{x\in H^d} e_p(qF(x)).$$

Multiplication of $F(x) - F(y)$ by all $2d$ variables clears denominators and gives a nonzero polynomial of total degree at most $2d + 1 = 2r - 1$. Since $2r - 1 < K$, the grid zero bound

therefore gives at most $(2r-1)K^{2d-1}$ equal-value pairs. For completeness, that bound follows by induction on the number of variables: separate the leading coefficient in the last variable, use the univariate root bound off its zero set, and induct on that coefficient on the remaining fibres. Additive orthogonality now yields

$$\mathbb{E}_{q\neq 0}|S_F(q)|^2 = \frac{pK^{-2d}\#\{(x,y) : F(x) = F(y)\} - 1}{p-1} \leq \frac{p(2r-1)}{(p-1)K}. \tag{MQ.17}$$

This uses no independence assumption for matrix entries, words, or crossings.

21.8 Quantitative theorem and limiting law

Theorem 21.4. *Let K be an odd prime and let p be prime with $K \mid p-1$. For $2r < K$ and $p^2 > (2r)^{K-1}$, uniform nonzero q satisfies*

$$\begin{aligned}\|\mu_{2r} - C_r\|_{L^2(q\neq 0)} \leq{}& \frac{C_r}{p-1} \\ &+ \frac{p}{p-1}(P_r - C_r)\sqrt{\frac{p(2r-1)}{(p-1)K}} \\ &+ \frac{p}{p-1}\frac{P_r\binom{r}{2}}{K} + \frac{2^{r+1}K^{r+1}}{p-1}.\end{aligned} \tag{MQ.18}$$

For odd $m < K$ under $p^2 > m^{K-1}$,

$$\sup_{q\neq 0}|\mu_m(q)| \leq \frac{2^{m/2+1}K^{m/2+1}}{p-1}. \tag{MQ.19}$$

Proof. *Replace the balanced tuples of (MQ.13) by matching assignments. After division by $2^r K^{r-1}$, (MQ.15) gives the third term of (MQ.18). Each noncrossing matching contributes exactly one; together they give $pC_r/(p-1)$, whose difference from C_r is the first term. For each crossing average over its 2^r orientations, apply (MQ.17), and apply the L^2 triangle inequality over the $P_r - C_r$ matchings. This is the second term. The last term is (MQ.14). Lemma 21.3 and (MQ.14) give (MQ.19).* □

Corollary 21.5 (global law only)**.** *Suppose $K \to \infty$ through odd primes, $p \equiv 1 \pmod K$ is prime, and $K = o(\log p)$. Every fixed moment of the empirical spectral measure of X_q converges in L^2 over nonzero q to the standard semicircle moment. The empirical measures converge weakly in probability to*

$$\rho_{\mathrm{sc}}(x)\,dx = \frac{\sqrt{4-x^2}}{2\pi}\mathbf{1}_{[-2,2]}(x)\,dx. \tag{MQ.20}$$

The averaged empirical measures converge weakly as well.

Proof. For every fixed m, eventually $m < K$ and (MQ.10) holds, while every fixed power of K divided by p tends to zero. Equations (MQ.18)–(MQ.19) give the moment limits. From any subsequence, select a further subsequence on which all integer moments converge almost surely, by diagonal extraction. Bounded second moments give tightness; bounded higher even moments give uniform integrability for each lower moment. Every weak subsequential limit therefore has the semicircle moments. Since $C_r \leq 4^r$, any measure with these moments is supported on $[-2,2]$; polynomial approximation on that compact interval proves uniqueness. The subsequence criterion gives weak convergence in probability. Bounded continuous tests then also converge in expectation, proving the averaged assertion. □

This does not assert eigenvalue rigidity, an edge law, microscopic spacing statistics, or estimates uniform for m growing with n.

21.9 Obstruction for one fixed cat map and remaining work

Let $T \in \mathrm{SL}_2(\mathbb{Z})$ be one fixed hyperbolic matrix with order K modulo p. Then $T^K - I$ vanishes modulo p. At least one of its integer entries is nonzero, and its absolute value is at most $C_T \Lambda_T^K$, for constants depending only on T. Hence

$$p \le C_T \Lambda_T^K, \qquad \log p \le K \log \Lambda_T + \log C_T. \tag{MQ.21}$$

Thus $K = o(\log p)$ is impossible along such an order sequence. On the split semisimple locus the eigenvalue order equals the matrix order, so that interpretation does not avoid the obstruction.

The new global law is a theorem about an arithmetic matrix-design regime, not a fixed-cat-map BGS theorem. The exact trace formula, antiunitary, and orbit assembly remain valid outside the cutoff. The next substantive step is to control non-balanced reciprocal words in polynomial regimes rather than eliminate them with an exponential norm bound. Microscopic BGS then still requires a specified desymmetrized propagator and Heisenberg-scale joint-trace estimates. Neither assertion is proved here.

22 Exact finite spectra for arbitrary hyperbolic matrices

The following finite character argument is independent of the Gaussian and normal-form arguments below. In particular it does not assume a split torus or a prime Hilbert-space dimension. This section reconciles the positive- and negative-trace notes using one sign convention.

22.1 A fixed-point-free cyclic action

Let $N \ge 3$ and $q \ge 3$ be odd integers, and suppose $A \in \mathrm{SL}(2, \mathbb{Z}/N\mathbb{Z})$ satisfies

$$A^q = I, \qquad \det(A^j - I) \in (\mathbb{Z}/N\mathbb{Z})^\times \quad (0 < j < q). \tag{22.1}$$

Use modular-centred Weyl operators $W(v)$ and $R\psi(x) = \psi(-x)$. The assignment $W(v) \mapsto W(Av)$ is a $*$-automorphism of the full matrix algebra, since it preserves the Weyl product and adjoint. It is implemented by a unitary U, unique up to a scalar; this follows by mapping a complete system of matrix units. No field hypothesis is required.

Theorem 22.1 (Uniform parity sector)**.** *Under* (22.1)*, $N \equiv c \pmod q$ for exactly one $c \in \{1, -1\}$. A scalar normalization of U gives $U^q = I$ and*

$$\mathrm{Tr}(RU^j) = 1 \quad \textit{for every } j, \qquad \mathrm{Tr}\, U^j = c \quad (q \nmid j).$$

On $\mathcal{H}_c = \mathrm{ran}((I - cR)/2)$, every qth root has the same multiplicity $n = (N - c)/(2q)$, and $\dim \mathcal{H}_c = (N - c)/2$. On the complementary parity sector the multiplicities are n at nontrivial roots and $n + c$ at the root 1.

Proof. Conjugation by U^j permutes the Weyl basis. Its trace is consequently the number of fixed Weyl indices, while the trace of conjugation by a unitary V is $|\mathrm{Tr}\, V|^2$. Thus

$$|\mathrm{Tr}\, U^j|^2 = 1 \quad (0 < j < q), \qquad |\mathrm{Tr}(RU^j)|^2 = 1 \quad \text{for all } j.$$

For the second identity, a vector with $A^j v = -v$ is zero: applying the relation q times gives $v = -v$ and 2 is a unit. The same argument modulo each prime divisor of N proves invertibility, including prime powers. The operators RUR and U implement the same covariance. Their scalar ratio is one because $\mathrm{Tr}\, U \neq 0$. Normalize $U^q = I$.

Let b_l be the difference between the even and odd multiplicities of the lth root. Fourier inversion for the virtual cyclic character $\mathrm{Tr}(RU^j)$ gives $b_l \in \mathbb{Z}$, $\sum_l b_l^2 = 1$, and $\sum_l b_l = \mathrm{Tr}\, R = 1$.

Exactly one coefficient is one. A further root-of-unity scalar puts it at $l = 0$. Hence $\mathrm{Tr}(RU^j) = 1$ for every j. For the full multiplicities m_l, Parseval yields

$$\sum_l m_l = N, \qquad \sum_l m_l^2 = (N^2 + q - 1)/q.$$

Write $N = qk+r$, $0 \le r < q$. The least sum of squares of q integers with this sum is $qk^2+2kr+r$: moving a pair which differs by two or more closer together reduces its squared sum. It follows that $r(q-r) \le q-1$. Integrality also forces $r^2 \equiv 1 \pmod q$, so $r \ne 0$; therefore $r = 1$ or $q-1$. For either residue the squared-sum minimum is attained. All m_l equal $(N-c)/q$ except one, which differs by c. Since $(N-c)/q$ is even and $b_0 = 1$, that exception must be at zero. Splitting by the already known parity differences gives the claimed formulas. □

22.2 An explicit sequence, including negative trace

Let now $A \in \mathrm{SL}(2,\mathbb{Z})$ be hyperbolic, put $a = |\operatorname{tr} A| \ge 3$, $s = \operatorname{sgn}(\operatorname{tr} A)$, and let $u_0 = 0, u_1 = 1, u_{j+1} = au_j - u_{j-1}$. For every odd prime $K > 3$ not dividing $a - 2s$, set

$$m = (K-1)/2, \qquad N_K = u_{m+1} + su_m, \qquad c_K = \left(\frac{a-2s}{K}\right). \tag{22.2}$$

Here the last expression is a Legendre symbol, not an eigenvalue of A. Then N_K is odd and

$$A^K - I = s^m N_K A^m (A - I). \tag{22.3}$$

Moreover $\gcd(N_K, a-2s) = 1$, so Theorem 22.1 applies with $q = K$ and sign $c = c_K$.

Here are details of the signed extension. Put $B = sA$, of trace a. Summing $A^j = s^j B^j$ symmetrically around m and using Cayley–Hamilton yields $\sum_{j=0}^{K-1} A^j = s^m N_K A^m$, proving (22.3). For $s = 1$, the polynomial $N_K(a)$ evaluates to K at a=2. For $s = -1$, it evaluates to $(-1)^m K$ at a=-2. These evaluations prove the gcd assertion under the stated exclusion. Modulo 2, the recurrence is alternating when a is even and has period three when a is odd; the only obstruction is $K = 3$, already excluded. If $A^j v = v$ modulo a prime dividing N_K with $0 < j < K$, Bezout applied to j,K and (22.3) gives $Av = v$. Invertibility of $A - I$ then gives $v = 0$. Thus all required determinants are units.

The sign can be evaluated without a square root modulo N_K. For positive trace the integer polynomial identity

$$N_K(t) = 2^{1-K} \sum_{\substack{1 \le j \le K \\ j \text{ odd}}} \binom{K}{j} (t+2)^{(K-j)/2} (t-2)^{(j-1)/2}$$

gives $N_K(t) \equiv (t-2)^{(K-1)/2} \pmod K$. For negative trace apply it at $t = -a$ and use $N_K^{(+)}(-a) = (-1)^m N_K^{(-)}(a)$. Euler's criterion proves (22.2). With $\lambda = (a + \sqrt{a^2-4})/2$ one also obtains

$$N_K = \frac{\lambda^{K/2} - s\lambda^{-K/2}}{\sqrt{a-2s}}, \qquad K \asymp \log N_K.$$

For trace four and $K = 11$, $N_K = 989 = 23 \cdot 43$, $c = -1$: the uniform sector is even, of dimension 495 and multiplicity 45. The discriminant 12 is a square modulo 23 and a nonsquare modulo 43. For trace -6 and $K = 7$, $N_K = 169 = 13^2$, $c = 1$: the uniform odd sector has dimension 84 and multiplicity 12. These examples require neither a split characteristic polynomial nor squarefreeness.

22.3 The characteristic for general integer covariance

The physical lift $Q_N(a,b) = (-1)^{ab}W_N(a,b)$ used for T is not invariant under every integer symplectic matrix. For general A use instead

$$Q_N^*(a,b) = (-1)^{ab+a+b}W_N(a,b).$$

The function $ab + a + b$ modulo two equals one on all three nonzero vectors, so it is invariant under $\mathrm{SL}(2,\mathbb{F}_2)$. Therefore

$$Q_N^*(v)Q_N^*(w) = e^{\pi i\omega(v,w)/N}Q_N^*(v+w), \qquad U_N Q_N^*(v)U_N^* = Q_N^*(Av).$$

This is an explicit quantization convention, not a claim that the old and new finite matrices coincide. The additional sign represents a fixed half-torus translation of symbols. Parity still corresponds to $x \mapsto -x$. Whether the locality and return-word arguments later in the paper transfer to this characteristic and a given A depends on their remaining inputs, which are not checked here. What is verified here is exactly the displayed Weyl product law with the same cocycle, exact A-covariance for every integer A, and the parity correspondence. The transfer is therefore conditional on (i) finite support towers for the A-orbit with the gap and noncancellation properties those arguments assume, and (ii) the finite-order and unit-determinant hypotheses on A modulo N, in the form of (22.1), wherever they are used. No verbatim transfer to an arbitrary such A is asserted. The associated classical map is $S = A^{\mathsf{T}}$.

Chapter 4 — The Pell cat map

23 Global spectral blocks on a fixed Pell cat map

Unlike the separate magnetic-quotient semicircle theorem, this result uses a fixed smooth Fourier observable, deterministic parameters, and generally composite Pell moduli. No assertion that infinitely many Pell numbers are prime is needed. The two limiting laws concern different regimes and are not interchangeable.

23.1 Arithmetic moduli and a fixed classical map

Put

$$\alpha = 1 + \sqrt{2}, \qquad \lambda = \alpha^2 = 3 + 2\sqrt{2}, \qquad T = \begin{pmatrix} 3 & 2 \\ 4 & 3 \end{pmatrix}.$$

The integral matrix T has determinant one and eigenvalues λ and its reciprocal, so it is one fixed hyperbolic toral automorphism. Also T is the identity modulo 2, avoiding the usual centered-Weyl parity ambiguity.

Define integers Q_j,P_j by

$$\alpha^j = Q_j + P_j\sqrt{2}, \quad Q_0 = 1,\ P_0 = 0, \quad Q_{j+1} = Q_j + 2P_j,\ P_{j+1} = Q_j + P_j.$$

Let K tend to infinity through odd primes and set

$$N = Q_K = \frac{\alpha^K - \alpha^{-K}}{2}, \qquad F = P_K.$$

Then

$$N^2 - 2F^2 = -1, \qquad N \asymp \lambda^{K/2}.$$

Every Q_j is odd, and $\gcd(F, N) = 1$. In $\mathbb{Z}/N\mathbb{Z}$ define

$$s = F^{-1}, \qquad s^2 = 2, \qquad h = 3 + 2s, \qquad q_0 = -F.$$

The identity $(1+s)^K = Q_K + P_K s = 1$ gives $h^K = 1$. Moreover, $h - 1 = 2(1+s)$ is a unit since $(1+s)(1-s) = -1$. Modulo each prime $\ell \mid N$, the order of h is exactly K: it divides the prime K and is not one. Hence $\ell \equiv 1 \pmod{K}$. In particular,

$$1 - h^j \text{ is a unit for } 1 \le j < K, \qquad 1 + h^j \text{ is a unit for every } j. \tag{PB.1}$$

The latter assertion uses odd K. Thus $H = \langle h \rangle$ has order K and acts freely on every nonzero residue, not just on the units. The signed orbits have size $2K$: a stabilizer would give $h^j = \pm 1$ modulo some prime $\ell \mid N$ at which the residue is nonzero, but modulo ℓ the element h has order exactly the odd prime K, while $-1 \neq 1$ has order two. Hence

$$n = \frac{N-1}{2K}$$

is an integer. The moduli need not be prime: for example $Q_{11} = 8119 = 23 \cdot 353$.

23.2 Operators and identification with a fixed observable

On $\mathbb{C}^{\mathbb{Z}/N\mathbb{Z}}$, with counting-measure inner product, write

$$W(a,b)f(x) = e_N(a(x - b/2))f(x-b), \qquad S_u f(x) = f(u^{-1}x), \qquad Rf(x) = f(-x).$$

Here $e_N(t) = e^{2\pi i t/N}$, and divisions in exponents mean inverses modulo N. The symmetrized observables and spectral projectors preserve the odd subspace $R = -1$. Individual Weyl translations do not, in general, preserve parity. For a character η of H, let E_η be the η eigenspace of S_h in that odd subspace, and let P_η denote its orthogonal projection. Its dimension is n.

Set

$$B_q = W(q,1) + W(q,1)^*, \qquad A_{q,\eta} = P_\eta B_q P_\eta|_{E_\eta}, \qquad X_{q,\eta} = \sqrt{K/2}\, A_{q,\eta}. \tag{PB.2}$$

In the theorem $q = q_0$ is deterministic and is a unit modulo N.

This is a compression of a fixed Fourier observable of the fixed cat map, not merely a modulus-dependent choice of an unrelated classical map. Indeed, over $\mathbb{Z}/N\mathbb{Z}$ put

$$M = \begin{pmatrix} -1/(2s) & 1 \\ -1/2 & -s \end{pmatrix}, \qquad T = M \operatorname{diag}(h, h^{-1}) M^{-1}. \tag{PB.3}$$

The matrix M has determinant one. Define $C_t f(x) = e_N(tx^2/2)f(x)$ and the unitary Fourier transform F_N with kernel $e_N(xy)/\sqrt{N}$. Direct substitution gives their Weyl covariance matrices

$$C_t : \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}, \quad S_u : \begin{pmatrix} u^{-1} & 0 \\ 0 & u \end{pmatrix}, \quad F_N : \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

Therefore

$$V = C_{s^{-1}} S_{1/2} F_N C_{2s}, \qquad U_N(T) = V S_h^{-1} V^* \tag{PB.4}$$

satisfies the exact finite Egorov identity

$$U_N(T) W(v) U_N(T)^* = W(Tv).$$

These formulas also prove unitarity without invoking a prime-modulus-only Weil construction. They preserve parity. A global phase convention for the cat propagator does not affect any compression or relative eigenphase below.

Take the fixed observable

$$B = W(1,0) + W(-1,0), \qquad Bf(x) = 2\cos(2\pi x/N) f(x). \tag{PB.5}$$

Under conjugation by V^*, its Weyl index is $M^{-1}(1,0) = (-s, 1/2)$. Conjugating further by S_2, which commutes with S_h, gives $(-s/2, 1) = (-F, 1)$. Thus the compressions of (PB.5) to every odd cat eigenspace are unitarily equivalent to $A_{q_0,\eta}$, with a harmless permutation of sector labels.

23.3 Main theorem

Theorem 23.1 (deterministic global Gaussian blocks)**.** *Let K run through odd primes, with N,Q_K,F,h,q_0 as above. For every fixed integer $m \geq 1$,*

$$\sup_{\eta} \left| \frac{1}{n} \operatorname{Tr} X_{q_0,\eta}^m - \gamma_m \right| \longrightarrow 0, \qquad \gamma_{2r} = (2r-1)!!, \quad \gamma_{2r+1} = 0. \tag{PB.6}$$

Consequently the empirical spectral measures converge weakly to the standard normal distribution, uniformly over all K odd-sector cat eigenspaces. There is no random q in this theorem. For fixed even m the proof gives an $O_m(\log K/K)$ error, apart from an exponentially small polynomial-in-K term. For fixed odd m the main word contribution is eventually absent.

The name Gaussian here describes the limiting **eigenvalue density**. It does not assert that the matrices form a Gaussian orthogonal ensemble.

23.4 Exact traces over the composite ring

The finite Weyl product remains

$$W(a,b)W(c,d) = e_N((ad-bc)/2) W(a+c, b+d).$$

In an expansion of $(B_q P_\eta)^m$, let $\epsilon_j \in \{1,-1\}$ and $h_j \in H$. Starting with $(g,a,b,c) = (1,0,0,0)$, use

$$a' = a + \epsilon/g, \quad b' = b + \epsilon g, \quad c' = c + \epsilon(ag - b/g)/2, \quad g' = gh. \tag{PB.7}$$

The word is $e_N(qc) W(qa, b) S_g$, with the additional sector weight $\overline{\eta(g)}$. For $g \neq 1$, (PB.1) gives a unique affine fixed point, so the trace has modulus one and phase $qab(1+g)/(2(1-g))$. For $g = 1$, since q is a unit, the trace is N if $a = b = 0$ and zero otherwise. The parity insertion replaces g by $-g$ and always has a unique fixed point by (PB.1).

For $g = 1$ write $t_1 = 1$ and $t_j = h_1 \cdots h_{j-1}$. The remaining labels are arbitrary in H, and the final h_m is determined. The main term is independent of η:

$$\mu_{m,\eta}(q) = \frac{N}{N-1} \frac{1}{2^{m/2} K^{m/2-1}} \sum_{\substack{t_1=1,\ t_2,\dots,t_m \in H;\ \epsilon \\ \sum \epsilon_j t_j = \sum \epsilon_j / t_j = 0}} e_N(q\Omega) + \mathcal{R}_{m,\eta}(q), \tag{PB.8}$$

$$\Omega = \frac{1}{2} \sum_{i<j} \epsilon_i \epsilon_j (t_j/t_i - t_i/t_j), \qquad |\mathcal{R}_{m,\eta}(q)| \leq \frac{2^{m/2+1} K^{m/2+1}}{N-1}. \tag{PB.9}$$

This is an exact identity plus an elementary absolute bound. No field-only root-counting theorem or random-entry hypothesis is used. The bound is the following word count. In $\frac{1}{n} \operatorname{Tr} X_{q,\eta}^m$ the expansion of $(B_q P_\eta)^m$ has $(2K)^m$ words, indexed by $\epsilon_j \in \{\pm 1\}$ and $h_j \in H$; each word carries the projector weight K^{-m}, a sector weight of modulus one, and the single parity factor $1/2$ coming from the projection onto the odd subspace (the dimension $n = (N-1)/(2K)$ of E_η). By the trace alternatives above, a word with $g \neq 1$, or with the parity insertion replacing g by $-g$, has trace of modulus one, and a word with $g = 1$ and $(a,b) \neq (0,0)$ has trace zero; its trace over

E_η is an average of such full-space traces, hence also of modulus at most one. The words with $g = 1$ and $(a, b) = (0, 0)$ are exactly the main sum in (PB.8). Therefore

$$|\mathcal{R}_{m,\eta}(q)| \le \frac{1}{n}\Big(\frac{K}{2}\Big)^{m/2} K^{-m}(2K)^m = \frac{2^{m/2+1}K^{m/2+1}}{N-1}.$$

23.5 Lifting reciprocal relations into a quadratic field

A tuple in (PB.8) gives $P(Z) = \sum_j \epsilon_j Z^{e_j}$, where $t_j = h^{e_j}$. Cancel opposite signs at equal labels first. If P is nonzero, its support has at most m points on $\mathbb{Z}/K\mathbb{Z}$. Cut at a largest gap and rotate all labels. The resulting polynomial has coefficient ℓ^1-norm at most m and degree

$$L \le K - \lceil K/m \rceil. \tag{PB.10}$$

Both reciprocal congruences survive this rotation.

Divide P by the monic polynomial $Z^2 - 6Z + 1$, obtaining an integer remainder $aZ + b$. Since $P(h) = P(h^{-1}) = 0$ and $h - h^{-1} = 4s$ is a unit modulo N, both a and b are divisible by N. Hence, for $\lambda = 3 + 2\sqrt{2}$,

$$N^2 \mid \mathrm{Norm}_{\mathbb{Q}(\sqrt{2})/\mathbb{Q}}\, P(\lambda).$$

If this norm is nonzero, its absolute value is at most

$$|P(\lambda)P(\lambda^{-1})| \le m^2\lambda^L. \tag{PB.11}$$

But $N^2 \sim \lambda^K/4$, so (PB.10)–(PB.11) are incompatible for all sufficiently large K at each fixed m. Thus the rotated P satisfies

$$P(\lambda) = 0. \tag{PB.12}$$

An elementary sufficient numerical cutoff is $\lambda^{K/m} > 16m^2$. A sharper entirely rational check uses L in (PB.10) and

$$m^2(2Q_{2L} + 3P_{2L}) < 2N^2, \tag{PB.13}$$

since $\sqrt{2} < 3/2$. The implementation uses (PB.13), not an uncertified decimal comparison.

Crucially, (PB.12) does **not** imply coefficientwise balance. The structural relation $h+h^{-1}-6 = 0$ is a nonbalanced reciprocal relation of length eight for every modulus in this family. An exclusion-of-all-relations programme would therefore be false even here.

23.6 A bounded-cluster lemma and counting, instead of exclusion

Lemma 23.2. *Let P be an integer polynomial of coefficient ℓ^1-norm at most m, with $P(\lambda) = 0$. Split its support at a gap $d > 2\log_\lambda m$. Then both pieces vanish separately at λ.*

Proof. Write $P = P_{\mathrm{low}} + Z^{a+d}P_{\mathrm{high}}$, with $\deg P_{\mathrm{low}} \le a$. If $P_{\mathrm{high}}(\lambda) \ne 0$, its algebraic-integer norm is a nonzero integer. Since $|P_{\mathrm{high}}(\lambda^{-1})| \le m$, we obtain $|P_{\mathrm{high}}(\lambda)| \ge 1/m$. Cancellation with the low part would imply $\lambda^{a+d}/m \le m\lambda^a$, a contradiction to the gap assumption. Hence $P_{\mathrm{high}}(\lambda) = 0$, and then $P_{\mathrm{low}}(\lambda) = 0$. An empty low part is included. □

Choose

$$B_m = \lceil 2\log_\lambda m \rceil + 1, \qquad D_m = (m-1)B_m.$$

Repeatedly applying the lemma partitions a nonzero relation into components of span at most D_m. Every nonzero component contains at least three signed occurrences; a two-term relation between positive distinct powers of λ is impossible. Reinsert the cancelled opposite-sign pairs as two-occurrence components of span zero.

If an m-tuple is not coefficientwise balanced, it therefore has at most $\lfloor (m-1)/2 \rfloor$ components. For each partition of the m positions, each sign pattern, and each bounded offset pattern, a

component has one translation parameter in $\mathbb{Z}/K\mathbb{Z}$. The component containing $t_1 = 1$ has that parameter fixed. It follows that the number of unbalanced tuples in (PB.8) is at most

$$C_m K^{\lfloor (m-1)/2 \rfloor - 1}. \tag{PB.14}$$

The constant depends only on m. For example, $C_m = (2m(2D_m + 1))^m$ suffices by overcounting partitions, signs and offsets. Cyclic rotation introduces no further factor K: the count is over cyclic component translations, with the component containing position 1 pinned.

For even $m = 2r$, (PB.14), divided by the normalization in (PB.8), is $O_m(1/K)$, uniformly in q and η. Odd m has an additional simplification. A polynomial vanishing at λ is divisible in $\mathbb{Z}[Z]$ by $Z^2 - 6Z + 1$, so $P(1)$ is divisible by -4. Yet the parity of $P(1)$ equals the parity of the number of signed occurrences. Consequently odd-m reciprocal tuples do not exist once the lifting argument applies.

23.7 All pairings survive for the fixed observable

For a balanced $2r$-tuple, match equal labels with opposite signs. Distinct pair labels give a unique matching; repeated labels contribute $O_r(1/K)$ normalized overcount. For a matching π, orient each pair by $\sigma_i \in \{1, -1\}$, and let its label be $x_i = h^{e_i}$, with $x_1 = 1$. The exact phase is

$$F_{\pi,\sigma}(x) = \sum_{\substack{i<j \\ \text{pairs } i,j \text{ cross}}} \sigma_i \sigma_j (x_j/x_i - x_i/x_j). \tag{PB.15}$$

Unlike a q-average, the deterministic value q_0 does not suppress crossings. Let δ be the representative of d modulo K in $[-(K-1)/2, (K-1)/2]$. The Pell recurrence gives

$$q_0(h^d - h^{-d}) \equiv -2P_{2\delta} \pmod{N}. \tag{PB.16}$$

For $|\delta| \le (K-1)/2 - L$,

$$|P_{2\delta}|/N \le C\lambda^{-L}. \tag{PB.17}$$

The implicit constant is absolute. A difference of two independent uniform pair labels is uniform on $\mathbb{Z}/K\mathbb{Z}$, also when one label is fixed. For each fixed pairing a union bound shows that only $O_r(L/K)$ assignments have a crossing difference in the excluded central window. On all other assignments, (PB.15)–(PB.17) give phase $1 + O_r(\lambda^{-L})$. Taking $L = \lceil \log_\lambda K \rceil$ proves

$$K^{-(r-1)} \sum_{x_2,\dots,x_r \in H} e_N(q_0 F_{\pi,\sigma}(x)) = 1 + O_r(\log K/K). \tag{PB.18}$$

There are $(2r-1)!!$ matchings, not just the Catalan number of noncrossing matchings. Equations (PB.8)–(PB.9), (PB.14), and (PB.18) prove (PB.6) and its stated even moment error. Odd moments follow from the parity argument and (PB.9).

Bounded second moments give tightness, and higher even moments give uniform integrability. The Gaussian moment sequence is determinate: its even moments imply a finite exponential-square moment for every exponent below $1/2$, by monotone expansion, and hence determine the characteristic function $e^{-t^2/2}$. The usual subsequence argument therefore gives weak convergence. Because all bounds are uniform in η, selecting a worst sector on any hypothetically nonconvergent subsequence proves the claimed uniformity.

23.8 An explicit fourth-moment certificate

For $m = 4$, a nonzero polynomial of coefficient ℓ^1-norm at most 4 cannot vanish at $\lambda > 3$: its leading term dominates all lower terms. Thus, once (PB.13) holds, all four-letter reciprocal tuples are balanced. Exact pairing overcount is six, giving uniformly in η,

$$\left| \mu_{4,\eta} - \frac{N}{N-1} \left(2 + S_K - \frac{3}{2K} \right) \right| \le \frac{8K^3}{N-1}, \tag{PB.19}$$

$$S_K = \frac{1}{K}\left(1 + 2\sum_{d=1}^{(K-1)/2} \cos(4\pi P_{2d}/N)\right). \tag{PB.20}$$

These are finite-size bounds, not fitted matrix histograms. The maintained companion reports high-precision cosines and an analytic series-tail bound, separately from floating-point rounding. It includes the full remainder in (PB.19), but does not provide rational interval enclosures.

There is also a nontrivial explicit asymptotic correction:

$$K(3 - \mu_{4,\eta}) \longrightarrow \frac{3}{2} + 2\sum_{j=0}^{\infty}\left[1 - \cos\left((4 - 2\sqrt{2})\pi\lambda^{-j}\right)\right]. \tag{PB.21}$$

To see this, reindex (PB.20) by $d = (K-1)/2 - j$. Then $P_{K-1-2j}/N \to \alpha^{-1-2j}/\sqrt{2}$. The inequality $1 - \cos x \le x^2/2$ gives a uniform summable geometric majorant. The remainder in (PB.19), even after multiplication by K, is exponentially small. This proves (PB.21), uniformly in the sector.

23.9 A finite-perturbation consequence for actual eigenphases

The following elementary comparison explains precisely how far the global observable theorem transfers to a propagator.

Lemma 23.3. *Let U_0 be unitary, let P be its spectral projection at 1, and let $\delta > 0$ be the distance from 1 to the other eigenvalues. Let $B = B^*$, $\|B\| \le b$, and $U_e = e^{ieB}U_0$. If $0 < b|e| \le \delta/16$, the cluster near 1 has rank $\operatorname{rank} P$. Write its principal eigenphases as θ_j, and the eigenvalues of $A = PBP$ on $\operatorname{ran} P$ as a_j. Their empirical measures satisfy*

$$W_2\left(\frac{1}{n}\sum_j \delta_{\theta_j/e}, \frac{1}{n}\sum_j \delta_{a_j}\right) \le \frac{13\pi b^2|e|}{\delta}. \tag{PB.22}$$

Here W_2 is quadratic Wasserstein distance between equal-weight real spectra.

Proof. Put $\Delta = U_e - U_0$ and $d = \|\Delta\| \le b|e|$. Resolvent integration on $|z-1| = \delta/2$ gives $\|P_e - P\| = \eta \le 4d/\delta \le 1/4$. The new projection is orthogonal. Set $H = (PP_eP)^{1/2}$ on $\operatorname{ran} P$ and $J = P_ePH^{-1}$. Then J is an isometry onto the perturbed cluster, $PJ = H$, $\|(I-P)J\| \le \eta$, and $\|J - P\| \le 2\eta$. Let $V = J^*U_eJ$. Spectral inclusion for the normal matrix U_0 gives $\|V - I\| \le d$: every cluster eigenvalue is within d of 1, since all other old eigenvalues are farther than $\delta/2$.

Writing $Z = (I-P)J$ and using invariance of the new cluster gives $(U_0 - I)Z = Z(V - I) - (I-P)\Delta J$. Hence $\|Z^*(U_0 - I)Z\| \le \eta(1+\eta)d$. Also $\|J^*\Delta J - P\Delta P\| \le 4\eta d$. Therefore $\|V - I - P\Delta P\| \le 24d^2/\delta$. Taylor's integral remainder for the two unitary exponentials, and $\delta \le 2$, yield

$$\|V - e^{ieA}\| \le 26e^2b^2/\delta. \tag{PB.23}$$

For two unitary matrices, the normalized squared Frobenius difference is a doubly stochastic average of squared eigenvalue distances, using the squared absolute overlaps of their eigenbases. Decomposing that matrix into permutation matrices gives a matching whose mean squared chord distance is at most the squared operator norm in (PB.23). Such a decomposition follows by repeatedly selecting a positive perfect matching (Hall's condition follows from row and column sums), subtracting its smallest entry, and iterating. All phases lie in one short arc. For angular differences at most π, arc distance is at most $\pi/2$ times chord distance. Dividing by $|e|$ proves (PB.22). □

For the cat propagator, rotate any of its K clusters to 1. Its gap is

$$\delta_K = 2\sin(\pi/K) \ge 4/K, \qquad b = 2.$$

Thus the empirical law of the rescaled relative eigenphases $\sqrt{K/2}\,\theta_j/e$ differs in W_2 from the corresponding observable block by at most

$$\frac{13\pi}{\sqrt{2}}|e|K^{3/2}. \tag{PB.24}$$

Corollary 23.4. *For any nonzero perturbation sequence $e = e_K$ satisfying $|e_K|K^{3/2} \to 0$, every odd-sector eigenphase cluster of $e^{ie_K B}U_N(T)$, rescaled as above, converges globally to the standard normal distribution, uniformly over the K clusters. The symmetrically kicked propagator $e^{ie_K B/2}U_N(T)e^{ie_K B/2}$ is isospectral and has the same conclusion.*

This does not exchange the small-perturbation and semiclassical limits without control: (PB.24) is the control, and its hypothesis is part of the claim. Fixed nonzero perturbations, microscopic within-cluster correlations, and full BGS universality are not established.

24 A harmonic-scale transition in the Pell cat map

Status: project proof, with explicit scope. This note derives a family of global observable laws in one fixed arithmetic quantum map. It does not prove microscopic BGS statistics. The limiting graphon moment construction is prior work; the claim here is its realization by growing harmonics of the Pell cat map. Priority for that realization has not been established.

24.1 Statement and normalization

Let K tend to infinity through odd primes. Write

$$\alpha = 1+\sqrt{2}, \quad \lambda = \alpha^2, \qquad N + F\sqrt{2} = \alpha^K, \qquad T = \begin{pmatrix} 3 & 2 \\ 4 & 3 \end{pmatrix}.$$

Use the parity-preserving quantization U_0 of this fixed matrix, normalized by $U_0^K = I$, as in the preceding Pell block proof. The odd parity space has dimension $D = (N-1)/2$. Its K eigenspaces E_η have equal dimension $n = (N-1)/(2K)$; let P_η be their projections.

For a positive integer a coprime to N, take the actual physical harmonic

$$B_a = W(a,0) + W(-a,0), \qquad (B_a u)(x) = 2\cos(2\pi a x/N)u(x).$$

Here W has the modular-centered convention of the Pell block proof. Define

$$X_{K,a,\eta} = \sqrt{K/2}\,(P_\eta B_a P_\eta)|_{E_\eta}, \qquad \mu_{m,K,\eta}(a) = n^{-1}\operatorname{Tr} X_{K,a,\eta}^m. \tag{H1}$$

Let $1 \le L_K \le \lfloor\sqrt{N}\rfloor$ be integers with

$$\frac{\log L_K}{\log N} \longrightarrow \beta \in [0,1/2], \qquad \mathcal{A}_K = \{1 \le a \le L_K : \gcd(a,N) = 1\}. \tag{H2}$$

Probability and L^2 below use the uniform distribution on $\mathcal{A}_K$. No randomness is added to the map or to its quantization.

Put $\rho = 1 - 2\beta$ and define on the unit circle

$$w_\beta(x,y) = \mathbf{1}\{\operatorname{dist}_{\mathbb{R}/\mathbb{Z}}(x,y) < \rho/2\}. \tag{H3}$$

Endpoint values on the distance boundary do not affect integrals. For a pair partition π of $2r$ positions, its crossing graph G_π has one vertex per pair and an edge precisely when the two pairs interlace. Define

$$t(G,w) = \int_{[0,1]^{V(G)}} \prod_{\{i,j\}\in E(G)} w(x_i,x_j)\,dx.$$

$$M_{2r}(\beta) = \sum_{\pi \in \mathcal{P}_2(2r)} t(G_\pi, w_\beta), \quad M_{2r+1}(\beta) = 0, \quad M_0 = 1. \tag{H4}$$

Theorem 24.1. *For every fixed nonnegative integer m,*

$$\left\| \sup_{\eta} |\mu_{m,K,\eta} - M_m(\beta)| \right\|_{L^2(\mathcal{A}_K)} \longrightarrow 0. \tag{H5}$$

There is a unique probability measure ν_β with moments (H4). The empirical spectral measures of (H1) converge weakly to ν_β in probability, uniformly over all K sectors. The averaged measures converge weakly too. At $\beta = 0$ the limit is the standard Gaussian; at $\beta = 1/2$ it is the variance-one semicircle.

For every $0 \le \beta < 1/2$, ν_β has unbounded support. Thus the compactly supported endpoint is a genuine change of support, despite weak continuity of the family.

The intermediate laws are not the ordinary q-Gaussian laws with $q = \rho$. In particular,

$$M_2 = 1, \qquad M_4 = 2 + \rho = 3 - 2\beta,$$
$$M_6 = 5 + 6\rho + \frac{15}{4}\rho^2 + \left(\frac{3}{2}\rho - 1\right)_+^2. \tag{H6}$$

Thus (H5) is an all-fixed-order statement, not an interpolation inferred from the fourth moment.

24.2 Why the harmonic enters quadratically

Set $s = F^{-1} \pmod N$, $h = 3 + 2s$, and $q_0 = -F$. In the diagonal coordinates of the Pell block proof, the fixed conjugation taking B_1 to $W(q_0, 1) + W(q_0, 1)^*$ takes B_a to

$$W(aq_0, a) + W(aq_0, a)^*.$$

The dilation $S_u f(x) = f(u^{-1}x)$ satisfies

$$S_u W(b, c) S_u^{-1} = W(b/u, uc).$$

Taking $u = a^{-1}$ converts this to $W(a^2 q_0, 1) + W(a^2 q_0, 1)^*$. This dilation commutes with S_h and parity, so it preserves every sector. Consequently (H1) is unitarily equivalent to the previously defined block $X_{q,\eta}$ with $q = a^2 q_0$, up to the fixed initial permutation of sector labels. It is not the block with $q = aq_0$.

For a residue d modulo K, let $\delta(d)$ be its signed representative in $[-(K-1)/2, (K-1)/2]$. Extend the even Pell sequence by $P_{-2d} = -P_{2d}$. The exact identity is

$$q_0(h^d - h^{-d}) \equiv -2P_{2\delta(d)} \pmod N, \qquad P_{2d} = \frac{\lambda^d - \lambda^{-d}}{2\sqrt{2}}. \tag{H7}$$

Thus a crossing contributes a phase quadratic in a.

24.3 The unit restriction has vanishing cost

Every prime $\ell \mid N$ admits an element h of exact order K: $h^K = 1$ and $h - 1$ is a unit, so reduction modulo ℓ cannot have order one. Therefore $\ell \equiv 1 \pmod{2K}$ and $\ell \ge 2K + 1$. Counting distinct prime divisors gives

$$\frac{\#([1, L_K] \setminus \mathcal{A}_K)}{L_K} \le \sum_{\ell \mid N} \frac{1}{\ell} \le \frac{\log N}{(2K+1)\log(2K+1)} = O(1/\log K). \tag{H8}$$

The first inequality uses $\lfloor L_K/\ell \rfloor \le L_K/\ell$ and is valid even for bounded L_K. We will perform phase averaging over the whole interval and condition afterwards. All averaged phase expressions are bounded in modulus; no unbounded-moment conditioning is implicit.

24.4 Reduction to pairings, uniformly in the harmonic

The all-order trace identity and largest-gap lifting argument of the Pell block proof apply uniformly to every unit q, hence to $q = a^2 q_0$. For completeness, a central signed m-word gives both reciprocal congruences. After cutting at its largest cyclic gap, its signed polynomial P has degree at most $K - \lceil K/m \rceil$ and coefficient norm at most m. Its remainder modulo $Z^2 - 6Z + 1$ is divisible coefficientwise by N. A nonzero quadratic norm would therefore satisfy

$$N^2 \le |\operatorname{Norm} P(\lambda)| \le m^2 \lambda^{K - \lceil K/m \rceil},$$

which is impossible for sufficiently large K at fixed m. Thus the relation lifts to $P(\lambda) = 0$.

A gap greater than $2 \log_\lambda m$ splits this relation into separately vanishing pieces: if the high piece were nonzero, its value at λ would have modulus at least $1/m$ by its integral quadratic norm, while the lower part cannot cancel it across such a gap. Each nonzero piece has bounded span and at least three signed occurrences. Reinsert cancelled opposite pairs as two-occurrence pieces. A nonbalanced $2r$-word then has at most $r - 1$ translation parameters; after pinning one label there are $O_r(K^{r-2})$ such words. The normalized contribution is $O_r(K^{-1})$, independently of their phases. For odd m, the equality $P(\lambda) = 0$ implies $4 \mid P(1)$, impossible for an odd number of signs.

Balanced words are counted by pair partitions, opposite orientations $\sigma_i \in \{-1, 1\}$, and labels $e_i \in \mathbb{Z}/K\mathbb{Z}$. Duplicated labels contribute $O_r(K^{-1})$ after normalization. Orient pairs by their first endpoints. For a crossing graph with r vertices, set

$$Z_{\pi,\sigma,K}(a) = K^{-(r-1)} \sum_{\substack{e_1 = 0 \\ e_2, \dots, e_r \bmod K}} e\Big(a^2 u_{\pi,\sigma,K}(e)\Big), \qquad e(t) = e^{2\pi i t},$$

$$u_{\pi,\sigma,K}(e) = -\frac{2}{N} \sum_{\substack{i<j \\ \{i,j\} \in E(G_\pi)}} \sigma_i \sigma_j P_{2\delta(e_j - e_i)}. \tag{H9}$$

The exact Weyl multiplication formula gives

$$\mu_{2r,K,\eta}(a) = \frac{N}{N-1} \sum_{\pi} 2^{-r} \sum_{\sigma} Z_{\pi,\sigma,K}(a) + O_r(K^{-1}) + O_r(K^{r+1}/N). \tag{H10}$$

The errors are uniform over all unit a and all sectors. Odd moments are uniformly $O_m(K^{m/2+1}/N)$. The common leading expression in (H10), rather than a union bound over K sectors, will give (H5).

24.5 An elementary quadratic-sum estimate

Lemma 24.2. *Suppose $u = u_K$ is real and*

$$0 < |u| \le N^{-c}, \qquad L_K \sqrt{|u|} \ge N^c \tag{H11}$$

for some fixed $c > 0$. Then

$$L_K^{-1} \sum_{a=1}^{L_K} e(ua^2) \longrightarrow 0$$

uniformly under (H11).

Proof. For $z_a = e(ua^2)$ on $1 \le a \le L$, zero elsewhere, Cauchy–Schwarz applied to $H \sum z_a = \sum_b \sum_{j=0}^{H-1} z_{b-j}$ gives, for $H \le L$,

$$\Big| L^{-1} \sum z_a \Big|^2 \le \frac{2}{H} + \frac{4}{HL} \sum_{j=1}^{H-1} \min\left(L, \frac{1}{2\|2uj\|_{\mathbb{R}/\mathbb{Z}}} \right). \tag{H12}$$

Take $H = \lfloor (4\sqrt{|u|})^{-1} \rfloor$. Eventually $2 \le H \le L$ and $\|2uj\| = 2|u|j$ for $1 \le j < H$. The geometric-sum bound and the harmonic sum therefore imply

$$\left| L^{-1} \sum z_a \right|^2 \ll \sqrt{|u|} + \frac{\log(2/|u|)}{L\sqrt{|u|}}. \tag{H13}$$

In our application $|u| \ge N^{-C}$ for fixed C, so (H11)–(H13) tend to zero. In fact (H11) and $L \le \sqrt{N}$ already imply such a polynomial lower bound. This proves the lemma without assuming independence or irrationality of u. Rational phases are included.

24.6 The largest crossing distance controls the phase

Lemma 24.3. *For every fixed pair partition and orientation,*

$$\|Z_{\pi,\sigma,K} - t(G_\pi, w_\beta)\|_{L^2([1,L_K])} \longrightarrow 0. \tag{H14}$$

Proof. If there are no edges, both expressions are one. Otherwise regard $x_i = e_i/K$ as points on the circle, with $x_1 = 0$, and let $d_{ij} = \operatorname{dist}(x_i, x_j)$ for graph edges. Fix a small $\varepsilon > 0$. Discard configurations for which an edge distance is within ε of 0, 1/2, or $\rho/2$, or two distinct edge distances differ by at most ε. Their normalized number is $O_r(\varepsilon + K^{-1})$.

Here is the geometric justification, including dependent edges. Each circle distance is piecewise affine in the labels. On each of finitely many cells, distance equality for two distinct edges is a nontrivial affine equation: their endpoint coefficient vectors differ, even when the edges share a vertex or one vertex is fixed at zero. Its ε-neighborhood has volume $O_r(\varepsilon)$ and grid count $O_r(\varepsilon K^{r-1} + K^{r-2})$. The same argument handles the three single-distance boundaries. No assertion of independent edge distances is made.

On the remaining configurations, let $d_* = \max d_{ij}$. Its edge is unique and is separated from every other edge by at least ε. Formula (H7) implies that the associated Pell term dominates the sum in (H9), regardless of orientations. More precisely,

$$|u_{\pi,\sigma,K}(e)| \asymp_r \lambda^{Kd_* - K/2} \asymp_r N^{2d_* - 1}, \tag{H15}$$

uniformly on these configurations for sufficiently large K. The ratio of all other terms to the largest is $O_r(\lambda^{-\varepsilon K})$. The exclusion near 1/2 ensures $|u| \ll_r N^{-2\varepsilon}$: there is no hidden reduction near a nonzero integer in applying Lemma 24.2.

If $d_* < \rho/2 - \varepsilon$, then $L_K^2 |u| \to 0$ uniformly, so $e(a^2 u) \to 1$ uniformly for $a \le L_K$. If $d_* > \rho/2 + \varepsilon$, then $L_K\sqrt{|u|} \ge N^{\varepsilon/2}$ eventually; (H13) makes its average over a tend to zero. Taking Riemann sums, then letting $\varepsilon \downarrow 0$, proves

$$\mathbb{E}_a Z_{\pi,\sigma,K} \longrightarrow t(G_\pi, w_\beta). \tag{H16}$$

Pinning $x_1 = 0$ does not change this integral, by translation invariance.

For the second moment use two independent label configurations e, f and expand

$$\mathbb{E}_a |Z_{\pi,\sigma,K}(a)|^2 = \mathbb{E}_{e,f} \mathbb{E}_a e\Big(a^2 [u_{\pi,\sigma,K}(e) - u_{\pi,\sigma,K}(f)]\Big). \tag{H17}$$

In addition discard ties between an edge distance in one copy and an edge distance in the other. They again have measure and grid proportion $O_r(\varepsilon + K^{-1})$, since the two copies have independent free coordinates. Outside this set there is a unique largest edge among both copies. Its coefficient is 1 or -1 and cannot cancel. The same low/high dichotomy shows that only configurations where **all** edges of both copies have distance below $\rho/2$ survive. Their limiting volume is $t(G_\pi, w_\beta)^2$. Together with (H16), this proves (H14). At $\beta = 0$ only the low region survives; at $\beta = 1/2$ only the high region survives for a graph with an edge. Thus both endpoints are included.

24.7 Completion of the spectral limit

Since $|Z_{\pi,\sigma,K}| \le 1$, (H8) transfers (H14) to $\mathcal{A}_K$. Sum over the finitely many pairings and orientations in (H10). Its uniform remainder proves (H5), including the supremum over sectors.

To justify the probability measure and weak convergence, note

$$0 \le M_{2r}(\beta) \le (2r-1)!!.$$

The averaged empirical measures have bounded second moments and hence are tight. Their uniformly bounded higher even moments, supplied by (H10), give uniform integrability for each lower moment. Any weak subsequential limit therefore has moments (H4), proving existence. The displayed Gaussian moment bound gives $\int e^{tx^2}\, d\nu_\beta(x) < \infty$ for $0 < t < 1/2$ by monotone expansion. It implies moment determinacy (expand the characteristic function under this domination). Thus the measure is unique.

For the random empirical laws, from any subsequence choose a further one on which the suprema in (H5) tend to zero almost surely for every integer m, by diagonal extraction. Tightness and higher-moment uniform integrability then give weak convergence for every choice of sector, including choices depending on a. A contradiction using a maximizing sector proves uniform weak convergence in probability, equivalently convergence of the supremum of bounded-Lipschitz distances. Boundedness of that distance also gives the averaged statement. At the endpoints (H4) counts respectively all pairings and only noncrossing pairings.

Corollary 24.4 (support transition and continuity)**.** *For $\rho > 0$, choose one circle interval I of length $b = \rho/2$. On $I \times I$ the kernel is one almost everywhere. For every crossing graph on r vertices, $t(G, w_\beta) \ge b^r$ by restricting all labels to I. Consequently*

$$b^r (2r-1)!! \le M_{2r}(\beta) \le (2r-1)!!. \tag{H22}$$

The $2r$-th root of the lower bound tends to infinity: the last $\lfloor r/2 \rfloor$ factors of $(2r-1)!!$ are at least r. A probability measure supported on $[-R, R]$ would instead have all these roots bounded by R. Hence the support is unbounded for $\beta < 1/2$. For $\beta = 1/2$ only noncrossing pairings survive, giving support $[-2, 2]$.

For each fixed graph, its density is continuous in β by dominated convergence, since each distance boundary has measure zero. Thus every fixed moment is continuous, uniformly bounded by the Gaussian moments. Tightness, uniform integrability, and moment determinacy prove weak continuity of $\beta \mapsto \nu_\beta$, including at the compact endpoint. This is a statement about the limiting measures, not finite-N extreme eigenvalue rigidity.

24.8 Sixth moment and the failure of an independent-crossing model

The fifteen pairings of six positions have respectively $0, 1, 2, 3$ crossings with multiplicities $5, 6, 3, 1$. The two-edge graphs are trees, whose densities are ρ^2 because w_β has constant degree ρ. The last graph is a triangle. Put $b = \rho/2$. Fix its first vertex at zero:

$$t(K_3, w_\beta) = \int_{-b}^{b} \int_{-b}^{b} \mathbf{1}\{\operatorname{dist}(x, y) < b\}\, dx\, dy = 3b^2 + (3b-1)_+^2. \tag{H18}$$

Indeed the central region $|x - y| < b$ has area $3b^2$; when $3b > 1$, the two wrapped corner triangles add $(3b-1)^2$. This proves (H6). An independent-crossing model would instead assign ρ^3 to the triangle. The difference is

$$t(K_3, w_\beta) - \rho^3 = \begin{cases} \rho^2(3/4 - \rho), & 0 \le \rho \le 2/3, \\ (1-\rho)^3, & 2/3 \le \rho \le 1. \end{cases} \tag{H19}$$

It is strictly positive for $0 < \rho < 1$. For example, at $\beta = 1/4$, $M_4 = 5/2$, $M_6 = 143/16$, whereas that q-Gaussian model gives $71/8$.

24.9 Weighted quantum fields and a noncommutativity diagnostic

There is also an ordered mixed-moment version on the full odd space. For a bounded real Riemann-integrable profile f on $[0,1)$, set

$$Y_{K,a}(f) = \frac{1}{\sqrt{2K}} \sum_{j=0}^{K-1} f(j/K) U_0^j B_a U_0^{-j}, \qquad \tau_-(A) = D^{-1} \operatorname{Tr}_- A.$$

For fixed profiles $f_1, \dots, f_{2r}$, their ordered mixed trace converges in $L^2(\mathcal{A}_K)$ to

$$\sum_{\pi \in \mathcal{P}_2(2r)} \int_{[0,1]^r} \prod_{\{u,v\} \in \pi} f_u(x_{\{u,v\}}) f_v(x_{\{u,v\}}) \prod_{\{i,j\} \in E(G_\pi)} w_\beta(x_i, x_j)\, dx. \tag{H20}$$

Odd mixed traces tend to zero. In proving this version, use all r labels without pinning one; keep the profiles inside the sums. The same discarded sets and quadratic-phase bounds apply with a fixed supremum-norm factor. The product profiles and indicator kernels are Riemann integrable, so the remaining grid sums converge. The noncentral full-space trace error is $O_{r,f}(K^r/N)$ and the unbalanced/repeated-label error is $O_{r,f}(K^{-1})$. This proves (H20); it is not an assumption of finite quantum independence.

At $\beta = 0$, (H20) is the commuting Gaussian field with covariance $\int fg$; at $\beta = 1/2$ it is the free semicircular field with that covariance. These descriptions concern tracial moments. For $f = \sqrt{2}\,\mathbf{1}_{[0,1/2)}$ and $g = \sqrt{2}\,\mathbf{1}_{[1/2,1)}$, (H20) gives

$$\tau_-([Y_{K,a}(f), Y_{K,a}(g)]^*[Y_{K,a}(f), Y_{K,a}(g)]) \longrightarrow 2(1-\rho^2) = 8\beta(1-\beta) \tag{H21}$$

in $L^2(\mathcal{A}_K)$. To check the coefficient, for any disjointly supported real profiles the limit is $2\int f(x)^2 g(y)^2 [1 - w_\beta(x,y)]\, dx\, dy$. The area of allowed pairs across the two half-circles is b^2. This gives (H21), and shows why the limiting intermediate fields cannot be treated as commuting random variables.

24.10 Prior work, verification, and the remaining BGS gap

Cébron, Oliveira Santos, and Youssef, *Graphon-Theoretic Approach to Central Limit Theorems for ϵ-Independence*, arXiv:2411.13062v2, Theorem 1.1 and Section 2, already give the graphon moment family (H4), including Gaussian and semicircular endpoints. Their weighted and multivariate extensions also precede this note. The present Weyl orbit is not assumed to satisfy their finite independence hypotheses. The additional argument here is the arithmetic harmonic realization (H7)–(H17). Its priority is unverified.

That paper's Proposition 2.6 assumes $\|w\|_\infty < 1$ to obtain a density. For $0 \le \beta < 1/2$, our kernel has essential supremum one, so that proposition does not automatically prove absolute continuity here. No density or finite-N spectral-edge assertion is being imported; the limiting support statement follows instead from (H22).

The dedicated verifier checks the physical harmonic/squared-phase identification against finite matrices, exact Pell and oriented-word identities, and the low-order crossing geometry. The companion numerical experiment samples harmonics with integer modular phases; sampling and floating-point output are not an asymptotic proof. Source hashes and actual run ranges belong to the dated review record.

The classical map is fixed, but the observable frequency and its distribution vary with N. These are global spectra of compressed observables and ordered moments of orbit sums. No local spacing law for U_0, perturbed-propagator universality, Heisenberg-time trace comparison, or absence of arithmetic symmetries follows. BGS remains open.

25 Harmonic-scale return laws for the kicked Pell cat map

Status: project proof. This is a global spectral theorem for the actual return propagator. It concerns a vanishing quantum kick and times of order the arithmetic period, not Heisenberg-scale correlations or BGS. The endpoint probability laws are known; priority for their arithmetic realization is not established. All asymptotic statements below keep the return multiplicity and kick parameter fixed.

25.1 1. Statement

Use the quantization and physical harmonics of the preceding harmonic observable theorem:

$$N + F\sqrt{2} = (1+\sqrt{2})^K, \qquad D = (N-1)/2, \qquad U_0^K = I, \qquad B_a(x) = 2\cos(2\pi a x/N).$$

Here K runs through odd primes. Let a be uniform on $\mathcal{A}_K = \{1 \le a \le L_K : \gcd(a,N) = 1\}$, where $1 \le L_K \le \lfloor\sqrt{N}\rfloor$ and $\log L_K/\log N \to \beta \in [0,1/2]$. Put

$$\epsilon = \frac{\tau}{\sqrt{2K}}, \qquad U_{\epsilon,a} = e^{i\epsilon B_a} U_0, \qquad t = \tau^2, \qquad w_\beta(x,y) = \mathbf{1}\{\mathrm{dist}(x,y) < (1-2\beta)/2\}. \tag{R1}$$

The normalized trace tr_- is on the full odd parity space.

For $m \ge 1$, let $\mathcal{C}_m = \{(r,s) : 0 \le r < s < m\}$. Given ℓ ordered labels and types $c_i = (r_i, s_i) \in \mathcal{C}_m$, arrange their 2ℓ endpoints $(r_i, i), (s_i, i)$ in lexicographic order (cycle first, label second). Join the two endpoints of each label. Write $G(c)$ for the crossing graph of these chords, retaining the label i on its vertex. Set

$$I_c(\beta) = \int_{0<x_1<\cdots<x_\ell<1} \prod_{\{i,j\}\in E(G(c))} w_\beta(x_i,x_j)\,dx,$$

$$F_{m,\beta}(t) = e^{-mt/2} \sum_{\ell=0}^{\infty} (-t)^\ell \sum_{c\in\mathcal{C}_m^\ell} I_c(\beta). \tag{R2}$$

The empty integral is one. For $m = 1$ there are no nonempty type lists. The series is absolutely convergent, since its ℓth absolute term is at most $(\binom{m}{2}t)^\ell/\ell!$.

Theorem 25.1. *For every fixed real τ and positive integer m,*

$$\left\| \mathrm{tr}_-\, U_{\epsilon,a}^{mK} - F_{m,\beta}(\tau^2) \right\|_{L^2(\mathcal{A}_K)} \longrightarrow 0. \tag{R3}$$

There is a unique probability measure $\mu_{\beta,t}$ on the unit circle with positive Fourier moments $F_{m,\beta}(t)$. The empirical eigenvalue measures of $U_{\epsilon,a}^K$ converge weakly to it in probability, and their averages converge weakly. In particular

$$F_{1,\beta}(t) = e^{-t/2}, \qquad F_{m,0}(t) = e^{-m^2t/2}, \qquad F_{m,1/2}(t) = \frac{e^{-mt/2}}{m} L_{m-1}^{(1)}(mt). \tag{R4}$$

Thus the endpoints are the wrapped normal law and the spectral law of free unitary Brownian motion at time t, respectively. The latter identification uses the known moment formula; the calculation of these moments from the quantum map is proved below. This is not the exponential of a semicircular random variable.

There is also a closed intermediate formula:

$$F_{2,\beta}(t) = e^{-t}\left(1 - te^{-(1-2\beta)t/2}\right), \qquad 1/6 \le \beta \le 1/2. \tag{R5}$$

The family $\mu_{\beta,t}$ is weakly continuous in (β,t).

25.2 2. Exact expansion and uniform truncation

Conjugation and dilation as in the harmonic theorem turn the physical kick into $W(q,1)+W(q,1)^*$ with $q=a^2q_0$, $q_0=-F$. The square of a is essential. The unperturbed operator becomes S_h^{-1}, $h=3+2F^{-1}$. Choose orbit vectors $v_j=(qh^j,h^{-j})$, $0\le j<K$; these have precisely the order in

$$U_{\epsilon,a}^{mK}=\left[\prod_{j=0}^{K-1}\exp\{i\epsilon(W(v_j)+W(-v_j))\}\right]^m. \tag{R6}$$

All products are read left to right with increasing j.

The absolutely convergent Fourier–Bessel identity is

$$e^{i\epsilon(W+W^*)}=\sum_{n\in\mathbb{Z}} i^nJ_n(2\epsilon)W^n,\qquad |J_n(2\epsilon)|\le I_n(2|\epsilon|). \tag{R7}$$

It follows directly by multiplying the power series for $e^{i\epsilon W}$ and $e^{i\epsilon W^{-1}}$; thus it applies equally to finite-order Weyl unitaries. Let $n_{r,j}$ be the coefficient chosen in cycle r, put $R=\sum_{r,j}|n_{r,j}|$, and $c_j=\sum_r n_{r,j}$. A word is a scalar phase times $W(\sum_j c_jv_j)$.

Individual Weyl operators do not preserve parity. Applying the full-space trace with the odd projector gives, nonetheless, the exact linear identity

$$D^{-1}\operatorname{Tr}\left(\frac{I-\mathcal{R}}{2}W(v)\right)=\frac{N\mathbf{1}_{\{v=0\}}-1}{N-1}. \tag{R8}$$

Indeed $\operatorname{Tr}W(v)=N\mathbf{1}_{\{v=0\}}$ and $\operatorname{Tr}(\mathcal{R}W(v))=1$ for the centered convention. Hence replacing every normalized word trace by its central indicator times its phase has total error at most

$$\frac{e^{2mK\delta}}{N-1}=o(1),\qquad \delta=|\epsilon|. \tag{R9}$$

Here $\sum_n I_n(2\delta)=e^{2\delta}$.

Let $R_0=\lfloor K^{2/3}\rfloor$. For $z\ge1$, the coefficient series gives

$$I_n(2\delta)\le e^{\delta^2}\frac{\delta^{|n|}}{|n|!},\qquad \sum_n I_n(2\delta)z^{|n|}\le e^{\delta^2}(2e^{\delta z}-1)\le e^{\delta^2+2\delta z}.$$

Thus the absolute weight of words with $R>R_0$ is at most

$$z^{-R_0}\exp(mK\delta^2+2mK\delta z).$$

For $\tau\neq0$ take $z=R_0/(2mK\delta)>1$, eventually. This bound is

$$\exp\{mK\delta^2+R_0-R_0\log(R_0/(2mK\delta))\}=\exp\{-(1/6+o(1))K^{2/3}\log K\}=o(1). \tag{R10}$$

For $\tau=0$ the expansion has only its constant term.

25.3 3. All-order suppression of nonzero aggregate relations

Lemma 25.2. *Uniformly in unit a, the absolute weight of central words with $R\le R_0$ and $c\neq0$ is $O_{m,\tau}(K^{-1})$.*

Proof. Centrality and the unit property of q give both congruences $\sum c_jh^j=\sum c_jh^{-j}=0$. Cancel zero coefficients and cut the support at a largest cyclic gap. Its size is at least $\lceil K/R\rceil$. The rotated polynomial P has coefficient norm at most R and degree at most $K-\lceil K/R\rceil$. Dividing

by the monic polynomial z^2-6z+1, the two congruences and the unit $h-h^{-1}$ show that both remainder coefficients are divisible by N. A nonzero quadratic norm would therefore obey

$$N^2 \le |\operatorname{Norm} P(\lambda)| \le R^2 \lambda^{K-\lceil K/R\rceil}, \qquad \lambda = 3+2\sqrt{2}. \tag{R11}$$

This is impossible uniformly for $R \le R_0$: since $N^2 = \lambda^K(1-\lambda^{-K})^2/4$, the ratio is at most $4R_0^2\lambda^{-\lceil K/R_0\rceil}/(1-\lambda^{-K})^2 = o(1)$. Consequently P is an integer multiple of z^2-6z+1.

Work now in $\mathbb{Z}[z]/(z^K-1)$. Rotation back is multiplication by a power of z, so the original coefficient vector satisfies

$$c = Mq', \qquad M = S^2-6S+I, \qquad q' \in \mathbb{Z}^K, \tag{R12}$$

where S is cyclic shift. The triangle inequality gives

$$\|Mq'\|_1 \ge 6\|Sq'\|_1 - \|S^2q'+q'\|_1 \ge 4\|q'\|_1. \tag{R13}$$

In particular M is injective; no counting multiplicities of preimages is needed.

For a prescribed c, summing absolute weights over its cycle assignments is bounded, by modified-Bessel convolution, by

$$\prod_{j=0}^{K-1} I_{c_j}(2m\delta) \le e^{Kb^2} b^{\|c\|_1}, \qquad b = m\delta < 1.$$

We may enlarge the truncated sum to all assignments with c in the integer image of M. Equations (R12)–(R13) imply the bound

$$e^{Kb^2} \sum_{q'\neq 0} b^{4\|q'\|_1} = e^{Kb^2}\left[\left(\frac{1+b^4}{1-b^4}\right)^K - 1\right] = O_{m,\tau}(K^{-1}). \tag{R14}$$

This proves the lemma. In contrast to fixed-order relation counting, (R14) controls the entire truncated exponential at once.

25.4 4. Balanced sites and the limiting series

After (R9)–(R14), only coefficientwise-balanced arrays $c=0$ remain. At a site j the total absolute coefficient majorant is $I_0(2m\delta)$. The inactive array has majorant $I_0(2\delta)^m$; therefore the active majorant is

$$A = I_0(2m\delta) - I_0(2\delta)^m = m(m-1)\delta^2 + O_m(\delta^4).$$

In particular, the total mass with ℓ active sites is at most

$$\binom{K}{\ell} A^\ell I_0(2\delta)^{m(K-\ell)} \le e^{O_{m,\tau}(1)} \frac{C_{m,\tau}^\ell}{\ell!}. \tag{R15}$$

The bound is summable uniformly in sufficiently large K.

An active balanced site has even total order. Its order-two arrays have one $+1$ and one -1 in two different cycles. The majorant of all order-at-least-four balanced arrays is $O_m(\delta^4)$, by its convergent coefficient series. Marking one such site and summing the remaining sites gives an error $O_m(K\delta^4)I_0(2m\delta)^{K-1} = O_{m,\tau}(K^{-1})$. Every remaining active site is consequently a chord with a type in $\mathcal{C}_m$ and one of two orientations. Its exact signed Bessel coefficient is $-J_1(2\epsilon)^2$, because $i^{-1}J_{-1}(2\epsilon) = iJ_1(2\epsilon)$. For ℓ such sites the common coefficient is

$$(-J_1(2\epsilon)^2)^\ell J_0(2\epsilon)^{mK-2\ell}. \tag{R16}$$

Fix ℓ, its type list and orientations. The Weyl phase is a sum of crossing contributions. Indexing chords in first-endpoint order, as in (H9) of the harmonic theorem, (R6) uses the inverse orbit and reverses the phase sign. If chords are instead indexed by time label, each edge also records which chord has the earlier first endpoint. These signs do not change the limiting argument. The phase lemma of that theorem extends to the ordered grid $j_1 < \cdots < j_\ell$ as follows. Remove neighborhoods of coincident labels, circle-distance boundaries and equal crossing distances, of normalized size $O_\ell(\eta + K^{-1})$. On the remaining configurations, the unique largest crossing distance dominates the signed Pell sum. Its harmonic average tends to one below the threshold $(1-2\beta)/2$ and to zero above it, by the quadratic-sum estimate (H11)–(H13). The same argument on two copies, additionally excluding ties between edges in distinct copies, gives variance tending to zero. Restriction to the simplex merely inserts a bounded indicator whose boundary has measure zero. Thus

$$K^{-\ell} \sum_{0 \le j_1 < \cdots < j_\ell < K} \mathrm{phase}(a; j, c, \sigma) \longrightarrow I_c(\beta) \quad \text{in } L^2(\mathcal{A}_K). \tag{R17}$$

For edgeless diagrams the phase is identically one, and $K^{-\ell}\binom{K}{\ell} \to 1/\ell!$, including $\ell = 0$. Conditioning on units has vanishing cost by (H8); all these normalized phase sums are bounded.

Now $KJ_1(2\epsilon)^2 \to t/2$ and $J_0(2\epsilon)^{mK-2\ell} \to e^{-mt/2}$ at fixed ℓ. The 2^ℓ orientations cancel the factors $2^{-\ell}$. Equations (R15)–(R17), with a fixed active-site cutoff followed by its removal, prove (R2)–(R3). The deterministic absolute errors are uniform in a; the fixed-diagram phase limits supply the L^2 convergence.

The Fourier moments of averaged empirical measures converge to (R2). Compactness of the circle supplies a subsequential probability limit; all its Fourier moments are specified, including the negative ones by complex conjugation. Trigonometric-polynomial approximation gives uniqueness and convergence of the full sequence. The same approximation and (R3) give weak convergence in probability. Dominated convergence in the simplex integrals and in (R2) gives continuity in β, t, uniformly on compact t intervals at each fixed m.

25.5 5. Endpoints: a direct noncrossing enumeration

At $\beta = 0$, all $I_c = 1/\ell!$. Summation in (R2) gives $e^{-mt/2}e^{-\binom{m}{2}t} = e^{-m^2t/2}$. At $\beta = 1/2$, only noncrossing type lists survive and each contributes $1/\ell!$. Let $A_{m,\ell}$ count these lists; $A_{m,0} = 1$.

For $\ell \ge 1$, delete the chord with the largest label, of type (r, s). Its endpoints split the remaining circular order into complete sets of $s - r$ and $m - s + r$ cycle rows. Noncrossing forces every other chord to stay in one of the two arcs. If p of the remaining labels lie in the first arc, their choice gives $\binom{\ell-1}{p}$; both relative orders retain the original definition. Conversely any two such noncrossing lists give exactly one list with the marked chord restored. Hence

$$A_{m,\ell} = \frac{m}{2} \sum_{j=1}^{m-1} \sum_{p=0}^{\ell-1} \binom{\ell-1}{p} A_{j,p} A_{m-j,\ell-1-p}. \tag{R18}$$

The coefficient is initially $m-j$, the number of (r, s) with difference j; symmetry under $j \leftrightarrow m-j$ replaces it by $m/2$.

Set $H_m(t) = \sum_\ell A_{m,\ell}(-t)^\ell/\ell!$. Equation (R18) gives $H_m' = -(m/2)\sum_{j=1}^{m-1} H_j H_{m-j}$, $H_m(0) = 1$. As formal series in z, $H = \sum_{m\ge1} H_m z^m$ obeys $\partial_t H = -zH\partial_z H$, $H(0, z) = z/(1-z)$, or equivalently

$$H = z(1+H)e^{-tH}.$$

Formal Lagrange inversion then yields

$$H_m(t) = \frac{1}{m}[u^{m-1}](1+u)^m e^{-mtu} = \frac{1}{m}\sum_{\ell=0}^{m-1} \binom{m}{\ell+1} \frac{(-mt)^\ell}{\ell!}. \tag{R19}$$

The last polynomial is $L^{(1)}_{m-1}(mt)/m$. This proves (R4), including $A_{m,\ell} = m^{\ell-1}\binom{m}{\ell+1}$ for $0 \le \ell < m$ and zero for $\ell \ge m$. For the conventional identification with free unitary Brownian motion, see N. Demni, *Lagrange inversion formula, Laguerre polynomials and the free unitary Brownian motion*, equation (1), arXiv:1606.03755v3. No stochastic-process convergence is inferred from the marginal law.

25.6 6. A closed second-return formula

For $m = 2$ every type is $(0, 1)$ and all distinct chords cross. Writing $b = (1 - 2\beta)/2$, we obtain

$$F_{2,\beta}(t) = e^{-t} \sum_{\ell=0}^{\infty} \frac{(-t)^\ell}{\ell!}\, t(K_\ell, w_\beta). \tag{R20}$$

When $0 < b < 1/3$, a finite set of circle points at pairwise distance less than b must lie in an arc of length b. To see this, put one point at zero. All others lie within clockwise distance b or counterclockwise distance b. If the maximal distances in these two directions are u, v, then $u + v \le 2b < 1 - b$; pairwise closeness therefore forces $u + v < b$. The covering arc has a unique first sample point, except on a null set. For ℓ independent uniform points, each candidate first point covers all others with probability $b^{\ell-1}$, and these events are disjoint. Consequently $t(K_\ell, w_\beta) = \ell b^{\ell-1}$ for $\ell \ge 1$. The formulas extend to $b = 1/3$ and $b = 0$ by continuity, interpreting the $\ell = 1$ value as one. Summing (R20) proves (R5).

25.7 7. What this removes, and what remains open

The earlier weighted-observable moment limit alone did not justify time-ordered exponentiation. Equations (R10)–(R15) provide the missing uniform summability for this arithmetic model. They also avoid extending the small-harmonic commuting normal form beyond its valid phase regime.

The new observable is the actual unitary $U^K_{\epsilon,a}$. Nevertheless $K \asymp \log N$, whereas the relevant Heisenberg scale is of order D. Here m is fixed, and constants and phase-cutoff complexities depend on m. Neither this proof nor its noncrossing endpoint controls $m \asymp D/K$, joint microscopic correlations, or fixed-strength classical perturbations. In particular the result supplies no positive BGS theorem. Lean formalization and an external referee assessment remain outstanding.

Chapter 5 — Return laws I (analytic and smooth kicks)

26 Fixed-harmonic returns from a convergent complex gas

Numbering convention: section and theorem references within this chapter use the source module's local numbering; source Section 1 is subsection 26.1. Equation references carry the prefix Ret.

Status. Project proof, checked by the author, with separately scoped exact finite checks and floating matrix experiments. Independent review and priority verification are outstanding. The starting repository revision is 9ed93cd1bc5e3362c1dbf520f0f391478dfb6969; this note also uses the uncommitted 32-class lattice proof in the even-period note. The target remains microscopic BGS for fixed smooth chaotic systems. The results below concern arithmetic return times and do not achieve that target.

26.1 Statements and the physical regime

Let

$$A = \begin{pmatrix} 3 & 4 \\ 2 & 3 \end{pmatrix}, \quad \lambda = 3 + 2\sqrt{2}, \quad b_0 = 0,\ b_1 = 1,\ b_{j+1} = 6b_j - b_{j-1}.$$

For odd $m \geq 3$ set $K = 2m$, $N = b_m$, $D = (N-1)/2$. Use centered Weyl operators with $W(v)W(w) = e_N(\omega(v,w)/2)W(v+w)$ and $U_0 W(v) U_0^{-1} = W(Av)$, normalized so that $U_0^K = I$. For a nonzero integer a with $\gcd(a, N) = 1$, define

$$U_{\epsilon,a} = e^{2i\epsilon\cos(2\pi a x/N)} U_0, \qquad C_a = 1 + \frac{4\pi a^2}{\lambda - 1}, \qquad s_a = \frac{1}{32\sqrt{C_a}}. \tag{Ret.G1}$$

Here tr_- denotes the normalized odd trace. In particular $a = 1$ is admissible for every K.

Theorem 26.1 (uniform return bound)**.** *For every positive integer r and real ϵ with $s = r|\epsilon| \leq s_a$,*

$$|\,\mathrm{tr}_-\, U_{\epsilon,a}^{rK}| \leq 34 e^{-Ks^2/2} + \frac{e^{2Ks}}{N-1}. \tag{Ret.G2}$$

The number of replicas r is not held fixed in this finite inequality. Every central aggregate and every Bessel charge is included.

Corollary 26.2 (all fixed returns and the global return law)**.** *Fix $a \neq 0$, restricting to the indicated unit subsequence if necessary. For every real sequence $\epsilon_K \to 0$ and every fixed integer $r \geq 1$,*

$$\mathrm{tr}_-\, U_{\epsilon_K,a}^{rK} - e^{-r^2 K \epsilon_K^2} \longrightarrow 0. \tag{Ret.G3}$$

If $K\epsilon_K^2 \to \infty$, the empirical eigenphase measure of $U_{\epsilon_K,a}^K$ converges weakly to uniform measure on the circle. If $K\epsilon_K^2 \to t < \infty$, its Fourier moments are $e^{-r^2 t}$, the wrapped centered Gaussian with variance $2t$. These are global laws for the return operator.

There is also a relative correction, not just decay to zero. Put

$$S_{a,K} = \sum_{\ell=1}^{m-1} \left[1 - \cos\left(4\pi a^2 \frac{b_{m-\ell}}{b_m}\right)\right], \quad S_a = \sum_{\ell \geq 1} [1 - \cos(4\pi a^2 \lambda^{-\ell})]. \tag{Ret.G4}$$

For example $S_1 = 1.622127498679229524490366806 8\ldots$. Both series and their comparison will be bounded below.

Theorem 26.3 (relative fourth-order transition)**.** *Fix $a \neq 0$. Let $r_K \geq 1$ be any sequence of positive integers, let ϵ_K be real, $x_K = r_K \epsilon_K$, and suppose $K x_K^4 \to u < \infty$. Then, along admissible moduli,*

$$\frac{\mathrm{tr}_-\, U_{\epsilon_K,a}^{r_K K}}{J_0(2x_K)^K} - \exp\left[-\frac{1 - r_K^{-2}}{3} S_a K x_K^4\right] \longrightarrow 0. \tag{Ret.G5}$$

The denominator is positive eventually. Formula (Ret.G5) is uniform in the replica count; it does not require r_K to converge. If r is fixed and $K\epsilon_K^4 \to v < \infty$, its correction is $\exp[-r^2(r^2-1)S_a v/3]$.

The symbol $2\cos(2\pi aq)$ is fixed and smooth when a is fixed. Nevertheless the associated classical kick strength is proportional to ϵ/N. These statements do not concern a fixed nonzero classical perturbation, whose quantum exponent would be of order N. The distinction remains essential even in (Ret.G5), where $r_K K$ may be large.

26.2 All ordered phases have a summable spatial kernel

Let $\boldsymbol{n}_j = (n_{1j}, \dots, n_{rj}) \in \mathbb{Z}^r$ be the indices at site j in the r consecutive periods, and let $c_j = \sum_\nu n_{\nu j}$. Define the replica matrix

$$H_{\nu\mu} = \begin{cases} 1, & \nu \le \mu, \\ -1, & \nu > \mu. \end{cases}$$

Multiplying Weyl operators in their original period order gives the phase

$$e_N\left(a^2 \sum_{i<j} b_{j-i} \boldsymbol{n}_i^{\mathsf{T}} H \boldsymbol{n}_j\right). \tag{Ret.G6}$$

Indeed equal-period pairs give the diagonal of H; for different periods the earlier occurrence of site i gives a plus sign and the earlier occurrence of site j gives a minus sign. This derivation does not assume that the aggregate is zero or central.

Since $b_{K-d} \equiv -b_d \pmod N$ and $b_m = N$, the phase in (Ret.G6) has the exact real representative

$$\prod_{i<j} e^{i\theta_{j-i} \boldsymbol{n}_i^{\mathsf{T}} H \boldsymbol{n}_j}, \qquad \theta_d = \begin{cases} 2\pi a^2 b_d/N, & d < m, \\ 0, & d = m, \\ -2\pi a^2 b_{K-d}/N, & d > m. \end{cases} \tag{Ret.G7}$$

At the antipodal separation the omitted angle is an integer multiple of 2π, including for all replica vectors. Because $b_{m-\ell}/b_m \le \lambda^{-\ell}$,

$$\sup_i \sum_{j \ne i} |\theta_{|j-i|}| \le \frac{4\pi a^2}{\lambda - 1} = C_a - 1. \tag{Ret.G8}$$

For the norm $l(\boldsymbol{n}) = \sum_\nu |n_\nu|$, $|\boldsymbol{n}^{\mathsf{T}} H \boldsymbol{p}| \le l(\boldsymbol{n}) l(\boldsymbol{p})$. Thus all these estimates are independent of r.

The normalized full trace selects precisely

$$\mathcal{C} = \{c : \sum_j c_j A^j(1,0) = 0 \bmod N\}.$$

The unit assumption on a is used here. The 32-class proof gives

$$\sum_{c \in \mathcal{C} \setminus \{0\}} z^{\|c\|_1} \le G_K(z) := e^{5Kz^8} - 1 + 2(13/50)^K + 31(3/5)^K \quad (0 \le z \le 3/10). \tag{Ret.G9}$$

Here is the 32-class argument. On parity-constant vectors the operator L (symbol $z + z^{-1} - 6$) is given by $B = \left(\begin{smallmatrix} -6 & 2 \\ 2 & -6 \end{smallmatrix}\right)$, and $\mathbf{Z}^2/B\mathbf{Z}^2$ has order $\det B = 32$ and maps injectively to $\mathcal{C}/L\mathbf{Z}^K$. A Hermite basis of B is $\left(\begin{smallmatrix} 16 & 10 \\ 0 & 2 \end{smallmatrix}\right)$, so the classes are represented by (r, s), $0 \le r < 16$, $0 \le s < 2$, with fractional preimages $p_{\text{even}} = -(3r + s)/16$, $p_{\text{odd}} = -(r + 3s)/16$. The noninteger classes have numerator pairs among $(0,8), (1,3), (1,5), (2,2), (2,6), (3,7), (4,4), (5,7), (6,6), (8,8)$. With $x = z^{1/4}$ and $H_j(x) = (x^j + x^{16-j})/(1 - x^{16})$, the Green-kernel bound $\|Lp\|_1 \ge 4\|p\|_1$ (the inverse of L has kernel $-\lambda^{-|n|}/(4\sqrt{2})$, of ℓ^1 norm $1/4$) bounds the (i,j) class by $[H_i(x) H_j(x)]^{K/2}$; at $x = 3/4$ the maximum is $H_1 H_3 = \frac{10040323357589760}{28926125721773329} < \frac{9}{25}$, so the 31 noninteger classes contribute at most $31(3/5)^K$. For the integer class, intervals starting at a fixed site have total activity, with $v = 2z^4/(1 - z^4)$,

$$B(z) = z^4 \frac{2v + 4v^2}{1 - 4v} = \frac{4z^8(1 + 3z^4)}{(1 - z^4)(1 - 9z^4)} \le 5z^8 \qquad (z \le 3/10);$$

disjoint intervals therefore contribute at most $e^{KB(z)}$, and configurations with no adjacent zeros at most $2(2\sqrt{v})^K \le 2(13/50)^K$, since $2\sqrt{v} < 13/50$. Subtracting the zero configuration gives (Ret.G9).

The difference between the full and odd normalized traces is at most $e^{2rK|\epsilon|}/(N-1)$. Individual Weyl words need not preserve parity; the correction uses the linear trace with the odd projector.

26.3 A complex-gas lemma with a proof of convergence

At a site with aggregate zero, the vacuum weight is $v = J_0(2\epsilon)^r$. Nonzero particles are vectors $\boldsymbol{n} \neq 0$ with $\sum_\nu n_\nu = 0$, of activity

$$w_j(\boldsymbol{n}) = v^{-1} \prod_\nu J_{n_\nu}(2\epsilon) e^{i\boldsymbol{h}_j \cdot \boldsymbol{n}}, \tag{Ret.G10}$$

where $\boldsymbol{h}_j$ is any real external field. The factors $i^{\sum n_\nu}$ are one. The activities may have either sign and may be complex. Two particles at the same site have factor $1 + f = 0$; at different sites $1 + f$ is the phase in (Ret.G7), always evaluated in increasing physical site order. Thus f is symmetric as a function of the two labeled particles. In particular $|1 + f| \leq 1$.

Here is the precise estimate that will be used. Suppose

$$|f((i, \boldsymbol{n}), (j, \boldsymbol{p}))| \leq c_{ij} l(\boldsymbol{n}) l(\boldsymbol{p}), \quad \sup_i \sum_j c_{ij} \leq C,$$

and

$$M = \sup_j \sum_{\boldsymbol{n} \neq 0} |w_j(\boldsymbol{n})| \, l(\boldsymbol{n}) e^{l(\boldsymbol{n})}, \qquad CM < 1.$$

For k allowed sites the gas partition function is nonzero, has the logarithm continued from zero activities, and satisfies

$$\left| \log Z - \sum_{j,\boldsymbol{n}} w_j(\boldsymbol{n}) \right| \leq \frac{1}{2} k C M^2. \tag{Ret.G11}$$

For clarity, this is an application of the classical Penrose tree-graph identity, not a newly claimed graph identity. Its complex version appears as Proposition 1 of Procacci and Yuhjtman, arXiv:1508.07379v1. The needed specialization and convergence argument follow.

Root a connected labeled graph at its first vertex, assign distance levels, and give each other vertex its smallest neighbor in the preceding level as parent. The resulting tree T determines an interval of graphs: one may add all edges within a level, and all edges from a vertex to preceding-level vertices with label no smaller than its parent. Conversely exactly these additions preserve the levels and smallest parents. These intervals partition the connected graphs. It follows algebraically that their sum of products of f is a sum over trees of

$$\prod_{e \in T} f_e \prod_{e \in R(T) \setminus T} (1 + f_e).$$

Taking absolute values bounds the connected sum by $\sum_T \prod_{e \in T} |f_e|$.

The rooted positive tree series, with its root particle fixed and its root activity omitted, obeys the iterative majorant

$$T_\gamma = \exp\left(\sum_{\gamma'} |w_{\gamma'}| \, |f_{\gamma\gamma'}| T_{\gamma'} \right).$$

Start the iteration at one. If $T_{\gamma'} \leq e^{l(\gamma')}$, the next iterate is at most $e^{CMl(\gamma)} \leq e^{l(\gamma)}$. Monotone convergence bounds the entire rooted series and gives

$$T_\gamma - 1 \leq CMl(\gamma) e^{l(\gamma)}.$$

In the connected logarithm, a graph with $n \geq 2$ vertices has weight $1/n!$, whereas the rooted series has $1/(n-1)!$. The ratio is $1/n \leq 1/2$. Summing its root activity therefore proves (Ret.G11).

For a fully justified logarithm, first truncate the charge set and multiply all activities by one complex parameter. The usual finite exponential formula partitions graphs into connected

components and identifies the Taylor series of the partition function with the exponential of the connected series near zero. The strict tree bound gives absolute convergence on a disk containing the closed unit disk. Analytic continuation establishes the identity and nonvanishing there. Finally pass to the countable charge set by absolute convergence; the same bounds are uniform in the cutoff. This also establishes the analytic version for complex ϵ used below.

In the present gas choose $c_{ii} = 1$ and $c_{ij} = |\theta_{|j-i|}|$ for different sites, so $C = C_a$. Every nonzero zero-sum vector has norm at least two, which handles the same-site hard core.

26.4 Uniform activity and one-site estimates

Write $\delta = |\epsilon|$, $s = r\delta \le s_a$. The Bessel series and modified-Bessel convolution give

$$M \le \frac{2e^2 s^2 e^{e^2 s^2}}{(1 - \delta^2 e^{\delta^2})^r} \le 16 s^2. \tag{Ret.G12}$$

To obtain the numerator, multiply each charge by $e^{|n_\nu|}$, majorize by $I_{|n_\nu|}(2e\delta)$, and differentiate their zero-coefficient convolution $I_0(2re\delta)$. Each series monomial has degree at least the sum of the absolute indices. Use $I_1(2se) \le se\, e^{s^2 e^2}$ and $|J_0(2\epsilon)| \ge 1 - \delta^2 e^{\delta^2}$. The last inequality follows for $s \le 1/32$ by the positive exponential series, using $r\delta^2 \le s^2$. In particular $C_a M \le 16/1024 < 1$.

The exact one-site sum is

$$v \left(1 + \sum_{\boldsymbol{n} \neq 0} w_j(\boldsymbol{n}) \right) = B_j := J_0 \left(2\epsilon \left| \sum_{\nu=1}^{r} e^{i h_{\nu j}} \right| \right). \tag{Ret.G13}$$

This follows by taking the zero Fourier coefficient of the product of r shifted factors $e^{2i\epsilon \cos \alpha}$. For real fields $|\sum_\nu e^{i h_{\nu j}}| \le r$. Taylor's formula, uniformly in these fields, gives

$$\left| r \log J_0(2\epsilon) + (B_j/v - 1) + \epsilon^2 \left| \sum_{\nu} e^{i h_{\nu j}} \right|^2 \right| \le 16 s^4. \tag{Ret.G14}$$

One direct estimate is to put $A = r\epsilon^2$ and $B = \epsilon^2 |\sum e^{i h_\nu}|^2$. The bounds $|A|, |B| \le s^2$, $|r \log J_0(2\epsilon) + A| \le s^4$, $|v^{-1} - 1 - A| \le 4s^4$, and $|B_j - 1 + B| \le s^4/3$ suffice. They follow from the absolutely convergent Bessel and logarithm series for $s \le 1/32$. The same estimate holds for complex ϵ, with absolute values on the remainders.

Combining (Ret.G11)–(Ret.G14), for k clean sites with specified fields, the corresponding conditional partition sum $\mathcal{Q}$ satisfies

$$\log |\mathcal{Q}| \le -\delta^2 \sum_{j \text{ clean}} \left| \sum_{\nu} e^{i h_{\nu j}} \right|^2 + 160 C_a K s^4. \tag{Ret.G15}$$

The harmless replacement of k by K is convenient. The constants account for $16ks^4$ and at most $128 C_a k s^4$ of connected-graph remainder.

26.5 Nonzero aggregates do not destroy the cancellation

Fix $c \in \mathcal{C}$, and let $B = \{j : c_j \neq 0\}$. Condition on the complete replica vector at every site in B. Write $S_B = \sum_{j \in B} l(\boldsymbol{n}_j)$. All their interactions with clean sites are real external fields. The entrywise bound $|H_{\nu\mu}| = 1$ and (Ret.G8) give

$$\sum_{j \notin B} \sum_{\nu} |h_{\nu j}| \le r(C_a - 1) S_B.$$

Also

$$\left|\sum_{\nu} e^{ih_\nu}\right|^2 \geq r^2 - 2r\sum_{\nu}|h_\nu|.$$

Thus (Ret.G15) bounds the clean sum by

$$\exp\left[-Ks^2 + 160C_aKs^4 + s^2|B| + 2(C_a-1)s^2S_B\right]. \tag{Ret.G16}$$

The phases internal to B, including $i^{\sum_j c_j}$, have unit modulus.

Let $\tau = 2(C_a-1)s^2$. For a dirty site of aggregate c_j, the weighted absolute mass is bounded by

$$\sum_{\sum n_\nu = c_j}\prod_{\nu}|J_{n_\nu}(2\epsilon)|e^{\tau\sum|n_\nu|} \leq I_{|c_j|}(2re^{\tau}\delta) \leq e^{s^2e^{2\tau}}\frac{(se^{\tau})^{|c_j|}}{|c_j|!}.$$

The first inequality follows termwise from $|J_n(2\epsilon)|e^{\tau|n|} \leq I_{|n|}(2e^{\tau}\delta)$. Since $|c_j| \geq 1$, the factor e^{s^2} in (Ret.G16) is absorbed by setting

$$z = s\exp\left[\tau + s^2(1+e^{2\tau})\right] \leq 2s.$$

Therefore the contribution of each aggregate is at most

$$e^{-Ks^2+160C_aKs^4}z^{\|c\|_1}. \tag{Ret.G17}$$

All charges at dirty sites were integrated, not truncated. All Fourier series converge absolutely at finite K.

Sum (Ret.G17) using (Ret.G9). The entire normalized full trace is bounded by

$$e^{-Ks^2+160C_aKs^4}[1+G_K(z)] \leq 34\exp[-Ks^2 + 160C_aKs^4 + 1280Ks^8].$$

For $s^2 \leq 1/(1024C_a)$ the positive terms are less than $Ks^2/2$. Adding the odd-projector error proves (Ret.G2). The same argument, omitting $c=0$, gives the sharper comparison

$$|\operatorname{tr}_- U_{\epsilon,a}^{rK} - Q_{r,K}(\epsilon)| \leq e^{-Ks^2+160C_aKs^4}G_K(2s) + \frac{e^{2Ks}}{N-1}, \tag{Ret.G18}$$

where $Q_{r,K}$ denotes the exact zero-aggregate contribution.

26.6 Quartic pressure, including a growing number of replicas

At zero external field the gas argument also proves that $Q_{r,K}$ is nonzero on $|r\epsilon| \leq s_a$. Its analytic logarithm, normalized to zero at $\epsilon = 0$, obeys

$$\left|K^{-1}\log Q_{r,K}(\epsilon) + (r\epsilon)^2\right| \leq 160C_a|r\epsilon|^4. \tag{Ret.G19}$$

It is an even function: every zero-aggregate configuration has total charge zero, so changing ϵ to $-\epsilon$ leaves its weight invariant.

The order-four coefficient has a finite description. An order-two particle is $e_\nu - e_\mu$, $\nu \neq \mu$, with leading activity $-\epsilon^2$. For r replicas let

$$q_r = \#\{(\alpha,\beta) : \alpha,\beta = e_\nu - e_\mu,\ \alpha^{\mathsf{T}}H\beta \neq 0\} = \frac{r^2(r^2-1)}{3}. \tag{Ret.G20}$$

To verify the count, for $\nu < \mu$ the row $(e_\nu - e_\mu)^{\mathsf{T}}H$ equals two on the interval $[\nu,\mu)$ and zero elsewhere. If that interval has length d, precisely $2d(r-d)$ oriented charges cross its boundary.

Summing both orientations of its $r-d$ placements gives $4\sum_{d=1}^{r-1} d(r-d)^2 = r^2(r^2-1)/3$. All nonzero pairings equal 2 or -2, in equal numbers.

The one-site part of the logarithm, with all charges at that site resummed, is $\log J_0(2r\epsilon)$. At order four, a connected contribution involving different sites has exactly two order-two particles. Their sum is $q_r[\cos(2\theta_{j-i})-1]\epsilon^4$. Graphs on three occupied sites start at order six. The two complementary separations $m-\ell, m+\ell$ have total multiplicity K; the antipodal phase is zero. Consequently

$$K^{-1}\log Q_{r,K}(\epsilon) = -(r\epsilon)^2 - \left[\frac{1}{4} + \frac{1-r^{-2}}{3}S_{a,K}\right](r\epsilon)^4 + O_a(|r\epsilon|^6). \tag{Ret.G21}$$

The remainder is uniform in both r and K. Indeed apply Cauchy's coefficient bound to the even analytic function in (Ret.G19), in the variable $x = r\epsilon$, on the circle $|x| = s_a$. For $|x| \le s_a/2$, the tail from degree six is bounded by $220000C_a^2|x|^6$. The strict activity criterion supplies analyticity on a neighborhood of that circle.

For the series in (Ret.G4),

$$\sum_{\ell>M}[1-\cos(4\pi a^2\lambda^{-\ell})] \le \frac{8\pi^2a^4\lambda^{-2(M+1)}}{1-\lambda^{-2}}.$$

The Binet formula and $|\cos x - \cos y| \le |x^2-y^2|/2$ for $x, y \ge 0$ also give

$$|S_{a,K} - S_a| \le \frac{16\pi^2a^4(m-1)\lambda^{-K}}{1-\lambda^{-K}} + \frac{8\pi^2a^4\lambda^{-K}}{1-\lambda^{-2}}. \tag{Ret.G22}$$

For $a \neq 0$, $S_a > 0$, since already the first angle is not an integer multiple of 2π.

If Kx_K^4 is bounded, then $K|x_K|^6 \to 0$. Subtract $\log J_0(2x_K)$ in (Ret.G21), use (Ret.G22), and exponentiate. This proves (Ret.G5) first for $Q_{r,K}$. To replace it by the physical odd trace, divide (Ret.G18) by $J_0(2x_K)^K$. The first error is bounded by a constant times $G_K(2|x_K|)$, which tends to zero since $K|x_K|^8 \to 0$. The odd error tends to zero because its logarithm is $-\frac{1}{2}K\log\lambda + 2K|x_K| + O(Kx_K^2) + O(1)$. This completes Theorem 26.3 without a fixed-r assumption.

For bounded $K\epsilon_K^2$ and fixed r, (Ret.G18)–(Ret.G19) give (Ret.G3). For $K\epsilon_K^2 \to \infty$, use (Ret.G2). A subsequence argument covers arbitrary sequences. Fourier approximation by trigonometric polynomials then proves Corollary 26.2, including the uniform circle limit.

26.7 Independent checks and a failed test that was corrected

The companion fixed_harmonic_return_gas.py performs the following finite checks:

- all 33,866 labeled simple graphs with two through six vertices, including exact interval cardinalities for every connected graph;
- 4,320 seeded ordered Weyl words, with one through six replicas, four moduli, and three harmonics, including constructed nonzero central aggregates and completely unrestricted words;
- all 49,192 oriented charge-species pairings through twelve replicas;
- eighteen high-precision checks of the conservative Bessel constants;
- physical matrices at $N = 1189$ in both parity sectors, two harmonics, and the first five return traces at three kick strengths.

The first attempt to extract the quartic coefficient from the odd trace alone failed. The reason was substantive: its finite-N noncentral projection correction has lower-order Taylor terms. Dividing it by ϵ^4 makes it grow as ϵ decreases. The corrected experiment combines the two physical parity blocks to obtain the exact normalized full trace and reports the odd trace

separately. At $K = 10$, (Ret.G9) excludes all nonzero aggregate terms through order four. Thus the full trace genuinely tests (Ret.G21) at that order.

All ten matrix cases show fourth-coefficient error decreasing by roughly a factor four when the kick is halved at the smaller strengths. This supports the derived coefficient; it is not an interval certificate or an independent proof of the uniform remainder. The test of the full-space coefficient does not erase the odd-projection term from (Ret.G2), (Ret.G5), or (Ret.G18).

The exact verifier invocation is reproduced as listing S1 in the Supplement.

26.8 Why this still does not reach microscopic BGS

The estimate covers $r \le s_a/|\epsilon|$. At the same time, removal of the first return requires $K\epsilon^2 \to \infty$. Therefore its largest directly controlled time obeys

$$rK \le s_a K/|\epsilon| = o_a(K^{3/2}) = o_a((\log N)^{3/2}). \tag{Ret.G23}$$

Here $K \asymp \log N$. The exact microscopic Fourier bridge requires joint trace information at times comparable to $D \asymp N$. To put $rK \asymp N$ inside the present radius would instead require $|\epsilon| = O_a(K/N)$, which forces $K\epsilon^2 \to 0$ and leaves the first arithmetic return intact. This is an incompatibility of the proved ranges, not a counterexample to BGS.

The fixed-classical-strength route is outside a different hypothesis: it requires ϵ of order N, rather than a small quantum kick. The gas expansion cannot be continued there by the bounds above. The local coefficient $J_0(2r\epsilon)$ itself acquires zeros when $r|\epsilon|$ is of order one, so zero-free pressure continuation across that range also needs a new argument.

The next missing input is a nonperturbative comparison of the required joint traces, including symmetry sectors, at Heisenberg times. Neither global equidistribution of U^K nor the relative law (Ret.G5) supplies that comparison. These implications remain open; the user's original target has not been relabeled as the results of this note.

27 Analytic and smooth position kicks: resonances and return asymptotics

Numbering convention: section and theorem references within this chapter use the source module's local numbering; source Section 1 is subsection 27.1. Equation references carry the prefix Ret.

Status. Project proof, checked by the author, with finite exact controls; independent review and priority verification remain open. Base revision: 9ed93cd1bc5e3362c1dbf520f0f391478dfb6969. The arithmetic setup and complex-gas lemma are those of the fixed-harmonic note. This is a further result about arithmetic returns, not microscopic BGS or a fixed nonzero classical perturbation.

27.1 Statement for a class of fixed smooth symbols

Let $K = 2m$, $m \ge 3$ odd, $N = b_m$, and A, U_0, D be as in (Ret.G1) of the preceding note. Fix $a \ne 0$, with $\gcd(a, N) = 1$ along the sequence being considered. In particular $a = 1$ needs no restriction.

Let f be a nonzero real even 2π-periodic function with

$$\widehat{f}_0 = 0, \qquad F_\rho := \sum_{n\in\mathbb{Z}} |\widehat{f}_n| e^{\rho|n|} < \infty \quad \text{for some } \rho > 0.$$

Every real analytic periodic function has this property after reducing the strip width if necessary. Define

$$U_{\epsilon,f,a} = e^{i\epsilon f(2\pi a x/N)} U_0, \qquad \sigma_f^2 = \frac{1}{2\pi}\int_0^{2\pi} f(\theta)^2\, d\theta > 0, \quad F_0 = \sum_n |\widehat{f}_n|.$$

The kick is a fixed smooth position symbol and preserves parity. Its classical strength is nevertheless proportional to ϵ/N.

Theorem 27.1. *There are constants $s_0 > 0$ and $C < \infty$, depending only on f, a, such that for every admissible K, every integer $r \geq 1$, and every real ϵ with $s = r|\epsilon| \leq s_0$,*

$$|\operatorname{tr}_- U_{\epsilon,f,a}^{rK}| \leq e^{-\sigma_f^2 K s^2/4} + (3/4)^K + \frac{e^{KsF_0}}{N-1}, \tag{Ret.A1}$$

and

$$\left|\operatorname{tr}_- U_{\epsilon,f,a}^{rK} - e^{-\sigma_f^2 K s^2/2}\right| \leq \frac{C}{\sqrt{K}} + (3/4)^K + \frac{e^{KsF_0}}{N-1}. \tag{Ret.A2}$$

We may, and do, choose $s_0 F_0 < \log \lambda/4$, so the last error decays exponentially in K uniformly in the stated range. These are uniform estimates over the replica count, not estimates with an unspecified constant depending on r.

For every $\epsilon_K \to 0$ and each fixed r,

$$\operatorname{tr}_- U_{\epsilon_K,f,a}^{rK} - e^{-\sigma_f^2 r^2 K \epsilon_K^2/2} \to 0. \tag{Ret.A3}$$

In particular $K\epsilon_K^2 \to \infty$ gives uniform circle measure as the global eigenphase law of $U_{\epsilon_K,f,a}^K$. For $K\epsilon_K^2 \to t < \infty$ the law is the wrapped Gaussian of variance $\sigma_f^2 t$.

For this class, σ_f^2 is also the classical Green–Kubo variance: all nonzero-time autocorrelations vanish by Fourier orthogonality. Indeed $A^j(1,0)$ is not parallel to $(1,0)$ for $j \neq 0$. No assertion of this simpler variance formula is made for arbitrary two-dimensional symbols or for coboundaries.

27.2 A weighted bound for the entire resonance lattice

Let $L = S + S^{-1} - 6I$ on the cyclic index set and

$$\mathcal{C} = \{c \in \mathbb{Z}^K : \sum_j c_j A^j(1,0) = 0 \bmod N\}, \qquad q(c) = |\{j : c_j \neq 0\}|.$$

The 32-class calculation gives $p = L^{-1}c \in (1/16)\mathbb{Z}^K$. Its fractional parts are constant on each parity. A noninteger class therefore has no two consecutive zero coordinates.

Weighted lattice lemma. For every $\alpha > 0$ there are finite $B_\alpha > 0$ and $\eta_\alpha > 0$, independent of K, such that

$$\sum_{c \in \mathcal{C}} \eta^{q(c)} e^{-\alpha \|c\|_1} \leq e^{B_\alpha K \eta^3} + (3/4)^K \quad (0 \leq \eta \leq \eta_\alpha). \tag{Ret.A4}$$

The third power counts occupied Fourier sites. It replaces a bound in which a mode of index six would incorrectly cost six powers of the kick.

Here are explicit admissible constants. Put

$$b_\alpha = \frac{2}{e^{\alpha/2} - 1}, \qquad h_\alpha(g) = \exp\left[-\frac{\alpha}{200}\lambda^{(g-1)/2}\right], \quad H_\alpha(R) = \sum_{g \geq 1} h_\alpha(g) R^g.$$

The last series converges for every finite R. One can take

$$\eta_\alpha = \frac{1}{2 b_\alpha H_\alpha(2)}, \qquad B_\alpha = 2 b_\alpha^3 H_\alpha(1)^2. \tag{Ret.A5}$$

27.2.1 Long homogeneous gaps require large rational amplitudes

Suppose the forcing $c = Lp$ vanishes strictly between two successive nonzero forcing sites $i < j$, with gap $g = j - i$. If the preimage has no consecutive zeros in this interval, the homogeneous state (p_i, p_{i+1}) is nonzero. Along the interval

$$p_{n+1} = 6p_n - p_{n-1}, \qquad p_{n+1}^2 - 6p_n p_{n+1} + p_n^2 = Q \neq 0.$$

Since the entries have denominator dividing 16, $|Q| \geq 1/256$. The invariant cannot vanish on a nonzero rational state because its two real slopes λ, λ^{-1} are irrational. Writing $p_n = u\lambda^n + v\lambda^{-n}$ gives $Q = -32uv$. Both coefficients at a specified state are bounded by twice the ℓ^1 norm of that state. The two endpoint states, separated by $g - 1$ iterations, consequently satisfy

$$|p_i| + |p_{i+1}| + |p_{j-1}| + |p_j| \geq \frac{1}{100}\lambda^{(g-1)/2}. \tag{Ret.A6}$$

For example their norm product is at least $\lambda^{g-1}/32768$, and the arithmetic-geometric mean inequality gives (Ret.A6).

The endpoint norms summed over disjoint forcing gaps count each coordinate of p at most four times. Also $\|Lp\|_1 \geq 4\|p\|_1$: on the cycle, L^{-1} is convolution with the periodization of the Green kernel $g_n = -\lambda^{-|n|}/(4\sqrt{2})$ of $z + z^{-1} - 6$, all of whose terms have one sign, and $\sum_{n\in\mathbf{Z}} |g_n| = \frac{1}{4\sqrt{2}}(1 + \frac{2}{\lambda - 1}) = \frac{1}{4}$, so $\|p\|_1 \leq \frac{1}{4}\|Lp\|_1$. For a nonwinding component, or for all cyclic gaps in a winding configuration, this proves

$$\|c\|_1 \geq \frac{1}{100} \sum_{\text{forcing gaps } g} \lambda^{(g-1)/2}. \tag{Ret.A7}$$

27.2.2 Nonwinding components start at three occupied forcing sites

If an integer preimage has consecutive zeros, divide it into its unique components, joining nonzero coordinates whose gap is at most two. Add one exterior zero at either end of each component. The forcing supports of different components are disjoint.

A finite nonzero component has at least three nonzero forcing entries. Indeed its forcing Laurent polynomial is divisible by $z + z^{-1} - 6$. One nonzero monomial cannot vanish at λ; two would require λ^d to be rational for some $d \geq 1$, which is impossible.

For a component with $q \geq 3$ forcing sites and successive gaps $g_1, \ldots, g_{q-1}$, split the exponential weight in (Ret.A4) into two equal parts. Sum the first part over all nonzero integer forcing values, and apply (Ret.A7) to the second part. Its total weight is bounded by

$$(\eta b_\alpha)^q \prod_{\nu=1}^{q-1} h_\alpha(g_\nu).$$

This remains an upper bound after dropping the integrality and component constraints on the candidate forcing values. For a fixed starting site, the sum over every component is at most

$$\sum_{q\geq 3} (\eta b_\alpha)^q H_\alpha(1)^{q-1} \leq B_\alpha \eta^3.$$

Allowing arbitrary subsets of components, rather than only disjoint influence intervals, bounds the nonwinding partition by $e^{B_\alpha K \eta^3}$.

27.2.3 Winding configurations include all noninteger classes

For a cyclic preimage with no consecutive zeros, let $q \geq 1$ be the number of nonzero forcing sites. Its cyclic gaps sum to K and obey (Ret.A7). Mark a forcing site as origin. There are K

choices for its position and each unmarked configuration is counted q times. Thus the winding contribution is at most

$$\sum_{q=1}^{K} \frac{K}{q} (\eta b_\alpha)^q [z^K] H_\alpha(z)^q \le K 2^{-K} \sum_{q \ge 1} \frac{(\eta b_\alpha H_\alpha(2))^q}{q} \le K 2^{-K} \log 2 \le (3/4)^K.$$

The counting is over forcing vectors, whose real preimage is unique; no additional multiplicity of 32 is introduced. Together with the preceding part this proves (Ret.A4), including every noninteger class and arbitrary Fourier indices.

27.3 Analytic Fourier coefficients and the zero-aggregate gas

Let

$$e^{i\epsilon f(\theta)} = \sum_{n \in \mathbb{Z}} k_n(\epsilon) e^{in\theta}.$$

These are the exact coefficients of the physical diagonal kick, including all lattice aliases. For complex ϵ,

$$\sum_n |k_n(\epsilon)| e^{\rho|n|} \le e^{|\epsilon| F_\rho}, \qquad k_0(\epsilon) = 1 - \tfrac{1}{2}\sigma_f^2 \epsilon^2 + O_f(|\epsilon|^3). \tag{Ret.A8}$$

The vacuum is nonzero in a fixed disk. For any $0 < \beta < \rho$, both

$$\sum_{n \ne 0} |k_n(\epsilon)| e^{\beta|n|}, \qquad \sum_{n \ne 0} |n|\, |k_n(\epsilon)| e^{\beta|n|}$$

are $O_{f,\beta}(|\epsilon|)$. The second estimate follows by leaving a positive gap between β and ρ.

At a clean site take vectors $\boldsymbol{n} \ne 0$ with $\sum_\nu n_\nu = 0$, vacuum $v = k_0(\epsilon)^r$, and activities

$$w_j(\boldsymbol{n}) = v^{-1} \prod_\nu k_{n_\nu}(\epsilon) e^{i \boldsymbol{h}_j \cdot \boldsymbol{n}}.$$

At least two coordinates must be nonzero. For $s = r|\epsilon|$ sufficiently small, (Ret.A8) therefore yields

$$M_\beta := \sup_j \sum_{\boldsymbol{n} \ne 0} |w_j(\boldsymbol{n})|\, l(\boldsymbol{n}) e^{\beta l(\boldsymbol{n})} \le C_f s^2. \tag{Ret.A9}$$

For an explicit intermediate bound, if A_β and B_β denote the two nonzero-mode sums above, the left side is at most

$$\frac{r B_\beta}{|k_0|} \left[\left(1 + \frac{A_\beta}{|k_0|} \right)^{r-1} - 1 \right].$$

This proves the uniformity in r.

The ordered phase kernel is exactly (Ret.G6)–(Ret.G8), with the same matrix H and the same summable row bound. Apply the tree argument of Section 3 of the preceding note with root majorant $e^{\beta l}$. Its criterion is $C_a M_\beta < \beta$, and its connected remainder is still bounded by $\frac{1}{2} K C_a M_\beta^2 = O_{f,a}(K s^4)$. All charges and repeated-particle hard-core terms are included.

The full one-site sum at external fields is

$$B_j = \frac{1}{2\pi} \int_0^{2\pi} \exp\left(i\epsilon \sum_{\nu=1}^{r} f(\theta + h_{\nu j}) \right) d\theta.$$

Its first moment vanishes. Taylor's formula and the gas estimate give for the partition sum on any specified clean sites

$$\log |\mathcal{Q}| \le -\frac{\epsilon^2}{2} \sum_{j \text{ clean}} \left\| \sum_\nu f(\cdot + h_{\nu j}) \right\|_2^2 + C_{f,a} K s^3. \tag{Ret.A10}$$

The circle L^2 norm is normalized by $1/(2\pi)$. The cubic remainder is uniform because the summed function has supremum norm at most $r\|f\|_\infty$. The corresponding complex estimate at zero external field is

$$\left|\log Q_{r,K}(\epsilon) + \tfrac{1}{2}\sigma_f^2 K(r\epsilon)^2\right| \le C_{f,a} K|r\epsilon|^3. \tag{Ret.A11}$$

27.4 Conditioning on all nonzero aggregates

For a fixed $c \in \mathcal{C}$, condition on every replica vector at the dirty sites $B = \{j : c_j \neq 0\}$. As before $S_B = \sum_{j\in B} l(\boldsymbol{n}_j)$ and $\sum_{j\notin B,\nu} |h_{\nu j}| \le r(C_a - 1)S_B$. Translation in L^2 gives

$$\left\|\sum_\nu f(\cdot + h_\nu)\right\|_2^2 \ge r^2\sigma_f^2 - 2r\|f\|_2\|f'\|_2 \sum_\nu |h_\nu|.$$

Therefore the conditional clean sum is at most

$$\exp\left[-\tfrac{1}{2}\sigma_f^2 K s^2 + C_{f,a} K s^3 + C_{f,a} s^2(|B| + S_B)\right]. \tag{Ret.A12}$$

Choose $\alpha = \rho/4$ and make s_0 small enough that the weight $e^{C_{f,a}s^2 l}$ leaves at least this analytic strip width. The convolution at a dirty site of aggregate $c_j \neq 0$, including its factor $e^{C_{f,a}s^2}$, is bounded by

$$C'_{f,a} s\, e^{-\alpha|c_j|}. \tag{Ret.A13}$$

Indeed the total weighted mass of all nonzero aggregate coefficients is bounded by $(|k_0| + A_{\alpha+C_{f,a}s^2})^r - |k_0|^r = O_{f,a}(s)$. The absence of a nonzero aggregate in the all-vacuum term is essential.

Put $\eta = C'_{f,a}s$ and use (Ret.A4). The normalized full trace is at most

$$e^{-\sigma_f^2 K s^2/2 + C_{f,a} K s^3}\left[e^{B_\alpha K \eta^3} + (3/4)^K\right].$$

Shrinking s_0 makes both cubic exponents smaller than $\sigma_f^2 K s^2/4$ and ensures $\eta \le \eta_\alpha$. This proves (Ret.A1) after the odd-projector correction is added.

For completeness the rate in (Ret.A2) follows without an uncontrolled change of limits. Equations (Ret.A11)–(Ret.A13) and (Ret.A4) imply, after possibly decreasing s_0 again,

$$|Q_{r,K} - e^{-\sigma_f^2 K s^2/2}| \le C_{f,a} K s^3 e^{-c_f K s^2},$$

and the entire nonzero-aggregate contribution is at most

$$C_{f,a} K s^3 e^{-c_f K s^2} + (3/4)^K$$

for a fixed $c_f > 0$. Use $|e^z - 1| \le |z|e^{|z|}$, with the cubic exponent absorbed by the negative quadratic exponent. Finally

$$\sup_{s\ge 0} K s^3 e^{-c_f K s^2} = K^{-1/2} \sup_{y\ge 0} y^3 e^{-c_f y^2} < \infty \cdot K^{-1/2}.$$

This proves (Ret.A2).

27.5 A cubic term and a finite alias that the first prediction missed

Take

$$f(\theta) = 2\cos\theta + 2b\cos(6\theta), \qquad b \neq 0.$$

Its one-site third moment vanishes and $\sigma_f^2 = 2(1+b^2)$. The ordinary finite-component resonance $(1,-6,1) = Le_j$ has ordered Weyl phase $e_N(-6a^2)$. Its two signs contribute the cubic activity $-2ib\,e_N(-6a^2)\epsilon^3$ per site.

At $K = 10$, however, this is not the whole finite cubic coefficient. The exact additional relation

$$6v_0 + 6v_4 + v_7 = v_9 + v_1 + v_3 + v_5 + v_7 = 0 \bmod N, \qquad v_j = A^j(1,0),$$

comes from the parity-orbit sum. Its aggregate has fractional preimage $15/16$ on even sites and $13/16$ on odd sites, modulo integers; it is a winding class. Its phase at $a = 1, N = 1189$ is $e_N(210)$. For $b = 1/5$, exact enumeration gives the cubic coefficient of $K^{-1}\log(\operatorname{Tr} U^K_{\epsilon,f,a}/N)$

$$-\frac{2i}{5}e_N(-6) - \frac{2i}{25}e_N(210) \quad (K = 10). \tag{Ret.A14}$$

The second term was absent from the preliminary prediction. It was found by the independent atom-word enumeration, and is explicitly retained here. At $K = 14, 18$ the same complete cubic enumeration has only the ordinary term.

More generally, grouping adjacent terms of the length-m parity sum using $v_{j-1} + v_{j+1} = 6v_j$ produces a winding relation with $(m+1)/2 = (K+2)/4$ occupied Fourier sites using only modes 1 and 6. Thus these finite aliases cannot simply be declared absent. They are covered collectively by the winding part of (Ret.A4).

The rate $K^{-1/2}$ in (Ret.A2) is in fact attained by this example. For each fixed real t and fixed admissible a,

$$\sqrt{K}\left[\operatorname{tr}_- U^K_{t/\sqrt{K},f,a} - e^{-(1+b^2)t^2}\right] \longrightarrow -2ib\,t^3 e^{-(1+b^2)t^2}. \tag{Ret.A15}$$

Here is a remainder argument for this last assertion. For one period and an integer preimage, the ordered action is $a^2\sum_j p_j(Lp)_j$ modulo N. Separated nonwinding components therefore multiply exactly, including their phases. Normalize each local Fourier coefficient by k_0. The component sum is analytic, uniformly bounded by $C|\epsilon|^3$ on a fixed complex disk by the proof of (Ret.A4). Cauchy's estimate leaves $O(|\epsilon|^4)$ after its cubic coefficient.

The only finite three-monomial relations with coefficients in $\{\pm1, \pm6\}$ are translates of $\pm(1,-6,1)$. To see this, evaluate $a_0 + a_1 z^d + a_2 z^e$ at λ, $0 < d < e$. The largest term forces $e = d+1$; the coefficient bounds then force $a_2 = \pm1$, $a_1 = \mp6$, and $a_0 = \pm\lambda^{d-1}$. Rationality gives $d = 1$ and $a_0 = \pm1$. Thus the cubic component coefficient is the one stated above. Terms with two or more components have absolute mass $O(K^2|\epsilon|^6 e^{CK|\epsilon|^3})$. Truncating component lengths to the circle changes the estimates only exponentially in K, by $H_\alpha(2) < \infty$. Consequently the full central trace is

$$k_0(\epsilon)^K\left[1 - 2ib\,e_N(-6a^2)K\epsilon^3 + O(K|\epsilon|^4 + K^2|\epsilon|^6 e^{CK|\epsilon|^3})\right] + O(e^{-cK}).$$

At $\epsilon = t/\sqrt{K}$, $k_0(\epsilon)^K = e^{-(1+b^2)t^2}(1 + O(K^{-1}))$. The odd correction and $e_N(-6a^2) - 1$ are exponentially small. This proves (Ret.A15), which is nonzero for $bt \neq 0$.

27.6 The first Edgeworth correction for every analytic position symbol

The cubic mechanism above is not restricted to two Fourier modes. Write $f_j(z) = f(2\pi\langle A^j(1,0), z\rangle)$ on the classical two-torus, with normalized area, and put

$$\mu_3 = \int f_0^3, \qquad C_{d,e} = \int f_0 f_d f_e, \qquad \kappa_3(f) = \mu_3 + 6\sum_{1\le d<e} C_{d,e}. \tag{Ret.A16}$$

The series is absolutely convergent, as proved below.

Theorem 27.2. *For every fixed $T < \infty$, uniformly over all integers $r \geq 1$ and all real $|t| \leq T$,*

$$\operatorname{tr}_- U^{rK}_{t/(r\sqrt{K}),f,a} = e^{-\sigma_f^2 t^2/2}\left[1 - \frac{i\kappa_3(f)t^3}{6\sqrt{K}}\right] + O_{f,a,T}(K^{-1}). \tag{Ret.A17}$$

This is an expansion of the normalized trace, not a local spectral correlation theorem. When r grows, the kick in (Ret.A17) decreases with r. It does not provide control of the same kick at all times.

27.6.1 A relative estimate for the conditioned background

Set $x = r\epsilon$ and let $Q_{r,K}$ be the zero-aggregate partition from (Ret.A11). The gas proof gives a nonvanishing analytic $Q_{r,K}$ on a fixed disk $|x| < s_*$, uniformly in r, K. Remove a set B of sites and impose the real fields generated by specified dirty replica charges of total absolute mass S_B. Let $Q_{B,\boldsymbol{h}}$ include the vacuum factors at the remaining sites. Then, after reducing s_*,

$$\left|\log \frac{Q_{B,\boldsymbol{h}}}{Q_{r,K}}\right| \leq C_{f,a}|x|^2(|B| + S_B). \tag{Ret.A18}$$

The logarithm is continued from $x = 0$. This is a local perturbation estimate; there is no factor K on the right.

Here are details needed for this strengthening of (Ret.A12). Differentiate the absolutely convergent connected series with respect to an activity. The derivative is a rooted connected series, bounded by $e^{\beta l}$ by the same tree majorant as in Section 3. Removing a site by varying its fugacity from one to zero therefore changes the logarithm by at most $C|x|^2$. For fields use the interpolation $w_j(\boldsymbol{n}; u) = w_j(\boldsymbol{n}; 0)e^{iu\boldsymbol{h}_j\cdot\boldsymbol{n}}$, $0 \leq u \leq 1$. Its logarithmic derivative has absolute sum bounded by

$$M_\beta \sum_j \max_\nu |h_{\nu j}| \leq C|x|^2(C_a - 1)S_B.$$

The last inequality follows entry by entry from $|H_{\nu\mu}| = 1$ and the row bound (Ret.G8). Finally the missing vacua contribute $|B|\,|r \log k_0(x/r)| \leq C|B||x|^2$. All interpolations obey the same strict tree criterion. This proves (Ret.A18) also for complex x; the imposed fields remain real.

At zero external field the exact one-site resummation is $k_0(x)$. Since the nonvacuum site activity is $O(|x|^2)$, (Ret.G11) also gives

$$\log Q_{r,K} = K \log k_0(x) + O_{f,a}(K|x|^4) = -\tfrac{1}{2}\sigma_f^2 K x^2 - \tfrac{i}{6}\mu_3 K x^3 + O_{f,a}(K|x|^4). \tag{Ret.A19}$$

This estimate is uniform on a fixed complex disk.

27.6.2 Isolating one finite component with a uniform remainder

Divide the full central trace by $Q_{r,K}$, and use its unique preimage decomposition. For each prescribed dirty vector, (Ret.A18) absorbs the background ratio into the Fourier weight $\exp(C|x|^2(|B| + S_B))$. Leaving analytic strip width as in (Ret.A13) gives the majorant

$$(C|x|)^{q(c)} e^{-\alpha\|c\|_1}. \tag{Ret.A20}$$

In particular the entire winding contribution to this relative trace is $O((3/4)^K)$ by Section 2.3.

For each possible start, sum the exact contributions of configurations with exactly one nonwinding component, including their conditional background ratios. This is an analytic function of x, uniformly $O(|x|^3)$ on a fixed disk by (Ret.A20) and Section 2.2. The connected-background ratio equals one at $x = 0$. Cauchy's estimate therefore leaves $O(|x|^4)$ after the cubic

term, uniformly in r, K and in the start. More explicitly, divide this analytic function by x^3, use its uniform bound on $|x| = s_*$, and apply the geometric-series estimate on $|x| \le s_*/2$.

For configurations with two or more components, the phases need not factor when $r > 1$. Their absolute values are one, and (Ret.A20) does factor over the disjoint forcing supports. Their total mass is at most

$$CK^2|x|^6 e^{CK|x|^3}.$$

No factorization of replica phases is being assumed here. We have proved

$$\frac{\operatorname{Tr} U_{\epsilon,f,a}^{rK}}{NQ_{r,K}} = 1 + Kx^3 d_{r,K} + O_{f,a}\Big(K|x|^4 + K^2|x|^6 e^{CK|x|^3} + (3/4)^K\Big). \tag{Ret.A21}$$

27.6.3 Exact cubic orientation count

A finite three-site forcing is a triple of nonzero integers satisfying

$$n_0 v_0 + n_d v_d + n_e v_e = 0 \quad \text{over } \mathbb{Z}^2, \qquad 0 < d < e. \tag{Ret.A22}$$

It is a single finite component: every component needs at least three forcing sites. Its one-period phase is $e^{i\vartheta}$, where

$$\vartheta = \frac{2\pi a^2 n_0 n_d b_d}{N}.$$

For a central triple, an even permutation of its three Weyl factors has this phase and an odd permutation has its inverse. Assign its three distinct physical sites independently to r replicas, then order by replica first and physical site second. The two orientation counts are exactly

$$\frac{r^3 + r^2}{2}, \qquad \frac{r^3 - r^2}{2}. \tag{Ret.A23}$$

With three distinct replica labels the orientations balance. With two equal labels their signed excess, summed over the possible equal pairs, is two for each pair of replica labels. With all labels equal the excess is one. Thus the excess is $2\binom{r}{2} + r = r^2$, proving (Ret.A23).

The resulting phase average is

$$\Psi_r(\vartheta) = \cos\vartheta + \frac{i}{r}\sin\vartheta.$$

Every finite component has K translates. The sum over all replica assignments is invariant under cyclically moving the physical origin: the corresponding time order is cyclically permuted, and conjugation by A preserves its central Weyl phase. Consequently

$$d_{r,K} = -i \sum_{\substack{1 \le d < e \le K-1 \\ n_0 v_0 + n_d v_d + n_e v_e = 0 \\ n_0 n_d n_e \neq 0}} \widehat{f}_{n_0} \widehat{f}_{n_d} \widehat{f}_{n_e} \Psi_r(\vartheta). \tag{Ret.A24}$$

The cutoff is the length of a finite influence interval. Winding triples, including (Ret.A14), belong to the separate remainder in (Ret.A21); they have not been included in (Ret.A24) or discarded.

27.6.4 Convergence to the classical third cumulant

For a triple in (Ret.A22), (Ret.A7) bounds $\lambda^{(d-1)/2}$ by $100\|c\|_1$. Hence

$$|n_0 n_d b_d| \le C\|c\|_1^4.$$

The analytic Fourier decay and the component estimate imply absolute summability even with this polynomial factor inserted. Since $|\Psi_r(\vartheta) - 1| \le 2|\vartheta|$ for every $r \ge 1$, the phase error in the

entire sum (Ret.A24) is $O_{f,a}(N^{-1})$, uniformly in r. For the length tail, insert the factor $2^{d+(e-d)}$ in the gap majorant. The finiteness of $H_\alpha(2)$ gives an $O_f(2^{-K})$ tail. Therefore

$$d_{r,K} = -i \sum_{1\le d<e} C_{d,e} + O_{f,a}(N^{-1} + 2^{-K}), \tag{Ret.A25}$$

uniformly over all replica counts. This also proves the absolute convergence asserted in (Ret.A16).

For explicit computation, the integral $C_{d,e}$ is precisely the Fourier sum in (Ret.A24) without Ψ_r. All its integer triples are multiples of

$$\frac{1}{g}(b_{e-d}, -b_e, b_d), \qquad g = \gcd(b_{e-d}, b_e, b_d). \tag{Ret.A26}$$

The two determinant equations prove this formula directly. Terms with exactly two equal time indices have zero integral, because the Fourier support of f_i^2 remains parallel to v_i, whereas v_j is not parallel to v_i for $j \neq i$. Expanding the cube of a classical time sum and using absolute convergence now proves

$$\lim_{L\to\infty} \frac{1}{L} \int \left(\sum_{j=0}^{L-1} f_j \right)^3 = \kappa_3(f).$$

Thus (Ret.A16) is a classical cumulant rate, with all combinatorial factors specified; this identification does not appeal to a separate classical central limit theorem.

Finally put $x = t/\sqrt{K}$ in (Ret.A19)–(Ret.A25). The two remainders $K|x|^4$ and $K^2|x|^6 e^{CK|x|^3}$ are $O_T(K^{-1})$. The cross terms in multiplying the two expansions have the same bound. The odd-projector correction is $O(e^{F_0 T\sqrt{K}}/N) = O_{f,T}(K^{-1})$. This proves (Ret.A17).

For $f = 2\cos\theta + 2b\cos 6\theta$, the only temporal term is $C_{1,2} = 2b$ and $\mu_3 = 0$, so $\kappa_3 = 12b$. Theorem 27.2 therefore contains (Ret.A15) and makes it uniform in the replica count with the stated rescaling. For the independent four-mode check with positive Fourier coefficients $(1, 1/3, 1/5, 1/7)$ at modes $(1, 2, 6, 35)$, the nonzero temporal terms are

$$C_{1,2} = 2/5, \quad C_{1,3} = C_{2,3} = 2/35, \quad \mu_3 = 2, \quad \kappa_3 = 178/35.$$

These examples separate one-site skewness from temporal skewness.

27.7 Removing analyticity in the central-limit window

Analyticity can be removed from the first-correction result, with explicit control of the constants needed in Fourier truncation. In this section suppose that f is real, even, nonzero and mean zero, and that

$$\mathcal{F}_{32}(f) := \sum_n (1 + |n|)^{32} |\widehat{f}_n| < \infty.$$

In particular every C^∞ position symbol qualifies.

Theorem 27.3. *For every fixed T, uniformly in $r \ge 1$ and $|t| \le T$,*

$$\operatorname{tr}_- U_{t/(r\sqrt{K}),f,a}^{rK} = e^{-\sigma_f^2 t^2/2} \left[1 - \frac{i\kappa_3(f) t^3}{6\sqrt{K}} \right] + O_{f,a,T}(K^{-7/12}). \tag{Ret.A27}$$

Here κ_3 is still the absolutely convergent series (Ret.A16). The exponent and the Fourier regularity requirement are convenient sufficient choices, not claimed optimal.

Put $f_M = \sum_{|n|\le M} \widehat{f}_n e^{in\theta}$ and give it strip width $\rho_M = 1/M$. Its weighted Fourier norms of orders zero and one are bounded by $e\mathcal{F}_{32}(f)$, uniformly in M. Thus the constants in the one-site

estimates, (Ret.A9), and (Ret.A18) are uniform; the tree criterion only requires $C|x|^2 < c/M$. For $\alpha = 1/(4M)$, the explicit lattice constants obey

$$b_\alpha \le CM, \quad H_\alpha(1) \le CM, \quad H_\alpha(2) \le CM, \quad B_\alpha \le CM^5, \quad \eta_\alpha \ge cM^{-2}. \tag{Ret.A28}$$

Indeed $e^{-y} \le (ey)^{-1}$ and $2/\sqrt{\lambda} < 1$ bound both series $H_\alpha(R)$ by convergent geometric series times C/α, for $R = 1, 2$.

Choose the uniform complex disk $|x| \le cM^{-2}$. It satisfies both the tree criterion and the required analytic strip margin in (Ret.A20). The one-component sum per start is $O(M^5|x|^3)$ there; Cauchy's estimate after the cubic term is $O(M^7|x|^4)$ on the disk of half that radius. The multi-component bound becomes $CK^2M^{10}|x|^6e^{CKM^5|x|^3}$. The polynomial factor in the phase estimate (Ret.A25) costs at most another M^4, by absorbing $\|c\|_1^4$ into half of the strip. Consequently at $x = t/\sqrt{K}$ the error in (Ret.A17) for f_M is at most

$$C_{f,a,T}\left[\frac{1+M^7}{K} + \frac{M^{10}}{K}e^{CM^5/\sqrt{K}} + \frac{M^9}{N\sqrt{K}} + \frac{M^5\,2^{-K}}{\sqrt{K}} + (3/4)^K + \frac{e^{F_0T\sqrt{K}}}{N-1}\right]. \tag{Ret.A29}$$

This estimate, rather than an unspecified constant depending on f_M, justifies the moving truncation.

For completeness, absolute convergence of (Ret.A16) for the present nonanalytic f follows directly from the gap bound. For any fixed nonzero charge triple of total absolute size R, (Ret.A7) allows at most $C\log^2(2+R)$ pairs (d, e). Summing this bound against the three Fourier coefficients proves absolute convergence. It also gives

$$|\kappa_3(f) - \kappa_3(f_M)| \le C_f M^{-16}, \qquad |\sigma_f^2 - \sigma_{f_M}^2| \le C_f M^{-32}. \tag{Ret.A30}$$

For example on the tail $\max|n_j| > M$, absorb the logarithmic factor into $(1+\max|n_j|)^{16}$, extract M^{-16}, and use the remaining order-32 Fourier moment. The onsite cubic tail has the stronger order-32 bound.

Finally the unitary telescoping inequality, applied to the actual diagonal kicks, gives

$$\left|\mathrm{tr}_-\, U_{\epsilon,f,a}^{rK} - \mathrm{tr}_-\, U_{\epsilon,f_M,a}^{rK}\right| \le rK|\epsilon|\|f - f_M\|_\infty \le T\sqrt{K}\,\mathcal{F}_{32}(f)M^{-32}. \tag{Ret.A31}$$

Choose $M = \lfloor K^{1/24} \rfloor$. For all sufficiently large K the complex-disk condition holds. The largest term in (Ret.A29) is $M^{10}/K = O(K^{-7/12})$; $M^5/\sqrt{K} \to 0$. The error in (Ret.A31) is $O(K^{-5/6})$. The variance and third-cumulant replacements from (Ret.A30) are smaller. This proves (Ret.A27).

This argument proves a bounded-t result for smooth symbols. It does not extend (Ret.A1) to arbitrary smooth f and every sequence $\epsilon_K \to 0$ with $K\epsilon_K^2 \to \infty$. The moving cutoff would have to satisfy incompatible estimates in some of those larger ranges, so no density argument is asserted there.

27.8 Checks and unresolved scope

analytic_kick_return_gas.py checks 1,152 exact rational homogeneous gaps, all 1,034 finite components in the indicated bounded alphabet and hull range, and 1,579 seeded cyclic rational-preimage configurations. It also exhausts 612,864 ordered cubic atom words across $K = 10, 14, 18$, including repeated physical kick sites. An independently organized expansion into unordered atom multisets with exact factorial weights checks another 2,532,760 configurations. It verifies (Ret.A23) for two Fourier alphabets and several replica counts, including the winding alias at $K = 10$. For the four-mode classical coefficient, (Ret.A7) proves that both temporal gaps are at most 11; exhaustive enumeration in that proved range gives the three correlations displayed above.

A further 5,152,304 unordered cubic configurations are classified by their exact rational preimages at $K = 10, 14, 18, 22, 34$. They confirm separately the zero aggregate, the finite-component coefficient (Ret.A24), and winding contributions. For these four modes, additional winding terms occur at $K = 14, 18$ as well as $K = 10$. They are absent at $K = 22, 34$ in the enumeration. At $K \geq 34$ their absence is proved without enumeration: a cubic forcing has total absolute size at most $3 \cdot 35$, so (Ret.A7) bounds each winding gap by 11 and $K \leq 3 \cdot 11$ would be necessary. These checks verify finite algebra and expose the extra term in (Ret.A14). They do not replace the analytic convergence arguments.

The exact verifier invocation is reproduced as listing S5 in the Supplement.

The analytic strip assumption is essential to the uniform bound (Ret.A1). Section 7 extends the central-limit window and its first correction to smooth position symbols, with a controlled moving Fourier truncation. Arbitrary functions of both position and momentum are not covered here. The analytic time bound remains $r|\epsilon| \leq s_0$. When $K\epsilon^2 \to \infty$, this gives $rK = o(K^{3/2})$, far below the Heisenberg scale $D \asymp N$. No implication from (Ret.A1)–(Ret.A3) to microscopic BGS is asserted.

Chapter 6 — Return laws II (general laws, obstructions, return words)

28 A return-trace comparison for general smooth phase-space symbols

Numbering convention: section and theorem references within this chapter use the source module's local numbering; source Section 1 is subsection 28.1. Equation references carry the prefix Ret.

Status. Project proof with exact finite algebra checks and separately labelled floating physical diagnostics. Limited independent internal analytic checks cover Sections 1–5 and 8–10; they are not external referee acceptance. The coupling and joint-law extension have been checked by the author. Priority verification and Lean formalization are outstanding. Base revision: 9ed93cd1bc5e3362c1dbf520f0f391478dfb6969.

This note removes the position-only restriction for a one-period comparison. Sections 7–9 now identify its Gaussian limit and extend the leading asymptotic to every replica count at the critical scale. This does not prove microscopic BGS or random-matrix correlations. The classical kick strength tends to zero.

28.1 Quantization and statement

Retain

$$A = \begin{pmatrix} 3 & 4 \\ 2 & 3 \end{pmatrix}, \quad \lambda = 3 + 2\sqrt{2}, \quad K = 2m, \quad m \geq 3 \text{ odd}, \quad N = b_m,$$

$$b_0 = 0, \quad b_1 = 1, \quad b_{j+1} = 6b_j - b_{j-1}.$$

Normalize the metaplectic operator by $U_0^K = I$. For integer frequency vectors use the balanced Weyl convention

$$\mathcal{W}_N(v) = (-1)^{v_1 v_2} W_N(v), \qquad \mathcal{W}_N(v)\mathcal{W}_N(w) = e^{\pi i \omega(v,w)/N} \mathcal{W}_N(v+w). \tag{Ret.P1}$$

W_N is the modular-centered convention of the preceding notes. Unlike W_N, $\mathcal{W}_N$ has signs under shifts by N. In particular $\mathcal{W}_N(2Nv) = I$. Since $A \equiv I \pmod 2$, exact Egorov holds in this convention too.

For a real smooth symbol on $\mathbb{T}^2$, write

$$f(z)=\sum_{v\in\mathbb{Z}^2}\widehat{f}_v e^{2\pi i\langle v,z\rangle},\quad F_j=\sum_v \|v\|_1^j|\widehat{f}_v|,\quad H_N=\mathrm{Op}_N(f)=\sum_v \widehat{f}_v\mathcal{W}_N(v),\quad U_{\epsilon,f}=e^{i\epsilon H_N}U_0.$$

In F_0 the weight is one, including at zero frequency. H_N is self-adjoint. No parity or mean-zero assumption is needed for the full trace. If $f(-z)=f(z)$, the odd block is invariant.

Let

$$\mathcal{P}_K=\{z:(A^{\mathsf{T}})^K z=z\pmod{\mathbb{Z}^2}\},\qquad Z_{K,f}(\epsilon)=\frac{1}{|\mathcal{P}_K|}\sum_{z\in\mathcal{P}_K}\exp\left(i\epsilon\sum_{j=0}^{K-1}f((A^{\mathsf{T}})^j z)\right). \tag{Ret.P2}$$

All fixed points are counted, including those of smaller primitive period, with equal weight. This finite group is explicitly

$$\mathcal{P}_K=\left\{\left(\frac{u}{8N},\frac{v}{4N}\right):0\le u<8N,\ 0\le v<4N\right\},\qquad |\mathcal{P}_K|=32N^2. \tag{Ret.P3}$$

Theorem 28.1 (finite Fourier support)**.** *Suppose $\widehat{f}_v=0$ for $\|v\|_1>M$, $M\ge 1$. Put $\delta=|\epsilon|$, $u=K\delta F_0$, $L=\lceil K/(64M)\rceil$, and $E_L(u)=\sum_{\ell\ge L}u^\ell/\ell!$. For every real ϵ,*

$$\left|\frac{\operatorname{Tr}U_{\epsilon,f}^K}{N}-Z_{K,f}(\epsilon)\right|\le\frac{2\pi e^u}{N}\left[K\delta F_2+(K\delta F_1)^2+K\delta^2F_1^2\right]+E_L(u). \tag{Ret.P4}$$

For an even symbol the normalized odd trace satisfies the same bound with the additional term $e^u/(N-1)$.

Theorem 28.2 (arbitrary fixed smooth symbol)**.** *For every real $f\in C^\infty(\mathbb{T}^2)$, every $T<\infty$, and $q>0$,*

$$\sup_{|t|\le T}\left|\frac{\operatorname{Tr}U_{t/\sqrt{K},f}^K}{N}-Z_{K,f}(t/\sqrt{K})\right|=O_{f,T,q}(K^{-q}). \tag{Ret.P5}$$

The same holds for the odd trace when f is even. This is superpolynomial accuracy in $K\asymp\log N$, not in N. For a fixed Fourier polynomial (Ret.P4) is stronger at this scale: the error is $N^{-1+o(1)}$. It also gives an exponentially small error in K for a sufficiently small fixed $\delta>0$, depending on M,F_0. These statements concern one arithmetic period K.

28.2 The first-order lattice and its 32 classes

On cyclic vector sequences define

$$(\mathcal{D}p)_j=p_j-Ap_{j+1},\qquad \mathcal{C}_K=\left\{c\in(\mathbb{Z}^2)^K:\sum_j A^jc_j=0\pmod N\right\}. \tag{Ret.P6}$$

The total-frequency map onto $(\mathbb{Z}/N\mathbb{Z})^2$ is surjective, so $\mathcal{C}_K$ has index N^2. Also $\mathcal{D}(\mathbb{Z}^{2K})\subset\mathcal{C}_K$, and

$$|\det\mathcal{D}|=|\det(I-A^K)|=32N^2.$$

Consequently

$$|\mathcal{C}_K/\mathcal{D}(\mathbb{Z}^{2K})|=32,\qquad p=\mathcal{D}^{-1}c\in(1/32)\mathbb{Z}^{2K}. \tag{Ret.P7}$$

The last assertion follows because a finite group of order 32 is annihilated by 32. Individual classes need not be classified.

There is a uniform cyclic inverse bound

$$\|p\|_1 \le 2\|\mathcal{D}p\|_1. \tag{Ret.P8}$$

Let P_+, P_- be the spectral projectors of A with eigenvalues λ, λ^{-1}. Extend the cyclic forcing periodically. The unstable inverse is $-\sum_{h\ge1}\lambda^{-h}P_+c_{j-h}$; the stable inverse is $\sum_{h\ge0}\lambda^{-h}P_-c_{j+h}$. These absolutely convergent sums give the unique cyclic inverse. Their induced ℓ^1 norm is at most

$$\frac{\|P_+\|_1}{\lambda-1}+\frac{\|P_-\|_1}{1-\lambda^{-1}} = 1+\frac{1}{\sqrt{2}} < 2.$$

If p is not integer, none of its vector coordinates is an integer vector: $p_j - Ap_{j+1} \in \mathbb{Z}^2$ and the integral inverse of A propagate integrality in both directions. By (Ret.P7), every coordinate then has norm at least $1/32$. Thus every noninteger class satisfies

$$\|c\|_1 \ge K/64. \tag{Ret.P9}$$

This counts total Fourier mass, not occupied sites.

To verify (Ret.P3), write $A^m = \alpha_m I + b_m B$, $B = A - 3I$, so $B^2 = 8I$ and $\alpha_m^2 - 8b_m^2 = 1$. Then

$$A^K - I = \begin{pmatrix} 16N^2 & 8\alpha_m N \\ 4\alpha_m N & 16N^2 \end{pmatrix}.$$

Every grid point in (Ret.P3) is fixed by $(A^{\mathsf{T}})^K$. The grid has the determinant cardinality and is therefore the whole group. This also proves $A^K - I \in 2NM_2(\mathbb{Z})$.

28.3 The local Weyl area and the parity sign

Take integer p and $c = \mathcal{D}p$. Put $z_j = A^j p_j$ for $0 \le j < K$, and $z_K = A^K p_0$. The time-ordered frequencies are $A^j c_j = z_j - z_{j+1}$. Their sum is $(I - A^K)p_0 \in 2N\mathbb{Z}^2$, so their final balanced Weyl operator is the identity. The ordered symplectic area telescopes to

$$\sum_{i<j}\omega(A^i c_i, A^j c_j) = -\omega(p_0, A^K p_0) + \sum_j \omega(p_j, Ap_{j+1}).$$

The first term is divisible by $2N$. The normalized trace of the entire word is therefore exactly

$$\exp\left(\frac{\pi i}{N}\mathcal{S}(p)\right), \qquad \mathcal{S}(p) = \sum_j \omega(p_j, Ap_{j+1}). \tag{Ret.P10}$$

Since $\mathcal{S}(p) = -\sum_j \omega(p_j, c_j)$, (Ret.P8) gives

$$|\mathcal{S}(p)| \le 2\|c\|_1^2, \qquad \left|e^{\pi i\mathcal{S}(p)/N} - 1\right| \le \frac{2\pi}{N}\|c\|_1^2. \tag{Ret.P11}$$

This bound does not grow with the largest frequency $A^j c_j$: a complete arithmetic return cancels that growth.

The convention matters. For $p_0 = (1,0)$, $p_1 = (0,1)$, and all other $p_j = 0$, the local area is 3. Using the modular inverse of two without $(-1)^{v_1 v_2}$ would give the negative of the small phase in (Ret.P10). The checker includes odd local areas at every tested period.

28.4 From the physical kick to scalar Fourier coefficients

Let $k_v(\epsilon)$ be the Fourier coefficients of $e^{i\epsilon f}$. They are not the coefficients of $e^{i\epsilon H_N}$: Fourier modes within a physical kick need not commute. For an ordered word of ℓ Weyl frequencies, (Ret.P1) gives

$$|\text{Weyl phase} - 1| \le \frac{\pi}{N}\sum_{i<j}\|v_i\|_1\|v_j\|_1.$$

Summing its exponential series against the absolute Fourier coefficients proves

$$\|e^{i\epsilon H_N} - \mathrm{Op}_N(e^{i\epsilon f})\| \le \frac{\pi\delta^2 F_1^2}{2N} e^{\delta F_0}. \tag{Ret.P12}$$

Both factors have norm at most $e^{\delta F_0}$. Replacing all K interaction-picture kicks therefore costs at most $\pi K\delta^2 F_1^2 e^u/(2N)$ in operator norm. Commuting the original Fourier modes without this estimate would not be justified.

The normalized trace of the resulting Fourier product selects $\mathcal{C}_K$. Finite-group Fourier orthogonality in (Ret.P2) selects exactly $\mathcal{D}(\mathbb{Z}^{2K})$: the dual kernel equations are $q_j = A^{\mathsf{T}} q_{j-1}$ cyclically. For an integer preimage, (Ret.P10) is the only remaining difference from the classical coefficient, whose phase is one.

The absolute Fourier mass is at most e^u. Expanding into Fourier atoms also gives the moment bound

$$\sum_{(c_j)} \prod_j |k_{c_j}(\epsilon)| \, \|c\|_1^2 \le [K\delta F_2 + (K\delta F_1)^2] e^u. \tag{Ret.P13}$$

The atom norm bounds the aggregate norm; the right side is the second moment of the positive exponential majorant. No derivative of an absolute Fourier coefficient is taken. Equations (Ret.P11)–(Ret.P13) give the N^{-1} part of (Ret.P4).

For the other 31 classes, each Taylor atom has norm at most M. By (Ret.P9), each such term has at least $L = \lceil K/(64M) \rceil$ atoms. The absolute mass at degree ℓ is at most $(K\delta F_0)^\ell/\ell!$, giving $E_L(u)$. All classes and Taylor degrees have been included. For the odd correction, a noncentral Weyl word contributes at most $1/(N-1)$ to the difference of normalized traces, and a central word has zero correction. The atom sum for the actual physical kick is e^u. This proves Theorem 28.1.

28.5 Smooth truncation and the time range

Truncate a real smooth f to $\|v\|_1 \le M$. The norms F_0, F_1, F_2 stay bounded, and for every p the Fourier tail is at most $C_{f,p} M^{-p}$. Unitary telescoping and the Lipschitz estimate for real exponentials give

$$\left| \frac{\operatorname{Tr} U^K_{t/\sqrt{K},f}}{N} - \frac{\operatorname{Tr} U^K_{t/\sqrt{K},f_M}}{N} \right| \le C_{f,p} T\sqrt{K}\, M^{-p},$$

$$|Z_{K,f}(t/\sqrt{K}) - Z_{K,f_M}(t/\sqrt{K})| \le C_{f,p} T\sqrt{K}\, M^{-p}. \tag{Ret.P14}$$

For the first inequality use Duhamel's formula for self-adjoint exponentials and the operator norm bound by the Fourier norm.

Choose $M = \lfloor K^{1/4} \rfloor$. Then $u = O_{f,T}(\sqrt{K})$ and $L \ge cK^{3/4}$. For $L > u$ the exponential Markov bound gives

$$E_L(u) \le (eu/L)^L \le e^{-c_{f,T} K^{3/4} \log K}$$

for large K. The other term in (Ret.P4) is $N^{-1} e^{O(\sqrt{K})} O(K)$. Choose $p > 4q + 2$ in (Ret.P14) to prove (Ret.P5). The odd correction is exponentially small as well.

For a fixed Fourier polynomial and a fixed $\delta F_0 < 1/(128eM)$, instead $eu/L \le 1/2$. Thus $E_L(u) \le 2^{-L}$ and the other error in (Ret.P4) also decays exponentially in K. The zero symbol is trivial. This proves the fixed-small-kick assertion.

For a classical coboundary $f = g - g\circ A^{\mathsf{T}}$, (Ret.P2) telescopes at every periodic point: $Z_{K,f} = 1$. Theorem 28.2 correctly retains a coherent first return for these symbols. No positive-variance claim is made for arbitrary phase-space symbols.

A classical limit theorem is a separate input to one-return asymptotics; it is not used in the proof of (Ret.P5). Section 7 supplies the missing passage by a direct coupling to Haar measure

and a verified Haar CLT. The following unsuccessful source leads are retained to identify what is not being used. The flow theorem of Cantrell and Sharp, arXiv:1805.05692v1, was inspected, but its Theorem 1.1 concerns flows and is not used as a theorem about these discrete cat maps. Its reference to Coelho–Parry, Section 6, is a source lead, not a checked proof input. The publisher record confirms that the paper discusses closed orbital measures, but the full section was not accessible in this run and is not cited as verified.

28.6 Verification and the remaining obstruction

periodic_point_trace_comparison.py checks the determinant, all 32 fractional classes, 960 seeded rational-preimage configurations, and 1,206 exact integer-word phases at $K = 6, 10, 14, 18, 22, 34$. It includes 603 odd local areas, testing the sign in (Ret.P1). The physical matrix diagnostic at $N = 35$ uses position, momentum, and mixed Fourier modes. Double-precision Egorov residuals are below $5 \cdot 10^{-15}$. It compares the physical trace with all 39,200 classical periodic points. Differences at $\epsilon = .08, .04, .02, .01$ are approximately $1.82 \cdot 10^{-3}, 2.38 \cdot 10^{-4}, 3.02 \cdot 10^{-5}, 3.79 \cdot 10^{-6}$. These are finite diagnostics, not interval bounds or replacements for the analytic proof.

The exact verifier invocation is reproduced as listing S3 in the Supplement.

The cancellation in (Ret.P10) uses one complete arithmetic period. Repeating the word introduces interactions between replica assignments. Two natural extensions have distinct, explicitly demonstrable failures.

First replace the classical period in (Ret.P2) by rK while keeping the quantum dimension $N = b_{K/2}$. For $f = 2\cos(2\pi z_1)$ the exact quadratic coefficients are

$$[\epsilon^2]\frac{\operatorname{Tr} U_{\epsilon,f}^{rK}}{N} = -Kr^2, \qquad [\epsilon^2]Z_{rK,f}(\epsilon) = -Kr. \tag{Ret.P15}$$

Here $Z_{rK,f}$ averages over the actual classical fixed points of $(A^{\mathsf{T}})^{rK}$, not the smaller grid $\mathcal{P}_K$. For the quantum coefficient each orbit frequency occurs r times; opposite frequencies give r^2 matching pairs. Within a single period the orbit vectors are distinct up to sign. Indeed $0 < b_d < N$ for $0 < d < m$, and $A^m = \alpha_m I \pmod N$ reduces $m < d < 2m$ to that range. At $d = m$ the scalar α_m is neither 1 nor -1 modulo N: $2N < \alpha_m < 3N$ and the Pell identity excludes $\alpha_m = 2N + 1$ and $\alpha_m = 3N - 1$ for the present $N \geq 35$. The classical period-rK grid has no such earlier repeated frequency. Writing $N' = b_{rK/2}$, a frequency difference has second coordinate $2b_d$ modulo $4N'$; it cannot vanish for $0 < d < rK$. The only possible multiple of N' occurs at $d = rK/2$, where $2b_d = 2N'$ is still nonzero modulo $4N'$. This proves (Ret.P15) by character orthogonality. The exact checker confirms these two counts at $K = 10$, $1 \leq r \leq 8$.

Second keep the classical period K and replace its kick by $r\epsilon$, which repairs the quadratic discrepancy. The quartic calculation in the fixed-harmonic note gives, for every fixed $r \geq 2$ and real t,

$$K\left[\frac{\operatorname{Tr} U_{t/(r\sqrt{K}),f}^{rK}}{N} - Z_{K,f}(t/\sqrt{K})\right] \longrightarrow -\frac{1 - r^{-2}}{3} S_1 t^4 e^{-t^2}, \qquad S_1 = \sum_{\ell \geq 1}(1 - \cos(4\pi\lambda^{-\ell})) > 0. \tag{Ret.P16}$$

To justify this consequence, use that note's uniform sixth-order remainder at $x = r\epsilon = t/\sqrt{K}$: it is $O(Kx^6) = O(K^{-2})$. Its nonzero-aggregate error is $O(Kx^8)$ plus exponential terms. Apply (Ret.P4) to the single-period classical comparison; its error at this scale is exponentially small. The two traces have the same Gaussian factor, and the displayed quartic difference gives (Ret.P16). Thus even this repaired extension of the superpolynomial estimate (Ret.P5) is false: its error generally starts at order K^{-1}. This is a failure of the proposed comparison, not a refutation of BGS.

Neither this comparison nor the all-replica position-symbol bounds control the joint traces

at $n \asymp N$ needed for microscopic BGS. A fixed nonzero classical kick instead requires quantum amplitude of order N, outside the useful range of (Ret.P4).

28.7 A centered-time coupling closes the classical limit

Write $T = A^{\mathsf{T}}$ and use the torus distance induced by the maximum norm. For a Lipschitz real function let $\mathrm{Lip}(f)$ refer to this distance. Haar measure is normalized to mass one.

Lemma 28.3 (coupling)**.** *There is a coupling of a uniform $q \in \mathcal{P}_K$ and a Haar point $z \in \mathbb{T}^2$ for which*

$$\left| \sum_{j=-m}^{m-1} f(T^j z) - \sum_{j=-m}^{m-1} f(T^j q) \right| \le 3\,\mathrm{Lip}(f). \tag{Ret.P17}$$

Consequently the Wasserstein distance of order one between the normalized Birkhoff sums for these two measures is at most $3\,\mathrm{Lip}(f)/\sqrt{K}$. In particular,

$$\left| Z_{K,f}(t/\sqrt{K}) - \int_{\mathbb{T}^2} e^{it\sum_{j=0}^{K-1} f(T^j z)/\sqrt{K}}\, dz \right| \le \frac{3|t|\,\mathrm{Lip}(f)}{\sqrt{K}}.$$

Partition the torus into centered rectangular cells of side lengths $1/(8N)$ and $1/(4N)$. Couple each point to its cell center. The cells have equal Haar mass; ties have measure zero. A lift of $z - q = \eta$ satisfies $|\eta_1| \le 1/(16N)$ and $|\eta_2| \le 1/(8N)$. The two spectral projectors of T have maximum-norm operator norm $(1+\sqrt{2})/2$. Hence

$$\|T^j\|_\infty \le (1+\sqrt{2})\lambda^{|j|}, \qquad \sum_{j=-m}^{m-1} \lambda^{|j|} = \frac{(\lambda+1)(\lambda^m-1)}{\lambda-1}.$$

Using $N = (\lambda^m - \lambda^{-m})/(4\sqrt{2})$, the sum of the distances is at most

$$\frac{1+\sqrt{2}}{8N}\,\frac{(\lambda+1)(\lambda^m-1)}{\lambda-1} \le \frac{1+\sqrt{2}}{1-\lambda^{-6}} < 3.$$

This proves (Ret.P17). The centered sum at a periodic point equals its forward period sum. The centered Haar sum has the forward sum's distribution by invariance. These two observations are necessary: the corresponding pointwise estimate for two forward sums need not be bounded uniformly.

For a real smooth mean-zero symbol define

$$\sigma_A^2(f) = \sum_{j\in\mathbb{Z}} \int f(z) f(T^j z)\, dz = \sum_{\mathcal{O}} \left| \sum_{v\in\mathcal{O}} \widehat{f}_v \right|^2. \tag{Ret.P18}$$

Here $\mathcal{O}$ runs over oriented nonzero A-orbits, so an orbit and its negative are counted separately. Absolute convergence follows because for each pair of nonzero integer frequencies at most one integer iterate identifies them; the total absolute correlation sum is at most F_0^2. This is the existing classical orbit-variance formula, not a new claim of priority; see the earlier derivation.

Theorem 28.4 (general one-return law)**.** *For every fixed real mean-zero f with $F_4 < \infty$, in particular every smooth f,*

$$\frac{\operatorname{Tr} U_{t/\sqrt{K},f}^K}{N} \longrightarrow e^{-\sigma_A^2(f)t^2/2}, \tag{Ret.P19}$$

uniformly on compact real t-intervals. The odd trace obeys the same law when f is even, and $\sigma_A^2(f) = 0$ is allowed.

The external input is Cohen–Conze, arXiv:1304.4556v2, Theorem 2.18: the Haar CLT for a totally ergodic toral action and a real mean-zero observable with absolutely convergent Fourier series. Its assumptions hold here: the eigenvalues $\lambda^{\pm 1}$ imply total ergodicity, $F_4 < \infty$ gives the required Fourier convergence, and intervals $\{0, \dots, K-1\}$ are a Følner sequence. The exact source and scope are recorded in the source check. For zero variance the limit also follows directly from the variance formula and convergence to zero in L^2. For a smooth symbol combine this input with (Ret.P17) and (Ret.P5). For finite F_4, truncate at $M = \lfloor K^{1/4} \rfloor$ instead: the two tail errors in (Ret.P14) are at most $T\sqrt{K}F_4M^{-4} = O(K^{-1/2})$, while (Ret.P4) still has a superpolynomially small right side, since F_0, F_1, F_2 are uniformly bounded. This suffices for the same limit. For compact uniformity, the Haar characteristic functions have Lipschitz constant at most F_0, by their second-moment bound. A finite mesh, pointwise convergence, and the two uniform comparisons finish the proof. No quantitative rate for the external Gaussian approximation is assumed. For a nonzero mean μ, apply the statement to $f - \mu$ and multiply the quantum trace by $e^{-i\mu t\sqrt{K}}$.

28.8 Pair multiplicity and a finite-dimensional unitary normal form

The next step adapts the averaging mechanism of the earlier odd-prime-period note, Section 4, to the present even arithmetic periods. The 32-class lattice makes it possible to prove the required pair bound directly, without assuming that different Fourier modes occupy different classical orbits.

Let $\mathcal{S} \subset \mathbb{Z}^2 \setminus \{0\}$ be a finite frequency set with $\|v\|_1 \le M$. Assume $K > 256M$ and put $L = 8M + 1$. Every nonzero key in $(\mathbb{Z}/N\mathbb{Z})^2$ has at most

$$B_M = [2(2L+1)|\mathcal{S}|]^2 \tag{Ret.P20}$$

ordered representations $A^i v + A^j w$ with $0 \le i, j < K$ and $v, w \in \mathcal{S}$. Equal times are included.

To prove this, fix one representation and compare another. Their difference is a labeled four-atom relation with total norm at most $4M$. Aggregate the atoms at each time into c. Equation (Ret.P9) forces its preimage to be integer, and (Ret.P8) gives $\|p\|_1 \le 8M$. There is a zero coordinate of p. Split its nonzero coordinates into maximal contiguous runs. Each run has length at most $8M$, and its forcing is supported on that run and the one preceding site. Forcing intervals from different runs are disjoint. After cutting the time circle at a zero preimage coordinate, each run gives an exact integer-frequency zero relation by telescoping. For a run crossing the original time origin use the lifted time indices; these agree with the original indices modulo N because $A^K = I \bmod N$.

Assign all atoms at a nonzero forcing site to its run. At any remaining occupied site the aggregate is zero; those atoms form a separate zero group. Each group contains at least two atoms. A nonempty run has nonzero forcing at both endpoints; a single nonzero integer frequency cannot form a zero relation. There are thus either one group of four atoms or two groups of two. In the latter case each group must match an atom from each representation: an internal match would make the fixed output key zero. It follows that both new times are in the union of the cyclic radius-L neighborhoods of the two fixed times. This union has at most $2(2L+1)$ sites. Counting the two new frequencies proves (Ret.P20). The argument includes coincident atoms, equal times, canceled aggregates, and a vanishing entire forcing vector. Equal times in the first representation do not permit an internal zero group for a nonzero key.

For $H = \mathrm{Op}_N(f)$ with this Fourier support, define

$$H_j = U_0^j H U_0^{-j}, \quad \bar{H} = \frac{1}{K}\sum_{j=0}^{K-1} H_j, \quad c_j = \frac{K-1-2j}{2K}, \quad G = \sum_{j=0}^{K-1} c_j H_j.$$

Both $G, \bar{H}$ are self-adjoint; $\bar{H}$ commutes with U_0, and $G - U_0 G U_0^{-1} = H - \bar{H}$. Let

$$a_d = \max_{v,w\in\mathcal{S}} |e^{2\pi i\omega(v,A^d w)/N} - 1|.$$

Taking the centered representative d modulo K, using $A^K = I \bmod N$, and summing geometric tails from $|d| = m$ gives

$$\sum_{d=0}^{K-1} a_d \le C_A M^2, \qquad \sum_{d=0}^{K-1} |c_d| a_d \le C_A M^2/K. \tag{Ret.P21}$$

More explicitly, if $\ell = m - \min(d, K-d)$, then $a_d \le C_A M^2 \lambda^{-\ell}$ and $|c_d| \le (\ell+1)/K$. The same estimates include $d = 0$.

Use the normalized Hilbert–Schmidt norm $\|X\|_{2,N}^2 = N^{-1}\operatorname{Tr}(X^*X)$. The triangle inequality and (Ret.P21) give

$$\|[G,H]\|_{\mathrm{op}} + \|[H,\bar H]\|_{\mathrm{op}} \le C_A F_0^2 M^2/K.$$

For the remaining commutator,

$$[G,\bar H] = \frac{1}{K^2}\sum_{i<j}(j-i)[H_i,H_j].$$

Balanced Weyl matrices in distinct residue classes are orthogonal. The zero output key has zero commutator coefficient. Cauchy–Schwarz on each nonzero key, followed by (Ret.P20), therefore yields

$$\begin{aligned}\|[G,\bar H]\|_{2,N}^2 &\le \frac{B_M F_0^4}{K^4}\sum_{d=1}^{K-1}(K-d)d^2a_d^2 \\ &\le \frac{C_A B_M F_0^4 M^2}{K}.\end{aligned} \tag{Ret.P22}$$

The last inequality uses $d(K-d) \le K^2/4$ and $a_d^2 \le 2a_d$. Since $|\mathcal{S}| \le (2M+1)^2$, it follows that $\|[G,\bar H]\|_{2,N} \le C_A F_0^2 M^4/\sqrt{K}$. This last bound is a Hilbert–Schmidt bound, not an operator-norm bound.

For self-adjoint X, Y, integration over a time triangle proves

$$\|e^{i\epsilon X}e^{i\epsilon Y} - e^{i\epsilon(X+Y)}\| \le \frac{\epsilon^2}{2}\|[X,Y]\|$$

in either of these unitarily invariant norms, for every real ϵ. Apply it first to $-G, H$ and then to $H-G, U_0 G U_0^{-1}$. The second commutator is $[H-G,\bar H]$. Hence

$$\left\|e^{-i\epsilon G}U_{\epsilon,f}e^{i\epsilon G} - e^{i\epsilon\bar H}U_0\right\|_{2,N} \le C_A F_0^2\epsilon^2\left(\frac{M^2}{K} + \frac{M^4}{\sqrt{K}}\right). \tag{Ret.P23}$$

No smallness assumption on $\epsilon\|G\|$ has been used. Constants in f can be removed exactly; they do not affect commutators.

28.9 Every replica at the critical scale

Theorem 28.5 (uniform replica comparison)**.** *For every fixed real symbol f with $F_4 < \infty$ and every $T < \infty$,*

$$\sup_{\substack{r\ge 1\\ |t|\le T}}\left|\frac{\operatorname{Tr} U_{t/(r\sqrt{K}),f}^{rK}}{N} - \frac{\operatorname{Tr} U_{t/\sqrt{K},f}^{K}}{N}\right| = O_{f,T}(K^{-1/2}). \tag{Ret.P24}$$

The result also holds for the normalized odd traces of even symbols. It concerns integer r, including r depending on K.

We first remove the bandwidth loss in (Ret.P23). Subtract the mean throughout this argument; restoring it multiplies both traces by the same phase. Split the nonzero Fourier frequencies into the shells $0 < \|v\|_1 \le 1$ and $2^{a-1} < \|v\|_1 \le 2^a$ for $a \ge 1$. Let f_a be the corresponding symbol and $\mu_a = \sum |\widehat{f_a}|$. The zero mode is excluded. For two shells of maximal radius M, with $K > 256M$, (Ret.P20) and Weyl orthogonality applied to $[G_a, \bar{H}_b] = K^{-1} \sum_{i,j} c_i [H_{a,i}, H_{b,j}]$ give

$$\|[G_a, \bar{H}_b]\|_{2,N}^2 \le \frac{B_M \mu_a^2 \mu_b^2}{K^2} \sum_{i,j} c_i^2 a_{j-i}^2 \le \frac{C_A B_M \mu_a^2 \mu_b^2 M^2}{K}.$$

Here $\sum_i c_i^2 \le K/4$ and $\sum_d a_d^2 \le 2 \sum_d a_d \le C_A M^2$. The same bound includes equal-time pairs. Thus the bilinear norm is at most $C_A \mu_a \mu_b \max(2^a, 2^b)^4 / \sqrt{K}$.

Take the largest dyadic $R < K/512$ and sum over shells of radius at most R. There is no logarithmic loss, since

$$\sum_{a,b} \mu_a \mu_b \max(2^a, 2^b)^4 \le 2 \left(\sum_a \mu_a \right) \left(\sum_b 2^{4b} \mu_b \right) \le C F_0 F_4.$$

The omitted commutators are $[G_{>R}, \bar{H}] + [G_{\le R}, \bar{H}_{>R}]$. Their operator norm is at most $CKF_0 \sum_{\|v\|_1 > R} |\widehat{f_v}| \le CK^{-3} F_0 F_4$. For the finitely many smaller K, enlarge the constant. The full operator-norm bounds also follow by summing (Ret.P21) first for each frequency pair and then against its Fourier coefficients. Therefore

$$\begin{aligned} \|[G, \bar{H}]\|_{2,N} &\le C_A F_0 F_4 / \sqrt{K}, \\ \|[G, H]\|_{\mathrm{op}} + \|[H, \bar{H}]\|_{\mathrm{op}} &\le C_A F_1^2 / K, \\ \left\| e^{-i\epsilon G} U_{\epsilon,f} e^{i\epsilon G} - e^{i\epsilon \bar{H}} U_0 \right\|_{2,N} &\le C_A \epsilon^2 (F_1^2/K + F_0 F_4 / \sqrt{K}). \end{aligned} \tag{Ret.P24a}$$

All Fourier passages are norm-convergent. No truncation of the actual quantum trace is required here.

Telescope rK unitary factors in (Ret.P24a), with $\epsilon = t/(r\sqrt{K})$. The reference product is exactly $e^{it\sqrt{K}\bar{H}}$; its normalized trace differs from the physical trace by at most $C_A T^2 r^{-1} (F_1^2/K + F_0 F_4/\sqrt{K})$. Comparing with $r = 1$ proves (Ret.P24). The gauge depends on ϵ but disappears from each trace. For even symbols all gauges preserve parity; the restriction of a normalized full Hilbert–Schmidt bound to the odd space costs at most $\sqrt{2N/(N-1)}$.

Combining Theorems 28.4 and 28.5 gives, for every fixed real mean-zero f with $F_4 < \infty$,

$$\sup_{\substack{r \ge 1 \\ |t| \le T}} \left| \frac{\operatorname{Tr} U_{t/(r\sqrt{K}),f}^{rK}}{N} - e^{-\sigma_A^2(f) t^2/2} \right| \to 0. \tag{Ret.P25}$$

In particular, if $\epsilon_K \sqrt{K} \to s \in \mathbb{R}$, the global empirical eigenphase distribution of $U_{\epsilon_K,f}^K$ converges to the wrapped Gaussian of variance $s^2 \sigma_A^2(f)$. Indeed each fixed integer moment tends to $e^{-r^2 s^2 \sigma_A^2(f)/2}$; negative moments are conjugates, and trigonometric polynomials are dense on the circle. For a nonzero mean μ, center the return operator by $e^{-iK\epsilon_K \mu}$. The odd-block conclusion requires even f.

The $K^{-1/2}$ estimate is for the comparison of quantum returns; the Gaussian limit itself is qualitative here. The nonzero K^{-1} correction in (Ret.P16) is consistent with (Ret.P24). This does not extend the large-variance decay estimate (Ret.A1) to all two-dimensional smooth symbols. The separate analytic phase-space argument covers general analytic symbols in that regime using a vector-lattice gas estimate. Neither argument determines local level spacings, or traces at the Heisenberg scale for one fixed perturbed classical system.

28.10 The zero-variance case and its next scale

Theorem 28.6 (smooth degeneracy)**.** *For real mean-zero smooth f, $\sigma_A^2(f) = 0$ if and only if $f = g - g \circ T$ for a real smooth g. If f is even, g can be chosen even. This is a smooth Fourier-coboundary criterion, a classical mechanism already used in the earlier smooth note. Here is the regularity argument for the current covector matrix A.*

Choose real eigen-coordinates ξ, ζ with $A : (\xi, \zeta) \mapsto (\lambda\xi, \lambda^{-1}\zeta)$ and let $|v|_* = \max(|\xi(v)|, |\zeta(v)|)$. No nonzero integer vector lies on either eigenline. On each oriented orbit select a representative v_0 minimizing this norm, paired with $-v_0$ on the negative orbit. Minimality implies $\lambda^{-1} \le |\xi(v_0)/\zeta(v_0)| \le \lambda$. Writing $v_j = A^j v_0$, one obtains $\lambda^{|j|-1}|v_0|_* \le |v_j|_* \le \lambda^{|j|}|v_0|_*$.

If the orbit sum of f vanishes, define its transfer coefficients by

$$\widehat{g}_{v_j} = \begin{cases} \sum_{\ell \le j} \widehat{f}_{v_\ell}, & j < 0, \\ -\sum_{\ell > j} \widehat{f}_{v_\ell}, & j \ge 0. \end{cases}$$

For every $s > 0$, summing the geometric weights on each half-orbit gives $\sum_j |v_j|_*^s |\widehat{g}_{v_j}| \le C_s \sum_j |v_j|_*^s |\widehat{f}_{v_j}|$. For example, on the positive half use $|v_j|_*^s \le \lambda^s \lambda^{-s(\ell-j)} |v_\ell|_*^s$ for $\ell > j \ge 0$, and interchange absolutely convergent sums. The negative half is identical with time reversed. Summing over all orbits proves smoothness of g by equivalence of norms. Its coefficient difference is exactly f, including at $j = 0$ because the orbit sum vanishes. Reality and parity are preserved by the construction. Conversely, the Fourier differences of a smooth coboundary telescope on every orbit. Formula (Ret.P18) proves the equivalence.

Let $X = \mathrm{Op}_N(g)$, $Y = \mathrm{Op}_N(g \circ T)$. The elementary two-factor Duhamel estimate gives

$$\left\| e^{-i\epsilon X} U_{\epsilon,f} e^{i\epsilon X} - U_0 \right\|_{\mathrm{op}} \le \frac{\epsilon^2}{2} \|[X, Y]\|_{\mathrm{op}} \le \frac{\pi \epsilon^2}{N} F_1(g) F_1(g \circ T). \tag{Ret.P26}$$

Consequently, at $\epsilon = t/(r\sqrt{K})$ the normalized rK-return trace differs from one by at most $C_g t^2/(rN)$. For $r = 1$ even the operator $U_{\epsilon,f}^K$ is within $C_g t^2/N$ of the identity. This is stronger than the degenerate case of the qualitative limit.

The next effect is described by the fixed smooth symbol

$$h_g = \frac{\{g, g \circ T\}}{4\pi}, \qquad \Gamma_A(g) = \sigma_A^2(h_g), \qquad \{a, b\} = \partial_1 a \, \partial_2 b - \partial_2 a \, \partial_1 b.$$

It is real and mean zero; it is even if g is even. For self-adjoint X, Y, put $Q = [X, Y]$. The uniform unitary loop bound is

$$\left\| e^{-i\epsilon X} e^{i\epsilon(X-Y)} e^{i\epsilon Y} - e^{-\epsilon^2 Q/2} \right\|_{\mathrm{op}} \le |\epsilon|^3 \left(\tfrac{1}{2} \|[X, Q]\| + \tfrac{1}{6} \|[Y, Q]\| \right). \tag{Ret.P27}$$

Indeed its logarithmic derivative is $-\int_0^u \mathrm{Ad}_{e^{-iuX} e^{iv(X-Y)}} Q \, dv$. Compare it with $-uQ$, the logarithmic derivative of $e^{-u^2 Q/2}$, using unitary conjugation, and integrate over u. The bound is valid for every real ϵ; it does not require $\epsilon \|X\|$ to be small.

The Weyl formula and $|\sin u - u| \le |u|^3/6$ imply $[X, Y] = -2i \, \mathrm{Op}_N(h_g)/N + O_g(N^{-3})$. Both double commutators in (Ret.P27) are $O_g(N^{-2})$ by two applications of $|2\sin(\pi\omega/N)| \le 2\pi|\omega|/N$ to the Fourier series. All weighted sums converge for smooth g, with no frequency exclusion. Thus

$$\left\| e^{-i\epsilon X} U_{\epsilon, g - g\circ T} e^{i\epsilon X} - U_{\epsilon^2/N, h_g} \right\|_{\mathrm{op}} \le C_g \left(\frac{|\epsilon|^3}{N^2} + \frac{\epsilon^2}{N^3} \right). \tag{Ret.P28}$$

This is the same local loop mechanism established in the earlier note; the present consequence concerns the even-period sequence and all replicas.

Theorem 28.7 (uniform second-order return law)**.** *For every fixed real smooth g and every finite T_0,*

$$\sup_{\substack{r\ge 1,\ 0\le t\le T_0\\ \epsilon=\pm\sqrt{tN/(r\sqrt{K})}}}\left|\frac{\operatorname{Tr} U_{\epsilon,g-g\circ T}^{rK}}{N}-e^{-\Gamma_A(g)t^2/2}\right|\to 0. \tag{Ret.P29}$$

Here r ranges over positive integers. The same holds on the odd block for even g. After rK steps the error in (Ret.P28) is at most

$$C_{g,T_0}\left(\frac{K^{1/4}}{\sqrt{rN}}+\frac{\sqrt{K}}{N^2}\right).$$

The effective kick is exactly $t/(r\sqrt{K})$, so (Ret.P25) applies to the fixed symbol h_g, uniformly in r. This proves (Ret.P29). No positivity of $\Gamma_A(g)$ for every nonconstant g is asserted.

For example, take $g=2\cos(2\pi z_1)$ and $v=(1,0)$. Then

$$h_g=2\pi\big(e_{v-Av}+e_{-v+Av}-e_{v+Av}-e_{-v-Av}\big),\qquad \Gamma_A(g)=16\pi^2. \tag{Ret.P30}$$

The two frequency pairs occupy different signed A-orbits: $\omega(v-Av,A(v-Av))=-8$ and $\omega(v+Av,A(v+Av))=16$. These invariants differ and are nonzero. The limiting second-order trace is therefore $e^{-8\pi^2t^2}$. The quantum amplitude in (Ret.P29) can grow, but its classical strength $|\epsilon|/N$ still tends to zero uniformly in the indicated range.

28.11 Joint traces of physical return operators

The normal form also controls the return operator itself. For a real symbol with $F_4<\infty$, let $X_K(f)=\sqrt{K}\,\bar H_f$. For fixed bounded t and every positive integer r,

$$\left\|U_{t/(r\sqrt{K}),f}^{rK}-e^{itX_K(f)}\right\|_{2,N}\le\frac{C_At^2}{r}\left(\frac{F_1^2}{K}+\frac{F_0F_4}{\sqrt{K}}\right). \tag{Ret.P31}$$

To remove the gauge from (Ret.P24a), apply the integrated commutator bound

$$\|e^{i\alpha G}e^{i\beta\bar H}e^{-i\alpha G}-e^{i\beta\bar H}\|_{2,N}\le|\alpha\beta|\|[G,\bar H]\|_{2,N}$$

with $\alpha=t/(r\sqrt{K})$ and $\beta=t\sqrt{K}$. Together with the telescoped normal form, this proves (Ret.P31). It does not require the gauge itself to be close to the identity.

For two fixed real symbols f,h, the same ordered-pair argument as in Section 9 gives

$$\|[X_K(f),X_K(h)]\|_{2,N}\le\frac{C_A}{\sqrt{K}}(F_0(f)F_4(h)+F_4(f)F_0(h)). \tag{Ret.P32}$$

For two low-frequency shells the coefficient in $[\bar H_f,\bar H_h]$ is K^{-2} at each ordered time pair. Thus its squared norm is bounded by $B_M\mu_a^2\nu_b^2K^{-4}\sum_{i,j}a_{j-i}^2\le C_AB_M\mu_a^2\nu_b^2M^2/K^3$. Multiply the norm by K for the two factors X_K, and sum the dyadic weights. The high-frequency remainder before multiplication by K is $O(F_0F_{\mathrm{tail}})$, hence $O(K^{-4})$ with the corresponding weighted norms. Constants can again be adjusted for bounded K. This proves (Ret.P32), including constants after their removal.

Theorem 28.8 (joint critical return law)**.** *Fix real mean-zero $f_1,\dots,f_J$ with finite fourth Fourier moments and real $t_1,\dots,t_J$. Write*

$$\Sigma_{ab}=\sum_{\mathcal O}\left(\sum_{v\in\mathcal O}\widehat{f_{a_v}}\right)\overline{\left(\sum_{v\in\mathcal O}\widehat{f_{b_v}}\right)}.$$

This real symmetric covariance is the polarization of (Ret.P18). Then, uniformly over positive integers $r_1, \dots, r_J$,

$$\frac{1}{N}\operatorname{Tr}\left(\prod_{j=1}^{J} U_{t_j/(r_j\sqrt{K}),f_j}^{r_jK}\right) \longrightarrow \exp\left(-\tfrac{1}{2}\sum_{a,b=1}^{J} t_a t_b \Sigma_{ab}\right). \tag{Ret.P33}$$

The product has the displayed order; every fixed order has the same limit. For the odd-block statement every f_j must be even.

Equation (Ret.P31) replaces the physical factors by $e^{it_jX_K(f_j)}$ with a total normalized Hilbert–Schmidt error $O(K^{-1/2})$. Two-factor Duhamel, iterated over this fixed number of factors, gives

$$\left\| \prod_j e^{it_jX_K(f_j)} - e^{iX_K(\sum_j t_j f_j)} \right\|_{2,N} \le \tfrac{1}{2}\sum_{a<b} |t_a t_b| \|[X_K(f_a), X_K(f_b)]\|_{2,N}.$$

Apply (Ret.P32), then (Ret.P31) with $r = 1, t = 1$ to the fixed symbol $\sum_j t_j f_j$, and finally Theorem 28.4. Its variance is the quadratic form in (Ret.P33). This proves the joint law without asserting convergence of unbounded moments from characteristic functions alone.

These are joint traces of a fixed finite family of arithmetic return operators with the explicitly scaled amplitudes. They are not the Heisenberg-scale joint traces of a fixed classical perturbation. For a fixed amplitude $\epsilon = t/\sqrt{K}$, the direct normal-form error after rK steps grows as $O_f(rt^2/\sqrt{K})$. Thus its useful fixed-amplitude time range is $r = o(\sqrt{K})$, whereas Heisenberg times require r of order N/K.

28.12 Verification scope of the continuation

The same verifier now includes seven exact centered-distance bounds, through $K = 262$, and exhausts 1,122,112 ordered two-atom representations at $K = 6, 10, 14, 34, 262$, including equal-time pairs. The $K = 262$ case contains 1,098,304 representations and satisfies the hypothesis $K > 256M$ with $M = 1$. The observed maximum nonzero-key multiplicity is 4 in every case; the proved bound is 23,104. The smaller cases are explicitly marked outside that theorem's range. The discrepancy between this conservative universal bound and these observations is not used to strengthen the theorem.

At $N = 35$, physical matrices check the exact averaging identity, five unitary normal-form inequalities, ten replica comparisons both with and without the gauge, and six joint physical-product comparisons. For the coboundary example, the generated Fourier coefficients and the orbit invariants in (Ret.P30) are checked exactly. Nine unitary loops, including negative and non-small amplitudes, and eight second-order replica comparisons are tested in double precision against their finite Duhamel bounds. No interval enclosure or numerical proof of a limit is claimed.

The independent internal review checked the 32-class and balanced-sign argument, the ordered-pair decomposition including equal times, the dyadic F_4 estimate, and the smooth transfer and second-order signs. The initial P4–P5 snapshot had SHA-256 6324e620fad6a8cb13888b80738de3d0153d6a1b3f3f74bf1897019717247fb0. The Section 10 audit referenced snapshot 6e79e11455a9b91226eda1b66cd01079ddf9344d1e18a43f01ab6e6e09b9f1f2. Subsequent amendments make the integer-replica quantifier explicit and add Section 11; that extension is not represented as independently audited. The source check documents the external classical CLT separately. Generated results remain in the reproducible build output. These proofs are included in the canonical manuscript through a generated TeX chapter; the analytic limit theorems are not Lean-formalized.

Chapter 5 — Return laws I (analytic and smooth kicks) (continued)

29 General analytic phase-space kicks and the regularity of return averages

Numbering convention: section and theorem references within this chapter use the source module's local numbering; source Section 1 is subsection 29.1. Equation references carry the prefix Ret.

Status. Project derivation with a limited independent internal check of the vector lattice, Weyl phases, analytic gas, and regularity proofs. Exact finite arithmetic checks and floating matrix diagnostics are separated in Section 10. External refereeing, priority verification, and Lean formalization remain open. No microscopic BGS universality is claimed.

Use the balanced quantization, arithmetic sequence and notation of the periodic-point comparison:

$$A = \begin{pmatrix} 3 & 4 \\ 2 & 3 \end{pmatrix}, \quad T = A^{\mathsf{T}}, \quad \lambda = 3 + 2\sqrt{2}, \quad K = 2m, \quad m \geq 3 \text{ odd}, \quad N = b_m, \quad U_0^K = I.$$

The scalar-site gas and its classical Penrose tree identity are proved in the fixed-harmonic note, Section 3. The argument below retains the additional onsite and antipodal phases that arise for a general two-dimensional symbol.

29.1 Analytic return theorem

Let f be a fixed real analytic mean-zero function on $\mathbb{T}^2$. Equivalently for our purposes, assume

$$F_\rho(f) = \sum_{v \in \mathbb{Z}^2} |\widehat{f}_v| e^{\rho \|v\|_1} < \infty \quad \text{for some } \rho > 0, \qquad \widehat{f}_0 = 0.$$

Let $\sigma^2 = \sigma_A^2(f)$ be the full orbit variance (Ret.P18).

Theorem 29.1. *If $\sigma^2 > 0$, there are $s_0, C, c > 0$, depending on f, such that for every admissible K, every positive integer r, and every real ϵ with $s = r|\epsilon| \leq s_0$,*

$$\left| \frac{1}{N} \operatorname{Tr} U_{\epsilon,f}^{rK} \right| \leq e^{-\sigma^2 K s^2/4} + Ce^{-cK}, \tag{Ret.R1}$$

$$\left| \frac{1}{N} \operatorname{Tr} U_{\epsilon,f}^{rK} - e^{-\sigma^2 K s^2/2} \right| \leq \frac{C}{\sqrt{K}} + Ce^{-cK}. \tag{Ret.R2}$$

For even f the same estimates hold on the normalized odd block. Constants are not uniform as f approaches the zero-variance subspace. If $\sigma^2 = 0$, the stronger coherent estimate is

$$\left| \frac{1}{N} \operatorname{Tr} U_{\epsilon,f}^{rK} - 1 \right| \leq \frac{C_f K s^2}{rN}. \tag{Ret.R3}$$

Thus for every $\epsilon_K \to 0$ and fixed positive integer r, the trace differs from $e^{-\sigma^2 r^2 K \epsilon_K^2/2}$ by $o(1)$. When $\sigma^2 > 0$, the global eigenphase law of $U_{\epsilon_K,f}^K$ is Haar measure on the circle if and only if $K\epsilon_K^2 \to \infty$. For $K\epsilon_K^2 \to v < \infty$ it is the wrapped Gaussian with variance $v\sigma^2$. For $\sigma^2 = 0$ the return operator tends to the identity in operator norm. These are global return laws at vanishing classical kick strength, not statements about local level spacings.

29.2 Analytic cohomology and orbit representatives

Choose the eigen-coordinate norm $|v|_*$ of Section 10 of the companion note. In each oriented nonzero A-orbit choose a representative of minimal $|\cdot|_*$, with paired representatives for negative orbits. Put

$$b_{v_{\mathcal O}} = \sum_{v\in\mathcal O} \widehat{f}_v, \qquad b(z) = \sum_{\mathcal O} b_{v_{\mathcal O}} e_{v_{\mathcal O}}(z).$$

Then there is a real analytic g such that

$$f = b + g - g\circ T, \qquad \int b^2 = \sum_{\mathcal O} |b_{v_{\mathcal O}}|^2 = \sigma^2. \tag{Ret.R4}$$

The nonzero frequencies of b meet each oriented A-orbit at most once. Both b and g are even when f is even.

For completeness, analyticity is preserved with a positive strip width. First use an exponential weight in the equivalent norm $|\cdot|_*$. Minimality gives $|v_{\mathcal O}|_* \le |v|_*$ on its orbit, so the weighted Fourier norm of b is at most that of f. Write $v_j = A^j v_0$ on one orbit and define

$$\widehat{g}_{v_j} = \begin{cases} \sum_{\ell\le j} \widehat{f}_{v_\ell}, & j < 0, \\ -\sum_{\ell> j} \widehat{f}_{v_\ell}, & j \ge 0. \end{cases}$$

The difference at $j = 0$ is $\widehat{f}_{v_0} - b_{v_0}$, as required. On each half-orbit the norms grow geometrically from their minimum. In particular $|v_j|_* \le |v_\ell|_*$ between the representative and v_ℓ, and $|\ell| \le C_A \log(2 + |v_\ell|_*)$. For $0 < \rho' < \rho$, summing the transfer coefficients with weight $e^{\rho'|v|_*}$ is therefore bounded by

$$C_A \sum_v |\widehat{f}_v| \log(2 + |v|_*) e^{\rho'|v|_*} \le C_{A,\rho,\rho'} \sum_v |\widehat{f}_v| e^{\rho|v|_*}.$$

This proves (Ret.R4) and analyticity after reverting to $\|\cdot\|_1$.

Two-factor unitary Duhamel applied to $e^{i\epsilon \operatorname{Op}(g)} e^{i\epsilon \operatorname{Op}(b)} e^{-i\epsilon \operatorname{Op}(g\circ T)}$ shows that $U_{\epsilon,f}$ is unitarily conjugate to $U_{\epsilon,b}$ up to operator error $C_f \epsilon^2/N$. Each commutator of two fixed smooth symbols costs at most $2\pi F_1(a) F_1(d)/N$. Consequently the rK-trace error is at most $C_f K s^2/(rN)$, with the same result on an invariant odd block. If $\sigma^2 = 0$, all coefficients of b vanish and (Ret.R3) follows. It remains to prove (Ret.R1)–(Ret.R2) for an analytic b with the stated orbit-representative property and $\int b^2 > 0$.

29.3 The entire vector resonance lattice

On cyclic vector sequences let

$$(\mathcal D p)_j = p_j - A p_{j+1}, \qquad \mathcal C_K = \left\{ c : \sum_j A^j c_j = 0 \pmod N \right\}, \quad q(c) = |\{j : c_j \ne 0\}|.$$

The companion note proves $p = \mathcal D^{-1} c \in (1/32)\mathbb Z^{2K}$ and $\|p\|_1 \le 2\|c\|_1$ for $c \in \mathcal C_K$. If one p_j is an integer vector, integrality propagates around the circle. Thus any noninteger class has no zero vector coordinate.

For a nonzero rational vector $p = (x, y)$ of denominator dividing 32,

$$Q(p) = \omega(p, Ap) = 2x^2 - 4y^2 \ne 0, \qquad |Q(p)| \ge 1/1024.$$

In coordinates $p = \xi(\sqrt2, 1) + \zeta(-\sqrt2, 1)$, $Q = -16\xi\zeta$ and $\|p\|_1 \ge 2\max(|\xi|, |\zeta|)$. If the forcing vanishes strictly between sites $i < j$, with gap $d = j - i$, and the intervening homogeneous state

is nonzero, its endpoint states are related by A^{d-1}. The arithmetic-geometric mean inequality gives

$$\|p_{i+1}\|_1 + \|p_j\|_1 \geq \lambda^{(d-1)/2}/32.$$

For the internal gaps of a finite nonzero run, or all cyclic gaps of a configuration with no zero coordinate, every endpoint coordinate is counted at most twice. Hence

$$\|c\|_1 \geq \frac{1}{128} \sum_{\text{forcing gaps } d} \lambda^{(d-1)/2}. \tag{Ret.R5}$$

Fix $\alpha > 0$ and define

$$b_\alpha = \sum_{v \neq 0} e^{-\alpha\|v\|_1/2} = \left(\frac{1+e^{-\alpha/2}}{1-e^{-\alpha/2}}\right)^2 - 1, \quad h_\alpha(d) = e^{-\alpha\lambda^{(d-1)/2}/256}, \quad H_\alpha(R) = \sum_{d\geq 1} h_\alpha(d) R^d.$$

The last series is finite for every finite $R > 0$. For $0 \leq \eta \leq [2b_\alpha H_\alpha(2)]^{-1}$,

$$\sum_{c \in \mathcal{C}_K} \eta^{q(c)} e^{-\alpha\|c\|_1} \leq \exp\bigl(2b_\alpha^2 H_\alpha(1) K\eta^2\bigr) + (3/4)^K. \tag{Ret.R6}$$

Indeed, if p has a zero vector, its nonzero coordinates split into maximal contiguous runs. Each run's forcing support is contained in the run and its preceding site and contains both boundary sites. The supports for distinct runs are disjoint. For $q \geq 2$ forcing sites and internal gaps $d_1, \ldots, d_{q-1}$, split the exponential weight in half. Summing one half over nonzero forcing vectors and using (Ret.R5) for the other half gives the bound $(\eta b_\alpha)^q \prod h_\alpha(d_i)$. The component sum with fixed starting site is at most

$$\sum_{q\geq 2} (\eta b_\alpha)^q H_\alpha(1)^{q-1} \leq 2b_\alpha^2 H_\alpha(1)\eta^2.$$

Allowing arbitrary component subsets gives the exponential in (Ret.R6).

If every p_j is nonzero, mark a nonzero forcing site. For q such sites the cyclic gaps sum to K, and every unmarked configuration is counted q times. Its total weight is bounded by

$$\sum_{q=1}^{K} \frac{K}{q} (\eta b_\alpha)^q [z^K] H_\alpha(z)^q \leq K 2^{-K} \sum_{q\geq 1} \frac{(\eta b_\alpha H_\alpha(2))^q}{q} \leq K 2^{-K} \log 2 \leq (3/4)^K.$$

This includes every noninteger class. Counting is over forcing vectors with unique real preimages, so no extra factor of 32 is needed.

The power η^2 in (Ret.R6) cannot in general be replaced by η^3: a run with forcing $(-A^d v, v)$ at two sites exists for every $d \geq 1$. The third power of the kick needed below will instead come from the Fourier-atom degree after the reduction (Ret.R4).

29.4 Exact replica phases for vector charges

Write $e^{i\epsilon b} = \sum_v k_v(\epsilon) e_v$. Replacing the physical kick $e^{i\epsilon \operatorname{Op}(b)}$ by $\operatorname{Op}(e^{i\epsilon b})$ costs at most $\pi\epsilon^2 F_1(b)^2 e^{|\epsilon| F_0(b)}/(2N)$ per kick. The total rK-operator error is bounded by

$$\frac{C_b K s^2}{rN} e^{KsF_0(b)}. \tag{Ret.R7}$$

For sufficiently small fixed s_0 this is $O_b(e^{-cK})$. The original physical operator is retained through this explicit bound; commuting different Fourier modes within a kick would be incorrect.

Let $v_{\nu j} \in \mathbb{Z}^2$ be its Fourier index in replica ν and at time j, and $c_j = \sum_\nu v_{\nu j}$. Regroup the original replica-ordered word by sites, keeping replica order within a site. Its exact crossing phase is

$$\exp\left[-\frac{2\pi i}{N}\sum_{i<j}\sum_{\nu<\mu}\omega(v_{\mu i}, A^{j-i}v_{\nu j})\right]. \tag{Ret.R8}$$

The regrouped word contains the onsite factors

$$\exp\left[\frac{\pi i}{N}\sum_{j}\sum_{\nu<\mu}\omega(v_{\nu j}, v_{\mu j})\right] \tag{Ret.R9}$$

and the remaining product $\prod_j \mathcal{W}_N(A^j c_j)$. Only this remaining product depends solely on the nonzero aggregates. Both (Ret.R8) and (Ret.R9) can depend on clean replica configurations. The sign in (Ret.R8) follows from each inversion: $\mathcal{W}(a)\mathcal{W}(d) = e^{-2\pi i\omega(d,a)/N}\mathcal{W}(d)\mathcal{W}(a)$.

In (Ret.R8), replace $j-i>m$ by $j-i-K$. This preserves the phase because $A^K = I \bmod N$. Its bilinear kernel has a uniformly bounded spatial row sum. Using the maximum matrix-entry norm, an admissible bound is

$$\sum_{d=1}^{K-1}\frac{2\pi}{N}\max|A^{\mathrm{center}(d)}| \le 2\pi\left(4+\frac{8}{\lambda-1}\right) = 8\pi\sqrt{2}. \tag{Ret.R10}$$

Here $\max|A^j| = 4b_{|j|}$ for $j \neq 0$. The bound on a particle interaction is this kernel times $l(\boldsymbol{v})l(\boldsymbol{w})$, where $l(\boldsymbol{v}) = \sum_\nu \|v_\nu\|_1$, independently of r.

The antipodal phase must not be set to one for general vector charges: $\omega((1,0), A^m(0,1)) = \alpha_m$, and $e^{2\pi i\alpha_m/N} \neq 1$. The position-only cancellation at $d=m$ does not apply here.

29.5 The clean complex gas, including its onsite correction

At a clean site $\sum_\nu v_\nu = 0$. The vacuum is $v_0 = k_0(\epsilon)^r$; every nonvacuum particle has at least two nonzero replica entries. Its activity is

$$w_j(\boldsymbol{v}) = v_0^{-1}\prod_\nu k_{v_\nu}(\epsilon)\exp\left[\frac{\pi i}{N}\sum_{\nu<\mu}\omega(v_\nu, v_\mu) + i\sum_\nu\langle h_{\nu j}, v_\nu\rangle\right].$$

The fields are real and arise from specified dirty profiles through (Ret.R8). The onsite factor has modulus one. For a fixed positive analytic weight β smaller than the strip width, the nonvacuum activity moment satisfies, uniformly in the fields,

$$\sup_j\sum_{\boldsymbol{v}\neq 0}|w_j(\boldsymbol{v})|\,l(\boldsymbol{v})e^{\beta l(\boldsymbol{v})} \le C_b s^2. \tag{Ret.R11}$$

This follows by summing the two required nonzero entries, using $\sum_{v\neq 0}|k_v|e^{\beta\|v\|} = O_b(|\epsilon|)$, the corresponding first-moment bound, and $|k_0(\epsilon)|^{-r} \le e^{C_b r\epsilon^2}$. The replica binomial factors are bounded by powers of r, absorbed in s.

Combine (Ret.R10), same-site exclusion, and (Ret.R11) with the complex tree estimate proved in the fixed-harmonic note. Its criterion is $(1+8\pi\sqrt{2})M_\beta < \beta$. On any set of clean sites the connected logarithm after the one-particle term is $O_b(Ks^4)$, uniformly in K, r and all real fields.

One must also control the single-site sum. Without (Ret.R9) it is

$$B_j^{\mathrm{cl}} = \int_{\mathbb{T}^2}\exp\left(i\epsilon\sum_{\nu=1}^{r} b(z + h_{\nu j}/(2\pi))\right)dz.$$

With (Ret.R9), denote it by B_j. Fourier first moments give

$$|B_j - B_j^{\mathrm{cl}}| \le \frac{\pi s^2 F_1(b)^2}{2N} e^{sF_0(b)}. \tag{Ret.R12}$$

To see this, bound the difference of the onsite phase from one by $\pi N^{-1}\sum_{\nu<\mu}\|v_\nu\|_1\|v_\mu\|_1$, drop the zero-sum constraint, and use $\sum_v \|v\|_1|k_v| \le |\epsilon|F_1(b)e^{|\epsilon|F_0(b)}$. The pair count is at most $r^2/2$.

Since b has mean zero, Taylor expansion and the gas remainder imply

$$\log|\mathcal{Q}_{B,\boldsymbol{h}}| \le -\frac{\epsilon^2}{2}\sum_{j\notin B}\Big\|\sum_\nu b(\cdot + h_{\nu j}/(2\pi))\Big\|_2^2 + C_b K(s^3 + s^2/N). \tag{Ret.R13}$$

Vacuum factors at the clean sites are included in $\mathcal{Q}$. For precision, the logarithmic gas formula is $\sum_{j\notin B}\log v_0 + \sum_{j\notin B}(B_j/v_0 - 1) + O(Ks^4)$. Because $B_j/v_0 - 1 = O(s^2)$, replacing its first two terms by $\sum \log B_j$ costs $O(Ks^4)$. This proves (Ret.R13). At no external field the full clean partition $Q_{r,K}$ is nonzero and

$$\left|\log Q_{r,K} + \tfrac{1}{2}\sigma^2 K(r\epsilon)^2\right| \le C_b K(s^3 + s^2/N). \tag{Ret.R14}$$

For dirty profiles of total absolute mass $S_B = \sum_{j\in B,\nu}\|v_{\nu j}\|_1$, the field row bound gives $\sum_{j\notin B}\max_\nu \|h_{\nu j}\|_\infty \le C_A S_B$. Translation in L^2 and (Ret.R13) consequently yield

$$|\mathcal{Q}_{B,\boldsymbol{h}}| \le \exp\left[-\tfrac{1}{2}\sigma^2 K s^2 + C_b K(s^3 + s^2/N) + C_b s^2(|B| + S_B)\right]. \tag{Ret.R15}$$

Indeed $\|\sum_\nu b(\cdot + a_\nu)\|_2^2 \ge r^2\sigma^2 - 2r\|b\|_2\|\nabla b\|_2\sum_\nu\|a_\nu\|$. This estimate requires no smallness of the dirty charges or their fields.

29.6 Why the finite-component contribution starts at cubic order

Let $D > 0$ dominate the constant multiplying $s^2(|B| + S_B)$ in (Ret.R15). Define a Fourier series with positive coefficients

$$B_s(z) = \sum_{v\neq 0}|\widehat{b}_v|e^{Ds^2\|v\|_1}e_v(z), \qquad a_c(s) = e^{Ds^2}\,[e^{sB_s}]_c \quad (c \neq 0). \tag{Ret.R16}$$

For a dirty site, the absolute sum over all its replica profiles, including the factor from (Ret.R15), is bounded by $a_c(s)$. The inequality follows by expanding each $k_v(\epsilon)$ into its original Fourier atoms and using the triangle inequality on their aggregate norm. Summing replica assignments gives $r|\epsilon| = s$. This also accounts for internal cancellations inside a dirty replica.

For some $\alpha > 0$ and small $s_* > 0$, $a_c(s) \le C_b s\, e^{-\alpha\|c\|_1}$ for $0 \le s \le s_*$. The same estimate with $|s|$ bounds its absolute Taylor series on a complex disk: choose Ds_*^2 smaller than half the available strip width. Each $a_c(s)$ has nonnegative Taylor coefficients.

Let $\mathcal{M}(s)$ be the sum, with one fixed starting site, of $\prod_{c_j\neq 0} a_{c_j}(s)$ over all finite nonzero integer preimage runs. The component proof of (Ret.R6) gives a normally convergent analytic series $\mathcal{M}(s) = O_b(s^2)$ on a fixed disk. Its coefficient of degree two is, however, zero. A term of that degree would use two original Fourier atoms of b at two distinct forcing sites. Their exact lifted relation would be $A^i v + A^j w = 0$ with $i \neq j$. The orbit-representative property, including the paired negative representatives, would force $A^{i-j}v = v$ for a nonzero integer vector. Hyperbolicity excludes this. Two opposite atoms at the same site have zero aggregate and do not form a nonzero forcing component.

Since the coefficients of $\mathcal{M}$ are nonnegative and its degrees zero through two vanish, for a fixed $0 < R < s_*$,

$$\mathcal{M}(s) \le (s/R)^3\mathcal{M}(R) \le C_b s^3 \quad (0 \le s \le R). \tag{Ret.R17}$$

This is a bound on the total original Fourier-atom degree, not a false assertion that every vector component occupies at least three sites. For example, the orbit-representative modes $(1,0),(0,1),(1,1)$ allow the two-site forcing

$$c_0=(1,1),\qquad c_1=(1,-1),\qquad c_0+Ac_1=0.$$

It uses three original atoms: one at site zero and two at site one.

The nonwinding sum over nonzero aggregates is at most $e^{K\mathcal{M}(s)}-1$. For winding configurations apply (Ret.R6) with $\eta=C_bs$ to get $(3/4)^K$ after reducing R. All scalarized central words and all 32 classes are therefore included. Writing their normalized full trace as $\mathcal{T}_{r,K}$, (Ret.R15) gives

$$|\mathcal{T}_{r,K}-Q_{r,K}|\le e^{-\sigma^2Ks^2/2+C_bK(s^3+s^2/N)}\left[e^{C_bKs^3}-1+(3/4)^K\right]. \tag{Ret.R18}$$

No factorization of the quantum phases of distinct dirty components is assumed; the sum is bounded after taking absolute values.

For sufficiently large K and sufficiently small fixed s_0, the cubic and N^{-1} terms are absorbed into a fixed fraction of the negative quadratic term. Equations (Ret.R14) and (Ret.R18) imply

$$|\mathcal{T}_{r,K}|\le e^{-\sigma^2Ks^2/4}+(3/4)^K,$$

$$|\mathcal{T}_{r,K}-e^{-\sigma^2Ks^2/2}|\le C_b(Ks^3+Ks^2/N)e^{-c_bKs^2}+(3/4)^K. \tag{Ret.R19}$$

The maximum of $Ks^3e^{-c_bKs^2}$ is $O_b(K^{-1/2})$; the other term is $O_b(N^{-1})$. Restore the physical kick by (Ret.R7) and f by (Ret.R4), reducing s_0 so that $s_0F_0(b)<\log\lambda/4$. The odd correction, when relevant, is at most $e^{KsF_0(b)}/(N-1)$ before the already controlled replacements. All these errors are exponentially small in K. Enlarging C handles the finitely many remaining small K. This proves (Ret.R1)–(Ret.R2).

The spectral consequences in Section 1 follow by testing each fixed integer Fourier moment. Necessity of $K\epsilon_K^2\to\infty$ for Haar measure follows already from the first moment and (Ret.R2). This proof of the analytic return law does not import a classical CLT.

29.7 A logarithmic Fourier bound for quantum period averages

For an absolutely summable Fourier series of mean zero define

$$X_K(f)=\frac{1}{\sqrt{K}}\sum_{j=0}^{K-1}U_0^j\,\mathrm{Op}_N(f)U_0^{-j}.$$

This section concerns these averaged observables. It does not lower the regularity assumption for the actual kicked propagator in (Ret.R1).

If a nonzero integer vector v has an early modular return $A^dv=v \bmod N$, $0<d<K$, consider the forcing $c_0=-v, c_d=v$. Its preimage has no zero coordinate. Otherwise it would be integer and would split into finite components. The only two forcing atoms would have to belong to the same component, and its lifted exact relation would imply $A^ev=v$ for some nonzero $|e|<K$. This is impossible for a hyperbolic A. Apply (Ret.R5) to the two cyclic gaps $d, K-d$ to obtain

$$2\|v\|_1\ge\frac{1}{128}\left(\lambda^{(d-1)/2}+\lambda^{(K-d-1)/2}\right),\qquad \|v\|_1\ge\frac{\lambda^{K/4}}{256\sqrt{\lambda}}. \tag{Ret.R20}$$

Thus an early return is possible only if $K\le C_A\log(2+\|v\|_1)$.

For one Fourier character, if all K orbit residues are distinct, Weyl orthogonality gives $\|X_K(e_v)\|_{2,N}=1$ exactly. If any two coincide, use (Ret.R20) and the elementary bound $\|X_K(e_v)\|_{2,N}\le\sqrt{K}$. Consequently, for every K and every $v\neq0$,

$$\|X_K(e_v)\|_{2,N}\le C_A\sqrt{\log(2+\|v\|_1)}. \tag{Ret.R21}$$

This includes central aliases and possible balanced signs among repeated residues; the upper bound does not treat all such signs as positive. By the triangle inequality,

$$\sup_K \|X_K(f)\|_{2,N} \le C_A \mathcal{L}_{1/2}(f), \qquad \mathcal{L}_\beta(f) = \sum_{v\neq 0} |\widehat{f}_v|[\log(2+\|v\|_1)]^\beta. \tag{Ret.R22}$$

Theorem 29.2 (low-regularity averaged quantum CLT)**.** *If f is real, mean zero and $\mathcal{L}_{1/2}(f) < \infty$, the empirical spectral law of $X_K(f)$ converges weakly to $N(0, \sigma_A^2(f))$. More generally, for every fixed finite family in this class and fixed real $t_1, \dots, t_J$,*

$$\frac{1}{N} \operatorname{Tr} \prod_{j=1}^{J} e^{it_j X_K(f_j)} \longrightarrow \exp\left[-\tfrac{1}{2}\sigma_A^2 \left(\sum_j t_j f_j \right) \right]. \tag{Ret.R23}$$

For even symbols the statement holds on the odd block as well. No convergence of arbitrary unbounded moments is asserted.

For a fixed Fourier polynomial, (Ret.R2), or (Ret.R3) when its variance vanishes, and the gauge-free estimate (Ret.P31) give the scalar statement. For finitely many such polynomials, (Ret.P32) and iterated unitary Duhamel give the joint statement. These uses of (Ret.P31)–(Ret.P32) require only their finite-dimensional estimates, not the classical CLT imported elsewhere in the companion note.

Now let f_M be a fixed finite Fourier cutoff. By (Ret.R22),

$$\sup_K \|X_K(f) - X_K(f_M)\|_{2,N} \le C_A \mathcal{L}_{1/2}(f - f_M) \to 0.$$

The corresponding exponential difference is bounded by $|t|$ times this norm. Telescope a fixed product of unitary exponentials to obtain the same approximation for its normalized trace. First take $K \to \infty$ at fixed cutoffs, then let the cutoffs grow. The covariance converges by absolute Fourier summability and the orbit formula (Ret.P18). This proves (Ret.R23) and, by characteristic functions, the weak spectral convergence. The transfer to the odd space costs the bounded norm factor $\sqrt{2N/(N-1)}$.

29.8 The exponent one half is sharp within these weighted Fourier classes

Theorem 29.3. *There is a single fixed real even continuous mean-zero function f for which $\mathcal{L}_\beta(f) < \infty$ for every $\beta < 1/2$, but the empirical laws of $X_K(f)$ are not tight. In particular replacing $1/2$ by any smaller exponent in Theorem 29.2 would be false. This is sharpness of a uniform class guarantee, not a necessary condition for every individual observable.*

For $j \ge 1$ put

$$K_j = 4 \cdot 2^{2^j} + 2, \qquad N_j = b_{K_j/2}, \qquad c_j = \frac{j}{\sqrt{K_j}}, \qquad f(z) = 2\sum_{j\ge 1} c_j \cos(2\pi N_j z_1). \tag{Ret.R24}$$

Every K_j is admissible. The coefficients are summable, so this defines one fixed continuous function, not a dimension-dependent symbol. Since $\log N_j \asymp K_j$,

$$\mathcal{L}_\beta(f) \le C_\beta \sum_j j K_j^{\beta - 1/2} < \infty \quad (\beta < 1/2).$$

At dimension N_j, the jth character is the identity: $\mathcal{W}_{N_j}(N_j, 0) = I$. It stays the identity under every Egorov iterate. Its contribution to $X_{K_j}(f)$ is therefore $2jI$.

Every earlier frequency $N_i(1,0)$, $i<j$, has K_j distinct orbit residues for all sufficiently large j. Indeed $N_i<\lambda^{K_i/2}$, whereas an early return would require the lower bound (Ret.R20). The difference K_j-2K_{j-1} tends to infinity. Thus the contribution Y_j of the earlier modes has

$$\|Y_j\|_{2,N_j}\le 2\sum_i c_i<\infty.$$

The later modes have operator norm at most $2\sqrt{K_j}\sum_{i>j}c_i$. The recurrence $K_{j+1}=(K_j-2)^2/4+2$ gives $K_{j+1}\ge K_j^2/5$; for $j\ge 2$ it also gives $c_{j+1}/c_j\le 1/2$. Hence their contribution E_j satisfies

$$\|E_j\|_{\mathrm{op}}\le 4(j+1)\sqrt{K_j/K_{j+1}}\le\frac{4\sqrt{5}(j+1)}{\sqrt{K_j}}\longrightarrow 0,\qquad X_{K_j}(f)=2jI+Y_j+E_j. \quad \text{(Ret.R25)}$$

The spectral Chebyshev inequality and the Hermitian min-max principle now show that the fraction of eigenvalues of $X_{K_j}(f)$ below j is at most $C/(j-\|E_j\|_{\mathrm{op}})^2$, which tends to zero. The empirical spectra escape to positive infinity along this subsequence. The same argument works on the odd block.

The classical variance is nevertheless finite and nonzero: $\sigma_A^2(f)=2\sum_j c_j^2$. Different positive multiples of $(1,0)$ belong to different oriented A-orbits, and nonzero-time classical correlations vanish. The failure comes from quantum aliases of the high Fourier modes. This continuous example is outside the smooth and analytic kick classes; it is not a counterexample to their return theorems or to a formulated fixed-system BGS conjecture.

29.9 The failure persists after subtracting the quantum mean

The previous example uses a divergent scalar alias. The loss of tightness below the exponent $1/2$ is not repaired by subtracting that scalar: there is also one fixed real even continuous function f in every class $\mathcal{L}_\beta$, $\beta<1/2$, for which

$$\widetilde{X}_K(f)=X_K(f)-\frac{1}{N}\operatorname{Tr}X_K(f)\,I \quad \text{(Ret.R26)}$$

has non-tight spectral laws.

Here is an explicit construction. Use the K_j of (Ret.R24) and set

$$L_j=15K_j,\quad M_j=b_{L_j/2},\quad d_j=M_j/5,\quad a_j=j/\sqrt{L_j},\qquad f(z)=2\sum_{j\ge1}a_j\cos(2\pi d_j z_1). \quad \text{(Ret.R27)}$$

The same summability argument proves the stated regularity. All L_j are admissible. The exact recurrence gives $25\mid b_{15}$. Since $15\mid L_j/2$ and $b_u\mid b_{uv}$ (apply the formula for A^{uv} to $A^u=\alpha_u I+b_u B$), we have $25\mid M_j$.

Fix one such dimension $M=25t$, where t is odd, and let $d=M/5$. On the physical M-dimensional space write

$$C=\mathcal{W}_M(d,0),\qquad S=\mathcal{W}_M(0,d).$$

They commute, have order five, and

$$\mathcal{W}_M(dv)=(-1)^{v_1v_2}C^{v_1}S^{v_2}.$$

Indeed their commutator is $e^{2\pi iM/25}=1$, and the balanced onsite factor is $e^{-\pi i(M/25)v_1v_2}=(-1)^{v_1v_2}$. Their 25 joint characters have equal multiplicity $M/25$: the normalized trace of C^uS^v is zero unless $u=v=0 \bmod 5$.

Since $A = I \bmod 2$, every $A^\ell(1,0)$ has even coordinate product. Modulo five its orbit has length six and consists of

$$(1,0),\ (3,2),\ (2,2),\ (-1,0),\ (-3,-2),\ (-2,-2).$$

Because $6 \mid L_j$, the averaged character at frequency $d_j(1,0)$ is the Hermitian contraction

$$H_j = \frac{1}{6}\sum_{\ell=0}^{5} C^{(A^\ell(1,0))_1} S^{(A^\ell(1,0))_2}, \qquad \frac{1}{M_j}\operatorname{Tr} H_j = 0. \tag{Ret.R28}$$

It equals the identity on the joint $C = S = 1$ subspace of dimension $M_j/25$. Thus the contribution of the jth real Fourier mode to $X_{L_j}(f)$ is $2jH_j$, rather than a scalar matrix.

The earlier frequencies have full modular period L_j for all sufficiently large j, by (Ret.R20). Their contribution Y_j has uniformly bounded normalized Hilbert–Schmidt norm and zero trace. The later frequencies contribute an operator E_j with norm tending to zero, by the same geometric-tail estimate as (Ret.R25). Hence

$$\widetilde{X}_{L_j}(f) = 2jH_j + D_j, \qquad D_j = Y_j + E_j - \frac{1}{M_j}\operatorname{Tr} E_j\, I, \qquad \sup_j \|D_j\|_{2,M_j} < \infty. \tag{Ret.R29}$$

The spectral subspace where $|D_j| > j$ has dimension at most $M_j\|D_j\|_{2,M_j}^2/j^2$. Its orthogonal complement intersects the $H_j = 1$ subspace in dimension at least the difference of these ranks. On this intersection the quadratic form of $\widetilde{X}_{L_j}(f)$ is at least j. The min-max principle therefore gives

$$\frac{1}{M_j}\#\{\text{eigenvalues of } \widetilde{X}_{L_j}(f) \text{ at least } j\} \geq \frac{1}{25} - \frac{C_f}{j^2}. \tag{Ret.R30}$$

This proves non-tightness even with the actual quantum mean removed.

The conclusion also holds after restriction to the odd space and subtraction of its own normalized trace. To count the relevant subspace, an orthonormal basis of $C = S = 1$ is

$$w_a = \frac{1}{\sqrt{5}}\sum_{\ell=0}^{4} |5a + \ell d\rangle, \qquad a \in \mathbb{Z}/t\mathbb{Z}.$$

Parity sends w_a to w_{-a}, so its odd dimension is $(t-1)/2$. Its relative dimension in the odd space is $(M/25 - 1)/(M - 1) \to 1/25$. The normalized odd trace of H_j is $O(M_j^{-1})$, since the parity-twisted trace of each Weyl operator has modulus one. Subtracting this trace changes $2jH_j$ by $O(j/M_j) = o(1)$. The odd Hilbert–Schmidt norm of Y_j is bounded; subtracting its odd trace preserves a uniform bound. The same intersection and min-max argument, with threshold $j/2$ if necessary, gives a positive limiting fraction of escaping eigenvalues.

29.10 Verification and scope

The companion verifier is phase_space_return_gas.py. Run from the repository root with NumPy available:

The exact verifier invocation is reproduced as listing S4 in the Supplement.

The recorded run passes the following controls:

- 320 exact replica-word reordering identities at $K = 6, 10, 14, 34$; explicit nontrivial antipodal and onsite phases;
- 3,000 rational endpoint inequalities, checked in $\mathbb{Q}(\sqrt{2})$ without floating point, and 120 finite two-forcing-site runs;
- 1,224 finite two-atom exclusions and an explicit three-atom component occupying only two forcing sites;

- 25 early-return controls through $K = 262$, with the stronger two-gap inequality checked exactly when a return occurs;
- exact recurrence and tail-ratio checks, central-alias controls at $K = 18, 66, 1026$, and noncentral period-six aliases at $L = 270, 990, 15390$. Older modes have the full target periods in the checked cases; the proof above establishes the eventual assertion;
- six finite Taylor-polynomial onsite matrix comparisons at $N = 35$, separately labeled as floating diagnostics. Commuting subgroup projectors at $M = 25, 75, 125$ reproduce their full and odd ranks; the largest matrix residual is below $4.4 \cdot 10^{-14}$.

The analytic lattice, conditional gas, cubic-degree mechanism, and both regularity counterexamples received limited independent internal review. That review led to two clarifications: the forcing support is contained in a run and its preceding site, and the zero-variance case of Theorem 29.2 uses (Ret.R3). This is not external referee acceptance or formal verification.

The actual-kick theorem requires analytic f, while the low-regularity theorem concerns $X_K(f)$. Uniformity in r in Theorem 29.1 requires $r|\epsilon| \le s_0$: it does not permit arbitrary time growth at a fixed kick. Neither global Gaussian return laws nor a regularity counterexample supply local spectral correlations for a fixed nonzero classical perturbation.

The subsequent general cat-map argument extends the analytic return theorem, the logarithmic regularity threshold and its centered counterexample to every fixed hyperbolic $A \in SL(2, \mathbb{Z})$, on explicitly specified arithmetic subsequences. The Gaussian CLT for classical toral sums is prior mathematics. Its source-level check is in the literature note. No claim of a first result in the literature follows from these project derivations.

Chapter 6 — Return laws II (general laws, obstructions, return words) (continued)

30 Return laws and a sharp regularity boundary for every hyperbolic toral automorphism

Numbering convention: section and theorem references within this chapter use the source module's local numbering; source Section 1 is subsection 30.1. Equation references carry the prefix Ret.

Status. Project derivation. The arithmetic, quantization, estimate transfer, regularity counterexample and Gevrey extension have limited independent internal review; the proof below records the changes needed from the previously checked special matrix. External priority, external refereeing, and Lean formalization are open. The statements concern arithmetic subsequences and global laws at vanishing classical perturbation. They do not establish microscopic BGS.

The analytic gas estimates used below are proved in the phase-space note. The normalized Hilbert–Schmidt normal form is proved in the periodic-point note. This note removes the restriction to the particular matrix in those notes.

30.1 The arithmetic subsequence and the statements

Fix any $A \in SL(2, \mathbb{Z})$ with $|t| > 2$, where $t = \operatorname{Tr} A$. Write

$$\Delta = t^2 - 4, \qquad \lambda = \frac{|t| + \sqrt{\Delta}}{2} > 1, \qquad b_0 = 0, \quad b_1 = 1, \quad b_{n+1} = tb_n - b_{n-1}.$$

Define the fixed positive integer q_A and allowed integers m by

	q_A	m
t even	1	positive odd integers
t odd	$2^{v_2(t^2-1)}$	3ℓ, ℓ positive odd

$$K = 2m, \qquad N = b_m/q_A. \tag{Ret.U1}$$

Retain only $N \geq 3$. These are odd positive integers and $N \asymp_A \lambda^{K/2}$, including when $t < 0$.

Use the Weyl convention and the unitary $U_{0,N}$ constructed in Section 2, with $U_{0,N}^K = I$. For real f let

$$U_{\epsilon,f} = e^{i\epsilon \operatorname{Op}_N(f)} U_{0,N}, \qquad X_K(f) = \frac{1}{\sqrt{K}} \sum_{j=0}^{K-1} U_{0,N}^j \operatorname{Op}_N(f) U_{0,N}^{-j}.$$

Let $T = A^{\mathsf{T}}$. For an absolutely summable mean-zero Fourier series put

$$\sigma_A^2(f) = \sum_{\mathcal{O}} \left| \sum_{v \in \mathcal{O}} \widehat{f}_v \right|^2, \qquad \mathcal{L}_\beta(f) = \sum_{v \neq 0} |\widehat{f}_v| [\log(2 + \|v\|_1)]^\beta, \tag{Ret.U2}$$

where $\mathcal{O}$ ranges over oriented nonzero A-orbits. The variance equals the absolutely convergent Green–Kubo sum for real f.

Theorem 30.1 (Gevrey return law)**.** *Suppose f is fixed, real, mean zero, and for some $\rho > 0$ and $0 < \alpha \leq 1$ satisfies*

$$F_{\rho,\alpha}(f) = \sum_v |\widehat{f}_v| e^{\rho \|v\|_1^\alpha} < \infty.$$

If $\sigma_A^2(f) > 0$, there are positive constants s_0, C, c, depending on A, f, ρ, α, such that for all the dimensions (Ret.U1), all positive integers r, and $s = r|\epsilon| \leq s_0$,

$$|\tau_N U_{\epsilon,f}^{rK}| \leq e^{-\sigma_A^2(f) K s^2/4} + C e^{-cK}, \tag{Ret.U3}$$

$$|\tau_N U_{\epsilon,f}^{rK} - e^{-\sigma_A^2(f) K s^2/2}| \leq C K^{-1/2} + C e^{-cK}. \tag{Ret.U4}$$

Here $\tau_N = N^{-1} \operatorname{Tr}$. If $\sigma_A^2(f) = 0$, instead

$$\|U_{\epsilon,f}^{rK} - I\|_{\mathrm{op}} \leq C_{A,f} K s^2/(rN). \tag{Ret.U5}$$

For even f, (Ret.U3)–(Ret.U5) also hold on the odd block with its own normalized trace. Constants are not uniform as A or f varies.

Consequently, for $\epsilon_K \to 0$ and positive variance the global return eigenphase law is Haar exactly when $K\epsilon_K^2 \to \infty$; when $K\epsilon_K^2 \to v < \infty$ it is wrapped Gaussian with variance $v\sigma_A^2(f)$. With $\hbar_N = (2\pi N)^{-1}$, the classical action parameter is $\kappa = \hbar_N \epsilon = \epsilon/(2\pi N)$.

Theorem 30.2 (averaged-observable CLT)**.** *For fixed real mean-zero $f_1, \ldots, f_J$ with $\mathcal{L}_{1/2}(f_j) < \infty$ and fixed $u_1, \ldots, u_J \in \mathbb{R}$,*

$$\tau_N \prod_{j=1}^J e^{i u_j X_K(f_j)} \longrightarrow e^{-\frac{1}{2}\sigma_A^2(\sum_j u_j f_j)}. \tag{Ret.U6}$$

In particular the empirical spectral law of $X_K(f)$ tends weakly to $N(0, \sigma_A^2(f))$. The same statements hold on the odd block when all symbols are even.

Theorem 30.3 (sharpness, including quantum centering)**.** *For every such A there is one fixed real even continuous mean-zero f with $\mathcal{L}_\beta(f) < \infty$ for every $\beta < 1/2$, such that the empirical spectral laws of*

$$X_K(f) - \tau_N X_K(f)\, I \tag{Ret.U7}$$

are not tight along (Ret.U1). The conclusion also holds after restriction to the odd block and subtraction of its own quantum mean. The exponent $1/2$ is therefore sharp as a sufficient guarantee for these weighted Fourier classes. It is not asserted to be a necessary condition for each individual observable.

30.2 Arithmetic and quantization for general A

The Cayley–Hamilton identities give

$$A^m = b_m A - b_{m-1} I, \qquad A^{2m} - I = b_m A^m (2A - tI), \qquad |\det(A^{2m} - I)| = \Delta b_m^2. \quad \text{(Ret.U8)}$$

For negative t, b_m has sign $(-1)^{m-1}$ times the corresponding positive-trace sequence, so every allowed b_m is positive. If t is even, parity of the recurrence gives b_m odd for odd m, and $2A - tI$ is an even matrix. If t is odd, $b_3 = t^2 - 1$ and

$$b_{3\ell} = b_3 u_\ell, \qquad u_0 = 0, \quad u_1 = 1, \quad u_{\ell+1} = (\operatorname{Tr} A^3) u_\ell - u_{\ell-1}.$$

Since $\operatorname{Tr} A^3 = t^3 - 3t$ is even, u_ℓ is odd for odd ℓ. This proves precisely the power of two removed in (Ret.U1). In particular $q_A \geq 8$ for odd t. It follows in both cases that

$$A^K = I \bmod 2N, \qquad |\det(A^K - I)| = D_A N^2, \quad D_A = \Delta q_A^2. \quad \text{(Ret.U9)}$$

The usual odd-N finite Weyl operators $W_N(v)$ use the inverse of two modulo N and depend only on v mod N. Introduce the quadratic refinement

$$q_*(v) = v_1 v_2 + v_1 + v_2 \bmod 2, \qquad \mathcal{W}_N^*(v) = (-1)^{q_*(v)} W_N(v). \quad \text{(Ret.U10)}$$

The function q_* equals one on every nonzero vector of $\mathbb{F}_2^2$; thus $q_*(Av) = q_*(v)$ for every integer unimodular A. Its polarization is the symplectic form modulo two. Consequently

$$\mathcal{W}_N^*(v)\mathcal{W}_N^*(w) = e^{\pi i \omega(v,w)/N} \mathcal{W}_N^*(v+w), \qquad \mathcal{W}_N^*(2Nv) = I. \quad \text{(Ret.U11)}$$

In contrast, $\mathcal{W}_N^*(Nv) = (-1)^{q_*(v)} I$. In particular the central character at $(N, 0)$ is $-I$.

The assignment $W_N(v) \mapsto W_N(Av)$ extends to a star automorphism of the full matrix algebra: Weyl multiplication is symplectically invariant, and the N^2 residue operators form an orthogonal basis. Every star automorphism of a finite full matrix algebra is implemented by a unitary. One elementary verification is to map a complete family of rank-one matrix units, choose unit vectors in the image diagonal projections, and fix their phases using the image off-diagonal units. This constructs $U_{0,N}$ with exact Egorov for W_N, and invariance of q_* gives exact Egorov for $\mathcal{W}_N^*$. Equation (Ret.U9) implies that $U_{0,N}^K$ commutes with the entire matrix algebra, so a scalar phase makes it the identity.

Parity $R|x\rangle = |-x\rangle$ commutes with $U_{0,N}$: its conjugate implements the same Weyl automorphism $v \mapsto -v$, so differs from R by a scalar; $\operatorname{Tr} R = 1$ fixes that scalar. Quantization of even symbols also commutes with parity. These statements require odd N, not primality of N.

The refinement $v_1 v_2$ used for the earlier special matrix is also invariant when $A = I$ mod 2. The present choice supplies a common convention for all A; its different central sign is retained below.

30.3 Uniform lattice and gas estimates

For cyclic vector sequences define

$$(\mathcal{D}_A p)_j = p_j - A p_{j+1}, \qquad \mathcal{C}_K = \{c \in \mathbb{Z}^{2K} : \sum_j A^j c_j = 0 \bmod N\}.$$

The map to $(\mathbb{Z}/N\mathbb{Z})^2$ is onto through c_0. Together with (Ret.U9), the determinant of $\mathcal{D}_A$ therefore gives

$$[\mathcal{C}_K : \mathcal{D}_A \mathbb{Z}^{2K}] = D_A, \qquad \mathcal{D}_A^{-1} c \in D_A^{-1} \mathbb{Z}^{2K}. \quad \text{(Ret.U12)}$$

For the second assertion, telescope to $(I - A^K)p_0 = Nz$. By (Ret.U8), $I - A^K$ has all entries divisible by $b_m = q_A N$. Its inverse formula shows that p_0 has denominator dividing Δq_A, hence dividing D_A. The recurrence and integrality of A^{-1} propagate this denominator.

Let P_s, P_u be the real spectral projections of A. For periodically extended forcing, its unique bounded preimage is

$$p_j = \sum_{\ell \geq 0} A^{\ell} P_s c_{j+\ell} - \sum_{\ell \geq 1} A^{-\ell} P_u c_{j-\ell}.$$

Both geometric series converge absolutely, even with negative eigenvalues. Summation over one period yields

$$\|p\|_1 \leq C_A \|c\|_1. \tag{Ret.U13}$$

No constant in this estimate grows with K.

The integral quadratic form $Q_A(v) = \omega(v, Av)$ does not vanish on a nonzero rational vector. A rational eigenline would give a rational algebraic-integer eigenvalue; its reciprocal is also an algebraic integer, forcing an eigenvalue 1 or -1, contrary to hyperbolicity. Thus $|Q_A(p)| \geq D_A^{-2}$ on the nonzero lattice in (Ret.U12). Eigen-coordinate decomposition and the arithmetic-geometric mean inequality now give, for a homogeneous gap of length d,

$$\|p_{\text{left}}\|_1 + \|p_{\text{right}}\|_1 \geq c_A \lambda^{(d-1)/2}.$$

As in (Ret.R5), every endpoint is counted at most twice. By (Ret.U13),

$$\|c\|_1 \geq \gamma_A \sum_{\text{gaps}} \lambda^{(d-1)/2}, \qquad \gamma_A > 0. \tag{Ret.U14}$$

This applies to internal gaps of finite nonzero integer runs and to all gaps of winding configurations. Noninteger preimages have no zero coordinate and hence belong to the latter class.

For $\alpha > 0$ put $h_{\alpha,A}(d) = \exp[-\alpha\gamma_A \lambda^{(d-1)/2}/2]$. Its generating series is entire. Splitting the exponential Fourier weight in half, the component enumeration from (Ret.R6) gives

$$\sum_{c \in \mathcal{C}_K} \eta^{q(c)} e^{-\alpha \|c\|_1} \leq e^{C_{\alpha,A} K \eta^2} + (3/4)^K \tag{Ret.U15}$$

for a sufficiently small fixed $\eta \geq 0$. The winding count is still $K 2^{-K} \sum_{a \geq 1} (\eta b_\alpha H_{\alpha,A}(2))^a / a$. It counts forcing vectors with unique preimages and already includes all D_A classes. There is no unaccounted multiplicity depending on K.

All exact reordering identities (Ret.R8)–(Ret.R9) use only (Ret.U11), symplecticity, and period reduction (Ret.U9). Their signs are unchanged. The centered interaction kernel satisfies

$$\frac{2\pi}{N} \sum_{d=1}^{K-1} \max |A^{\text{center}(d)}| \leq C_A, \tag{Ret.U16}$$

since $\max |A^j| \leq C_A \lambda^{|j|}$ and $N \asymp_A \lambda^{K/2}$. No antipodal phase is discarded.

Analytic cohomology is unchanged: choose minimal representatives in the adapted eigen-coordinate norm, paired for negative orbits, to write $f = b + g - g \circ T$ with analytic b, g and $\int b^2 = \sigma_A^2(f)$. The Fourier support of b meets each oriented orbit at most once. The quantum conjugacy error over rK steps is $C_{A,f} K s^2/(rN)$, proving (Ret.U5) when $b = 0$.

For $b \neq 0$, (Ret.U16) and the original onsite estimate (Ret.R12) give the conditional clean-gas bound (Ret.R15), with constants depending on A, b. Use the positive analytic Fourier-atom majorant (Ret.R16) and replace (Ret.R6) by (Ret.U15). Its finite-component generating function has no degree-two coefficient: two atoms at distinct forcing sites would contradict the orbit-representative property. It is therefore $O_{A,b}(s^3)$ on a fixed disk. These are exactly the inputs to (Ret.R18)–(Ret.R19); all integer and winding classes have been retained.

Finally the physical scalarization error remains $C_{A,b}Ks^2e^{KsF_0(b)}/(rN)$, by (Ret.U11). Choose s_0 small enough that $s_0F_0(b) < \frac{1}{4}\log\lambda$. This error, the conjugacy error, and the odd-trace correction are exponentially small in K. Equations (Ret.U3)–(Ret.U4) follow. This proof uses no unproved classical CLT. It proves the analytic case $\alpha = 1$ of Theorem 30.1. Section 6 supplies the changes for every $0 < \alpha < 1$.

30.4 The weighted normal form and low-regularity limit

Here are the inputs for carrying the polynomial argument of (Ret.P31)–(Ret.P32) to general A. For frequencies of norm at most M, equality between two ordered pair sums modulo N gives at most four forcing atoms of total norm at most $4M$. If $K > C_AM$, (Ret.U12)–(Ret.U13) force the preimage to be integer and to have a zero coordinate. Every nonzero run has length at most C_AM.

The resulting local grouping argument of (Ret.P20) therefore bounds the number of representations of each nonzero output key by

$$C_AM^2|S|^2, \tag{Ret.U17}$$

where S is the frequency support. The zero output key contributes no commutator: the symplectic area of opposite residues vanishes modulo N. Centered kernel summation gives the other inputs,

$$\sum_d a_d(v,w) \le C_A\|v\|_1\|w\|_1, \qquad \sum_d |c_d|a_d(v,w) \le C_A\|v\|_1\|w\|_1/K,$$

where $a_d = |e^{2\pi i\omega(v,A^dw)/N} - 1|$ and $c_d = (K-1-2d)/(2K)$ with $0 \le d < K$. The second estimate uses cancellation of c_d near the antipodal time and exponential decay of a_d away from it.

For Fourier shells of radius M, $|S| = O(M^2)$. The Cauchy–Schwarz argument with (Ret.U17) and the subsequent dyadic sum are consequently the same as in (Ret.P24a): with $F_a(f) = \sum_v(1+\|v\|_1)^a|\widehat{f}_v|$,

$$\|[G_f, \overline{H}_h]\|_{2,N} \le C_AK^{-1/2}(F_0(f)F_4(h) + F_4(f)F_0(h)).$$

The operator commutator estimates and unitary Duhamel therefore give (Ret.P31)–(Ret.P32) with constants depending on A. Together with (Ret.U4), or (Ret.U5) at zero variance, they prove (Ret.U6) for fixed Fourier polynomials.

For an early return $A^dv = v \bmod N$, $0 < d < K$, apply (Ret.U14) to the two forces $-v, v$. Their preimage cannot have a zero: an integer finite component would give $A^ev = v$ for a nonzero $|e| < K$. Thus both cyclic gaps are included, and

$$\|v\|_1 \ge c_A\lambda^{K/4}, \qquad K \le C_A\log(2+\|v\|_1). \tag{Ret.U18}$$

Weyl orthogonality gives $\|X_K(e_v)\|_{2,N} = 1$ when all orbit residues are distinct, and the elementary bound $\sqrt{K}$ suffices otherwise. Hence

$$\|X_K(e_v)\|_{2,N} \le C_A\sqrt{\log(2+\|v\|_1)}, \qquad \sup_K \|X_K(f)\|_{2,N} \le C_A\mathcal{L}_{1/2}(f). \tag{Ret.U19}$$

Finite Fourier cutoffs now approximate the unitary characteristic traces uniformly in K. First taking $K \to \infty$ at fixed cutoffs, then removing the cutoffs, proves (Ret.U6). The orbit variance is continuous in the absolute Fourier norm. On the odd block the normalized Hilbert–Schmidt norm costs only the bounded factor $\sqrt{2N/(N-1)}$. This proves Theorem 30.2.

30.5 A centered counterexample for every A

The needed fixed odd prime always exists. The integer

$$B = b_5 = t^4 - 3t^2 + 1$$

is odd and greater than five. If $5 \mid B$, then $t^2 = 4+5a$ for some integer a, giving $B = 5+25a+25a^2$; thus $v_5(B) = 1$. Consequently B has a prime divisor $p \neq 5$, and p is odd. Since $B = 5 \bmod \Delta$, this prime does not divide Δ.

Modulo p, (Ret.U8) gives $A^5 = \alpha I$ with $\alpha^2 = 1$. Therefore the order h of A modulo p divides ten. It cannot divide two: over a field of odd characteristic, $A^2 = I$ and $\det A = 1$ would force $A = I$ or $-I$, and then $p \mid \Delta$. Thus $h = 5$ or 10. Moreover

$$p^2 \mid b_{5p}. \tag{Ret.U20}$$

Indeed $b_{5p} = b_5 u_p$ for the recurrence with trace $\operatorname{Tr} A^5$. Modulo p, this trace is 2α and its solution is $u_\ell = \ell\alpha^{\ell-1}$. In particular $u_p = 0 \bmod p$.

Let $c = 5p$ for even t, and $c = 15p$ for odd t. Set

$$\begin{gathered} L_j = c(4 \cdot 2^{2^j} + 2), \quad M_j = b_{L_j/2}/q_A, \quad d_j = M_j/p, \quad a_j = j/\sqrt{L_j}, \\ f(z) = 2\sum_{j\geq 1} a_j \cos(2\pi d_j z_1). \end{gathered} \tag{Ret.U21}$$

All L_j belong to (Ret.U1). The divisibility $b_u \mid b_{u\ell}$, which follows from the same recurrence identity, and (Ret.U20) show that $p^2 \mid M_j$; division by the power of two q_A does not remove this factor. The series defines one fixed real even continuous function. Since $\log d_j \asymp_A L_j$ and the L_j grow doubly exponentially, $\mathcal{L}_\beta(f) < \infty$ for all $\beta < 1/2$. Different positive multiples of $(1,0)$ belong to different A-orbits, so its classical variance is $2\sum_j a_j^2 > 0$.

At $M = M_j = p^2 t_j$, put $d = M/p$ and use the ordinary clock and shift operators

$$C|x\rangle = e^{2\pi i x/p}|x\rangle, \qquad S|x\rangle = |x+d\rangle.$$

They commute, both have order p, and all p^2 joint characters have equal multiplicity M/p^2, by Weyl trace orthogonality. The integer t_j is odd. For every integer vector v,

$$\mathcal{W}_M^*(dv) = (-1)^{q_*(v)} C^{v_1} S^{v_2}.$$

Every $A^\ell(1,0)$ is primitive, hence nonzero modulo two, so its sign here is minus one. Define the Hermitian contraction

$$H_j = \frac{1}{h}\sum_{\ell=0}^{h-1} \operatorname{Re}\left(C^{(A^\ell(1,0))_1} S^{(A^\ell(1,0))_2}\right). \tag{Ret.U22}$$

Its full trace is zero because all these frequencies are nonzero modulo p, and $H_j = 1$ on the joint $C = S = 1$ subspace. Since $h \mid L_j$, the contribution of the jth mode to $X_{L_j}(f)$ is exactly $-2jH_j$.

By (Ret.U18), every earlier frequency has the full L_j modular period for all sufficiently large j. Their contribution Y_j has zero trace and uniformly bounded normalized Hilbert–Schmidt norm. The later modes contribute E_j with $\|E_j\|_{\mathrm{op}} \to 0$: the ratio a_{j+1}/a_j is eventually at most one half, and $\sqrt{L_j}a_{j+1} = (j+1)\sqrt{L_j/L_{j+1}} \to 0$. Thus the centered operator is $-2jH_j + D_j$ with $\sup_j \|D_j\|_{2,M_j} < \infty$.

Intersect the $H_j = 1$ subspace with the spectral subspace $|D_j| \leq j$. The intersection has dimension at least $M_j/p^2 - M_j\|D_j\|_{2,M_j}^2/j^2$. Its quadratic form is at most $-j$, so min-max gives

$$\frac{1}{M_j}\#\{\text{eigenvalues of the centered } X_{L_j}(f) \text{ at most } -j\} \geq \frac{1}{p^2} - \frac{C_{A,f}}{j^2}. \tag{Ret.U23}$$

The resulting sequence is not tight.

For the odd block, a basis of the joint vacuum is $p^{-1/2}\sum_{\ell=0}^{p-1}|pa+\ell d\rangle$, $a\in\mathbb{Z}/t_j\mathbb{Z}$. Parity sends a to $-a$, giving odd dimension $(t_j-1)/2$ and asymptotic relative dimension $1/p^2$. The odd trace of H_j is $O(M_j^{-1})$. Subtracting this mean costs $O(j/M_j)=o(1)$; the remaining centered perturbation still has bounded odd Hilbert–Schmidt norm. The same min-max argument, with threshold $-j/2$, proves the odd version. This completes Theorem 30.3.

30.6 Removing spatial analyticity: every finite Gevrey order

The linear charge penalty in (Ret.R15) is unnecessarily costly for nonanalytic symbols. The interaction is a unit-modulus phase, and its difference from one is bounded by two. Retaining this saturation permits the stated stretched-exponential Fourier weights.

Write $l(\boldsymbol{v})=\sum_\nu\|v_\nu\|_1$ and $L(\boldsymbol{v})=1+\log(1+l(\boldsymbol{v}))$ for a nonvacuum clean particle. By (Ret.U16), its Mayer interaction with another particle satisfies, away from same-site exclusion,

$$|e^{i\theta_d(\boldsymbol{v},\boldsymbol{w})}-1|\le\min\{2,C_A\lambda^{-|d-m|}l(\boldsymbol{v})l(\boldsymbol{w})\}.$$

Split the geometric sum where its argument crosses two to get

$$\sum_d|e^{i\theta_d(\boldsymbol{v},\boldsymbol{w})}-1|\le C_AL(\boldsymbol{v})L(\boldsymbol{w}). \tag{Ret.U24}$$

Indeed the left side is bounded by $C_A[1+\log(1+l(\boldsymbol{v})l(\boldsymbol{w}))]$, and this is at most a constant times the displayed product. The same-site exclusion adds at most one and obeys the same bound.

For any fixed $\beta>0$ the absolute clean activity has moment

$$M_\beta=\sum_{\boldsymbol{v}\neq0}|w(\boldsymbol{v})|L(\boldsymbol{v})e^{\beta L(\boldsymbol{v})}\le C_{b,\beta}s^2. \tag{Ret.U25}$$

Only a fixed polynomial Fourier moment is required here. To verify this, bound the weight by $C_\beta(1+l)^{\beta+1}$ and use $1+\sum_\nu\|v_\nu\|\le\prod_\nu(1+\|v_\nu\|)$. Each nonvacuum clean profile has at least two nonzero replica entries. The weighted sum of one such entry is $O_b(|\epsilon|)$; the binomial factors are absorbed by $r|\epsilon|=s$. Vacuum normalization contributes at most $e^{C_br\epsilon^2}$. Gevrey symbols have all the polynomial moments used here.

Absolute activities are identical at all available clean sites: the onsite factor and the external fields only change their phase. It is therefore valid to sum the spatial kernel in (Ret.U24) before summing the second particle's mark. For the rooted tree criterion choose the size function $a_s(\boldsymbol{v})=Ds^2L(\boldsymbol{v})$. For a sufficiently large fixed D and then small s_0, $Ds^2\le\beta$ and (Ret.U24)–(Ret.U25) give

$$\sum_{\text{sites, marks }\boldsymbol{w}}|w(\boldsymbol{w})|\,|e^{i\theta}-1|\,e^{a_s(\boldsymbol{w})}\le a_s(\boldsymbol{v}).$$

The tree identity already proved in the phase-space argument applies. Its connected logarithm after the one-particle term is bounded by

$$K\sum_{\boldsymbol{v}}|w(\boldsymbol{v})|(e^{Ds^2L(\boldsymbol{v})}-1)\le C_{A,b}Ks^4. \tag{Ret.U26}$$

The s^2 in the chosen tree size is essential to this fourth-order remainder; using only a fixed size βL would be insufficient.

The dirty-site estimate improves as well. For a real smooth b, $\|b(\cdot+h)-b\|_2\le C_b\min(1,\|h\|)$. The field from a dirty site i at a clean site j is bounded by $C_A\lambda^{-|\delta_{ij}-m|}l_i$, where $\delta_{ij}=(j-i \bmod K)\in\{0,\dots,K-1\}$. The inequality $\min(1,\sum_i x_i)\le\sum_i\min(1,x_i)$ and the same capped geometric summation yield

$$\sum_{\text{clean }j}\max_\nu\min(1,\|h_{\nu j}\|)\le C_A\sum_{\text{dirty }i}L_i.$$

The variance lower bound used in (Ret.R13) now gives the conditional partition estimate

$$|\mathcal{Q}_{B,\boldsymbol{h}}| \le \exp\left[-\tfrac{1}{2}\sigma_A^2(b)Ks^2 + C_{A,b}K(s^3+s^2/N) + C_{A,b}s^2\sum_{i\in B}L_i\right]. \tag{Ret.U27}$$

The deleted-site cost is included because $L_i \ge 1$. This logarithmic profile penalty replaces the linear-mass penalty in (Ret.R15), uniformly in the replica count.

It can be absorbed into the positive atom majorant

$$B_s(z) = \sum_{v\neq 0} |\widehat{b}_v|(1+\|v\|_1)^{Ds^2}e_v(z), \qquad a_c(s) = e^{Ds^2}[e^{sB_s}]_c, \quad c\neq 0, \tag{Ret.U28}$$

after increasing D if necessary. The log of one plus the total profile mass is at most the sum of the logs of one plus the original atom norms, including atoms that cancel within a replica. The remaining constant per dirty site is the factor outside the exponential.

For $0<\alpha\le 1$, the weight $e^{\rho\|v\|^\alpha}$ is submultiplicative on frequency sums. Polynomial weights in (Ret.U28) can be absorbed by decreasing ρ. Thus $a_c(s)$ is normally analytic on a fixed complex disk, has nonnegative Taylor coefficients, and obeys

$$a_c(s) \le C_{A,b}se^{-\rho'\|c\|_1^\alpha} \quad (0\le s\le s_*). \tag{Ret.U29}$$

The corresponding full lattice bound is also available. Subadditivity of the power α applied to the Green series in Section 3 gives

$$\sum_j \|p_j\|_1^\alpha \le C_{A,\alpha}\sum_j \|c_j\|_1^\alpha.$$

Raising the homogeneous endpoint inequality to this power and counting endpoints at most twice yields

$$\sum_j \|c_j\|_1^\alpha \ge \gamma_{A,\alpha}\sum_{\text{gaps}} \lambda^{\alpha(d-1)/2}. \tag{Ret.U30}$$

Split each stretched-exponential forcing weight in half. The vector sum $\sum_{v\neq 0} e^{-\rho'\|v\|^\alpha/2}$ is finite, and the gap series with weights $\exp[-\rho'\gamma_{A,\alpha}\lambda^{\alpha(d-1)/2}/2]$ is entire. The finite-component and winding enumeration therefore gives (Ret.U15) with $\sum\|c_j\|^\alpha$ in the exponent. All arithmetic classes remain included.

Finally the minimal-orbit transfer construction preserves a positive stretched-exponential Fourier weight with the same α: the logarithmic number of intervening orbit sites is absorbed by a decrease of ρ. Thus b, g have the assumed regularity. Combine (Ret.U27)–(Ret.U30), the unchanged vanishing of the degree-two component coefficient, and the clean estimate (Ret.U26). This recovers (Ret.R18)–(Ret.R19), with exponential error in K, and proves (Ret.U3)–(Ret.U5) for every $0<\alpha\le 1$.

The conclusion covers every fixed finite Gevrey order $1/\alpha$. It does not automatically cover all smooth symbols: smoothness alone does not imply any stretched-exponential Fourier bound. No constant is asserted to remain bounded as $\alpha\downarrow 0$.

For example, the same nonanalytic function

$$f(z) = 2\sum_{n\ge 1} e^{-\sqrt{n}}\cos(2\pi n z_1), \qquad \sigma_A^2(f) = 2\sum_{n\ge 1} e^{-2\sqrt{n}} \tag{Ret.U31}$$

works for every hyperbolic A. Its Fourier coefficients satisfy the assumption with $\alpha = 1/2$ and any $0<\rho<1$, but with no analytic exponential weight. Distinct multiples of $(1,0)$ belong to different A-orbits, which gives the displayed variance.

30.7 Verification and the remaining boundary

Run the companion verifier from the repository root with NumPy available:

The exact verifier invocation is reproduced as listing S2 in the Supplement.

The verification program passed on 2026-09-12. It enumerates 72 hyperbolic unimodular matrices with entries between minus four and four. Their reductions cover all six matrices in $SL(2, \mathbb{F}_2)$. The 200 retained dimension cases include positive and negative, odd and even traces; 16 occurrences of $N = 1$ are explicitly excluded. There are 800 exact cyclic-preimage checks and 3,200 exact Weyl-phase checks. Prime-square lifting is checked at all 28 integer traces $3 \le |t| \le 16$, and eight large-modulus cases check the centered counterexample's own and earlier periods.

Another 72 rational-arithmetic checks sum an infinite dyadic envelope of the capped interaction kernel exactly, including particle masses as large as 2^{200}. They test the logarithmic bound; the tree criterion, Gevrey quasi-norm bound, and limiting theorems are proved analytically above.

Independent physical intertwiners are constructed at $N = 3, 15, 35, 323$. Unitarity, exact-Egorov targets, parity and the normalized return period are tested separately from integer arithmetic. The largest double-precision matrix residual is below $1.4 \cdot 10^{-10}$. These are numerical diagnostics, not interval-certified matrix identities.

Limited independent internal review checked the arithmetic subsequence, universal refinement, composite-modulus construction, negative traces, the transfer of the normal-form estimates, and the centered prime-square counterexample. A separate pass checked the capped interaction and conditional-field estimates used in Section 6. The review required explicit exclusion of $N = 1$, retention of the negative central sign, and the $a_s = Ds^2L$ tree size for the fourth-order connected remainder. The final proofs include these points.

Generality over A means every fixed hyperbolic element of $SL(2, \mathbb{Z})$ on the specified subsequence; it does not mean all dimensions or all nonlinear chaotic systems. Generality over Gevrey order means every fixed $0 < \alpha \le 1$, with constants that may deteriorate as α tends to zero. The proof does not extend by taking that endpoint. Theorem 30.2 has a lower regularity requirement because it concerns averaged observables rather than the actual kicked propagator.

For the BGS programme the remaining target is still local, desymmetrized spectral correlation at Heisenberg times for a fixed nonzero classical perturbation. Here K is logarithmic in N, the replica amplitude obeys $r|\epsilon| \le s_0$, and the classical action parameter is $\kappa = \epsilon/(2\pi N)$. The estimates above do not supply that missing comparison. External priority has not been established. These proofs are included in the canonical manuscript through a generated TeX chapter; the analytic limit theorems are not Lean-formalized.

31 Short quantum returns, intensity loss, and local spectral limits

Numbering convention: section and theorem references within this chapter use the source module's local numbering; source Section 1 is subsection 31.1. Equation references carry the prefix Ret.

Status: project proof with adversarial checking by the author. Finite computations are separate evidence. The positivity mechanism has close published antecedents; priority is unverified. Cat-map applications depend on the existing return-law proofs, whose limited internal-review status is unchanged. BGS remains open.

31.1 Local processes and a uniform necessary condition

Let U_D be a deterministic or random unitary of dimension $D \to \infty$. Write its eigenphases, with multiplicities, as $\theta_j \in \mathbb{R}/\mathbb{Z}$ and $t_D(m) = D^{-1} \operatorname{Tr} U_D^m$. For an independent uniform circle origin X_D, define the locally finite counting measure

$$\Xi_D = \sum_{j=1}^{D} \sum_{\ell \in \mathbb{Z}} \delta_{D(\theta_j - X_D + \ell)}. \tag{Ret.S1}$$

This is stationary under real translations and has intensity one at each finite dimension. Local-process convergence means convergence in law in the vague topology, allowing multiplicities. Ergodicity means that every event invariant under all real translations has probability zero or one.

Theorem 31.1. *If Ξ_D converges to a stationary ergodic process Ξ of intensity one, then for every integer $M_D \geq 1$ with $M_D = o(D)$,*

$$\mathbb{E} \max_{1 \leq m \leq M_D} |t_D(m)| \longrightarrow 0. \tag{Ret.S2}$$

Consequently, for any $1 \leq K_D = o(D)$ the empirical eigenvalue law of $U_D^{K_D}$ tends to Haar measure in probability. For deterministic spectra this is ordinary weak convergence. No second-moment or pair-correlation convergence hypothesis is imposed.

Proof. Choose a measurable maximizing integer $q_D \in [1, M_D]$, breaking ties by the smallest index. It may depend on the spectrum. Put $\zeta_D = \overline{t_D(q_D)}/|t_D(q_D)|$ when the denominator is nonzero, and $\zeta_D = 1$ otherwise. Introduce the mark

$$Z_D = \zeta_D e^{2\pi i q_D X_D}. \tag{Ret.S3}$$

Conditionally on the spectrum this mark is Haar on the unit circle. Joint tightness gives a subsequential limit (Ξ, Z) along every subsequence. Replacing X_D by $X_D + a/D$ translates Ξ_D by $-a$ and multiplies its mark by $e^{2\pi i q_D a/D}$. The latter tends uniformly to one. The joint limit is thus invariant under translations of Ξ leaving Z fixed.

Such an invariant coupling with an ergodic first marginal is a product coupling. Indeed, for any bounded real $b(Z)$ the conditional expectation $\mathbb{E}[b(Z) \mid \Xi]$ is invariant under translations, hence constant almost surely. A countable determining family of b proves independence. In particular $\mathbb{E}[Z\langle \Xi, \psi \rangle] = 0$ for integrable local linear statistics, since Z is Haar.

Choose $\psi \geq 0$ continuous and compactly supported with $\int \psi > 0$, and use $\widehat{\psi}(\xi) = \int \psi(y) e^{-2\pi i \xi y}\, dy$. Both $\mathbb{E}\langle \Xi_D, \psi \rangle$ and $\mathbb{E}\langle \Xi, \psi \rangle$ equal $\int \psi$. Nonnegative random variables converging in law with convergent means are uniformly integrable: apply convergence to $y \wedge R$, subtract from the means to control $(y - R)_+$, and then let $R \to \infty$. This permits passage to the marked first moment; no corresponding assertion about squared local counts is used.

An exact change of variables in the circle-origin integral gives

$$\mathbb{E}[Z_D \langle \Xi_D, \psi \rangle \mid U_D] = |t_D(q_D)| \widehat{\psi}(q_D/D). \tag{Ret.S4}$$

The substitution is $y = D(\theta_j - X_D + \ell)$; integer q_D makes the winding phase one. Since $q_D/D \to 0$ uniformly, the expectation of the right side is $(\int \psi)\mathbb{E}|t_D(q_D)| + o(1)$, whereas the left side tends to zero. Every subsequence has such a further subsequence, proving (Ret.S2). Apply (Ret.S2) to each fixed multiple of K_D and approximate continuous circle functions by trigonometric polynomials to obtain the Haar conclusion.

Strengthening (all slowly varying tests). Under the hypotheses of Theorem 31.1, let

$$\mathfrak{d}_M(\nu, dx) = \sup_{\substack{f : \mathbb{R}/\mathbb{Z} \to \mathbb{R} \\ \|f\|_\infty \leq 1,\ \operatorname{Lip}(f) \leq M}} \left| \int f\, d\nu - \int f\, dx \right|.$$

For the empirical phase measure $\nu_D = D^{-1}\sum_j \delta_{\theta_j}$,

$$\mathbb{E}\mathfrak{d}_{M_D}(\nu_D, dx) \to 0 \qquad (M_D = o(D),\ M_D \geq 1). \tag{Ret.S2a}$$

Choose a spectrum-dependent real test f_D within D^{-1} of the supremum, with sign giving a nonnegative discrepancy. Such a measurable choice exists by taking the first suitable member of a countable dense subset of the compact Lipschitz ball. Replace (Ret.S3) by the bounded real mark $Z_D = f_D(X_D) - \int f_D \in [-2, 2]$. It has mean zero conditionally on the spectrum and changes by at most $M_D|a|/D$ under a local translation. The same invariant-coupling and uniform-integrability proof applies. The replacement for (Ret.S4) is

$$\mathbb{E}[Z_D\langle \Xi_D, \psi\rangle \mid U_D] = \left(\int \psi\right)\left(\int f_D\, d\nu_D - \int f_D\right) + E_D, \qquad |E_D| \leq \frac{M_D}{D}\int |y|\psi(y)\, dy.$$

The left side tends to zero, proving (Ret.S2a).

In particular, for every $0 < w_D \leq 1$ with $Dw_D \to \infty$, the mesoscopic empirical density $\rho_{D,w}(x) = w^{-1}\nu_D([x, x+w])$ satisfies

$$\mathbb{E}\|\rho_{D,w_D} - 1\|_{L^1} \to 0. \tag{Ret.S2b}$$

Indeed, duality against any real a with $|a| \leq 1$ converts this density error to a phase-measure discrepancy tested by $f_a(\theta) = w^{-1}\int_{\theta-w}^{\theta} a(x)dx$. This test is bounded by one and has Lipschitz constant at most $2/w$. Apply (Ret.S2a) with $M_D = 2/w_D$. Thus an ergodic intensity-one local limit forces uniform density in spatial L^1 on every window containing a diverging number of mean spacings. No rate of mixing or quantitative rate of local-process convergence was assumed.

31.2 The conditional intensity encoded by a return law

Suppose the spectra are deterministic, $K_D = o(D)$, and their K_D-folded laws converge to a probability measure μ. Suppose a local-process subsequence converges with intensity one. Attach the unrotated mark $Y_D = K_D X_D \bmod 1$. Every joint limit (Ξ, Y) is invariant under translating Ξ with Y fixed, and Y is uniform. The same first-moment argument for all Fourier characters shows that μ must be absolutely continuous and

$$\mathbb{E}[\langle \Xi, \psi\rangle \mid Y = y] = \rho(y)\int \psi, \qquad d\mu(y) = \rho(y)\, dy. \tag{Ret.S5}$$

To verify the absolute continuity, the measure $B \mapsto \mathbb{E}[\mathbf{1}_{Y\in B}\langle \Xi, \psi\rangle]$ is absolutely continuous with respect to the uniform law of Y. Its Fourier coefficients equal those of $(\int \psi)\mu$ by (Ret.S4) without phase alignment. Uniqueness of Fourier coefficients proves both assertions. A nonconstant return density therefore forces nonergodicity of any local limit preserving intensity one. A singular return law precludes such an intensity-preserving limit. No conditional spacing law is identified by (Ret.S5).

31.2.1 Intensity loss: a sharp compatibility theorem

Theorem 31.2. *Let $1 \leq K_D = o(D)$ be integers, and suppose that the empirical law of $U_D^{K_D}$ converges weakly in probability to a deterministic probability measure μ on the circle. Write its Lebesgue decomposition as $\mu = \rho(y)\, dy + \mu_{\mathrm{s}}$. Let Ξ be any subsequential local-process limit; intensity preservation and ergodicity are not assumed. There is a joint limit with the uniform mark $Y_D = K_D X_D \bmod 1$ whose conditional first moment has the form*

$$\mathbb{E}[\langle \Xi, \psi\rangle \mid Y = y] = h(y)\int \psi, \qquad 0 \leq h(y) \leq \rho(y) \quad a.e. \tag{Ret.S19}$$

for every continuous compactly supported ψ. In particular, writing α for the intensity of Ξ,

$$\alpha \le \int \rho\, dy, \qquad \mathbb{P}\{\Xi = 0\} \ge |\{y : \rho(y) = 0\}|. \tag{Ret.S20}$$

If Ξ is ergodic, the stronger bound is

$$\alpha \le \operatorname{ess\,inf} \rho. \tag{Ret.S21}$$

Thus a purely singular return law forces convergence of the entire local process sequence to the empty process, not just failure of intensity one.

Proof. Finite processes have uniformly bounded expected mass on every compact set, so their laws are tight in the vague topology of locally finite counting measures. The compact circle mark preserves joint tightness. As in Section 1, any joint limit is invariant under translating the point process with the mark fixed. Its joint first-moment measure on $\mathbb{T} \times \mathbb{R}$ is locally finite: truncation and weak convergence give $\mathbb{E}\langle \Xi, \psi\rangle \le \int \psi$ for $\psi \ge 0$. Translation invariance makes this measure $\eta(dy)\, dx$. The uniform marginal of Y implies $\eta \ll dy$; write $\eta = h(y)dy$. This proves the conditional first-moment formula without assuming that moments converge.

For nonnegative continuous b on the circle and $\psi \ge 0$, the exact finite identity is

$$\mathbb{E}[b(Y_D)\langle \Xi_D, \psi\rangle \mid U_D] = \frac{1}{D} \sum_j \int \psi(z) b(K_D\theta_j - K_D z/D)\, dz.$$

Uniform continuity of b, $K_D/D \to 0$, and the assumed folded-law convergence show that its expectation tends to $(\int \psi) \int b\, d\mu$. On the other hand, apply weak convergence to the bounded continuous function $b(Y_D)\min(\langle \Xi_D, \psi\rangle, R)$, then increase R. It follows that $\int bh\, dy \le \int b\, d\mu$. Thus $h(y)dy \le \mu$, equivalently $h \le \rho$ a.e.; the singular part cannot supply a conditional intensity at a uniform circle mark. This is an inequality, not the unjustified equality obtained by exchanging unbounded first moments and a weak limit.

Integrating proves the first bound in (Ret.S20). Where $\rho = 0$, every nonnegative compact local count has conditional expectation zero; a countable family of windows shows that the whole conditional process is empty there. If the first marginal is ergodic, the invariant coupling is a product by the argument of Section 1, so $h(y) = \alpha$ a.e. and (Ret.S21) follows. For purely singular μ, every subsequential limit has intensity zero and is empty almost surely; tightness upgrades this to convergence of the sequence.

There is also a version requiring no specified return law. If $\Xi_D \Rightarrow \Xi$ with ergodic intensity $\alpha \in [0, 1]$, then

$$\limsup_D \mathbb{E} \max_{1 \le m \le M_D} |t_D(m)| \le 1 - \alpha \qquad (M_D = o(D)). \tag{Ret.S22}$$

Use the aligned mark (Ret.S3) and put $N_D = \langle \Xi_D, \psi\rangle$, $I = \int \psi > 0$. Independence in the joint limit gives $\mathbb{E}[Z_D \min(N_D, R)] \to 0$ for fixed R. The remaining modulus is at most $\mathbb{E}(N_D - R)_+ = I - \mathbb{E}\min(N_D, R)$. First take the upper limit in D, then $R \to \infty$; the bound is $(1-\alpha)I$. Equation (Ret.S4) proves (Ret.S22). This explicitly quantifies the first-moment loss that Theorem 31.1 excludes.

Sharpness among abstract spectra. For every probability measure μ, every $1 \le K_D = o(D)$, and every $0 \le \alpha \le \operatorname{ess\,inf} \rho$, there are deterministic simple D-point spectra with folded law tending to μ and ergodic local limit of intensity α. For $\alpha > 0$ the limit can be the uniformly shifted lattice of spacing $1/\alpha$; for $\alpha = 0$ it is empty. A random construction can instead give the Poisson process of intensity α. These are abstract spectra, not new cat-map quantizations.

Here is an explicit construction and its limiting proof. If $\alpha = 1$, then $\mu = dy$ and a D-point clock (or independent uniform points) suffices. Otherwise put $n_D = \lfloor \alpha D \rfloor$, $m_D = D - n_D$, and $\sigma = (\mu - \alpha\, dy)/(1 - \alpha)$. Choose $B_D = \lfloor \sqrt{m_D/K_D} \rfloor$ for all sufficiently large D. Then $B_D \to \infty$, $J_D = K_D B_D = o(D)$, and $J_D/m_D \to 0$. Choose points $y_1, \dots, y_{B_D}$ whose equally weighted

empirical laws tend to σ; quantiles of a representative on $[0,1)$ suffice. Place clouds at the J_D circle centers

$$(b+y_j)/K_D, \qquad 0 \le b < K_D,\ 1 \le j \le B_D.$$

Distribute m_D levels among them with occupancies differing by at most one. Together with an n_D-point clock these give D levels. Choose all cloud levels distinct, and different from clock levels, within circle distance D^{-3} of their centers. Such a finite choice always exists.

Folding the clock gives Haar in the limit when $\alpha > 0$: its distinct folded grid has $n_D/\gcd(n_D, K_D) \ge n_D/K_D \to \infty$ sites. The cloud folded law tends to σ, since occupancy rounding has total variation error at most J_D/m_D and folded displacements are at most $K_D D^{-3}$. Their weights tend to α and $1-\alpha$, giving μ. For a local window of length L the probability of hitting any cloud is at most

$$J_D(L/D + 2D^{-3}) \longrightarrow 0. \tag{Ret.S23}$$

The remaining clock converges to the stated ergodic lattice, or is absent when $\alpha = 0$. Replacing the clock by n_D independent uniform circle points gives Poisson intensity α by the elementary multinomial rare-event limit; their folded empirical measure tends to Haar in probability. When $K_D \to \infty$, the full phase laws in either construction also tend to Haar: averaging a continuous test over the K_D translates is a uniformly convergent Riemann sum. Thus the example need not violate macroscopic equidistribution.

31.3 A finite positive-Fourier pair inequality

For Schwartz ϕ define the ordered factorial statistic

$$C_D(U;\phi) = D^{-1} \sum_{j \ne l} \sum_{n \in \mathbb{Z}} \phi(D(\theta_j - \theta_l + n)). \tag{Ret.S6}$$

Put $\phi_L(x) = e^{-\pi x^2/L^2}$ for $L > 0$. Poisson summation gives exactly

$$C_D(U;\phi_L) = L \sum_{m \in \mathbb{Z}} e^{-\pi L^2 m^2/D^2} |t_D(m)|^2 - \sum_{n \in \mathbb{Z}} e^{-\pi D^2 n^2/L^2}. \tag{Ret.S7}$$

All sums converge absolutely. For any distinct positive integers $q_1, \dots, q_R$,

$$C_D(U;\phi_L) \ge L \left(1 + 2 \sum_{r=1}^{R} e^{-\pi L^2 q_r^2/D^2} |t_D(q_r)|^2 \right) - \sum_{n \in \mathbb{Z}} e^{-\pi D^2 n^2/L^2}. \tag{Ret.S8}$$

The subtracted sum is the diagonal term; at fixed L it tends to one. Dropping that term is false even in dimension one. Positivity also permits any probability average over spectra.

If $K_D = o(D)$ and $t_D(rK_D) \to a_r$ for every fixed r, first taking the limit at finite R, then increasing R, gives

$$\liminf_D C_D(U;\phi_L) \ge L \left(1 + 2 \sum_{r \ge 1} |a_r|^2 \right) - 1. \tag{Ret.S9}$$

An infinite sum means divergence to positive infinity. There is no exchange of an uncontrolled infinite sum and a limit. If the folded law has an L^2 density ρ, the parenthesis is $\int \rho^2$ by Parseval. For random spectra, L^2 trace convergence implies the averaged version; convergence in probability implies each finite-R lower bound in probability.

Corollary 31.3. *If $C_D(U;\phi) \to \int \phi\rho_2$ for all Schwartz tests, where ρ_2 is bounded and $\rho_2(x) \to 1$ at infinity, then every sublinear-frequency maximum in (Ret.S2) tends to zero for deterministic spectra. Indeed, a retained modulus $a > 0$ would imply $\int \phi_L \rho_2 \ge L(1+2a^2) - 1$ for every fixed L. But dominated convergence after $x = Ly$ gives $L^{-1} \int \phi_L \rho_2 \to 1$ as $L \to \infty$. This is a separate argument from Theorem 31.1; weak process convergence was not silently promoted to convergence of pair moments.*

For the Poisson target $C_{\mathrm{P}}(\phi_L) = L$. For the sine-kernel target $\rho_{2,\mathrm{sine}}(x) = 1-(\sin(\pi x)/(\pi x))^2$,

$$C_{\mathrm{sine}}(\phi_L) = L - \operatorname{erf}(\sqrt{\pi}L) + \frac{1-e^{-\pi L^2}}{\pi L}, \qquad 0 \le C_{\mathrm{sine}}(\phi_L) - L + 1 \le \frac{1}{\pi L}. \tag{Ret.S10}$$

The transform of the squared sinc is $(1-|\xi|)_+$; integration gives the formula. The bound follows from $\operatorname{erfc}(x) \le e^{-x^2}/(\sqrt{\pi}x)$, obtained by bounding $\int_x^\infty e^{-u^2}du$ by $x^{-1}\int_x^\infty ue^{-u^2}du$.

Under convergence to these respective Gaussian pair targets, (Ret.S8) implies, for every fixed $\delta > 0$,

$$\limsup_D \sum_{1\le m\le \delta D} |t_D(m)|^2 \le \begin{cases} \sqrt{\pi e/2}\,\delta, & \text{Poisson,} \\ (e/2)\,\delta^2, & \text{sine kernel.} \end{cases} \tag{Ret.S11}$$

For Poisson minimize $e^{\pi L^2\delta^2}/(2L)$ at $L^2 = (2\pi\delta^2)^{-1}$. For the sine kernel minimize $e^{\pi L^2\delta^2}/(2\pi L^2)$ at $L^2 = (\pi\delta^2)^{-1}$. The same argument applies to averaged pair convergence and squared traces. These are necessary budgets, not sufficient criteria.

31.4 Consequences for the cat-map programme

Use the hypotheses and convention of the general return theorem: a fixed hyperbolic matrix, a fixed real mean-zero Gevrey symbol, and its explicit arithmetic subsequence with $K \asymp \log N$. Take $D = N$, or $D = (N-1)/2$ on the odd space for an even symbol. These are the stated sectors, not an assertion that every possible arithmetic symmetry has been removed.

For $\epsilon_K \to 0$, positive $\sigma_A^2(f)$, and $K\epsilon_K^2 \to v < \infty$, the existing return theorem gives

$$t_D(rK) \longrightarrow e^{-r^2 v\sigma_A^2(f)/2} \quad (r \text{ fixed}). \tag{Ret.S12}$$

Its small-amplitude condition holds eventually for each fixed r. Thus Theorem 31.1 excludes any ergodic stationary intensity-one local limit. For $v > 0$, (Ret.S9) gives the explicit pair obstruction

$$\liminf C_D(U_{\epsilon_K,f};\phi_L) \ge L\Theta(v\sigma_A^2(f)) - 1, \qquad \Theta(a) = \sum_{r\in\mathbb{Z}} e^{-ar^2} > 1. \tag{Ret.S13}$$

Already $L > e^{v\sigma_A^2(f)}/2$ separates the bound using only $r = 1$ from both Poisson and sine-kernel pair targets. A necessary condition for either an ergodic intensity-one local limit or the decorrelating pair limits in Corollary 31.3 is therefore

$$K\epsilon_K^2 \longrightarrow \infty. \tag{Ret.S14}$$

Otherwise take a subsequence with a finite limit v and apply (Ret.S12). With $\hbar_N = (2\pi N)^{-1}$ and $\kappa_K = \hbar_N\epsilon_K$, this is $|\kappa_K|N\sqrt{K} \to \infty$ within the stated shrinking-quantum-kick regime. It is not a sufficient threshold or a theorem about fixed nonzero classical strength.

For $v = 0$, all return moments tend to one and (Ret.S9) diverges for each L. In fact the local process tends to the empty process. Fix $0 < \eta < 1/2$ and let $b_D(\eta)$ be the fraction of folded phases outside distance η from zero. The probability that a window of unfolded length L meets a level is bounded by

$$2\eta + KL/D + Lb_D(\eta). \tag{Ret.S15}$$

Good levels lie in K arcs of total length 2η; enlarge them by the window length. Bad levels contribute at most their expected window count. Since the folded law tends to a point mass, first let $D \to \infty$, then $\eta \downarrow 0$. The finite mean density one is lost in this weak limit. For zero $\sigma_A^2(f)$ and $\epsilon_K \to 0$, the existing operator-norm return estimate gives the same point-mass return and empty local limit.

In the sampled-harmonic family of pell_harmonic_returns.md, the L^2 limit $t_D(K) \to e^{-\tau^2/2}$ holds for every $\beta \in [0,1/2]$ and fixed τ. The harmonic-averaged local process therefore has

no ergodic intensity-one limit. Averaged Poisson and sine-kernel pair limits are excluded; a fixed $L > e^{\tau^2}/2$ also gives a positive pair-test gap in probability. This includes the free-unitary-Brownian return endpoint.

If $\|\widetilde{U}_D - U_D\|_{\mathrm{op}} = \eta_D$ and $K_D\eta_D \to 0$, telescoping unitary powers gives, for every fixed positive integer r,

$$|D^{-1}\operatorname{Tr}\widetilde{U}_D^{rK_D} - t_D(rK_D)| \le rK_D\eta_D \longrightarrow 0. \tag{Ret.S16}$$

The exclusions persist under these $o(1/K_D)$ perturbations, which may be larger than the mean spacing. Equality of classical principal maps alone is not asserted to imply the required norm estimate.

31.4.1 A support gap forces an empty ergodic limit

In the harmonic family, (Ret.S22) and its first return give the bound $\alpha \le 1 - e^{-t/2}$ for every ergodic harmonic-averaged local limit, where $t = \tau^2$ and any $\beta \in [0, 1/2]$ is allowed. This conclusion does not assume intensity preservation.

At $\beta = 1/2$, the full return law is the known free unitary Brownian law. For $0 < t < 4$ it is supported on $|\vartheta| \le b(t)$ in radian coordinates, where

$$b(t) = \tfrac{1}{2}\sqrt{t(4-t)} + \arccos(1 - t/2) < \pi.$$

The support statement and moment normalization are verified in Demni–Hmidi, Sections 1, 4 and 5, as recorded in the source comparison. Theorem 31.2 therefore implies, for every harmonic-averaged subsequential local limit,

$$\mathbb{P}\{\Xi = 0\} \ge g(t) := 1 - \frac{b(t)}{\pi} > 0, \qquad 0 < t < 4. \tag{Ret.S24}$$

If that limit is ergodic, it is empty almost surely. This is stronger than excluding an intensity-one ergodic limit, but does not identify the nonempty components or assert that the full sequence tends to the empty process. The application retains all assumptions of the project harmonic return theorem, including averaging over the prescribed sampled harmonics.

For example, $g(1) \approx 0.3910$, $g(2) \approx 0.1817$, and $g(3) \approx 0.05767$. Differentiation gives $b'(t) = \frac{1}{2}\sqrt{(4-t)/t}$, hence $g(4-s) = s^{3/2}/(6\pi) + O(s^{5/2})$ as $s \downarrow 0$. These describe a lower bound on the empty component, not its exact weight. The strict support-gap argument makes no claim at $t = 4$.

31.5 Sharp boundaries and unfolding

A D-point clock has $t_D(D) = 1$, while its local process is the uniformly shifted integer lattice: stationary, ergodic under real translations, and intensity one. Thus $o(D)$ cannot be replaced by $O(D)$ in Theorem 31.1. The clock also has $t_D(m) = 0$ for $1 \le m < D$, and its coprime powers permute the spectrum. Necessary short-return conditions are therefore not sufficient for Poisson or sine-kernel statistics.

Ergodicity is essential. For $D = Kn$, $n \to \infty$, place

$$\theta_{b,j} = K^{-1}(b + j/(2n)), \qquad 0 \le b < K, \quad 0 \le j < n. \tag{Ret.S17}$$

Here K can grow. Each unfolded length-n cell is half filled by a spacing-$1/2$ lattice and half empty. The local limit is an equal mixture of that stationary lattice and the empty process, with intensity one but without ergodicity. Boundary windows have probability $O(L/n)$. The folded law has density $2\mathbf{1}_{[0,1/2]}$, and $|t_D(K)| \to 2/\pi$. The Gaussian pair limit is $\sum_{j\neq 0} e^{-\pi j^2/(4L^2)} = 2L - 1 + O(Le^{-4\pi L^2})$. This realizes (Ret.S5) and the leading density-square bound sharply.

Intensity preservation is also essential: coincident spectra have unit return moments but an empty local limit of intensity zero.

Global mean-spacing unfolding is part of the statement. For $v > 0$ the wrapped Gaussian return density ρ is smooth and strictly positive. Its CDF lift $F(y) = \int_0^y \rho(u)du$ satisfies $F(y+1) = F(y) + 1$, and

$$H_K(\theta) = K^{-1}F(K\theta) \tag{Ret.S18}$$

is an increasing degree-one circle map. Its Lipschitz constant and inverse Lipschitz constant are bounded independently of K. The transformed folded law is Haar, so this density-dependent unfolding removes this particular obstruction. It does not prove local universality after the change.

The effect on the local process can be identified exactly, provided its joint law with the slow phase is known. Write $S_c\xi$ for the point measure obtained by multiplying all locations by $c > 0$. Let ρ be any fixed positive C^1 circle density, $F' = \rho$, $F(0) = 0$, $F(y+1) = F(y) + 1$, and use H_K in (Ret.S18). If $(\Xi_D, K_D X_D \bmod 1) \Rightarrow (\Xi, Y)$ with conditional process laws P_y, then the spectra transformed by H_{K_D} have local limit

$$Q = \int_0^1 \rho(y)(S_{\rho(y)})_* P_y \, dy. \tag{Ret.S25}$$

The weighting factor $\rho(y)$ is indispensable: a uniform origin in the transformed coordinate has pullback density $H_K'(x) = \rho(Kx)$.

Proof. Set the new origin to $H_K(X_D)$. Conditional on this origin, each local displacement $z = D(\theta_j - X_D + \ell)$ is changed to $D[H_K(X_D + z/D) - H_K(X_D)]$. On bounded z sets Taylor's theorem gives

$$|D[H_K(X_D + z/D) - H_K(X_D)] - \rho(KX_D)z| \le \|\rho'\|_\infty K z^2/(2D) = o(1).$$

The positive upper and lower bounds on ρ keep the relevant point locations in a bounded set. Tightness of their counts and uniform continuity of compact test functions identify the limiting dilation. Changing the origin density multiplies expectations of bounded continuous process tests by the bounded continuous mark weight $\rho(Y_D)$. Joint weak convergence proves (Ret.S25), without a pair-moment assumption.

Same returns, different unfolded statistics. Start with any D-point spectra whose local processes tend to an ergodic intensity-one law P. Transform their phases by $H_{K_D}^{-1}$. The new spectra have folded law $\rho(y)dy$, while their local process law tends to

$$\int_0^1 (S_{1/\rho(y)})_* P \, dy. \tag{Ret.S26}$$

Indeed Theorem 31.1 makes the original folded laws Haar, and the inverse CDF sends Haar to $\rho\, dy$. The old local process and its slow uniform mark become independent by ergodicity. Apply the preceding change-of-origin argument with inverse derivative $1/\rho(F^{-1}(v))$; the substitution $v = F(y)$ cancels the origin weight and yields (Ret.S26). Each conditional component has intensity $\rho(y)$, and the total intensity is one. Applying H_{K_D} back recovers the original finite spectra exactly.

The starting spectra may be a clock, independent uniform points, or Haar unitary eigenphases. Their respective local limits are the shifted lattice, Poisson process, and sine process. The latter is the standard CUE kernel limit and is ergodic; its precise source is in the linked source comparison. Consequently the same entire family of fixed return moments, including a prescribed positive wrapped-Gaussian profile, is compatible with all three different locally unfolded laws. This establishes non-identifiability from return laws among abstract spectra. It supplies no dynamical realization of the positive Poisson or sine examples in the maintained cat-map family.

31.6 Prior work, checks, and the remaining gap

Aistleitner, Lachmann and Pausinger, *Pair correlations and equidistribution*, Theorems 1–2, establish the Poissonian implication to equidistribution and a density-square lower bound. Steinerberger, *Poissonian Pair Correlation in Higher Dimensions*, Theorem 2, uses a positive Fourier kernel for quantitative low-frequency exponential-sum bounds. Those principles are prior work. Priority for the marked first-moment formulation here and its specific cat-map applications has not been established.

The sine process satisfies the ergodicity hypothesis by Soshnikov, *Determinantal random point fields*, Theorem 7. Its kernel is the Fourier projection onto $[-1/2, 1/2]$, a translation-invariant, locally trace-class positive contraction in the stated Hermitian setting. The Poisson process is mixing by independence on disjoint bounded sets. Other ensemble targets require their own ergodicity input for Theorem 31.1, or the explicit pair-density hypotheses of Corollary 31.3.

The companion Python verifier checks exact quarter-circle Fourier and marked convolution identities, then independently compares floating Gaussian pair and trace formulas. It also checks the sine integral and scope counterexamples. These finite checks do not prove process convergence or the cat-map return laws. See the scoped research review. The intensity-loss and unfolding extension has a separate proof and computation review. The positive BGS target still requires fixed-classical-strength, Heisenberg-time desymmetrized joint-trace estimates.

Chapter 7 — Floquet, coboundary and smooth resonance fields

32 The arithmetic Floquet normal form and critical recurrence law

Put

$$\alpha = 1 + \sqrt{2}, \quad \lambda = \alpha^2, \quad T = \begin{pmatrix} 3 & 2 \\ 4 & 3 \end{pmatrix}.$$

Let K tend to infinity through odd primes, and define positive integers N,F by

$$N + F\sqrt{2} = \alpha^K, \qquad N^2 - 2F^2 = -1.$$

The moduli N may be composite. On the odd subspace of the standard N-dimensional torus quantization, let U_N be the phase-normalized quantization of T described in Subsection 32.1, and let

$$B_N f(x) = 2\cos(2\pi x/N) f(x), \quad U_{\varepsilon,N} = e^{i\varepsilon B_N} U_{N,\mathrm{Fl}}, \quad d_N = (N-1)/2.$$

Theorem 32.1 (critical resonance law)**.** *For every fixed real τ and every integer m, with $\varepsilon = \tau/\sqrt{2K}$,*

$$\lim_{K \longrightarrow \infty} \frac{1}{d_N} \operatorname{Tr}_- U_{\varepsilon,N}^{mK} = \exp(-m^2\tau^2/2). \tag{FN.A}$$

Consequently the empirical eigenvalue measure of $U_{\varepsilon,N}^K$ converges weakly on the unit circle to the wrapped normal law. For τ nonzero its angular density on [-π,π) is

$$w_\tau(\theta) = \frac{1}{\sqrt{2\pi}|\tau|} \sum_{\ell \in \mathbb{Z}} \exp\left[-\frac{(\theta + 2\pi\ell)^2}{2\tau^2} \right]. \tag{FN.B}$$

For $\tau = 0$ the limit is the point mass at 1. In particular the normalized resonance echo obeys

$$d_N^{-2} \left| \operatorname{Tr}_- U_{\varepsilon,N}^{mK} \right|^2 \longrightarrow e^{-m^2\tau^2}. \tag{FN.C}$$

This is a deterministic limit: there is no parameter average in (FN.A).

The principal finite estimate, proved before any limit is taken, is

$$\left\| V_\varepsilon U_{\varepsilon,N} V_\varepsilon^* - e^{i\varepsilon\overline{B}} U_{N,\mathrm{Fl}} \right\|_{2,-} \le \frac{12\varepsilon^2}{\sqrt{K}}, \qquad K \ge 11, \tag{FN.D}$$

for an explicit unitary V_ε. Here the norm is normalized Hilbert–Schmidt norm on the entire odd subspace. It does not control an individual level or an operator-norm spectral gap. The finite comparison error in (FN.A) is at most $6|m|\tau^2/\sqrt{K}$; an additional Gaussian characteristic-function convergence error is $o(1)$, not assigned that rate.

32.1 Arithmetic and the fixed classical observable

The recurrence $(N_j, F_j) = (N_{j-1} + 2F_{j-1}, N_{j-1} + F_{j-1})$, starting at $(1, 0)$, shows that N is odd. In $\mathbb{Z}/N\mathbb{Z}$, put

$$s = F^{-1}, \qquad s^2 = 2, \qquad h = 3 + 2s, \qquad q = -F.$$

Since $(1+s)^K = N + Fs = 1$, we have $h^K = 1$. Also $h - 1 = 2(1+s)$ is a unit. Modulo every prime $\ell \mid N$, the order of h is precisely K: it divides the prime K and is not one. Hence $1 - h^j$ is a unit for $1 \le j < K$, and $1 + h^j$ is a unit for every j, because K is odd. In particular, $\ell \equiv 1 \pmod{K}$ and $N \equiv 1 \pmod{2K}$. Neither squarefreeness nor primality of N is required.

Use the counting-measure inner product and centered Weyl operators

$$W(a,b)f(x) = e_N(a(x - b/2))f(x - b), \qquad S_u f(x) = f(u^{-1}x), \qquad Rf(x) = f(-x).$$

Divisions in Weyl exponents mean inverses in $\mathbb{Z}/N\mathbb{Z}$. The multiplication and covariance formulas are

$$W(a,b)W(c,d) = e_N((ad - bc)/2)W(a + c, b + d),$$
$$S_u W(a,b) S_u^* = W(u^{-1}a, ub), \qquad RW(v)R = W(-v).$$

Individual Weyl translations need not preserve parity; their symmetrized sums do. On the odd subspace take

$$U_0 = S_h^{-1}, \qquad B = W(q,1) + W(q,1)^*, \qquad U_\varepsilon = e^{i\varepsilon B} U_0. \tag{FN.1}$$

In these coordinates $U_0^K = I$ exactly.

For completeness, this is unitarily equivalent to the fixed-map formulation of Theorem 32.1, including the kick, not merely its compressed observable. Let $C_t f(x) = e_N(tx^2/2)f(x)$, and let F_N have kernel $e_N(xy)/\sqrt{N}$. Set

$$M = \begin{pmatrix} -1/(2s) & 1 \\ -1/2 & -s \end{pmatrix}, \qquad V = C_{s^{-1}} S_{1/2} F_N C_{2s}.$$

Its factors have Weyl covariance matrices respectively

$$\begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}, \quad \begin{pmatrix} u^{-1} & 0 \\ 0 & u \end{pmatrix}, \quad \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

Multiplication gives covariance M, $\det M = 1$, and $T = M\,\mathrm{diag}(h, h^{-1})M^{-1}$. Define $U_{N,\mathrm{Fl}} = VU_0V^*$. The index of $V^*W(1,0)V$ is $M^{-1}(1,0) = (-s, 1/2)$. Further conjugation by S_2, which commutes with U_0, changes this to $(-s/2, 1) = (-F, 1) \pmod{N}$. Thus S_2V^* conjugates the entire physical kicked propagator to (FN.1). These unitaries all preserve parity.

With the usual semiclassical convention $e^{i\,\mathrm{action}/\hbar}$, the classical kick amplitude associated with ε is proportional to $\hbar\varepsilon$. Here $\hbar \asymp 1/N$. At the critical scale the amplitude is of order $1/(N\sqrt{K})$, not a fixed classical perturbation.

32.2 Two summable phase estimates

Write $C(A) = U_0 A U_0^{*}$, and

$$B_j = C^j(B) = W(v_j) + W(-v_j), \qquad v_j = (qh^j, h^{-j}), \qquad \overline{B} = K^{-1}\sum_{j=0}^{K-1} B_j. \tag{FN.2}$$

The conjugation C shifts the label cyclically, so $C(\bar{B}) = \bar{B}$. Define

$$a_j = \left| e_N(q(h^j - h^{-j})) - 1 \right|, \qquad c_j = \frac{K-1-2j}{2K}, \qquad G = \sum_{j=0}^{K-1} c_j B_j. \tag{FN.3}$$

Then $a_0 = 0$, $a_{K-j} = a_j$. Direct subtraction of the coefficients gives the exact cohomological equation

$$G - C(G) = B - \overline{B}. \tag{FN.4}$$

The coefficients have zero mean. Importantly, they are small at the half-period at which the large commutator phases accumulate.

Let P_t denote the second Pell sequence, $\alpha^t = Q_t + P_t\sqrt{2}$. For the signed representative δ of j mod K one has

$$q(h^j - h^{-j}) = -2P_{2\delta} \pmod N. \tag{FN.5}$$

Indeed $h^\delta = (3+2s)^\delta = Q_{2\delta} + P_{2\delta}s$, and $Fs = 1$. For positive $\delta = (K-1)/2 - \ell$,

$$\frac{P_{2\delta}}{N} = \frac{\lambda^{-\ell}}{\sqrt{2}\,\alpha}\,\frac{1-\lambda^{-2\delta}}{1-\lambda^{-K}}. \tag{FN.6}$$

Put $r = \lambda^{-1}$, $A = 4\pi/(\sqrt{2}\,\alpha)$. Using $|e^{it} - 1| \le |t|$ and summing both signed representatives proves

$$L_K := \sum_{j=0}^{K-1} a_j \le \frac{2A}{(1-r)(1-\lambda^{-K})} = \frac{2\sqrt{2}\pi}{1-\lambda^{-K}} < 9 \quad (K \ge 3). \tag{FN.7}$$

There is a sharper weighted estimate. At j=(K-1)/2-ell the weights at j and K-j have absolute values ell/K and (ell+1)/K. Therefore

$$\sum_j |c_j| a_j \le \frac{A}{K(1-\lambda^{-K})} \sum_{\ell \ge 0} (2\ell+1) r^\ell = \frac{2\pi}{K(1-\lambda^{-K})}. \tag{FN.8}$$

The final identity follows from $A(1+r)/(1-r)^2 = 2\pi$. The estimates are analytic inequalities; the regression's decimal phase values are only finite checks of them.

32.3 Quadratic lifting and orthogonal commutator supports

Lemma 32.2 (lifting)**.** *Fix m. For all sufficiently large K, every signed m-term reciprocal relation*

$$\sum_{\nu=1}^{m} \epsilon_\nu h^{j_\nu} = \sum_{\nu=1}^{m} \epsilon_\nu h^{-j_\nu} = 0 \pmod N, \qquad \epsilon_\nu \in \{-1, 1\}, \tag{FN.9}$$

can be cyclically rotated to a polynomial P with $P(\lambda) = 0$. Its coefficient ℓ^1-norm is at most m, and its degree is at most $K - \lceil K/m \rceil$. For $m = 4$, this holds for every odd prime $K \ge 11$, and the relation is then coefficientwise balanced.

Proof. Cancel opposite signs at identical labels. If anything remains, cut the cyclic support at a largest gap and rotate. There are at most m support points, giving the stated degree bound. Both congruences in (FN.9) survive. Divide P by $Z^2 - 6Z + 1$ over the integers. Its remainder $aZ + b$ vanishes at h and h^{-1}; their difference $4s$ is a unit. Thus $N \mid a$ and $N \mid b$. A nonzero quadratic norm of $P(\lambda)$ would be an integer divisible by N^2, but

$$|P(\lambda)P(\lambda^{-1})| \le m^2 \lambda^{K-\lceil K/m \rceil} < N^2 \tag{FN.10}$$

for sufficiently large K at fixed m. This is impossible, so $P(\lambda) = 0$. For $m = 4$, a sufficient condition is $\lambda^{K/4}(1 - \lambda^{-K})^2 > 64$. It holds for $K \ge 11$: $\lambda > 5$, $5^{11/4} > 80$, and $1 - 5^{-11} > 9/10$ give a strict lower bound of 64.8. A nonzero polynomial of coefficient ℓ^1-norm at most four cannot vanish at $\lambda > 3$, since its leading term strictly dominates the sum of all lower terms. Thus in this case the polynomial was zero. End of proof.

It follows that the $2K(K-1)$ vectors

$$\epsilon v_t + \delta v_s, \qquad 0 \le t < s < K, \quad \epsilon, \delta \in \{-1, 1\}, \tag{FN.11}$$

are nonzero and distinct for $K \ge 11$. Any collision would give a balanced four-letter relation, forcing the original unordered signed pair. The matrices $W(v)$ form an orthonormal basis for the full normalized Hilbert–Schmidt inner product, also for composite odd N, because $\operatorname{Tr} W(v) = N\mathbf{1}_{v=0}$.

Each of the four Weyl coefficients in $[B_t, B_s]$ has modulus a_{s-t}. Different unordered time pairs have disjoint support by (FN.11). Since $c_t - c_s = (s-t)/K$, exact orthogonality gives

$$\|[G, \overline{B}]\|_{2,N}^2 = \frac{4}{K^4} \sum_{j=1}^{K-1} (K-j) j^2 a_j^2 = \frac{2}{K^3} \sum_{j=1}^{K-1} j(K-j) a_j^2 \le \frac{9}{K}. \tag{FN.12}$$

Here $\|A\|_{2,N}^2 = \operatorname{Tr}(A^*A)/N$. The middle identity uses $a_{K-j} = a_j$; the last uses $j(K-j) \le K^2/4$, $a_j^2 \le 2a_j$, and (FN.7).

The operator norm needs no orthogonality. Four Weyl coefficients, (FN.7), and (FN.8) give

$$\|[G, B]\|_{\mathrm{op}} < \frac{26}{K}, \qquad \|[B, \overline{B}]\|_{\mathrm{op}} < \frac{36}{K}. \tag{FN.13}$$

For a parity-preserving A, positivity of A*A implies

$$\|A\|_{2,-} \le \sqrt{\frac{2N}{N-1}}\, \|A\|_{2,N}. \tag{FN.14}$$

No orthogonality of Weyl operators restricted to the odd subspace is assumed.

32.4 A finite normal form without a small generator

For finite Hermitian A,D, real ε, and any unitarily invariant norm,

$$\|e^{i\varepsilon A} e^{i\varepsilon D} - e^{i\varepsilon(A+D)}\| \le \frac{\varepsilon^2}{2} \|[A, D]\|. \tag{FN.15}$$

This is the usual two-factor commutator estimate, not a new general product formula. Differentiate $e^{i(\varepsilon - t)(A+D)} e^{itA} e^{itD}$. After factoring out unitaries, the integrand is $e^{itA} D e^{-itA} - D$. Integrate its derivative once more; the triangular integration domain has area $\varepsilon^2/2$. Negative ε follows by replacing A, D with $-A, -D$. In particular, $|\varepsilon| \|A\|$ need not be small.

Set $V_\varepsilon = e^{-i\varepsilon G}$. By (FN.4),

$$V_\varepsilon U_\varepsilon V_\varepsilon^* = e^{-i\varepsilon G} e^{i\varepsilon B} e^{i\varepsilon C(G)} U_0.$$

Apply (FN.15) first to $-G, B$, then to $B-G, C(G)$. Their total sum is $\bar{B}$, and $[B-G, C(G)] = [B-G, \bar{B}]$. Equations (FN.12)–(FN.14) give

$$\begin{aligned} \|V_\varepsilon U_\varepsilon V_\varepsilon^* - e^{i\varepsilon\overline{B}} U_0\|_{2,-} &\le \frac{\varepsilon^2}{2}\left(\frac{62}{K} + 3\sqrt{\frac{2N}{(N-1)K}}\right) \\ &\le \frac{12\varepsilon^2}{\sqrt{K}}, \qquad K \ge 11. \end{aligned} \tag{FN.16}$$

For the last step it even suffices to use $N \ge 7$ and $31/\sqrt{11} + (3/2)\sqrt{7/3} < 12$. This proves (FN.D) for all real ε. G may have operator norm of order K. Expanding its exponential in powers of ε and requiring εK small would incorrectly discard the critical regime. Estimate (FN.15) avoids that requirement.

Since $\bar{B}$ commutes with U_0 and $U_0^K = I$, telescoping unitary powers in the same norm yields, for any integer m,

$$\left|\tau_-(U_\varepsilon^{mK}) - \tau_-(e^{im\varepsilon K\overline{B}})\right| \le 12|m|\varepsilon^2\sqrt{K}, \qquad \tau_-(A) = d_N^{-1}\,\mathrm{Tr}_-\,A. \tag{FN.17}$$

For negative m use adjoints. Cauchy–Schwarz for the normalized trace justifies the passage from the norm to traces.

32.5 A self-contained global law for the averaged observable

The remaining input is established here directly on the whole odd subspace; no unproved claim about the old arithmetic-moment corpus is imported. Define

$$X_K = \sqrt{K/2}\,\overline{B} = (2K)^{-1/2}\sum_{j=0}^{K-1} B_j.$$

Lemma 32.3 (Gaussian average)**.** *For each fixed integer $r \ge 1$,*

$$\tau_-(X_K^{2r}) = (2r-1)!! + O_r(K^{-1}) + O_r(K^r/N), \tag{FN.18}$$

and for each fixed odd m, eventually

$$|\tau_-(X_K^m)| \le (2K)^{m/2}/(N-1). \tag{FN.19}$$

The empirical measures of X_K converge weakly and in quadratic Wasserstein distance to the standard normal law.

Proof. In the full Weyl representation $\mathrm{Tr}(RW(v)) = 1$ for every v: the unique fixed point $x = -b/2$ of RW(a,b) contributes phase 1. Thus the normalized odd trace of a word $e_N(\phi)$W(v) is

$$e_N(\phi)\frac{N\mathbf{1}_{v=0} - 1}{N-1}. \tag{FN.20}$$

Expanding X_K^m, the noncentral part is bounded by $(2K)^{m/2}/(N-1)$. The central part is the weighted count of the reciprocal relations (FN.9), normalized by $[N/(N-1)](2K)^{-m/2}$.

Here is the count including structurally forced nonbalanced relations. For a polynomial of coefficient ℓ^1-norm at most m vanishing at λ, a gap $d > 2\log_\lambda m$ separates two independently vanishing pieces. Write $P = P_{\mathrm{low}} + Z^{a+d}P_{\mathrm{high}}$, with $\deg P_{\mathrm{low}} \le a$ and nonnegative exponents in P_{high}. If $P_{\mathrm{high}}(\lambda) \ne 0$, its algebraic-integer norm is a nonzero integer. Since $|P_{\mathrm{high}}(\lambda^{-1})| \le m$, we have $|P_{\mathrm{high}}(\lambda)| \ge 1/m$. Cancellation with $|P_{\mathrm{low}}(\lambda)| \le m\lambda^a$ would force $\lambda^d \le m^2$, a contradiction.

Repeatedly splitting gives components with span bounded in terms of m, for example by $D_m = (m-1)(\lceil 2\log_\lambda m\rceil + 1)$. Every nonzero component has at least three signed occurrences:

two distinct positive powers cannot cancel with signs of magnitude one. Restore previously cancelled opposite pairs as two-occurrence components. A nonbalanced 2r-letter word therefore has at most r-1 components. For each of finitely many partitions of the 2r positions, signs and bounded relative offsets, each component has at most K translations on the cyclic time set. There are only $O_r(K^{r-1})$ such words. There is no additional factor K from choosing a cut: the count is over the component translations themselves, and the largest-gap cut was used only to prove bounded width. Their normalized contribution is $O_r(1/K)$.

For odd m, a lifted P is an integer multiple of $Z^2 - 6Z + 1$, so $P(1)$ is divisible by -4. But $P(1)$ is the sum of an odd number of signs. Thus no central word exists once Lemma 32.2 applies, proving (FN.19).

A balanced $2r$-word pairs identical time labels with opposite signs. Assignments with repeated pair labels cause only $O_r(K^{r-1})$ overcount. Each matching of the $2r$ positions has K^r pair-label choices and 2^r orientations. Its Weyl phase is a product, over crossing pairs, of factors $e_N(q(h^a - h^{-a}))$ or their conjugates. Summing the signed symplectic products in word order shows that disjoint or nested pairs cancel, while a crossing pair retains one full symplectic product.

Each crossing difference of independent uniform pair labels is uniform modulo K. Independence of the different crossing factors is neither true in general nor needed. The inequality $|\prod_l u_l - 1| \le \sum_l |u_l - 1|$, for $|u_l| = 1$, and (FN.7) give average phase error at most nine times the number of crossings divided by K. Hence each matching contributes $1 + O_r(1/K)$ after normalization. There are $(2r-1)!!$ matchings. Together with the nonbalanced count and (FN.20), this proves (FN.18).

The second-moment bound gives tightness. Higher even moments give uniform integrability when passing moments to subsequential limits. The Gaussian moment sequence is determinate, for example by Carleman's criterion, since $((2r-1)!!)^{1/(2r)} \le \sqrt{2r}$. Every subsequential limit is standard normal. Convergence of the second moments upgrades weak convergence to quadratic Wasserstein convergence. End of proof.

32.6 Critical and subcritical propagator spectra

For $\varepsilon = \tau/\sqrt{2K}$, (FN.17) becomes

$$\left|\tau_-(U_\varepsilon^{mK}) - \tau_-(e^{im\tau X_K})\right| \le \frac{6|m|\tau^2}{\sqrt{K}}. \tag{FN.21}$$

Lemma 32.3 and bounded continuity of exp(im τ x) give (FN.A). Integer Fourier coefficients determine measures on the circle: trigonometric polynomials are dense in continuous functions there. Applying the characteristic function of a normal variable proves (FN.B), and taking absolute squares proves (FN.C). No eigenphase labelling across the old degenerate sectors is used or asserted.

There is also a genuine extension of the earlier subcritical global transfer, but for **folded phases** rather than individually isolated clusters. Let ε_K be nonzero and $\tau_K = \varepsilon_K\sqrt{2K}$ tend to zero. If $e^{i\theta_l}$ are the eigenvalues of $U_{\varepsilon,K}$, define

$$y_l = \frac{\operatorname{Arg}(e^{iK\theta_l})}{\tau_K}, \qquad \operatorname{Arg} \in [-\pi, \pi).$$

Corollary 32.4 (subcritical folded Gaussian law)**.** *The empirical measures of y_l converge in quadratic Wasserstein distance to the standard normal law whenever $\varepsilon_K\sqrt{K}$ tends to zero.*

Proof with a quantitative tail control. For two unitary matrices, the squared absolute overlaps of their orthonormal eigenbases form a doubly stochastic matrix. The corresponding

average squared chord distance equals their normalized squared Hilbert–Schmidt difference. Decomposing this matrix into permutation matrices gives an eigenvalue matching with no larger average cost. The decomposition can be obtained by Hall's matching criterion and successive subtraction; it uses no eigenvalue simplicity. Thus (FN.16), telescoped K times, matches the powered spectrum to $e^{i\tau_K x_l}$, where x_l are the eigenvalues of X_K, with root mean square chord error at most $6\tau_K^2/\sqrt{K}$.

If $|\tau_K x_l| \le \pi/2$, principal argument differences obey $|\operatorname{Arg} u - \tau_K x_l| \le 4|u - e^{i\tau_K x_l}|$. On the complementary set, $|\operatorname{Arg} u/\tau_K - x_l| \le 3|x_l|$. Consequently

$$\begin{aligned} W_2(\operatorname{law}(y), \operatorname{law}(X_K)) &\le \frac{24|\tau_K|}{\sqrt{K}} + 3\sqrt{\tau_-\left(X_K^2 \mathbf{1}_{|X_K|>\pi/(2|\tau_K|)}\right)} \\ &\le \frac{24|\tau_K|}{\sqrt{K}} + \frac{6|\tau_K|}{\pi}\sqrt{\tau_-(X_K^4)}. \end{aligned} \tag{FN.22}$$

The fourth moment stays bounded by Lemma 32.3, so the right side tends to zero. Combine with the W_2 conclusion of that lemma. End of proof.

The former isolated-cluster argument required $\varepsilon_K K^{3/2}$ tending to zero. Corollary 32.4 permits an additional factor K in the kick scale, but its observable is a folded spectral measure. It must not be restated as an individual-cluster rank or no-overlap theorem.

32.7 A compact-test microscopic obstruction

A negative statement about microscopic correlations needs less information than a positive universality theorem. The recurrence law above supplies such a statement through Fourier positivity. This section concerns the whole odd spectrum, with its multiplicities and the **global mean-density unfolding**. It is not a claim about a further symmetry sector or an unfolding which first removes a nonconstant mesoscopic intensity profile.

Let $x_1,\dots,x_d$ be the eigenphases of a d-dimensional unitary U, measured in turns, so its eigenvalues are $e^{2\pi i x_j}$. For a compactly supported continuous test ϕ define the ordered factorial pair statistic

$$C_d(U;\phi) = \frac{1}{d}\sum_{a\ne b}\sum_{n\in\mathbb{Z}} \phi\big(d(x_a - x_b + n)\big). \tag{FN.23}$$

Distinct indices, not distinct eigenvalues, are required in this definition. For $L > 0$ use the compact triangular test

$$\phi_L(s) = \frac{1}{L}(1 - |s|/L)_+, \quad \int \phi_L = 1, \quad \phi_L(0) = 1/L, \quad \widehat{\phi}_L(u) = \operatorname{sinc}^2(Lu) \ge 0, \tag{FN.24}$$

where $\operatorname{sinc}(v) = \sin(\pi v)/(\pi v)$, with value 1 at zero. Fourier transforms use $e^{-2\pi i u s}$. Absolute convergence of the periodized Fourier series (the coefficients decay quadratically) gives the exact identity

$$C_d(U;\phi_L) = d^{-2}\sum_{t\in\mathbb{Z}} \widehat{\phi}_L(t/d)(|\operatorname{Tr} U^t|^2 - d). \tag{FN.25}$$

If $d > L$, the total diagonal subtraction is exactly 1/L, because $\sum_n \phi_L(dn) = \phi_L(0)$. For any distinct positive integers $r_1,\dots,r_M$, positivity therefore gives

$$C_d(U;\phi_L) \ge 1 - \frac{1}{L} + 2\sum_{j=1}^{M} \operatorname{sinc}^2(Lr_j/d)|d^{-1}\operatorname{Tr} U^{r_j}|^2. \tag{FN.26}$$

No control or truncation of the remaining high-frequency traces is needed for this lower bound; their contribution is nonnegative.

There is an equivalent counting interpretation. Let N_L(y) count the phases in the circular interval [y,y+L/d), with y uniform in [0,1). Then $E_y\, N_L$=L and an overlap-of-intervals calculation gives

$$\mathrm{Var}_y\, N_L = L^2\big(C_d(U;\phi_L) - 1\big) + L. \tag{FN.27}$$

This identity is also verified independently by integrating the piecewise-constant counting function, rather than reusing the pair sum.

Theorem 32.5 (obstruction and necessary kick scale)**.** *For the kicked Pell family of Theorem 32.1, put $\tau_K = \varepsilon_K\sqrt{2K}$. If τ_K tends to a finite nonzero τ, then for every fixed $L > 0$,*

$$\liminf_{K\to\infty} C_{d_N}(U_{\varepsilon_K,N};\phi_L) \geq \Theta(\tau) - \frac{1}{L}, \qquad \Theta(\tau) = \sum_{m\in\mathbb{Z}} e^{-m^2\tau^2} > 1, \tag{FN.28}$$

and

$$\liminf_{K\to\infty} \mathrm{Var}_y\, N_L \geq L^2\big(\Theta(\tau) - 1\big). \tag{FN.29}$$

If $\tau_K \to 0$, then $C_{d_N}(U_{\varepsilon_K,N};\phi_L) \to +\infty$ for every fixed L. Consequently, convergence of the globally unfolded pair measures to a bounded density $\rho(s)$ with $\rho(s) \to 1$ as $|s| \to \infty$ requires

$$|\varepsilon_K|\sqrt{K} \longrightarrow \infty. \tag{FN.30}$$

This is a necessary condition only, not a sufficient universality criterion.

Proof. The recurrence trace limit remains valid when $\tau_K \to \tau$ varies with K. Lemma 32.3 bounds $\tau_-(|X_K|)$, and $|\tau_-(e^{im\tau_K X_K}) - \tau_-(e^{im\tau X_K})| \leq |m|\,|\tau_K - \tau|\,\tau_-(|X_K|)$. The error in (FN.21) is uniform for bounded τ_K. Thus the normalized mK-trace tends to $e^{-m^2\tau^2/2}$ for each fixed m, including $\tau = 0$.

Apply (FN.26) with $r_j = jK$. Since $K/d_N \to 0$, the sinc factors tend to one. At each fixed M, the resulting lower bound is $1 - 1/L + 2\sum_{m=1}^{M} e^{-m^2\tau^2}$. Let M tend to infinity *after* taking the lower limit in K. This proves (FN.28), or divergence when $\tau = 0$. Equation (FN.27) gives (FN.29).

Suppose the pair measures converge vaguely to ρ(s)ds, where ρ is bounded and clusters to unit intensity. For every fixed L the limiting triangular statistic would be

$$A_L = \int_{-1}^{1} (1 - |u|)\rho(Lu)\, du.$$

Dominated convergence gives $A_L \to 1$ as $L \to \infty$. But (FN.28) requires $A_L \geq \Theta(\tau) - 1/L$ for every L, a contradiction. For $\tau = 0$, a finite locally bounded target already contradicts the fixed-L divergence. If $|\varepsilon_K|\sqrt{K}$ does not tend to infinity, a bounded subsequence of τ_K has a further subsequence converging to a finite limit. The same contradiction applies. End of proof.

The use of a compact test is essential for the logical scope. A Gaussian test without an independently controlled tail would only obstruct a Schwartz-test limit, not automatically vague local convergence. If the pair convergence definition uses only smooth compact tests, positive measures acquire locally uniform mass bounds by testing a bump which is one on a given compact set; approximation then extends convergence to the continuous triangular tests. Alternatively, start directly with a smooth nonnegative compact f of integral one, take ϕ=f*tilde(f), and scale it: $\widehat{\phi} = |\widehat{f}|^2 \geq 0$ and the same proof uses $\phi(0)$/L instead of 1/L. No unproved high-frequency-tail assumption is hidden in Theorem 32.5.

The Poisson pair density ρ=1 and the sine-kernel density $\rho_2(s) = 1 - \mathrm{sinc}^2(s)$ are both excluded. More explicitly, keeping only the first resonance gives

$$\liminf C_{d_N}(U_{\varepsilon_K,N};\phi_L) \geq 1 + 2e^{-\tau^2} - 1/L. \tag{FN.31}$$

Since $\int \phi_L \rho_2 \le 1$, taking $L = e^{\tau^2}$ yields a gap of at least $e^{-\tau^2}$ from the sine-kernel prediction. At $\tau = 1$, already L=2 gives lower bound 1.2357588823…, greater than both that prediction and the Poisson value 1.

The standard GOE bulk pair density is excluded by the bounded-clustering part of the theorem, not by an assumed pointwise bound $\rho \le 1$. In unit intensity normalization it is the even continuation of

$$\rho_1(s) = 1 - S(s)^2 - S'(s) \int_s^\infty S(t)\, dt, \qquad S(s) = \operatorname{sinc}(s), \quad s \ge 0. \tag{FN.32}$$

This standard formula is recorded in Brezin–Hikami, equation (181) after unfolding. It is bounded at zero, and integration by parts in the sine tail shows $\rho_1(s) = 1 + O(s^{-2})$ at infinity. The same elementary criterion therefore applies. The random-matrix formulas, Fourier positivity and counting-variance identity are standard ingredients; no priority is claimed for those general tools.

This changes the route toward a positive BGS result: the critical and subcritical globally unfolded odd spectra considered here are not merely unproved universal families; they are **excluded** from the bounded clustering pair limits just specified. The statement does not settle nearest-neighbour distributions individually, density-adapted re-unfolding, a further desymmetrized block, or fixed nonzero classical kicks. It is not a refutation of a correctly qualified generic BGS conjecture.

The finite verifier keeps forty configurations (K=3,5,7, several τ and L) and two synthetic controls. At K=7,τ = 1,L=2 it numerically gives C_d=1.30936229449… and Var N_L=3.23744917797…, while the finite lower bound retaining K and 2K is 1.21861711382… . These are floating-point checks of finite identities, not the proof of (FN.28). In particular the asymptotic bound is not incorrectly enforced on small K.

Chapter 8 — Weighted echoes and Edgeworth corrections

33 Weighted echoes, a resonance comb, and fixed-polynomial corrections

This section consolidates the weighted-observable, echo, resonance-comb, and polynomial-correction notes. All statements use the fixed Pell map and odd prime K, with $N + F\sqrt{2} = (1+\sqrt{2})^K$, $D = (N-1)/2$, $B_j = U_0^j B U_0^{-j}$, and $\tau_- = D^{-1} \operatorname{Tr}_-$. The parameter τ below is a real kick amplitude, not a trace symbol. The preceding normal form and lifting lemmas supply the stated inputs; no independence of the finite Weyl operators is assumed.

33.1 Bounded arrays and the ordered Gaussian law

For real arrays f define

$$X_K(f) = (2K)^{-1/2} \sum_{j<K} f_j B_j, \qquad Q_K(f,g) = K^{-1} \sum_{j<K} f_j g_j.$$

For fixed r and arrays uniformly bounded by M, the ordered mixed moments obey

$$\tau_-(X_K(f_1)\cdots X_K(f_{2r})) = \sum_{\pi \in \mathcal{P}_2(2r)} \prod_{\{a,b\}\in\pi} Q_K(f_a, f_b) + O_{r,M}(K^{-1}) + O_{r,M}(K^r/N). \tag{33.1}$$

For each fixed odd m, after the lifting threshold,

$$|\tau_-(X_K(f_1)\cdots X_K(f_m))| \le M^m (2K)^{m/2}/(N-1).$$

Thus convergence of a finite set of empirical Gram matrices gives the Wick moments of a commuting real Gaussian vector with that covariance.

Proof. The exact parity-projected word trace is $e_N(\phi)(N\mathbf{1}_{v=0}-1)/(N-1)$. It bounds the noncentral contribution as displayed. The largest-gap norm lift and bounded-cluster argument confine nonpaired words to $O_r(K^{r-1})$ choices. Odd central words are impossible: a lifted polynomial is divisible by z^2-6z+1, so its value at one is divisible by four and cannot be a sum of an odd number of signs. For each matching of a balanced word, pair-label summation with the phase replaced by one gives exactly the product of Q's. Repeated labels cost $O_{r,M}(K^{-1})$. Its crossing phase is a product of $z_{a-b}^{\pm 1}$. Each individual difference is uniform, and $\sum_d |z_d-1|<9$; therefore $|\prod u_e-1|\le\sum_e|u_e-1|$ bounds its weighted mean error by $9M^{2r}$ times the number of crossings divided by K. Distinct crossings need not be independent. Summation proves the formula. □

The exact four-letter support theorem also gives

$$\|[X_K(f),X_K(g)]\|_{2,-}\le\frac{6}{\sqrt{K}}\sqrt{\frac{2N}{N-1}}\|f\|_\infty\|g\|_\infty,\qquad K\ge 11. \tag{33.2}$$

Indeed, the coefficient of $[B_t,B_s]$ is $(f_tg_s-f_sg_t)/(2K)$, bounded by $\|f\|_\infty\|g\|_\infty/K$. The four Weyl terms for different unordered pairs have disjoint support. Their full-space squared norm is therefore at most $4\|f\|_\infty^2\|g\|_\infty^2K^{-2}\sum_{j=1}^{K-1}(K-j)a_j^2$. Use $a_{K-j}=a_j$, $a_j^2\le 2a_j$, and $\sum a_j<9$, then restrict the positive quadratic form to the odd sector. This proves (33.2); it is not an operator-norm assertion.

33.2 The interaction-picture echo

Let $U_\varepsilon=e^{i\varepsilon B}U_0$, $\varepsilon=\tau/\sqrt{2K}$ and $E_K(t)=U_\varepsilon^tU_0^{-t}$. Write $t=mK+s$, $0\le s<K$, including negative t by the usual floor convention. Define $w_{t,K}(j)=m+\mathbf{1}_{j<s}$. The centered coefficients $c_j=(K-1-2j)/(2K)$ satisfy the exact identity

$$c_j+t/K-c_{(j-t)\bmod K}=w_{t,K}(j).$$

Inserting the finite normal form, telescoping t unitary factors, and combining the three exponentials with (33.2) gives

$$\|E_K(t)-e^{i\tau X_K(w_{t,K})}\|_{2,-}\le(11C+2)\tau^2/\sqrt{K}\qquad(|t|\le CK). \tag{33.3}$$

To detail the constant, the original normal-form error contributes $6|t/K|\tau^2/\sqrt{K}$. The comparison has three factors with arrays c, the constant t/K, and the negative shifted c. The two Duhamel errors sum to at most $c_N(3|t/K|+3/4)\tau^2/\sqrt{K}$, where $c_N=\sqrt{2N/(N-1)}\le\sqrt{7/3}$. This yields the displayed bound. The same argument uses adjoints at negative times.

Let $\mathcal{B}(a)$, $0\le a\le 1$, be one standard Brownian motion and, for $\alpha=m+a$ with $0\le a<1$, put

$$Y(\alpha)=m\mathcal{B}(1)+\mathcal{B}(a).$$

For $\beta=n+b$ its covariance is $mn+mb+na+\min(a,b)$. The arrays $w_{t,K}$ have precisely this limiting Gram matrix when $t/K\to\alpha$. Therefore for every fixed finite collection of integer exponents r_l and times $t_l/K\to\alpha_l$,

$$\tau_-\left(\prod_l E_K(t_l)^{r_l}\right)\longrightarrow\exp\left[-\frac{\tau^2}{2}\sum_{l,j}r_lr_j\,\mathrm{Cov}(Y(\alpha_l),Y(\alpha_j))\right].$$

The order of the factors on the left is retained. Commutator estimates allow them to be combined in the limit; there is no common eigenbasis at finite K. In particular the phase variance of one

echo is $\tau^2[m^2+(2m+1)a]$. The same Brownian interval is reused on each cycle: fresh independent noise would give a different variance. This is a tracial finite-dimensional-distribution statement, not convergence of quantum measurement trajectories in a path space.

33.3 The full bounded recurrence window

For $1 \le s < K$, the fixed-point equations for $W(v)U_0^s$ and $RW(v)U_0^s$ have unique solutions, so $|\tau_-(W(v)U_0^s)| \le 1/D$. Hence, with $X_K = X_K(1)$,

$$|\tau_-(X_K^l U_0^s)| \le (2K)^{l/2}/D.$$

For $|t| \le CK$, $K \nmid t$, let $A_0 = C|\tau|$. Taylor's integral remainder of degree $2r-1$ and the normal form yield

$$|\tau_-(U_\varepsilon^t)| \le \frac{6C\tau^2}{\sqrt{K}} + \frac{1}{D}\sum_{l<2r}\frac{(A_0\sqrt{2K})^l}{l!} + \frac{A_0^{2r}}{(2r)!}\tau_-(X_K^{2r}).$$

First let K grow at fixed r, then r grow. The last limiting bound is $A_0^{2r}/(2^r r!) \to 0$. Thus nonresonant traces vanish uniformly in this window, whereas $\tau_-(U_\varepsilon^{mK}) \to e^{-m^2\tau^2/2}$. If $e^{i\theta_l}$ are the actual eigenvalues, the empirical measure of $(e^{i\theta_l}, e^{iK\theta_l})$ converges to Haar measure times the wrapped normal law: its Fourier coefficients are traces at $a+bK$. This is a two-scale law of one eigenvalue, not independence of two levels.

33.4 Minimal Pell relations and the first polynomial correction

For an integer polynomial P, write $\|P\|_1$ for its coefficient norm. If $P \neq 0$ and $P(\lambda) = 0$, then $\|P\|_1 \ge 8$, with equality only for $\pm z^j(z^2-6z+1)$. Indeed $P(1)$ is divisible by four, so its norm is even; if it were at most six the leading term would dominate since $\lambda > 5$. At norm eight the leading coefficient is one in absolute value. After a sign change write $P = z^L + cz^{L-1} + R$, $\|R\|_1 = 7-|c|$. Evaluation gives $|(6+c)\lambda - 1| \le 7-|c|$, forcing $c=-6$ and then $R = z^{L-2}$. Thus the first genuinely nonpaired central words have length eight. The bounded-cluster argument improves their normalized contribution at any fixed even order to $O_r(K^{-3})$.

For a fixed simple crossing graph, expand the phase product through first order in $z_d - 1$. Any two distinct edges form a forest, so their label differences are independent uniform variables. The remainder is therefore $O(K^{-2})$. A matching with c crossings contributes $1 + c(S_K - 1) + O(K^{-2})$, $S_K = K^{-1}\sum_d z_d$. The total crossing count is $A_r = \binom{2r}{4}(2r-5)!!$: choose four interlaced endpoints and match the rest. Four repeated positions cause one excess matching; their six balanced sign assignments contribute $-3A_r/(2K) + O(K^{-2})$ to the moment: the matching-indexed sum counts each such word with one excess matching, so the excess is subtracted. The crossings contribute $A_r(S_K - 1) = -(\gamma - 3/2)A_r/K + O(K^{-2})$, and the two corrections add to $-\gamma A_r/K$. Since $S_K = 1 - (\gamma - 3/2)/K + O((KN)^{-1})$, it follows that

$$\tau_-(X_K^{2r}) = (2r-1)!! - \frac{\gamma}{K}\binom{2r}{4}(2r-5)!! + O_r(K^{-2}) + O_r(K^r/N). \tag{33.4}$$

Here

$$\gamma = \frac{3}{2} + 2\sum_{j\ge 0}\left[1 - \cos\left(\frac{4\pi}{\sqrt{2}(1+\sqrt{2})}\lambda^{-j}\right)\right].$$

For every fixed real polynomial f and standard Gaussian Z,

$$\tau_-(f(X_K)) = \mathbb{E}f(Z) - \frac{\gamma}{24K}\mathbb{E}f^{(4)}(Z) + O_f(K^{-2}) + O_f(K^{\deg f/2}/N).$$

This is a fixed-polynomial expansion, not a distribution-function Edgeworth theorem. Constants are not controlled as the polynomial degree grows.

For completeness, here is the stronger crossing-graph expansion, with the notation it needs. For a fixed simple graph G on the pair labels, let the labels t_v be independent and uniform in $\mathbf{Z}/K\mathbf{Z}$, orient each edge, choose signs $\sigma_e \in \{-1, 1\}$, and put $Z_G = \prod_{e=(u,v)} z_{t_v - t_u}^{\sigma_e}$, where z_d are the crossing phases above; $\mathbb{E}$ is the average over the labels. Then

$$\mathbb{E} Z_G = S_K^{|E(G)|} + O_G(K^{-3} + N^{-1}).$$

Indeed, expand $\prod_e (1 + (z_e^{\sigma_e} - 1))$. Every subset of at most two edges is a forest, so its edge differences are independent and uniform, and its expectation is exactly $(S_K - 1)$ per edge, resp. $(S_K - 1)^2$ per pair; this matches the binomial expansion of $S_K^{|E(G)|}$ through second order, the remaining binomial terms being $O_G(K^{-3})$ since $S_K - 1 = O(K^{-1})$. A subset of at least three edges either contains a three-edge forest, whose expected absolute defect product is at most $(9/K)^3$ by $\sum_d |z_d - 1| < 9$, or is a triangle. For a triangle the bound is on the defect product, not on the unit-modulus phases themselves: the three differences sum to zero modulo K, their signed representatives therefore sum to 0 or $\pm K$ (for $\pm K$ all three have the same sign, since each is at most $(K-1)/2$ in absolute value), in both cases $\sum_i |\delta_i| \le K$, i.e. $\sum \ell \ge (K-3)/2$ for $\ell(d) = (K-1)/2 - |\delta|$; the phase-rate bound $|z_d - 1| \le A\lambda^{-\ell(d)}/(1 - \lambda^{-K})$ with $A = 4\pi/(\sqrt{2}(1+\sqrt{2}))$ then gives a defect product $O(\lambda^{-K/2}) = O(N^{-1})$, uniformly in the labels, and if one difference is zero the product is zero. All remaining defect factors are bounded by two. This proves the stated refinement.

At order eight, the minimal relation is $(3, -4) + (3, 4) - 6(1, 0) = 0$. Among its 56 orders the seven half-symplectic phases $-12, -8, -4, 0, 4, 8, 12$ each occur eight times. Both signs and K time shifts therefore give the isolated nonpaired central contribution

$$\frac{N}{N-1} K^{-3} [1 + 2\cos(8\pi/N) + 2\cos(16\pi/N) + 2\cos(24\pi/N)].$$

It is $7/K^3 + O((K^3 N)^{-1})$, in addition to all balanced contributions; it is not the whole eighth moment. The sufficient eight-letter lifting threshold holds for odd primes $K \ge 29$. Its remaining odd-parity trace error is at most $16K^4/(N-1)$.

33.5 Subcritical microscopic concentration without a Gaussian input

For a circle spectrum define $a^2 = D^{-1} \sum_l \operatorname{dist}(\theta_l, K^{-1}\mathbb{Z})^2$. Arcs of halfwidth $w < 1/(2K)$ about the roots contain at least $D(1 - a^2/w^2)_+$ points. Counting gaps and applying Cauchy–Schwarz to cells of length x/D gives

$$\frac{1}{D} \#\{S_l > x\} \le \frac{2a^2}{w^2} + \frac{K}{D} + \frac{2Kw}{x}, \qquad R_D(x) \ge \frac{(1 - a^2/w^2)_+^2}{K/D + 2Kw/x} - 1.$$

An eigenvalue matching for normal matrices follows by decomposing their squared-eigenbasis-overlap matrix into permutations. Chord distance bounds circle distance, and the finite normal form gives

$$a \le \frac{3\varepsilon^2 + (|\varepsilon|/\pi)\sqrt{N/(N-1)}}{\sqrt{K}}.$$

Consequently $|\varepsilon|\sqrt{K} \to 0$ implies $Ka \to 0$, collapse of the gap law to δ_0, and divergence of $R_D(x)$ for each fixed x. Set $w = (Ka)^{2/3}/K$ for the gap estimate and $w = 2a$ for the pair estimate; the case $a = 0$ is direct. These conclusions need no Gaussian limit and refer to the original phases, not folded phases or isolated old eigenspaces.

Chapter 7 — Floquet, coboundary and smooth resonance fields (continued)

34 A second-order recurrence scale for a coboundary kick

Set

$$T = \begin{pmatrix} 3 & 2 \\ 4 & 3 \end{pmatrix}, \qquad \alpha = 1 + \sqrt{2}, \quad \lambda = 3 + 2\sqrt{2}, \quad N + F\sqrt{2} = \alpha^K,$$

where K tends to infinity through odd primes. Then N is odd, $N^2 - 2F^2 = -1$, and $N \asymp \lambda^{K/2}$. The moduli need not be prime. Use the phase-normalized cat quantization $\widetilde{U}_0$ from the finite Floquet construction: $\widetilde{U}_0$ is the operator denoted $U_{N,\mathrm{Fl}} = VU_0V^*$ there, not the auxiliary $U_0 = S_h^{-1}$ of (FN.1); the tilde distinguishes the two throughout this coboundary section. It satisfies $\widetilde{U}_0 W(v)\widetilde{U}_0^* = W(Tv)$ and $\widetilde{U}_0^K = I$. Let

$$W(a,b)f(x) = e_N(a(x - b/2))f(x - b), \qquad B_v = W(v) + W(-v).$$

All symmetrized observables and $\widetilde{U}_0$ preserve odd parity. Write $d = (N-1)/2$ and $\tau_-(M) = d^{-1}\,\mathrm{Tr}_-\, M$.

The perturbation is now a different, fixed Fourier observable:

$$A = B_{(1,0)}, \quad D = \widetilde{U}_0 A \widetilde{U}_0^* = B_{(3,4)}, \quad H = A - D, \quad V_\varepsilon = e^{i\varepsilon H}\widetilde{U}_0. \tag{CB.1}$$

It is a coboundary: $\sum_{j=0}^{K-1} \widetilde{U}_0^j H \widetilde{U}_0^{-j} = 0$ exactly. The real classical Fourier symbol of H is independent of N. The products of the two coordinates of (1,0), (3,4), (2,4), and (4,4) are even, so the centered finite-Weyl convention introduces no N-dependent sign for these fixed Fourier modes. As throughout the project, the quantum exponent ε corresponds to a classical action amplitude proportional to ε/N, not ε itself.

Put

$$s_N = \sin(4\pi/N) > 0, \quad \zeta = \varepsilon^2 s_N, \quad t_K = 2\zeta\sqrt{K}.$$

Theorem 34.1 (second-order critical law)**.** *If $t_K \to t \in [0,\infty)$, then for every fixed integer m,*

$$\tau_-(V_{\varepsilon_K}^{mK}) \longrightarrow e^{-m^2t^2/2}. \tag{CB.2}$$

Thus the empirical spectrum of $V_{\varepsilon_K}^K$ tends to the wrapped normal law of variance t^2, or to the point mass at 1 when t=0. The critical quantum scale is

$$|\varepsilon_K| = \left(\frac{t}{2s_N\sqrt{K}}\right)^{1/2} \sim \sqrt{\frac{t}{8\pi}}\,\sqrt{N}K^{-1/4} \quad (t > 0). \tag{CB.3}$$

This can grow without bound, even though the classical amplitude tends to zero. The sign of ε does not change the limit in (CB.2).

Theorem 34.2 (negative microscopic consequences)**.** *For the original propagator in (CB.1), on the whole odd space and with global mean-spacing normalization, the following hold.*

If $|\varepsilon_K|K^{1/4}/\sqrt{N} \to 0$, the empirical nearest-neighbour spacing measures tend weakly to δ_0, and the normalized number of ordered distinct pairs at distance at most x mean spacings tends to infinity for every fixed $x > 0$.

For any subsequence on which $|\varepsilon_K|K^{1/4}/\sqrt{N}$ is bounded, the pair measures cannot converge to a bounded density R(s) with $R(s) \to 1$ as $|s| \to \infty$. In particular, escape from this obstruction requires

$$|\varepsilon_K|K^{1/4}/\sqrt{N} \longrightarrow \infty. \tag{CB.4}$$

Condition (CB.4) is only necessary. No positive universality theorem beyond it is proved. These statements concern global unfolding of the specified odd spectrum, not arbitrary clusterwise unfolding or an assertion that all arithmetic symmetries are resolved. They do not refute fixed-system BGS.

34.1 A second-order loop estimate valid at large quantum exponents

Lemma 34.3. *For finite Hermitian A,D and any real ε,*

$$\left\| e^{-i\varepsilon A} e^{i\varepsilon(A-D)} e^{i\varepsilon D} - e^{-\varepsilon^2[A,D]/2} \right\|_{\mathrm{op}} \le |\varepsilon|^3 \left(\frac{1}{2} \|[A,[A,D]]\|_{\mathrm{op}} + \frac{1}{6} \|[D,[A,D]]\|_{\mathrm{op}} \right). \tag{CB.5}$$

There is no hypothesis $|\varepsilon|\|A\| \ll 1$.

Proof. Write $Q = [A, D]$ and $L(u) = e^{-iuA} e^{iu(A-D)} e^{iuD}$. Direct differentiation gives

$$L'(u)L(u)^* = -\int_0^u \mathrm{Ad}_{e^{-iuA} e^{iv(A-D)}}\, Q\, dv.$$

The logarithmic derivative of $E(u) = e^{-u^2 Q/2}$ is -uQ. By differentiating unitary conjugation and integrating,

$$\| \mathrm{Ad}_{e^{-iuA} e^{iv(A-D)}}\, Q - Q \| \le u\|[A,Q]\| + v\|[A-D,Q]\| \quad (0 \le v \le u).$$

Duhamel comparison of the two unitary evolutions, integrated over $0 \le v \le u \le \varepsilon$, is at most $\varepsilon^3(\|[A,Q]\|/3 + \|[A-D,Q]\|/6)$. The triangle inequality proves (CB.5) for nonnegative ε. Replace A,D by -A,-D for the other sign. This argument is a norm estimate, not a truncation of a formal BCH series. End of proof.

Let u=(2,4), z=(4,4) and C=B_u-B_z. Since $W(v)W(w) = e_N(\omega(v,w)/2)W(v+w)$ and $\omega((1,0),(3,4)) = 4$, exact multiplication yields

$$[A, D] = -2is_N C. \tag{CB.6}$$

Each double commutator in (CB.5) has at most eight Weyl terms of coefficient modulus $4s_N^2$. Therefore both norms are at most $32s_N^2$, and

$$\left\| e^{-i\varepsilon A} V_\varepsilon e^{i\varepsilon A} - e^{i\zeta C} \widetilde{U}_0 \right\|_{\mathrm{op}} \le R_N(\varepsilon) := \frac{64}{3} |\varepsilon|^3 s_N^2. \tag{CB.7}$$

The same bound holds in normalized odd Hilbert–Schmidt norm. At the scale (CB.3), $KR_N = O_t(K^{1/4}/\sqrt{N})$; this tends to zero. The small quantity is controlled by nested commutators, not by the norm of ε A.

34.2 A two-species arithmetic lemma: collisions must not be discarded

Identify an integer vector v=(a,b) with $\psi(v) = b + \sqrt{2}a$. Then

$$\psi(Tv) = \lambda\psi(v), \quad \psi(u) = 4 + 2\sqrt{2}, \quad \psi(z) = \sqrt{2}\psi(u), \quad \mathrm{Norm}(\psi(u)) = 8.$$

Since N is odd, multiplication by $\psi(u)$ is invertible in $(\mathbb{Z}/N\mathbb{Z})[\sqrt{2}]$. The two species of time translates are therefore faithfully represented by λ^j and $\sqrt{2}\lambda^j$. Also $T^K = I$ modulo N, as follows from $Q_{2K} = 2N^2 + 1$ and $P_{2K} = 2NF$. Rotating all labels modulo K preserves a vector relation.

Lemma 34.4 (lifting)**.** *A signed m-atom modular zero relation in these two species, after cancellation of identical opposite atoms and rotation across a largest cyclic gap, has the form*

$$P(\lambda) = \sum_\nu \epsilon_\nu b_\nu \lambda^{j_\nu}, \quad b_\nu \in \{1, \sqrt{2}\}, \quad 0 \le j_\nu \le K - \lceil K/m \rceil.$$

For sufficiently large K at fixed m this expression is zero in $\mathbb{Z}[\sqrt{2}]$, not just modulo N. For m=4, $K \ge 11$ suffices.

Proof. There are at most m occupied time labels, hence a cyclic gap of at least ceil(K/m). Modular vanishing and the invertibility just proved imply $P(\lambda) \in N\mathbb{Z}[\sqrt{2}]$. Its nonzero algebraic norm would thus have absolute value at least N^2, whereas the two real embeddings give

$$|\operatorname{Norm} P(\lambda)| \le 2m^2 \lambda^{K-\lceil K/m \rceil}.$$

Since $N^2 = \lambda^K(1-\lambda^{-K})^2/4$, the latter is smaller than N^2 for large K at fixed m. For m=4 it suffices that $\lambda^{\lceil K/4 \rceil}(1-\lambda^{-K})^2 > 128$. At $K \ge 11$ this follows from $\lambda > 11/2$, $\lceil K/4 \rceil \ge 3$, and $1-\lambda^{-K} > 9/10$, since $(11/2)^3(9/10)^2 > 128$. End of proof.

A four-atom relation in characteristic zero need **not** balance within individual species. Apart from cancellation, the sole type is

$$z_{j+1} - u_{j+1} = z_j + u_j, \qquad v_j := T^j v, \tag{CB.8}$$

up to translation and multiplication by -1. The identity is already true over the integers, since $T(z-u) = z+u$.

Here is a direct completeness proof. Look at the maximal exponent E after cancellation. With one atom at E, its coefficient has modulus at least 1, whereas the three lower atoms sum to at most $3\sqrt{2}/\lambda < 1$ times λ^E. With three atoms at E, their nonzero coefficient has modulus at least $2-\sqrt{2} > \sqrt{2}/\lambda$, which the remaining lower atom cannot cancel. With four atoms at E, linear independence of 1 and $\sqrt{2}$ forces balance. With two atoms at E, the only possible nonzero coefficient small enough is $\pm(\sqrt{2}-1)$. The lower atoms must both be at E-1: otherwise at that scale their modulus is at most $\sqrt{2}(1+\lambda^{-1}) < 1+\sqrt{2}$. At E-1 they must be one of each species with equal signs because $\lambda(\sqrt{2}-1) = 1+\sqrt{2}$. This is exactly (CB.8). Relations with fewer occurrences reduce to the same cases; there is no nontrivial relation with at most three atoms.

Corollary 34.5 (multiplicity two)**.** *For $K \ge 11$, form all signed unordered pairs with distinct time labels:*

$$\epsilon v_t + \delta w_s, \quad 0 \le t < s < K, \quad v, w \in \{u, z\}, \quad \epsilon, \delta \in \{-1, 1\}.$$

There are $8K(K-1)$ such pairs. Their vector keys are nonzero, have multiplicity at most two, and have precisely

$$4K \text{ double keys}, \quad 8K(K-2) \text{ single keys}, \quad 8K^2 - 12K \text{ distinct keys}. \tag{CB.9}$$

The double classes are the two identities

$$z_{j+1} - z_j = u_{j+1} + u_j, \qquad z_{j+1} - u_j = u_{j+1} + z_j,$$

with their negatives, for j modulo K. The third partition of (CB.8) uses equal-time pairs and is not in the domain. Each displayed signed pair belongs to only one class: its species, the two adjacent cyclic labels, and its signs determine the other pair. Thus there are no triple classes or overlapping equivalence chains. Lemma 34.4 ensures that no additional modular relation is missed. K=3 is a useful negative control: it has six triple keys, and the range restriction must not be removed.

34.3 Weighted phase control and a finite normal form for C

Write $C_j = \widetilde{U}_0^j C \widetilde{U}_0^{-j}$, $\bar{C} = K^{-1}\sum_j C_j$, $G_C = \sum_j c_j C_j$, where $c_j = (K-1-2j)/(2K)$. Then

$$G_C - \widetilde{U}_0 G_C \widetilde{U}_0^* = C - \bar{C}, \qquad [\bar{C}, \widetilde{U}_0] = 0. \tag{CB.10}$$

Define a phase majorant, including both species,

$$a_j = \max_{v,w \in \{u,z\}} |e_N(\omega(v, T^j w)) - 1|.$$

One has $a_{K-j} = a_j$. In particular, no independence assumption about the several phase factors is being made.

For a centered representative δ of j modulo K,

$$\omega(v, T^{\delta} w) = (2v_1 w_1 - v_2 w_2) P_{2\delta} + \det(v, w) Q_{2\delta}.$$

For the four choices of species these are $-8P_{2\delta}$, $16P_{2\delta}$, and $\pm 8Q_{2\delta}$, respectively. Their absolute values are at most $8\lambda^{|\delta|}$. If $|\delta| = (K-1)/2 - \ell > 0$, this gives

$$a_j \le \frac{32\pi}{\alpha(1-\lambda^{-K})} \lambda^{-\ell}.$$

At zero, $a_0 \le 16\pi/N$. Summing both signs, including their weights ℓ/K and $(\ell+1)/K$, proves the explicit estimates

$$\begin{aligned} L_K := \sum_j a_j &\le \frac{32\pi}{1-\lambda^{-K}} + \frac{16\pi}{N} < 128, \\ K \sum_j |c_j| a_j &\le \frac{16\sqrt{2}\pi}{1-\lambda^{-K}} + \frac{8\pi K}{N} < 128. \end{aligned} \tag{CB.11}$$

For $K \ge 3$, use $\lambda^{-K} < 1/10$, $N \ge 2K+1$, $\pi < 22/7$, and $\sqrt{2} < 3/2$ to check the last inequalities. These constants are deliberately conservative.

The commutator $[C_t, C_s]$ has sixteen Weyl terms, each of modulus at most a_{s-t}. Corollary 34.5 says that each key in the sum over $t < s$ has at most two preimages. Weyl matrices are orthonormal in the *full* normalized Hilbert–Schmidt inner product, also over composite odd N. Cauchy–Schwarz inside each key class therefore gives

$$\begin{aligned} \|[G_C, \bar{C}]\|_{2,N}^2 &\le \frac{32}{K^4} \sum_{j=1}^{K-1} (K-j) j^2 a_j^2 \\ &= \frac{16}{K^3} \sum_{j=1}^{K-1} j(K-j) a_j^2 \le \frac{1024}{K}. \end{aligned} \tag{CB.12}$$

The equality between the two sums uses $a_{K-j} = a_j$; the final bound uses $a_j^2 \le 2a_j$ and (CB.11). Norm restriction to the odd subspace obeys $\|M\|_{2,-} \le \sqrt{2N/(N-1)}\|M\|_{2,N}$ by positivity. We have not assumed orthogonality of restricted Weyl matrices.

The triangle inequality in operator norm also gives

$$\|[G_C, C]\|_{\mathrm{op}} \le 2048/K, \qquad \|[C, \bar{C}]\|_{\mathrm{op}} \le 2048/K. \tag{CB.13}$$

The usual two-exponential Duhamel inequality is $\|e^{ixA}e^{ixD} - e^{ix(A+D)}\| \le x^2\|[A,D]\|/2$ in any unitarily invariant norm. Applying it first to -G_C,C, then to C-G_C and $\widetilde{U}_0 G_C \widetilde{U}_0^*$, using (CB.10)–(CB.13), proves

$$\left\| e^{-i\zeta G_C}(e^{i\zeta C}\widetilde{U}_0)e^{i\zeta G_C} - e^{i\zeta \bar{C}}\widetilde{U}_0 \right\|_{2,-} \le \frac{1000\zeta^2}{\sqrt{K}} \quad (K \ge 11). \tag{CB.14}$$

Indeed the less rounded bound is $\zeta^2(2048/K + 16\sqrt{2N/(N-1)}/\sqrt{K})$, which is smaller than (CB.14) already using $K \ge 11$ and $N \ge 7$. This is another estimate valid for every real zeta, not a formal expansion requiring a small generator.

Combining the two gauges and telescoping unitary powers gives

$$\left| \tau_-(V_\varepsilon^{mK}) - \tau_-(e^{im\zeta K\bar{C}}) \right| \le |m| K R_N(\varepsilon) + 1000|m|\zeta^2\sqrt{K}. \tag{CB.15}$$

34.4 The two-species Gaussian limit

Lemma 34.6. *Put $X_K = (\sqrt{K}/2)\bar{C}$. For every fixed $r \geq 1$,*

$$\tau_-(X_K^{2r}) = (2r-1)!! + O_r(K^{-1}) + O_r(K^r/N). \tag{CB.16}$$

For every fixed odd m, eventually $|\tau_-(X_K^m)| \leq (4K)^{m/2}/(N-1)$. Consequently its empirical spectral measures converge weakly and in quadratic Wasserstein distance to standard normal.

Proof. There are 4K signed Weyl atoms, with coefficient +1 for species u and -1 for species z. For a word with central phase $e_N(\phi)$ and total index v, the exact odd trace is

$$e_N(\phi)\frac{N\mathbf{1}_{v=0}-1}{N-1}. \tag{CB.17}$$

This follows from $\operatorname{Tr} W(v) = N\mathbf{1}_{v=0}$ and $\operatorname{Tr}(RW(v)) = 1$. The noncentral contribution after normalization $(4K)^{-m/2}$ is at most $(4K)^{m/2}/(N-1)$.

Apply Lemma 34.4 to central words. The following bounded-cluster argument handles the unavoidable identities (CB.8), rather than incorrectly excluding them. In a nonzero polynomial relation with at most m signed atoms and coefficients 1 or $\sqrt{2}$, a gap d with $\lambda^d > 2m^2$ separates independently vanishing components. To see this, write $P = P_{\text{low}} + Z^{a+d}P_{\text{high}}$ with degree of P_{low} at most a. If $P_{\text{high}}(\lambda) \neq 0$, its nonzero integer norm and the bound $|P_{\text{high}}(\lambda^{-1})^{\text{conjugate}}| \leq \sqrt{2}m$ give $|P_{\text{high}}(\lambda)| \geq 1/(\sqrt{2}m)$. This contradicts cancellation by $|P_{\text{low}}(\lambda)| \leq \sqrt{2}m\lambda^a$. Iterating produces components with spans bounded solely in m.

Cancel opposite identical atoms first and restore these pairs after splitting. Every nonzero component then contains at least three occurrences. A two-atom relation cannot change species, because its two algebraic norms would be 1 and -2, and cannot change exponent, because $\lambda > 1$. A nonbalanced 2r-word thus has at most r-1 components. There are only finitely many partitions of its positions, species, signs and bounded offset patterns at fixed r. Each component has K possible cyclic translations, so there are $O_r(K^{r-1})$ such words. Choosing the largest-gap cut introduces no additional K factor: this count is over the actual translated components. Their normalized contribution is $O_r(1/K)$.

In a balanced word, pair identical species-time labels with opposite signs. For each pairing there are K^r time choices, 2^r species choices and 2^r sign choices. Their product is precisely the normalization $(4K)^r$. Repeated pair labels cause at most $O_r(K^{r-1})$ overcount. Signs from species z occur twice in each pair and hence multiply to +1. The Weyl phase is a product of crossing factors or their conjugates. For fixed species, each crossing time difference is uniform on the cyclic group. By (CB.11) and $|\prod u_j - 1| \leq \sum |u_j - 1|$, its average error is at most 128 times the number of crossings divided by K. No independence of the crossing factors is assumed. Summing over the $(2r-1)!!$ pairings gives (CB.16).

For odd m, reduce a lifted relation in $\mathbb{Z}[\sqrt{2}]$ modulo 2. Here $\lambda = 3 + 2\sqrt{2}$ reduces to 1. The coefficients of both 1 and $\sqrt{2}$ in the relation must be even. Both species counts would therefore be even, contradicting odd m. Thus no odd central word exists after lifting.

Bounded second moments yield tightness, higher even moments yield uniform integrability, and the Gaussian moment sequence is determinate, since its Carleman sum diverges. Therefore all weak subsequential limits are normal. Convergence of the second moments gives the claimed Wasserstein upgrade. End of proof.

Since $\zeta K\bar{C} = t_K X_K$, (CB.15) and Lemma 34.6 prove Theorem 34.1. More explicitly, the finite comparison error is bounded by

$$|m|\left(KR_N(\varepsilon) + \frac{250t_K^2}{\sqrt{K}}\right). \tag{CB.18}$$

For bounded t_K, $KR_N = O(K^{1/4}/\sqrt{N})$ follows directly from $\varepsilon^2 = t_K/(2s_N\sqrt{K})$ and $s_N \le 4\pi/N$. The remaining Gaussian characteristic-function error is $o(1)$; no universal $K^{-1/2}$ rate is asserted for that separate step.

34.5 Microscopic consequences without conflating unfolding conventions

For completeness we give the two spectral implications used in Theorem 34.2. First, the 4K Weyl atoms in C_j are distinct, so $\mathrm{Tr}(\bar{C}^2)/N = 4/K$. Distinctness follows within species from the unit conditions on $1 \pm h^j$ in the Pell construction. Between species, norm 1 versus -2 would require N to divide 3, which is impossible for $K \ge 3$. Consequently

$$\tau_-(\bar{C}^2) \le \frac{8N}{K(N-1)}.$$

The squared-overlap matrix between orthonormal eigenbases of two unitaries is doubly stochastic. Decomposing it into permutations yields an eigenvalue matching whose mean squared chord cost is bounded by normalized squared Hilbert–Schmidt distance. Circle distance in $\mathbb{R}/\mathbb{Z}$ is at most one quarter of chord distance. Using this matching, (CB.7),(CB.14), and $[\widetilde{U}_0, \bar{C}] = 0$, the RMS distance a of the spectrum of V_ε from the K-point lattice satisfies

$$a \le \frac{R_N(\varepsilon)}{4} + \frac{250\zeta^2}{\sqrt{K}} + \frac{|\zeta|}{2\pi}\sqrt{\frac{8N}{K(N-1)}}. \tag{CB.19}$$

When t_K tends to zero, Ka tends to zero, and K/d tends to zero.

For an arbitrary d-point circle multiset with this RMS lattice distance, placing arcs of half-width $w < 1/(2K)$ around the lattice and using Markov's inequality gives, for its unfolded cyclic gaps S_l,

$$d^{-1}\#\{l : S_l > x\} \le 2a^2/w^2 + K/d + 2Kw/x. \tag{CB.20}$$

There are at most da^2/w^2 bad points; at most twice as many gaps touch one, at most K other gaps leave an arc, and the within-arc gaps have total length at most 2Kw. This proves (CB.20). Partitioning the arcs into cells of length at most x/d and applying Cauchy–Schwarz to their occupancies also gives the ordered close-pair bound

$$R_2(x) \ge \frac{\max(0, 1 - a^2/w^2)^2}{K/d + 2Kw/x} - 1. \tag{CB.21}$$

Take $w = (Ka)^{2/3}/K$ for (CB.20), and w=2a for (CB.21). These prove subcritical spacing collapse and divergent close-pair counts. The a=0 case follows by counting at most K nonzero gaps and using $R_2(x) \ge d/K - 1$. The average spacing remains exactly one: weak convergence to δ_0 does not imply convergence of first moments, which are carried by rare long gaps.

For bounded nonzero critical parameter a different argument is necessary. Let $\phi_L(s) = L^{-1}(1 - |s|/L)_+$, a compactly supported nonnegative function with integral one and Fourier transform $\mathrm{sinc}^2(L\xi)$. For the eigenangles θ_l in $\mathbb{R}/\mathbb{Z}$, define

$$C_d(\phi_L) = \frac{1}{d}\sum_{l \ne j}\sum_{q \in \mathbb{Z}} \phi_L(d(\theta_l - \theta_j + q)).$$

Poisson summation, including the zero Fourier mode and the removed diagonal, proves, for $d > L$,

$$C_d(\phi_L) \ge 1 - \frac{1}{L} + 2\sum_{m=1}^{M} \mathrm{sinc}^2(LmK/d)|\tau_-(V_\varepsilon^{mK})|^2. \tag{CB.22}$$

All omitted Fourier coefficients are nonnegative. Take K to infinity at fixed M, use (CB.2), then let M grow. For $t > 0$ this gives

$$\liminf C_d(\phi_L) \geq \Theta(t) - 1/L, \qquad \Theta(t) = \sum_{m\in\mathbb{Z}} e^{-m^2t^2} > 1. \tag{CB.23}$$

For t=0 the lower bound is infinite. If the pair measures converged on compact tests to a bounded density R(s) tending to 1 at infinity, then $\int \phi_L R \to 1$ as L grows. This contradicts (CB.23). A bounded subsequence of $|\varepsilon|K^{1/4}/\sqrt{N}$ has a further subsequence with finite limiting t_K, so (CB.4) follows. This is a necessary-condition argument, not an exchange of a growing test support with the semiclassical limit.

An equivalent moving-window consequence is

$$\liminf \mathrm{Var}_\theta[\#\{l : \theta_l \in [\theta, \theta + L/d)\}] \geq L^2(\Theta(t) - 1).$$

Indeed the finite variance is $L^2(C_d(\phi_L) - 1) + L$. The underlying positive Fourier test was already established in the earlier Floquet note; it is not claimed as a new result of this pass.

35 The exact fourth moment of the generated two-species observable

This is an analytic verification addendum to `pell_coboundary_resonance.md`. It uses that note's proved four-atom classification, not a new universality assumption. Let $K \geq 11$ be an odd prime, with its Pell modulus N, and put

$$u = (2, 4), \quad z = (4, 4), \quad X_K = (4K)^{-1/2} \sum_{j=0}^{K-1} (B_{T^j u} - B_{T^j z}), \qquad \mu_{4,N} = N^{-1} \operatorname{Tr} X_K^4.$$

The trace in this definition is on the full N-dimensional space. Odd-space normalization and its noncentral error are stated separately below.

35.1 Closed finite formula

Write $c_a = \cos(a\pi/N)$ and

$$g_j = \cos^2(8\pi P_{2j}/N) + \cos^2(16\pi P_{2j}/N) + 2\cos^2(8\pi Q_{2j}/N).$$

Then the exact formula is

$$\boxed{\mu_{4,N} = 1 + \frac{1}{4K} + \frac{1}{K} \sum_{j=1}^{(K-1)/2} g_j + \frac{c_{32}c_{16} + c_{24}^2 + 2c_8^2}{K}.} \tag{CF.F1}$$

This is a real trigonometric identity, not a floating-point assertion.

Proof. Expand X_K^2 in the orthonormal Weyl basis. Its coefficient at zero is 1. Doubling one signed atom gives 4K distinct keys, each of coefficient $1/(4K)$, contributing $1/(4K)$ to the squared norm. For a distinct-time unordered signed pair the coefficient is the product of its species signs multiplied by $\cos(\pi\omega/N)/(2K)$. Squaring these contributions before combining coincident keys gives $K^{-2}\sum_{j=1}^{K-1}(K-j)g_j$. The equality $g_{K-j} = g_j$ reduces this to the sum in (CF.F1). Determinants of the original even integer vectors are even; centered modular division by two therefore has the same phase as the ordinary half of this determinant. No sign is lost at the cyclic seam.

The complete four-atom classification gives exactly two distinct-time double classes at each cyclic time, with their negatives. The same-species class $z_{j+1} - z_j = u_{j+1} + u_j$ has coefficient products $c_{32}c_{16}/(4K^2)$; its 2K instances contribute the cross term $c_{32}c_{16}/K$. The mixed class $z_{j+1} - u_j = u_{j+1} + z_j$ similarly contributes c_{24}^2/K. The signs of both coefficients within each class agree, including the minus signs in the mixed class.

There remain the 4K equal-time cross-species pairs. They form 2K double classes by $z_{j+1} - u_{j+1} = z_j + u_j$, with combined coefficient $-c_8/K$. Their contribution is $2c_8^2/K$. The classification also excludes all collisions of doubled-atom keys with these classes or with distinct-time keys. Summing the disjoint classes proves (CF.F1). End of proof.

35.2 Explicit limiting correction and finite remainder

Let $r = \lambda^{-1}$ and set

$$a = \frac{8\pi}{\sqrt{2}\alpha}, \qquad b = \frac{8\pi}{\alpha}.$$

Define

$$\gamma_C = -\frac{9}{4} + \sum_{\ell=0}^{\infty}[\sin^2(ar^\ell) + \sin^2(2ar^\ell) + 2\sin^2(br^\ell)]. \tag{CF.F2}$$

The series is absolutely convergent. Its numerical value begins $4.1785026500210028\ldots$. Define

$$E_K = \frac{(3a+2b)(r^K + r^{(K+1)/2})}{(1-r^K)(1-r)} + \frac{(5a^2+2b^2)r^{K-1}}{1-r^2} + \frac{1344\pi^2}{N^2}.$$

An explicit finite bound is

$$|K(3 - \mu_{4,N}) - \gamma_C| \le E_K. \tag{CF.F3}$$

In particular $\mu_{4,N} = 3 - \gamma_C/K + O(1/(KN))$ along this family.

To prove the estimate, reindex $j = (K-1)/2 - \ell$. The exact ratios are

$$\frac{P_{2j}}{N} = \frac{r^\ell}{\sqrt{2}\alpha}\frac{1-\lambda^{-2j}}{1-r^K}, \qquad \frac{Q_{2j}}{N} = \frac{r^\ell}{\alpha}\frac{1+\lambda^{-2j}}{1-r^K}.$$

Use $|\sin^2 x - \sin^2 y| \le |x-y|$. Summing the argument errors gives the first term of E_K. The omitted series begins at $\ell = (K-1)/2$ and is bounded by the second term, using $\sin^2 x \le x^2$. Finally

$$|c_{32}c_{16} + c_{24}^2 + 2c_8^2 - 4| \le 1344\pi^2/N^2.$$

Since $\sum_j g_j = 2K - 2$ minus the finite sine-square sum, substitution in (CF.F1) yields precisely the constant $-9/4$ in (CF.F2) and proves (CF.F3).

For the odd-space fourth moment $\mu_{4,-}$, the exact parity insertion formula gives the separate bound

$$\left|\mu_{4,-} - \frac{N}{N-1}\mu_{4,N}\right| \le \frac{16K^2}{N-1}. \tag{CF.F4}$$

Thus $\mu_{4,-} = 3 - \gamma_C/K + O(K^2/N)$. The much larger odd-trace error must not be replaced by the full-trace error in (CF.F3).

36 Finite Fourier alphabets and cohomological variance

Fix $T = \left(\begin{smallmatrix} 3 & 2 \\ 4 & 3 \end{smallmatrix}\right)$, $\alpha = 1 + \sqrt{2}$, $\lambda = \alpha^2$, and $N + F\sqrt{2} = \alpha^K$ for odd prime K. Use $U_0^K = I$, $U_0 W(v) U_0^* = W(Tv)$ and $W(a,b)f(x) = e_N(a(x - b/2))f(x-b)$ as in the finite construction. Let $B_v = W(v) + W(-v)$. Write $\tau_N = \mathrm{Tr}\,/N$, $\tau_- = \mathrm{Tr}_-\,/d$, $d = (N-1)/2$, and $c_N = \sqrt{2N/(N-1)}$.

36.1 Uniform finite-alphabet arithmetic

Choose nonzero integral representatives $v_1, \dots, v_S$ from distinct signed T-orbits: $v_a \neq \pm T^j v_b$ for a different from b and every integer j. Put $\beta_{(a,b)} = b + \sqrt{2}a$, so $\beta_{Tv} = \lambda\beta_v$, and let $M \geq 1$ bound both real embeddings of all β values.

Lifting lemma. If

$$\lambda^{\lceil K/m \rceil}(1 - \lambda^{-K})^2 > 4m^2M^2, \tag{FA.1}$$

a signed modular zero relation of m orbit atoms, after cyclic rotation, is a characteristic-zero relation with all exponents between 0 and $K - \lceil K/m \rceil$.

Indeed cut at the largest of the gaps between at most m occupied labels. Rotation is valid modulo N since $T^K = I$. The resulting algebraic integer P is in $N\mathbb{Z}[\sqrt{2}]$, since its two coordinates vanish modulo N. If nonzero its integer norm is divisible by N^2, but its two real embeddings bound that norm by $m^2M^2\lambda^{K-\lceil K/m \rceil} < N^2$. Here $N^2 = \lambda^K(1 - \lambda^{-K})^2/4$. This proves the claim also for composite N, without invertibility of the chosen β values.

A gap B satisfying $\lambda^B > m^2M^2$ splits a lifted relation into independently vanishing pieces. Write it as $P_{\text{low}} + \lambda^{a+B}P_{\text{high}}$ with low exponents at most a and high exponents nonnegative. A nonzero high part has absolute value at least $1/(mM)$ because its conjugate is bounded by mM and its norm is a nonzero integer. It cannot be cancelled by a low part bounded by $mM\lambda^a$.

There is no one-atom zero relation. A two-atom relation only cancels an identical species and exponent with the opposite sign, by the orbit assumption and $\lambda > 1$.

Bounded multiplicity lemma. Choose integer B with $\lambda^B > 16M^2$. Put $D_* = 3B$ and $J = [2S(2D_* + 1)]^2$. Once (FA.1) holds for m=4, every signed unordered pair key with distinct time labels is nonzero and has at most J representations.

For two equal keys cancel identical opposite atoms. A trivial cancellation gives the same pair. Otherwise four occurrences remain, and cannot split into proper zero components: the only possible two-plus-two decomposition would itself be cancellation. The gap lemma confines all four to one cyclic interval of length at most D_*. Pin an atom in the first pair. Each atom of any other representation lies within circular distance D_* of it, giving at most $2S(2D_* + 1)$ choices per atom. Squaring gives J, also when this neighbourhood covers the circle. A zero pair would lift to a forbidden two-atom relation. This bound allows structural collisions; it is not claimed sharp.

36.2 Ordered Gaussian moments and the variance kernel

Let $Y_{s,K} = (2K)^{-1/2} \sum_{j<K} B_{T^j v_s}$. Their ordered joint moments converge to those of S independent standard real Gaussians. In particular, for r in a fixed bounded subset of $\mathbb{R}^S$,

$$\tau_-\big((\sum_s r_s Y_{s,K})^{2q}\big) = (2q-1)!!\|r\|_2^{2q} + O(K^{-1}) + O(K^q/N). \tag{FA.2}$$

Odd moments of fixed order m are $O(K^{-1/2}) + O(K^{m/2}/N)$, not in general exponentially small. The same assertions hold for the full normalized trace.

Proof. The exact odd trace of a Weyl word is

$$\tau_-(e_N(\phi)W(v)) = e_N(\phi)\frac{N\mathbf{1}_{v=0} - 1}{N - 1}. \tag{FA.3}$$

The subtracted parity term in the mth moment is bounded by $(\sqrt{2K}\|r\|_1)^m/(N-1)$. Lift central words, cancel opposite identical atoms, and split the rest at large gaps. Nontrivial components have at least three occurrences and bounded span at fixed m. Restore the cancelled pairs as two-occurrence components. A nonpaired 2q-word then has at most q-1 components, while a word of length 2q+1 has at most q. There are finitely many partitions, species, signs and bounded offsets; each component has K translations. Their normalized contributions are respectively

$O(1/K)$ and $O(1/\sqrt{K})$. The cyclic cut introduces no extra K: translations were already counted on the original cyclic label set.

The remaining words pair identical species-time atoms with opposite signs. Each pairing gives the Wick covariance product; repeated labels cause only $O(1/K)$ normalized overcount. The phase is a product of crossing factors $e_N(\omega(v_s, T^j v_t))$ and conjugates. Their uniform sum bound (FA.5) below, uniformity of each pair-label difference, and $|\prod z_i - 1| \le \sum |z_i - 1|$ give an $O(1/K)$ phase error. No independence of crossing phases is required. This proves the ordered mixed version of (FA.2). Tightness, uniform integrability from higher even moments and Gaussian moment determinacy give weak convergence of each real linear combination, also for bounded coefficient vectors converging along subsequences.

For a fixed finite real cosine symbol $f = \sum_v a_v B_v$, first combine v and -v, then group frequencies by signed T-orbits O. Write $c_O = \sum_{v \in O} a_v$. Exactly,

$$\bar{f} = \frac{1}{K} \sum_{j<K} U_0^j f U_0^{-j} = \sum_O c_O \frac{1}{K} \sum_{j<K} B_{T^j v_O}.$$

Consequently $\sqrt{K}\,\bar{f}$ has Gaussian trace limit of variance

$$\sigma_f^2 = 2 \sum_O c_O^2. \tag{FA.4}$$

Fourier orthogonality identifies this with the classical Green–Kubo sum $\sum_{n \in \mathbb{Z}} \int f\, \mathcal{C}^n f$, where $\mathcal{C}$ acts on Fourier indices by T. Only finitely many terms are nonzero for finite support. The kernel consists exactly of finite Fourier coboundaries: on each orbit, if a finite sequence a_n has sum zero, then $b_n = \sum_{j \le n} a_j$ has finite support and $a_n = b_n - b_{n-1}$. Conversely a difference has zero sum. The classical interpretation is prior mathematics, not a novelty claim.

36.3 Explicit phase and commutator bounds

Define

$$a_j = \max_{s,t} |e_N(\omega(v_s, T^j v_t)) - 1|, \quad L_K = \sum_j a_j, \quad W_K = K \sum_j |c_j| a_j, \quad c_j = \frac{K - 1 - 2j}{2K}.$$

Species exchange gives $a_{K-j} = a_j$. The embeddings give $|\omega(v_s, T^\delta v_t)| \le M^2 \lambda^{|\delta|}$ for centered δ. With $r = \lambda^{-1}$ and $|\delta| = (K-1)/2 - \ell > 0$, $a_j \le 4\pi M^2 r^\ell / [\alpha(1 - r^K)]$; also $a_0 \le 2\pi M^2 / N$. The two weights are ℓ/K and $(\ell+1)/K$. Summing geometric series gives

$$L_K \le \frac{4\pi M^2}{1 - r^K} + \frac{2\pi M^2}{N}, \qquad W_K \le \frac{2\sqrt{2}\pi M^2}{1 - r^K} + \frac{\pi M^2 K}{N}. \tag{FA.5}$$

Both are less than $32M^2$ for $K \ge 3$. The identity needed for the weighted sum is $(1 + r)/[\alpha(1 - r)^2] = 1/\sqrt{2}$.

For $Q = \sum_s r_s B_{v_s}$, set $Q_j = U_0^j Q U_0^{-j}$, $\bar{Q} = K^{-1} \sum Q_j$, $G = \sum c_j Q_j$ and $l = \|r\|_1$. Then $G - U_0 G U_0^* = Q - \bar{Q}$, and

$$\|[G, Q]\|_{\mathrm{op}} \le 4l^2 W_K / K, \quad \|[Q, \bar{Q}]\|_{\mathrm{op}} \le 4l^2 L_K / K.$$

The distinct-time expansion of $[G, \bar{Q}]$ has coefficient $(s - t)/K^2$. Cauchy–Schwarz within each of the at most J representations of a key, followed by full-space Weyl orthogonality, yields the useful sharper bound

$$\|[G, \bar{Q}]\|_{2,N}^2 \le \frac{4J\|r\|_2^4}{K^4} \sum_{j=1}^{K-1} (K - j) j^2 a_j^2 \le \frac{J\|r\|_2^4 L_K}{K}. \tag{FA.6}$$

The last inequality uses symmetry, $a_j^2 \le 2a_j$ and $j(K-j) \le K^2/4$. The weaker bound with $S^2\|r\|_\infty^4$ in place of $\|r\|_2^4$ is also valid. Restriction to odd space costs c_N by positivity; restricted Weyl orthogonality is not assumed.

Finite normal form. For all real x, once four-atom lifting holds,

$$\begin{aligned}&\|e^{-ixG}(e^{ixQ}U_0)e^{ixG} - e^{ix\bar{Q}}U_0\|_{2,-}\\ &\quad\le x^2\left[\frac{2l^2(W_K+L_K)}{K} + \frac{c_N\|r\|_2^2\sqrt{JL_K}}{2\sqrt{K}}\right].\end{aligned} \tag{FA.7}$$

Join -G,Q and then Q-G,$U_0GU_0^*$ using the two-factor Duhamel inequality. The second commutator is $[Q-G,\bar{Q}]$. Equations (FA.5)–(FA.6) prove (FA.7), without requiring xG small. On full space replace c_N by one.

At $x = \tau/\sqrt{K}$ this gives

$$\tau_-((e^{i\tau f/\sqrt{K}}U_0)^{mK}) \longrightarrow e^{-m^2\tau^2\sigma_f^2/2} \tag{FA.8}$$

for every fixed finite real cosine symbol, not only orbit representatives. Indeed write $f = f_{\mathrm{rep}} + g - \mathcal{C}g$ with fixed finite g. The triple $e^{-ixg}e^{ixf}e^{ix\mathcal{C}g}$ has sum f_{rep}; its commutators $[-g,f]$ and $[f-g,\mathcal{C}g]$ are $O_f(1/N)$ in norm because their frequencies are fixed. Its error is $O_f(x^2/N)$, hence $O_f(1/N)$ over K steps. Apply (FA.7) to f_{rep} and (FA.2) to its average.

36.4 Further scope

General real Fourier symbols on the full quantum space are covered by replacing r_sB_v with $c_sW(v) + \bar{c}_sW(-v)$. In the proof of (FA.2), the two orientations of a pair sum to $2\,\mathrm{Re}(c_a\bar{c}_b)$; the scalar variance is $2\sum|c_O|^2$. All norm estimates use absolute coefficients and remain valid. On odd space retain inversion-even symbols. The standard physical centered operator is $(-1)^{ab}W(a,b)$. Since $T = I$ modulo 2, this sign is constant on each orbit and does not change the variance or the covariance. It must nevertheless be included when identifying a physical Fourier symbol.

The smooth-symbol extension, with a justified growing cutoff, is proved in `pell_smooth_observable_recurrence.md`. The distinct-orbit tool also proves the mixed-kick crossover. No constants here supply Heisenberg-scale joint traces or positive BGS universality.

37 Infinite Fourier tails and a smooth recurrence law

Let $f : \mathbb{T}^2 \to \mathbb{R}$ have mean zero and Fourier coefficients satisfying, for some $p > 5$,

$$|\widehat{f}(v)| \le C_f(1+|v|)^{-p}. \tag{SR.S1}$$

In particular every real mean-zero C^6 function satisfies (SR.S1), with p=6, by applying three Laplacians in Fourier space. Use the fixed index action $T = \left(\begin{smallmatrix}3&2\\4&3\end{smallmatrix}\right)$ and its Pell quantizations, as in the finite-alphabet proof. The dual classical action is $\mathcal{C}f(x) = f(T^{\mathsf{T}}x)$. Define the physical centered quantization by the absolutely convergent series

$$A_N(f) = \sum_{v=(a,b)\in\mathbb{Z}^2} \widehat{f}(v)(-1)^{ab}W_N(v), \qquad U_{N,f}(\tau) = e^{i\tau A_N(f)/\sqrt{K}}U_0. \tag{SR.S2}$$

The factor $(-1)^{ab}$ converts modular-half Weyl operators to the physical centered convention. Since T is the identity modulo two, it is constant on each T-orbit. Egorov covariance and Hermiticity are exact.

Let O run through the oriented nonzero T-orbits in $\mathbb{Z}^2$ and put

$$\sigma_f^2 = \sum_O \left| \sum_{v \in O} \widehat{f}(v) \right|^2 = \sum_{n \in \mathbb{Z}} \int_{\mathbb{T}^2} f(x)(\mathcal{C}^n f)(x)\, dx. \tag{SR.S3}$$

Both series converge absolutely: each ordered pair of nonzero frequencies can correspond to at most one shift n, and the Fourier coefficients are absolutely summable. Formula (SR.S3) is nonnegative. For real inversion-even f it is equivalently $2\sum_{\text{signed } O}(\sum_{v\in O} a_v)^2$ in cosine coefficients with one representative of each pair v,-v.

Theorem 37.1 (smooth recurrence law)**.** *For every fixed real τ and integer m,*

$$\boxed{\frac{1}{N}\operatorname{Tr} U_{N,f}(\tau)^{mK} \longrightarrow e^{-m^2\tau^2\sigma_f^2/2}.} \tag{SR.S4}$$

If f is inversion-even, the same conclusion holds on the odd subspace with normalized trace τ_- and dimension $(N-1)/2$. The Kth-power empirical spectral measure tends to the wrapped normal law of variance $\tau^2\sigma_f^2$. Zero variance gives a point mass, not a nondegenerate normal law. The theorem applies to each fixed f, not uniformly over an unbounded smoothness ball.

37.1 Fourier tails and bounded-orbit reduction

Take $f_R = \sum_{0<|v|\le R} \widehat{f}(v) e^{2\pi i v\cdot x}$. The quantization tail has operator norm at most its Fourier l1 norm, hence

$$\|A_N(f - f_R)\|_{\text{op}} \le C_f' R^{2-p}. \tag{SR.S5}$$

For K steps at strength $\tau/\sqrt{K}$, telescoping thus costs $C_{f,m,\tau}\sqrt{K}R^{2-p}$. A fixed cutoff is insufficient for this propagator comparison. This is why a growing cutoff must be justified.

Choose one minimal-norm integral representative in every signed orbit meeting the ball of radius R. The representative lies in that ball, and all its two real embeddings are bounded by $M_R = 2\max(1, R)$. For any two frequencies in that orbit and ball, the difference of their indices is $O(\log(2R))$. Indeed their β values have absolute values between $1/M_R$ and M_R, since their nonzero integer norms have absolute value at least one. Their ratio is a power of λ. The same bound holds for negative powers by exchanging the two points.

Intermediate orbit points between two such endpoints stay in the ball: $|T^t v|^2 = A\lambda^{2t} + B\lambda^{-2t} + C$, with A,B positive, is convex in real t. Thus the orbitwise telescoping construction gives exactly

$$A_N(f_R) = Q_R + g_R - \mathrm{Ad}_{U_0}(g_R), \tag{SR.S6}$$

where Q_R uses one signed-orbit representative, has coefficient l1 norm bounded in terms of f, has at most $S_R = O(R^2)$ species, and $\|g_R\|_{\text{Fourier l1}} = O_f(\log(2R))$. Its frequencies have size $O(R)$. These assertions also hold for complex Fourier coefficients paired with their conjugates, since the physical parity sign is constant on orbits.

Commutators of two Fourier sums with frequencies of size O(R) have norm at most CR^2/N times the product of their coefficient l1 norms. This follows from the exact Weyl commutator and $|\sin x| \le |x|$; it remains true for modular half-exponents, including odd determinants. Using the two-factor Duhamel estimate twice in (SR.S6), the gauge e^{-ixg_R} removes the coboundary with error $O_f(x^2R^2\log^2(2R)/N)$, for every real x.

37.2 Uniform normal form for the growing representative alphabet

In the finite-alphabet lemma take $B_R = O(\log(2R))$ with $\lambda^{B_R} > 16M_R^2$, and $J_R = [2S_R(6B_R + 1)]^2 = O(R^4\log^2(2R))$. Its four-atom lifting condition holds for all sufficiently large K whenever R is a fixed power of K, since $\lambda^{\lceil K/4\rceil}$ dominates R^2. The phase bounds give $L_K, W_K = O(R^2)$.

The sharper Hilbert–Schmidt estimate uses $\|r\|_2^2$, not $S_R\|r\|_\infty^2$. Since $\|r\|_2 \le \|r\|_1 = O_f(1)$, formula (7) of the finite-alphabet proof, at $x = \tau/\sqrt{K}$ and after mK steps, costs

$$O_{f,m,\tau}\left(\frac{R^2}{K} + \frac{R^3\log(2R)}{\sqrt{K}}\right). \tag{SR.S7}$$

The period average of the coboundary vanishes exactly. Thus the target after both gauges is $\exp(im\tau\sqrt{K}\,\overline{A_N(f_R)})$. Combining (SR.S5)–(SR.S7) yields the fully specified finite comparison

$$\begin{aligned}&\left|\tau\big(U_{N,f}(\tau_0)^{mK}\big) - \tau\big(e^{im\tau_0\sqrt{K}\overline{A_N(f_R)}}\big)\right| \\ &\quad\le C_{f,m,\tau_0}\left[\sqrt{K}R^{2-p} + \frac{R^2}{K} + \frac{R^3\log(2R)}{\sqrt{K}} + \frac{R^2\log^2(2R)}{N}\right].\end{aligned} \tag{SR.S8}$$

Here τ without a subscript on an operator means either normalized trace; τ_0 is the fixed real kick parameter. This notation avoids confusing the trace with that parameter. Choose $R = K^{1/(p+1)}$. Then the right side is

$$O_{f,m,\tau_0}\left(K^{-(p-5)/(2(p+1))}\log K\right), \tag{SR.S9}$$

up to an exponentially small term. For p=6 the exponent is 1/14. This is a comparison rate, not a claimed rate for the final Gaussian limit.

37.3 Exact second moments close the approximation argument

One must not insert the growing alphabet directly into a fixed-moment CLT without controlling its constants. Instead use a separate uniform second moment and a two-limit argument.

For any real Fourier polynomial h supported in the radius-R ball, once the two-atom lifting condition holds and $\log R = o(K)$,

$$\tau_N\left((\sqrt{K}\,\overline{A_N(h)})^2\right) = \sum_O\left|\sum_{v\in O}\widehat{h}(v)\right|^2 = \sigma_h^2 \le \|\widehat{h}\|_1^2. \tag{SR.S10}$$

To verify it, expand the squared average in the full Weyl basis. A congruence $T^j v = T^k w$ lifts, after cutting between its two labels, to a true signed orbit relation. The true time offset is $O(\log R)$, so it has exactly one representative modulo K; each such offset supplies K ordered time pairs. No extra alias occurs. This proves the equality. Physical parity signs are constant on each orbit and cancel in the absolute square. For an even h, positivity bounds the odd norm by c_N times the full norm.

Fix a cutoff L and let $R = K^{1/(p+1)}$. Apply (SR.S10) to $h = f_R - f_L$. Uniformly for sufficiently large K, its averaged observable differs from that of f_L in normalized Hilbert–Schmidt norm by at most $c_N\|\widehat{f - f_L}\|_1$ (drop c_N on full space). The Duhamel inequality for exponentials in this norm then bounds the characteristic-function difference by $|m\tau_0|c_N\|\widehat{f - f_L}\|_1$. For fixed L the finite-alphabet theorem supplies the Gaussian limit with variance $\sigma_{f_L}^2$. Finally let L grow. Equation (SR.S3) and absolute summability imply $\sigma_{f_L}^2 \to \sigma_f^2$; the tail bound tends to zero. This proves (SR.S4) together with (SR.S8), without a growing-order moment argument or an unjustified exchange of cutoffs.

The same approximation also proves the joint characteristic-function trace limit of finitely many averaged smooth observables: use telescoping of a fixed product of unitary exponentials, (SR.S10), and the ordered finite-alphabet limit. The covariance is the polarized Green–Kubo form. This does not assert that the unaveraged finite operators commute or possess a joint eigenvalue law.

37.4 Consequences and limitations

The theorem includes generic finite Fourier symbols, smooth symbols with infinitely many frequencies, and degenerate variance. The familiar classical coboundary criterion concerns the kernel of (SR.S3); it is not a newly invented criterion. No constant classical kick is covered: the action amplitude in (SR.S2) is proportional to $1/(N\sqrt{K})$. The time is $mK = O(\log N)$ for fixed m, not the Heisenberg scale O(N). Neither the variance formula nor the wrapped-normal Kth-power density proves local spectral universality.

The companion exact test independently groups integer Fourier modes into signed orbits and compares (SR.S10) with modular Weyl coefficient accumulation. Small-modulus aliasing is retained as a negative control. Finite numerical agreement is not substituted for the cutoff proof above.

38 Smooth residual kicks and a quantified compensation

Use the physical Weyl convention, fixed index action T, Pell moduli and normalized traces of the smooth recurrence theorem. Let g be real C^7, let f be real mean-zero C^6, and put

$$d = \mathcal{C}g = g \circ T^{\mathsf{T}}, \qquad r_g = \frac{1}{4\pi}\{g, d\}, \qquad \{g, d\} = \partial_1 g \partial_2 d - \partial_2 g \partial_1 d. \tag{SC.R1}$$

The symbol r_g has mean zero and is C^6. Write $A = A_N(g)$, $D = A_N(d) = U_0 A U_0^*$, $F = A_N(f)$, $R = A_N(r_g)$, and consider

$$\mathcal{V}_{\varepsilon,\eta} = e^{i[\varepsilon(A-D)+\eta F]} U_0. \tag{SC.R2}$$

Set $a_K = \sqrt{K}\eta_K$ and $t_K = \sqrt{K}\varepsilon_K^2/N$.

Theorem 38.1 (smooth joint crossover)**.** *If $a_K \to a \in \mathbb{R}$ and $t_K \to t \geq 0$, then for every fixed integer m,*

$$\boxed{\tau_N(\mathcal{V}_{\varepsilon_K,\eta_K}^{mK}) \longrightarrow \exp[-m^2\sigma_{af+tr_g}^2/2].} \tag{SC.R3}$$

For inversion-even f and g the same holds on the odd sector. The variance is

$$\sigma_{af+tr_g}^2 = a^2\sigma_f^2 + 2at\,\Gamma(f, r_g) + t^2\sigma_{r_g}^2, \tag{SC.R4}$$

where Gamma is the polarized Green–Kubo form. Its cross term need not vanish. Thus the sum of two variances from the special three-orbit example is not a universal rule. The resulting Kth-power law is wrapped normal, including its degenerate zero-variance case. No independence of finite spectra is assumed.

Theorem 38.2 (compensated pinning)**.** *Define the distinct, engineered family*

$$\mathcal{V}_\varepsilon^{\mathrm{comp}} = \exp\{i[\varepsilon(A - D) - \varepsilon^2 R/N]\}U_0. \tag{SC.R5}$$

There are explicit finite constants C_g, D_g, independent of N,K, such that for every real ε

$$\boxed{\|e^{-i\varepsilon A}\mathcal{V}_\varepsilon^{\mathrm{comp}} e^{i\varepsilon A} - U_0\|_{\mathrm{op}} \leq C_g|\varepsilon|^3/N^2 + D_g\varepsilon^2/N^3.} \tag{SC.R6}$$

Consequently, if $|\varepsilon_K| = o(N^{2/3}K^{-1/3})$, all fixed recurrence traces tend to one. The globally unfolded neighbouring spacings tend weakly to δ_0 and normalized close-pair counts diverge at every fixed positive microscopic distance, on full space or either applicable parity sector. This statement preserves multiplicities and uses global mean spacing. It is not a universality claim after clusterwise unfolding.

38.1 Exact commutator estimates from Fourier moments

For h define $M_j(h) = \sum_v |v|^j |\widehat{h}(v)|$. The moments through j=3 of g and d are finite under the stated regularity. The physical Weyl multiplication is exactly $W^{\mathrm{ph}}(v)W^{\mathrm{ph}}(w) = e^{\pi i\omega(v,w)/N}W^{\mathrm{ph}}(v+w)$, even when indices must be reduced modulo N. Therefore

$$\|[A_N(h), A_N(k)]\|_{\mathrm{op}} \le (2\pi/N)M_1(h)M_1(k). \tag{SC.R7}$$

The Fourier coefficient of r_g is $-\pi\sum_{v+w=z}\omega(v,w)\widehat{g}(v)\widehat{d}(w)$. Since $|\sin x - x| \le |x|^3/6$ for all real x,

$$\|[A,D] + (2i/N)R\|_{\mathrm{op}} \le \frac{\pi^3 M_3(g)M_3(d)}{3N^3}. \tag{SC.R8}$$

This fixes both the sign and the factor $1/(4\pi)$ in (SC.R1). Moreover, with $H_* = M_2(g)M_1(d) + M_1(g)M_2(d)$,

$$\|[A,[A,D]]\| \le L_A/N^2, \quad \|[D,[A,D]]\| \le L_D/N^2, \quad L_A = 4\pi^2 M_1(g)H_*, \quad L_D = 4\pi^2 M_1(d)H_*. \tag{SC.R9}$$

For example a term with three frequencies u,v,w is bounded by $4\pi^2|u||v||w|(|v|+|w|)/N^2$ times its coefficient product. Summing proves (SC.R9), so no unbounded pseudodifferential remainder is being assumed.

38.2 Uniform loop and the mixed error budget

The three-factor unitary Duhamel lemma in the residual note gives, for all real ε,

$$\|e^{-i\varepsilon A}e^{i\varepsilon(A-D)}e^{i\varepsilon D} - e^{-\varepsilon^2[A,D]/2}\| \le |\varepsilon|^3(L_A/2 + L_D/6)/N^2.$$

Combining with (SC.R8) yields

$$\|e^{-i\varepsilon A}e^{i\varepsilon(A-D)}e^{i\varepsilon D} - e^{i\varepsilon^2 R/N}\| \le C_3|\varepsilon|^3/N^2 + C_2\varepsilon^2/N^3, \tag{SC.R10}$$

where $C_3 = L_A/2 + L_D/6$ and $C_2 = \pi^3 M_3(g)M_3(d)/6$. The estimates do not require $|\varepsilon|\|A\|$ small.

Split (SC.R2) into $e^{i\eta F}e^{i\varepsilon(A-D)}U_0$ by the two-factor bound. Its cost is at most $\pi|\varepsilon\eta|M_1(f)(M_1(g) + M_1(d))/N$. Under the gauge, the first factor becomes $e^{i\eta\,\mathrm{Ad}_{e^{-i\varepsilon A}}F}$. Replacing it by $e^{i\eta F}$ costs at most $2\pi|\varepsilon\eta|M_1(g)M_1(f)/N$ by integrating the conjugation. Finally join it to $e^{i\varepsilon^2 R/N}$ using (SC.R7). Hence

$$\|e^{-i\varepsilon A}\mathcal{V}_{\varepsilon,\eta}e^{i\varepsilon A} - e^{iA_N(\eta f + \varepsilon^2 r_g/N)}U_0\| \le E_N, \tag{SC.R11}$$

$$E_N = C_3|\varepsilon|^3/N^2 + C_2\varepsilon^2/N^3 + \pi M_1(f)(3M_1(g) + M_1(d))|\varepsilon\eta|/N + \pi M_1(f)M_1(r_g)|\eta|\varepsilon^2/N^2.$$

The moment $M_1(r_g)$ is finite; directly it is at most πH_*. For bounded a_K,t_K, $KE_N = O(K^{1/4}/\sqrt{N}) + O(1/N)$ and tends to zero.

The effective symbol is $(a_K f + t_K r_g)/\sqrt{K}$. The smooth recurrence proof is uniform over bounded coefficients in the fixed two-dimensional span of f and r_g: its Fourier bounds are uniform, and its fixed-cutoff characteristic-function argument allows converging coefficient vectors. It therefore applies without the invalid telescoping estimate $\sqrt{K}|a_K - a|$. Telescoping (SC.R11) mK times and then using that uniform proof establishes (SC.R3)–(SC.R4).

38.3 Why compensation extends the pinning range

For (SC.R5) take $f = r_g$ and $\eta = -\varepsilon^2/N$ in the preceding factorization. After replacing the gauged R by R, the two exponentials involving R cancel exactly. There is no joining error and no term of order ε^4/N^3. Thus (SC.R6) holds with

$$C_g = C_3 + \pi M_1(r_g)(3M_1(g) + M_1(d)), \qquad D_g = C_2.$$

If $\varepsilon = o(N^{2/3}K^{-1/3})$, K times the bound tends to zero. For the second term this also follows directly from $K\varepsilon^2/N^3 = o(K^{1/3}/N^{5/3})$; the zero or very small ε case needs no division by ε.

Normal-matrix spectral inclusion puts every eigenvalue of the gauged propagator within this operator-norm error of a Kth root of unity. Circle distance in R/Z is at most one quarter of chord distance. Hence its RMS lattice distance q obeys Kq tending to zero; K/d also tends to zero on each specified spectral space. The lattice concentration lemma (source file: `pell_microscopic_clustering.md`), whose proof is purely geometric, gives the spacing and pair-count conclusions in Theorem 38.2. Alternatively unitary telescoping immediately proves the recurrence-trace statement.

This is a larger quantum-exponent range than $\sqrt{N}K^{-1/4}$, but only for the explicitly compensated family, not the original unchanged perturbation. With $\kappa = \varepsilon/N$, its exponent is $N[\kappa(g - \mathcal{C}g) - \kappa^2 r_g]$. Thus the extra term is a specified second-order classical correction. The classical strength still tends to zero, of order $o(N^{-1/3}K^{-1/3})$, and is not a fixed perturbation.

38.4 Consistency example and verification

For $g = 2\cos(2\pi x_1)$, direct differentiation gives $r_g = 4\pi(B_{(2,4)} - B_{(4,4)})$ at the symbol level. Its variance is $64\pi^2$, recovering the special residual parameter b=8pi t. For an ordinary mode f on the original orbit the covariance is zero, recovering (SC.R4) as a sum of two terms. In contrast f=r_g gives $\sigma^2_{af+tr_g} = (a+t)^2\sigma^2_{r_g}$ and the cancellation line a=-t. An ordinary term cannot universally be treated as independent noise.

The verifier uses a finite multimode real g including an odd-ab frequency, constructs r_g by exact rational Fourier convolution before multiplying by π, and checks the physical Weyl sign, Egorov covariance, (SC.R8)–(SC.R11), and compensated pinning by dense matrix calculations. These are numerical falsification tests of finite identities and bounds, not an asymptotic proof. The analytic regularity and cutoff arguments above are not formalized in Lean. Neither compensation nor the known residual mechanism is presented as a newly invented general principle.

38.5 Recovered averaged Heisenberg-window consequence

For odd $K \geq 3$, let $D = Kn$ and suppose a unitary spectrum has exactly n phases in each arc of radius $\eta \leq 1/(4K)$ around j/K. For $M \geq 1$ consecutive integer times I, set $L = \max_{\ell\in I}|\ell|$. Writing $H_h = \sum_{r=1}^{h} r^{-1}$, the exact finite estimate is

$$\left|\frac{1}{M}\sum_{\ell\in I}\frac{|\operatorname{Tr}U^\ell|^2}{D} - n\right| \leq 8\pi^2 nL^2\eta^2 + \frac{2D}{M}H_{(K-1)/2}.$$

Indeed, ordered pairs within a cluster contribute between $n - 8\pi^2 nL^2\eta^2$ and n, by $1-\cos t \leq t^2/2$. For two clusters at cyclic distance h, phase separation is at least $h/(2K)$, so the averaged geometric sum is at most $K/(Mh)$. Summing over ordered clusters and their n^2 ordered level pairs gives the second term. The argument requires no independence.

For the inversion-even compensated family in (SC.R5), put $D = d_N = (N-1)/2$ and $\delta_N = C_g|\varepsilon|^3/N^2 + D_g\varepsilon^2/N^3$. If $\varepsilon = o(N^{1/3})$, then $N\delta_N \to 0$. The norm-small straight homotopy from U_0 to the gauged propagator preserves each contour projection rank in disjoint disks around

the K roots. Thus each cluster has rank $n = d_N/K$ and radius at most $\eta = \delta_N/4$ in $\mathbb{R}/\mathbb{Z}$. For fixed $0 < a < b$ and $I_N = \mathbb{Z} \cap [ad_N, bd_N]$, the bound gives

$$\frac{1}{\#I_N} \sum_{\ell \in I_N} \frac{|\operatorname{Tr}_-((\mathcal{V}_\varepsilon^{\mathrm{comp}})^\ell)|^2}{d_N} = \frac{d_N}{K}(1 + o(1)).$$

Here the trace denotes the trace of the ℓth power on the odd sector. The relative error is $O((d_N\delta_N)^2 + K \log K/d_N)$; the Pell sequence has $K = O(\log N)$. This is an obstruction for the engineered vanishing-strength family, conditional on its parent operator estimate. It supplies no fixed-strength BGS limit. The full proof and finite verification are in `compensated_heisenberg_trace_control.md`.

39 Mixed first- and second-order resonance fields

Use the same T, K, N, Weyl convention and odd trace as in the finite Fourier theorem. Let $A = B_{(1,0)}$, $D = U_0 A U_0^* = B_{(3,4)}$, $C = B_{(2,4)} - B_{(4,4)}$, and define

$$\mathcal{U}_{\varepsilon,\eta} = \exp\{i[\varepsilon(A - D) + \eta A]\}U_0. \tag{MX.M1}$$

This is a single exponential, not a split-kick model. Put

$$s_N = \sin(4\pi/N) > 0, \quad \zeta = \varepsilon^2 s_N, \quad a_K = \sqrt{2K}\eta_K, \quad b_K = 2\sqrt{K}\varepsilon_K^2 s_N.$$

Theorem 39.1. *If $a_K \to a \in \mathbb{R}$ and $b_K \to b \geq 0$, then*

$$\boxed{\tau_-(\mathcal{U}_{\varepsilon_K,\eta_K}^{mK}) \longrightarrow e^{-m^2(a^2+b^2)/2}} \qquad (m \in \mathbb{Z} \text{ fixed}). \tag{MX.M2}$$

Thus the Kth-power spectrum tends to the wrapped normal law of variance $a^2 + b^2$. Independence describes limiting tracial Gaussian variables, not independence of finite matrices or their eigenvalues.

39.1 Finite proof including the mixed errors

The exact identity $[A, D] = -2is_N C$ implies $\|[A, D]\| \leq 8s_N$ and $\|[A, C]\| \leq 16s_N$. The two-factor Duhamel bound splits (MX.M1) into $e^{i\eta A}e^{i\varepsilon(A-D)}U_0$ at cost $4|\varepsilon\eta|s_N$. Conjugation by $e^{-i\varepsilon A}$ commutes with the first factor. The uniform residual loop estimate has remainder $R_N = (64/3)|\varepsilon|^3 s_N^2$. Finally joining $e^{i\eta A}e^{i\zeta C}$ costs $8|\eta\zeta|s_N$. Consequently, with $Q = \eta A + \zeta C$,

$$\|e^{-i\varepsilon A}\mathcal{U}_{\varepsilon,\eta}e^{i\varepsilon A} - e^{iQ}U_0\|_{\mathrm{op}} \leq \Delta_N, \quad \Delta_N = R_N + 4|\varepsilon\eta|s_N + 8|\eta\zeta|s_N. \tag{MX.M3}$$

No smallness of ε A is needed; all comparisons integrate unitary evolutions rather than truncate BCH at a growing exponent.

The three frequencies v=(1,0),u=(2,4),z=(4,4) have quadratic norms -2,8,-16, so represent different signed T-orbits. Their embeddings are bounded by10. In the finite-alphabet theorem one may take

$$S = 3, \quad M = 10, \quad B = 5, \quad D_* = 15, \quad J = 34596, \quad L_K, W_K < 3200.$$

Four-atom lifting holds for every prime $K \geq 23$ because $\lambda^{\lceil K/4 \rceil}(1 - \lambda^{-K})^2 > 5^6(9/10)^2 > 6400$. These deliberately conservative constants are analytic, not fitted.

Let $Q_j = U_0^j Q U_0^{-j}$, $\bar{Q} = K^{-1}\sum Q_j$, $G_Q = \sum c_j Q_j$ and $l = |\eta| + 2|\zeta|$. The sharper version of (7) in that note gives the normal-form error

$$F_K = \frac{2l^2(W_K + L_K)}{K} + \frac{c_N(\eta^2 + 2\zeta^2)\sqrt{JL_K}}{2\sqrt{K}}.$$

After the two gauges, unitary telescoping gives the finite estimate

$$\left|\tau_-(\mathcal{U}_{\varepsilon,\eta}^{mK}) - \tau_-(e^{imK\bar{Q}})\right| \le |m|K(\Delta_N + F_K). \tag{MX.M4}$$

For bounded a_K,b_K, the right side is $O(K^{-1/2}) + O(K^{1/4}/\sqrt{N})$. Indeed $|\eta| = |a_K|/\sqrt{2K}$ and $|\varepsilon| = \sqrt{b_K/(2s_N\sqrt{K})}$. The zero b_K case is continuous.

If Y_0, Y_1, Y_2 denote the normalized three orbit sums, exactly

$$K\bar{Q} = a_K Y_0 + (b_K/\sqrt{2})(Y_1 - Y_2). \tag{MX.M5}$$

Their joint trace limit is three independent standard Gaussians by the finite-alphabet theorem; hence (MX.M5) tends to a normal variable of variance $a^2 + b^2$. Its bounded continuous characteristic function and (MX.M4) prove (MX.M2). The characteristic-function convergence error is separately o(1); no rate for it is inferred from one fourth moment.

The relation $u = Tv - v$ gives real three-atom zero words. Thus the exponentially small odd-moment claim of a one-species model cannot be imported. The general $O(K^{-1/2})$ bound, and the retained finite nonzero mixed third moments, are the appropriate statements here.

39.2 How precisely must a residual perturbation be tuned?

Write $\eta_K = \varepsilon_K\delta_K$, so the fixed-shape symbol is $\varepsilon_K[(A-D)+\delta_K A]$. At a nonzero residual critical limit b,

$$|\varepsilon_K| \sim \sqrt{b/(8\pi)}\sqrt{N}K^{-1/4}, \qquad |a_K| \sim \sqrt{b/(4\pi)}|\delta_K|\sqrt{N}K^{1/4}.$$

The relative contamination scale is therefore

$$\boxed{|\delta_K| \asymp N^{-1/2}K^{-1/4}.} \tag{MX.M6}$$

Below it the ordinary contribution vanishes; at it a finite additional variance survives. Above it this bounded-parameter theorem does not predict positive universality. This quantifies fine tuning, not genericity of the residual mechanism. The sign of ε affects a_K when δ is held fixed but not the leading pure residual term b_K.

39.3 A combined necessary escape criterion

Let $\rho_K^2 = a_K^2 + b_K^2 = 2K\eta_K^2 + 4K\varepsilon_K^4 s_N^2$. Keep the whole odd spectrum, multiplicities and global mean-spacing normalization. If ρ_K has a bounded subsequence, extract a further one with a_K,b_K convergent. For $\phi_L(x) = L^{-1}(1-|x|/L)_+$ and the periodized ordered distinct-pair test C_d, Poisson summation gives, for $d > L$,

$$C_d(\phi_L) \ge 1 - 1/L + 2\sum_{m=1}^{M_0} \mathrm{sinc}^2(LmK/d)|\tau_-(\mathcal{U}_{\varepsilon,\eta}^{mK})|^2, \quad \mathrm{sinc}(x) = \frac{\sin \pi x}{\pi x}. \tag{MX.M7}$$

All omitted Fourier terms are nonnegative. Take K to infinity at fixed L,M_0, then M_0 to infinity. Equation (MX.M2) gives $\liminf C_d(\phi_L) \ge \sum_{m\in\mathbb{Z}} e^{-m^2\rho^2} - 1/L$. This exceeds one as L grows for every finite positive ρ, and diverges at ρ=0. A bounded limiting pair density R(x) with R(x) tending to one at infinity would instead satisfy $\int \phi_L R \to 1$. Thus escape requires

$$\boxed{2K\eta_K^2 + 4K\varepsilon_K^4 \sin^2(4\pi/N) \to \infty.} \tag{MX.M8}$$

It is only necessary, not sufficient. The Fourier test is from the earlier project; the new input is the mixed recurrence law. L grows after, not simultaneously with, the semiclassical limit for fixed compact tests.

When ρ_K tends to zero there is also spacing collapse. Distinctness of the 6K signed atoms gives $\tau_N(\bar{Q}^2) = 2(\eta^2 + 2\zeta^2)/K$. Spectral matching and arc-versus-chord comparison bound the RMS distance q_K to the K-point lattice by

$$q_K \le (\Delta_N + F_K)/4 + \frac{c_N}{2\pi}\sqrt{2(\eta^2 + 2\zeta^2)/K}.$$

Therefore $Kq_K \to 0$, and $K/d \to 0$. The independent geometric concentration lemma (source file: `pell_microscopic_clustering.md`) proves weak convergence of globally normalized neighbouring spacings to δ_0 and divergent normalized close-pair counts at every fixed positive distance.

39.4 An exact finite dependence diagnostic

For odd prime $K \ge 23$, the full-space trace of the normalized orbit sums in (MX.M9) below satisfies

$$Y_i = (2K)^{-1/2} \sum_{j=0}^{K-1} B_{T^j v_i}, \qquad v_0 = (1,0), \quad v_1 = (2,4), \quad v_2 = (4,4), \tag{MX.M9}$$

$$\tau_N(Y_0^2 Y_1) = \tau_N(Y_0^2 Y_2) = \sqrt{\frac{2}{K}}\cos\!\left(\frac{4\pi}{N}\right). \tag{MX.M10}$$

After quadratic-norm lifting, a central word of two original-orbit atoms and one of the other species reduces to two powers of λ whose sum or difference is a unit translate of the corresponding relative frequency. Their algebraic norms are -4 and 8. The norms of $1 \pm \lambda^j$ are $2 \pm (\lambda^j + \lambda^{-j})$, so only adjacent times occur. The identities $v_1 = Tv_0 - v_0$ and $v_2 = Tv_0 + v_0$ give $4K$ ordered signed words, with phase sum $4K\cos(4\pi/N)$. Division by $(2K)^{3/2}$ proves the formulas. Thus the finite variables are not independent or exactly jointly Gaussian, although these third moments tend to zero. Their equality can cancel in particular scalar combinations and must not be replaced by an assumption that each third moment vanishes.

40 Compensated kicks on dimension-proportional time windows

This is a consequence of the preceding smooth operator estimate, not a separate positive universality theorem. Let g be a fixed real C^7 symbol, $d = \mathcal{C}g$, $r_g = \{g, d\}/(4\pi)$, and write $A = \mathrm{Op}_N(g)$, $D_0 = \mathrm{Op}_N(d)$, $R_0 = \mathrm{Op}_N(r_g)$. Define the explicitly compensated family

$$V_\varepsilon = \exp(i[\varepsilon(A - D_0) - \varepsilon^2 R_0/N])U_0, \qquad V' = e^{-i\varepsilon A} V_\varepsilon e^{i\varepsilon A}.$$

The compensation changes the symbol of the kick; it is part of the model, not a conclusion about the uncompensated family. The established uniform remainder is

$$\|V' - U_0\|_{\mathrm{op}} \le \delta_N = C_g|\varepsilon|^3/N^2 + D_g\varepsilon^2/N^3.$$

For every integer ℓ, telescoping unitaries or adjoints gives

$$\|(V')^\ell - U_0^\ell\|_{\mathrm{op}} \le \min(2, |\ell|\delta_N).$$

On the odd space, provided g is even, $\tau_-(U_0^\ell) = \mathbf{1}_{K|\ell}$. On the full space the normalized trace is $1/N$ at nonmultiples, not zero; at multiples it equals one. Consequently

$$\varepsilon_N = o(N^{1/3}) \quad \Longrightarrow \quad \sup_{|\ell| \le cN} |\tau_-(V_{\varepsilon_N}^\ell) - \mathbf{1}_{K|\ell}| \longrightarrow 0$$

for each fixed c. The quantum exponent may grow, but the corresponding classical strength $\varepsilon_N/(2\pi N)$ still vanishes. This is persistence of returns at the Heisenberg scale.

For the conventional form factor $\mathcal{F}_N(\ell) = D^{-1}|\operatorname{Tr}_- V^\ell|^2$, $D = (N-1)/2$, nonresonant times satisfy

$$\mathcal{F}_N(\ell) \le D\ell^2\delta_N^2.$$

The stronger sufficient condition $\varepsilon_N = o(N^{1/6})$ makes this uniformly vanish for $0 < |\ell| \le cN$, $K \nmid \ell$. At multiples of K, $\mathcal{F}_N(\ell)/D \to 1$. The extra factor D cannot be discarded by citing convergence of normalized traces alone. For spectral clustering only $K\delta_N \to 0$ is needed; a sufficient range is $|\varepsilon_N| = o(N^{2/3}K^{-1/3})$. Thus recurrence traces, conventional form factors, and microscopic concentration have genuinely different error budgets. None gives positive random-matrix statistics.

Chapter 9 — Locality at fixed classical strength

41 Fixed classical strength: locality and all-orders comparison

Let

$$T = \begin{pmatrix} 3 & 2 \\ 4 & 3 \end{pmatrix}, \qquad M = T^{\mathsf{T}}, \qquad \mathcal{C}g = g \circ M.$$

On the two-torus choose a real even smooth function g with

$$\operatorname{supp} g \cap \operatorname{supp}(\mathcal{C}g) = \varnothing, \qquad f = g - \mathcal{C}g. \tag{LC.1}$$

There is an infinite-dimensional real vector space of such generators, constructed in Subsection 41.2, containing nonzero functions. Let N be odd, let $U_{N,\mathrm{loc}}$ be the cat quantization with the exact covariance below, and write

$$G_N = \mathrm{Op}_N(g), \quad D_N = \mathrm{Op}_N(\mathcal{C}g), \quad A_{\kappa_0,N} = e^{i\kappa_0 N G_N}, \quad V_{\kappa_0,N} = e^{i\kappa_0 N(G_N - D_N)}U_{N,\mathrm{loc}}. \tag{LC.2}$$

The real parameter κ_0 is fixed as $N \to \infty$. With the conventions below it represents a classical Hamiltonian time $\kappa_0/(2\pi)$, not a classical perturbation that shrinks with N.

Theorem 41.1 (all-orders comparison)**.** *For each fixed bounded set of κ_0 values and each $R > 0$,*

$$\delta_N := \|V_{\kappa_0,N} - A_{\kappa_0,N}U_{N,\mathrm{loc}}A^*_{\kappa_0,N}\|_{\mathrm{op}} = O_{g,\kappa_0,R}(N^{-R}). \tag{LC.3}$$

For every fixed $p > 0$ the same all-orders estimate holds uniformly after raising both unitaries to any integer power $|t| \le N^p$. For the explicit Gevrey-2 bump in Subsection 41.2, constants $c, C > 0$ also give

$$\delta_N \le C{\kappa_0}^2 N^2 e^{-cN^{1/3}} \tag{LC.4}$$

for sufficiently large N. This conservative exponent is not claimed optimal. It gives vanishing error even for $|t| \le e^{cN^{1/3}/2}$.

Theorem 41.2 (classical interpretation)**.** *Let Φ_g^t be the Hamiltonian flow specified by $\frac{d}{dt}(a \circ \Phi_g^t) = \{g, a\} \circ \Phi_g^t$, with $\{g, a\} = g_q a_p - g_p a_q$. The map quantized by (LC.2) is*

$$M_{\kappa_0} = M \circ \Phi_f^{\kappa_0/(2\pi)} = \Phi_g^{-\kappa_0/(2\pi)} \circ M \circ \Phi_g^{\kappa_0/(2\pi)}. \tag{LC.5}$$

It is Anosov and mixing for area measure. For any nonzero example g and all sufficiently small fixed $\kappa_0 \neq 0$, it is nonlinear and differs from M. Its conjugacy to M is essential, not a symmetry that may be omitted when interpreting its spectral statistics.

Theorem 41.3 (microscopic stability along Pell moduli)**.** *Let K tend to infinity through odd primes, put $N + F\sqrt{2} = (1+\sqrt{2})^K$, and restrict to the odd parity space of dimension $D = (N-1)/2 = Kn$. For fixed κ_0 and sufficiently large N, the spectrum consists of K clusters, each containing exactly n eigenvalues, whose angular halfwidth a_N is $O(N^{-R})$ for every R. After a common rotation their centres are the Kth roots of unity.*

Under unfolding by the mean density D, the spacing measure tends weakly to δ_0. More quantitatively, for every fixed finite $p \geq 1$,

$$W_p\left(\mu_{\mathrm{gaps}}, (1-1/n)\delta_0 + (1/n)\delta_n\right) \leq 2Da_N. \tag{LC.6}$$

For each fixed $x > 0$, eventually the normalized ordered close-pair count is **exactly** $n-1$. The local point process viewed from a uniformly random origin converges vaguely in probability to the empty process, despite having mean intensity one at every finite N.

Spacing collapse, divergence of close-pair counts, and this empty local limit remain true under any increasing degree-one unfolding F_N satisfying $\|F_N'\|_\infty \leq CN^A$ for fixed C, A. Thus the present conclusion cannot be removed by an unfolding with polynomially bounded distortion. No assertion is made about unfoldings whose derivatives grow faster than every power of N.

41.1 Quantization and a finite locality estimate

For an integer frequency $v = (a, b)$ use the **physical integer Fourier lift**

$$(Q_N(a,b)u)(x) = e^{2\pi i a(x-b/2)/N} u(x - b \bmod N).$$

The half in this exponent is ordinary division, not the inverse of two in $\mathbb{Z}/N\mathbb{Z}$. For the modular-centred convention W_N used in the earlier notes, $Q_N(a,b) = (-1)^{ab} W_N(a,b)$. In particular,

$$Q_N(v)Q_N(w) = e^{\pi i \omega(v,w)/N} Q_N(v+w), \qquad \omega(v,w) = v_1 w_2 - v_2 w_1. \tag{LC.7}$$

Each $Q_N(v)$ is unitary, $Q_N(v)^* = Q_N(-v)$, and parity sends $Q_N(v)$ to $Q_N(-v)$. Set

$$\mathrm{Op}_N(a) = \sum_{v \in \mathbb{Z}^2} \widehat{a}(v) Q_N(v), \quad \mathcal{M}_r(a) = \sum_v |\widehat{a}(v)|(1+|v|_1)^r.$$

The series converges absolutely in operator norm for smooth a; $\|\mathrm{Op}_N(a)\| \leq \mathcal{M}_0(a)$, and real symbols give Hermitian matrices. Frequency aliases cause no problem in this definition.

Since T is the identity modulo two, its finite Weyl implementer also has exact physical covariance

$$U_{N,\mathrm{loc}} Q_N(v) U_{N,\mathrm{loc}}^* = Q_N(Tv), \qquad U_{N,\mathrm{loc}} \mathrm{Op}_N(a) U_{N,\mathrm{loc}}^* = \mathrm{Op}_N(a \circ M). \tag{LC.8}$$

One such implementer has the kernel

$$U_{N,\mathrm{loc}}(x,y) = \frac{c_N}{\sqrt{N}} e_N\Big((3x^2 - 2xy + 3y^2)/8\Big), \quad |c_N| = 1, \tag{LC.9}$$

where division by eight is modular. Orthogonality of additive characters proves unitarity; direct substitution on the two Weyl generators proves (LC.8). The kernel is parity-preserving. Its global phase can be fixed later.

Lemma 41.4 (disjoint-symbol product)**.** *If $a, b \in C^\infty(\mathbb{T}^2)$ have disjoint closed supports, then for every integer $r \geq 1$,*

$$\|\mathrm{Op}_N(a)\,\mathrm{Op}_N(b)\|_{\mathrm{op}} \leq \frac{\pi^r}{r! N^r} \mathcal{M}_r(a) \mathcal{M}_r(b). \tag{LC.10}$$

Proof. Expand (LC.7) and Taylor-expand $e^{i\pi\omega/N}$ through degree $r-1$. For real z the integral Taylor remainder satisfies $|e^{iz}-\sum_{j<r}(iz)^j/j!|\le|z|^r/r!$, with no restriction on z. For each $j<r$ the coefficient sum

$$\sum_{v,w}\widehat{a}(v)\widehat{b}(w)\omega(v,w)^jQ_N(v+w)$$

is the quantization of

$$\frac{1}{(-4\pi^2)^j}\sum_{l=0}^{j}(-1)^l\binom{j}{l}(\partial_q^{j-l}\partial_p^la)(\partial_q^l\partial_p^{j-l}b). \tag{LC.11}$$

All these functions vanish, including ab at $j=0$, because derivatives are supported in the respective original closed supports. Their Fourier cancellation occurs before applying Q_N, so aliases do not invalidate it. The remaining absolutely convergent sum is bounded using $|\omega(v,w)|\le|v|_1|w|_1$ and $\|Q_N(v+w)\|=1$. This proves (LC.10).

For real a,b, their two products are adjoints, so

$$\|[\mathrm{Op}_N(a),\mathrm{Op}_N(b)]\|\le\frac{2\pi^r}{r!N^r}\mathcal{M}_r(a)\mathcal{M}_r(b). \tag{LC.12}$$

This all-orders statement is stronger than just Poisson commutation.

41.2 An explicit infinite-dimensional family and a Gevrey bound

Put $r_0=1/16$ and let S be the union of the closed torus squares of halfwidth r_0 centred at $(1/4,0)$ and $(3/4,0)$. If $(q,p)\in S$, the second coordinate $2q+3p$ of $M(q,p)$ has distance at least

$$1/2-5r_0=3/16>r_0 \tag{LC.13}$$

from the integers. Therefore $S\cap M^{-1}S=\varnothing$. Every real even smooth function supported in S satisfies (LC.1). This is an infinite-dimensional space. For nonzero g, f is nonzero because it equals g on S. Also $\int f=0$, regardless of the mean of g.

Here is the fixed example used numerically. Write $\langle x\rangle$ for the representative of x mod 1 in $[-1/2,1/2)$, set

$$\beta(u)=\begin{cases}\exp(1-1/(1-u^2)), & |u|<1,\\ 0, & |u|\ge1,\end{cases}\qquad u_\pm(q)=\langle q\mp1/4\rangle/r_0,$$

and choose $\eta=1/3$. Then

$$\begin{aligned}A(q)&=\beta(u_+(q))(1+\eta u_+(q))+\beta(u_-(q))(1-\eta u_-(q)),\\ B(p)&=\beta(\langle p\rangle/r_0),\qquad g(q,p)=A(q)B(p).\end{aligned} \tag{LC.14}$$

This g is real, even and nonzero. The unequal tilts of its two lobes avoid a spurious half-period translation symmetry in the numerical example; no absence of all other symmetries is claimed.

For completeness, its Gevrey regularity and the nonoptimal stretched exponential bound need no black-box pseudodifferential theorem. If $D_t=1-t^2>0$, use the complex disk $|z-t|\le D_t/8$. On this disk

$$|(1-z^2)-D_t|\le17D_t/64,\qquad \mathrm{Re}(1-z^2)^{-1}\ge\frac{3008}{6561D_t}>\frac{1}{3D_t}.$$

Cauchy's estimate and maximization of $D_t^{-j}e^{-1/(3D_t)}$ give

$$|\beta^{(j)}(t)|\le e\,24^j(j!)^2. \tag{LC.15}$$

All derivatives extend by zero at ± 1. Multiplication by $1 \pm \eta t$ changes the prefactor to at most $2e$, with the same base 24.

A concrete Fourier-moment estimate follows. Put $L_0 = 24/r_0$, $B_0 = 9L_0/\pi$ and

$$A_0 = 8r_0 + (8/3)r_0 e L_0^2, \qquad B_1 = 2r_0 + (2/3)r_0 e L_0^2.$$

Integrating the one-dimensional Fourier coefficients by parts $j+2$ times and using $\sum_{k\neq 0} k^{-2} = \pi^2/3$ proves

$$\sum_k (1+|k|)^j |\widehat{A}(k)| \le A_0 B_0^j (j!)^2, \quad \sum_k (1+|k|)^j |\widehat{B}(k)| \le B_1 B_0^j (j!)^2. \tag{LC.16}$$

One uses $(j+2)^2(j+1)^2 \le 4\,9^j$. Expanding the sum of the two frequency weights gives

$$\mathcal{M}_j(g) \le A_0 B_1 (2B_0)^j (j!)^2, \quad \mathcal{M}_j(\mathcal{C}g) \le A_0 B_1 (14 B_0)^j (j!)^2, \tag{LC.17}$$

since $\|T\|_1 = 7$. These explicit constants are intentionally coarse and are not useful finite-size error bars for the numerical matrices below.

More generally, suppose $\mathcal{M}_j(g) \le A_g B_g^j (j!)^s$ and $\mathcal{M}_j(\mathcal{C}g) \le A_d B_d^j (j!)^s$ for some $s > 1$. The comparison proved in Subsection 41.3 has the bound

$$\delta_N \le {\kappa_0}^2 N^2 A_g A_d (C/N)^j (j!)^{2s-1}, \qquad C = \pi B_g B_d. \tag{LC.18}$$

For $j = \lfloor (N/(2C))^{1/(2s-1)} \rfloor \ge 1$, use $j! \le j^j$ to bound the last factor by 2^{-j}. This proves a bound $C' {\kappa_0}^2 N^2 \exp(-cN^{1/(2s-1)})$, and (LC.4) follows for $s = 2$. The general factorial-growth/optimal-truncation mechanism is established Gevrey analysis; this section does not claim to originate it or to improve the best Gevrey remainders in other quantization settings.

41.3 Fixed-strength conjugacy, classical dynamics, and long times

For Hermitian H, J and real t, the two-factor Duhamel inequality gives

$$\|e^{it(H+J)} - e^{itH} e^{itJ}\| \le (t^2/2)\|[H, J]\|. \tag{LC.19}$$

One proof differentiates $e^{i(t-u)(H+J)} e^{iuH} e^{iuJ}$ and integrates the conjugation difference of J twice. Unitary invariance of the norm leaves a triangular integration domain of area $t^2/2$. In particular, $|t|\|H\|$ need not be small.

By (LC.8),

$$A_{\kappa_0,N} U_{N,\mathrm{loc}} A_{\kappa_0,N}^* = e^{i\kappa_0 N G_N} e^{-i\kappa_0 N D_N} U_{N,\mathrm{loc}}.$$

Apply (LC.19) with $t = \kappa_0 N$, $H = G_N$, $J = -D_N$, and then (LC.12). For every integer $r \ge 1$,

$$\boxed{\delta_N \le \frac{{\kappa_0}^2 \pi^r}{r!} N^{2-r} \mathcal{M}_r(g) \mathcal{M}_r(\mathcal{C}g).} \tag{LC.20}$$

This proves Theorem 41.1 and (LC.18). Notice that we have not set the finite commutator to zero, nor silently replaced the one-exponential kick by a product of two kicks.

For integer t, telescoping products of unitaries, or their adjoints when $t < 0$, yields

$$\|V_{\kappa_0,N}^t - A_{\kappa_0,N} U_{N,\mathrm{loc}}^t A_{\kappa_0,N}^*\| \le |t| \delta_N. \tag{LC.21}$$

Choosing r larger than any fixed polynomial time exponent proves the all-polynomial-time assertion. The same estimate proves the stretched exponential-time assertion from (LC.4). Normalized traces differ by at most $|t|\delta_N$; their absolute squares differ by at most $2|t|\delta_N$. These statements control Heisenberg times $t \asymp D$, not just $O(\log N)$.

To establish (LC.5), note that (LC.1) implies $\{g, \mathcal{C}g\} = 0$, so the two Hamiltonian flows commute and

$$\Phi^t_{\mathcal{C}g} = M^{-1}\Phi^t_g M, \qquad \Phi^t_f = \Phi^t_g \Phi^{-t}_{\mathcal{C}g}.$$

Consequently $M\Phi^t_f = \Phi^{-t}_g M \Phi^t_g$. This is an exact classical identity for every finite t. A smooth symplectic conjugacy transports the invariant hyperbolic splitting, with bounded derivatives in both directions, so it preserves uniform hyperbolicity. It also transports area measure; mixing follows from mixing of M. The latter can be checked on Fourier characters: for nonzero integer v, $|T^n v| \to \infty$, and hence fixed Fourier modes eventually have zero correlation. Density extends the result to square-integrable observables.

The normalization of classical time is checked rather than presumed: (LC.7) gives

$$[\mathrm{Op}_N(g), \mathrm{Op}_N(a)] = -\frac{i}{2\pi N}\,\mathrm{Op}_N(\{g, a\}) + O(N^{-3})$$

for fixed smooth symbols. Differentiating unitary conjugation and using Duhamel on a fixed compact time interval gives the pullback by $\Phi_g^{\kappa_0/(2\pi)}$, with a remainder tending to zero. Thus (LC.2) is a quantization of the fixed map in (LC.5).

For nonzero g the smooth function f is nonconstant. Its flow moves some point for every sufficiently small nonzero time. The complement of the support of f has nonempty interior (the support occupies less than the whole torus), and the flow is exactly the identity there. Accordingly M_{κ_0} equals M on an open set but is not equal to it everywhere for small nonzero κ_0. An affine torus map agreeing with M on an open set agrees everywhere. Therefore these fixed maps are genuinely nonlinear, although smoothly conjugate to the same linear map.

41.4 Arithmetic multiplicities and individual cluster control

The following finite arithmetic facts can be established without primality of N. Write $\alpha = 1+\sqrt{2}$, $\lambda = \alpha^2$ and $N + F\sqrt{2} = \alpha^K$ with K an odd prime. The Pell identity gives $N^2 - 2F^2 = -1$, hence F is invertible modulo N. Set

$$s = F^{-1} \pmod N, \quad s^2 = 2, \quad h = 3 + 2s.$$

Since $(1+s)^K = N + Fs = 1$, $h^K = 1$. Also $h - 1 = 2(1+s)$ is a unit. Modulo each prime divisor of N, the order of h is exactly K. Thus $1 - h^j$ is a unit for $1 \le j < K$, and $1 + h^j$ is a unit for every j.

The symplectic matrix

$$P = \begin{pmatrix} 1 & -1/(2s) \\ s & 1/2 \end{pmatrix}$$

satisfies $T = P\,\mathrm{diag}(h, h^{-1})P^{-1}$. Its lower-left entry is a unit, so its finite modular Weyl implementer has the same type of quadratic kernel as (LC.9) and preserves parity. Conjugating S_h^{-1} by this implementer constructs a phase-normalized $U_{N,\mathrm{loc}}$ with $U_{N,\mathrm{loc}}^K = I$. Here $S_h u(x) = u(h^{-1}x)$. The unit conditions above show that every nonzero residue has an orbit of length K, and that an orbit and its negative are distinct. On the odd space, there are $n = (N-1)/(2K)$ cyclic orbits, each contributing every Kth root once. Hence the multiplicity is exactly n for each root. A different global phase merely rotates all centres and changes none of the spacing conclusions.

Let $W = A_{\kappa_0,N} U_{N,\mathrm{loc}} A^*_{\kappa_0,N}$ on the odd space. Restriction cannot increase (LC.20). For large N take $\delta_N < \sin(\pi/K)$. Spectral stability for a normal matrix places each eigenvalue of V within δ_N of a Kth root. To verify multiplicities, let $E = VW^*$ and use its principal Hermitian logarithm, $E = e^{iH}$. Then $\|H\| \le 2\arcsin(\delta_N/2)$, and the path $e^{iuH}W$, $0 \le u \le 1$, stays within

δ_N of W. The disjoint slightly enlarged root disks have constant Riesz ranks along this path. Thus every disk contains exactly n eigenvalues. Their circle-coordinate halfwidth is bounded by

$$a_N = \pi^{-1} \arcsin(\delta_N/2) = O(N^{-\infty}). \tag{LC.22}$$

This is an operator-norm, individual-cluster statement, unlike the earlier normalized Hilbert–Schmidt comparison. It still makes no claim about individual eigenvectors inside a nearly degenerate cluster.

41.5 Gaps, local counts, and robustness against unfolding

Sort the D eigenangles cyclically, retaining multiplicities, and put $S_l = D(\theta_{l+1} - \theta_l)$, with the cyclic final gap. Exactly $D - K$ gaps are within clusters and at most $2Da_N$. The other K gaps are in $[n - 2Da_N, n + 2Da_N]$. Matching these two classes to zeros and copies of n proves (LC.6). Since $1/n = K/D \to 0$, the reference measures in (LC.6) converge weakly to δ_0, and so do the actual gap measures. Their means are nevertheless exactly one at each N: rare large gaps carry that mean.

Define the ordered close-pair count

$$R_N(x) = \frac{1}{D}\#\{(l, j) : l \neq j,\ D \operatorname{dist}_{\mathbb{T}}(\theta_l, \theta_j) \leq x\}.$$

For fixed $x > 0$, eventually $2Da_N < x$ and $D(1/K - 2a_N) > x$. Every within-cluster ordered pair is counted and no cross-cluster pair is counted. Therefore

$$\boxed{R_N(x) = n - 1 \quad \text{eventually.}} \tag{LC.23}$$

For a fixed compactly supported continuously differentiable test ϕ, its periodized factorial statistic likewise obeys

$$\mathcal{R}_N(\phi) = (n-1)\phi(0) + O\big((n-1)Da_N\|\phi'\|_\infty\big). \tag{LC.24}$$

The remainder is superpolynomially small. This formula is not asserted with this rate for an arbitrary continuous test.

Choose an origin uniformly on the circle. A window of unfolded length L can meet a cluster only if its origin belongs to one of K arcs of length at most $L/D + 2a_N$. Hence

$$\mathbb{P}\{\text{window contains a level}\} \leq KL/D + 2Ka_N \longrightarrow 0. \tag{LC.25}$$

The same argument applies to any fixed compact observation window, giving vague convergence in probability of the translated local point measures to the empty measure. Their expected count is still L; uniform integrability fails. This is a point-process statement, not an inference from diverging second moments alone.

Now let $F_N : \mathbb{R} \to \mathbb{R}$ be increasing with $F_N(x + 1) = F_N(x) + 1$ and Lipschitz constant at most CN^A. Image clusters have angular diameter at most $2CN^Aa_N$. There are at most K gaps not internal to these clusters. Internal unfolded gaps tend uniformly to zero, proving spacing collapse. All $n(n - 1)$ ordered pairs in each original cluster are still close, so $R_N(x) \geq n - 1$ eventually. Finally (LC.25) is replaced by

$$\mathbb{P}\{\text{image window contains a level}\} \leq KL/D + 2KCN^Aa_N \longrightarrow 0. \tag{LC.26}$$

No lower bound on F_N' is used, and image clusters need not remain uniformly separated. Thus (LC.23) need not remain an equality after unfolding; the lower bound and empty-process conclusion are the correct statements.

41.6 Flat directions, hidden structure, and a falsification control

For every generator in (LC.1), $\{g, \mathcal{C}g\} = 0$. In the previous smooth notation, $\Gamma(g) = \sigma^2(\{g, \mathcal{C}g\}/(4\pi)) = 0$. Thus the second-order exceptional set contains an explicit infinite-dimensional linear subspace containing nonzero smooth generators. This does not contradict openness and density of $\Gamma > 0$; an infinite-dimensional exceptional subspace can still have empty interior. Subtracting the mean of g places it in the zero-mean class without changing f, Γ, or the quantum commutator. One should apply the support argument to the original bump, not falsely describe its mean-subtracted version as compactly supported.

Poisson commutation **alone** is insufficient for (LC.3). For a negative control take $a = b_{(1,0)} + b_{(0,1)}$, where $b_v = 2\cos(2\pi v \cdot x)$, and $b = a^2$. Of course $\{a, b\} = 0$. Yet the cubic Weyl coefficient

$$\sum_{v+w=k} \widehat{a}(v)\widehat{b}(w)\omega(v,w)^3$$

is nonzero: at frequencies $(-2,-1), (-2,1), (-1,-2), (-1,2)$ its values are $-6, 6, 6, -6$, respectively, and the four opposite frequencies give the same values. The commutator therefore has a nonzero term of order N^{-3}. For a sufficiently large odd N these finite frequencies are distinct modulo N, so this is not a formal nonzero coefficient that disappears under quantization. Both implementations retain this negative control. The all-orders cancellation in (LC.11), not just its first term, is essential.

Finally, if J_N commutes with $U_{N,\mathrm{loc}}$ and $\|J_N\| \le 1$, then

$$\|[V_{\kappa_0,N}, A_{\kappa_0,N} J_N A^*_{\kappa_0,N}]\| \le 2\delta_N. \tag{LC.27}$$

The arithmetic commutant is therefore transported to an all-orders approximate commutant. These systems do not evade arithmetic structure by being nonlinear. We do not claim their complete symmetry reduction, simple spectrum, or generic random-matrix behaviour. Obtaining corresponding positive universality for fixed maps not conjugate to arithmetic models is not accomplished by this construction.

Chapter 10 — Resonance rigidity

42 Microscopic stability above the mean-spacing error scale

This addendum strengthens the consequences of fixed-strength locality. The finite covering bounds below are elementary. Their role is to make the perturbation and unfolding thresholds explicit; no priority claim is made. In particular, operator-norm accuracy better than the mean spacing is **not** necessary for the negative local-statistics conclusion. The unperturbed reference still has arithmetic clusters. Nothing here proves positive BGS for a nonarithmetic chaotic system.

42.1 A finite covering theorem

Let $x_1, \dots, x_D$ be a multiset on $\mathbb{R}/\mathbb{Z}$, including multiplicities, covered by at most K arcs whose total length is at most ℓ. Assume $0 \le \ell < 1$. No uniform placement or equal occupancy of the arcs is assumed. Write S_j for the cyclic gaps multiplied by D, and set

$$R_D(x) = \frac{1}{D}\#\{(i,j) : i \ne j,\ D\,\mathrm{dist}(x_i, x_j) \le x\}.$$

For every $x > 0$ and $0 < L < D$,

$$\frac{1}{D}\#\{j : S_j > x\} \le \frac{K}{D} + \frac{\ell}{x}, \tag{RB.R1}$$

$$R_D(x) \geq \frac{1}{K/D + \ell/x} - 1, \tag{RB.R2}$$

and, for a uniformly distributed origin z,

$$\mathbb{P}\{[z, z + L/D) \text{ contains a point}\} \leq KL/D + \ell. \tag{RB.R3}$$

The last upper bound can of course be replaced by its minimum with one. The denominator in (RB.R2) is positive for a nonempty multiset; $K \geq 1$.

Proof. Merge overlapping arcs first. There are at most K components, with total length at most ℓ, and a point outside their union can be chosen as the seam of the circle. There are at most K gaps connecting different components. The lengths of the remaining gaps sum to at most ℓ. Each gap counted in (RB.R1) has length greater than x/D, proving (RB.R1), without any lower bound on individual gaps.

Divide each covering arc into consecutive cells of length at most x/D, using at most $1 + D\ell_i/x$ cells for a component of length ℓ_i. A zero-length component needs one cell. Assign shared endpoints to one cell only, and let m_c be the cell occupancies. All ordered distinct pairs inside a cell are counted by $R_D(x)$. Cauchy–Schwarz gives

$$\frac{1}{D}\sum_c m_c(m_c - 1) \geq \frac{D}{\#\text{cells}} - 1 \geq \frac{1}{K/D + \ell/x} - 1.$$

This proves (RB.R2). For (RB.R3), the possible origins are covered by the original arcs enlarged backwards by L/D; their total length is at most $\ell + KL/D$. End of proof.

Consequently, along any sequence with $K/D \to 0$ and $\ell \to 0$, the gap measures converge weakly to δ_0, every fixed-threshold $R_D(x)$ diverges, and the local point measure at a uniform origin converges vaguely in probability to the empty measure. The last conclusion follows from (RB.R3) for a compact window containing the support of any test function. It does not rely on an inference from second moments, nor on uniform integrability. The expected number in a window is still exactly L at every finite dimension.

42.2 Perturbations much larger than the mean spacing

Use the notation of the locality theorem, with odd Pell dimension $D = (N-1)/2 = Kn$, the natural single-kick unitary V_N, its conjugated arithmetic reference W_N, and

$$\delta_N = \|V_N - W_N\| = O(N^{-\infty}).$$

Let $\widetilde{V}_N$ be any other unitary satisfying

$$\|\widetilde{V}_N - V_N\| \leq \eta_N, \qquad K\eta_N \to 0. \tag{RB.R4}$$

The same estimate applies to the parity-restricted matrices; $\widetilde{V}_N$ need only be defined on that fixed D-dimensional space. Then all three negative local-statistics conclusions above hold for $\widetilde{V}_N$.

Proof. Set $b_N = \delta_N + \eta_N$. The triangle inequality gives $\|\widetilde{V}_N - W_N\| \leq b_N$. For large N, $b_N < \sin(\pi/K)$. Normal spectral stability and the unitary homotopy argument in the locality theorem put exactly n eigenvalues in each root cluster of circle-coordinate halfwidth

$$a'_N \leq \pi^{-1} \arcsin(b_N/2).$$

Thus the spectrum has a cover with total length

$$\ell_N \leq \frac{2K}{\pi} \arcsin(b_N/2) = O(Kb_N) \to 0. \tag{RB.R5}$$

Apply (RB.R1)–(RB.R3). End of proof.

This statement is uniform over *all* unitary perturbations with the norm bound (RB.R4); their eigenvectors, internal cluster splitting, and probability law are unrestricted. In particular, for any fixed $b > 0$, the sufficient condition $\eta_N = O(N^{-b})$ holds since $K = O(\log N)$. When $0 < b < 1$, this allowed error is parametrically larger than $1/D$.

For example, a unitary perturbation with norm $O(N^{-1/2})$ still obeys

$$\frac{1}{D}\#\{j : S_j > x\} = O_x(K/D + K/\sqrt{N}), \qquad R_D(x) = \Omega_x(\sqrt{N}/K), \tag{RB.R6}$$

where the lower bound is meant for all sufficiently large N. The exact identity $R_D(x) = n - 1$ from the unperturbed locality theorem is not retained under this larger error: individual clusters may have widths larger than a single mean spacing. The covering proof is what replaces that identity.

This gives a precise quantization-stability statement **only** for a second quantization meeting (RB.R4). The same classical principal canonical map by itself does not imply such an operator-norm bound; leading Fourier-integral amplitudes or subprincipal conventions must not be silently identified.

42.3 Necessary distortion of any unfolding that restores a nonempty limit

Let F_N be an increasing degree-one map of the circle, with Lipschitz constant A_N. The image of a cover of total length ℓ_N is covered by at most K arcs with total length at most $A_N\ell_N$. Their positions and occupancies may be arbitrary. Therefore all three collapse conclusions persist whenever

$$A_N\ell_N \to 0. \tag{RB.R7}$$

This is sharper than requiring that every cluster have diameter smaller than $1/D$ after unfolding.

For a converse necessary bound, suppose the probability that an image window of a fixed unfolded length L contains a level tends to some $p_L > 0$. By (RB.R3) and $K/D \to 0$,

$$\liminf A_N\ell_N \geq p_L. \tag{RB.R8}$$

This is a necessary condition, not a sufficient recipe for a desired point process. If one uses the original equal-width cover, then $\ell_N \leq 2Ka_N$, giving

$$\liminf 2KA_Na_N \geq p_L. \tag{RB.R9}$$

For the explicit Gevrey-2 fixed-strength example and fixed $\kappa_0 \neq 0$, there is a constant C_{κ_0} such that $a_N \leq C_{\kappa_0}N^2e^{-cN^{1/3}}$. Thus any such unfolding must satisfy

$$A_N \geq c_{L,\kappa_0}\frac{e^{cN^{1/3}}}{KN^2} \quad \text{for all sufficiently large } N. \tag{RB.R10}$$

To see the quantifiers, first make the left probability at least $p_L/2$ and $KL/D \leq p_L/4$; then (RB.R3) gives $2KA_Na_N \geq p_L/4$. Substitute the displayed upper bound on a_N. If a cluster has zero width, no finite Lipschitz unfolding can split its coincident levels; (RB.R8) remains the appropriate general necessary statement. The constant in (RB.R10) is not claimed numerically sharp.

43 Poisson rigidity and a quantitative quantum lower bound

Let $S \in \mathrm{SL}(2,\mathbb{Z})$ satisfy $|\operatorname{tr} S| > 2$ and act on $\mathbb{T}^2$ with normalized Haar measure. For real C^1 functions put $\{f,h\} = f_qh_p - f_ph_q$.

Theorem 43.1 (regular-gradient rigidity)**.** *If $g \in C^1(\mathbb{T}^2)$ is nonconstant and its critical set $\{dg = 0\}$ has measure zero, then*

$$\{g, g \circ S\} \not\equiv 0. \tag{AR.R1}$$

Consequently every real analytic g satisfying $\{g, g \circ S\} = 0$ is constant. No parity assumption is needed. Equivalently a nonconstant C^1 observable commuting with its next iterate must have a positive-measure critical set.

The analytic conclusion uses the standard measure-zero theorem for zeros of a nontrivial real analytic function, not a new unique-continuation theorem. A short primary reference is Boris Mityagin, *The Zero Set of a Real Analytic Function*, arXiv:1512.07276, Proposition 0.

43.1 Proof

Suppose that $\{g, g \circ S\} = 0$ and $dg \neq 0$ almost everywhere. Remove the union of all forward and backward images of the null critical set. On the remaining invariant full-measure set define a measurable projective line

$$L(x) = [\nabla g(x)] \in \mathbb{RP}^1.$$

The chain rule and the zero determinant of the two gradients give

$$L(Sx) = S^{-\mathsf{T}} L(x). \tag{AR.R2}$$

Both gradients in this assertion are nonzero; this is exactly where the critical-set hypothesis is required.

Push Haar measure forward by L. Equation (AR.R2) gives an invariant probability measure for the hyperbolic projective map $P = S^{-\mathsf{T}}$. Every such probability is supported on its two eigenlines. Indeed, in an eigenbasis, the slope away from those lines is multiplied by a fixed number c with $c > 1$ after possibly replacing P by its inverse. The disjoint sets

$$\{c^n \leq |z| < c^{n+1}\}, \qquad n \in \mathbb{Z},$$

all have the same measure, so they each have measure zero. Their union is the complement of the two eigenlines. Negative eigenvalues cause no problem: their ratio is positive, or one may square the projective map.

The toral automorphism S is ergodic. This can be seen directly: if an L^2 function is invariant, its Fourier coefficients are constant along each orbit of S^{T} on $\mathbb{Z}^2 \setminus \{0\}$. Each such orbit is infinite, so square summability forces every nonzero coefficient to vanish. The sets where L equals the two eigenlines are invariant by (AR.R2); hence one of them has full measure. Thus ∇g lies in a single fixed eigencovector line almost everywhere.

Let w be a nonzero vector annihilated by that covector. Its slope is irrational: a rational eigendirection would give a rational eigenvalue of the monic polynomial $z^2 - (\operatorname{tr} S)z + 1$, hence an integer eigenvalue dividing one, contrary to hyperbolicity. We have $w \cdot \nabla g = 0$ almost everywhere, and therefore everywhere by continuity. Fourier integration by parts yields

$$(w \cdot v)\widehat{g}(v) = 0 \quad (v \in \mathbb{Z}^2).$$

For every nonzero integer v the factor $w \cdot v$ is nonzero. All nonconstant Fourier coefficients vanish and g is constant, the desired contradiction.

For a nonconstant real analytic g at least one of its partial derivatives is not identically zero. Its zero set has measure zero by the analytic zero-set theorem, and the critical set is a subset. The preceding argument proves the analytic conclusion. This finishes the proof.

43.2 A quantum consequence, and an important non-consequence

Specialize now to the project's $T = \left(\begin{smallmatrix} 3 & 2 \\ 4 & 3 \end{smallmatrix}\right)$, $S = T^{\mathsf{T}}$ and odd-dimensional physical Weyl quantization

$$Q_N(a,b) = (-1)^{ab} W_N(a,b), \qquad Q_N(v)Q_N(w) = e^{\pi i \omega(v,w)/N} Q_N(v+w), \qquad \hbar_N = (2\pi N)^{-1}.$$

For a fixed smooth real g set $d = g \circ S$ and $h = \{g, d\}$. Then

$$[\mathrm{Op}_N(g), \mathrm{Op}_N(d)] = -i\hbar_N \,\mathrm{Op}_N(h) + O_g(N^{-3}) \quad \text{in operator norm.} \tag{AR.R3}$$

For clarity the remainder is elementary: in the exact Fourier commutator expand $2i\sin(\pi\omega/N)$ through its linear term and use $|\sin u - u| \le |u|^3/6$. The norm bound is $\pi^3 M_3(g) M_3(d)/(3N^3)$, where $M_j(a) = \sum_v |\widehat{a}(v)|\,|v|_2^j$ with the Euclidean norm $|v|_2$; the estimate uses $|\omega(v,w)| \le |v|_2 |w|_2$. The symbol M_j is local to this section and is not the $\mathcal{M}_r$ of the locality section.

The normalized Hilbert–Schmidt norm obeys, for every fixed smooth h,

$$\|\,\mathrm{Op}_N(h)\|_{2,N} \longrightarrow \|h\|_{L^2}, \qquad \|B\|_{2,N}^2 = N^{-1}\,\mathrm{Tr}(B^*B).$$

One proof truncates its absolutely convergent Fourier series at a fixed radius; for sufficiently large N the finite Weyl matrices are exactly orthonormal, and the remaining norm is bounded by the Fourier l1 tail. Alternatively, modular aliases involve at least one frequency of size at least N/2 and their contribution is smaller than every inverse power for a fixed smooth h. Together with (AR.R3),

$$\boxed{\lim_{N\to\infty} \hbar_N^{-1} \|[\mathrm{Op}_N(g), \mathrm{Op}_N(g\circ S)]\|_{2,N} = \|\{g, g\circ S\}\|_{L^2}.} \tag{AR.R4}$$

For nonconstant analytic g the right side is strictly positive by Theorem 43.1. In particular this pair of quantized observables cannot commute to every inverse power of N. Since $\|B\|_{\mathrm{op}} \ge \|B\|_{2,N}$, it also has an operator-norm lower bound c_g/N for all sufficiently large N.

For $g = b_{(1,0)} = 2\cos(2\pi q)$, direct algebra gives

$$\|\{g, g\circ S\}\|_2 = 32\pi^2, \quad \hbar_N^{-1} \|[\mathrm{Op}_N(g), \mathrm{Op}_N(g\circ S)]\|_{2,N} = 8\pi N |\sin(4\pi/N)|$$

whenever the indicated finite frequencies are distinct modulo N. This is retained as a numerical normalization check.

The boundary matters. The theorem proves $\{g, g\circ S\} \neq 0$, not $\sigma^2(\{g, g\circ S\}) > 0$. A nonzero bracket could still be a coboundary. Also (AR.R4) does not exclude every conceivable unitary conjugacy at an isolated nonzero kick strength: exponentials can have cancellations that an individual commutator does not detect. The result excludes the all-orders commuting-observable mechanism for a nonconstant analytic generator. It is not positive BGS universality, nor a general classification of analytic spectral statistics.

43.3 Relation to the smooth flat construction

The all-orders note constructs nonconstant smooth functions supported in two small boxes whose images are disjoint from those boxes. Their derivatives vanish on a nonempty open set, so their critical sets have positive measure. They are precisely outside the regular-gradient hypothesis above. This yields an explicit smooth-versus-analytic distinction without an appeal to a formal infinite BCH or Moyal series.

Numerical checks and their scope are in `pell_flat_quantum_check.py`. The projective measure and analytic zero-set arguments are analytic proofs, not claims certified by finite matrices. No assertion of priority follows from targeted searches failing to locate the exact statement.

44 The sharp resonance minimum on a hyperbolic Fourier orbit

This section generalizes the Pell-only sharp theorem (source file: `pell_single_orbit_sharp_constant.md`) already recorded in the project. That earlier theorem already solves real autocorrelation realizability for the Pell matrix; this is not counted as a new achievement here. The present proof applies to **every** hyperbolic matrix $A \in \mathrm{SL}(2,\mathbb{Z})$ and supplies independent exact checks for four matrices, a different uniform positivity argument, and numerical real spectral factors. It is a project proof; priority and external referee acceptance are not claimed. Its spectral-factorization step is the classical Fejer–Riesz theorem, not a new factorization theorem.

We may assume $\operatorname{tr} A > 2$: replacing A by -A leaves its action on even cosine observables unchanged. Write $\lambda > 1$ for its expanding eigenvalue, $b_v(x) = 2\cos(2\pi v \cdot x)$, and $\mathcal{C}_A b_v = b_{Av}$. In particular $\lambda \geq (3+\sqrt{5})/2 > 2$. For a real even trigonometric polynomial h of zero mean define

$$\sigma_A^2(h) = 2\sum_{\mathcal{O}} \left(\sum_{w \in \mathcal{O}/\{\pm 1\}} a_w(h) \right)^2, \qquad \Gamma_A(g) = \sigma_A^2\left(\frac{\{g, \mathcal{C}_A g\}}{4\pi} \right),$$

where a_w is the coefficient of b_w and $\mathcal{O}$ ranges over the signed A-orbits. This is the finite-Fourier Green–Kubo form.

Fix $v \in \mathbb{Z}^2 \setminus \{0\}$ and consider

$$g = \sum_{j \in \mathbb{Z}} c_j b_{A^j v}, \qquad c_j \in \mathbb{R}, \qquad C_r = \sum_j c_j c_{j+r}, \qquad d_r = \det(v, A^r v).$$

The coefficients have finite support. We write its diameter as at most L: after an integer translation all nonzero coefficients lie in $0 \leq j \leq L$. The number d_1 is nonzero: otherwise v is a rational eigendirection of a hyperbolic determinant-one integral matrix, which is impossible. Set

$$M = \lfloor L/2 \rfloor, \qquad S_M = \sum_{k=0}^{M} d_{2k+1}^{-2}, \qquad S_\infty = \sum_{k=0}^{\infty} d_{2k+1}^{-2}.$$

44.1 The sharp variational theorem

Theorem 44.1. *For every L and every prescribed $C_0 = c > 0$,*

$$\boxed{\min_{\substack{\operatorname{diam}(\operatorname{supp} c_j) \leq L \\ \sum_j c_j^2 = c}} \Gamma_A(g) = \frac{4\pi^2 c^2}{S_M}.} \tag{SH.S1}$$

The minimum is attained by a real sequence supported on the even indices $0, 2, \ldots, 2M$. All minimizing autocorrelation sequences are specified by

$$C_{2k} = c\,\frac{\sum_{j=k}^{M} d_{2j+1}^{-2}}{S_M} \quad (0 \leq k \leq M), \qquad C_{2k+1} = 0. \tag{SH.S2}$$

Terms beyond the support are zero. There may be more than one coefficient sequence with these autocorrelations; uniqueness of the coefficients is not asserted.

Consequently the infimum over all nonzero finite single-orbit polynomials is

$$\boxed{\inf_g \frac{\Gamma_A(g)}{\|g\|_2^4} = \frac{\pi^2}{S_\infty}.} \tag{SH.S3}$$

No finite sequence attains (SH.S3), because $S_M < S_\infty$ for every finite M. This statement concerns all finite support lengths, not the existence or regularity of a minimizer in an infinite-frequency function space.

44.2 Step 1: the exact orbit identity

Direct differentiation yields

$$\frac{\{b_u, b_w\}}{4\pi} = \pi \det(u, w)(b_{u-w} - b_{u+w}).$$

Combining the terms from times j and j+r in $\{g, \mathcal{C}_A g\}$ gives, for $r \geq 1$, coefficient $\pi d_r(c_j c_{j+r-1} - c_{j+r}c_{j-1})$ on the orbit represented by $v - A^r v$, and its negative on that represented by $v + A^r v$. The orbit sums of these coefficients are $\pm\pi d_r(C_{r-1} - C_{r+1})$.

It is necessary to check that these generated orbits do not coalesce. Define $Q_A(w) = \det(w, Aw)$, invariant under $w \mapsto \pm Aw$. Since $I \pm A^r$ commutes with A,

$$Q_A(v \pm A^r v) = \det(I \pm A^r) Q_A(v) = (2 \pm (\lambda^r + \lambda^{-r})) Q_A(v). \tag{SH.S4}$$

For $r \geq 1$ these values have opposite signs for the two choices. Within each sign their absolute values are strictly increasing with r. All generated orbits in question are therefore distinct. The variance formula gives exactly

$$\boxed{\Gamma_A(g) = 4\pi^2 \sum_{r \geq 1} d_r^2 (C_{r-1} - C_{r+1})^2.} \tag{SH.S5}$$

This proves the identity for a general A; no special quadratic-ring coordinate system for the Pell example is needed.

44.3 Step 2: a lower bound with an exact equality condition

The even autocorrelations telescope:

$$C_0 = \sum_{k=0}^{M} (C_{2k} - C_{2k+2}).$$

Cauchy–Schwarz with weights d_{2k+1}^2 gives

$$C_0^2 \leq S_M \sum_{k=0}^{M} d_{2k+1}^2 (C_{2k} - C_{2k+2})^2.$$

The omitted even-r terms in (SH.S5) are nonnegative. Therefore (SH.S1)'s right side is a lower bound. Equality in the weighted inequality requires

$$C_{2k} - C_{2k+2} = \frac{C_0}{S_M d_{2k+1}^2}.$$

Equality in the omitted terms requires all successive odd autocorrelations to be equal, hence zero by finite support. These are precisely (SH.S2).

A lower bound alone does **not** solve the problem: the desired sequence of C's must be realizable as the autocorrelation of real finite coefficients. That constraint is checked next.

44.4 Step 3: positive spectral density and a real spectral factor

Cayley–Hamilton gives

$$\frac{d_r}{d_1} = \frac{\lambda^r - \lambda^{-r}}{\lambda - \lambda^{-1}}.$$

Let $q = \lambda^{-4} < 1/16$. The positive inverse weights $u_k = d_{2k+1}^{-2}$ satisfy $u_{k+1} < qu_k$. Define the normalized targets

$$a_k = \frac{\sum_{j=k}^{M} u_j}{S_M}, \qquad a_0 = 1.$$

Termwise comparison of the tails gives $0 \le a_k \le q^k$. Hence the real Laurent polynomial

$$P_M(z) = 1 + \sum_{k=1}^{M} a_k(z^k + z^{-k})$$

is strictly positive on the unit circle, with the explicit bound

$$P_M(e^{i\theta}) \ge 1 - \frac{2q}{1-q} > 0. \tag{SH.S6}$$

The classical Fejer–Riesz theorem provides a polynomial $p_M(z) = \sum_{j=0}^{M} b_j z^j$ with $P_M(e^{i\theta}) = |p_M(e^{i\theta})|^2$. It can be chosen with real coefficients. To see the real property directly, consider $z^M P_M(z)$: its coefficients are real and its zeros come in conjugate and reciprocal-conjugate pairs, with no zero on the unit circle. Choosing one complete conjugation-invariant side of each reciprocal pair gives a real factor, up to a real normalization. This is also a consequence of uniqueness of the outer factor with positive value at zero.

Set $c_{2j} = \sqrt{c}\, b_j$ and $c_{2j+1} = 0$. Comparing the Laurent coefficients in $|p_M|^2 = P_M$ shows that its autocorrelations are exactly (SH.S2), including $\sum c_j^2 = c$. Thus the lower bound is attained. This completes (SH.S1), and the monotone limit in M proves (SH.S3).

A primary account of the factorization theorem is Dritschel and Rovnyak, *The Operator Fejer–Riesz Theorem*, arXiv:0903.3639, Section 1. Only its scalar, one-variable theorem is used here.

44.5 Explicit examples and a tail estimate

For the Pell matrix $A = \left(\begin{smallmatrix} 3 & 2 \\ 4 & 3 \end{smallmatrix}\right)$, $v = (1, 0)$, we have $d_1 = 4$ and

$$d_r = 4s_r, \qquad s_0 = 0, \quad s_1 = 1, \quad s_{r+1} = 6s_r - s_{r-1}.$$

For L=2 or 3, $s_3 = 35$, so the sharp ratio to the single-mode value is

$$\frac{1}{d_1^2 S_1} = \frac{1225}{1226}. \tag{SH.S7}$$

The minimizing coefficients can be chosen at times 0 and 2. Put $a = 1/1226$,

$$c_0 = \sqrt{\frac{1 + \sqrt{1 - 4a^2}}{2}}, \qquad c_2 = \frac{a}{c_0}, \qquad c_1 = 0.$$

They have $C_0 = 1$ and $C_2 = a$, and attain (SH.S1). Thus the assertion that a single mode minimizes Gamma at fixed L2 norm would already be false at L=2.

The all-length infimum has ratio

$$\frac{1}{d_1^2 S_\infty} = 0.99918363250315720691404018094106030 6\ldots. \tag{SH.S8}$$

The displayed number is high-precision arithmetic, not interval-certified rounding. An analytic positive tail bound is

$$0 < S_\infty - S_M \le \frac{d_{2M+3}^{-2}}{1 - \lambda^{-4}} \le \frac{\lambda^{-4(M+1)}}{d_1^2(1 - \lambda^{-4})}. \tag{SH.S9}$$

Inverting S_M and S_M plus this upper bound brackets the sharp constant in exact real arithmetic.

For the trace-three cat matrix $\left(\begin{smallmatrix} 2 & 1 \\ 1 & 1 \end{smallmatrix}\right)$ and $v = (1, 0)$, the L=2 ratio is 64/65. The phenomenon and its proof are not peculiar to the Pell recurrence.

45 Sharp multiorbit coercivity and an exact radius-three counterexample

The single-orbit theorem cannot simply be added across orbits: generated frequencies can coincide and their coefficients interfere. This section handles those coincidences exactly on the complete first Fourier box for the Pell matrix $T = \left(\begin{smallmatrix} 3 & 2 \\ 4 & 3 \end{smallmatrix}\right)$. It is a finite-dimensional algebraic theorem, not positive BGS or a priority claim. Its scope includes mixtures of distinct frequency orbits.

Write $b_v(x) = 2\cos(2\pi v \cdot x)$, $\mathcal{C}b_v = b_{Tv}$, and use the same Green–Kubo form and $\Gamma(g) = \sigma^2(\{g, \mathcal{C}g\}/(4\pi))$ as in the general single-orbit note. Every real, even, mean-zero trigonometric polynomial with frequencies $\|v\|_\infty \le 1$ has the unique form

$$g = ab_{(0,1)} + bb_{(1,-1)} + cb_{(1,0)} + db_{(1,1)}, \qquad a, b, c, d \in \mathbb{R}. \tag{FM.B1}$$

Here b without a subscript is a real coefficient. The frequencies $(1, -1)$ and $(1, 1)$ share an orbit since $T(1, -1) = (1, 1)$; the other two belong to different signed orbits.

Theorem 45.1. *On this whole four-dimensional space,*

$$\boxed{\Gamma(g) \ge 4\pi^2 \|g\|_{L^2}^4.} \tag{FM.B2}$$

The constant is sharp. Among nonzero symbols equality occurs exactly for a scalar multiple of one of $b_{(0,1)}, b_{(1,-1)}, b_{(1,1)}$. In particular, there is no nonzero second-order flat generator in this Fourier box.

45.1 Exact quartic identity

Set $Q = \Gamma(g)/\pi^2$. Direct orbit reduction gives

$$\begin{aligned} Q = {} & 16a^4 + 16b^4 + 64c^4 + 16d^4 \\ & + 104a^2(b^2 + d^2) + 140a^2c^2 + 200c^2(b^2 + d^2) \\ & + 608b^2d^2 + 8a^2bd + 4bc^2d. \end{aligned} \tag{FM.B3}$$

The last two terms show why the four modes cannot be treated as independent. Nevertheless the difference from the claimed bound is a sum of squares:

$$\begin{aligned} Q - 16(a^2 + b^2 + c^2 + d^2)^2 = {} & 48c^4 + 108a^2c^2 + 68a^2(b^2 + d^2) \\ & + 4(ab + ad)^2 + 166c^2(b^2 + d^2) \\ & + 2(bc + cd)^2 + 576b^2d^2. \end{aligned} \tag{FM.B4}$$

Each summand is nonnegative. Cosine orthogonality gives $\|g\|_2^2 = 2(a^2 + b^2 + c^2 + d^2)$, proving (FM.B2). Equality in (FM.B4) first forces $c = 0$, then $bd = 0$ and $a^2(b^2 + d^2) = 0$. The nonzero possibilities are precisely the three asserted axes. Conversely those axes give equality in (FM.B3). Thus both sharpness and the complete equality set are established.

45.2 Auditable orbit reduction behind the quartic

The basic differentiation identity is

$$\{b_u, b_w\}/(4\pi) = \pi \det(u, w)(b_{u-w} - b_{u+w}).$$

Apply it to the sixteen ordered pairs of modes in (FM.B1), replacing the second frequency w by Tw. For each resulting frequency identify v and -v, then sum along its T-orbit. The following

table lists every nonzero orbit sum divided by π; call these polynomials $P_{\mathcal{O}}$. Then $Q = 2\sum_{\mathcal{O}} P_{\mathcal{O}}^2$, whose expansion is (FM.B3).

Orbit representative	$P_{\mathcal{O}}$	Orbit representative	$P_{\mathcal{O}}$
$(-6,-6)$	$-12bd$	$(-4,-8)$	$12bd$
$(-4,-4)$	$-4c^2$	$(-4,-3)$	$-7bc$
$(-4,3)$	$-7cd$	$(-3,-3)$	$-6ac$
$(-3,-2)$	$-5ab$	$(-3,2)$	$-5ad$
$(-2,-5)$	$7bc$	$(-2,-4)$	$2a^2+4c^2$
$(-2,-2)$	$-2a^2$	$(-2,-1)$	$-bc$
$(-2,0)$	$-2b^2-2d^2$	$(-2,1)$	$-cd$
$(-2,5)$	$7cd$	$(-1,-4)$	$5ab$
$(-1,-3)$	$3ac$	$(-1,-2)$	$ab+ad$
$(-1,0)$	$-ab-ad$	$(-1,3)$	$3ac$
$(-1,4)$	$5ad$	$(0,-2)$	$2b^2+2d^2$
$(0,-1)$	$bc+cd$		

Representatives are selected by minimizing $E(v) = 2v_1^2 + v_2^2$ along the signed T-orbit, with a lexicographic tie rule. This procedure is exact: writing $v_2 + \sqrt{2}v_1$ and its conjugate shows that $E(T^j v) = A\lambda^{2j} + B\lambda^{-2j}$ with A,B strictly positive for nonzero integral v. It is strictly convex as a function of real j. Integer neighbour descent therefore reaches its global integer minimum, with at most two adjacent ties. This also gives a finite way to check that the displayed representatives are distinct orbit classes. No unexplained bound on orbit search depth is imported into the proof.

The maintained Python script constructs this table in the original integral coordinates and verifies (FM.B3)–(FM.B4) coefficient by coefficient over the integers. The independent C++ quadratic-ring implementation agrees at all 80 nonzero coefficient vectors in $\{-1,0,1\}^4$. The finite evaluations are cross-checks; the coefficientwise identities and orbit reduction establish the theorem for arbitrary real coefficients.

45.3 An exact obstruction to extending the same constant

The constant in (FM.B2) cannot hold on every larger Fourier box. Define a sixteen-mode polynomial g_* by the following nonzero cosine coefficients:

Frequency	Coefficient	Frequency	Coefficient
$(0,1)$	-5	$(0,2)$	-7
$(0,3)$	4	$(1,-3)$	1
$(1,-2)$	5	$(1,-1)$	-10
$(1,1)$	10	$(1,2)$	-5
$(1,3)$	-1	$(2,-2)$	2
$(2,-1)$	2	$(2,0)$	-6
$(2,1)$	2	$(2,2)$	2
$(3,-1)$	1	$(3,1)$	-1

All frequencies lie in the radius-three box. Exact integer Poisson and orbit reduction gives

$$\sum_v c_v^2 = 396, \qquad \frac{\Gamma(g_*)}{\pi^2} = 1890144, \qquad \boxed{\frac{\Gamma(g_*)}{\|g_*\|_2^4} = \frac{6563}{2178}\pi^2 < 4\pi^2.} \tag{FM.B5}$$

The complete 106 nonzero integer orbit sums are recorded by the checker; their squared sum times two is 1890144. Original-coordinate Python and independent quadratic-ring C++ both obtain

this exact integer. Thus (FM.B5) is an exact finite arithmetic witness, not a fitted floating-point minimum. It does not claim to be the optimal radius-three polynomial or a zero of Gamma.

46 Quantitative transversality and the absence of a bandwidth-only bound

This section consolidates the previously recorded one-mode transversality argument. It treats arbitrary smooth backgrounds and does not classify every zero of the second resonance. The last proposition explains why uniform polynomial coercivity on larger Fourier boxes is a different, false target.

Let $A \in \mathrm{SL}(2,\mathbb{Z})$ be hyperbolic and let g be real, even, smooth and of mean zero. For $h_g = \{g, \mathcal{C}_A g\}/(4\pi)$ let

$$Q(g) = \sqrt{2}\Big(\sum_{w \in \mathcal{O}/\{\pm 1\}} a_w(h_g)\Big)_{\mathcal{O}}, \qquad \Gamma_A(g) = \|Q(g)\|_{\ell^2}^2.$$

The weighted Fourier absolute sums make Q a continuous homogeneous quadratic map. For a nonzero integer mode $b = b_v$, put

$$\gamma_b = \Gamma_A(b) = 4\pi^2 \det(v, Av)^2 > 0.$$

The two generated frequencies $v - Av, v + Av$ have opposite nonzero values of the invariant form $q_A(w) = \det(w, Aw)$; hence their orbit sums cannot cancel. A rational eigenline of A would have an integral eigenvalue with integral inverse, necessarily ± 1, so $\det(v, Av) \neq 0$.

Theorem 46.1 (Uniform one-mode sublevel estimate)**.** *For every such g and every $\epsilon > 0$,*

$$\mathrm{Leb}\{t \in \mathbb{R} : \Gamma_A(g + tb) \le \epsilon \gamma_b\} \le 2\sqrt{2}\,\epsilon^{1/4}.$$

The constant is independent of the background's number of Fourier modes, coefficients and frequency radius. It is a length estimate in the explicitly normalized scalar coefficient t, not a lower bound at a prescribed t.

Proof. Project $Q(g + tb)$ onto $Q(b)$. The normalized projection is a monic real quadratic $p(t) = t^2 + \beta t + \eta$, and Cauchy–Schwarz gives $\Gamma_A(g + tb) \ge \gamma_b p(t)^2$. Completing the square reduces $|p(t)| \le \delta$ to $|x^2 - a| \le \delta$. If $a \le \delta$ this is empty or an interval of length at most $2\sqrt{2\delta}$. If $a > \delta$, the total length is $2(\sqrt{a+\delta} - \sqrt{a-\delta}) \le 2\sqrt{2\delta}$. Take $\delta = \sqrt{\epsilon}$. □

If T has a density bounded by M, the corresponding probability is at most $\min(1, 2\sqrt{2}M\epsilon^{1/4})$. This also holds conditionally on an arbitrary random background whenever the selected coefficient's conditional density is bounded by M. No independence of all the other coefficients is required. For $Z = \Gamma_A(g + Tb)/\gamma_b$, integrating the sublevel estimate at its transition point $(2\sqrt{2}M)^{-4}$ gives

$$\mathbb{E} Z^{-p} \le \frac{(2\sqrt{2}M)^{4p}}{1 - 4p}, \qquad 0 < p < \tfrac{1}{4}.$$

The exponent threshold is exact for a background-uniform statement: at g=0, $Z = T^4$, so the negative moment diverges for $p \ge 1/4$ when T has a density bounded below near zero. This statement is about a classical variance functional, not eigenvalue statistics of a randomly perturbed propagator.

Proposition 46.2 (A deterministic three-point certificate)**.** *For every r>0,*

$$\max\{\Gamma_A(g - rb), \Gamma_A(g), \Gamma_A(g + rb)\} \ge \tfrac{1}{4} r^4 \gamma_b.$$

If $\Gamma_A(g) = 0$, then $\max\{\Gamma_A(g - rb), \Gamma_A(g + rb)\} \ge r^4 \gamma_b$.

Proof. For the monic quadratic above, $p(r) + p(-r) - 2p(0) = 2r^2$. At least one of its three absolute values is at least $r^2/2$. For the stronger assertion use $Q(g + rb) + Q(g - rb) = 2r^2 Q(b)$ when Q(g)=0 and the triangle inequality. □

Proposition 46.3 (No polynomial-bandwidth coercivity on all backgrounds)**.** *For the Pell matrix T, let $\mathcal{P}_R$ be the real even mean-zero Fourier polynomials with $\|v\|_\infty \le R$. There are $g_R \in \mathcal{P}_R$ with norm bounded below away from zero and bounded in every fixed smooth seminorm, such that*

$$0 < \Gamma_T(g_R) = O_L(R^{-L}) \quad \text{for every } L > 0.$$

Thus no constants c,p>0 give $\Gamma_T(g) \ge cR^{-p}\|g\|_2^4$ for all nondegenerate g in all these Fourier boxes.

Proof. Choose a nonzero mean-zero smooth generator g of the separated-support construction, so $\{g, \mathcal{C}_T g\} = 0$. Let $P_R g$ be its rectangular Fourier partial sum and $e_R = P_R g - g$. Smoothness implies $M_1(e_R) = O_L(R^{-L})$ for every L and $M_1(P_R g) \le M_1(g)$. The exact Fourier bracket bound $\|\widehat{\{a, b\}/(4\pi)}\|_1 \le \pi M_1(a) M_1(b)$ gives

$$\left\| \widehat{\{P_R g, \mathcal{C}_T P_R g\}/(4\pi)} \right\|_1 \le 2\pi \|T\|_{\mathrm{op}} M_1(g) M_1(e_R).$$

For a real even h, $\sigma_T^2(h) \le \frac{1}{2}\|\widehat{h}\|_1^2$, so $\Gamma_T(P_R g)$ is superpolynomially small. Partial sums converge in every smooth seminorm and in L^2.

To ensure strict positivity, choose one nonzero mode v of radius one. The polynomial $z \mapsto \Gamma_T(P_R g + z b_v)$ has degree four and positive leading coefficient $\Gamma_T(b_v)$. At least one of the five numbers $e^{-R}, 2e^{-R}, 3e^{-R}, 4e^{-R}, 5e^{-R}$ is not a root. Choose it as z_R. Then $g_R = P_R g + z_R b_v$ has strictly positive Gamma; this exponentially small change preserves every displayed estimate. □

The certificate for the fixed radius-eight space remains consistent with this proposition: its positive coercivity constant depends on that fixed radius. The unrestricted finite-Fourier nondegeneracy theorem is proved in Section 48; its positivity does not imply a bandwidth-uniform lower bound.

47 Arbitrary-phase nondegeneracy and a local Hessian obstruction

We study the second-order cohomological variance $\Gamma_A(g) = \sigma_A^2(\{g, g \circ A^{\mathsf{T}}\}/(4\pi))$ for hyperbolic two-torus automorphisms. For the fixed Pell matrix we give a finite, independently replayable proof that it is positive for every nonconstant real trigonometric polynomial of frequency radius at most eight. All 288 real parameters are arbitrary: evenness and Fourier-phase alignment are not assumed. An exact finite quantum variance identity connects this certificate to a period-averaged observable, with an explicit non-aliasing hypothesis. A separate analytic theorem detects nonzero variance from the two hyperbolic curvatures at a fixed critical point. It implies nondegeneracy, without a bandwidth restriction, on the cone of even symbols with one-sign cosine coefficients, for every hyperbolic integral determinant-one matrix.

These are project proofs and exact finite certificates, not a claim of world priority or positive BGS universality. The Fourier-orbit formula for asymptotic variance is established theory; we prove the specialized identities needed here. The certificate is stronger than sampling real coefficients but is not a Lean formalization. The unbounded-radius mixed-sign problem remains unresolved by these arguments. No uniform long-time quantum comparison is inferred from a positive classical variance alone.

47.1 Normalization and the cohomological question

Let $A \in \mathrm{SL}(2,\mathbb{Z})$ be hyperbolic, $S = A^{\mathsf{T}}$, and $\mathcal{C}_A f = f \circ S$. On the unit torus put

$$e_v(x) = e^{2\pi i v\cdot x}, \qquad b_v = e_v + e_{-v}, \qquad \{f,h\} = f_{x_1}h_{x_2} - f_{x_2}h_{x_1}.$$

For a real smooth mean-zero function h, define

$$\sigma_A^2(h) = \sum_{\mathcal{O}} \left| \sum_{v\in\mathcal{O}} \widehat{h}(v) \right|^2, \qquad \Gamma_A(g) = \sigma_A^2\Big(\frac{\{g, \mathcal{C}_A g\}}{4\pi} \Big), \tag{ND.1}$$

where $\mathcal{O}$ ranges over **oriented** A-orbits in $\mathbb{Z}^2 \setminus \{0\}$. An orbit and its negative are distinct: a nonzero integer vector cannot be periodic under $-A^j$ for a hyperbolic A. For real h they contribute conjugate sums. For even $h = \sum a_v b_v$, (ND.1) therefore reduces to twice the sum of the squared cosine sums on signed orbits. Constants in g affect neither the bracket nor Γ_A.

For finite Fourier support, character orthogonality gives exactly

$$\sum_{n\in\mathbb{Z}} \int h(x) h(S^n x)\, dx = \sum_{\mathcal{O}} \left| \sum_{v\in\mathcal{O}} \widehat{h}(v) \right|^2. \tag{ND.2}$$

The double absolute sum is bounded by $\|\widehat{h}\|_1^2$, so the same identity holds for smooth h by absolute convergence. This is the usual Green–Kubo/orbit-sum variance, not the instantaneous L^2 norm. Such spectral and orbit-sum descriptions are prior theory, including Cohen–Conze [1].

For a finite polynomial h, $\sigma_A^2(h) = 0$ if and only if $h = u - \mathcal{C}_A u$ for a finite polynomial u. Indeed, on each orbit, let h_j be the finite coefficient sequence. If its sum is zero, the partial sums $u_j = \sum_{k\le j} h_k$ have finite support and $h_j = u_j - u_{j-1}$. Conversely a finite difference has zero sum. Consequently proving only $\{g, \mathcal{C}_A g\} \neq 0$ does not settle the nondegeneracy question.

For Sections 2–3 fix

$$T = \begin{pmatrix} 3 & 2 \\ 4 & 3 \end{pmatrix}, \qquad \lambda = 3 + 2\sqrt{2}.$$

Let $\mathcal{P}_R$ be the real mean-zero trigonometric polynomials supported in $\|v\|_\infty \le R$. Its real dimension is $4R(R+1)$.

47.2 An exact finite theorem for arbitrary Fourier phases

Theorem 47.1 (the full radius-eight box)**.** *For every nonzero $g \in \mathcal{P}_8$,*

$$\boxed{\Gamma_T(g) > 0.} \tag{ND.3}$$

In particular (ND.3) holds at every smaller integer radius. There is no restriction on the signs or phases of the Fourier coefficients. This proves that the second-order bracket is not a finite Fourier coboundary throughout this 288-dimensional real space. The assertion is not extrapolated to all radii.

47.2.1 2.1 Exact polynomial equations

Choose

$$V_R = \{(a,b) : 0 \le a \le R,\ -R \le b \le R,\ a > 0 \text{ or } b > 0\}.$$

Write, uniquely,

$$g = \sum_{v\in V_R} (z_v e_v + \overline{z_v} e_{-v}), \qquad z_v \in \mathbb{C}, \qquad \|g\|_2^2 = 2\sum_v |z_v|^2. \tag{ND.4}$$

There are $n_R = 2R(R+1)$ complex variables. The exact Fourier bracket is

$$\{g, \widehat{\mathcal{C}_T g\}/}(4\pi)(k) = -\pi \sum_{u+Tv=k} \det(u, Tv)\widehat{g}(u)\widehat{g}(v). \tag{ND.5}$$

Combining the coefficients along one of the two orientations of each signed orbit gives integer-coefficient quadratic polynomials $Q_j(z, \bar{z})$ with

$$\Gamma_T(g) = 2\pi^2 \sum_j |Q_j(z, \bar{z})|^2. \tag{ND.6}$$

Thus (ND.3) is the assertion that all the equations $Q_j = 0$ have only the zero solution subject to the genuine complex-conjugation relation. The latter is not replaced by treating z and $\bar{z}$ as independent complex variables.

47.2.2 2.2 Why the orbit classification terminates and preserves phases

For $v = (a, b)$ put $\beta = b + \sqrt{2}a$, $\beta' = b - \sqrt{2}a$. Then

$$E(T^j v) := 2(T^j v)_1^2 + (T^j v)_2^2 = \tfrac{1}{2}(\beta^2 \lambda^{2j} + \beta'^2 \lambda^{-2j}). \tag{ND.7}$$

Both coefficients are positive for $v \neq 0$. This is strictly convex as a function of real j, tends to infinity at both ends, and has at most two adjacent integer minimizers. Integer neighbour descent reaches the global minimum. At equal energies we use a fixed lexicographic rule, including the choice of a sign for the vector. Energy decreases in the non-tie steps, so the algorithm cannot loop indefinitely.

Crucially, the program records the orientation bit ϵ in $w = \epsilon T^j v$. A negative orientation contributes to the conjugate orbit, not a second independent copy of the same complex coefficient. Adjacent minimum ties, for example the orbit containing $(1, -2)$, are retained as explicit normalization tests. Omitting this bit can produce an incorrect arbitrary-phase variance even when all even-symbol tests pass.

The independent C++ constructor works in the quadratic ring with vector $b + a\sqrt{2}$ and enumerates an orbit window $-16 \leq j \leq 16$. It fails if a minimizer is at the boundary. Interior minimizers have the global meaning just established by (ND.7); the finite window is not an unproved coverage assumption. All 1,475 nonzero radius-eight equations agree exactly, term by term, with the independently constructed Python equations.

47.2.3 2.3 A small proof language and its soundness

The proof records known zero coefficients, known zero products, and temporary nonzero assumptions. Only the following four rules are admitted.

P (product). After terms killed by established facts are deleted, an equation has one cross monomial $az_i^\epsilon z_j^\eta = 0$, with $a \neq 0$ and $i \neq j$. Record $z_i z_j = 0$, equivalently that at least one factor is zero. Here the exponent sign denotes the variable or its conjugate, not an inverse.

S (squares). A sole nonzero multiple of z_i^2, $\bar{z}_i^2$, or $|z_i|^2$ implies $z_i = 0$. A sum $\sum_i a_i |z_i|^2 = 0$ with all nonzero a_i of one sign implies that every listed variable vanishes. A sum of ordinary complex squares is *not* admitted by the second rule.

D (nonzero division). A recorded zero product and a currently assumed nonzero factor imply that the other factor is zero. Division without this explicit assumption is rejected.

F (refutation). Temporarily assume $z_i \neq 0$, replay the supplied child proof with inherited established facts, and require an actual contradiction. If the child succeeds, conclude $z_i = 0$ outside the branch. An empty child or a branch that merely makes partial progress is rejected.

Each rule is an elementary implication over $\mathbb{C}$ with its usual conjugation. Induction on the sequential deductions and refutations proves soundness of a complete certificate. The search need not be complete: failure to eliminate all variables is reported as UNKNOWN, never as a counterexample. The validator replays the proof and does not trust the generator's status.

The radius-eight certificate contains 5,704 elementary deductions: 3,293 P rules, 1,077 S rules, 1,229 D rules and 105 F rules. Every one of its 144 complex variables is eliminated. Refutations in this particular certificate have depth one. Both Python and C++ independently reconstruct the coefficient system and replay the deductions. Four malformed Python objects and two malformed C++ instruction streams are required to fail closed. This completes a finite computer-assisted proof of (ND.3), with the analytic normalization (ND.5)–(ND.7) and logical soundness above as explicit dependencies.

Radius	Real dimension	Equations	P	S	D	F
1	8	23	0	3	0	0
3	48	212	81	35	16	4
8	288	1,475	3,293	1,077	1,229	105

The radius-eight coefficient table has SHA-256 `7462bc1a522a57741028efd6642710fa50aa271b27f21bfc6a683f578aee53f4`. The plain-text proof has SHA-256 `ed26c33904551e11a5185d55702d781b4f2555d7a14153ceaf6e8a0b5f71d94a`. The delivered manifest provides all source and result hashes. Hash agreement identifies a proof object; it does not by itself prove that object sound.

The certificate objects are in the repository. The plain-text proof itself (the 5,704 P/S/D/F deductions, with rule counts 3,293/1,077/1,229/105) is `evidence/executions/20261006/corpus-revision/final/checks/certificates-mpuvq_28/resonance_nondegeneracy_radius8.proof` (git blob `17a70f2b42166c0e8a339050d38f3b1883483720`; its SHA-256 is the plain-text proof hash above). The accompanying certificate record, stating radius 8, real dimension 288, 144 complex variables and 1,475 orbit equations with the coefficient-table hash above, is `evidence/executions/20261006/corpus-revision/final/checks/certificates-mpuvq_28/resonance_nondegeneracy_radius8.json` (git blob `dcbe0a7d1a208c6596d37fd4178d20f9cbd07a52`). The replay receipt `evidence/executions/20261006/corpus-revision/final/checks/certificates-mpuvq_28/resonance_certificate_replay.json` (git blob `3621e53121a776596d936829b7a6b75bd97975d2`) records PASS with the same rule counts and both hashes. The generator and the replay checker are `computations/verification/resonance_nondegeneracy_certificate.py` (git blob `6503d86b52b713bfb042031a7e5a97e6f62e8cc8`) and `computations/verification/check_resonance_certificate.py` (git blob `67b715a3888baa2da7d288a5e3d577ebc3d501c1`).

47.3 A human-readable arbitrary-phase example and a quantitative margin

For $R = 1$, in the order $(0,1),(1,-1),(1,0),(1,1)$, three of the exact chosen orbit coefficients are

$$Q_{(-4,-4)} = -4\bar{z}_2^2, \quad Q_{(-2,-4)} = 2\bar{z}_0^2 + 4|z_2|^2, \quad Q_{(0,-2)} = 2|z_1|^2 + 2|z_3|^2. \tag{ND.8}$$

Their successive vanishing immediately forces all four variables to vanish. This is not restricted to a real coefficient vector.

A useful, though not sharp, bound follows without the larger certificate:

$$\Gamma_T(g) \geq \frac{8\pi^2}{17}\|g\|_2^4, \qquad g \in \mathcal{P}_1. \tag{ND.9}$$

Indeed, put $x = |z_0|^2$, $y = |z_2|^2$, $w = |z_1|^2 + |z_3|^2$. Equation (ND.8) and the reverse triangle inequality give $\Gamma_T(g) \geq 8\pi^2[(x-2y)^2 + 4y^2 + w^2]$. Cauchy–Schwarz gives $(x+y+w)^2 \leq$

$(17/4)[(x-2y)^2+4y^2+w^2]$, and (ND.4) proves (ND.9). The sharper even-symbol constant from the earlier project is not silently transferred to arbitrary Fourier phases.

47.3.1 A constructive margin for the radius-eight certificate

Compactness already gives some $c_8 > 0$ with $\Gamma_T(g) \geq c_8\|g\|_2^4$. There is also a completely explicit, deliberately very poor lower bound:

$$\boxed{\Gamma_T(g) \geq \frac{\pi^2}{41472}\, 2^{-2^{715}} \|g\|_2^4, \qquad g \in \mathcal{P}_8.} \tag{ND.10}$$

This number has no useful interpretation as a physical crossover estimate; its purpose is to remove an unspecified compactness constant. It is not claimed to approximate the sharp minimum.

Here is the quantitative compilation rule. Normalize $\max_i |z_i| = 1$ and suppose $|Q_j| \leq \delta$. Replace an exact zero fact by an upper bound $2^L\delta^{p-kb}$ for its absolute value. The symbol b appears only under a temporary assumption $|z_i| \geq \delta^b$. Every deleted monomial is bounded by a known coefficient bound times $|z_j| \leq 1$, or by a known product bound. In the residual row, use the triangle inequality, the minimum of the base exponents, the maximum of their b slopes, and a power-of-two upper bound on the sum of absolute integer coefficients. P records the resulting product bound. S takes a square root, rounding its power-of-two coefficient upward. D subtracts b from the exponent using the explicit nonzero lower bound.

At the end of a refutation child, the pivot has bound $2^L\delta^{p-kb}$ with $p > 0$. Choose $b = p/[2(k+1)]$ and require $\delta \leq 2^{-B}$ with $B \geq 2(L+1)/p$. Then this bound is strictly less than δ^b, contradicting the temporary assumption. Therefore the pivot has the unconditional bound $|z_i| < \delta^b$. Proceed through the root proof and finally enlarge B so that every coefficient is at most $1/2$. This contradicts the normalization. All exponents and comparisons in the compiler are exact rational arithmetic; there is no numerical interval test on the coefficients of g.

For the maintained radius-eight proof the generated integer B has 714 binary digits, so the simple upper bound $B \leq 2^{714}$ suffices. The preceding argument gives $\max_j |Q_j| > 2^{-B}$ on $\max_i |z_i| = 1$. Since $\|g\|_2^4 \leq 4n_R^2(\max_i |z_i|)^4$, (ND.6) yields $\Gamma_T(g) \geq [\pi^2/(2n_R^2)]2^{-2B}\|g\|_2^4$; with $n_8 = 144$ this implies (ND.10). The exact, much less conveniently printed integer and every intermediate rational bound are retained in the margin report. The original margin report was not deposited; it has been reconstructed from the deposited certificate by an exact-rational-arithmetic implementation of the compilation rule above (`computations/verification/resonance_nondegeneracy_margin.py`), and the reconstructed report is deposited at `evidence/executions/20261006/corpus-revision/final/checks/certificates-mpuvq_28/resonance_nondegeneracy_margin.json`. The reconstructed integer has 712 binary digits, so $B \leq 2^{714}$ is confirmed and the explicit constant in (ND.10) is recheckable from deposited materials; two independent implementations of the compiler produce the identical reconstructed integer. The unconditional compactness bound $\Gamma_T(g) \geq c_8\|g\|_2^4$ for some $c_8 > 0$ does not depend on the margin report.

47.4 An invariant Hessian obstruction for every hyperbolic matrix

The finite certificate settles a large but fixed frequency box. A separate analytic criterion has no bandwidth restriction and is not specific to T. Write

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \quad t = a + d, \quad \Delta = t^2 - 4 > 0, \quad B_A = \begin{pmatrix} c & (d-a)/2 \\ (d-a)/2 & -b \end{pmatrix}.$$

The integral quadratic form

$$q_A(v) = v^{\mathsf{T}} B_A v = \det(v, Av)$$

is A-invariant. Define, for a symmetric matrix H,

$$\mathcal{E}_A(H) = (\operatorname{tr}(B_A H))^2 + \Delta \det H, \qquad \mathcal{L}_A = c\partial_{x_1}^2 + (d-a)\partial_{x_1}\partial_{x_2} - b\partial_{x_2}^2. \tag{ND.11}$$

Theorem 47.2 (fixed critical-point obstruction)**.** *Let g be real smooth. If x_0 is a fixed point of $S = A^{\mathsf{T}}$ and $dg(x_0) = 0$, then*

$$\boxed{\Gamma_A(g) = 0 \quad \Longrightarrow \quad \mathcal{E}_A(\operatorname{Hess} g(x_0)) = 0.} \tag{ND.12}$$

In particular, if the two pure second derivatives in the stable and unstable coordinate directions are both nonzero at that point, $\Gamma_A(g) > 0$. Every even g is critical at the fixed point zero, but evenness is not a hypothesis of Theorem 47.2. A nondegenerate extremum at a fixed point is a sufficient condition, not a necessary one.

Proof. Put $h = \{g, \mathcal{C}_A g\}/(4\pi)$. At a fixed point, $e_{Av}(x_0) = e_v(x_0)$, and $q_A(Av) = q_A(v)$. Grouping the absolutely convergent twice-differentiated Fourier series by oriented orbits gives

$$\mathcal{L}_A h(x_0) = -4\pi^2 \sum_{\mathcal{O}} q_A(v_{\mathcal{O}}) e_{v_{\mathcal{O}}}(x_0) \sum_{v \in \mathcal{O}} \widehat{h}(v). \tag{ND.13}$$

If the variance is zero, every orbit sum is zero, so (ND.13) vanishes. No regularity of a putative transfer function has been assumed.

Only the quadratic Taylor jet of g contributes to the second derivative of its bracket at a critical point. For $g_2 = (H_{11}x_1^2 + 2H_{12}x_1x_2 + H_{22}x_2^2)/2$, direct differentiation yields

$$\mathcal{L}_A\{g_2, g_2 \circ A^{\mathsf{T}}\} = t\mathcal{E}_A(H). \tag{ND.14}$$

Equivalently, diagonalize S in an area-one eigenbasis. If its eigenvalues are μ, μ^{-1}, set $\gamma = \mu - \mu^{-1}$; then $\gamma^2 = \Delta$, $\mathcal{L}_A = -\gamma\partial_u\partial_s$, and

$$\mathcal{E}_A(H) = \Delta H_{uu} H_{ss}. \tag{ND.15}$$

The mixed quadratic coefficient of the bracket is $(\mu^{-2} - \mu^2)H_{uu}H_{ss}$, giving (ND.14). This remains true for either sign of the trace. Since $t \neq 0$, (ND.13)–(ND.14) prove (ND.12). End of proof.

The identity (ND.14) was also checked as a polynomial identity with formal matrix entries and Hessian entries. Before the determinant-one specialization, its discriminant is $t^2 - 4\det A$. Two hundred integer instances provide a separate regression, not the general proof.

The converse of (ND.12) is false. For the Pell matrix,

$$g = 4b_{(1,0)} - b_{(2,0)}, \qquad \operatorname{Hess} g(0) = 0, \qquad \Gamma_T(g) = 25600\pi^2 > 0. \tag{ND.16}$$

The exact Fourier computation is retained as a negative control. A vanishing Hessian obstruction is therefore not reported as a zero of the variance.

47.4.1 A quantitative smooth version

The same obstruction has an explicit lower bound when one controls a weighted Fourier tail. Put $C_A = |c| + |d-a| + |b| > 0$, let $s > 0$, and set

$$\mathcal{H}_s(h) = \sum_v |\widehat{h}(v)|(1 + |v|_\infty)^{s+2}, \qquad J = \frac{|\mathcal{L}_A h(x_0)|}{4\pi^2} = \frac{|t\mathcal{E}_A(H)|}{16\pi^3}.$$

If $J > 0$, choose an integer

$$R \geq \max\{1, (2C_A\mathcal{H}_s(h)/J)^{1/s}\}.$$

Then

$$\Gamma_A(g) \geq \frac{J^2}{32C_A^2R^6} > 0. \tag{ND.17}$$

To prove it, choose each orbit's representative of minimum frequency radius. There are at most $8R^2$ oriented orbits meeting the radius-R box, each with $|q_A| \le C_A R^2$. Cauchy–Schwarz bounds their part of (ND.13) divided by $4\pi^2$ by $\sqrt{8}C_A R^3\sqrt{\Gamma}$. For all other orbits, every frequency has radius greater than R, and the sum is bounded by $C_A R^{-s}\mathcal{H}_s(h) \le J/2$. Rearranging proves (ND.17). This bound is for a fixed regular symbol with a nonzero obstruction. It is not uniform over all mixed-sign Fourier polynomials.

47.5 A one-sign Fourier theorem without a bandwidth restriction

Theorem 47.3. *Let $A \in \mathrm{SL}(2,\mathbb{Z})$ be hyperbolic. If the nonconstant coefficients of a real even smooth mean-zero function $g = \sum_v c_v b_v$ all have the same sign and at least one is nonzero, then*

$$\Gamma_A(g) > 0. \tag{ND.18}$$

This is a statement about cosine coefficients, not pointwise nonnegativity of a mean-zero function. There is no restriction on the number of signed orbits.

Proof. Change the overall sign if necessary, so $c_v \ge 0$, and put

$$M = \sum_v c_v v v^{\mathsf{T}} = -\operatorname{Hess} g(0)/(8\pi^2).$$

Polarization of q_A gives the exact discriminant identity

$$B_A(u,v)^2 - q_A(u)q_A(v) = \frac{\Delta}{4}\det(u,v)^2. \tag{ND.19}$$

Expanding $\mathcal{E}_A(M)$ and $\det M$ shows

$$\mathcal{E}_A(M) \ge \left(\sum_v c_v |q_A(v)|\right)^2 > 0. \tag{ND.20}$$

For clarity, each cross coefficient in the difference between the two sides is zero or positive: if $q_A(u)q_A(v) \ge 0$, its extra term is $\Delta\det(u,v)^2$; if the signs differ, it is $4B_A(u,v)^2$ by (ND.19). The diagonal terms agree. Also $q_A(v) \ne 0$ for a nonzero integer v: otherwise A would have a rational eigenline and a rational algebraic-unit eigenvalue, hence eigenvalue 1 or -1, contrary to hyperbolicity. Absolute smooth convergence justifies the sums. Equation (ND.12), or (ND.14), now proves (ND.18). End of proof.

For an even finite Fourier polynomial, there is a useful exact identity behind this criterion. If $a_{\mathcal{O}}$ denotes the cosine sum of h on a signed orbit, then

$$\boxed{\sum_{\mathcal{O}} q_A(v_{\mathcal{O}})a_{\mathcal{O}}(h) = -2\pi t\mathcal{E}_A(M).} \tag{ND.21}$$

It follows from (ND.13)–(ND.14) using $\mathcal{L}_A b_v(0) = -8\pi^2 q_A(v)$ and $\mathcal{E}_A(-8\pi^2 M) = 64\pi^4\mathcal{E}_A(M)$. Thus this test excludes a *nonzero coboundary bracket*, not just an identically zero bracket.

A coarse finite-bandwidth constant is explicit. Define

$$m_A = \gcd(|c|, |d-a|, |b|), \quad C_A = |c| + |d-a| + |b|, \quad D_A = 1 + \max\{|a|+|b|, |c|+|d|\}.$$

For the one-sign class supported in $\|v\|_\infty \le R$,

$$\boxed{\Gamma_A(g) \ge \frac{\pi^2 t^2 m_A^4}{2C_A^2 D_A^6}\, R^{-6}\|g\|_2^4.} \tag{ND.22}$$

Indeed, all generated frequencies lie in the radius-$D_A R$ box. Choosing representatives from those generated frequencies gives $\sum_{\mathcal{O}} q_A(v_{\mathcal{O}})^2 \le 4C_A^2(D_A R)^6$. Cauchy–Schwarz in (ND.21), (ND.1), (ND.20), and $|q_A(v)| \ge m_A$, $\|g\|_2^2 = 2\sum c_v^2$, prove (ND.22). For the Pell matrix this specializes to

$$\Gamma_T(g) \ge \frac{\pi^2}{32768R^6}\|g\|_2^4. \tag{ND.23}$$

No sharpness is claimed. There is no conflict with superpolynomially small positive variances of general Fourier approximations to smooth flat symbols: those approximations need not have one-sign coefficients.

47.6 An exact finite quantum variance consequence

The elementary second-moment bridge already occurs in earlier project notes. We reprove it here with an explicit integer sufficient bound so its application to Theorem 47.1 is auditable without a Gaussian-limit theorem. Let K be any odd positive integer greater than one, write

$$N + F\sqrt{2} = (1+\sqrt{2})^K, \qquad T = \begin{pmatrix} 3 & 2 \\ 4 & 3 \end{pmatrix},$$

and use the physical Weyl operators

$$Q_N(a,b) = (-1)^{ab} W_N(a,b), \qquad Q_N(v)Q_N(w) = e^{\pi i \det(v,w)/N} Q_N(v+w).$$

Here W_N uses modular division by two and is an orthonormal operator basis in normalized full-space Hilbert–Schmidt norm. Take $U_{N,\mathrm{rig}}$ with the exact covariance $U_{N,\mathrm{rig}} Q_N(v) U_{N,\mathrm{rig}}^* = Q_N(Tv)$. Such a unitary is explicitly furnished by the cat kernel in the earlier project; its phase is immaterial here. Primality of K is not needed in this section.

Put $H = (K-1)/2$ and let r_1, r_2 be the absolute row sums of T^H. For a real mean-zero polynomial h supported in $\|v\|_\infty \le M$, assume

$$N^2 > M^2(4 + 4r_1 + 2r_2). \tag{ND.24}$$

Define

$$X_{K,N}(h) = \frac{1}{\sqrt{K}} \sum_{j=0}^{K-1} U_{N,\mathrm{rig}}^j \,\mathrm{Op}_N(h) U_{N,\mathrm{rig}}^{-j}.$$

Then the following equality is **exact at finite N** on the full space:

$$\boxed{\frac{1}{N}\operatorname{Tr} X_{K,N}(h)^2 = \sigma_T^2(h).} \tag{ND.25}$$

No parity assumption is made. The equality is not an assertion about individual eigenphase spacings of a kicked propagator.

Proof. The Pell norm gives $N^2 - 2F^2 = -1$, whence

$$T^K = \begin{pmatrix} 2N^2+1 & 2NF \\ 4NF & 2N^2+1 \end{pmatrix} \equiv I \pmod{2N}. \tag{ND.26}$$

Thus $Q_N(T^K v) = Q_N(v)$, including its physical sign, and adjoint covariance has period K. Differences of times can be reduced to $-H \le s \le H$.

Let u, v lie in the frequency box and suppose $u + T^s v \equiv 0 \pmod N$. Use the indefinite integral norm $q(a,b) = b^2 - 2a^2$, invariant under T and nonzero on every nonzero integer vector. Since the row absolute sums of T^s are bounded by those of T^H,

$$|q(u+T^s v)| \le |q(u)| + |q(v)| + 2|u_2(T^s v)_2 - 2u_1(T^s v)_1| \le M^2(4 + 4r_1 + 2r_2).$$

If $u + T^s v$ were nonzero, its norm would be a nonzero integer multiple of N^2, contradicting (ND.24). Hence the congruence lifts to the exact relation $u = -T^s v$ in $\mathbb{Z}^2$.

Expand the square in (ND.25). Cyclicity of trace and covariance reduce its normalized value to

$$\sum_{s=0}^{K-1} \frac{1}{N} \operatorname{Tr}(\operatorname{Op}_N(h) \operatorname{Op}_N(\mathcal{C}_T^s h)).$$

Weyl traces vanish except at the zero modular frequency. Every surviving relation lifts as above; its two Weyl factors are adjoints and have product I, with no residual phase. There is no double counting of an integer relation since a nonzero integer vector has no nontrivial T-period. Conversely, any integer orbit relation between supported frequencies reduces modulo K and lifts back to the same centered relation. Thus the sum is exactly the oriented-orbit expression (ND.1). End of proof.

For $g \in \mathcal{P}_8$, its symbol $h_g = \{g, \mathcal{C}_T g\}/(4\pi)$ is supported in the radius-64 box. At $K = 13$, $N = 47321$, the two sides of the sufficient inequality (ND.24) are the exact integers

$$N^2 = 2239277041 > 935895040 = 64^2(4 + 4r_1 + 2r_2).$$

Therefore (ND.25) and Theorem 47.1 give a strictly positive quantum averaged-symbol variance for **every nonzero** such g at that module, not just the sampled coefficients. The same conclusion holds at every odd K satisfying (ND.24); for each fixed M the condition holds eventually because its right side grows like $\lambda^{K/2}$ and N^2 grows like λ^K.

The actual finite quantum commutator differs from that symbol by a controlled remainder. Put $G_N = \operatorname{Op}_N(g)$, $D_N = U_{N,\mathrm{rig}} G_N U_{N,\mathrm{rig}}^*$ and $R_N = (iN/2)[G_N, D_N]$. The sine remainder gives

$$\|R_N - \operatorname{Op}_N(h_g)\|_{\mathrm{op}} \le \frac{\pi^3}{6N^2} M_3(g) M_3(\mathcal{C}_T g), \qquad M_3(a) = \sum_v |\widehat{a}(v)|\, |v|_2^3.$$

Here M_3 is local to this section (Euclidean frequency weight); it is not the $\mathcal{M}_r$ of the locality section. Consequently, with normalized Hilbert–Schmidt norm,

$$\left| \left\| \frac{1}{\sqrt{K}} \sum_{j=0}^{K-1} U_{N,\mathrm{rig}}^j R_N U_{N,\mathrm{rig}}^{-j} \right\|_{2,N} - \sqrt{\Gamma_T(g)} \right| \le \frac{\pi^3 \sqrt{K}}{6N^2} M_3(g) M_3(\mathcal{C}_T g). \tag{ND.27}$$

This proves nonvanishing of the averaged commutator for each fixed nonzero radius-eight symbol for all sufficiently large admissible N. It does not assert exact equality between the commutator variance and the classical variance at finite N.

Twelve exact Gaussian-integer coefficient checks compare (ND.25) with a direct modular Weyl-coefficient squared norm, including dimensions 47321 and 1607521. No dense matrices of those sizes are constructed. A small-module negative control, $K = 3, N = 7$, violates the non-aliasing condition and gives different finite quantum and classical values; this mismatch is retained rather than removing hypothesis (ND.24).

Chapter 11 — Temporal resonance and spectral modules

48 Temporal Poisson brackets in dynamical cohomology

This section proves an unrestricted finite-Fourier theorem and a complete classification of quadratic temporal cancellations. The conclusions concern classical dynamical cohomology. Neither microscopic BGS universality nor priority over all external literature is asserted.

48.1 Definitions and the Fourier criterion

Let $A \in \mathrm{SL}(2,\mathbb{Z})$, $|\operatorname{tr} A| > 2$, let $S = A^{\mathsf{T}}$ act on $\mathbb{T}^2$, and write $Cf = f \circ S$. For a nonconstant real trigonometric polynomial g, subtract its mean and put

$$V = \{v : \widehat{g}(v) \neq 0\} \subset \mathbb{Z}^2 \setminus \{0\}, \qquad h_r = \frac{\{g, C^r g\}}{4\pi} \quad (r \geq 1).$$

Here $V = -V$, but Fourier coefficients need not be real. Our Poisson bracket is $\{f, q\} = \partial_1 f\, \partial_2 q - \partial_2 f\, \partial_1 q$ and $e_v(x) = \exp(2\pi i v \cdot x)$. Consequently

$$\widehat{h}_r(k) = -\pi \sum_{u + A^r v = k} \det(u, A^r v)\widehat{g}(u)\widehat{g}(v). \tag{48.1}$$

In particular $\widehat{h}_r(0) = 0$.

For a mean-zero trigonometric polynomial f, let

$$H_f(\mathcal{O}) = \sum_{k \in \mathcal{O}} \widehat{f}(k), \qquad \mathcal{V}_A(f) = \sum_{\mathcal{O}} |H_f(\mathcal{O})|^2,$$

where $\mathcal{O}$ runs over oriented A-orbits, not orbits modulo sign. These are finite sums. Expanding the squared norm and using Fourier orthogonality gives

$$\lim_{n\to\infty} \frac{1}{n} \left\| \sum_{j=0}^{n-1} C^j f \right\|_2^2 = \sum_{q \in \mathbb{Z}} \langle C^q f, f \rangle = \mathcal{V}_A(f).$$

Only finitely many correlations are nonzero: a nonzero integer frequency cannot return to itself under a nonzero power of A. Moreover $\mathcal{V}_A(f) = 0$ if and only if $f = (C - I)F$ for a trigonometric polynomial F, if and only if it is such a coboundary with $F \in L^2$. Indeed, on each orbit the Fourier equation is a difference equation on $\mathbb{Z}$ with finitely supported forcing. Square-summability forces both constant tails to vanish, hence the forcing has zero sum. Conversely a zero-sum finite forcing has a finite cumulative-sum solution. This also proves the criterion without invoking a regularity theorem for transfer functions.

48.2 A support lemma uniform in the temporal lag

Lemma 48.1 (Isolated highest-lag frequency)**.** *For every finite nonempty $V = -V \subset \mathbb{Z}^2 \setminus \{0\}$ and every $r \geq 1$, there are $u, v \in V$ and $k = u + A^r v$ such that this ordered decomposition is unique, $D = \det(u, A^r v) \neq 0$, and*

$$k \notin \bigcup_{j=1}^{r-1} (V + A^j V), \qquad A^q k \notin \bigcup_{j=1}^{r} (V + A^j V) \quad (q \in \mathbb{Z} \setminus \{0\}).$$

Proof. Set $s = \operatorname{sign}(\operatorname{tr} A)$, $B = sA$, $t = |\operatorname{tr} A| \geq 3$, and

$$\lambda = \frac{t + \sqrt{t^2 - 4}}{2}, \qquad \Lambda = \lambda^r.$$

We use $\lambda - \lambda^{-1} > 2$ and $\lambda^2 - 2\lambda - 1 = (t-2)\lambda - 2 > 0$. Choose independent real eigen-covectors ξ, η of B, with factors λ, λ^{-1}. Neither vanishes on a nonzero integer vector: $t^2 - 4$ lies strictly between $(t-1)^2$ and t^2, so the eigendirections are irrational. Set $X = \max_V |\xi| > 0$ and $Y = \max_V |\eta| > 0$. Choose $v_X, u_Y \in V$ attaining $\xi(v_X) = X$, $\eta(u_Y) = Y$, and put $a = \xi(u_Y)/X$, $b = \eta(v_X)/Y$. Choose $\sigma \in \{1, -1\}$ with $\sigma(a + b) \geq 0$ and define

$$L(z) = \xi(z)/X + \Lambda\sigma\eta(z)/Y.$$

The trial points $\sigma u_Y \in V$ and $B^r v_X \in B^r V$ show that the sum of the two separate maxima of L is at least $2\Lambda + |a+b|$. Select the lexicographic maximizers of (L, ξ) on V and $B^r V$. The two forms are independent, so both maximizers are unique. Write them as u and $B^r w$, put $v = s^r w$, and let $k = u + B^r w = u + A^r v$. Then

$$L(k) \geq 2\Lambda. \tag{48.2}$$

An alternative ordered decomposition of k would sum two lexicographically bounded coordinate pairs to the sum of their maxima. Both pairs must attain their separate maxima. This proves uniqueness, including ties in L.

Every point of $V + A^j V$ satisfies $|\xi(z)| \leq (1+\lambda^j)X$ and $|\eta(z)| \leq (1+\lambda^{-j})Y$. In particular k satisfies these bounds with $j = r$. If $A^q k \in V + A^j V$ for $q \geq 1$, $j \leq r$, the first bound implies $|\xi(k)| \leq (1+\Lambda)X/\lambda$. Using k's own second bound,

$$L(k) \leq \Lambda + \Lambda/\lambda + 1 + 1/\lambda < 2\Lambda.$$

The gap is minimized at $r = 1$ and is then $\lambda - \lambda^{-1} - 2 > 0$. For $q \leq -1$ and $j \geq 1$ the second bound instead implies $|\eta(k)| \leq (1+\lambda^{-1})Y/\lambda$. Thus

$$L(k) \leq 1 + \Lambda + \Lambda/\lambda + \Lambda/\lambda^2 < 2\Lambda.$$

Here too the gap is minimized at $r = 1$ and equals $\lambda - \lambda^{-1} - 2$. This excludes all nonzero iterates, without a finite orbit cutoff and for either sign of the trace. For $1 \leq j < r$, membership of k itself would give

$$L(k) \leq 1 + \Lambda + \lambda^j + \lambda^{r-j} \leq 1 + \Lambda + \lambda + \lambda^{r-1} < 2\Lambda.$$

The middle inequality follows from $(\lambda^{j-1} - 1)(\lambda^{r-j-1} - 1) \geq 0$. The last gap is at least $\lambda^2 - 2\lambda - 1 > 0$ since $r \geq 2$.

It remains to exclude $D = 0$. If $u = cB^r w$, put $R = |c|$, $a_0 = |\xi(w)|/X$, $b_0 = |\eta(w)|/Y$. Membership of u, w in V gives $a_0, b_0 \leq 1$, $R\Lambda a_0 \leq 1$, $R\Lambda^{-1} b_0 \leq 1$. For $R > 0$,

$$|L(k)| \leq (1+R)\{\min(\Lambda, R^{-1}) + \min(1, \Lambda/R)\} \leq \Lambda + 2 + \Lambda^{-1} < 2\Lambda.$$

To verify the middle inequality, divide into $R \leq \Lambda^{-1}$, $\Lambda^{-1} \leq R \leq \Lambda$, and $R \geq \Lambda$. The expressions are respectively $(1+R)(\Lambda+1)$, $2 + R + R^{-1}$, and $(1+R^{-1})(\Lambda+1)$, maximized at the indicated boundary points. For $R = 0$ the bound $\Lambda + 1$ suffices. All these conclusions contradict (48.2). □

48.3 Independence and a covariance determinant bound

Theorem 48.2. *The classes $[h_1], [h_2], \ldots$ are linearly independent modulo L^2 C-coboundaries. If $a_m \neq 0$ and m is the highest lag in a finite real combination, the witness in Lemma 48.1 gives*

$$\mathcal{V}_A\left(\sum_{r=1}^{m} a_r h_r\right) \geq 2\pi^2 a_m^2 D_m^2 |\widehat{g}(u_m)\widehat{g}(v_m)|^2 > 0. \tag{48.3}$$

Proof. On the orbit of k_m the entire combination has just one coefficient, $-\pi a_m D_m \widehat{g}(u_m)\widehat{g}(v_m)$, by the support lemma and (48.1). It is nonzero. The orbit of $-k_m$ is distinct, since $A^q k_m = -k_m$ would require an eigenvalue -1 of A^q, impossible for $q \neq 0$, while $q = 0$ would force $k_m = 0$. Reality supplies the conjugate coefficient on that second orbit, proving the factor two and the theorem. □

Let G_m be the covariance matrix $(G_m)_{ij} = \sum_{\mathcal{O}} H_{h_i}(\mathcal{O})\overline{H_{h_j}(\mathcal{O})}$. It is real positive definite. More precisely,

$$\det G_m \geq (2\pi^2)^m \prod_{r=1}^{m} D_r^2 |\widehat{g}(u_r)\widehat{g}(v_r)|^2. \tag{48.4}$$

Indeed the chosen orbits for distinct lags, including their negatives, are pairwise distinct. Their orbit-sum rows have zeros in columns $j < r$ and nonzero diagonal entries $z_r = -\pi D_r \widehat{g}(u_r)\widehat{g}(v_r)$. The Cauchy–Binet formula includes 2^m triangular minors obtained by choosing one member of each conjugate pair; each has squared modulus $\prod_r |z_r|^2$. All other terms are nonnegative.

48.4 A complete classification of quadratic temporal cancellations

Theorem 48.3. *Given real c_{ij}, $0 \le i < j \le m$, define*

$$H = \frac{1}{4\pi} \sum_{0 \le i < j \le m} c_{ij} \{C^i g, C^j g\}, \qquad a_r = \sum_{i=0}^{m-r} c_{i,i+r}.$$

Then H is an L^2 coboundary if and only if $a_1 = \cdots = a_m = 0$. The coefficient-to-cohomology map has rank m and kernel dimension $m(m-1)/2$.

Proof. Since $\det S = 1$, the Poisson bracket commutes with C, and $\{C^i g, C^{i+r} g\}/(4\pi) = C^i h_r$. Therefore

$$H = \sum_{r=1}^{m} a_r h_r + (C - I)F, \qquad F = \sum_{r=1}^{m} \sum_{i=0}^{m-r} c_{i,i+r} \sum_{\ell=0}^{i-1} C^\ell h_r.$$

An empty sum is zero. The independence theorem proves necessity; this formula proves sufficiency with a polynomial transfer. The map $c \mapsto a$ is surjective onto $\mathbb{R}^m$, which proves the rank and nullity. A kernel basis is given by $E_{i,i+r} - E_{0,r}$, $1 \le i \le m - r$. □

Corollary 48.4 (Finite temporal filters)**.** *For real polynomials P, Q, with $P \neq 0$,*

$$\begin{aligned} &\{P(C)g, Q(C)g\}/(4\pi) \text{ is a coboundary} \\ &\iff P(z^{-1})Q(z) = P(z)Q(z^{-1}) \\ &\iff Q/P \in \mathbb{R}(z + z^{-1}). \end{aligned}$$

Proof. Extend coefficient lists by zeros. The coefficient of z^r in the difference of the two Laurent polynomials is $\sum_i (p_i q_{i+r} - q_i p_{i+r})$, exactly the diagonal sum in the preceding theorem. The constant coefficient is zero and negative coefficients are the negatives of the positive ones. For the last equivalence put $w = z + z^{-1}$. The extension $\mathbb{R}(z)/\mathbb{R}(w)$ has basis $1, z$, and its involution sends z to $w - z$. An invariant element $a(w) + zb(w)$ must have $b = 0$. □

The criterion permits nonzero brackets with zero variance: for $P = z$ and $Q = 1 + z^2$ it gives

$$\{Cg, g + C^2 g\}/(4\pi) = (C - I)h_1.$$

Thus nonvanishing of a Poisson bracket must not be confused with a nonzero cohomology class.

Corollary 48.5 (Autocorrelation determines the second-order class)**.** *Write $h(f) = \{f, Cf\}/(4\pi)$. For two real finite filters,*

$$[h(P(C)g)] = [h(Q(C)g)] \quad \iff \quad P(z)P(z^{-1}) = Q(z)Q(z^{-1}).$$

Proof. Let $b_j(P) = \sum_i p_i p_{i+j}$ for $j \ge 0$. Applying the cancellation formula to P and zP gives the coefficient $b_{r-1}(P) - b_{r+1}(P)$ at $[h_r]$. Equality of classes therefore makes the difference of the two autocorrelation sequences constant separately on each parity. Both sequences vanish eventually, so their difference is zero. The converse follows from the same formula. □

For example $P = 1 + 5z + 6z^2$ and $Q = 2 + 7z + 3z^2$ are not related by a sign, shift or reversal, but both have autocorrelations $(62, 35, 6)$ and give the same class $56[h_1] + 35[h_2] + 6[h_3]$. This is equality of a second-order class, not equality of full quantum spectra or of all perturbative orders.

48.5 Finite bandwidth and loss of coercivity

The case $r = 1$ proves $\Gamma_A(g) := \mathcal{V}_A(h_1) > 0$ without a bandwidth or phase restriction. If $M = |V|$ and $|\widehat{g}(v)| \geq \alpha\|g\|_2/\sqrt{M}$ on V, then the integer $D_1 \neq 0$ in (48.3) yields

$$\Gamma_A(g) \geq 2\pi^2\alpha^4 M^{-2}\|g\|_2^4.$$

The amplitude hypothesis cannot be dropped. Let $\mathcal{P}_R$ be the real mean-zero Fourier polynomials of max-norm bandwidth at most $R \geq 1$ and $c_R(A) = \inf_{0\neq g\in\mathcal{P}_R} \Gamma_A(g)/\|g\|_2^4$. Compactness of the finite-dimensional unit sphere gives $c_R(A) > 0$. Nevertheless, for every $s > 1$ there are constants depending on A, s with

$$0 < c_R(A) \leq C_s \exp(-c_s R^{1/s}). \tag{48.5}$$

To see this, choose a small box U around a point outside $Sx = \pm x$ so that $U \cup (-U)$ is disjoint from its inverse image under S. Take a nonzero real even Gevrey-s bump supported there, and subtract its mean to obtain b. Then $\{b, Cb\} = 0$, since the derivative supports are disjoint. Such bumps are obtained from $\exp(-(1-x^2)^{-1/(s-1)})$ on $(-1, 1)$, extended by zero, by products, translation and rescaling. Their derivatives obey $CL^n(n!)^s$. For completeness, this derivative estimate follows from Cauchy's formula on a complex disk of radius proportional to the distance d to the endpoint: the bound is $n!(cd)^{-n}\exp(-c'd^{-1/(s-1)})$. Maximizing in d gives $CL^n(n!)^s$; all derivatives tend to zero at the endpoint. Integration by parts and optimization in n imply $|\widehat{b}(v)| \leq C\exp(-c|v|^{1/s})$. For $b_R = P_R b$ and $M_1(f) = \sum_v |v||\widehat{f}(v)|$, this gives $M_1(b - b_R) \leq C\exp(-cR^{1/s})$ after absorbing polynomial factors. The determinant bound in the Fourier bracket implies

$$\left\|\widehat{\{b_R, Cb_R\}/(4\pi)}\right\|_1 \leq \pi\|A\|_{\mathrm{op}}\{M_1(b_R) + M_1(b)\}M_1(b_R - b).$$

Orbit sums satisfy $\mathcal{V}_A(f) \leq \|\widehat{f}\|_1^2$. Also $\|b_R\|_2$ stays bounded below. This proves (48.5), using strict positivity for every nonzero b_R. No nondegeneracy theorem for arbitrary infinite analytic Fourier support is inferred from these finite-band statements.

48.6 Arbitrarily high-order collective soft modes

The following result distinguishes finite-dimensional nondegeneracy from coercivity uniform in the number of temporal lags. Let $t = |\operatorname{tr} A|$, $\delta = \sqrt{t^2 - 4}$, $\lambda = (t + \delta)/2$, and $g_0 = a_0 e_v + \overline{a_0} e_{-v}$, where $v \neq 0$ is integral and $a_0 \neq 0$.

Lemma 48.6 (Exact single-mode orthogonality)**.** *For $D_j = \det(v, A^j v)$, the lag classes generated by g_0 have covariance*

$$\langle [h_j(g_0)], [h_k(g_0)]\rangle = 4\pi^2|a_0|^4 D_j^2\, \mathbf{1}_{j=k}, \qquad D_j^2 = D_1^2\frac{(\lambda^j - \lambda^{-j})^2}{\delta^2}.$$

Proof. The four Fourier frequencies of $h_j(g_0)$ are $\pm(v + A^j v)$ and $\pm(v - A^j v)$; each coefficient has modulus $\pi|D_j||a_0|^2$. The invariant $q_A(w) = \det(w, Aw)$ satisfies

$$q_A(v \pm A^j v) = q_A(v)(2 \pm \operatorname{tr} A^j).$$

The numbers $|\operatorname{tr} A^j| = \lambda^j + \lambda^{-j}$ strictly increase and exceed two. These invariants distinguish the two types and all different positive lags, including negative trace. Opposite frequencies cannot share an oriented orbit. The covariance formula follows. Cayley–Hamilton gives $D_{j+1} = \operatorname{tr}(A)D_j - D_{j-1}$ with $D_0 = 0$, which proves the squared closed form. □

Theorem 48.7 (Soft modes of every even differential order)**.** *Fix an integer $q \geq 1$ and set*

$$P_q(z) = (z^2 - tz + 1)^q, \qquad g_q = P_q(C)g_0, \qquad d = 2q.$$

For any fixed $R \geq d$, consider the n cohomology vectors $\lambda^{-r}[h_r(g_q)]$, $r = R+1, \dots, R+n$, and their covariance matrix $\widetilde{G}_{R,n}$. There are positive constants, independent of n, such that

$$c\, n^{-4q} \leq \lambda_{\min}(\widetilde{G}_{R,n}) \leq C\, n^{-4q}.$$

Thus the finite matrices are all positive definite but have collective almost-null directions of arbitrarily high algebraic order.

Proof. Write $P_q(z) = \sum_{i=0}^{d} p_i z^i$ and $b_k = \sum_i p_i p_{i+k}$, with $b_{-k} = b_k$ and $b_k = 0$ outside $[-d, d]$. For $r > d$, symplectic transport and reduction modulo coboundaries give the exact identity

$$[h_r(g_q)] = \sum_{k=-d}^{d} b_k [h_{r+k}(g_0)].$$

Let $w_k = b_k \lambda^k$ and $C_0 = 4\pi^2 |a_0|^4 D_1^2/\delta^2$. By the preceding lemma, for a real coefficient vector $x = (x_r)$ the squared norm of its combination is

$$x^{\mathsf{T}} \widetilde{G}_{R,n} x = C_0 \sum_j (1 - \lambda^{-2j})^2 \left| \sum_{r=R+1}^{R+n} x_r w_{j-r} \right|^2 .$$

All relevant j satisfy $j \geq R + 1 - d \geq 1$. Hence, in the order of quadratic forms,

$$C_0 (1 - \lambda^{-2(R+1-d)})^2 T_n(F_q) \;\leq\; \widetilde{G}_{R,n} \;\leq\; C_0 T_n(F_q), \tag{48.6}$$

where $T_n(F_q)$ is the Toeplitz matrix with symbol $F_q(\theta) = |\sum_k w_k e^{ik\theta}|^2$ and Haar normalization $d\theta/(2\pi)$. The autocorrelation identity gives

$$\sum_k w_k e^{ik\theta} = P_q(z) P_q(z^{-1}), \qquad z = \lambda e^{i\theta}.$$

Since $(z^2 - tz + 1)(z^{-2} - tz^{-1} + 1) = (z + z^{-1} - t)^2$, putting $u(\theta) = 2 - 2\cos\theta$ yields the exact factorization

$$\boxed{F_q(\theta) = u(\theta)^{2q} (\delta^2 + u(\theta))^{2q}.} \tag{48.7}$$

Indeed $|z + z^{-1} - t|^2 = t^2(\cos\theta - 1)^2 + \delta^2 \sin^2\theta = u(\delta^2 + u)$.

It remains to bound the smallest eigenvalue. Since $0 \leq u \leq 4$, the factor $(\delta^2 + u)^{2q}$ is bounded above and below by positive constants. For $p(\theta) = \sum_{j=1}^{n} x_j e^{ij\theta}$ with $\|x\|_2 = 1$, Jensen's inequality and the Dirichlet tridiagonal Laplacian give

$$\int u^{2q} |p|^2 \frac{d\theta}{2\pi} \geq \left(\int u |p|^2 \frac{d\theta}{2\pi} \right)^{2q} \geq \left(4 \sin^2 \frac{\pi}{2(n+1)} \right)^{2q}.$$

For an upper bound choose $x_j = \phi(j/(n+1))$, where $\phi(x) = x^{2q}(1-x)^{2q}$ on $[0,1]$ and is zero elsewhere. This extension has bounded weak derivative of order $2q$. Its finite differences of order $2q$ are bounded by $C_q (n+1)^{-2q}$, including across the endpoints. Parseval identifies the last integral, before normalization, with the sum of squares of these differences. That sum is at most $C_q n^{1-4q}$, while $\sum_j |x_j|^2 \geq c_q n$ for all sufficiently large n. This proves the matching $O(n^{-4q})$ Rayleigh bound; finitely many smaller n are absorbed in the constants. Combining with (48.6) proves the theorem for the actual finite-lag covariance matrices, not only for a limiting Toeplitz model. □

The normalization λ^{-r} removes the natural exponential growth of individual brackets. The theorem does not assert an exact null vector at any finite n, a random-matrix law, or a uniform-in-time quantum correspondence. It identifies a quantified obstruction to replacing finite-lag positivity by a lag-uniform lower bound.

48.7 A finite quantum consequence and its time boundary

Use the physical Weyl lift

$$(Q_N(a,b)u)(x) = e^{2\pi i a(x-b/2)/N} u(x-b \bmod N), \qquad Q_N(v)Q_N(w) = e^{\pi i \det(v,w)/N} Q_N(v+w).$$

Write $\mathrm{Op}_N(f) = \sum_v \widehat{f}(v) Q_N(v)$ and, for real a_r with highest nonzero index m, form the Hermitian operator

$$T_{n,N} = \sum_{\ell=0}^{n-1} \sum_{r=1}^{m} a_r \frac{iN}{2} [\mathrm{Op}_N(C^\ell g), \mathrm{Op}_N(C^{\ell+r} g)].$$

These are defined as transported commutators. Identifying them with time evolution by a quantum cat map additionally requires exact Egorov in the chosen boundary sector; it is not assumed for every A in a fixed sector.

Proposition 48.8. *Let $R = \max_{v\in V} \|v\|_\infty$, $a = \|A\|_\infty$. If*

$$2Ra^{n-1}(1+a^m) < N,$$

then the witness of Lemma 48.1 at lag m gives

$$\frac{\mathrm{Tr}(T_{n,N}^2)}{nN} \geq 2a_m^2 N^2 \sin^2(\pi D_m/N) |\widehat{g}(u_m)\widehat{g}(v_m)|^2.$$

It is strictly positive when N does not divide D_m.

Proof. All occurring frequencies lie in the open max-norm box of radius $N/2$, so distinct integer frequencies remain distinct modulo N and their Weyl operators are orthogonal for the normalized Hilbert–Schmidt inner product. The support lemma isolates the $2n$ different frequencies $\pm A^\ell k_m$, $0 \leq \ell < n$, in the entire transported sum. Their coefficient moduli are $|a_m|N|\sin(\pi D_m/N)|\,|\widehat{g}(u_m)\widehat{g}(v_m)|$, since determinants are preserved by A. Summing their squared moduli proves the bound. Hermiticity identifies the squared Hilbert–Schmidt norm with the trace in the statement. □

For fixed A, R, m the sufficient no-alias condition allows only $n = O(\log N)$, not the Heisenberg scale $n \asymp N$. This is an observable variance bound, not a joint-trace estimate for eigenphase correlations. The distinction is essential: at $N = 3$, $A = \left(\begin{smallmatrix} 4 & 1 \\ 3 & 1 \end{smallmatrix}\right)$ and $g = e_{(1,0)} + e_{(-1,0)}$, the classical determinant is 3, but $Q_3(A(1,0)) = Q_3(1,0)$ and the first quantum commutator vanishes. The no-alias condition fails in this negative control.

Chapter 10 — Resonance rigidity (continued)

49 Smooth resonance rigidity on finitely many hyperbolic Fourier orbits

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Status: project proof with adversarial checking by the author and separate finite exact regressions. Not independently refereed, not Lean-certified, and no external priority or BGS solution is claimed. The finite-Fourier independence theorem and common real all-pass invariance are existing repository inputs. The extension here treats any finite union of infinite signed frequency orbits, with arbitrary phases, using smoothness alone.

49.1 Contract

Let $A \in \mathrm{SL}(2,\mathbb{Z})$ be hyperbolic, and let $Cf(x) = f(A^{\mathsf{T}}x)$ on the two-dimensional torus with normalized Lebesgue measure. Write

$$e_k(x) = e^{2\pi i k\cdot x}, \quad h_r(g) = \frac{\{g, C^r g\}}{4\pi}, \qquad \{f,h\} = f_{x_1}h_{x_2} - f_{x_2}h_{x_1}.$$

Oriented orbits are $A^{\mathbb{Z}}k$; signed orbits also identify k and $-k$. These notions must not be interchanged in the variance. For a smooth mean-zero f, put

$$\mathcal{V}_A(f) = \sum_{\mathcal{O}} \left| \sum_{k\in\mathcal{O}} \widehat{f}(k) \right|^2 . \tag{FO.1}$$

Theorem 49.1. *Let g be a nonconstant real C^∞ function whose nonzero Fourier support is contained in finitely many signed A-orbits. Then, for every $R \geq 1$ and every nonzero real vector $(a_1,\dots,a_R)$,*

$$\boxed{\mathcal{V}_A\left(\sum_{r=1}^{R} a_r h_r(g)\right) > 0.} \tag{FO.2}$$

Consequently the classes $[h_r(g)]$, $r \geq 1$, are linearly independent modulo L^2 C-coboundaries. In particular $\Gamma_A(g) = \mathcal{V}_A(h_1(g)) > 0$. Complex linear independence follows by separating real and imaginary parts.

There is no restriction on the number of coefficients on each orbit, their phases, the sign of $\operatorname{tr} A$, or an analytic strip width. No uniform positive lower bound over all g or all orbit families is asserted. Constants in g do not affect the conclusion and will be subtracted.

49.2 Variance and coboundaries

For a smooth mean-zero f, absolute Fourier summability gives

$$\sum_{q\in\mathbb{Z}} \sum_{k\neq 0} |\widehat{f}(k)\widehat{f}(A^q k)| \leq \left(\sum_{k\neq 0} |\widehat{f}(k)| \right)^2 < \infty.$$

A hyperbolic A has no periodic nonzero integer frequency, so each ordered pair on one orbit occurs once. Fourier orthogonality and Cesaro summation give

$$\mathcal{V}_A(f) = \lim_{n\to\infty} \frac{1}{n} \left\| \sum_{j=0}^{n-1} C^j f \right\|_2^2 . \tag{FO.3}$$

If $f = u - Cu$ with $u \in L^2$, its equation on an orbit is $f_j = u_j - u_{j-1}$. Square summability makes u_j tend to zero at both ends; the absolutely convergent sum of f_j is zero. A nonzero orbit sum therefore excludes an L^2 coboundary. Equation (FO.2) is stronger than pointwise nonvanishing.

49.3 Eventual separation of every pair channel

Choose $v_1,\dots,v_d \in \mathbb{Z}^2 \setminus \{0\}$, distinct modulo signed A-orbits, and define

$$\mathcal{S} = \bigcup_{i=1}^{d} \{\pm A^j v_i : j \in \mathbb{Z}\}.$$

Lemma 49.2 (large-gap separation)**.** *There exists n_0, depending on A and this finite list, such that the following holds. If $n > n_0$, i, j are any indices, $\epsilon \in \{+1, -1\}$, and*

$$v_i + \epsilon A^n v_j = \zeta A^t (v_k + \delta A^m v_l), \quad m \geq 0, \quad t \in \mathbb{Z}, \quad \delta, \zeta \in \{+1, -1\}, \tag{FO.4}$$

then $i = k$, $j = l$, $m = n$, $t = 0$, $\zeta = 1$, and $\delta = \epsilon$. Also $D_{ij}(n) = \det(v_i, A^n v_j) \neq 0$ for every $n > n_0$.

Every sum of two elements of $\mathcal{S}$ can be put in the right-hand form by a common signed iterate and an interchange. Thus the statement covers all input exponents, without a guessed finite time cutoff.

Proof. Put $s = \operatorname{sign}(\operatorname{tr} A)$, $B = sA$, and let $\lambda > 1$ be the expanding eigenvalue of B. Choose independent real eigen-covectors ξ, η with factors λ and λ^{-1}. Neither vanishes on a nonzero integer vector: a rational invariant line of an integral unimodular hyperbolic matrix would give a rational algebraic-unit eigenvalue, necessarily ± 1.

The quadratic form $Q(z) = \xi(z)\eta(z)$ obeys $Q(Az) = Q(z)$. For fixed nonzero integer vectors u, v,

$$Q(u + \epsilon A^n v) = Q(u) + Q(v) + \epsilon s^n \{\eta(u)\xi(v)\lambda^n + \xi(u)\eta(v)\lambda^{-n}\}. \tag{FO.5}$$

The leading coefficient is nonzero. Its absolute value is therefore bounded above and below by positive multiples of λ^n for sufficiently large n.

Suppose (FO.4) has nontrivial solutions with unbounded n. Pass to a subsequence with fixed $i, j, k, l, \epsilon, \delta, \zeta$. Applying Q to (FO.4) removes t. Formula (FO.5) forces $m \to \infty$ and makes $m - n$ a bounded integer. Pass to a subsequence with $m = n + b$, with b fixed. Applying ξ and taking absolute values in (FO.4) gives

$$\lambda^t = \frac{|\xi(v_i + \epsilon A^n v_j)|}{|\xi(v_k + \delta A^{n+b} v_l)|}. \tag{FO.6}$$

The numerator is comparable to λ^n and the denominator to λ^{n+b}. Thus t is also a bounded integer; pass to a subsequence with t fixed. Now (FO.4) becomes the exact vector equation

$$v_i - \zeta A^t v_k = A^n(\zeta\delta A^{b+t} v_l - \epsilon v_j). \tag{FO.7}$$

Both parenthesized vectors are fixed integer vectors, even when $b+t < 0$, since A is unimodular. If the right parenthesis were nonzero, its expanding covector would be nonzero and the right side would be unbounded as $n \to \infty$. Thus both sides vanish. The first equality and the distinct signed-orbit representatives imply $i = k$. Then $v_i = \zeta A^t v_i$ forces $t = 0$ and $\zeta = 1$ by hyperbolicity. The second equality implies $j = l$, $b = 0$, and $\delta = \epsilon$. This is exactly the trivial solution, contradicting the choice of nontrivial solutions. Finitely many index and sign choices make the upper bound n_0 uniform over this fixed list.

Finally $\det(u, A^n v)$ is a nonzero multiple of $s^n[\xi(u)\eta(v)\lambda^{-n} - \eta(u)\xi(v)\lambda^n]$. Its leading coefficient is nonzero, so it cannot vanish for arbitrarily large n. Increase n_0 for the finitely many pairs. This proves the lemma for either trace sign.

Caution. Equality of Q alone is not a criterion for orbit equivalence. Equation (FO.7), not just comparison of quadratic invariants, is essential.

49.4 Vanishing variance forces polynomial pair products

Write, omitting empty orbit families,

$$g=\sum_{i=1}^{d}\sum_{j\in\mathbb{Z}}(c_{i,j}e_{A^jv_i}+\overline{c_{i,j}}e_{-A^jv_i}),\quad P_i(z)=\sum_{j\in\mathbb{Z}}c_{i,j}z^j. \tag{FO.8}$$

Smoothness implies every exponential moment: for every $L>0$, $\sum_j|c_{i,j}|e^{L|j|}<\infty$. Indeed $|A^jv_i|$ is bounded below by a positive multiple of $\lambda^{|j|}$, and smooth Fourier coefficients decay faster than every inverse power. Thus each nonzero P_i is holomorphic on $\mathbb{C}^*=\mathbb{C}\setminus\{0\}$.

Introduce the two different correlations

$$S_{ij}(n)=\sum_t c_{i,t}c_{j,t+n},\qquad K_{ij}(n)=\sum_t c_{i,t}\overline{c_{j,t+n}}. \tag{FO.9}$$

For $n>n_0$, Lemma 49.2 and direct Fourier differentiation give

$$\begin{aligned}J_{A^{\mathbb{Z}}(v_i+A^nv_j)}(h_r)&=-\pi D_{ij}(n)[S_{ij}(n-r)-S_{ij}(n+r)],\\ J_{A^{\mathbb{Z}}(v_i-A^nv_j)}(h_r)&=\pi D_{ij}(n)[K_{ij}(n-r)-K_{ij}(n+r)].\end{aligned} \tag{FO.10}$$

For the second row, the two ordered input pairs contribute

$$+\pi D_{ij}(n)c_{i,t}\overline{c_{j,t+n-r}},\qquad -\pi D_{ij}(n)\overline{c_{j,t+n}}c_{i,t-r}.$$

Summing over t proves the formula. The first row is the same computation without conjugation and with the opposite sign. Lemma 49.2 excludes any additional input families or gap values. All rearrangements are justified by smooth weighted Fourier summability.

Assume, for contradiction, that the variance in (FO.2) is zero for a nonzero real vector a. Every orbit sum of the combination is zero. For $X=S$ and $X=K$, and every i,j, (FO.10) implies

$$B_{ij}^X(n):=\sum_{r=1}^{R}a_r[X_{ij}(n-r)-X_{ij}(n+r)]=0\quad(n>n_0). \tag{FO.11}$$

The symmetries $S_{ij}(-n)=S_{ji}(n)$ and $K_{ij}(-n)=\overline{K_{ji}(n)}$ also make B_{ij} vanish for all sufficiently negative n. Hence all these B sequences are finitely supported.

Put $P^*(z)=\overline{P(1/\bar z)}$ and $L_a(z)=\sum_{r=1}^{R}a_r(z^{-r}-z^r)$, a nonzero Laurent polynomial. The generating identities are

$$\begin{aligned}P_i(z)P_j(1/z)&=\sum_n S_{ij}(n)z^{-n},\\ P_i(z)P_j^*(z)&=\sum_n K_{ij}(n)z^{-n}.\end{aligned} \tag{FO.12}$$

Multiplication by L_a produces the finite Laurent series B in (FO.11). Thus each product in (FO.12) is rational and holomorphic on $\mathbb{C}^*$. It is consequently a Laurent polynomial. Both kinds of product are required: Hermitian products alone would not force the common factor below to have real Laurent coefficients.

49.5 A common real all-pass factor

Lemma 49.3 (pair-product factorization)**.** *Suppose nonzero functions $P_1, \dots, P_d$ are holomorphic on $\mathbb{C}^*$, and all $P_i(z)P_j(1/z)$ and $P_iP_j^*$ are Laurent polynomials. Then*

$$P_i(z) = R(z)Q_i(z), \tag{FO.13}$$

where every Q_i is a Laurent polynomial (complex coefficients allowed), R is holomorphic and zero-free on $\mathbb{C}^*$, has real Laurent coefficients, and

$$R(z)R(1/z) = R(z)R^*(z) = 1. \tag{FO.14}$$

Proof. The ratios P_i/P_1 are rational, by dividing $P_i(z)P_1(1/z)$ by $P_1(z)P_1(1/z)$. Clear denominators and divide the resulting polynomial vector by its common divisor. This gives Laurent polynomials q_i with greatest common divisor 1 and a meromorphic F with $P_i = Fq_i$. Bezout polynomials b_i satisfy $\sum_i b_i q_i = 1$, so $F = \sum_i b_i P_i$ is holomorphic on $\mathbb{C}^*$.

The zeros of F form a finite set with finite multiplicities: they are zeros of the nonzero Laurent polynomial $P_1P_1^*$. Divide F by the ordinary polynomial formed from these zeros, and multiply every q_i by that polynomial. The resulting F_0 is holomorphic and zero-free on $\mathbb{C}^*$.

Using a nonzero q_i, both $F_0(z)F_0(1/z)$ and $F_0(z)F_0^*(z)$ are rational. They are holomorphic and zero-free on $\mathbb{C}^*$, so each is a nonzero constant times an integer power of z. The first is unchanged by $z \mapsto 1/z$ and is therefore a constant c. The second is strictly positive on the unit circle, so its integer power is zero; it is a positive constant d. Evaluation at $z = 1$ gives $c = F_0(1)^2$ and $d = |F_0(1)|^2$.

Normalize $R = F_0/F_0(1)$, absorbing $F_0(1)$ into the polynomial factors. This gives (FO.14). Dividing the two identities gives $R(1/z) = R^*(z)$, or $R(w) = \overline{R(\bar{w})}$. Hence its Laurent coefficients are real. This proves the lemma. Notice that R need not be a monomial: smoothness allows nontrivial entire all-pass filters.

49.6 Reduce to the existing finite-Fourier theorem

Apply Lemma 49.3 to (FO.12). Expand $R(z) = \sum_j p_j z^j$. Its real Laurent coefficients have every exponential moment, by Cauchy estimates on arbitrarily large and small circles. Define the real trigonometric polynomial q by placing the coefficients of Q_i along $A^{\mathbb{Z}}v_i$ and their complex conjugates along the distinct negative orbits. Then

$$g = T_R q := \sum_{j \in \mathbb{Z}} p_j C^j q. \tag{FO.15}$$

This follows coefficient by coefficient from (FO.13). The sum converges in every smooth seminorm. The polynomial q is nonconstant since g is.

For completeness, the needed all-pass invariance is elementary here. Define $h_s(q)$ also for negative s, and set $h_0 = 0$. Symplectic invariance gives

$$h_r(T_R q) = \sum_{i,j} p_i p_j C^i h_{r+j-i}(q).$$

The Fourier ℓ^1 norm of $h_s(q)$ grows at most exponentially in $|s|$, whereas the p_j have every exponential moment. All sums converge absolutely, and orbit summation commutes with rearrangement. Since $J(C^i f) = J(f)$ and $R(z)R(1/z) = 1$ implies $\sum_i p_i p_{i+n} = \delta_{n,0}$,

$$\boxed{J(h_r(g)) = J(h_r(q)) \quad \text{for every } r.} \tag{FO.16}$$

This is the existing common real entire-all-pass invariance mechanism. The existing finite-Fourier all-lag theorem, applied to the nonconstant real polynomial q, now contradicts the assumed zero variance of the nonzero combination in (FO.2). This proves Theorem 49.1.

The internal finite-Fourier input is pinned at the base revision: `manuscript/unified_sections/temporal_resonance.tex`, isolated highest-lag frequency lemma and independence theorem. Its $r = 1$ version is also in `mathematics/extremal_resonance_nondegeneracy.md`. The earlier all-pass invariance theorem is in `mathematics/entire_temporal_gauge.md`. The large-gap separation mechanism is also already used in `mathematics/phase_regularized_resonance.md`, Section 2. Lemma 49.2 gives an alternative qualitative proof; it is not a priority claim for this mechanism. The contribution here is the smooth finite-orbit extension and the common factor normal form, relative to these inspected inputs.

49.7 Two additional consequences and a sharp boundary of this route

Finite invariant range. Write $q_A(k) = \det(k, Ak) \neq 0$ for $k \in \mathbb{Z}^2 \setminus \{0\}$. For each finite H, the set $\{k : 0 < |q_A(k)| \leq H\}$ is a finite union of signed A-orbits. Indeed q_A is a nonzero constant multiple of $\xi(k)\eta(k)$. Choose an iterate on each orbit with $1 \leq |\xi(k)|/|\eta(k)| < \lambda^2$. A bound on the product bounds both coordinates and hence the Euclidean norm. Only finitely many integer representatives remain.

The theorem applies to every smooth nonconstant g whose Fourier support has bounded $|q_A(k)|$. Any smooth, and in particular any analytic, zero-variance counterexample must occupy infinitely many signed orbits and have unbounded $|q_A(k)|$ on its Fourier support.

Finite-tail normal form. The proof also shows: if for one nonzero finite real temporal coefficient vector all sufficiently large same-sign and opposite-sign pair-channel sums vanish, then $g = T_R q$, with q a real Fourier polynomial and R a common real entire all-pass filter. Conversely this form has finite tails in all these pair-channel sums, by (FO.12)–(FO.14). This is a normal form for eventual pair-channel cancellation, not for zero variance.

No uniform isolation over arbitrary families. For any n and $v \neq 0$,

$$v - A^n v = 2v + (-v - A^n v). \tag{FO.17}$$

With positive trace and n growing, the added vectors $-v - A^n v$ have distinct invariant values $q_A(v)(2 + \operatorname{tr} A^n)$. Thus arbitrarily late new-family collisions can be introduced. There is no family-independent n_0. A countable union can have infinitely many such late collisions, even if its Fourier coefficients are chosen to decay analytically. This exhibits failure of the isolation argument, not a zero-variance analytic counterexample.

49.8 Remaining obligations

The theorem does not treat arbitrary analytic observables with infinitely many signed Fourier orbits. No limiting passage through orbit truncations is provided: positivity without a uniform quantitative lower bound is insufficient. The finite-tail characterization does not assert that general smooth g has finite tails; that was a consequence of a hypothetical zero combination.

This result is classical cohomological nondegeneracy. It proves no Heisenberg-time joint-trace or mixed-resolvent cancellation for the frozen quantum map. It does not imply GOE, GUE, COE, CUE, or microscopic BGS. External priority is unestablished. Independent review should focus on Lemma 49.2, both channel identities in (FO.10), and the reality step in Lemma 49.3.

Chapter 11 — Temporal resonance and spectral modules (continued)

50 Spectral modules, filter roots, and exact quantum cycles

This section strengthens the temporal support theorem in five directions. The classical statements concern observable covariance, not eigenvalue statistics. The finite-field theorem is exact at every time, including times of order the Hilbert-space dimension; it is not a local spectral law.

50.1 All temporal translates form a free, stable module

Retain A, C, g, h_r and the isolated frequencies k_r of Lemma 48.1. On each oriented frequency orbit choose a representative $k_{\mathcal{O}}$ and define the finite Laurent polynomial

$$H_r(\mathcal{O}, z) = \sum_{\ell \in \mathbb{Z}} \widehat{h}_r(A^\ell k_{\mathcal{O}}) z^\ell.$$

Only finitely many orbits and coefficients occur for fixed m. Let $M_m(z) = (H_r(\mathcal{O}, z))_{\mathcal{O}, 1 \le r \le m}$ and $S_m(\theta) = M_m(e^{i\theta})^* M_m(e^{i\theta})$. Set

$$D_m^* = (2\pi^2)^m \prod_{r=1}^{m} \det(u_r, A^r v_r)^2 |\widehat{g}(u_r)\widehat{g}(v_r)|^2, \qquad L_m = \sum_{r=1}^{m} \|\widehat{h}_r\|_1^2.$$

Both are strictly positive.

Theorem 50.1 (Uniform spectral positivity and temporal Riesz bounds)**.** *For all real θ,*

$$\det S_m(\theta) \ge D_m^*, \qquad \frac{D_m^*}{L_m^{m-1}} I_m \le S_m(\theta) \le L_m I_m.$$

Consequently for every finitely supported complex coefficient array $(b_{r,j})$, with arbitrary integer times j,

$$\frac{D_m^*}{L_m^{m-1}} \sum_{r,j} |b_{r,j}|^2 \le \left\| \sum_{r=1}^{m} \sum_{j} b_{r,j} C^j h_r \right\|_2^2 \le L_m \sum_{r,j} |b_{r,j}|^2. \tag{50.1}$$

The family $h_1, h_2, \dots$ is free over $\mathbb{C}[z, z^{-1}]$ under $z \cdot f = Cf$. In particular all functions $\{C^i g, C^j g\}$, indexed by integer $i < j$, are linearly independent over $\mathbb{C}$ in finite combinations.

Proof. Use k_r and $-k_r$ themselves as the representatives of their pairwise distinct orbits. In the rows of M_m corresponding to this pair, columns $j < r$ vanish, while column r is the constant $z_r = -\pi \det(u_r, A^r v_r)\widehat{g}(u_r)\widehat{g}(v_r)$ or its conjugate. Selecting one row from each pair gives 2^m triangular minors with squared absolute determinant $\prod_r |z_r|^2$, independently of θ. Cauchy–Binet proves the determinant bound. The triangle inequality gives $\operatorname{Tr} S_m(\theta) \le L_m$, so every eigenvalue is at most L_m and the smallest is at least D_m^*/L_m^{m-1}.

Transport by C^j shifts the Fourier exponent on an orbit by j. Parseval on each orbit identifies the middle member of (50.1) with $\int \|M_m(e^{i\theta}) b(e^{i\theta})\|^2 d\theta/(2\pi)$, where $b_r(z) = \sum_j b_{r,j} z^j$. A second application of Parseval proves the bounds and module freeness. The last assertion uses $\{C^i g, C^j g\} = 4\pi C^i h_{j-i}$. □

The constants depend on the largest lag m and the symbol g but not on the length or location of the time window. Thus the assertion does not contradict the soft-mode theorem, which increases the number and size of the lags. It also shows that no nonzero finite scalar combination of the h_r is an L^2 twisted coboundary $(C - e^{i\theta_0})F$: its orbit polynomial on the isolated highest-lag orbit is a nonzero constant. Solving the corresponding one-dimensional difference equation would require a nonsquare-summable tail.

50.2 Complete coercivity classification for filtered single modes

Let $g_0 = a_0 e_v + \overline{a_0} e_{-v}$ as before. Fix any nonzero real polynomial $P(z) = \sum_{i=0}^{d} p_i z^i$, put $g = P(C)g_0$, and take $R \geq d$. For $r = R+1, \dots, R+n$, form the covariance matrix of $\lambda^{-r}[h_r(g)]$, denoted $\widetilde{G}_{P,R,n}$. Let $\nu_P(w)$ be the root multiplicity of P at w (zero if not a root), and define

$$\mu_P(\theta) = \nu_P(\lambda e^{i\theta}) + \nu_P(\lambda^{-1} e^{-i\theta}), \qquad M_P = \max_{\theta \in \mathbb{R}} \mu_P(\theta).$$

The maximum is zero when there are no roots on either indicated circle.

Theorem 50.2 (Root-circle classification)**.** *With constants depending on A, v, a_0, P, R but not on n,*

$$\lambda_{\min}(\widetilde{G}_{P,R,n}) \asymp n^{-2M_P}.$$

In particular lag-uniform coercivity holds if and only if P has no roots on $|z| = \lambda$ or $|z| = \lambda^{-1}$.

Proof. Set $b_k = \sum_i p_i p_{i+k}$, extended evenly and by zero, and $w_k = b_k \lambda^k$. The proof of (48.6) applies unchanged and gives

$$C_0(1 - \lambda^{-2(R+1-d)})^2 T_n(F_P) \leq \widetilde{G}_{P,R,n} \leq C_0 T_n(F_P), \qquad F_P(\theta) = |P(\lambda e^{i\theta}) P(\lambda^{-1} e^{-i\theta})|^2.$$

Here $C_0 = 4\pi^2 |a_0|^4 \det(v, Av)^2/(t^2-4) > 0$. The zero order of F_P at θ_0 is exactly $2\mu_P(\theta_0)$. We give a direct finite-section estimate so that no external Toeplitz asymptotic theorem is needed.

If there are no zeros, continuity and positivity bound F_P above and below. Otherwise let Z be its finite zero set, of cardinality J, and $M = M_P$. There is $c > 0$ such that $F_P(\theta) \geq c \operatorname{dist}(\theta, Z)^{2M}$, using circular angular distance. For a degree-$(n-1)$ Fourier polynomial p with normalized L^2 norm one, $|p(\theta)|^2 \leq n$. The union of the radius $\varepsilon = \pi/(4Jn)$ arcs about Z therefore carries at most $1/4$ of its squared mass. Outside these arcs $F_P \geq c\varepsilon^{2M}$. This gives a lower bound $c' n^{-2M}$ for every Rayleigh quotient.

Choose a zero θ_0 of maximal order and factor the finite Laurent polynomial $P(\lambda z)P(\lambda^{-1}z^{-1})$ by $(z - e^{i\theta_0})^M$. The remaining Laurent polynomial is bounded on the unit circle. Use the coefficients $x_j = e^{-ij\theta_0}\phi(j/(n+1))$, where $\phi(x) = x^M(1-x)^M$ on $[0,1]$ and zero otherwise. Its zero extension has bounded weak derivative of order M. The twisted Mth finite differences of x have squared sum $O(n^{1-2M})$, while $\|x\|^2 \geq c_M n$. Parseval gives the upper bound $O(n^{-2M})$. For a real symmetric matrix a complex trial vector is harmless: its Rayleigh quotient is a weighted average of those of its real and imaginary parts. The quadratic-form comparison completes the proof. □

This specializes the familiar zero-order mechanism for Toeplitz eigenvalues to an exact dynamical Gram representation. The general Toeplitz phenomenon is prior theory, not a new universality mechanism; see Böttcher–Widom, *From Toeplitz Eigenvalues through Green's Kernels to Higher-Order Wirtinger–Sobolev Inequalities*, arXiv:math/0412269.

There is also an exact repair for collective soft modes. Choose finitely many filters P_a, take $R \geq \max_a \deg P_a$, and add their covariance matrices with the same coefficient vector for their separate observations. Their symbols add. Uniform coercivity is equivalent to absence of a common zero of the F_{P_a}. If common zeros exist, the rate is n^{-2M}, where

$$M = \max_{\theta:\, F_{P_a}(\theta) = 0 \ \forall a} \min_a \mu_{P_a}(\theta).$$

The preceding lower and upper proofs apply to this positive sum; there can be no cancellation of its leading nonnegative terms. For example $P_\pm(z) = (z^2 \mp tz + 1)^q$ separately give n^{-4q}, but with $u = 2 - 2\cos\theta$ their sum obeys

$$F_+(\theta) + F_-(\theta) = u^{2q}(t^2 - 4 + u)^{2q} + (4-u)^{2q}(t^2-u)^{2q} \geq 2^{2q+1}(t^2-4)^{2q} > 0.$$

Thus two explicit individually soft observations have a jointly uniform positive lower covariance bound at every lag-window size. This statement adds covariances of separate observations, not the covariance of their sum.

50.3 The high-lag classification for every finite Fourier symbol

The single-orbit restriction can be removed. Decompose the nonzero Fourier support of an arbitrary real trigonometric polynomial into oriented A-orbits. Choose one integer representative w_a from each, shifting it so that

$$g=\sum_{a=1}^{J}P_a(C)e_{w_a},\qquad P_a(z)=\sum_{i=0}^{d_a}c_{a,i}z^i,\qquad P_a(0)\neq 0.$$

Representatives and filters can be paired by $w_{-a}=-w_a$ and $P_{-a}=\overline{P_a}$ (coefficient conjugation). Put $d=\max_a d_a$ and let P_g be the monic greatest common divisor of the P_a in $\mathbb{C}[z]$. It has real coefficients, and changing representatives only multiplies filters by Laurent monomials, which does not change its nonzero roots.

Lemma 50.3 (Four-atom separation)**.** *For a fixed finite nonzero integer alphabet $V=-V$, there is an integer G such that every relation $\sum_{i=1}^{4}A^{n_i}v_i=0$, $v_i\in V$, splits into two vanishing subrelations across any consecutive time gap larger than G. If $r,s>3G$ and*

$$u+A^{r}v=A^{j}(u'+A^{s}v'),\qquad u,v,u',v'\in V,$$

then necessarily $u=A^{j}u'$ and $A^{r}v=A^{j+s}v'$.

Proof. Set $B=\operatorname{sign}(\operatorname{tr}A)A$, and choose its eigen-covectors ξ,η. There is $\kappa\neq 0$ with $Q_B(x)=\det(x,Bx)=\kappa\xi(x)\eta(x)$. Let $X=\max_V|\xi|$, $Y=\max_V|\eta|$, and choose G so that $16|\kappa|XY\lambda^{-G}<1$. Suppose a gap separates times at most a from times at least b, $b-a>G$. The sum w of the terms on the left is integral, and, using minus the sum on the right for its contracting coordinate,

$$|\xi(w)|\le 4X\lambda^{a},\qquad |\eta(w)|\le 4Y\lambda^{-b}.$$

Thus the integer $Q_B(w)$ has absolute value less than one. It vanishes, and irrationality of the eigenlines implies $w=0$. The other subrelation also vanishes. Signs from negative eigenvalues are harmless in these bounds. The same argument shows that cancellation of two alphabet atoms can occur only when their time difference has absolute value at most G.

For the second assertion, order the four times $0,r,j,j+s$. Their span exceeds $3G$, so one of the three gaps exceeds G. Neither side can be a singleton because no atom is zero. Hence they split into two cancelling pairs. A pair cannot consist of the two atoms of the same side of the original identity, since $r,s>3G$. Crossed pairing would require $|j+s|\le G$ and $|r-j|\le G$, hence $r+s\le 2G$, also impossible. Only the asserted parallel pairing remains. □

For definiteness the constant in this lemma can be bounded using integers. Write $B=(b_{ij})$, $t=\operatorname{tr}B$, $\Delta=t^2-4$. The covectors $\xi(x,y)=2b_{21}x+(t-2b_{11}+\sqrt{\Delta})y$ and its conjugate satisfy $\xi\eta=4b_{21}Q_B$; here $b_{21}\neq 0$. Taking $M=\max_{(x,y)\in V}\{2|b_{21}||x|+(|t-2b_{11}|+t)|y|\}$, any integer G with $2^{G}>4M^2/|b_{21}|$ suffices, since $\lambda>2$.

Choose eigenvectors e_+,e_- of B and write $w_a=\xi_a e_+ +\eta_a e_-$. Every ξ_a,η_a is nonzero. Set $D_e=\det(e_+,e_-)$ and

$$F_g(\theta)=\pi^2D_e^2\left(\sum_a\eta_a^2|P_a(\lambda^{-1}e^{-i\theta})|^2\right)\left(\sum_a\xi_a^2|P_a(\lambda e^{i\theta})|^2\right).$$

This expression is independent of eigenvector rescaling.

Theorem 50.4 (Complete finite-symbol high-lag criterion)**.** *There is $R_0(A,g)$ such that, for every fixed $R\ge R_0$, the covariance matrix $\widetilde{G}_{g,R,n}$ of $\lambda^{-r}[h_r(g)]$, $r=R+1,\dots,R+n$, satisfies*

$$c\,T_n(F_g)\le\widetilde{G}_{g,R,n}\le C\,T_n(F_g)\quad(n\ge 1)$$

with positive constants independent of n. Therefore

$$\lambda_{\min}(\widetilde{G}_{g,R,n}) \asymp n^{-2M_g}, \qquad M_g = \max_{\theta}\{\nu_{P_g}(\lambda e^{i\theta}) + \nu_{P_g}(\lambda^{-1}e^{-i\theta})\}.$$

In particular uniform high-lag coercivity holds precisely when the common finite temporal divisor P_g has no root on the two hyperbolic circles. This covers every finite real Fourier symbol, with arbitrary phases and any number of frequency orbits.

Proof. Apply Lemma 50.3 to the actual Fourier support V. For sufficiently large positive r, s, equality of two contributing frequency orbits is possible only by parallel matching of their two atoms. In the chosen orbit coordinates, this says that the orbit is uniquely indexed by an ordered pair (a, b) and an effective lag $\ell = r + j - i$. Thus its Fourier sum is

$$-\pi D_{ab}(\ell)\, b_{ab}(\ell - r), \qquad D_{ab}(\ell) = \det(w_a, A^{\ell} w_b), \quad b_{ab}(k) = \sum_i c_{a,i} c_{b,i+k}.$$

The coefficients have been extended by zero; $|k| \le d$. Consequently the exact covariance, before normalization, is

$$(G_{g,R,n})_{rs} = \pi^2 \sum_{a,b,\ell} D_{ab}(\ell)^2 b_{ab}(\ell - r)\overline{b_{ab}(\ell - s)}.$$

If $s_A = \operatorname{sign}(\operatorname{tr} A)$, diagonalization gives

$$D_{ab}(\ell) = s_A^{\ell}\{-D_e \eta_a \xi_b \lambda^{\ell} + D_e \xi_a \eta_b \lambda^{-\ell}\}.$$

The leading coefficient $\gamma_{ab} = -D_e \eta_a \xi_b$ is nonzero. Choose R_0 so that all $r, s > 3G$, all $\ell \ge R + 1 - d > 0$, and

$$\left|\frac{D_e \xi_a \eta_b}{\gamma_{ab}}\right| \lambda^{-2\ell} \le \tfrac{1}{2} \quad \text{for every } a, b.$$

Then the squared residual factors are between $1/4$ and $9/4$. For $w_{ab,k} = b_{ab}(k)\lambda^k$, the normalized quadratic form is a sum of their convolution norms with these positive residual weights. Parseval gives the claimed comparison with constants $1/4$ and $9/4$ and symbol

$$\pi^2 \sum_{a,b} \gamma_{ab}^2 |P_a(\lambda^{-1}e^{-i\theta}) P_b(\lambda e^{i\theta})|^2 = F_g(\theta).$$

There is no missing conjugation in b_{ab}: it arises from the bilinear Poisson bracket. Conjugation enters only in the covariance and absolute square. The reality pairs make F_g even and the covariance real.

Each positive sum in F_g vanishes precisely when all of its filters vanish. Its vanishing multiplicity is twice their minimum root multiplicity, namely that of their gcd. The product therefore has zero order $2M_g$ at its most degenerate zero. The multi-zero finite-section estimate in Theorem 50.2 applies to this symbol as well: it is a finite nonnegative trigonometric polynomial; near a maximal zero it is bounded above by a constant times $|e^{i\theta} - e^{i\theta_0}|^{2M_g}$. The same concentration lower bound and modulated finite-difference trial vector prove the asserted rate. □

For example, if one of the orbit filters is a nonzero constant, then $P_g = 1$ and the high-lag covariance is uniformly coercive. More generally, adding a second orbit with a coprime temporal filter removes the common-root obstruction within a single observable. This is stronger than merely adding covariance matrices of separate observations, but relies on the exact large-lag orbit separation just proved. No claim about infinite Fourier support or nonlinear kicked spectral statistics follows from this criterion.

Corollary 50.5 (One-harmonic repair and its crossover scale)**.** *Let $v \neq 0$ be integral, $g_v = e_v + e_{-v}$, $q \geq 1$, and*

$$g_\varepsilon = (C^2 - tC + I)^q g_v + \varepsilon(e_{2v} + e_{-2v}), \qquad 0 \leq \varepsilon \leq 1.$$

For a sufficiently large fixed starting lag R, there are positive constants c, C, independent of both n and ε, such that

$$c\,(n^{-2q} + \varepsilon^2)^2 \leq \lambda_{\min}(\widetilde{G}_{g_\varepsilon,R,n}) \leq C\,(n^{-2q} + \varepsilon^2)^2.$$

Thus a single additional harmonic changes the arbitrarily soft n^{-4q} family into a uniformly coercive one. The crossover occurs at $n \asymp \varepsilon^{-1/q}$, and the large-window floor has order ε^4.

Proof. The oriented orbits of v and $2v$ are distinct, as $Q_B(2v) = 4Q_B(v) \neq Q_B(v)$. The four filters in Theorem 50.4 are $P_q, P_q, \varepsilon, \varepsilon$; the same separation and residual bounds work for all ε, including zero, by using the larger fixed support. Set $u = 2 - 2\cos\theta$, $B(\theta) = u(t^2 - 4 + u)$ and $C_0 = 4\pi^2 \det(v, Av)^2/(t^2 - 4)$. The exact Toeplitz symbol is

$$F_{g_\varepsilon}(\theta) = C_0\{B(\theta)^{2q} + 4\varepsilon^2(\lambda^{2q} + \lambda^{-2q})B(\theta)^q + 16\varepsilon^4\}.$$

Since $t^2 - 4 > 0$ and $0 \leq u \leq 4$, this is bounded above and below by positive constant multiples of $(u^q + \varepsilon^2)^2$, uniformly in ε. For every normalized degree-$(n-1)$ polynomial p, Jensen's inequality bounds its integrals against u^q and u^{2q} below by the corresponding powers of the Dirichlet Laplacian minimum $4\sin^2(\pi/(2(n+1)))$. This gives the required lower estimate. The trial coefficients from the soft-mode theorem have normalized qth and $2q$th finite-difference squared norms $O(n^{-2q})$ and $O(n^{-4q})$. They yield the upper estimate including both cross and constant terms. The exact finite-covariance comparison completes the proof. □

This repair is a theorem about the covariance of one perturbed classical observable. It does not claim that this harmonic perturbation proves BGS for the corresponding quantum propagator.

50.4 An exact all-time quantum module over a finite field

Let p be an odd prime and $A \in \mathrm{SL}(2, \mathbb{F}_p)$ satisfy $(\operatorname{tr} A)^2 \neq 4$. Assume its exact order $K \geq 3$ is odd. Choose $v \in \mathbb{F}_p^2$ with $\det(v, Av) \neq 0$. Use modular-centred Weyl matrices $W_p(w)$, so

$$W_p(u)W_p(w) = e_p(\omega(u,w)/2)W_p(u+w), \qquad W_p(w)^* = W_p(-w),$$
$$p^{-1}\operatorname{Tr}(W_p(u)^* W_p(w)) = \mathbf{1}_{u=w}.$$

Here $1/2$ is the inverse of two in $\mathbb{F}_p$, not ordinary division. The map $\alpha(W_p(w)) = W_p(Aw)$ extends to a $*$-automorphism of the full matrix algebra, hence is implemented by a unitary. This fixes the finite quantization convention; it does not assert physical-lift Egorov in every boundary sector for every integral matrix. Put $G = cW_p(v) + \bar{c}W_p(-v)$, $c \neq 0$, and

$$H_r = \frac{ip}{2}[G, \alpha^r G], \quad 1 \leq r \leq m = (K-1)/2, \quad d_r = \det(v, A^r v) \in \mathbb{F}_p.$$

Write $\widetilde{d}_r$ for any integer representative of d_r and $\tau_p = p^{-1}\operatorname{Tr}$.

Theorem 50.6 (Cyclic module and variance at every time)**.** *For any finitely supported complex coefficients $b_{r,j}$, $j \in \mathbb{Z}$,*

$$\tau_p \left| \sum_{r=1}^{m} \sum_{j} b_{r,j} \alpha^j H_r \right|^2 = 4p^2|c|^4 \sum_{r=1}^{m} \sin^2(\pi \widetilde{d}_r/p) \sum_{a=0}^{K-1} \left| \sum_{j \equiv a \bmod K} b_{r,j} \right|^2. \tag{50.2}$$

All d_r are nonzero. The Km operators $\alpha^a H_r$, $0 \le a < K$, $1 \le r \le m$, are pairwise orthogonal and nonzero. The kernel of the temporal Laurent-polynomial map is exactly $(z^K-1)\mathbb{C}[z,z^{-1}]^m$.

For $n = qK + s$, $0 \le s < K$, in particular,

$$\tau_p \left| \sum_{j=0}^{n-1} \alpha^j H_r \right|^2 = 4p^2|c|^4 \sin^2(\pi \widetilde{d}_r/p) \left\{ \frac{n^2}{K} + s\left(1 - \frac{s}{K}\right) \right\}. \tag{50.3}$$

This holds for every integer $n \ge 1$, without a no-alias time restriction.

Proof. Over a splitting field, A has distinct eigenvalues ζ, ζ^{-1} of exact odd order K. Every nonzero vector has orbit length K, and no such orbit contains both w and $-w$: otherwise a power of ζ would be -1. Let $Q(w) = \det(w, Aw)$ and $t_r = \zeta^r + \zeta^{-r}$. Then

$$Q(v \pm A^r v) = (2 \pm t_r) Q(v).$$

For $1 \le r \le (K-1)/2$, all the $2m$ values $2 \pm t_r$ are nonzero and distinct. Equality $t_r = t_s$ implies $r \equiv \pm s \pmod K$, while $t_r = -t_s$ would imply $\zeta^r = -\zeta^{\pm s}$, impossible at odd order. Thus the $4m$ frequencies $\pm(v + A^r v)$, $\pm(v - A^r v)$ lie in pairwise distinct orbits. The condition $Q(v) \ne 0$ also gives $d_r = Q(v)(\zeta^r - \zeta^{-r})/(\zeta - \zeta^{-1}) \ne 0$.

Expanding the Weyl commutator shows that H_r has exactly these four frequencies, each with coefficient modulus $p|c|^2|\sin(\pi \widetilde{d}_r/p)|$. The modular phase may differ by a sign from ordinary half-division, but its squared modulus is the displayed one. Transport permutes each orbit without an additional phase in this convention. Since the orbits are disjoint, collecting coefficients by their residue class in time and applying Weyl orthogonality proves (50.2). The only polynomial kernel is reduction to zero modulo $z^K - 1$. For the constant time coefficients there are s residue classes visited $q+1$ times and $K - s$ classes visited q times; their squared counts sum to $Kq^2 + (2q+1)s = n^2/K + s(1 - s/K)$. □

For $n \le K$ the brace equals n, giving exact linear variance. For $n \gg K$ it is $n^2/K + O(K)$, exposing coherent recurrences. When n is of order p the formula still holds, regardless of whether K is comparable to p; linear variance at that scale additionally requires $K \ge n$. No infinite sequence with $K \asymp p$ for one fixed integral A is inferred. The theorem is about an unperturbed finite quantum map and commutator observables, not the local eigenphase statistics of a kicked propagator. The odd-order hypothesis is essential: at $p = 7$ and $A = \left(\begin{smallmatrix} 3 & -1 \\ 1 & 0 \end{smallmatrix}\right)$ the order is 8 and opposite frequencies share an orbit, invalidating the displayed orthogonality and variance formula. This is an explicit negative control.

50.5 Phase completion at even order and an exact rank transition

The failure at even order is not unavoidable. It depends on the Fourier phase of the real initial observable, and can be removed without randomness. Retain the regular-semisimple assumptions over $\mathbb{F}_p$, but allow any order $K \ge 3$, and put $m = \lfloor (K-1)/2 \rfloor$.

Theorem 50.7 (Balanced-phase completion)**.** *If K is even, assume $c = a + ib$ with $a, b \in \mathbb{R}$ and $a^2 = b^2 > 0$. Then* (50.2) *and* (50.3) *hold unchanged for all $1 \le r \le m$. Thus the cyclic module is free of rank m over $\mathbb{C}[z,z^{-1}]/(z^K-1)$ for every regular semisimple order, using an arbitrary nonzero phase at odd order and a balanced phase at even order.*

More generally, at even order let $ab \ne 0$, set $s_r = \sin(2\pi \overline{2} \widetilde{d}_r/p)$, where $\overline{2} = (p+1)/2$, and define $J_r = H_r/(p s_r)$. If $B_{r,u} = \sum_{j \equiv u \bmod K} b_{r,j}$, then

$$16 \min(a^4, b^4) \sum_{r,u} |B_{r,u}|^2 \le \tau_p \left| \sum_{r,j} b_{r,j} \alpha^j J_r \right|^2 \le 16 \max(a^4, b^4) \sum_{r,u} |B_{r,u}|^2.$$

The dimension of the span of all $\alpha^j H_r$ is $K(K-2)/2$ for $ab \neq 0$, but is exactly $K(K-2)/8$ for a nonzero purely real or purely imaginary c.

Proof. Write $K = 2L$. Then $A^L = -I$. The invariant $Q(w) = \det(w, Aw)$ groups the lags r and $L-r$, for $1 \le r < L/2$, into pairs. Different such pairs have disjoint label orbits. Indeed $t_r = t_s$ implies $r = \pm s$ modulo K, while $t_r = -t_s$ now implies $r = L \pm s$ modulo K. Within a pair choose $E = W_p(v + A^r v)$ and $F = W_p(v - A^r v)$ as orbit seeds. Their K-element orbits are disjoint. Expanding the commutators gives, in the cyclic group algebra, with z acting by α,

$$J_r = -A_c(z)E + B_c(z)F, \qquad z^r J_{L-r} = B_c(z)E - A'_c(z)F,$$

where

$$A_c(z) = c^2 + \overline{c}^{\,2} z^L, \qquad A'_c(z) = \overline{c}^{\,2} + c^2 z^L, \qquad B_c(z) = |c|^2(1 + z^L).$$

Here $d_{L-r} = d_r$, so the two sine normalizations agree. The frequency identities used are $v + A^{L-r}v = -A^{-r}(v - A^r v)$ and $v - A^{L-r}v = A^{-r}(v + A^r v)$.

A unitary Fourier transform on each cyclic orbit reduces the block to

$$\begin{pmatrix} -A_c(z) & B_c(z) \\ B_c(z) & -A'_c(z) \end{pmatrix}, \qquad z^K = 1.$$

For $z^L = 1$, its singular values are $4a^2$ and $4b^2$. For $z^L = -1$, both singular values are $4|ab|$. Multiplication of the second generator by z^r is unitary and does not change a bound on coefficient norms.

If L is even, the remaining lag $r = L/2$ is a separate block: all four frequencies are on the single orbit of $E = W_p(v + A^r v)$, since $v - A^r v = A^{-r}(v + A^r v)$. Its orbit polynomial is

$$-c^2 - \overline{c}^{\,2} z^{2r} + |c|^2(z^r + z^{-r}).$$

At $z^r = 1, -1, i, -i$ its values are respectively $4b^2, -4a^2, -4iab, -4iab$. Its squared singular values therefore satisfy the same bounds. These blocks exhaust the half-period lags. When $a^2 = b^2$, every block is a scalar multiple of a unitary, with squared singular value $16a^4 = 4|c|^4$. Substituting $ps_r b_{r,j}$ for coefficients of J_r proves the claimed exact formula for H_r.

For $ab \neq 0$ all blocks are invertible at all characters, giving $Km = 2L(L-1)$. For a purely real or imaginary phase each two-generator block has rank one on the L characters with $z^L = 1$ and rank zero on the others. Its total rank is L. If present, the exceptional one-generator block has rank $K/4$. Counting $(L-1)/2$ pairs when L is odd, or $(L-2)/2$ pairs and one exceptional block when L is even, gives $L(L-1)/2 = K(K-2)/8$. □

This is a phase-controlled rank transition in the module of commutator observables. It is not a transition theorem for eigenvalue statistics. The purely real even-order negative control remains valid; the balanced choice $c = 1 + i$ removes precisely that phase-dependent degeneracy.

Chapter 6 — Return laws II (general laws, obstructions, return words) (continued)

51 Noncommuting return words: an explicit higher-order exceptional family

A nonzero Poisson bracket need not have positive Green–Kubo variance. This section constructs an explicit nonzero bracket which is a smooth coboundary and, more strongly, proves all-orders quantum conjugacy at fixed classical strength for its natural single-exponential kick.

The conjugating operator is a product of local propagators, not the exponential of the linear coboundary generator. This separates three genuinely different conditions:

$$\{g, \mathcal{C}g\} = 0, \qquad \sigma^2(\{g, \mathcal{C}g\}) = 0, \qquad V_{\kappa,N} = J_{\kappa,N} U_N J_{\kappa,N}^* + O(N^{-\infty}).$$

The construction satisfies the last two but not the first. It does not refute any previously stated theorem requiring analyticity or Fourier support in a fixed finite box. The generators are smooth nonanalytic functions with open critical regions. Arithmetic symmetries are transported, not eliminated.

The elementary telescoping identity for a group-valued coboundary is standard. No novelty is claimed for that identity, for unitary conjugacy preserving a spectrum, or for microlocal locality. The project's result is their explicit combination for one fixed local Hamiltonian kick and a noncommuting generator, with a finite operator-norm remainder. A targeted source search has not established priority for this exact example; no first-ever or prize-level claim is made.

51.1 Quantization and a finite return-word theorem

Put

$$T = \begin{pmatrix} 3 & 2 \\ 4 & 3 \end{pmatrix}, \quad S = T^{\mathsf{T}}, \quad \mathcal{C}f = f \circ S, \quad \hbar_N = (2\pi N)^{-1}.$$

Use physical integer Weyl operators satisfying

$$Q_N(v)Q_N(w) = e^{\pi i \det(v,w)/N} Q_N(v+w), \quad U_N Q_N(v) U_N^* = Q_N(Tv).$$

The latter convention is valid for this T, which is the identity modulo two. Write $C_N(X) = U_N X U_N^*$ and $\mathrm{Op}_N(f) = \sum_v \widehat{f}(v) Q_N(v)$. All symbols in this section are real and smooth; their quantizations are Hermitian. Even symbols preserve parity. A fixed finite family is used throughout.

Let $L \geq 2$ be a fixed integer. Let E be a closed subset of the torus such that $E_j = S^{-j}E$, $0 \leq j < L$, are pairwise disjoint. Choose symbols $h_0, \ldots, h_{L-1}$ supported in E, and put

$$H_j = \mathrm{Op}_N(h_j), \quad e = \kappa/\hbar_N, \quad W_j = e^{ieH_j}, \quad G_j = W_j W_{j-1} \cdots W_0.$$

Assume the exact return identity

$$W_{L-1} \cdots W_0 = I. \tag{RW.1}$$

The assumption is imposed for the parameters and quantizations being used; it is automatically true for every e and every N for an anti-palindromic list $h_{L-1-j} = -h_j$. It is not inferred from the weaker relation $\sum h_j = 0$. Define

$$f = \sum_{j=0}^{L-1} \mathcal{C}^j h_j, \qquad J_N = \prod_{j=0}^{L-2} C_N^j(G_j), \qquad V_N = e^{ie\,\mathrm{Op}_N(f)} U_N, \tag{RW.2}$$

where the product defining J is in increasing order of j from left to right.

Return-word theorem. For all pairs (N, κ) satisfying (RW.1), uniformly with $|\kappa| \leq B$ for each fixed B,

$$\|V_N - J_N U_N J_N^*\|_{\mathrm{op}} = O(N^{-\infty}). \tag{RW.3}$$

For every fixed p the same estimate holds after taking any integer power $|t| \leq N^p$. The statement holds on the full space, and on each parity sector if all the local symbols are even.

Proof. Locality gives, for i and j at different sites and odd r,

$$\|[\mathrm{Op}_N(\mathcal{C}^i h_a), \mathrm{Op}_N(\mathcal{C}^j h_b)]\| \leq \frac{2\pi^r}{r! N^r} B_r^2, \qquad B_r = \max_{i,a} M_r(\mathcal{C}^i h_a), \tag{RW.4}$$

with $M_r(a) = \sum_v |\widehat{a}(v)||v|_2^r$. The symbol M_r is local to this section (Euclidean frequency weight); it is not the $\mathcal{M}_r$ of Section 41, which uses the weight $(1 + |v|_1)^r$. Each finite Taylor coefficient of the exact sine commutator is a mixed local derivative and is zero on account of the disjoint closed supports. A real Taylor remainder proves (RW.4) without an infinite-series convergence assumption.

For Hermitian A,B, integration of two unitary conjugations gives

$$\|[e^{ieA}, e^{ieB}]\| \le e^2 \|[A, B]\|. \tag{RW.5}$$

Consequently elementary factors at distinct sites can be interchanged with this error; factors at the same site are never interchanged. Reorder the factors of $J_N C_N(J_N^*)$ by their site. The resulting site factors are

$$G_0 = W_0, \quad G_j G_{j-1}^* = W_j \ (1 \le j < L-1), \quad G_{L-2}^* = W_{L-1}, \tag{RW.6}$$

where the last identity uses (RW.1). Thus the reordered product is $\prod_{j=0}^{L-1} C_N^j(W_j)$. The original word has at most $L(L-1)$ elementary exponential factors. Fewer than $L^4/2$ interchanges suffice, so (RW.4)–(RW.5) control the accumulated error independently of the size of eH.

The standard two-factor Duhamel estimate, iterated L times, gives

$$\left\| e^{ie\sum_j C_N^j H_j} - \prod_j e^{ieC_N^j H_j} \right\| \le \frac{e^2}{2} \sum_{i<j} \|[C_N^i H_i, C_N^j H_j]\|.$$

Combining these inequalities, for $L \ge 2$ one may use the deliberately conservative explicit bound

$$\boxed{\|V_N - J_N U_N J_N^*\| \le \frac{8\kappa^2 \pi^{r+2} L^4}{r!} B_r^2 N^{2-r}, \qquad r \text{ odd.}} \tag{RW.7}$$

Choosing r as large as desired proves (RW.3). Telescoping unitary powers costs at most $|t|$ and proves the time statement. Negative t uses adjoints. No spectral modification and no N-dependent counterterm was introduced. End of proof.

For an anti-palindromic word, the corresponding identity of classical Hamiltonian pullbacks holds exactly. Pullbacks belonging to distinct sites commute, whereas their order at each site is retained. Thus (RW.6) also establishes an exact smooth symplectic conjugacy of the associated classical map to S. In particular the map is Anosov. This is an exceptional arithmetic family, not a demonstration of random-matrix universality.

51.2 Four sites with a nonzero second bracket

Choose real even a,b supported in E, with $p = \{a, b\} \ne 0$, and use

$$(h_0, h_1, h_2, h_3) = (a, b-a, a-b, -a).$$

This is an anti-palindromic word. Its linear coboundary generator is

$$g = a + \mathcal{C}b + \mathcal{C}^2 a, \qquad f = g - \mathcal{C}g = a + \mathcal{C}(b-a) + \mathcal{C}^2(a-b) - \mathcal{C}^3 a. \tag{RW.8}$$

Subtracting the mean of g changes neither f nor the bracket below, and only changes an exponential gauge by a scalar. Thus the example also belongs to the mean-zero generator class. Only the overlaps at sites 1 and 2 contribute to the Poisson bracket:

$$\boxed{\{g, \mathcal{C}g\} = \mathcal{C}^2 p - \mathcal{C}p = (I - \mathcal{C})(-\mathcal{C}p) \ne 0.} \tag{RW.9}$$

Nonvanishing follows because p is nonzero and its two translated supports are disjoint. The displayed transfer function proves exactly

$$\Gamma_T(g) = 0. \tag{RW.10}$$

Thus the implication "nonzero consecutive bracket implies nonzero second resonance" is refuted on the smooth class by an explicit construction, not merely left as an unproved inference.

Put $A = \mathrm{Op}_N(a)$, $B = \mathrm{Op}_N(b)$, $X = e^{ieA}$, and $Y = e^{ie(B-A)}$. The actual gauge is

$$\boxed{J_N = X\, C_N(YX)\, C_N^2(X).} \tag{RW.11}$$

The same-site factors Y and X need not commute. Replacing YX by e^{ieB} would replace J by a different gauge and is not justified. Its derivative at e=0 nevertheless has the generator $A + C_N(B) + C_N^2(A)$, matching (RW.8).

With $X_h = (h_p, -h_q)$ and flow Ψ_h^t, the smooth conjugacy can be written explicitly:

$$\Psi_\kappa = \Phi_a^\kappa \circ \Phi_{\mathcal{C}(b-a)}^\kappa \circ \Phi_{\mathcal{C}a}^\kappa \circ \Phi_{\mathcal{C}^2 a}^\kappa, \qquad S \circ \Phi_f^{-\kappa} = \Psi_\kappa S \Psi_\kappa^{-1}. \tag{RW.12}$$

Convention: in this section Ψ_h^t is the flow of $X_h = (h_p, -h_q)$, so $\frac{d}{dt}(a \circ \Psi_h^t) = (X_h \cdot \nabla a) \circ \Psi_h^t = -\{h, a\} \circ \Psi_h^t$; it is the inverse of the flow denoted Φ^t in Section 41, where $\frac{d}{dt}(a \circ \Phi_g^t) = \{g, a\} \circ \Phi_g^t$. Formula (RW.12) uses the present convention. The parameter κ here is the classical time itself, through $e = \kappa/\hbar_N$; it is not the rescaled time $\kappa_0/(2\pi)$ of Section 41. The rightmost classical map acts first. Formula (RW.12) is the reversed composition of the pullbacks quantized by (RW.11); it can also be obtained by the exact classical version of (RW.6). The map differs from S for all sufficiently small nonzero κ since f is nonconstant, and agrees with S on an open set. It is therefore nonaffine in this range.

An additional useful separation is

$$\lim_{N\to\infty} N \|[\mathrm{Op}_N(g), \mathrm{Op}_N(\mathcal{C}g)]\|_{2,N} = \frac{\|\{g, \mathcal{C}g\}\|_2}{2\pi} > 0. \tag{RW.13}$$

This follows from the cubic Fourier commutator remainder and the fixed-symbol Hilbert–Schmidt limit. Equations (RW.3) and (RW.13) coexist. A large enough principal commutator rules out the old *commuting-observable proof*, not every nonlinear unitary gauge. Neither (RW.9) nor (RW.13) contradicts the analytic rigidity theorem: the functions used here are not analytic.

On any Pell sequence with $U_N^K = I$, the average of the actual commutator is smaller than every inverse power, too. The two same-site contributions are $C_N([B, A]) + C_N^2([A, B])$ and telescope under the period average; every remaining contribution is a cross-site commutator controlled by (RW.4). This conclusion concerns the averaged commutator, not its nonzero local norm.

51.2.1 Extension to every hyperbolic integral toral automorphism

The construction is not specific to trace six. For any hyperbolic $A \in \mathrm{SL}(2, \mathbb{Z})$ use $S = A^{\mathsf{T}}$ and the invariant characteristic $Q_N^*(v) = (-1)^{v_1 v_2 + v_1 + v_2} W_N(v)$ described in the finite spectral theorem. Choose z outside the finite sets $S^j z = \pm z$ for $1 \le j < L$ and $2z = 0$. Small symmetric neighborhoods of z and -z then give the required disjoint tower. Real even local a,b with nonzero Poisson bracket exist inside these neighborhoods by the same bump-and-coordinate construction. Equations (RW.8)–(RW.13) remain unchanged, with this S and quantization. Thus every such A admits a noncommuting smooth generator with nonzero consecutive bracket, zero second variance, and fixed-strength all-orders quantum conjugacy under a product gauge. On the prime-period sequence of the general spectral theorem, use its prescribed uniform parity sector, which may be even. The cluster and local-statistics conclusions then follow there. This assertion does not replace A by its square when its trace is negative.

51.3 An explicit support and nonzero bracket

Let $r = 1/96$ and $z = (5/11, 0)$. Let E be the two closed torus squares of halfwidth r centered at z and -z. The four sets $E, S^{-1}E, S^{-2}E, S^{-3}E$ are pairwise disjoint. This is a finite rational geometry assertion, not a visual assertion from a coarse plot. It suffices to check $E \cap S^j E = \varnothing$, $j = 1, 2, 3$.

For a matrix $M = S^j$ and sign $\varepsilon \in \{\pm 1\}$, all possible integer translations n of a collision lie in the rectangle

$$|n_i - (Mz - \varepsilon z)_i| \le r \left(1 + \sum_k |M_{ik}|\right).$$

For every candidate n, a separating normal w from $(1, 0), (0, 1), (-M_{21}, M_{11}), (-M_{22}, M_{12})$ satisfies

$$|w \cdot (n - Mz + \varepsilon z)| > r \left(\|w\|_1 + \sum_k |w \cdot M e_k|\right). \tag{RW.14}$$

For j=1,2 the candidate rectangle contains no integer point. For j=3 there are 30 candidates; all are excluded by (RW.14) with exact rational arithmetic. The full list of normals, both sides and positive gaps is part of the certificate. An independent C++ check clears all denominators by 1056 and enumerates a wider integer box, then applies the same necessary inequalities. The two base lobes have exact separation margin 37/528. The certificate is that exact integer verifier, `computations/cpp/return_word_certificate.cpp` (git blob `ad80a3832555fac3a0afea2013a60a3de6522113`): it enumerates the candidate translations, applies (RW.14) with the four normals above to each candidate, and fails unless the number of separation obligations is exactly 30. Its recorded output, `evidence/executions/20261006/corpus-revision/final/checks/cpp_return_word_certificate.log` (git blob `c024cac38feddc565c60f3de288fc30b7928836f`), states status PASS with `geometry_obligations` = 30. The normals and gaps are generated by the verifier; they are not printed as a separate list.

Put $\beta(t) = e^{1-(1-t^2)^{-1}}$ for $|t| < 1$ and zero otherwise. With periodic lifts near the two lobes define

$$\chi_\pm(q, p) = \beta((q \mp 5/11)/r)\beta(p/r), \quad a = \chi_+ + \chi_-, \quad b = \frac{q - 5/11}{r}\chi_+ - \frac{q + 5/11}{r}\chi_-. \tag{RW.15}$$

All expressions are periodically extended. Both a,b are real and even. On the positive lobe b=ua, $u = (q - 5/11)/r$, hence $\{a, b\} = -aa_p/r$. At $(q, p) = (5/11, r/2)$ this is $-\beta(1/2)\beta'(1/2)/r^2 > 0$, proving the required nonzero bracket without numerical integration. The total area of the four support levels is at most $32r^2 = 1/288 < 1$, so their complement contains an open set.

51.4 Exact nodal quantization without the half-period restriction

For arbitrary smooth periodic functions $a(q), b(p)$, physical Weyl quantization of the separable symbol has the exact finite formula

$$\begin{aligned}[\mathrm{Op}_N(a(q)b(p))]_{xy} &= a((x + y)/(2N))\, B_{x-y} \\ &\quad + a((x + y)/(2N) - 1/2)\, B_{x-y+N}, \\ B_l &= \frac{1}{2N} \sum_{z=0}^{2N-1} b(z/(2N)) e^{-2\pi i l z/(2N)}.\end{aligned} \tag{RW.16}$$

Subscripts of B are reduced modulo 2N. To prove it, group the integer momentum frequency as $x - y + kN$ in the Weyl kernel and split k into its even and odd parts. Summation over the position frequency is exact; the two residue classes are exactly the indicated 2N-point Fourier coefficients. Absolute Fourier convergence justifies regrouping.

Unlike the earlier N-point formula, (RW.16) does not require a half-periodic position factor. It therefore applies to both symbols in (RW.15) without an unrecorded Fourier cutoff. It is a finite representation identity; floating point evaluation of its entries is not interval-certified.

51.5 Verification and what the finite data do not show

The exact geometry certificate has 30 rational separation obligations. The independent integer C++ verifier also checks the nonabelian prefix identity for eleven anti-palindromic words and the exact coefficients (0,-1,1,0) in (RW.9). The numerical block control uses four orthogonal sites with two noncommuting 2-by-2 Hermitian matrices. It verifies (RW.11) to rounding accuracy and rejects the incorrect linear gauge $e^{ie\,\mathrm{Op}(g)}$. This finite block model is a falsifier of factor order, not a torus theorem.

The 2N-node formula is compared with independent finite Weyl sums containing both odd position frequencies and aliases. The invalid N-node formula is retained as a negative control. Classical ODEs compare both sides of (RW.12) at the four support levels and at both signs of the strength.

Dense torus matrices are tested at N=63,127,255,511. These are mostly not Pell moduli and are operator-comparison tests, not demonstrations of a Pell limit. At kappa=0.02 some finite gauge errors are close to the maximal norm 2. The supports are narrow and the estimates in (RW.7) are asymptotic; no monotone convergence or useful finite-size rate is inferred from these data. Small fixed strengths provide nonvacuous finite-bound controls. An initial ODE-rich batch exceeded its time cap; the final documented smaller ODE family completed. The exact analytic support and word arguments do not depend on that reduction.

The local cluster, factorial correlation and Heisenberg-window consequences of all-orders operator-norm conjugacy now apply to this noncommuting example as well. They remain negative results for a smoothly arithmetic-conjugate family. Theorem (RW.3) supplies no positive BGS law, no symmetry elimination, and no classification of the zero set of Gamma for analytic or finite Fourier symbols. Its actual advance is the explicit nonzero-coboundary bracket and the return-word gauge which replaces the commuting-generator mechanism.

Appendix B — Provenance and verification

52 Verification, provenance, and the remaining positive problem

The mathematical manuscript, its exact certificates, finite matrix checks, software integration and external literature support are distinct layers. Agreement of two programs checks implementations of a finite assertion; it does not prove an asymptotic statement outside its certified range.

52.1 Repeated checks in this consolidation

The nondegeneracy package was reconstructed from the delivered archive and its complete 14-command verification rerun. It rebuilt the C++ interpreter, regenerated the radius-one, radius-three and radius-eight orbit equations, replayed the proofs independently, and recomputed the quantitative margins, Hessian identities and finite quantum variance controls. The radius-eight proof contains 5704 steps for 1475 orbit equations in 144 complex coefficients, i.e. 288 real Fourier parameters. Source fingerprints were checked across the run. The small proof language remains an ordinary exact executable certificate, not a Lean kernel proof.

A further 36-command historical component run completed with zero failures: eight C++ compilations and their independent executables, finite-field subresultant and reflection checks,

Frobenius amplification, elliptic spike controls, the exact interval computation of the fourth-moment constant, Pell Gaussian/Floquet/weighted-echo tests, robust covering inequalities, sharp-orbit and four-mode checks, and smooth/mixed-kick regressions. This is a named component suite, not an assertion that every historical script, every branch and every external theorem has been reverified. The original sources were not edited to obtain successful outputs. The same 36 commands were then replayed from a hash-indexed frozen input archive in a new isolated directory. All 36 passed again, with 362 scientific input files unchanged. This second execution checks portability and reproducibility; it does not count as 36 new mathematical assertions.

The exact interval computation reconfirms

$$\gamma \geq 5.61422069215119807162796524101779463077061685008385734751838 0,$$
$$\gamma \leq 5.61422069215119807162796524101779463077061685008385734752284 9.$$

The earlier decimal beginning 5.612 is not the value of the stated series. That old transcription error had already been recognized; the present run independently confirms the correction, rather than counting its discovery again. The two-species constant in the coboundary fourth moment is a different number and is not substituted for this one.

The new return-word tests have three independent components. Exact integer C++ code checks the free-word cancellation on eleven anti-palindromic words and all thirty rational support-separation obligations. A four-site block model with noncommuting two-by-two matrices verifies the product gauge and rejects the linear-exponential substitute. Finally physical torus Weyl matrices constructed by the exact two-grid nodal identity test the actual single exponential. The latter matrices use floating-point arithmetic. Their large finite errors at stronger kicks are retained; no decay exponent is fitted or inferred from them. Classical trajectory comparisons test the composition order and sign, but do not replace the exact support/flow proof.

The signed-trace arithmetic was separately checked for a family of positive and negative integer traces and prime periods, together with direct small matrix spectra on both uniform parity sectors. A deliberately excluded ramified period is retained as a negative control: $t = -3, K = 5$ gives $N = 5$ and $\gcd(N, |t| + 2) = 5$, violating invertibility of $A - I$. An early test fixture incorrectly included this case among admissible spectra; the hypothesis check now rejects it. The theorem was not weakened to make the fixture pass.

52.2 Proof-review findings and conservative corrections

Several apparent advances in the historical messages were duplicate versions of already present lemmas. In particular the regular-gradient analytic rigidity argument and the Pell-only sharp finite-span extremum are credited to their first preserved project statements. Their generalizations and new consequences are separately identified. Competing normalizations of the classical kick differ by 2π; this manuscript records the conversion rather than using the same parameter symbol with incompatible meanings.

The new noncommuting construction does not invalidate the proved nondegeneracy theorems. Its generators have infinite Fourier support, mixed Fourier signs and open critical sets. It lies outside the radius-eight certificate, the one-sign criterion, the regular-gradient analytic class, and the nonzero-Hessian sufficient condition. It does settle a stronger exceptional mechanism: $\{g, \mathcal{C}g\} \neq 0$ can coexist with $\Gamma(g) = 0$ and a nonlinear all-orders product gauge. The invalid implication from a nonzero bracket to a positive orbit variance is never used in the consolidated theorem chain.

Our proof review included domains, parity restrictions, Weyl sign lifts, normalizations of the trace and form factor, finite-to-infinite limits, exact versus approximate conjugacy, and finite-certificate soundness. External-source-dependent arithmetic statements remain marked by their named inputs. This is not a claim of complete independent referee acceptance of every

proof in the historical corpus. The accompanying audit records precisely which statements were rederived, which were checked by finite regression, and which rely on preserved external inputs.

52.3 Literature boundary

Cohen–Conze's classical cohomological variance statements require their regularity and action hypotheses; their classical CLT is not used in place of a quantum proof. Fejer–Riesz supplies a spectral factor only after positivity has been established. Prosen–Znidaric's residual/fidelity mechanism is a direct precedent, not a theorem about the particular fixed-strength eigenphase clusters constructed here. Keating–Mezzadri's nonlinear arithmetic pseudo-symmetries prevent equating mere nonlinearity with removal of arithmetic structure. General matrix-cocycle Livsic theory is background; the return-word factorization in this paper is proved explicitly and does not import a Livsic theorem with unchecked hypotheses.

The current source pass read the scalar factorization statement and its root-pair proof, the residual commutator formulas, and the classical orbit/coboundary criterion in the primary texts. The bidegree-$(1,1)$ subgroup-curve input was checked directly against Vyugin–Makarychev, Theorem 2, including the condition $\deg P(x,0) \geq 1$ and the subgroup-size range. Some other older finite-field sources could not be reacquired in full; their previously recorded input statements and conditional deductions are preserved, not newly certified. A title-only retrieval in the old record returned an unrelated paper and has been excluded. Targeted searches failing to find our exact construction do not establish its originality. No priority conclusion rests on a missing search result.

52.4 Consolidation and reproducibility contract

The source map records the archive and SHA-256 of every selected input. The older main arithmetic article, the distinct row-support refinements, the compressed-observable branch, Pell recurrence and smooth extensions, locality/robustness, sharpness/rigidity, and the full nondegeneracy package are represented in one document. Repeated abstracts, duplicated conclusions and obsolete numerical assertions are not repeated as independent results. The selected proof sources are preserved alongside the typeset sections; local equation tags are prefixed by their source unit to avoid duplicate cross-references. Machine-readable records contain commands, return codes, source and log hashes, timeouts and negative controls.

The PDF was built from the delivered TeX sources; syntax and cross-reference validation and page-image review are separate from mathematical correctness. The September 8 revision successfully compiled all 16 active Lean modules with Lean 4.33.1. Native finite certificates retain their native-evaluation trust boundary; the new signed-orbit kernels use ordinary proofs. No claim of a remote merge, successful GitHub Actions run or binary upload follows from a local Git commit. The delivery manifest identifies the last read remote base and the safe integration patch. Concurrent repository work must be preserved; force-updating the main branch is neither required nor authorized by the publication procedure.

52.5 The remaining mathematical target

The finite-field energy/support reductions and the positive spectral universality problem are separate. For the latter, one needs a fixed nonarithmetic quantum family, a precise symmetry resolution and unfolding, and joint trace or equivalent local-resolvent control at the Heisenberg scale. A global Gaussian observable law, a finite polynomial correction, an open-dense nondegeneracy statement, and an exceptional all-orders conjugacy do not supply that result. The noncommuting return-word theorem makes this separation sharper: even a nonzero principal commutator can conceal an exact cohomological cancellation and a product gauge. The positive

BGS conclusion is not proved in this manuscript.

For the fixed deterministic family the comparison is a signed weighted sum. The TRACE-010 continuation excludes its simultaneous unaveraged all-order absolute CUE strengthening: absolute pair agreement would force the normalized absolute cubic discrepancy to grow at least as $\sqrt{N}/1024$ on a fixed positive Fourier box. The absolute sufficient implication remains correct, including averages taken before the modulus. The fixed-map chapter records the viable signed target and the limits of its reference variance and formal phase controls.

Chapter 7 — Floquet, coboundary and smooth resonance fields (continued)

53 Uniform synthesis and the full resonance lattice

Let $A \in \mathrm{SL}(2,\mathbb{Z})$ be hyperbolic, $Cf = f \circ A^{\mathsf{T}}$, $\mu = \operatorname{sgn}(\operatorname{tr} A)\lambda$, and $\lambda > 1$. For a real observable, modulo constants, write

$$M_s(g) = \sum_{v \neq 0} |v|^s |\widehat{g}(v)|, \qquad h_r(g) = \frac{\{g, C^r g\}}{4\pi}.$$

Assume $M_1(g) < \infty$. The brackets have absolutely summable Fourier coefficients and zero mean. For representatives $v_{\mathcal{O}}$ of the oriented nonzero A-orbits define

$$J_\omega(f)_{\mathcal{O}} = \sum_{j \in \mathbb{Z}} \widehat{f}(A^j v_{\mathcal{O}}) e^{ij\omega}.$$

Then $J_\omega(Cf) = e^{i\omega} J_\omega(f)$ and $\|f\|_2^2 = \int \|J_\omega(f)\|_2^2 \, d\omega/(2\pi)$. The variable ω is dual to simultaneous time translation. The variable θ below is dual to lag. Neither is a quantum eigenphase.

Lemma 53.1 (Two visits)**.** *For fixed nonzero integer u, v, an oriented A-orbit contains $u + A^r v$ for at most two positive integers r, omitting the zero vector.*

Proof. In real eigen-coordinates the invariant $q_A(k) = \det(k, Ak)$ gives

$$q_A(u + A^r v) = q_A(u) + q_A(v) + a\mu^r + b\mu^{-r}, \qquad ab \neq 0.$$

Both eigen-coordinates of a nonzero integer vector are nonzero: otherwise A would have a rational eigenvalue, whereas $(\operatorname{tr} A)^2 - 4$ is not a square. A fixed value of q_A therefore gives a quadratic equation in the distinct numbers μ^r. □

Put $\kappa_A = \sup_{r \geq 1} \lambda^{-r} \|A^r\|_{\mathrm{op}} < \infty$.

Theorem 53.2 (Uniform bilinear synthesis)**.** *The operator*

$$B_{g,f}(\omega)x = \sum_{r \geq 1} x_r \lambda^{-r} J_\omega\Big(\frac{\{g, C^r f\}}{4\pi}\Big)$$

extends to $\ell^2(\mathbb{N})$ and satisfies

$$\sup_\omega \|B_{g,f}(\omega)\| \leq \sqrt{2}\pi\kappa_A M_1(g) M_1(f).$$

Proof. For one ordered Fourier pair the synthesis matrix has one entry per column and at most two per row, by Lemma 53.1. The entries have modulus at most $\pi\kappa_A|u||v|$. Rowwise Cauchy–Schwarz and the triangle inequality over Fourier pairs prove the estimate. The Fourier sum converges in operator norm, uniformly in ω. □

Let $S_g = B_{g,g}$ and $G_g = S_g^* S_g$. Bilinearity gives

$$\sup_{\omega} \|S_g - S_f\| \le \sqrt{2}\pi\kappa_A (M_1(g) + M_1(f)) M_1(g - f).$$

Thus finite Fourier truncation converges uniformly in operator norm, not merely entrywise in a fixed lag window.

Choose eigen-covectors ξ, η normalized by $\det(u, v) = \xi(u)\eta(v) - \eta(u)\xi(v)$. Use representatives v_a for *all* oriented orbits, paired by $v_{-a} = -v_a$, and put

$$\begin{aligned} P_a(z) &= \sum_j \widehat{g}(A^j v_a) z^j, \\ X_g(\theta) &= \sum_a \xi(v_a)^2 |P_a(\lambda e^{i\theta})|^2, \\ Y_g(\theta) &= \sum_a \eta(v_a)^2 |P_a(\lambda^{-1} e^{i\theta})|^2. \end{aligned}$$

Both functions are continuous, nonnegative and even. For example, $\sum_{a,j} |\xi(v_a)| \lambda^j |\widehat{g}(A^j v_a)| \le \|\xi\| M_1(g)$; the stable estimate is analogous.

Theorem 53.3 (Two-variable tail symbol)**.** *With T denoting the Toeplitz operator on positive lag indices,*

$$G_g(\omega) = T(F_g(\omega, \cdot)) + K_g(\omega), \qquad F_g(\omega, \theta) = \pi^2 X_g(\theta) Y_g(\omega - \theta).$$

The remainder is compact, self-adjoint and operator-norm continuous in ω. It is trace class for finite Fourier polynomials.

Proof. For finite support, sufficiently large pair gaps have unique ordered orbit labels (a, b, k) for $v_a + A^k v_b$. This rests on the following gap lemma. Identify integer vectors with a fixed fractional lattice $\mathcal{L}$ in the real quadratic field of μ, preserved by the unit μ; write σ, σ' for the two embeddings, so $|\sigma(\mu)| = \lambda$ and $|\sigma'(\mu)| = \lambda^{-1}$. Bound both embeddings of the finite representative set by M and the absolute norm of a nonzero element of $\mathcal{L}$ below by $\delta > 0$. A *four-atom relation* is a vanishing signed sum of four atoms $\mu^{e_i}\beta_i$, with β_i from the representative set.

Lemma 53.4 (Gap splitting)**.** *Let B satisfy $\lambda^B > 16M^2/\delta$. If the sorted exponents of a four-atom relation have a consecutive gap of at least B, the relation splits into two separately vanishing pieces. No one-atom piece vanishes. Consequently, if both internal pair gaps in a relation $v_a + A^k v_b = A^t(v_{a'} + A^{k'} v_{b'})$ exceed $3B$, then $(a, b, k) = (a', b', k')$ and $t = 0$.*

Proof. Split at the gap into a low piece L and a high piece $H = \mu^s H_0$, where s is the smallest high exponent and H_0 is a sum of at most four atoms with nonnegative relative exponents. Since $|\sigma'(\mu)| = \lambda^{-1}$, $|\sigma'(H_0)| \le 4M$. If $H_0 \neq 0$, then H_0 is a nonzero element of $\mathcal{L}$, so $|\sigma(H_0)| = |N(H_0)|/|\sigma'(H_0)| \ge \delta/(4M)$ and $|\sigma(H)| \ge \lambda^s \delta/(4M)$. On the other hand $|\sigma(L)| \le 4M\lambda^{s-B}$, and $H + L = 0$ would force $\lambda^B \le 16M^2/\delta$, contrary to the choice of B. Hence $H_0 = 0$ and $L = 0$. A single atom $\mu^e\beta$ is nonzero, since $\beta \neq 0$ and μ is a unit, so no singleton piece vanishes.

For the pair relation, the three consecutive gaps of the four sorted exponents sum to at least $\max(k, k') > 3B$, so some gap exceeds B. Splitting at such a gap, an end gap would isolate a singleton, which cannot vanish; hence the pieces have two atoms each. (Within a pair the lower atom precedes the upper one, so a mixed low piece consists of the two lower atoms.) If one piece were a whole pair, its vanishing would identify two distinct oriented orbits, or one orbit at a positive self-shift; both are excluded, since each orbit has a unique representative and the hyperbolic A has no periodic nonzero vector. Thus the low piece gives $\beta_a = \mu^t \beta_{a'}$, forcing $a = a'$ and $t = 0$, and the high piece gives $\mu^k \beta_b = \mu^{k'} \beta_{b'}$, forcing $b = b'$ and $k = k'$ by the same argument. □

This excludes all integer translates.

At such a label the exact output of $S_g(\omega)$ is

$$\pi \operatorname{sgn}(\operatorname{tr} A)^k \big(\eta(v_a)\xi(v_b) - \lambda^{-2k}\xi(v_a)\eta(v_b)\big)(R_{ab}(\omega,\cdot) * x)_k,$$

where

$$R_{ab}(\omega, z) = P_a(\lambda^{-1}e^{i\omega}z^{-1})P_b(\lambda z).$$

Indeed a pair at times i, j and lag r has $k = r+j-i$, simultaneous shift i, and factor $\lambda^{-r} = \lambda^{j-i}\lambda^{-k}$. The constant squared row weights give the stated symbol. Remaining weights decay as $O(\lambda^{-2k})$, so finite filters give a trace-class correction. The finitely many excluded columns add finite rank.

For general g, Theorem 53.2 gives uniform convergence of the Gram operators under truncation. The boundary sums give uniform convergence of the symbols on the two-torus. Compact operators are norm-closed, proving the assertion and continuity of the remainder. □

Theorem 53.5 (Full Riesz criterion)**.** *There are R and $c > 0$ such that every finite array supported on $r > R$ obeys*

$$\begin{aligned} c\sum_{r,l} |x_{r,l}|^2 &\le \left\| \sum_{r,l} x_{r,l}\lambda^{-r} C^l h_r(g) \right\|_2^2 \\ &\le 2\pi^2 \kappa_A^2 M_1(g)^4 \sum_{r,l} |x_{r,l}|^2 \end{aligned}$$

if and only if X_g, Y_g have no zeros. The limiting lower tail edge is

$$c_\infty(g) = \pi^2 \min X_g \min Y_g.$$

Proof. Compactness and continuity imply $\sup_\omega \|E_R K_g(\omega) E_R\| \to 0$; use a finite norm net of the compact image of K_g. Toeplitz positivity gives the lower edge. Shifted long plane waves at minimizing θ and $\omega - \theta$ give the matching upper edge. Finally apply Parseval to the simultaneous-translation variable. If the edge is zero, localization near the minimizing ω by a long trigonometric polynomial rules out a positive uniform lower bound. □

For every real inversion-even background, a sine quadrature gives

$$X_{g+\varepsilon i(e_v - e_{-v})} = X_g + 2\varepsilon^2 \xi(v)^2, \qquad Y_{g+\varepsilon i(e_v - e_{-v})} = Y_g + 2\varepsilon^2 \eta(v)^2.$$

Hence for every nonzero ε,

$$c_\infty \ge \frac{4\pi^2 \varepsilon^4 \det(v, Av)^2}{(\operatorname{tr} A)^2 - 4} > 0.$$

Only a phase quadrature is added when v is already supported. For an arbitrary real $M_{1+\delta}$ background, $\delta > 0$, varying the one conjugate coefficient pair by $(u + iw, u - iw)$ fails only on a subset of two rectifiable curves in the (u, w) plane: a zero requires both real and imaginary coefficient-filter components to equal $-u, -w$ on one of the two radii. Thus almost every arbitrarily small such change gives the full Riesz property. The property is open and dense in the real M_1 space and in C^∞, using finite-polynomial approximation for density. No quantum Heisenberg-scale comparison follows from these classical bounds.

54 A two-window phase crossover

Fix $q \ge 1$, $R \ge 2q$, $v \in \mathbb{Z}^2 \setminus \{0\}$ and put $t = |\operatorname{tr} A|$. Define one fixed-support family

$$g_\varepsilon = (C^2 - tC + I)^q (e_v + e_{-v}) + \varepsilon i (e_v - e_{-v}).$$

Let $L_{R,n,m}(\varepsilon)$ be the nm-dimensional Gram matrix in $L^2(\mathbb{T}^2)$ of

$$\lambda^{-r}C^l h_r(g_\varepsilon), \qquad R+1 \le r \le R+n, \quad 1 \le l \le m.$$

The first index is lag, the second simultaneous time translation.

Theorem 54.1 (Two-window lower edge)**.** *For $a = \min(n,m)$ there are positive constants depending only on A, v, q, R such that, uniformly in $n, m \ge 1$ and real ε,*

$$\lambda_{\min}(L_{R,n,m}(\varepsilon)) \asymp (n^{-2q}+\varepsilon^2)(a^{-2q}+\varepsilon^2).$$

This is a two-sided order estimate, not a claimed exact leading constant.

Proof. Write $u(\theta) = 2-2\cos\theta$ and $H(\theta) = u(\theta)(t^2-4+u(\theta))$. The exact symbol of Theorem 53.3 is

$$F_\varepsilon(\omega,\theta) = 4\pi^2\xi(v)^2\eta(v)^2\big(\lambda^{2q}H(\theta)^q+\varepsilon^2\big) \cdot \big(\lambda^{-2q}H(\omega-\theta)^q+\varepsilon^2\big).$$

Indeed the characteristic polynomial obeys

$$|((\lambda e^{i\theta})^2 - t\lambda e^{i\theta}+1)^q|^2 = \lambda^{2q}H(\theta)^q,$$

and the reciprocal-radius formula is analogous. The opposite orientations cancel the sine perturbation's mixed terms.

For this single-orbit family the exact output row weight, relative to the symbol, is $(1-\lambda^{-2k})^2$. The temporal filter has degree $2q$, so $k \ge R+1-2q \ge 1$. Thus the actual finite Gram matrix lies between the Toeplitz form and $(1-\lambda^{-2(R+1-2q)})^2$ times that form. This is a relative Loewner comparison, with no additive error near a closing gap. Since $H \asymp u$, it suffices to treat the symbol $(u(\theta)^q+\varepsilon^2)(u(\omega-\theta)^q+\varepsilon^2)$.

Extend an n by m coefficient array by zero to $\mathbb{Z}^2$. Let D_1 and D_2 be first differences in directions $(1,0)$ and $(-1,1)$. The form is exactly

$$\|D_1^q D_2^q x\|_2^2 + \varepsilon^2(\|D_1^q x\|_2^2 + \|D_2^q x\|_2^2) + \varepsilon^4\|x\|_2^2. \tag{TW.1}$$

All boundary terms are included. A sequence supported on s consecutive sites has first Dirichlet difference eigenvalue $4\sin^2(\pi/(2(s+1)))$. Fourier Jensen therefore gives $\|D^q y\|_2^2 \ge c_q s^{-2q}\|y\|_2^2$. Horizontal chains have length n, diagonal chains length at most a. After D_1^q the latter length is at most $\min(n+q,m) \le (q+1)a$. Applying these inequalities successively proves

$$\|D_1^q D_2^q x\|_2^2 \ge c_q n^{-2q}a^{-2q}\|x\|_2^2,$$

as well as the two single-difference lower bounds. Inserting them in (TW.1) proves the lower estimate.

For the upper estimate take

$$x_{j,l} = f(j/(n+1))f(l/(m+1)), \qquad f(s) = s^{2q}(1-s)^{2q}\mathbf{1}_{[0,1]}(s).$$

Vanishing derivatives through order $2q-1$ at both endpoints give the finite difference bounds, including boundary crossings. If E_j, E_l are shifts, $D_2 = E_j^{-1}(D_l - D_j)$. The binomial expansion gives

$$\begin{aligned}\|D_1^q x\|_2 &\le C_q n^{-q}\|x\|_2,\\ \|D_2^q x\|_2 &\le C_q(n^{-1}+m^{-1})^q\|x\|_2,\\ \|D_1^q D_2^q x\|_2 &\le C_q n^{-q}(n^{-1}+m^{-1})^q\|x\|_2.\end{aligned}$$

Use this trial vector in (TW.1). The finitely many small window lengths are absorbed in the constants. □

When $m \geq n$, the edge has order $(n^{-2q} + \varepsilon^2)^2$, with threshold $|\varepsilon| \asymp n^{-q}$. When $m \ll n$, the two thresholds are n^{-q} and m^{-q}, and the three orders are respectively $n^{-2q}m^{-2q}$, $\varepsilon^2 m^{-2q}$ and ε^4. In particular a single simultaneous time, $m = 1$, gives order n^{-2q} at zero perturbation, whereas $m \geq n$ gives n^{-4q}. Setting the dynamical frequency to zero before specifying the time window would miss this proved distinction. No identification with a quantum spectral form factor is made.

Chapter 10 — Resonance rigidity (continued)

55 Rank-controlled polynomial rigidity

This section is independent of the spectral-statistics conjecture. Its statements are self-contained; priority relative to polynomial sphere-map and filter-bank literature is not asserted.

For $P(z) = \sum_{j=0}^{n} v_j z^j \in \mathbb{C}^d[z]$, let $r = \dim \operatorname{span}\{v_j\} \geq 1$. Call $\rho > 0$ flat when $\|P(\rho e^{i\theta})\|^2$ is independent of θ.

Theorem 55.1 (Sharp multiradius rigidity)**.** *Either the nonzero coefficient vectors are pairwise Hermitian-orthogonal, so every positive radius is flat, or there are at most $r - 1$ distinct positive flat radii. The bound is independent of the polynomial degree.*

Proof. A nonzero real polynomial with k nonzero monomials has at most $k - 1$ distinct positive zeros: divide by the lowest monomial, differentiate and apply Rolle's theorem inductively. The same bound holds for complex coefficients by taking a scalar-rotated real part which is not identically zero.

If the coefficients are not pairwise orthogonal, choose the largest $L \geq 1$ with some $\langle v_j, v_{j+L} \rangle \neq 0$. The angular coefficient at lag L is $\rho^L B(\rho^2)$, where

$$B(x) = \sum_{j=0}^{n-L} \langle v_j, v_{j+L} \rangle x^j.$$

Let $j_1 < \cdots < j_k$ be its nonzero positions. The cross-Gram matrix

$$M_{ab} = \langle v_{j_a}, v_{j_b+L} \rangle$$

is triangular with nonzero diagonal: when $b > a$, the separation exceeds L and the entry is zero. Hence $k = \operatorname{rank} M \leq r$. Every flat radius gives a positive zero of the nonzero k-term polynomial B, so there are at most $k - 1 \leq r - 1$. If no such L exists, orthogonality makes the squared norm radial at every radius. □

The bound is sharp at arbitrary prescribed radii. Given $\rho_1, \ldots, \rho_{d-1} > 0$ distinct, write

$$B(x) = \prod_{s=1}^{d-1} (x - \rho_s^2) = \sum_{j=0}^{d-1} a_j x^j$$

and define the d components

$$P_j(z) = \sqrt{|a_j|}\, z^j (1 + \operatorname{sgn}(a_j) z).$$

Their coefficient span has rank d, and exactly

$$\|P(\rho e^{i\theta})\|^2 = (1 + \rho^2) \sum_j |a_j| \rho^{2j} + 2\rho \cos\theta\, B(\rho^2).$$

Thus precisely the prescribed $d - 1$ positive radii are flat. For radii $1, 2$ one may take $P = (2(1 + z), \sqrt{5} z(1 - z), z^2(1 + z))$.

Theorem 55.2 (Concentric complex spheres)**.** *Let $P : \mathbb{C}^m \to \mathbb{C}^d$ be a nonzero holomorphic polynomial map, with coefficient-span rank r. If $\|P\|$ is constant on each of r distinct positive concentric spheres, then*

$$\|P(z)\|^2 = \sum_k c_k \|z\|^{2k}, \qquad c_k \geq 0.$$

Writing $P(z) = \sum_\alpha v_\alpha z^\alpha$, one has

$$\langle v_\alpha, v_\beta \rangle = 0 \quad (\alpha \neq \beta), \qquad \|v_\alpha\|^2 = c_{|\alpha|} \frac{|\alpha|!}{\alpha!}.$$

Consequently P is, up to a target-unitary change of coordinates, an orthogonal sum of scaled homogeneous Veronese maps, and

$$r = \sum_{k : c_k > 0} \binom{m+k-1}{k}.$$

Proof. Decompose $P = \sum_k P_k$ into homogeneous degrees. Restriction to the complex line $z = wu$, for each unit vector u, and Theorem 55.1 give $\langle P_k(u), P_l(u) \rangle = 0$ for $k \neq l$. Real scaling extends this identity to all z. Expansion in the independent monomials $z^\alpha \overline{z}^\beta$ makes the coefficient ranges of different homogeneous pieces orthogonal.

There is a unit vector at which every nonzero homogeneous piece is nonzero, since a finite union of proper polynomial zero sets cannot cover the sphere. Thus the number of nonzero degrees is at most r. On the unit sphere, the r prescribed radii give linear equations

$$\sum_k \rho_s^{2k} \|P_k(u)\|^2 = C_s.$$

The generalized Vandermonde matrix has full column rank: a dependence would be a sparse polynomial with too many positive zeros, contradicting the lemma used above. Hence $\|P_k(u)\|^2 = c_k$ is constant in u. Homogeneity gives $\|P_k(z)\|^2 = c_k(\sum_j |z_j|^2)^k$; coefficient comparison with the multinomial expansion proves the assertions. □

The centers in this theorem are fixed, and the maps are polynomial. No extension to rational maps with poles or to infinite series is assumed.

55.1 A complete single-orbit flat-tail classification

For a finite complex filter $p = A_0 + iB_0$ with coefficientwise real A_0, B_0, apply Theorem 55.1 to (A_0, B_0). Flatness of $|A_0|^2 + |B_0|^2$ at two distinct radii forces at most two nonzero coefficients, orthogonal as vectors in $\mathbb{R}^2$. For a real observable on one signed A-orbit its tail symbol is

$$4\pi^2 \xi(v)^2 \eta(v)^2 N_\lambda(\theta) N_{\lambda^{-1}}(\theta), \quad N_\rho = |A_0(\rho e^{i\theta})|^2 + |B_0(\rho e^{i\theta})|^2.$$

The two factors are in fact separately constant. Each N_ρ, as a function of $\zeta = e^{i\theta}$, is a Laurent polynomial satisfying $N_\rho(\zeta) = N_\rho(\zeta^{-1})$: for real-coefficient A_0, B_0,

$$N_\rho(\zeta) = A_0(\rho\zeta)A_0(\rho\zeta^{-1}) + B_0(\rho\zeta)B_0(\rho\zeta^{-1}).$$

If a product of two Laurent polynomials is a nonzero constant, each factor is a unit of $\mathbf{C}[\zeta, \zeta^{-1}]$, hence a monomial $c\zeta^n$: the exponent spans of the factors add under multiplication, and the product has span zero. Inversion invariance forces $n = 0$. Thus N_λ and $N_{\lambda^{-1}}$ are separately constant (for a nonzero filter the constant product is positive; the monomial pair $N_1 = \zeta$, $N_2 = \zeta^{-1}$ shows that inversion invariance is essential here). Flatness of the product symbol therefore gives flatness at the two radii λ, λ^{-1}, and the symbol is flat exactly for one temporal tap, or two taps in phase quadrature.

For taps $c_0 = ae^{i\phi}$ and $c_d = ibe^{i\phi}$, $a, b \in \mathbb{R}$, put $D_r = \det(v, A^r v)$. The unnormalized dynamical Gram matrix is exactly diagonal for $r, s > d$, with

$$\Gamma_{rr} = 4\pi^2\big((a^4 + b^4)D_r^2 + a^2b^2(D_{r+d}^2 + D_{r-d}^2)\big).$$

Indeed the same-sign channel has taps $e^{2i\phi}(iab, a^2 - b^2, iab)$, while the opposite-sign channel has $(iab, a^2 + b^2, -iab)$. At lag difference d the cross terms are purely imaginary and cancel with the conjugate orientations; at difference $2d$ the two channel types cancel each other. Other off-diagonal overlaps are absent. This covariance identity does not imply Gaussianity or independence.

Chapter 7 — Floquet, coboundary and smooth resonance fields (continued)

56 Balanced phase and quantum controllability

This section proves an exact finite-dimensional Lie-algebra statement. Controllability allows varying controls; it is not a local-statistics theorem for one fixed kicked propagator. The underlying Lie-algebra mechanism is a standard type of quantum-control criterion, and priority is not asserted.

Let $N \geq 3$ be odd and use modular-centred Weyl operators, so

$$W(v)W(w) = e_N(2^{-1}\det(v, w))W(v + w), \qquad W(v)^N = I,$$

where 2^{-1} is the inverse modulo N. Suppose $\det(v, w)$ is a unit modulo N, and define

$$H_0 = (1 + i)W(v) + (1 - i)W(-v), \qquad H_1 = cW(w) + \overline{c}W(-w), \quad c \neq 0.$$

Theorem 56.1 (An explicit generating pair)**.** *The real Lie algebra generated by iH_0, iH_1 is $\mathfrak{su}(N)$. Composite odd dimensions are included.*

Proof. Put $\zeta = e^{2\pi i/N}$ and $d = \det(v, w)$. Weyl commutation gives a basis in which $W(v)$ has simple eigenvalues ζ^j and $W(w)$ maps the jth basis vector to the $(j + d)$th with a nonzero weight. Thus H_1 has a connected cyclic edge graph. The eigenvalues of H_0 are

$$E_j = 2\sqrt{2}\cos(2\pi j/N + \pi/4).$$

They are distinct: a nontrivial cosine equality would require $j + k + N/4$ to be an integer multiple of N. The edge gaps are

$$\Delta_j = -4\sqrt{2}\sin(\pi d/N)\sin(2\pi j/N + \pi d/N + \pi/4).$$

Their squares are pairwise distinct. Equality of squared sines implies either $2(j - k) = 0 \pmod N$, hence $j = k$, or $4(j + k + d) = (2a - 1)N$, impossible because the right side is odd. The gaps are nonzero because the E_j are distinct.

On the skew-Hermitian subspace of edge j, $\mathrm{ad}(iH_0)^2$ has eigenvalue $-\Delta_j^2$. Real polynomial interpolation isolates each edge component of iH_1. Commutation with iH_0 gives its other real quadrature. Their commutators give diagonal differences, and commutators along paths in the connected cycle give all off-diagonal generators. These span $\mathfrak{su}(N)$. □

Let $U_N W(v)U_N^* = W(Av)$ and $D = \det(v, Av) \neq 0$. For every odd N coprime to D, the theorem applies to $H_0, U_N H_0 U_N^*$. For the Pell matrix $A = \left(\begin{smallmatrix} 3 & 2 \\ 4 & 3 \end{smallmatrix}\right)$ and $v = (1, 0)$, $D = 4$, so every odd dimension is admissible. This matrix is identity modulo two; the physical integer lift has the same covariance along this orbit.

Theorem 56.2 (One parameter in all dimensions)**.** *Let K_N be arbitrary specified traceless Hermitian backgrounds. For each admissible N, a nonzero real polynomial of degree at most $(N^2-1)^2$ certifies that*

$$\mathrm{Lie}_{\mathbb{R}}\big(i(K_N+\varepsilon H_0), iU_N(K_N+\varepsilon H_0)U_N^*\big) = \mathfrak{su}(N)$$

outside its roots. Hence one real ε, outside a countable exceptional set, works simultaneously in every admissible dimension. Such choices exist arbitrarily close to zero.

Proof. Write $D_N = N^2-1$. The spaces spanned by left-nested Lie words of length at most L strictly increase until the generated algebra is reached. If one step fails to increase, the space is invariant under both generator adjoints and no later step can increase it. Thus at the pure H_0 pair choose D_N independent homogeneous words of lengths at most D_N. Evaluate these words on the perturbed pair. Their coordinate determinant in a fixed real basis of $\mathfrak{su}(N)$ is polynomial in ε. Its leading coefficient is the nonzero determinant of the pure-pair words; its degree is at most the sum of their lengths, hence at most D_N^2. Take the countable union of the finite root sets. Roots of this particular certificate need not be genuinely uncontrollable parameters. □

For a real even classical background, the balanced harmonic

$$b = (1+i)e_v + (1-i)e_{-v}$$

also gives the full classical Riesz property of Theorem 53.5. Its even part adds $\varepsilon(e_v+e_{-v})$ and its sine part adds $\varepsilon i(e_v-e_{-v})$. Hence

$$c_\infty(g+\varepsilon b) \geq \frac{4\pi^2\varepsilon^4 D^2}{(\operatorname{tr} A)^2-4}.$$

For the Pell example this is $2\pi^2\varepsilon^4$. Thus an arbitrarily small fixed harmonic perturbation can simultaneously regularize the entire classical resonance lattice and give full quantum control in every admissible odd dimension, outside the stated countable set.

There is no dimension-uniform synthesis-time estimate here. Nor is there a mixing-rate estimate for random controls, or a Heisenberg-scale estimate for a fixed propagator. In fact if an unperturbed U_N has scalar K_Nth power, the lag commutators

$$\frac{iN}{2}[G, U_N^r G U_N^{-r}]$$

are K_N-periodic and vanish at $r = K_N$. Any Gram matrix of more than $K_N - 1$ such lag observables is singular, regardless of nonzero column normalizations. A classical Gram block bounded below by cI cannot be approximated in operator norm within c by that singular quantum block. This is an obstruction to one transfer mechanism, not a conclusion about an actually perturbed propagator whose exact period has been destroyed.

Chapter 11 — Temporal resonance and spectral modules (continued)

57 Infinite-orbit independence and entire temporal gauges

Let a real observable with $M_1(g) < \infty$ be supported on one signed orbit:

$$g = \sum_{j\in\mathbb{Z}} (c_j e_{A^j v} + \overline{c_j} e_{-A^j v}), \qquad v \neq 0.$$

Put $D_k = \det(v, A^k v)$ and define the absolutely summable correlations

$$S_n = \sum_j c_j c_{j+n}, \qquad C_n = \sum_j c_j \overline{c_{j+n}}.$$

Then $S_{-n} = S_n$, $C_{-n} = \overline{C_n}$. Direct Fourier multiplication, with $J = J_0$, gives for $k \geq 1$

$$J(h_r)_{[v+A^k v]} = -\pi D_k (S_{k-r} - S_{k+r}),$$
$$J(h_r)_{[v-A^k v]} = \pi D_k (C_{k-r} - C_{k+r}).$$

Negative orientations contribute conjugates. The invariant $q_A(v \pm A^k v) = q_A(v)(2 \pm \operatorname{tr} A^k)$ distinguishes all positive gaps and sign types, and a vector and its negative lie in different oriented orbits. Thus for finite real a_r, exactly

$$\mathcal{V}_A\left(\sum_r a_r h_r\right) = 2\pi^2 \sum_{k \geq 1} D_k^2 (|A_k|^2 + |B_k|^2),$$

where $A_k = \sum_r a_r (S_{k-r} - S_{k+r})$ and $B_k = \sum_r a_r (C_{k-r} - C_{k+r})$.

Theorem 57.1 (Infinite-support single-orbit independence)**.** *For every nonzero g in this class, the classes $[h_r(g)]$, $r \geq 1$, are linearly independent modulo L^2 coboundaries.*

Proof. If the displayed variance vanishes, $B_k = 0$ for $k > 0$ and $B_{-k} = -\overline{B_k} = 0$. Set $P(e^{i\theta}) = \sum_j c_j e^{ij\theta}$ and $s(\theta) = \sum_r a_r \sin(r\theta)$. Uniform convergence gives

$$\sum_k B_k e^{-ik\theta} = -2is(\theta)|P(e^{i\theta})|^2.$$

The left side is constant. Evaluation at $\theta = 0$ makes that constant zero. The nonzero continuous function P is nonzero on an interval; there s vanishes, forcing the trigonometric polynomial s to vanish identically. Complex relations separate into two real ones. A nonzero orbit sum excludes an L^2 coboundary by the Fourier difference equation. □

This argument does not select a maximum of an infinite Fourier support and does not extend to arbitrary unions of Fourier orbits.

57.1 A classified smooth gauge group

Let P be holomorphic on $\mathbb{C} \setminus \{0\}$, have real Laurent coefficients p_j, and obey $|P(e^{i\theta})| = 1$. Then

$$T_P = \sum_{j \in \mathbb{Z}} p_j C^j$$

is unitary on L^2 and a continuous automorphism of every weighted Fourier space $M_s < \infty$ and of C^∞.

Theorem 57.2 (Entire real all-pass filters)**.** *Exactly these filters have the form*

$$P(z) = \sigma z^n \exp H(z), \qquad H(z) = \sum_{k \geq 1} a_k (z^k - z^{-k}),$$

where $\sigma \in \{\pm 1\}$, $n \in \mathbb{Z}$, $a_k \in \mathbb{R}$, and H is holomorphic on the punctured plane. For every real M_1 observable,

$$J(h_r(T_P g)) = J(h_r(g)) \quad \text{for all } r.$$

The difference is an actual L^2 coboundary, with a smooth transfer function when g is smooth.

Proof. Reality and the boundary modulus identity give $P(z)P(1/z) = 1$ by analytic continuation. Hence P has no zeros. Remove its winding monomial and its sign $P(1)$, and take a global logarithm H normalized by $H(1) = 0$. Reality and reciprocity give real Laurent coefficients and $H(z) + H(1/z) = 0$, proving the classification. Cauchy estimates give $\sum_j |p_j| R^{|j|} < \infty$ for every $R > 1$. Composition by C^j grows at most exponentially on each fixed smooth seminorm or weighted Fourier norm. The operator series and its inverse therefore converge in each of those norms. The adjoint is $T_{P(1/z)}$, so the operator is unitary.

Bilinearity and symplectic covariance yield

$$h_r(T_P g) = \sum_{i,j} p_i p_j C^i h_{r+j-i}(g).$$

The series converges in Fourier ℓ^1, since bracket norms grow at most exponentially in the relative lag. The autocorrelations $\sum_i p_i p_{i+n} = \delta_{n0}$ follow from $P(z)P(1/z) = 1$. Define $T_i = \sum_{l=0}^{i-1} C^l$ for $i > 0$, $T_0 = 0$, and $T_i = -\sum_{l=i}^{-1} C^l$ for $i < 0$. Then

$$h_r(T_P g) - h_r(g) = (C - I) \sum_{i,j} p_i p_j T_i h_{r+j-i}(g).$$

The transfer series converges in L^2 because $\|T_i\| \le |i|$, and in every smooth seminorm for smooth g. This proves both assertions. □

For example $T_a = \exp(a(C - C^{-1}))$, $a \in \mathbb{R}$, is a nontrivial smooth unitary group. It preserves every ordinary two-point correlation and the whole zero-frequency resonance synthesis operator. It does not preserve all nonzero dynamical-frequency matrices: its two-variable symbol multiplier is

$$\exp\bigl(2a(\lambda - \lambda^{-1})(\cos\theta - \cos(\omega - \theta))\bigr).$$

This equals one at $\omega = 0$, but not in general. The rational all-pass resolvent need not preserve C^∞; the entire filters above are different because their coefficients have every exponential moment.

57.2 A preserved hierarchy with a changing cubic moment

For $A = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 1 \end{smallmatrix}\right)$, put $v = (1, 0)$, $w = Av$, $z = v + w$, $B_u = e_u + e_{-u}$, and $g_* = B_v + 2B_w + B_z$. Exact Fourier multiplication gives

$$\int g_*^3 = 24, \qquad \int g_*^2 C g_* = 4, \qquad \int g_* (C g_*)^2 = 16.$$

For $g_a = T_a g_*$, the entire hierarchy in Theorem 57.2 is unchanged, while

$$\int g_a^3 = 24 - 36a + O(a^2).$$

The derivative is $3 \int g_*^2 (C - C^{-1}) g_* = -36$. For each fixed smooth symbol, direct dominated convergence in its Fourier triple expansion gives

$$\lim_{N \to \infty} N^{-1} \operatorname{Tr} \operatorname{Op}_N(g_a)^3 = \int g_a^3.$$

Thus the preserved quadratic data do not determine all quantum observable spectral moments. This does not prove different unfolded microscopic universality classes, or identify the eigenphase law of a kicked map.

A finite control makes the frequency distinction explicit. For $b = B_v$, $P = 1 - C/2$, $Q = C - I/2$, the dynamical cross-spectra of the first two brackets at $\omega = \pi$ are respectively $-12\pi^2$ and $+12\pi^2$. Indeed, writing $b_r = h_r(b)$,

$$h_r(Pb) = (1 + C/4) b_r - \tfrac{1}{2}(C b_{r-1} + b_{r+1}),$$
$$h_r(Qb) = (C + 1/4) b_r - \tfrac{1}{2}(C b_{r-1} + b_{r+1}).$$

The single-mode orbit weights are $4\pi^2 \det(v, A^r v)^2$. The stronger all-frequency invariance claim is therefore false.

Theorem 57.3 (A Gevrey obstruction)**.** *If a nontrivial entire real all-pass filter is applied to one nonzero harmonic $e_v + e_{-v}$, the result is smooth but satisfies no Fourier estimate $|\widehat{g}(k)| \le C\exp(-c|k|^\alpha)$ with $\alpha > 0$. The only such filters retaining this decay are $P(z) = \pm z^n$.*

Proof. The assumed decay and hyperbolicity give $|p_j| \le C' \exp(-c'\lambda^{\alpha|j|})$, hence Gaussian decay in j. Gaussian summation bounds $\log \max_{|z|=\rho} |P(z)|$ by $O(1+|\log\rho|^2)$ at both ends. Reciprocity then bounds $|\operatorname{Re} H(\rho e^{i\theta})|$ by the same order. Its kth angular Fourier coefficient is $a_k(\rho^k - \rho^{-k})/2$. Letting $\rho \to \infty$ forces every $a_k = 0$. □

The statement concerns the image of a single harmonic under an entire filter. It does not remove finite-polynomial homometric ambiguities, whose transfer ratios are rational, and does not prove general analytic identifiability from quadratic data.

Chapter 7 — Floquet, coboundary and smooth resonance fields (continued)

58 Weighted-Wiener stability of the isolated resonance mode

The finite-Fourier theorem above has an explicit infinite-support stability region. This is useful because finite support and arbitrary smooth support are qualitatively different: smooth exact coboundary cancellations exist, but they cannot approach a fixed finite polynomial too closely in a suitable weighted Wiener topology while preserving zero variance.

For a Fourier series f put

$$M_0(f) = \sum_{v\in\mathbb{Z}^2} \|v\|_1 |\widehat{f}(v)|, \qquad M_A(f) = \sum_{v\in\mathbb{Z}^2} \|Av\|_1 |\widehat{f}(v)|. \tag{W.1}$$

Constants are invisible to these seminorms and to the Poisson bracket. Write

$$B_A(f,g) = \frac{\{f, \mathcal{C}_A g\}}{4\pi}, \qquad h_g = B_A(g,g).$$

The Fourier formula is

$$\widehat{B_A(f,g)}(k) = -\pi \sum_{u+Av=k} \det(u, Av)\widehat{f}(u)\widehat{g}(v). \tag{W.2}$$

Since $|\det(u,w)| \le \|u\|_1\|w\|_1$, absolute summation gives

$$\|\widehat{B_A(f,g)}\|_{\ell^1} \le \pi M_0(f) M_A(g). \tag{58.1}$$

Thus all orbit sums are absolutely convergent whenever the two weighted moments are finite.

Theorem 58.1 (Explicit weighted-Wiener stability)**.** *Let p be a nonconstant real trigonometric polynomial. Choose the isolated finite-support witness u, v supplied by Lemma 48.1, for lag one, and put*

$$d = |\det(u, Av)|, \qquad D = d\,|\widehat{p}(u)\widehat{p}(v)| > 0.$$

For a real weighted-Wiener perturbation $g = p + r$, define

$$P_0 = M_0(p), \quad P_A = M_A(p), \quad \varepsilon_0 = M_0(r), \quad \varepsilon_A = M_A(r)$$

and

$$E = P_0\varepsilon_A + \varepsilon_0 P_A + \varepsilon_0\varepsilon_A.$$

If $E < D$, then

$$\Gamma_A(g) \geq 2\pi^2(D-E)^2 > 0. \tag{58.2}$$

Proof. Bilinearity and (58.1) give

$$\|\widehat{h_g - h_p}\|_{\ell^1} \leq \pi E.$$

If $k = u + Av$, the isolated-mode theorem says that the oriented A-orbit of k contains exactly one Fourier coefficient of h_p, with modulus πD. Hence the same orbit sum for h_g has modulus at least $\pi(D-E)$. For real g the distinct orbit of $-k$ has the conjugate sum, so those two orbits alone give (58.2). □

A one-parameter radius follows immediately. If

$$P = P_0 + P_A, \qquad \varepsilon = \max\{M_0(r), M_A(r)\},$$

then $E \leq P\varepsilon + \varepsilon^2$. Therefore every perturbation with

$$\varepsilon < \varepsilon_*(p; u, v) := \frac{\sqrt{P^2 + 4D} - P}{2} \tag{58.3}$$

has positive resonance variance. In particular every nonconstant real trigonometric polynomial is an interior point of the nondegenerate set in this weighted Wiener topology.

For analytic tails the criterion is fully explicit. If $|\widehat{r}(v)| \leq Ce^{-\rho\|v\|_1}$ and the tail is supported on $\|v\|_1 > R$, then

$$M_0(r) \leq 4C \sum_{n>R} n^2 e^{-\rho n}, \qquad M_A(r) \leq \|A\|_{1\to 1} M_0(r). \tag{W.3}$$

Substitution into the theorem gives a computable analytic-tail condition. This proves nondegeneracy on an infinite-dimensional open neighbourhood of every finite polynomial; it does not prove that every nonconstant analytic observable lies in one of these neighbourhoods. The latter global coverage question remains open.

Chapter 10 — Resonance rigidity (continued)

58.1 Generic infinite-support nondegeneracy

Let X_A be the real Banach space obtained by adjoining the constant Fourier mode to the two weighted moments in (W.1):

$$\|f\|_{X_A} = |\widehat{f}(0)| + M_0(f) + M_A(f).$$

On absolutely summable Fourier sequences define the orbit-sum map

$$(\mathcal{P}h)_{\mathcal{O}} = \sum_{k\in\mathcal{O}} \widehat{h}(k).$$

Because the oriented frequency orbits partition $\mathbb{Z}^2 \setminus \{0\}$,

$$\|\mathcal{P}h\|_{\ell^2(\mathcal{O})} \leq \|\mathcal{P}h\|_{\ell^1(\mathcal{O})} \leq \|\widehat{h}\|_{\ell^1}. \tag{58.4}$$

Moreover

$$h_g - h_f = B_A(g-f, g) + B_A(f, g-f),$$

so (58.1) yields

$$\|\widehat{h_g - h_f}\|_{\ell^1} \le \pi\{M_0(g-f)M_A(g) + M_0(f)M_A(g-f)\}. \tag{58.5}$$

Thus $g \mapsto \mathcal{P}h_g$ is continuous from X_A to $\ell^2(\mathcal{O})$ and $\Gamma_A(g) = \|\mathcal{P}h_g\|_2^2$ is continuous.

Theorem 58.2 (Generic weighted-Wiener nondegeneracy)**.** *Modulo constants, the set*

$$\{g \in X_A : \Gamma_A(g) > 0\}$$

is open and dense. Its exact-cancellation complement is closed and nowhere dense.

Proof. Openness follows from continuity. For density, let p_R be the Fourier truncation of g to $\|v\|_1 \le R$. Absolute convergence of the weighted moments gives $p_R \to g$ in X_A. If g is nonconstant then p_R is nonconstant for all sufficiently large R, and the unrestricted finite-Fourier theorem gives $\Gamma_A(p_R) > 0$. Constants are approximated by adding an arbitrarily small nonzero cosine mode. Hence the positive set is dense; its closed complement has empty interior. □

For $\rho > 0$ let

$$X_\rho = \left\{ f : \sum_v e^{\rho\|v\|_1} |\widehat{f}(v)| < \infty \right\}.$$

Since

$$M_0(f) \le C_\rho \|f\|_{X_\rho}, \qquad M_A(f) \le \|A\|_{1\to 1} C_\rho \|f\|_{X_\rho}, \qquad C_\rho = \sup_{n\ge 1} n e^{-\rho n} < \infty,$$

the embedding $X_\rho \hookrightarrow X_A$ is continuous and finite trigonometric polynomials are dense in X_ρ. Therefore $\{\Gamma_A > 0\}$ is open and dense in every fixed analytic Wiener strip as well.

There is also a natural Sobolev version. For $s > 2$, Cauchy–Schwarz gives

$$M_0(f) \le \|f\|_{H^s} \left(\sum_{v\in\mathbb{Z}^2} \|v\|_1^2 \langle v\rangle^{-2s} \right)^{1/2}, \qquad M_A(f) \le \|A\|_{1\to 1} M_0(f),$$

and the lattice sum converges because its radial shell is $O(n^{3-2s})$. Hence $H^s(\mathbb{T}^2)$ embeds continuously in X_A and trigonometric polynomials are dense there.

Corollary 58.3 (Sobolev and smooth genericity)**.** *For every $s > 2$, modulo constants,*

$$\{g \in H^s(\mathbb{T}^2;\mathbb{R}) : \Gamma_A(g) > 0\}$$

is open and dense; the zero-variance set is closed and nowhere dense. The same conclusion holds in $C^\infty(\mathbb{T}^2)$ with its Fréchet topology.

Thus the known smooth exact-cancellation examples are topologically exceptional even though they remain valid. The separated-support example is compactly supported Gevrey-s with $s > 1$, not analytic, while the nontrivial entire temporal all-pass gauges of one harmonic are proved elsewhere to lie in no finite Gevrey class. An all-members theorem for arbitrary nonconstant analytic observables, or an analytic exact-cancellation counterexample, remains open.

59 Analytic rigidity of zero resonance mechanisms

The genericity theorem leaves open the stronger question whether every nonconstant real-analytic observable has positive resonance variance. Two rigidity results substantially narrow any possible counterexample. First, pointwise vanishing of the Poisson bracket is impossible in the analytic category. Second, if the variance nevertheless vanishes, the cohomological transfer inherits analyticity from the bracket.

Theorem 59.1 (Analytic pointwise-bracket rigidity)**.** *Let $S \in \mathrm{SL}(2,\mathbb{Z})$ be hyperbolic and let $Cg = g \circ S$. If a real-analytic $g : \mathbb{T}^2 \to \mathbb{R}$ satisfies*

$$\{g, Cg\} \equiv 0,$$

then g is constant.

Proof. Suppose g is nonconstant and let $R = \{x : dg_x \neq 0\}$. At least one first derivative of g is a nonzero real-analytic function, so its zero set has measure zero; hence R is open and has full measure. On R define the level-set tangent line

$$L(x) = \ker dg_x \in \mathbb{RP}^1.$$

Whenever $x, Sx \in R$, the bracket identity says that the two nonzero covectors dg_x and $dg_{Sx} \circ S$ are linearly dependent. Thus

$$L(Sx) = SL(x). \tag{59.1}$$

The invariant full-measure set $R_\infty = \bigcap_{n\in\mathbb{Z}} S^{-n}R$ contains almost every recurrent point by Poincaré recurrence. If $S^{n_j}x \to x$ for such a point, then continuity gives $L(S^{n_j}x) \to L(x)$, while (59.1) gives $L(S^{n_j}x) = S^{n_j}L(x)$.

Let E^s, E^u be the stable and unstable eigenlines. The projective action of a hyperbolic matrix has precisely these fixed points, and every line other than E^s converges under positive iteration to E^u. Recurrence therefore forces $L(x) \in \{E^s, E^u\}$ for almost every regular x. By continuity this holds everywhere on R. On each connected component of R the line field is constant, because its target is the discrete two-point set $\{E^s, E^u\}$. On one nonempty component U there is an eigenvector e with

$$D_e g = 0 \quad \text{on } U.$$

The left side is real analytic, hence vanishes on all of the connected torus. An eigenline of a hyperbolic integral determinant-one matrix has irrational slope: a rational eigenline would give a rational algebraic-integer eigenvalue and therefore an integer unit ± 1. The linear flow in the direction e is consequently dense. Since g is constant along that flow, continuity makes g constant on the torus, a contradiction. □

We now make the remaining cohomological alternative analytic as well. Put $A = S^{\mathsf{T}}$ for the frequency action and $B = \mathrm{sign}(\operatorname{tr} A)A$. Choose real eigen-covectors ξ, η of B with factors $\lambda > 1$ and λ^{-1} and define

$$N_A(k) = |\xi(k)| + |\eta(k)|.$$

There are constants $0 < c_A \le C_A$ such that $c_A\|k\|_1 \le N_A(k) \le C_A\|k\|_1$. The original matrix A, including its sign, continues to define frequency orbits. With these covectors fixed, put

$$m_A = \inf_{k\in\mathbb{Z}^2\setminus\{0\}} N_A(k) \ge c_A > 0, \qquad d = \frac{\lambda + \lambda^{-1}}{2} - 1 = \frac{|\operatorname{tr} A|}{2} - 1 > 0.$$

Theorem 59.2 (Analytic orbit-sum coboundary regularity)**.** *Let h have zero mean and satisfy*

$$|\widehat{h}(k)| \le M e^{-\rho N_A(k)} \qquad (k \neq 0).$$

If every oriented A-orbit has zero coefficient sum, then there is an analytic transfer u, unique among analytic transfers modulo constants, such that

$$h = u - Cu,$$

and

$$|\widehat{u}(k)| \le M' e^{-\rho N_A(k)}, \qquad M' = \frac{M}{1 - e^{-\rho d m_A}}. \tag{59.2}$$

Thus the full adapted Fourier exponent is preserved. If h is real-valued, u may be chosen real-valued.

Proof. Fix an orbit $k_j = A^j k_0$ and write $X_j = |\xi(k_j)|$, $Y_j = |\eta(k_j)|$. Then $X_j = \lambda^j X_0$ and $Y_j = \lambda^{-j} Y_0$. The sign of A disappears only inside these absolute values. Both coordinates are positive because the eigenlines contain no nonzero integer vector. Choose the unique switching index j_* with $X_{j_*} \geq Y_{j_*}$ and $X_{j_*-1} < Y_{j_*-1}$. For $j \geq j_*$ and $n \geq 0$,

$$\frac{N_A(k_{j+n})}{N_A(k_j)} = \frac{\lambda^n X_j + \lambda^{-n} Y_j}{X_j + Y_j} \geq \frac{\lambda^n + \lambda^{-n}}{2} \geq 1 + nd.$$

Indeed, the difference in the first inequality is $(X_j - Y_j)(\lambda^n - \lambda^{-n})/[2(X_j + Y_j)] \geq 0$. For $a_n = (\lambda^n + \lambda^{-n})/2$, the identity

$$a_{n+2} - 2a_{n+1} + a_n = \tfrac{1}{2}\big(\lambda^n(\lambda - 1)^2 + \lambda^{-n}(\lambda^{-1} - 1)^2\big) \geq 0$$

gives $a_n \geq a_0 + n(a_1 - a_0) = 1 + nd$. Interchanging X_j, Y_j gives, for $j < j_*$,

$$N_A(k_{j-n}) \geq (1 + nd) N_A(k_j).$$

The zero orbit sum allows the same transfer coefficient to be written as

$$u_j = \sum_{\ell \leq j} h_\ell = -\sum_{\ell > j} h_\ell.$$

Use the forward tail on the expanding side and the backward tail on the contracting side. Either tail is bounded by

$$M \sum_{n \geq 0} e^{-\rho(1+nd)N_A(k_j)} \leq \frac{M e^{-\rho N_A(k_j)}}{1 - e^{-\rho d m_A}}.$$

This proves (59.2) on every orbit. The recurrence $h_j = u_j - u_{j-1}$ is precisely the Fourier form of $h = u - Cu$. If two decaying transfers existed, their difference would be C-invariant; constant coefficients along every infinite nonzero frequency orbit are incompatible with exponential decay, so only the zero frequency can remain. □

For completeness, the orbit-sum variance identity extends to mean-zero analytic h by absolute summability:

$$\sum_{q \in \mathbb{Z}} |\langle C^q h, h \rangle| \leq \sum_{\mathcal{O}} \left(\sum_{k \in \mathcal{O}} |\widehat{h}(k)| \right)^2 \leq \left(\sum_{k \neq 0} |\widehat{h}(k)| \right)^2 < \infty.$$

There are no periodic nonzero frequency orbits, so each pair in an orbit appears exactly once in the correlation sum. Rearrangement and the normalized squared norm of the Birkhoff sum give $\mathcal{V}_A(h) = \sum_{\mathcal{O}} |\sum_{k \in \mathcal{O}} \widehat{h}(k)|^2$. In particular its vanishing forces each orbit sum to vanish. The analytic bracket $h = \{g, Cg\}/(4\pi)$ has zero mean by integration by parts.

Corollary 59.3 (Reduced all-analytic frontier)**.** *If a nonconstant real-analytic g satisfies $\Gamma_A(g) = 0$, then there is a real-analytic u such that*

$$\frac{\{g, Cg\}}{4\pi} = u - Cu, \qquad \{g, Cg\} \not\equiv 0. \tag{59.3}$$

Thus an analytic zero-variance counterexample, if one exists, must solve the nontrivial analytic nonlinear cohomological equation (59.3)*. The compactly supported Gevrey separated-support construction cannot realize this mechanism.*

In ordinary Fourier coordinates, an estimate $|\widehat{h}(k)| \leq M e^{-r\|k\|_1}$ yields, for example,

$$|\widehat{u}(k)| \leq M' e^{-r c_A \|k\|_1 / C_A}.$$

These results stop short of the all-members analytic variance theorem. They exclude the pointwise-zero mechanism and show that loss of transfer regularity is not the remaining obstruction.

Final consolidation addendum

60 Final consolidation addendum

This addendum records the corrected canonical repository state after the recovery and cleanup pass of 6 September 2026, followed by the research continuation above. It supersedes delivery-state sentences in the preceding audit; it does not silently change the hypotheses of earlier theorems. The historical repository-shape record is `record/final_consolidation_audit_20260906.md`. Claim status is controlled by `record/claim_ledger.md`, with the explicit proof and correction scopes of the maintained continuation notes.

Three previously omitted arithmetic packets were recovered from exact historical Git blobs: the Cayley subgroup-curve route, the generic elliptic product-row reduction, and the specialization-safe projective Bezout rank criterion. The first is distinct from the bilinear curve used in the maintained direct Cayley theorem. In the recovered route the polynomial has bidegree $(2, 2)$. Specializing Makarychev–Vyugin Theorem 2 with $m = n = 2$ and $g = h = 1$ gives the coefficient

$$12mng(m + n) = 192,$$

with the stated subgroup-size hypotheses. This auxiliary route does not replace the sharper maintained bound $E_+(A) < 34K^{8/3}$ from the bilinear specialization [9, 10].

The projective Bezout criterion is not the retracted two-sided use of the older dehomogenized D'/G' invariants. Its matrix nullity equals the homogeneous gcd degree after specialization, giving an exact determinantal support description. Growing-K control of the resulting integral support remains open; no polynomial largest-prime theorem is inferred. The generic elliptic product-row construction reduces the cross-row problem to a genus-one fibre product, but no uniform $O(1)$ intersection theorem is claimed.

The maintained checkout has one publication surface, `manuscript/BGS_unified.tex`. Obsolete standalone article copies and the one-shot scripts that regenerated them are absent from canonical `main`. The active computational stack is Python/C++/Lean/TeX; Julia is retired and the structural audit rejects tracked Julia sources or project environments anywhere in the checkout. Maintained C++ verifiers belong to the CMake graph and the top-level Lean module is the import surface for active `Bgs.*` certificates. Draft Lean sources are not counted as compiled analytic proofs.

Divergent historical research branches are preserved by Git branch refs, commit SHAs, and tree SHAs rather than by copying their source trees under `archive/branch-snapshots`. That directory is metadata-only. The original 204-file consolidation ZIP was recovered from the user's delivery library and verified by SHA-256; its byte-level identity is recorded in `archive/delivery-20260906/README.md`. Historical logs and artifacts remain provenance and are not relabelled as executions of a later commit.

The earlier corrected canonical TeX graph was reconstructed and passed three pdfTeX passes after a boundary-only arithmetic repair. That is historical build evidence, not a fresh execution claim for the continuation inputs and not independent mathematical referee acceptance. Current sources, command exit codes and output hashes must be judged from their own execution record.

The new results distinguish two Fourier variables, retain both oriented orbits, and separate classical covariance from quantum spectral statistics. They include a uniform weighted-Fourier synthesis bound, a full translated resonance-lattice criterion, a two-window phase crossover, sharp polynomial rank rigidity, explicit quantum controllability, a classified smooth temporal

gauge, an explicit weighted-Wiener stability neighbourhood for every nonconstant finite Fourier symbol, open-dense nondegeneracy in Sobolev, smooth and fixed analytic Wiener classes, and an analytic rigidity reduction showing that any all-analytic zero-variance counterexample must have a nonzero analytic bracket which is itself an analytic dynamical coboundary. Their proofs do not establish priority or a complete Lean formalization. The corrected resultant arguments distinguish squarefreeness of a gcd from simplicity in an input and retain ambient degrees when an input degree drops. The former all-frequency all-pass promotion is excluded by an explicit negative control.

GitHub Actions attempts observed during the earlier finalization did not receive a runner, so no cloud-CI success is inferred from those runs. This infrastructure fact is separate from scientific status. In particular no positive microscopic BGS/GOE/COE universality theorem is claimed here: the missing obligation is appropriately desymmetrized Heisenberg-scale joint trace or local-resolvent control, together with the required high-frequency tail. Controllability with variable controls is not that spectral theorem.

Chapter 13 — Shells, Husimi tomography and aperture minimax

61 Holomorphic shell rigidity, normal jets, and an annular obstruction

The finite-rank argument in this section removes the polynomial hypothesis from the earlier shell-identification result. The domain must include the centre of the measured spheres. We also count normal multiplicities and give an explicit obstruction on a punctured domain. These are project derivations; external priority and a proof-assistant formalization are not claimed. They do not assert microscopic spectral universality.

61.1 The analytic dilation determinant

Lemma 61.1 (Confluent dilation independence)**.** *Let $f_1, \dots, f_r$ be linearly independent holomorphic functions on $D_R = \{z : |z| < R\}$. Let $0 < c_1 < \cdots < c_s$, and let positive integers m_i satisfy $\sum_i m_i = r$. The determinant of the $r \times r$ matrix*

$$\mathcal{D}(w)_{(i,k),j} = w^k f_j^{(k)}(c_i w), \qquad 0 \le k < m_i,$$

is not identically zero on $|w| < R/c_s$.

Proof. The finite-dimensional space spanned by the f_j admits a basis with strictly increasing orders of vanishing at zero. This follows by successively eliminating the first nonzero Taylor coefficient; the process cannot leave a nonzero function with every Taylor coefficient zero. A constant invertible change of basis therefore gives

$$f_j(w) = a_j w^{n_j} + O(w^{n_j+1}), \qquad 0 \le n_1 < \cdots < n_r, \quad a_j \neq 0.$$

Writing $(n)_k = n(n-1)\cdots(n-k+1)$, with $(n)_k = 0$ for $k > n$, yields

$$\det \mathcal{D}(w) = \left(\prod_j a_j \right) w^{\sum_j n_j} \det\left[(n_j)_k c_i^{n_j - k} \right]_{(i,k),j} + O\big(w^{1+\sum_j n_j}\big).$$

The constant determinant is nonzero. Otherwise a nonzero polynomial with at most r monomials would have positive zeros c_i of total multiplicity at least r. Such a polynomial has at most

$r-1$ positive zeros counted with multiplicity: divide by its lowest power, differentiate, and apply Rolle's theorem inductively. For complex coefficients take a nonzero real part after a common scalar rotation; all the specified zeros and derivatives persist. Thus the displayed leading coefficient is nonzero. The constant change of basis only multiplies the determinant by a nonzero constant. □

61.2 A rank bound without a degree bound

A holomorphic kernel on $D_{R_1} \times D_{R_2}$ has separated rank r if r is the least number of terms in an expression

$$K(z,w) = \sum_{j=1}^{r} a_j(z) b_j(w).$$

In a minimal expression both lists of functions are linearly independent. For $0 < c < R_1 R_2$, say that K has multiplicity at least m on $zw = c$ if

$$\partial_w^k K(z, c/z) = 0 \quad (0 \le k < m)$$

on the annulus $c/R_2 < |z| < R_1$.

Theorem 61.2 (Finite-rank contact bound)**.** *A nonzero holomorphic kernel of finite separated rank r has total multiplicity at most $r-1$ on distinct positive product curves $zw = c_i$.*

Proof. If the total were at least r, reduce some multiplicities to make their sum exactly r. On the common nonempty annulus $c_s/R_2 < |z| < R_1$, the column $(a_j(z))_j$ is annihilated by the matrix with rows $(b_j^{(k)}(c_i/z))_j$. Multiplying row (i,k) by z^{-k} gives the matrix in Lemma 61.1 with $w = 1/z$. Its determinant is a nonzero holomorphic function on $|w| < R_2/c_s$. It cannot vanish throughout the open subannulus $1/R_1 < |w| < R_2/c_s$. On a nonempty open set the matrix is therefore invertible, so every a_j vanishes there. The identity theorem contradicts minimality of the separated representation. □

For a real-analytic diagonal $H(z) = K(z, \overline{z})$, normal vanishing to order m on $|z| = \rho > 0$ means

$$\partial_t^k H(te^{i\theta})\Big|_{t=\rho} = 0 \quad \text{for every } \theta \text{ and } 0 \le k < m.$$

It is equivalent to the contact condition above with $c = \rho^2$. To verify the direction we need, expand $K(z, s/z) = \sum_{j \ge 0} A_j(z)(s-c)^j$ locally near $s = c$. The zeroth normal condition and the identity theorem give $A_0 = 0$. Inductively, once $A_0, \dots, A_{j-1}$ vanish, the jth normal derivative is $j!(2\rho)^j A_j(\rho e^{i\theta})$. The same identity theorem gives $A_j = 0$. These identities continue to the full product-curve annulus. The reverse direction follows by the same expansion. Consequently the sum of positive zero-shell multiplicities of a nonzero rank-r Hermitian analytic kernel is at most $r-1$.

61.3 Intensity identification in any complex dimension

Theorem 61.3 (Shell-and-jet identification)**.** *Let $P : B_R \subset \mathbb{C}^m \to \mathbb{C}^d$ and $Q : B_R \to \mathbb{C}^e$ be holomorphic, with finite d, e. Let distinct radii $0 < \rho_i < R$ and positive integers m_i satisfy $\sum_i m_i \ge d + e$. Assume that the normal derivatives of $\|P(tu)\|^2 - \|Q(tu)\|^2$ of orders $0, \dots, m_i - 1$ vanish at $t = \rho_i$ for every unit vector u. Then*

$$\langle P(z), P(w) \rangle = \langle Q(z), Q(w) \rangle \qquad (z, w \in B_R).$$

There is a fixed linear isometry from the span of $P(B_R)$ onto the span of $Q(B_R)$ taking $P(z)$ to $Q(z)$. In particular, if $d = e$, it extends to $U \in U(d)$ and $Q = UP$ throughout the ball.

Proof. Restrict to a complex line $z = \zeta u$ and polarize the norm difference in $\zeta, \overline{\zeta}$. This is a holomorphic kernel of separated rank at most $d + e$. Theorem 61.2, with the normal-jet equivalence, forces the restricted diagonal to vanish identically. Every point of the ball lies on such a line, so the norm difference is zero everywhere. Expanding at zero, independence of the real-analytic monomials $z^{\alpha}\overline{z}^{\beta}$ makes all polarized coefficients zero. The polarized identity extends to the ball. It makes the map $\sum_j t_j P(z_j) \mapsto \sum_j t_j Q(z_j)$ well-defined and norm preserving; extending orthonormal bases proves the last assertion. □

In particular, $2d$ distinct spheres suffice for two d-output holomorphic maps, with no polynomial, rationality, or finite-Taylor-support hypothesis. Alternatively one sphere and its normal derivatives of orders 0 through $2d - 1$ suffice. Mixed allocations of the same total multiplicity are allowed. The actual separated rank of the norm-difference kernel can replace $d + e$. All measurements here are functions on whole spheres, not finite angular sample sets.

Proposition 61.4 (Sharpness, including repeated shells)**.** *In complex dimension one the count* $2d$ *cannot be decreased, even for polynomials, and even if the* $2d - 1$ *conditions are prescribed as arbitrary positive shell multiplicities.*

Proof. Let $\sum_i m_i = 2d - 1$ and write

$$B(x) = \prod_i (x - \rho_i^2)^{m_i} = \sum_{j=0}^{2d-1} \beta_j x^j.$$

Its coefficients are nonzero and alternate in sign, with exactly d of each sign. Assign the components $\sqrt{\beta_j} z^j$ with $\beta_j > 0$ to P, and $\sqrt{-\beta_j} z^j$ with $\beta_j < 0$ to Q. Then $\|P(z)\|^2 - \|Q(z)\|^2 = B(|z|^2)$ has exactly the prescribed normal multiplicities, but is not identically zero. No constant unitary relates the maps. Multiplying both maps by e^z gives nonpolynomial entire sharp examples. □

The same identification holds for rational maps with poles off the measured spheres: multiply both maps by a common polynomial denominator nonzero there. It also holds for meromorphic maps whenever such a holomorphic common denominator is available. Multiplication by its squared modulus preserves normal zero multiplicities. A finite-order pole at the centre can therefore be cleared; an essential singularity cannot in general be ignored.

61.4 Why the centre hypothesis matters

The following explicit scalar construction has equal intensity on infinitely many concentric circles without a constant phase relation. It is holomorphic on $\mathbb{C}^*$, not at the centre. For $q = 1/4$, define the normally convergent product

$$\Theta(z; q) = \prod_{j=0}^{\infty} (1 - q^j z)(1 - q^{j+1}/z), \qquad z \neq 0.$$

Reindexing the product gives $\Theta(qz; q) = -z^{-1}\Theta(z; q)$. Its zeros are exactly $q^{\mathbb{Z}}$. Take $a = 2, b = 3, c = 5, d = 6/5$, so $ab = cd$, and set

$$P(z) = \Theta(az; q)\Theta(bz; q)\Theta(c/z; q)\Theta(d/z; q),$$
$$Q(z) = P(1/z).$$

Both functions have real Laurent coefficients. The functional equation gives

$$P(qz) = q^{-2}z^{-4}P(z), \qquad Q(qz) = q^{-2}z^{-4}Q(z).$$

Thus $R = P/Q$ is a meromorphic q-periodic function and $R(1/z) = 1/R(z)$. On $|z| = q^{n/2}$, $\overline{z} = q^n/z$, so

$$\overline{R(z)} = R(\overline{z}) = R(q^n/z) = 1/R(z).$$

Away from zeros and poles this proves $|R| = 1$. Continuity of $|P|^2 - |Q|^2$ extends the equality to the whole circle. The ratio is not constant: at $z = 1/3$, P is zero whereas the four arguments defining Q are $5/3, 2/5, 6, 9$, none in $q^{\mathbb{Z}}$. Hence $Q(1/3) \neq 0$. This supplies infinitely many equal-intensity shells for scalar functions on a punctured domain. It does not contradict Theorem 61.3.

61.5 Relation to earlier work and verification scope

The scalar two-circle uniqueness case is already contained in Perez's Theorem 2.2 [20], which treats the Nevanlinna class and boundary radial values; holomorphic functions on a larger disk are a special case. We do not claim that scalar statement as new. Al Helal, Lebl and Nandi [21], Theorems 1.4 and 3.5–3.15, study rational sphere maps and prove degree-dependent classification results. Their norm-to-unitary step is established prior mathematics, also proved above. The extension relative to this repository is the confluent finite-rank analytic contact argument and its vector-valued shell-and-jet consequences, not the unitary Gram argument, the generalized Vandermonde principle, or the theta functional equation. A literature search and a verified citation do not establish priority.

The accompanying verifier checks confluent determinants in two exact rational implementations, leading Taylor jets of genuinely nonpolynomial functions, prescribed sharp examples and negative controls. Theta-product evaluation is separately labelled high-precision numerical evidence. None of those finite checks replaces the analytic proof. The following section gives a separate quantitative instability construction, rather than inferring a stability estimate from exact uniqueness.

62 Exact identification does not imply a power-law stability bound

The scalar two-circle uniqueness theorem is prior work of Perez [20]. We now distinguish that exact theorem, and the preceding finite-rank extension, from quantitative recovery with errors. All norms below are on a fixed disk; the measurement circles and the target point are fixed.

Theorem 62.1 (A two-circle instability sequence)**.** *For every integer $M \geq 1$ there are scalar polynomials p_M, q_M of degree at most $8M + 2$ such that*

$$\tfrac{1}{2} \leq \|p_M\|_{H^\infty(D_2)} < 1, \qquad \tfrac{1}{2} \leq \|q_M\|_{H^\infty(D_2)} < 1,$$

and, for $H_M = |p_M|^2 - |q_M|^2$,

$$H_M(z) = 0 \qquad (|z| = 1), \tag{62.1}$$

$$\sup_{|z|=1/2} |H_M(z)| \leq 2^{-4M^2-18M-3}, \tag{62.2}$$

$$|H_M(3/4)| \geq \frac{1}{40}(3/8)^{8M+2}. \tag{62.3}$$

In particular there are no constants $C, \alpha > 0$ such that

$$\big|\, |p(3/4)|^2 - |q(3/4)|^2 \,\big| \leq C \left(\max_{\rho \in \{1/2,1\}} \sup_{\theta} \big|\, |p(\rho e^{i\theta})|^2 - |q(\rho e^{i\theta})|^2 \,\big| \right)^{\alpha}$$

for all scalar holomorphic p, q bounded by one on D_2. This fails even when both functions are polynomials and each has disk supremum at least $1/2$.

Proof. Put $b = 1/4$, $L = 4M + 1$, and define the Laurent polynomials

$$A_M(z) = \sum_{n=-M}^{M} b^{2n^2+n} z^{4n+1}, \qquad B_M(z) = A_M(1/z),$$
$$p_M(z) = \frac{z^L A_M(z)}{4\,2^L}, \qquad q_M(z) = \frac{z^L B_M(z)}{4\,2^L}.$$

The latter two are ordinary polynomials of degree at most $8M + 2$. All coefficients are positive. At $|z| = 2$ the Laurent sums are bounded by

$$A_\infty(2) = 2\sum_{n\in\mathbb{Z}} 2^{-4n^2+2n} \le 2\left(1 + 2\sum_{n\ge 1} 2^{-2n}\right) = \frac{10}{3} < 4.$$

Changing n to $-n-1$ gives $A_\infty(1/2) = A_\infty(2)$, so this bound also applies to $B_M(2)$. The maximum principle proves the disk upper bound. The $n = 0$ term of $A_M(2)$ is 2 and the $n = -1$ term of $A_M(1/2)$ is 2 for $M \ge 1$, proving the lower bound for both disk suprema.

Reindexing the finite sum leaves exactly two boundary terms:

$$D_M(z) := A_M(bz) - b^{-2}z^{-4}A_M(z) = b^{2M^2+5M+1}z^{4M+1} - b^{2M^2-M-2}z^{-4M-3}. \tag{62.4}$$

Consequently, on $|z| = 2$,

$$|D_M(z)| \le 2^{-4M^2-6M-1} + 2^{-4M^2-2M+1} \le 2^{-4M^2-2M+2}.$$

On $|z| = 1$, real coefficients give $B_M(z) = \overline{A_M(z)}$, which proves exact outer-circle equality. On $|z| = 1/2$ one has $\overline{z} = b/z$, so

$$\overline{A_M(z)} = b^{-2}z^4 B_M(z) + D_M(1/z),$$
$$\overline{B_M(z)} = b^{-2}z^4\{A_M(z) - D_M(z/b)\}.$$

Multiplication and subtraction eliminate the common term. As $|b^{-2}z^4| = 1$, $|A_M(z)|, |B_M(z)| < 4$, and both $1/z$ and z/b have modulus two, this bounds the unnormalized intensity difference by 2^{-4M^2-2M+5}. The squared normalization factor on this circle is 2^{-4L-4}, proving the stated inner-circle estimate. It is uniform in the angle; no angular sampling is involved.

At $x = 3/4$, positivity of coefficients and three terms of B_M give

$$B_M(x) \ge \frac{4}{3} + \frac{27}{256} + \frac{16}{243} > \frac{3}{2}.$$

For a checkable infinite-sum upper bound on $A_M(x)$ put $T_n = b^{2n^2+n}x^{4n+1}$. Consecutive ratios bound both omitted tails and give

$$A_M(x) \le A_\infty(x) \le \sum_{n=-2}^{1} T_n + \frac{T_2}{1-b^{11}x^4} + \frac{T_{-3}}{1-b^{13}x^{-4}} = \frac{86380343555303460998075}{64074215137581349797888} < \frac{27}{20}.$$

Thus $B_M(x)^2 - A_M(x)^2 > 2/5$. Restoring the squared normalization proves $|H_M(x)| \ge (3/8)^{2L}/40$. The logarithm of the ratio of this lower bound to any fixed positive power of the measurement upper bound tends to positive infinity quadratically in M, proving the absence of a uniform Hölder bound. □

The theorem concerns absolute intensity errors, not phase error at a fixed nonvanishing reference function. Its examples do not keep the measured intensities bounded below. Additional nonvanishing or finite-dimensional conditioning assumptions are not ruled out as routes to better stability.

Corollary 62.2 (A lower bound on any uniform modulus)**.** *Let $\Omega(\delta)$ be the supremum of the target intensity error at $3/4$ over pairs bounded by one on D_2 with measurement error at most δ on the two circles. Then $\Omega(\delta) \to 0$ as $\delta \downarrow 0$, but, for some positive constants c, C and all sufficiently small δ,*

$$\Omega(\delta) \geq c \exp\{-C\sqrt{\log(1/\delta)}\}.$$

No matching upper rate is asserted.

Proof. If the limit were not zero, Montel's theorem would provide locally uniformly convergent subsequences of a violating sequence of bounded pairs. The limits have equal intensity on both entire circles, hence are constant-phase multiples by Perez's theorem or Theorem 61.3. Their target intensity errors therefore tend to zero, a contradiction. For the lower bound take $M = \lceil \frac{1}{2}\sqrt{\log_2(1/\delta)} \rceil$ in the theorem. Its measurement upper bound is at most $2^{-4M^2} \leq \delta$, while its target lower bound has logarithm bounded below by a negative constant times $1 + \sqrt{\log(1/\delta)}$. □

Severe ill-posedness of infinite-dimensional phase retrieval is not a new phenomenon: Alaifari and Grohs [22] prove quadratic-exponential instability in a Gabor setting. Our construction concerns a different, explicit observation geometry, namely two complete circles inside a fixed disk and an interior target. The elementary recurrence (62.4) and exact rational bounds are the input here, not a claim to have discovered general phase-retrieval instability. The accompanying finite verifier re-expands the full angular intensity polynomial and checks its coefficient ℓ^1 bound independently of this recurrence. Neither the inverse-problem result nor its numerical illustration is a BGS spectral-statistics theorem.

63 Finite-rank bosonic tomography from complete Husimi shells

The preceding analytic theorem has a direct quantum-information consequence. The finite parameter is the rank of the density operator, not a photon-number cutoff or the dimension of the Hilbert space. Exact injectivity, finite-copy estimation, and stability in trace distance must nevertheless be separated.

63.1 The infinite-dimensional measurement model

For s bosonic modes let $\mathcal{H} = \ell^2(\mathbb{N}^s)$, with occupation basis $|\nu\rangle$, and define the normalized coherent vector

$$|z\rangle = e^{-\|z\|^2/2} \sum_{\nu \in \mathbb{N}^s} \frac{z^\nu}{\sqrt{\nu!}} |\nu\rangle, \qquad z \in \mathbb{C}^s.$$

For a positive trace-one operator ρ, use the Husimi convention

$$Q_\rho(z) = \pi^{-s} \langle z|\rho|z\rangle.$$

These coherent-state probabilities and their role in tomography are standard; for a quantitative continuous-variable reconstruction framework see Landon-Cardinal, Govia and Clerk [24]. We do not infer a sampling or numerical reconstruction algorithm from exact identifiability.

For a vector $\psi \in \mathcal{H}$, the function

$$f_\psi(z) = \sum_\nu \overline{\psi_\nu} \frac{z^\nu}{\sqrt{\nu!}}$$

is entire, since Cauchy–Schwarz bounds its modulus by $\|\psi\| e^{\|z\|^2/2}$, uniformly on compact sets. If $\rho = \sum_{j=1}^a \lambda_j |\psi_j\rangle\langle\psi_j|$ has finite rank,

$$\pi^s e^{\|z\|^2} Q_\rho(z) = \sum_{j=1}^a \lambda_j |f_{\psi_j}(z)|^2. \tag{H.1}$$

No finite mean energy is required for this identity or for the next theorem.

Theorem 63.1 (Rank-bounded shell tomography)**.** *Let ρ, σ be bosonic density operators of ranks $a, b < \infty$. If their Husimi functions have equal radial derivatives of orders $0, \dots, m_i - 1$ on each of the distinct complete spheres $\|z\| = \rho_i > 0$, and $\sum_i m_i \geq a + b$, then $\rho = \sigma$. In particular, any $2r$ distinct positive-radius spheres distinguish all states of rank at most r from one another, without a photon-number cutoff.*

Proof. Multiplication by $\pi^s e^{\|z\|^2}$ preserves zero normal multiplicities. Apply Theorem 61.3 to the maps with components $\sqrt{\lambda_j} f_{\psi_j}$ and the corresponding components for σ. The two polarized kernels coincide. The coefficient of $z^\nu \overline{w}^\mu / \sqrt{\nu!\mu!}$ is the corresponding density matrix element, so equality of the entire kernels gives equality of all matrix elements in the occupation basis. □

The conclusion is rank-r completeness, in the terminology distinguishing it from rank-r strict completeness [23]. The latter would distinguish a rank-r state from every state of arbitrary rank; it is not implied here. Each shell supplies a full angular function. The theorem does not replace that continuum by $2r$ scalar observations or claim finite sample complexity. Indeed no finite collection of coherent-state projectors distinguishes all pure states in $\mathcal{H}$: its vector span has an infinite-dimensional orthogonal complement, containing two orthogonal unit vectors with all those measurement probabilities zero.

63.2 Normalized sharp examples and the unknown-rank caveat

Let $m \geq 1$ and define two diagonal, positive, normalized states

$$\rho_m^{\mathrm{e}} = 2^{1-m} \sum_{\substack{0 \leq j \leq m \\ j \text{ even}}} \binom{m}{j} |j\rangle\langle j|, \quad \rho_m^{\mathrm{o}} = 2^{1-m} \sum_{\substack{0 \leq j \leq m \\ j \text{ odd}}} \binom{m}{j} |j\rangle\langle j|.$$

The even and odd binomial sums both equal 2^{m-1}. Their supports are orthogonal, and their trace distance is therefore one. Direct evaluation gives

$$Q_{\rho_m^{\mathrm{e}}}(z) - Q_{\rho_m^{\mathrm{o}}}(z) = \frac{2^{1-m}}{\pi} e^{-|z|^2} L_m(|z|^2), \qquad L_m(x) = \sum_{j=0}^{m} (-1)^j \binom{m}{j} \frac{x^j}{j!}. \tag{H.2}$$

The polynomial L_m has m distinct positive roots. A self-contained argument uses Rodrigues' identity $L_m(x) = e^x (d/dx)^m (e^{-x} x^m)/m!$. Integration by parts makes it orthogonal, with weight e^{-x} on $(0, \infty)$, to every polynomial of degree below m. If it had fewer than m positive sign changes, multiplying by the product of its sign-changing linear factors would give a nonzero integral of one sign, contradicting this orthogonality. Its degree is m, proving the assertion.

Taking $m = 2r - 1$ in (H.2) gives two rank-r states indistinguishable on $2r - 1$ distinct complete circles, despite trace distance one. Thus $2r$ is a sharp universal guarantee over radius choices. This example does not claim failure for every prescribed configuration of $2r - 1$ radii. Taking $m = 2r$ instead gives ranks $r + 1$ and r with identical data on $2r$ circles. Consequently the universal rank-r completeness guarantee cannot be relabelled strict completeness when the competitor's rank is unknown.

63.3 Trace-distance stability needs more than exact shell data

For states define $D(\rho, \sigma) = \frac{1}{2}\|\rho - \sigma\|_1$. Without an energy bound no uniform trace-distance modulus exists, even on pure states and with arbitrarily many fixed bounded measurement circles.

The orthogonal number states $|n\rangle$ and $|n+1\rangle$ have distance one, while on $|z| \le R$ their Husimi difference is bounded by

$$\frac{R^{2n}}{\pi n!}\left(1+\frac{R^2}{n+1}\right) \longrightarrow 0.$$

There is a stronger conditioned example. Fix $E > 0$ and take integers $n > 2E$. Set $u = E/n$ and

$$|\psi_n^{\pm}\rangle = \sqrt{1-u}|0\rangle \pm \sqrt{u}|n\rangle, \qquad \rho_n^{\pm} = |\psi_n^{\pm}\rangle\langle\psi_n^{\pm}|.$$

Both states have rank one, trace one and mean photon number exactly E. Their trace distance and data difference obey

$$D(\rho_n^+, \rho_n^-) = 2\sqrt{u(1-u)}, \tag{63.1}$$

$$\sup_{|z|\le R} |Q_{\rho_n^+}(z) - Q_{\rho_n^-}(z)| \le \frac{4}{\pi}\sqrt{u(1-u)}\frac{R^n}{\sqrt{n!}}. \tag{63.2}$$

The first formula follows by diagonalizing the two-by-two difference; the second follows by expanding the coherent-state amplitudes, whose diagonal terms cancel. For all sufficiently large n, uniformly on the same disk,

$$Q_{\rho_n^{\pm}}(z) \ge \frac{e^{-R^2}}{4\pi}.$$

Indeed the amplitude before its Gaussian factor is at least $\sqrt{1-u} - \sqrt{u}R^n/\sqrt{n!} \ge 1/2$. Thus even a fixed mean-energy bound and a positive lower bound on the measured intensities do not yield a uniform Holder estimate for trace-distance recovery. This is a global state error, not the interior intensity error in Theorem 62.1.

Proposition 63.2 (A compact inverse with a slow necessary modulus)**.** *Fix finitely many complete positive-radius circles, at least $2r$ in number, and let $\Omega_{r,E}(\delta)$ be the supremum of trace distance over pairs of rank-at-most-r states with mean photon number at most E and Husimi data error at most δ on these circles. Then $\Omega_{r,E}(\delta) \to 0$ as $\delta \downarrow 0$. For some $c > 0$ and all sufficiently small δ,*

$$\Omega_{r,E}(\delta) \ge c\sqrt{\frac{\log\log(1/\delta)}{\log(1/\delta)}}. \tag{H.3}$$

No matching upper rate is asserted.

Proof. The energy constraint makes the state set compact in trace norm. For a direct verification, let P_K project onto occupations $0, \dots, K$. Its omitted probability is at most $E/(K+1)$. By decomposing a state into unit vectors, using the two-by-two rank-one calculation and Cauchy–Schwarz, one obtains $\|\rho - P_K\rho P_K\|_1 \le 2\sqrt{E/(K+1)}$. The finite-dimensional compressions are totally bounded. Positivity, trace one, rank at most r, and the lower-semicontinuous energy bound are closed conditions, proving compactness. The Husimi map is continuous into the sup norm because $|Q_\rho(z) - Q_\sigma(z)| \le \pi^{-1}\|\rho - \sigma\|_1$. It is injective on this set by Theorem 63.1, so its inverse on its image is uniformly continuous.

For the lower bound put $L = \log(1/\delta)$ and $n = \lceil 4L/\log L \rceil$. The elementary estimate $\log(n!) \ge n\log n - n + 1$ shows that the right side of (63.2) is at most e^{-L} for large L and fixed R, E. Equation (63.1) is at least $\sqrt{2E/n} \ge c\sqrt{\log L/L}$. These rank-one pairs lie in every rank-at-most-r class, completing the proof. □

All assertions concern ideal complete angular data; no finite-copy confidence statement follows. The pure-state two-circle case follows from prior scalar phase retrieval [20]; bounded-rank and strictly-complete tomography are established subjects [23]. The contribution relative to this project is the no-cutoff finite-rank shell implication, its normalized Laguerre ambiguities, and the explicit conditioned instability comparison. Priority for this precise combination has not been established. These results do not determine the eigenphase statistics of a chaotic propagator.

64 Aperture-dependent minimax tomography with coherent-state tests

This section gives matching deterministic and statistical rates for a specified measurement model. A copy is measured with the binary POVM $\{\Pi_z, I - \Pi_z\}$, where $\Pi_z = |z\rangle\langle z|$ is a normalized multimode coherent-state projector. Classical adaptation of z is allowed; collective measurements, squeezing, and other measurement effects are not. This is not the continuous-outcome heterodyne POVM. Unlike the earlier shell injectivity theorem, the upper bounds below use points throughout a ball and do not assume bounded state rank. A photon cutoff is used in the estimator, with a proved bias bound, not imposed on the unknown state.

64.1 State class, data metric, and main statements

Fix a number of modes $s \geq 1$, a moment exponent $\beta > 0$, and $E > 0$. Let $\mathcal{N} = \sum_{j=1}^{s} a_j^* a_j$ be the total photon-number operator and set

$$\mathcal{S}_{s,\beta,E} = \{\rho \geq 0 : \operatorname{Tr}\rho = 1, \operatorname{Tr}(\rho \mathcal{N}^\beta) \leq E\}.$$

Thus E bounds the moment itself; at $\beta = 1$ it is the mean photon number. Write $p_\rho(z) = \langle z|\rho|z\rangle = \pi^s Q_\rho(z)$ and $D(\rho,\sigma) = \frac{1}{2}\|\rho - \sigma\|_1$. For $0 < R \leq \infty$ define

$$d_R(\rho,\sigma) = \sup_{\|z\|\leq R} |p_\rho(z) - p_\sigma(z)|, \quad \Omega_E(R,\delta) = \sup_{\substack{\rho,\sigma\in\mathcal{S}_{s,\beta,E} \\ d_R(\rho,\sigma)\leq\delta}} D(\rho,\sigma).$$

An infinite radius means all of $\mathbb{C}^s$. Use the conventions $\log_+ x = \max(0, \log x)$ and $\log_+(x/\infty) = 0$, and put

$$G_R(T) = 1 + \log_+(T/R^2), \qquad K_R(T) = T/G_R(T). \tag{A.1}$$

All asymptotic constants below may depend on the fixed s, β, E, R_0, but are uniform for $R \geq R_0 > 0$, including radii that vary with T. No uniform claim as the number of modes tends to infinity is made.

Theorem 64.1 (Sharp deterministic modulus on a ball)**.** *For $T = \log(1/\delta)$ sufficiently large,*

$$\boxed{c\,K_R(T)^{-\beta/2} \leq \Omega_E(R,\delta) \leq C\,K_R(T)^{-\beta/2}.} \tag{A.2}$$

The lower bound already holds for pairs of pure states of moment exactly E. The upper bound holds for states of arbitrary, possibly infinite rank.

For B copies, let $\mathfrak{R}_B(R)$ be the infimum, over all adaptive binary coherent-test protocols and density-operator estimators, of $\sup_{\rho\in\mathcal{S}_{s,\beta,E}} \mathbb{E}_\rho D(\widehat{\rho},\rho)$. At each round the chosen amplitude has norm at most R and a fresh copy is measured. The protocol may depend on B, s, β, E, R.

Theorem 64.2 (Sharp minimax rate, including classical adaptation)**.** *For all sufficiently large integers B,*

$$\boxed{c\,K_R(\log B)^{-\beta/2} \leq \mathfrak{R}_B(R) \leq C\,K_R(\log B)^{-\beta/2}.} \tag{A.3}$$

A nonadaptive tensor-Chebyshev design attains the upper order. The lower bound holds even if the unknown state is promised pure, and even against randomized adaptive choices of coherent amplitudes.

In particular, at fixed finite R and $\beta = 1$ the rate is $\sqrt{\log\log B/\log B}$; with arbitrary amplitudes it is $(\log B)^{-1/2}$. The transition is described by the whole factor G_R, not just these endpoints. For instance $R^2 = \log B/(\log\log B)^a$, with fixed $a > 0$, gives order $\sqrt{\log\log\log B/\log B}$ at $\beta = 1$. A bounded-rank promise with any rank bound at least one does not improve these minimax orders, when the estimate may be an arbitrary density operator. These are rates up to fixed multiplicative constants, not sharp leading constants or claims about optimal runtime.

64.2 A finite interpolation estimate

Let P_K project onto total occupations at most K, and let $d_K = \binom{K+s}{s}$. For $0 < S \le R^2$ put $a = \sqrt{S/(2s)}$. For an integer $L \ge \max(2s, 16S)$ use the $(L+1)^{2s}$ nodes whose real coordinates are

$$x_j = a \cos \frac{(j+1/2)\pi}{L+1}, \qquad 0 \le j \le L. \tag{A.4}$$

Each complex node lies in the ball of squared radius S. Define

$$\begin{aligned} A_K(S) &= \max\{1, K!(2s/S)^K\}, \\ C_{s,L} &= 4^s (L+1)^{2s} 3^L, \\ \tau_{s,L}(S) &= 4\sqrt{\binom{L+2s}{2s-1}} \frac{(2S)^{(L+1)/2}}{\sqrt{(L+1)!}}. \end{aligned} \tag{A.5}$$

Lemma 64.3 (Low coefficients from a finite probability grid)**.** *Suppose $L \ge 2K$ and two states obey $|p_\rho - p_\sigma| \le \eta$ at all nodes in* (A.4)*. Then, for all occupation multiindices $|\mu|, |\nu| \le K$,*

$$|(\rho - \sigma)_{\mu\nu}| \le C_{s,L} A_K(S) \{e^S \eta + \tau_{s,L}(S)\}. \tag{A.6}$$

There is also an explicit linear reconstruction from noisy values of a single state, with the same bound on each low matrix entry if the nodal probability errors are at most η.

Proof. For $\Delta = \rho - \sigma$ define

$$F_\Delta(z) = e^{\|z\|^2} p_\Delta(z) = \sum_{\mu,\nu} \frac{\Delta_{\mu\nu} \overline{z}^\mu z^\nu}{\sqrt{\mu!\nu!}}. \tag{A.7}$$

The entries of Δ have modulus at most two. For total degree j, Cauchy–Schwarz and the multinomial identity bound the absolute sum on $\|z\|^2 \le S$ by

$$2\sqrt{\binom{j+2s-1}{2s-1}} \frac{(2S)^{j/2}}{\sqrt{j!}}.$$

The ratio of consecutive summands without the leading two is $\sqrt{2S(j+2s)}/(j+1) \le 1/2$ for $j \ge L$ under the stated restrictions. Consequently the total-degree-L Taylor truncation T_L has uniform error at most $\tau_{s,L}(S)$. The same bound works for a single state.

In $2s$ real coordinates T_L is a polynomial of total degree at most L. Tensor discrete Chebyshev orthogonality at the chosen nodes recovers its Chebyshev coefficients exactly. Keep only indices of total degree at most L; all others are zero for T_L. The nodal error in F_Δ is bounded by $e^S \eta + \tau_{s,L}(S)$. Each recovered coefficient error is at most 4^s times this bound, because every tensor cosine has modulus at most one and each nonconstant one-dimensional coefficient has prefactor two.

The sum of absolute monomial coefficients of the Chebyshev polynomial T_j is at most 3^j, by its recurrence. Replacing each real coordinate by $(z_\ell + \overline{z}_\ell)/2$ or $(z_\ell - \overline{z}_\ell)/(2i)$ does not increase the sum of absolute coefficients of a homogeneous monomial. Scaling by a contributes exactly a^{-d} to coefficients of total degree d. There are at most $(L+1)^{2s}$ retained Chebyshev terms. It follows that the coefficient error at $\overline{z}^\mu z^\nu$ is bounded by $C_{s,L} a^{-|\mu|-|\nu|} (e^S \eta + \tau_{s,L})$.

Finally, $\mu! \le |\mu|!$. The sequence $\sqrt{j!}\, a^{-j}$ is log-convex, so its maximum for $0 \le j \le K$ is attained at an endpoint. Multiplying two such maxima bounds $\sqrt{\mu!\nu!} a^{-|\mu|-|\nu|}$ by $A_K(S)$. The coefficients of T_L below total degree $2K$ are the exact corresponding coefficients of (A.7). This proves the estimate and describes the reconstruction. Hermitian symmetrization cannot enlarge the largest entrywise error relative to the true Hermitian block. □

Lemma 64.4 (State reconstruction and controlled cutoff bias)**.** *If the reconstructed Hermitian block M has entrywise error at most b relative to $P_K\rho P_K$, take its positive part and normalize its trace, using the vacuum state if the positive part is zero. The resulting state $\widehat{\rho}$, supported on P_K, satisfies*

$$D(\widehat{\rho},\rho) \le \tfrac{3}{2}\sqrt{E/(K+1)^{\beta}} + 2d_K^{3/2}b \tag{A.8}$$

whenever $E/(K+1)^{\beta} \le 1/2$. For two states in the moment class,

$$D(\rho,\sigma) \le 2\sqrt{E/(K+1)^{\beta}} + \tfrac{1}{2}d_K^{3/2}\max_{|\mu|,|\nu|\le K}|\Delta_{\mu\nu}|. \tag{A.9}$$

Proof. The omitted probability $q = \mathrm{Tr}((I-P_K)\rho)$ is at most $E/(K+1)^{\beta}$. Decomposing ρ into rank-one states and applying Cauchy–Schwarz gives $\|\rho - P_K\rho P_K\|_1 \le 2\sqrt{q}$. Entrywise error b implies trace-norm block error $h \le d_K^{3/2}b$ by the Hilbert–Schmidt inequality. If $A = M_+$, positivity of the true block implies $\mathrm{Tr}\, M_- \le h$, hence $\|A - P_K\rho P_K\|_1 \le 2h$. If $A \ne 0$, normalizing it changes trace norm by $|1 - \mathrm{Tr}\, A| \le q + 2h$. Thus $D \le \sqrt{q} + q/2 + 2h$, which proves (A.8). If $A = 0$, then $h \ge 1 - q \ge 1/2$, so the displayed bound is at least one and remains valid for the vacuum fallback. Applying the cutoff bound to both states and the block trace-norm bound proves (A.9). □

64.3 Balancing the aperture and the cutoff

Set

$$\Phi_R(K) = K\{1 + \log_+(K/R^2)\}, \quad S = \min(R^2, K), \quad L = 128sK. \tag{A.10}$$

There is a constant c_s such that the noise multiplier in (A.6) obeys

$$C_{s,L}A_K(S)e^S \le C_sK^{2s}\exp(c_s\Phi_R(K)). \tag{A.11}$$

This follows from $K! \le K^K$, $S \le K$, and $\log(K/S) = \log_+(K/R^2)$. For the bias term one has the stronger uniform estimate

$$C_{s,L}A_K(S)\tau_{s,L}(S) \le C_sK^{3s-1/2}2^{-(64s-1)K}. \tag{A.12}$$

Indeed, use $n! \ge (n/e)^n$ in (A.5). Apart from the displayed polynomial factors, the left side is at most

$$(2s)^K\left(\frac{18e}{128s}\right)^{64sK}(S/K)^{(64s-1)K} \le (2s)^{-(64s-1)K} \le 2^{-(64s-1)K},$$

since $e < 3$ and $54/128 < 1/2$. The deliberately large $128s$ is a safe analytic choice, not an optimized numerical parameter.

The elementary inverse-scale estimate is

$$(1-e^{-1})T \le \Phi_R(K_R(T)) \le T. \tag{A.13}$$

For $T/R^2 \le 1$ it is immediate. Otherwise set $g = 1 + \log(T/R^2) \ge 1$. Then $K_R(T)/R^2 \ge 1$ and the middle quantity equals $T(1 - \log g/g)$; the maximum of $\log g/g$ on $g \ge 1$ is e^{-1}. Also $\Phi_R(cK) \le c\Phi_R(K)$ for $0 < c \le 1$, and the reversed inequality holds for $c \ge 1$. Integer rounding changes the resulting large-T scales only by fixed factors, uniformly for $R \ge R_0 > 0$.

To obtain the deterministic upper bound, take $K = \lfloor cK_R(T)\rfloor$ with a sufficiently small fixed $c > 0$. Equations (A.11)–(A.13) make the noise contribution, with $\eta = \delta = e^{-T}$, at most a polynomial in T times $e^{-T/2}$. The bias is exponentially small in K by (A.12). Equation (A.9) gives $CK^{-\beta/2}$, as claimed. Only a finite subset of the ball was used.

64.4 The two-level lower bound, uniformly over amplitudes

For large integers n let $u = E/n^{\beta} \le 1/2$ and use states

$$|\psi_n^{\pm}\rangle = \sqrt{1-u}|0,\dots,0\rangle \pm \sqrt{u}|n,0,\dots,0\rangle, \qquad \rho_n^{\pm} = |\psi_n^{\pm}\rangle\langle\psi_n^{\pm}|. \tag{A.14}$$

Their moment is exactly E and

$$D(\rho_n^{+},\rho_n^{-}) = 2\sqrt{u(1-u)} \ge \sqrt{2E}\, n^{-\beta/2}. \tag{A.15}$$

Their coherent amplitudes are the sum and difference of two terms of moduli

$$a_z = \sqrt{1-u}\, e^{-\|z\|^2/2}, \qquad b_z = \sqrt{u}\, e^{-\|z\|^2/2}\frac{|z_1|^n}{\sqrt{n!}}.$$

Their relative phase is $n \arg z_1$. Maximizing $e^{-x}x^{n/2}$ gives

$$|p_+(z) - p_-(z)| \le 4a_z b_z, \quad a_z b_z \le \sqrt{u}\, 2^{-n/2}, \quad b_z^2 \le u/2. \tag{A.16}$$

Here $e^{-n}n^n/n! \le 1/e < 1/2$ for $n \ge 1$; the sequence is decreasing because $(1+1/n)^n < e$. Other modes only decrease both amplitudes. It follows that

$$d_R(\rho_n^{+},\rho_n^{-}) \le 4\sqrt{u}\min\left\{2^{-n/2}, \frac{R^n}{\sqrt{n!}}\right\}. \tag{A.17}$$

Using $n! \ge (n/e)^n$, there is an absolute $c_0 > 0$ such that the right side is at most $4\exp(-c_0\Phi_R(n))$. To check uniformity, write $t = \log_+(n/R^2)$. The two exponents are $(\log 2)n/2$ and $(t-1)n/2$; their maximum bounds a positive constant times $(1+t)n$. Choose $n = \lceil CK_R(T)\rceil$ with sufficiently large fixed C. Then (A.13) makes (A.17) at most e^{-T} for large T, and (A.15) proves the lower bound in Theorem 64.1.

64.5 Adaptive experiments cannot beat the lower rate

For Bernoulli parameters p_+, p_- define the unnormalized squared Hellinger distance

$$h^2(p_+,p_-) = (\sqrt{p_+} - \sqrt{p_-})^2 + (\sqrt{1-p_+} - \sqrt{1-p_-})^2.$$

For all large n, uniformly in $z \in \mathbb{C}^s$, one has $p_{\pm}(z) \le 1 - u/4$. Indeed (A.16) bounds these probabilities by $1 - u + u/2 + 2\sqrt{u}\,2^{-n/2}$, and the last term is at most $u/4$ for large n, since $u = E/n^{\beta}$. The difference of the square roots of the first probabilities is at most $2\min(a_z, b_z)$. For the complementary probabilities, rationalizing their square-root difference and using $1 - p_{\pm} \ge u/4$ gives an upper bound $64a_z^2b_z^2/u$ for its square. Therefore

$$\begin{aligned} h^2(p_+(z),p_-(z)) &\le 4\min(a_z^2, b_z^2) + 64a_z^2b_z^2/u, \\ \sup_{\|z\|\le R} h^2(p_+(z),p_-(z)) &\le 68\min\left\{2^{-n/2}, \frac{R^{2n}}{n!}\right\} \le 68e^{-c_1\Phi_R(n)} \end{aligned} \tag{A.18}$$

for an absolute $c_1 > 0$. For the first of the two uniform bounds use $\min(a_z^2, b_z^2) \le a_z b_z$; for the second use $\min(a_z^2, b_z^2) \le b_z^2$ and drop Gaussian factors. The last inequality follows by comparing $(\log 2)/2$ and $t - 1$ for $t = \log_+(n/R^2)$.

For completeness, the same estimate controls an adaptive transcript. Conditional on any common history the protocol uses the same distribution of actions under the two hypotheses. Its conditional Hellinger affinity is at least $1 - \eta/2$ when the bound in (A.18) is η. Backward integration through the B rounds shows total affinity at least $(1-\eta/2)^B \ge 1 - B\eta/2$. Thus transcript Hellinger distance squared is at most $B\eta$, and total variation is at most $\sqrt{B\eta}$. This

argument includes randomized actions by conditioning also on the random seed, and does not suppose nonadaptive or independent action choices.

Take $n = \lceil CK_R(\log B)\rceil$ with C large enough that $\eta \le 68B^{-3}$. For large B the transcript total variation is at most $1/2$. Given any state estimate, classify it by the closer of ρ_n^+ and ρ_n^-. A classification error forces trace-distance loss at least half the distance in (A.15). The two-point testing bound therefore gives maximum expected loss at least one eighth of that distance. This proves the adaptive minimax lower bound.

64.6 A nonadaptive estimator attains the rate

Set $T = \log B$ and $K = \lfloor cK_R(T)\rfloor$, with $c > 0$ chosen sufficiently small depending on s. Use (A.10) and repeat every grid setting $b = \lfloor B/(L+1)^{2s}\rfloor$ times. Unused copies may be discarded. Hoeffding's bound and a union bound show that, with probability at least $1-B^{-2}$, every empirical probability has error at most

$$\eta_B = \sqrt{\frac{\log(2(L+1)^{2s}B^2)}{2b}} = \exp(-T/2 + O_s(\log T)). \tag{A.19}$$

This concentration follows directly from the Bernoulli moment-generating function bound $\mathbb{E}e^{t(X-\mathbb{E}X)} \le e^{t^2/8}$ and Markov's inequality; no assumption on the unknown rank is used. By decreasing c, (A.11) amplifies this by at most $e^{T/4}$ up to polynomial factors. Use the linear reconstruction in Lemma 64.3 and the positive-part estimator in Lemma 64.4. On the concentration event its error is at most $CK^{-\beta/2}$, since the analytic bias remains exponentially small by (A.12). On the complement its error is at most one. This proves the expected upper bound, completing Theorem 64.2.

The estimator has a finite classical output matrix of dimension d_K. Its grid has $(128sK+1)^{2s}$ settings, which is polynomial in $\log B$ for fixed s but not dimension-independent. Large interpolation condition numbers require sufficient arithmetic precision. Rounding settings to within ϵ perturbs a probability by at most ϵ, because $\|\Pi_z - \Pi_w\|_{\mathrm{op}} = \sqrt{1-e^{-\|z-w\|^2}} \le \|z-w\|$. Thus logarithmically many precision digits in B suffice for the asymptotic scheme. This does not make the exponential copy requirement practical.

64.7 Copy-complexity and measurement-model consequences

Let $B_*(\varepsilon, R)$ be the least budget with minimax expected error at most ε. For fixed s, β, E, R_0 and $\varepsilon \downarrow 0$,

$$\boxed{\log B_*(\varepsilon, R) \asymp \varepsilon^{-2/\beta}\{1 + \log_+(\varepsilon^{-2/\beta}/R^2)\}, \qquad R \ge R_0.} \tag{A.20}$$

This is equivalent to (A.3) via (A.13); constants are not uniform in E or β. For a mean-energy bound and fixed finite R, the copy complexity is $\exp(\Theta(\varepsilon^{-2}\log(1/\varepsilon)))$; for arbitrary coherent amplitudes it is $\exp(\Theta(\varepsilon^{-2}))$. Classical adaptation does not change either order. Rank one already forces the lower bounds.

This is a restriction imposed by the allowed measurements, not by quantum mechanics alone. With arbitrary binary projectors a polynomial-copy scheme at fixed s, β, E is elementary. Choose $K = O(\varepsilon^{-2/\beta})$ to control the cutoff bias, measure all occupation projectors and the projectors onto $(|j\rangle + |k\rangle)/\sqrt{2}$ and $(|j\rangle + i|k\rangle)/\sqrt{2}$ within P_K. Estimating these $O(d_K^2)$ probabilities to $O(\varepsilon/d_K^{3/2})$ recovers the low block in trace norm and needs $O(d_K^5\varepsilon^{-2}\log d_K)$ copies for fixed success probability. Normalization as above completes the estimate. This deliberately nonoptimal construction is sufficient to demonstrate a separation from coherent tests. Sharper general continuous-variable learning bounds are prior work [27]. The coherent-test lower bound must not be applied to these other projectors, to homodyne measurements, to collective protocols, or to all Gaussian measurement schemes.

Interpolation-based continuous-variable tomography is established prior work [24]; the proposed contribution here is the matching trace-distance minimax law with an aperture-dependent radius and a sole photon-moment constraint, including adaptive lower bounds and the fixed-mode extension. Exact injectivity from finitely many complete shells is a different theorem. In particular, (A.2) does not supply a matching upper rate for the fixed-shell modulus in (H.3). These results neither prove nor assume microscopic BGS universality. External priority and independent referee acceptance remain unestablished.

65 The same sharp law for censored and full heterodyne detection

We now change the measurement, rather than silently identifying binary coherent tests with heterodyne detection. Standard multimode heterodyne measurement produces $Z \in \mathbb{C}^s$ with Lebesgue density $Q_\rho(z) = \pi^{-s}\langle z|\rho|z\rangle$. Its radius-$R$ censored version reports Z when $\|Z\| \le R$ and a single symbol $\star$ otherwise. Every copy is counted, including outcomes labelled $\star$; the data are not postselected or renormalized. For $R = \infty$ this is ordinary heterodyne. Let $\mathfrak{H}_B(R)$ be its minimax expected trace-distance error on the same class $\mathcal{S}_{s,\beta,E}$, optimized over all estimators from B independent outcomes.

Theorem 65.1 (Heterodyne minimax law)**.** *Fix $s \ge 1$, $\beta, E, R_0 > 0$. Uniformly for $R \ge R_0$, including $R = \infty$ and radii depending on B, one has*

$$\boxed{c\left(\frac{1+\log_+((\log B)/R^2)}{\log B}\right)^{\beta/2} \le \mathfrak{H}_B(R) \le C\left(\frac{1+\log_+((\log B)/R^2)}{\log B}\right)^{\beta/2}} \tag{B.1}$$

for all sufficiently large B. The lower bound holds on pure states. The upper bound needs no rank constraint. Constants may depend on the fixed s, β, E, R_0, not on R or B.

65.1 Upper bound from local counts and stable regularization

No independent repetitions at prescribed points are available in heterodyne sampling. Instead estimate the density locally, using the very same samples at each of the finitely many interpolation nodes. For every state,

$$|Q_\rho(z) - Q_\rho(w)| \le \pi^{-s}\|z - w\|, \tag{B.2}$$

since the operator norm distance of the two coherent projectors is $\sqrt{1 - e^{-\|z-w\|^2}} \le \|z - w\|$.

Put $T = \log B$, choose $K = \lfloor cK_R(T)\rfloor$ and $S = \min(R^2/4, K)$, and use the Chebyshev grid with $L = 128sK$ from the previous section. Every node z_j has norm at most $R/2$. Let $h = B^{-1/(4s+2)}$ and $v_{2s} = \pi^s/s!$, the unit-ball volume in $2s$ real dimensions. For large B one has $h < R_0/2$. Therefore the balls $B(z_j, h)$ lie within the recorded region, and their membership is known even for censored observations. Define

$$\widehat{Q}_j = \frac{1}{Bv_{2s}h^{2s}} \sum_{b=1}^{B} \mathbf{1}_{\{Z_b \in B(z_j,h)\}}. \tag{B.3}$$

The bias is at most $\pi^{-s}h$ by (B.2). Hoeffding's inequality and the union bound, without independence between nodes, give simultaneously with probability at least $1 - B^{-2}$,

$$|\pi^s\widehat{Q}_j - p_\rho(z_j)| \le h + \frac{s!}{h^{2s}}\sqrt{\frac{\log(2(L+1)^{2s}B^2)}{2B}} = \exp\!\left(-\frac{T}{4s+2} + O_s(\log T)\right). \tag{B.4}$$

Replacing R by $R/2$ in the analytic interpolation estimate changes $\Phi_R(K)$ by at most $K\log 4$, hence by a fixed factor. Choose $c > 0$ small enough that the noise amplification is at most

$\exp(T/(8s+4))$ up to polynomial factors. Lemmas 64.3 and 64.4, and the analytic bias bound (A.12), then give expected loss at most $CK_R(T)^{-\beta/2}$. The estimate is the normalized positive part of a finite reconstructed block; it is a genuine density operator. This proves the upper bound in (B.1).

65.2 An explicit exponentially indistinguishable pure-state pair

Use (A.14), now with even $n = 2k$. With a_z, b_z as in the preceding section,

$$\bigl(\sqrt{Q_+(z)} - \sqrt{Q_-(z)}\bigr)^2 \le 4\pi^{-s}\min(a_z^2, b_z^2) \le 4\pi^{-s} a_z b_z.$$

Integration over the whole space separates the vacuum factors in modes $2, \dots, s$ and gives the finite, explicit bound

$$\begin{aligned} h^2(Q_+, Q_-) &\le 4\sqrt{u(1-u)}\,\frac{\Gamma(n/2+1)}{\sqrt{n!}} \\ &= \frac{4\sqrt{u(1-u)}}{\sqrt{\binom{2k}{k}}} \le 4\sqrt{u(n+1)}\,2^{-n/2}. \end{aligned} \tag{B.5}$$

The last inequality follows from $\binom{2k}{k} \ge 2^{2k}/(2k+1)$: the central binomial coefficient is the largest of $2k+1$ coefficients whose sum is 2^{2k}. This is an upper bound on statistical distinguishability, not merely a small difference of mean measurements.

For a ball centred at the origin the two probabilities of $\star$ are exactly equal. Indeed the difference of the two densities is proportional to $\cos(n \arg z_1)$, whose angular integral vanishes on any such ball. The censored Hellinger distance therefore has only the inside-ball contribution. Bounding its integrand by $4\pi^{-s} b_z^2$ and integrating the other modes over their whole spaces gives

$$h_R^2(Q_+, Q_-) \le \frac{4u}{n!}\int_0^{R^2} e^{-x} x^n\, dx \le \frac{4uR^{2n+2}}{(n+1)!}. \tag{B.6}$$

For $R^2 \le n$ this is at most $4uR^{2n}/n!$. Together with (B.5) and $n! \ge (n/e)^n$, it follows that

$$h_R^2(Q_+, Q_-) \le C_E\sqrt{n+1}\,\exp(-c\Phi_R(n)) \tag{B.7}$$

for an absolute $c > 0$ and large n, uniformly for every $R > 0$. For $R^2 > n$ the global estimate suffices; for $R^2 \le n$ compare the two exponent bounds, as in (A.18).

For B independent outcomes, product affinity gives $h^2(\mathbb{P}_+^{\otimes B}, \mathbb{P}_-^{\otimes B}) \le B h_R^2$. Choose the least even integer $n \ge CK_R(\log B)$ with a sufficiently large constant. Then (B.7) makes the transcript total variation at most $1/2$ for all large B. The two states remain separated in trace distance by at least $\sqrt{2E}n^{-\beta/2}$. The same two-point testing argument as before gives risk at least one eighth of that distance. This proves the lower bound and completes Theorem 65.1.

65.3 Consequences and comparison with prior tomography bounds

Both the censored-heterodyne model and the adaptive binary coherent-test model have the same minimax order, although they are different statistical experiments and our proofs use different upper estimators and Hellinger bounds. The copy-complexity law (A.20) holds for both. For fixed mean energy, ordinary heterodyne therefore requires and suffices with $\exp(\Theta(\varepsilon^{-2}))$ copies in this worst-case trace-distance problem. For a fixed finite recorded ball, the corresponding formula is

$$B_*(\varepsilon, R) = \exp\bigl(\Theta(\varepsilon^{-2}\log(1/\varepsilon))\bigr) \quad (\beta = 1,\ 0 < R < \infty \text{ fixed}).$$

No statement here covers arbitrary squeezed Gaussian measurements, homodyne detection, or measurements jointly acting on several copies. Adaptive passive transformations followed by uncensored heterodyne only relabel the complete outcome and cannot evade its lower bound.

Exponential heterodyne upper bounds at a hard photon cutoff were already reported by Gandhari et al. [25]; their paper explicitly leaves optimal lower bounds open. General moment-constrained tomography with more general measurements is studied by Mele et al. [27]. These are not priority claims for the difficulty of heterodyne inversion. The result proved here is a matching trace-distance rate under only one photon-moment bound, with no state truncation assumption and a uniform finite-aperture transition.

A source-level distinction matters in comparing Becker et al. [26]. The explicit heterodyne Theorem 4 in the journal version (Theorem 11 in arXiv v2) retains the cutoff-dependent norm $\widetilde{\Sigma}_r^{(\alpha)}(M,R)$ in its sample bound. It must not be replaced by a uniform constant. The overview Theorem 2 of the journal version (Theorem 3 in arXiv v2), read as a uniform polynomial-in-accuracy bound for full trace-norm recovery using ordinary heterodyne under only a fixed moment bound, is incompatible with the explicit pair (B.5). For an integer moment order β our pair also satisfies $\mathrm{Tr}(\rho_n^{\pm}(I+\mathcal{N})^{\beta}) \le 1+2^{\beta}E$ uniformly in n, so changing between the two moment conventions cannot remove the conflict. This narrow comparison does not retract the explicit cutoff-dependent theorem, the homodyne results, or statements about a fixed finite set of polynomial observables. No author correspondence or accepted correction is claimed; the comparison and our proof require independent review.

66 Rare-phase discrimination: a sharp exponent and radial transition

We refine the pure-state lower-bound family from the preceding minimax sections. The results give its exact exponential distinguishability scale, a universal radial profile, optimal fixed-test Chernoff constants, and a finite copy-complexity separation. Restricted-measurement data hiding itself is prior work [28, 29]. Here the states are nonorthogonal and approach the vacuum while one photon moment stays fixed. No result about all Gaussian measurements or microscopic BGS statistics is asserted.

66.1 Balance radius and angular overlap

Fix $\beta, E > 0$, let $u = E/n^{\beta} < 1/2$, and put

$$|\psi_n^{\pm}\rangle = \sqrt{1-u}|0\rangle \pm \sqrt{u}|n\rangle, \quad x_n = \left(\frac{n!(1-u)}{u}\right)^{1/n}, \quad b_n = (1-u)e^{-x_n}. \tag{C.1}$$

The states have moment E and trace distance $2\sqrt{u(1-u)}$. The squared radius x_n is where their vacuum and number-state coherent amplitudes have equal modulus. Stirling's formula gives

$$x_n = \frac{n}{e} + \frac{(\beta+\frac{1}{2})\log n - \log E + \frac{1}{2}\log(2\pi)}{e} + o(1). \tag{C.2}$$

Indeed the logarithm of the expression inside the nth root is $n\log n - n + (\beta+\frac{1}{2})\log n - \log E + \frac{1}{2}\log(2\pi) + \log(1-u) + O(1/n)$. Expanding the exponential proves the displayed formula; the contribution of $\log(1-u)$ is $o(1)$.

Define

$$F(t) = \frac{1}{2\pi}\int_0^{2\pi} \left(\left|1+\sqrt{t}e^{i\varphi}\right| - \left|1-\sqrt{t}e^{i\varphi}\right|\right)^2 d\varphi. \tag{C.3}$$

Then $F(t) = tF(t^{-1})$ for $t > 0$, and

$$\min(1,t) \le F(t) \le 4\min(1,t). \tag{C.4}$$

For the lower bound rationalize the difference of square roots: its squared denominator is at most $4(1+t)$, so the integrand is at least $4t\cos^2\varphi/(1+t)$. Its average is $2t/(1+t) \geq \min(1,t)$. The upper bound follows from the reverse triangle inequality in both orders.

Let $\mathsf{E}(m) = \int_0^{\pi/2} \sqrt{1 - m\sin^2\theta}\, d\theta$ be the complete elliptic integral with parameter m. Multiplying the two moduli and changing angular variables yields

$$F(t) = 2(1+t)\left[1 - \frac{2}{\pi}\mathsf{E}\Big(\frac{4t}{(1+t)^2}\Big)\right]. \tag{C.5}$$

An exact series, useful to avoid numerical cancellation at small t, is

$$F(t) = 2t - 2\sum_{k\geq 1} c_k t^{2k}, \qquad c_k = \binom{1/2}{k}^2 = \frac{\binom{2k}{k}^2}{16^k(2k-1)^2}, \quad 0 \leq t \leq 1. \tag{C.6}$$

To prove it for $t < 1$, multiply the absolutely convergent binomial series for $(1 - te^{i\varphi})^{1/2}$ and its conjugate and average. Only equal indices remain. Continuity and monotone convergence of the nonnegative series prove the endpoint. In particular $F(1) = 4 - 8/\pi$.

66.2 Exact radial formula and a universal boundary profile

Use unnormalized squared Hellinger distance $h^2(P,Q) = \int(\sqrt{p} - \sqrt{q})^2$, including atoms. Let $h^2_{n,R}$ refer to the radius-R censored heterodyne laws, with the outside flag retained. Their outside probabilities agree by angular integration. Polar coordinates therefore give the exact identity

$$\boxed{h^2_{n,R} = (1-u)\int_0^{R^2} e^{-x} F\big((x/x_n)^n\big)\, dx.} \tag{C.7}$$

The angular frequency $n\arg z$ disappears upon averaging since n is an integer. The formula includes ordinary heterodyne at $R = \infty$.

Theorem 66.1 (Radial transition profile)**.** *If $R_n^2 - x_n \to \ell \in \mathbb{R} \cup \{-\infty, +\infty\}$, then*

$$\frac{h^2_{n,R_n}}{b_n} \longrightarrow \mathcal{C}(\ell) := \int_{-\infty}^{\ell} e^{-y} F(e^{ey})\, dy. \tag{C.8}$$

Here $\mathcal{C}(-\infty) = 0$ and $0 < \mathcal{C}_* := \mathcal{C}(+\infty) < \infty$. For ordinary heterodyne,

$$h^2_{n,\infty} \sim \mathcal{C}_* E^{1/e}(2\pi)^{-1/(2e)} n^{-(\beta+1/2)/e} e^{-n/e}. \tag{C.9}$$

Proof. Put $x = x_n + y$ in (C.7). The normalized integrand is $e^{-y}F((1+y/x_n)^n)$ for $y \geq -x_n$ and zero otherwise. For fixed y it converges to $e^{-y}F(e^{ey})$. For $y \geq 0$ it is bounded by $4e^{-y}$. Choose $1 < \kappa < e$. For large n, $n/x_n \geq \kappa$; for $y < 0$, $(1+y/x_n)^n \leq e^{\kappa y}$, so the integrand is bounded by $4e^{(\kappa-1)y}$. These bounds are integrable on the two half-lines. Dominated convergence, also with the moving upper indicator, proves (C.8). Positivity follows from (C.4). Finally substitute (C.2) into b_n. □

Writing $a = 1/e$, the constant has the exact scalar representation

$$\mathcal{C}_* = 2a\left\{\frac{1}{1-a} + \frac{1}{a} - \sum_{k\geq 1} c_k\left(\frac{1}{2k-a} + \frac{1}{2k+a-1}\right)\right\}. \tag{C.10}$$

This follows by substituting $t = e^{ey}$, splitting the Mellin integral at one, and using the symmetry and series for F. Truncation after M terms is an upper estimate with error at most $a/(3M^3)$. Indeed, induction using the central-binomial recurrence gives $\binom{2k}{k}^2(3k+1) \leq 16^k$, and $(3k+$

$1)(2k-1)^2 \geq 4k^3$. Thus $c_k \leq 1/(4k^3)$. The sum of the two reciprocal denominators is at most $2/k$, and summing k^{-4} by an integral proves the claimed tail bound. Outward-rounded integer arithmetic at $M = 8192$, independently reproduced in Python and C++, encloses the constant as

$$2.9086102383180 < \mathcal{C}_* < 2.9086102383183. \tag{C.11}$$

The interval certificate uses a rational enclosure of e from its factorial series, not floating-point values of the elliptic integral.

66.3 Optimal coherent tests and the aperture exponent

Let $H^{\mathrm{bin}}_{n,R}$ be the supremum of squared Hellinger distances over binary coherent-projector tests with $|z| \leq R$. For $x = |z|^2$ set

$$A(x) = (1-u)e^{-x}, \quad B(x) = ue^{-x}x^n/n!, \quad m_{n,R} = \max_{0\leq x\leq R^2} \min(A(x), B(x)).$$

For large n one has $x_n < n$. The minimum is $B(x)$ below x_n and $A(x)$ above x_n, so its maximum is attained at $\min(R^2, x_n)$. An aligned phase, $\cos(n \arg z) = 1$, gives a click contribution exactly $4\min(A(x), B(x))$. For the complement, the estimate in (A.18) gives at most $64A(x)B(x)/u$. Consequently

$$4m_{n,R} \leq H^{\mathrm{bin}}_{n,R} \leq (4 + 64/u)m_{n,R}, \qquad 0 \leq H^{\mathrm{bin}}_{n,R} - 4m_{n,R} \leq 64\, 2^{-n}. \tag{C.12}$$

For the second inequality maximize $e^{-2x}x^n/n!$ at $x = n/2$ and use $n! \geq (n/e)^n$. Thus $H^{\mathrm{bin}}_{n,\infty} \sim 4b_n$. If $R_n^2 - x_n \to \ell \in \mathbb{R}$, then

$$\frac{H^{\mathrm{bin}}_{n,R_n}}{b_n} \longrightarrow \begin{cases} 4e^{(e-1)\ell}, & \ell \leq 0, \\ 4, & \ell \geq 0. \end{cases} \tag{C.13}$$

The additive error is exponentially smaller than b_n. For $\ell < 0$, $B(x_n + \ell)/b_n = e^{-\ell}(1 + \ell/x_n)^n$ proves the formula.

Theorem 66.2 (Exact exponential discrimination threshold)**.** *If $R_n^2/n \to \alpha \in (0, \infty]$, each of h^2_{n,R_n} and H^{bin}_{n,R_n} satisfies*

$$-\frac{1}{n}\log H_n \longrightarrow I(\alpha), \qquad I(\alpha) = \begin{cases} \alpha - 1 - \log\alpha, & 0 < \alpha \leq e^{-1}, \\ e^{-1}, & \alpha \geq e^{-1}. \end{cases} \tag{C.14}$$

If $\log B_n/n \to b < I(\alpha)$, no processing of the heterodyne outcomes or adaptive sequence of the allowed binary tests has equal-prior success bounded above $1/2$ in the limit. If $b > I(\alpha)$, likelihood testing of heterodyne data, or repetitions of one suitable coherent test, succeeds with probability tending to one. No conclusion at equality is asserted by this exponent statement alone.

Proof. At $x = \alpha n < x_n$, Stirling gives $n^{-1}\log B(x) \to -\alpha + 1 + \log\alpha$, whereas $n^{-1}\log A(x_n) \to -1/e$. These statements extend to sequences of radii, including equality at $1/e$. The polynomial factors in (C.12) do not change the logarithmic limit. By (C.4), the heterodyne integral is between one and four times the integral of $\min(A, B)$. Its integral is at most $(x_n + 1)m_{n,R}$, or $R^2 m_{n,R}$ if $R^2 < x_n$. A fixed-width interval immediately below the maximizing point gives a lower estimate within a fixed factor when $R^2/n \to \alpha > 0$; the logarithmic derivatives are bounded there. This proves its exponential limit.

If $b < I(\alpha)$, the transcript Hellinger bound B_nH_n tends to zero, also for adaptive binary choices by conditional affinity. Hence total variation tends to zero. If $b > I(\alpha)$, repeat a test with distance at least half the supremum. Its product affinity is at most $\exp(-B_nH_n/4) \to 0$. The equal-prior likelihood error is at most half the affinity, since $\min(p, q) \leq \sqrt{pq}$. The same proof applies to the fixed heterodyne experiment. □

The informative annulus is at squared radius n/e, not at the number-state density peak n. At full aperture the exact exponent $1/e$ improves the convenient lower-bound exponent $(\log 2)/2$ without changing the previously proved minimax order. At the balancing point with aligned phase, $p_- = 0$ and $p_+ = 4b_n$. The rule that reports plus on any click and minus otherwise has success

$$1 - \tfrac{1}{2}(1 - 4b_n)^B \;\longrightarrow\; 1 - \tfrac{1}{2}e^{-4c} \quad \text{if } Bb_n \to c. \tag{C.15}$$

This is a specified protocol, not an assertion of adaptive optimality at the critical constant scale.

66.4 Sharp Chernoff information for fixed coherent tests

For classical distributions define $\xi(P, Q) = -\log \inf_{0<s<1} \sum p^s q^{1-s}$ and $\mathcal{A} = \sum \sqrt{pq} = 1 - h^2/2$. Cauchy–Schwarz and Holder's inequality give $\mathcal{A}^2 \le \sum p^s q^{1-s} \le 1$ for all s. If the two laws are interchanged by an involution preserving their dominating measure, log-convexity and symmetry in s imply $\xi = -\log \mathcal{A}$. The heterodyne pair has this symmetry under $z \mapsto e^{i\pi/n}\overline{z}$. Therefore at full aperture

$$\xi_n^{\mathrm{het}} \sim \tfrac{1}{2}\mathcal{C}_* b_n, \qquad \sup_z \xi(P^+_{n,z}, P^-_{n,z}) \sim 4b_n. \tag{C.16}$$

For the second statement the universal bound is $\xi \le -2\log(1 - h^2/2)$ and (C.12) supplies the uniform upper equivalent. The aligned balancing test gives exactly $-\log(1 - 4b_n)$, proving the matching lower equivalent. Thus the ratio of the optimal fixed binary-test information to ordinary heterodyne information tends to $8/\mathcal{C}_*$. This comparison concerns one fixed test optimized for a known pair; it is not a minimax tomography constant or an adaptive Chernoff theorem.

66.5 The full critical heterodyne likelihood experiment

The Hellinger constant also arises from an entire limiting likelihood law, not just from a single overlap integral. Define, for $y \in \mathbb{R}$ and $\varphi \in [0, 2\pi)$,

$$\lambda_\pm = 1 + e^{ey} \pm 2e^{ey/2}\cos\varphi, \qquad L = \log(\lambda_+/\lambda_-).$$

Values on the isolated zero sets do not affect the integrals. For real v put

$$\Psi(v) = \int_{\mathbb{R}} \int_0^{2\pi} e^{-y}\{\lambda_-(e^{ivL} - 1) - iv(\lambda_+ - \lambda_-)\}\frac{d\varphi}{2\pi}\, dy. \tag{C.17}$$

This is an absolutely convergent compensated integral. Indeed for $r > 0$,

$$|r^{iv} - 1 - iv(r - 1)| \le C_v(\sqrt{r} - 1)^2.$$

The quotient is bounded near $r = 1$ by Taylor expansion, near zero by a constant, and near infinity by its leading linear terms. Multiplying by λ_- and using the angular overlap bounds gives an integrable majorant from the proof of the radial-profile theorem.

Let Z_n be the log likelihood ratio of all B_n independent heterodyne outcomes under the minus hypothesis, and suppose $B_n b_n \to c \ge 0$. Then

$$Z_n \implies Z_c, \qquad \mathbb{E}e^{ivZ_c} = e^{c\Psi(v)}, \qquad P^{\mathrm{het}}_{\mathrm{succ}} \longrightarrow 1 - \tfrac{1}{2}\mathbb{E}\min(1, e^{Z_c}). \tag{C.18}$$

To prove this, write the one-copy characteristic function minus one as $\sum q_-[(q_+/q_-)^{iv} - 1]$. Subtract $iv\sum(q_+ - q_-) = 0$, then rescale $x = x_n + y$ exactly as in the radial calculation. Dominated convergence gives $b_n\Psi(v) + o(b_n)$. Taking the B_nth power proves characteristic-function convergence. Continuity at $v = 0$ follows from the same bound with C_v locally bounded, so the limit is a probability law. Its dependence on every $c \ge 0$ makes the resulting convolution semigroup infinitely divisible. Finally, the equal-prior Bayes error is exactly $\frac{1}{2}\mathbb{E}_-\min(1, e^{Z_n})$;

this test function is bounded and continuous, which justifies the last limit without assuming uniform integrability of an unbounded likelihood ratio.

For $R_n^2 - x_n \to \ell$ the same formulas hold with the y integral cut off at ℓ. The outside flag has likelihood ratio one, and its probabilities are equal. The identity $\Psi(-i/2) = -\mathcal{C}_*/2$ follows directly from the convergent integral; it is not inferred by an unjustified complex extension of weak convergence. This statement closes the known-pair critical heterodyne experiment, not global tomography or the critical optimum of adaptive binary coherent protocols. The Poisson limit of the explicitly chosen binary protocol in (C.15) remains a separate result.

66.6 A finite separation at mean photon number one

Take $n = 128$ and $E = \beta = 1$:

$$|\psi^{\pm}\rangle = \sqrt{127/128}|0\rangle \pm \frac{1}{\sqrt{128}}|128\rangle, \quad D(\rho^+, \rho^-) = \frac{\sqrt{127}}{64}. \tag{C.19}$$

The task is binary discrimination of two known hypotheses with equal prior, not full tomography. Their B-copy overlap is $(63/64)^B$. Diagonalizing the two-dimensional span of these pure vectors gives unrestricted optimal success

$$P_{\mathrm{opt}}(B) = \tfrac{1}{2}\left(1 + \sqrt{1 - (63/64)^{2B}}\right). \tag{C.20}$$

The rational inequalities $(63/64)^{18} > 3/4$ and $(63/64)^{20} < 3/4$ show that precisely ten copies first suffice for success at least $3/4$. This does not assume an efficient implementation of the optimal joint measurement.

For ordinary heterodyne, (B.5) gives $h^2 \le \sqrt{127}/(32\sqrt{\binom{128}{64}})$. Success at least $3/4$ requires total variation at least $1/2$, while product Hellinger gives $\mathrm{TV} \le \sqrt{Bh^2}$. Consequently

$$\boxed{B \ge \frac{8\sqrt{\binom{128}{64}}}{\sqrt{127}} > 3 \cdot 10^{18}.} \tag{C.21}$$

If only $|z| \le 1$ and the outside flag are recorded, the necessary bound is

$$\boxed{B \ge 8\,129! > 3 \cdot 10^{218}.} \tag{C.22}$$

Both comparisons are verified with integers. All copies, including outside flags, are counted. They are necessary finite-copy bounds, not exact finite heterodyne optima. State-preparation cost is not included. No such lower bound is asserted for homodyne or general Gaussian processing.

66.7 Verification and scope

Integer interval arithmetic encloses (C.10); independent C++ arithmetic reproduces a contained interval using direct central-binomial coefficients instead of the Python recurrence. Elliptic and radial quadrature are separate high-precision diagnostics. Exact integer tests also check the finite-copy bounds and the nine-versus-ten-copy threshold. They do not replace the dominated-convergence or testing proofs.

The single-mode even/odd thermal-state hiding construction of Lami concerns orthogonal mixed states and a stronger class of Gaussian operations [28]. The family and limiting regime here differ. Proposed contributions are the fixed-moment rare-phase profile, exact aperture exponent, sharp fixed-test information comparison and explicit finite separation. External priority and independent referee acceptance remain unestablished.

67 Input tomography after Gaussian noise: limits of collective measurements

We now remove the restriction on the output measurements and instead fix a known noisy quantum channel. The task is reconstruction of its *input* state from B independently prepared copies of its output. Arbitrary joint POVMs on those outputs are allowed. The environment is inaccessible and no encoding or error-correcting intervention before the channel is included. Tomography of the output itself is a different problem.

Let $\mathcal{S}_{s,\beta,E}$ be the moment class of Section 64. A fixed product of phase-insensitive Gaussian channels is specified, in the Weyl convention $D(z) = \exp(za^* - \overline{z}a)$ and $\chi_\rho(z) = \mathrm{Tr}(\rho D(z))$, by

$$\chi_{\Phi(\rho)}(z) = \chi_\rho(\sqrt{\boldsymbol{\kappa}}\, z) \exp\left(-\tfrac{1}{2} \sum_{j=1}^{s} \nu_j |z_j|^2 \right), \quad \kappa_j > 0, \quad \nu_j \geq |1 - \kappa_j|. \tag{D.1}$$

The inequalities are the physical complete-positivity constraints in this normalization. Pure loss has $(\kappa, \nu) = (\eta, 1-\eta)$, the quantum-limited amplifier has $(g, g-1)$, and additive noise has $(1, \nu)$. The Weyl/Kraus representations used below are standard and are given, for example, by Ivan, Sabapathy and Simon [30]; they are not novelty claims of this project. Assume at least one $\nu_j > 0$.

Theorem 67.1 (Sharp Gaussian-input minimax rate)**.** *For fixed s, β, E and a fixed channel* (D.1) *with at least one noisy mode, the minimax expected input trace-distance risk over all collective output measurements satisfies*

$$\boxed{c(\log B)^{-\beta/2} \leq \inf_{\text{output POVM}, \widehat{\rho}} \sup_{\rho \in \mathcal{S}_{s,\beta,E}} \mathbb{E} D(\widehat{\rho}, \rho) \leq C(\log B)^{-\beta/2}.} \tag{D.2}$$

The upper bound is attained by nonadaptive randomized product homodyne measurements and an explicit finite-matrix estimator. The lower bound already holds on pure inputs. For the deterministic inverse modulus one has

$$\sup_{\substack{\rho,\sigma \in \mathcal{S}_{s,\beta,E} \\ D(\Phi\rho, \Phi\sigma) \leq \delta}} D(\rho, \sigma) \asymp (\log(1/\delta))^{-\beta/2}. \tag{D.3}$$

Constants depend on the fixed channel and moment parameters; in particular no uniform claim as the noise tends to zero is made. Complete erasure of a mode, $\kappa_j = 0$, is excluded and would prevent consistency for arbitrary inputs.

67.1 Uniform decay of low Fock-block Weyl kernels

Let P_K project onto total occupation at most $K \geq 1$ and $d_K = \binom{K+s}{s}$. For a single mode, normal ordering gives the polynomial representation of $\langle j|D(z)|k\rangle$. Its absolute value is at most

$$e^{-x/2} \frac{x^{(j+k)/2}}{\sqrt{j!k!}} \sum_{\ell \geq 0} \frac{(jk/x)^\ell}{\ell!} \leq e^{-x/2 + K^2/x} \frac{x^K}{K!}, \qquad x = |z|^2 \geq K, \quad j, k \leq K. \tag{D.4}$$

Indeed the polynomial coefficient relative to its leading term is $(j)_\ell (k)_\ell / \ell! \leq (jk)^\ell / \ell!$. Also $x^m/m!$ is increasing for $m \leq K \leq x$, which proves the second bound. For $x \geq 16K$, the last expression is at most $e^{-x/4}$. To verify this, use $K! \geq (K/e)^K$ and set $y = x/K \geq 16$; the needed inequality is $1/y + 1 + \log y \leq y/4$. Its difference is increasing on this range and positive at 16.

In s modes, at least one component has $|z_j|^2 \geq \|z\|^2/s$; all other displacement matrix elements have modulus at most one. Consequently, for $|\mu|, |\nu| \leq K$ and $\|z\|^2 \geq 16sK$,

$$|\langle \mu | D(-z) | \nu \rangle| \leq e^{-\|z\|^2/(4s)}. \tag{D.5}$$

With $L = 32sK$, the corresponding tail integral is bounded by

$$\tau_{s,K} = \pi^{-s} \int_{\|z\|^2 > L} e^{-\|z\|^2/(4s)}\, d^{2s}z = (4s)^s e^{-8K} \sum_{j=0}^{s-1} \frac{(8K)^j}{j!}. \tag{D.6}$$

This bound is independent of the unknown state and controls the inversion kernel, not a falsely assumed cutoff of the input characteristic function.

67.2 A constructive noisy-input inverse

Weyl inversion gives, for every trace-class state,

$$\rho_{\mu\nu} = \pi^{-s} \int_{\mathbb{C}^s} \chi_\rho(z) \langle \mu | D(-z) | \nu \rangle\, d^{2s}z. \tag{D.7}$$

One may verify the formula first on matrix units by Gaussian monomial integration, using normal ordering, and then pass to trace-class states: finite-rank approximants converge uniformly in characteristic functions since $|\operatorname{Tr}((\rho - \rho_m)D(z))| \le \|\rho - \rho_m\|_1$, and the fixed low kernel is absolutely integrable. Thus the use of (D.7) requires no hidden regularity of the state.

Set

$$a = \max_j \frac{\nu_j}{2\kappa_j}, \qquad V = L^s/s!, \qquad A_K = V e^{aL}.$$

On the ball $\|z\|^2 \le L$, recover $\chi_\rho(z) = \exp(\sum_j \nu_j |z_j|^2/(2\kappa_j))\chi_{\Phi\rho}(z/\sqrt{\boldsymbol{\kappa}})$. If $D(\Phi\rho, \Phi\sigma) \le \delta$, their input low-block entries differ by at most $2\delta A_K + 2\tau_{s,K}$. The projection bias and trace-norm block estimate from Lemma 64.4 therefore give

$$D(\rho, \sigma) \le 2\sqrt{E/(K+1)^\beta} + d_K^{3/2}(\delta A_K + \tau_{s,K}). \tag{D.8}$$

Choosing $K = \lfloor c \log(1/\delta) \rfloor$ with $c > 0$ small enough proves the upper half of (D.3).

For statistical reconstruction draw each z uniformly in that ball and put $w_j = z_j/\sqrt{\kappa_j}$. Measure on the jth output mode the quadrature $X_{\theta_j} = q_j \cos\theta_j + p_j \sin\theta_j$ with $\theta_j = \arg w_j - \pi/2$. If its outcome is x_j, then

$$Y = \exp\left(i\sqrt{2} \sum_j |w_j| x_j \right), \quad |Y| = 1, \quad \mathbb{E}(Y \mid z) = \chi_{\Phi\rho}(w).$$

At $w_j = 0$ the angle can be chosen arbitrarily. Use the estimator

$$M_{\mu\nu} = \frac{V}{B} \sum_{b=1}^{B} e^{\sum_j \nu_j |z_{b,j}|^2/(2\kappa_j)} Y_b \langle \mu | D(-z_b) | \nu \rangle. \tag{D.9}$$

Its expectation is the truncated integral, and each summand has modulus at most A_K. Hoeffding bounds for the real and imaginary parts and a union bound over d_K^2 entries imply, with probability at least $1 - B^{-2}$,

$$\max_{\mu,\nu} |M_{\mu\nu} - \rho_{\mu\nu}| \le 4A_K \sqrt{\frac{\log(4d_K^2 B^2)}{B}} + \tau_{s,K}. \tag{D.10}$$

Symmetrize, take the positive part, and normalize as in Lemma 64.4. The resulting state has trace-distance error bounded by the cutoff bias plus $2d_K^{3/2}$ times the entrywise error. Choose $K = \lfloor c \log B \rfloor$ with $32sac < 1/4$. The stochastic and tail terms then decay faster than $K^{-\beta/2}$. On the complement of the concentration event the loss is at most one. This proves the expected upper bound in (D.2).

If all noises vanish, physicality forces $\kappa_j = 1$. The same construction with $a = 0$ and $K = O(\varepsilon^{-2/\beta})$ has polynomial copy cost, for example $O(K^{5s}\varepsilon^{-2} \log K)$ at fixed success probability. We do not claim that this noiseless upper bound is optimal.

67.3 A lower bound against every collective output measurement

First consider pure loss with $0 < \eta < 1$. Use the input pair of (C.1), in one mode, with $u = E/n^\beta$. The output is a common diagonal binomial mixture plus opposite coherences between levels 0 and n. On their active two-dimensional subspace write

$$a_0 = 1 - u + u(1-\eta)^n, \quad b_0 = u\eta^n, \quad c_0 = \sqrt{u(1-u)}\eta^{n/2}, \quad s_0 = a_0 + b_0.$$

The blocks are $\left(\begin{smallmatrix} a_0 & \pm c_0 \\ \pm c_0 & b_0 \end{smallmatrix}\right)$, with identical orthogonal remainder. Their root fidelity is exactly

$$F = 1 - s_0 + \sqrt{s_0^2 - 4c_0^2}, \qquad 1 - F \le 4u\eta^n, \quad D(\Phi\rho_+, \Phi\rho_-) = 2\sqrt{u(1-u)}\eta^{n/2}. \tag{D.11}$$

For the fidelity formula, use $F(X,Y)^2 = \mathrm{Tr}(XY) + 2\sqrt{\det X \det Y}$ for positive 2-by-2 blocks. Rationalizing the square root and using $s_0 \ge 1 - u$ gives the displayed inequality. Fidelity is multiplicative on tensor products and $D \le \sqrt{1 - F^2}$; these facts also follow by purifying the states and using the pure-state trace-distance formula. Hence for B copies,

$$D((\Phi\rho_+)^{\otimes B}, (\Phi\rho_-)^{\otimes B}) \le \sqrt{8Bu\eta^n}. \tag{D.12}$$

Choosing $n = \lceil C \log B \rceil$ with C large makes this at most $1/2$. The two-point testing argument gives expected input loss at least $\frac{1}{8}D(\rho_+, \rho_-) \ge c(\log B)^{-\beta/2}$ for every joint POVM. Choosing $n = O(\log(1/\delta))$ in the one-copy trace-distance formula proves the lower deterministic modulus.

For a quantum-limited amplifier of gain $g > 1$, put $q = 1 - 1/g$. Its isometry is

$$|n\rangle \mapsto \sum_{\ell \ge 0} g^{-(n+1)/2} q^{\ell/2} \sqrt{\binom{n+\ell}{\ell}} |n+\ell\rangle|\ell\rangle.$$

Thus the image T_n of $|0\rangle\langle n|$ has trace norm

$$\|T_n\|_1 = g^{-n/2-1} \sum_{\ell \ge 0} q^\ell \sqrt{\binom{n+\ell}{\ell}}.$$

Choose $v = q(1+q)/2$, so $q^2 < v < q$. Cauchy–Schwarz and $\sum_\ell \binom{n+\ell}{\ell} v^\ell = (1-v)^{-n-1}$ give

$$\|T_n\|_1 \le C_g \theta_g^n, \qquad \theta_g = [g(1-v)]^{-1/2} < 1, \quad C_g = g^{-1}(1-v)^{-1/2}(1 - q^2/v)^{-1/2}. \tag{D.13}$$

The input pair therefore has output trace distance at most $2\sqrt{u(1-u)}C_g\theta_g^n$. Tensor telescoping bounds the B-copy distance by B times this quantity. Taking $n = O(\log B)$ proves the same collective lower rate. The one-copy bound proves its deterministic analogue.

A channel with $\kappa < 1$ is pure loss followed by additive Gaussian noise; with $\kappa > 1$ it is a quantum-limited amplifier followed by additive noise. The extra noise has parameter $\nu - |1-\kappa| \ge 0$. For $\kappa = 1, \nu > 0$, put $g = 1 + \nu/2$: loss of transmissivity $1/g$ followed by gain g has the required characteristic function. Thus every noisy mode factors through one of the hard channels above, and contractivity preserves the lower bounds. Place the pair in that mode and vacuum in the others; tensoring common outputs changes neither trace distance nor fidelity. This completes Theorem 67.1.

67.4 Exact critical law for collective discrimination after loss

The pure-loss bound can be sharpened to the limiting optimal testing curve. Let $u = E/n^\beta$, fix $0 < \eta < 1$, and keep a_0, b_0, c_0, s_0 as above. Set

$$d_0 = a_0 b_0 - c_0^2 = u^2[\eta(1-\eta)]^n, \quad r_0 = \sqrt{(a_0 - b_0)^2 + 4c_0^2}, \quad \ell_0 = (s_0 - r_0)/2 \le 2d_0/s_0.$$

Replace each active block by $s_0|\phi_\pm\rangle\langle\phi_\pm|$, where $\phi_\pm$ is its dominant eigenvector, and leave the common orthogonal remainder unchanged. Call the resulting states $\tau_\pm$ and the true outputs $\sigma_\pm$. Then $D(\sigma_\pm, \tau_\pm) = \ell_0$ and

$$\gamma := |\langle\phi_+|\phi_-\rangle|^2 = \frac{(a_0 - b_0)^2}{r_0^2}.$$

The block labels in B copies are orthogonal and have the same law under both hypotheses. Conditional on $M \sim \mathrm{Bin}(B, s_0)$ active copies, the pure-state trace distance is $\sqrt{1-\gamma^M}$. Therefore

$$\left| D(\sigma_+^{\otimes B}, \sigma_-^{\otimes B}) - \mathbb{E}\sqrt{1-\gamma^M} \right| \le 2B\ell_0 \le 4Bd_0/s_0. \tag{D.14}$$

This is an explicit finite bound, not an asymptotic substitution.

If $Bu\eta^n \to c \in (0, \infty)$, its remainder tends to zero, $-Bs_0 \log\gamma \to 4c$, and $\mathrm{Var}(M \log\gamma) \to 0$. Indeed $1 - \gamma = 4c_0^2/r_0^2 \sim 4u\eta^n$ and $s_0 \to 1$; the variance is at most $B(1-s_0)(\log\gamma)^2 = o(1)$. Bounded convergence in probability in (D.14) yields

$$\boxed{P_{\mathrm{succ}}^{\mathrm{collective}} \longrightarrow \frac{1 + \sqrt{1 - e^{-4c}}}{2}.} \tag{D.15}$$

The same limit at $c = 0$ follows from (D.12); for diverging c, the exact formula also gives $1 - F \ge 2c_0^2/s_0$, so product fidelity tends to zero and success tends to one by $D \ge 1 - F$. In particular, for fixed $p \in (1/2, 1)$ the optimal known-pair copy threshold has the sharp equivalent

$$B_p \sim \frac{-\log(4p(1-p))}{4u\eta^n}. \tag{D.16}$$

Monotonicity in B and the strictly increasing continuous limiting curve justify the inversion. This is not a sharp leading constant for global minimax tomography over all states.

67.5 A finite half-loss separation with certified upper and lower bounds

For (C.19), before noise precisely ten copies attain optimal success $3/4$. After $\eta = 1/2$, write $h = 2^{-128}$ and

$$u = 1/128, \quad a_0 = 1 - u + uh, \quad b_0 = uh, \quad c_0^2 = u(1-u)h.$$

All quantities needed for the following certificate are rational. The optimal output copy threshold obeys

$$\boxed{3 \cdot 10^{39} < B_{3/4} \le 4 \cdot 10^{39}.} \tag{D.17}$$

Here $B_{3/4}$ permits arbitrary joint POVMs on the outputs.

For the lower bound let $B_- = 3 \cdot 10^{39}$ and

$$g_* := \frac{4c_0^2}{2s_0 - 4c_0^2/s_0}.$$

The inequality $\sqrt{1-x} \ge 1 - x$ implies $1 - F \le g_*$. The exact rational check $2B_- g_*/(1-g_*) < 5/18 < \log(4/3)$, where the second inequality follows by integrating or truncating the alternating logarithm series, gives $F^{2B_-} > 3/4$. Thus output trace distance is less than $1/2$ and success is less than $3/4$.

For the upper bound let $B_+ = 4 \cdot 10^{39}$, $m_0 = \lfloor 9B_+ s_0/10 \rfloor$ and $v_0 = 4c_0^2/r_0^2 = 1 - \gamma$. The following are exact rational inequalities:

$$m_0 v_0 > 8/25, \qquad \frac{B_+ s_0(1-s_0)}{(B_+ s_0 - m_0)^2} < 10^{-3}, \qquad \frac{4B_+ d_0}{s_0} < 10^{-3}.$$

Chebyshev's inequality makes $M \geq m_0$ have probability at least 0.999. Also $\gamma^{m_0} \leq e^{-m_0 v_0} < e^{-8/25} < 7399/10000$; the last bound follows from $1+(8/25)+(8/25)^2/2+(8/25)^3/6 > 10000/7399$. Hence $\sqrt{1-\gamma^{m_0}} > 0.51$. Equation (D.14) gives distance greater than $0.999 \cdot 0.51 - 0.001 > 1/2$, proving the upper copy threshold. Python rational arithmetic and independent C++ rational arithmetic verify all six certificate inequalities. This includes neither state-preparation cost nor an efficient implementation of the optimal POVM.

67.6 What has and has not been established

At fixed nonzero Gaussian noise, (D.2) is equivalent to copy cost $\exp(\Theta(\varepsilon^{-2/\beta}))$ at fixed mode and moment parameters. The lower bound now concerns quantum information lost to an inaccessible environment, not merely a restriction to heterodyne detection. Randomized homodyne achieves the upper order; collective measurements cannot improve it. The sharp full-minimax transition as noise vanishes, efficient implementation constants, and joint large-mode asymptotics are not proved.

Noisy homodyne deconvolution and minimax Wigner-function estimation are established subjects [31, 32]. Their analytic/Sobolev classes and Wigner-function losses must not be identified with the sole photon-moment constraint and input trace distance here. General continuous-variable state-learning under moment conditions is also prior work [27]. The proposed claim is the matching collective-output inverse rate on the stated class, with the explicit pure-loss critical law and finite certificate. External priority and independent referee acceptance remain unestablished.

68 Uniform two-state discrimination through pure loss

The aperture and heterodyne minimax theorems above concern measurement restrictions. A separate question is how a known bosonic loss channel changes the intrinsic distinguishability of the hard two-level pair used in those lower bounds. The following result gives a joint small-signal/loss limit in which the transmissivity may vary with the photon number and copy budget. It is a binary discrimination theorem, not a full-tomography minimax theorem.

Let

$$|\psi_n^{\pm}\rangle = \sqrt{1-u}|0\rangle \pm \sqrt{u}|n\rangle, \qquad n \geq 1, \quad 0 < u \leq \frac{1}{16},$$

and let the one-mode pure-loss channel of transmissivity $0 < \eta \leq 1$ produce states $\sigma_{\pm}$. For $B \geq 1$ copies put $t = Bu\eta^n$.

Theorem 68.1 (Uniform pure-loss discrimination law)**.** *For every admissible u, η, n, B,*

$$\boxed{\left| D(\sigma_+^{\otimes B}, \sigma_-^{\otimes B}) - \sqrt{1-e^{-4t}} \right| \leq 5ut + \sqrt{u(43t + 8\sqrt{t})}.} \tag{PL.1}$$

Here $D(\rho,\tau) = \frac{1}{2}\|\rho - \tau\|_1$.

Proof. All loss events except the zero- and n-photon components are diagonal and identical under the two hypotheses. On $\mathrm{span}\{|0\rangle, |n\rangle\}$ the subnormalized blocks are

$$M_{\pm} = \begin{pmatrix} a & \pm c \\ \pm c & b \end{pmatrix}, \quad a = 1 - u + u(1-\eta)^n, \quad b = u\eta^n, \quad c^2 = u(1-u)\eta^n.$$

Put $q = u\eta^n$, $s = a + b$,

$$r = \sqrt{(a-b)^2 + 4c^2}, \qquad d = ab - c^2.$$

Direct algebra gives

$$d = u^2[\eta(1-\eta)]^n, \qquad 1-u \le s \le 1, \qquad r^2 \ge (1-2u)^2.$$

Let $\ell = (s-r)/2$ be the smaller block eigenvalue. Replace each block by s times its top normalized eigenvector, leaving the common orthogonal remainder unchanged. The trace distance from one original state to its replacement is exactly ℓ. Tensor-product telescoping and the triangle inequality therefore change the binary B-copy trace distance by at most $2B\ell$. Since

$$\ell = \frac{2d}{s+r} \le \frac{2d}{s},$$

we obtain

$$2B\ell \le \frac{4Bu^2\eta^n}{1-u} < 5ut. \tag{PL.2}$$

The squared overlap of the two top eigenvectors is

$$\gamma = \frac{(a-b)^2}{r^2}.$$

In the replacement model the number M of tensor factors in the non-common block has law $\mathrm{Bin}(B,s)$. Orthogonality of the block labels gives exactly

$$D_{\mathrm{repl}} = \mathbb{E}\sqrt{1-\gamma^M}. \tag{PL.3}$$

Set $\delta = 1-\gamma$ and $L = -\log\gamma$. With $z = (1-\eta)^n$, $e = \eta^n$ and $h = 1-z-e \in [0,1]$,

$$r^2 = 1 - 2uh + u^2(1-z+e)^2.$$

Using $u \le 1/16$ gives

$$\delta = \frac{4q(1-u)}{r^2}, \qquad \delta \le 5q, \qquad |\delta - 4q| \le 16uq. \tag{PL.4}$$

Because $\delta \le 5/16$ and $0 \le -\log(1-\delta) - \delta \le \delta^2/[2(1-\delta)]$,

$$|L - 4q| \le 35uq, \qquad L \le 8q. \tag{PL.5}$$

Moreover $1 - s = uh \le u$, hence

$$|BsL - 4t| \le 39ut \le 43ut. \tag{PL.6}$$

The binomial fluctuation satisfies

$$\mathbb{E}|ML - BsL| \le L\sqrt{Bs(1-s)} \le 8u\sqrt{t}. \tag{PL.7}$$

Finally, for $x, y \ge 0$, $|\sqrt{1-e^{-x}} - \sqrt{1-e^{-y}}| \le \sqrt{|x-y|}$. Apply this to $x = ML$, $y = 4t$, use Jensen's inequality, and combine the resulting bound with (PL.2)–(PL.7). This proves (PL.1). □

Corollary 68.2 (Joint weak-loss/small-signal transition)**.** *If $u_j \to 0$ and $B_j u_j \eta_j^{n_j} \to c \in [0,\infty)$, then*

$$P_{\mathrm{succ}}^{\mathrm{opt}} \longrightarrow \frac{1+\sqrt{1-e^{-4c}}}{2}.$$

For any fixed $p \in (1/2, 1)$, the least copy budget attaining asymptotic success p satisfies

$$B_p \sim \frac{-\log(4p(1-p))}{4u\eta^n}, \qquad \frac{B_p(\eta)}{B_p(1)} \sim \eta^{-n}. \tag{PL.8}$$

If $\eta_n \to 1$, the loss penalty tends respectively to 1, e^{γ}, or infinity when $n(1-\eta_n)$ tends to 0, to $\gamma \in (0,\infty)$, or to infinity.

For the moment-constrained pair $u = E/n^\beta$ this becomes

$$B_p \sim \frac{-\log(4p(1-p))}{4E}\, n^\beta \eta_n^{-n}. \tag{PL.9}$$

The theorem resolves the joint loss limit for this explicit hard pair. It does not identify the minimax law of unrestricted state tomography under a loss channel, and it has no implication for microscopic BGS/GOE/COE spectral statistics.

Chapter 11 — Temporal resonance and spectral modules (continued)

69 Spectral resolution: two exact tests for a proposed BGS argument

The shell theorems determine finite-rank analytic data exactly, but not with uniform microscopic resolution. This section returns to propagator spectra. It gives necessary safeguards for a universality proof, not a proof of BGS.

Proposition 69.1 (Simple spectra invisible to a long initial trace window)**.** *For $N = 2M$, let $U_{N,\mathrm{test}}$ have the Nth roots of unity as eigenvalues and let V_N have the $2M$ distinct eigenvalues*

$$e^{2\pi ij/M + i\pi/(3M)}, \quad e^{2\pi ij/M - i\pi/(3M)}, \qquad 0 \le j < M.$$

Then $\operatorname{Tr} U_{N,\mathrm{test}}^k = \operatorname{Tr} V_N^k = 0$ for every $1 \le k < M$. Nevertheless, with global mean-spacing unfolding, their cyclic nearest-neighbour spacing measures are exactly

$$\mu_{U_{N,\mathrm{test}}} = \delta_1, \qquad \mu_{V_N} = \tfrac{1}{2}\delta_{2/3} + \tfrac{1}{2}\delta_{4/3}.$$

Their Wasserstein distance of order one is $1/3$, independently of N. For their spectral determinants $D_U(z) = \det(I - zU_{N,\mathrm{test}})$ and $D_V(z) = \det(I - zV_N)$,

$$D_U(z) = 1 - z^{2M}, \qquad D_V(z) = 1 - z^M + z^{2M}.$$

In particular $\sup_{|z|=r} |D_U(z) - D_V(z)| \le 3r^M$ for $0 < r < 1$.

Proof. Summing an M-term geometric progression gives the trace assertion and, more generally, $N^{-1}\operatorname{Tr} V_N^{kM} = \cos(k\pi/3)$. The two phases in each cell have unfolded distance $2/3$, and the next gap has length $4/3$. Their determinants are the products of $1 - e^{i\pi/3}z^M$ and $1 - e^{-i\pi/3}z^M$. The claimed difference bound follows from $D_V - D_U = -z^M + 2z^{2M}$. The distance to the point mass at one is the mean absolute deviation, namely $1/3$. □

For the normalized Carathéodory transform

$$G_U(z) = N^{-1}\operatorname{Tr}\{(I + zU)(I - zU)^{-1}\} = 1 + 2\sum_{k\ge 1} z^k N^{-1}\operatorname{Tr} U^k,$$

the same trace formula gives

$$\sup_{|z|=r} |G_{U_{N,\mathrm{test}}}(z) - G_{V_N}(z)| \le \frac{4r^M}{1 - r^M}.$$

Consequently the determinants approach each other and these transforms both approach one on the circles $r_N = \exp(-s_N/N)$ whenever $s_N \to \infty$. Their microscopic spacing measures remain

separated. This is an explicit resolution obstruction even with simple spectra and bounded unfolded gaps. It does not say that either clock construction is a quantization of a chaotic map. It says that short traces and unquantified mesoscopic resolvent convergence cannot, as abstract spectral hypotheses, identify the local law. At $r_N = \exp(-s/N)$ with fixed s, the quantity $r_N^M = e^{-s/2}$ no longer vanishes; the missing boundary-layer scale is precisely of order $1/N$.

Proposition 69.2 (First-order Egorov alone cannot force local repulsion)**.** *Suppose $U_{N,\mathrm{test}}$ is any unitary quantization satisfying, for fixed observables a,*

$$\|U_{N,\mathrm{test}}^* \operatorname{Op}_N(a) U_{N,\mathrm{test}} - \operatorname{Op}_N(a \circ T)\| = O_a(N^{-1}), \qquad \sup_N \|\operatorname{Op}_N(a)\| < \infty.$$

Fix an integer $L \geq 2$ and take $M_N = \lfloor N/L \rfloor$, for $N \geq L$. There are unitaries $\widetilde{U_{N,\mathrm{test}}}$ with $\|\widetilde{U_{N,\mathrm{test}}} - U_{N,\mathrm{test}}\| = O_L(N^{-1})$, the same displayed Egorov estimate, and at least $N - M_N$ zero cyclic spacings. There is also a simple-spectrum version whose limiting spacing measures still have an atom of mass at least $1 - 1/L$ at zero, whenever a weak limit is taken.

Proof. Use the spectral theorem to round every eigenangle to a nearest M_Nth root of unity. The change in operator norm is at most $2\sin(\pi/(2M_N)) \leq \pi/M_N = O_L(N^{-1})$. There are at most M_N occupied roots, hence at least $N - M_N$ zero spacings. For any bounded operator B the conjugation difference is at most $2\|B\|\|\widetilde{U_{N,\mathrm{test}}} - U_{N,\mathrm{test}}\|$, which preserves the assumed Egorov bound.

For the simple-spectrum version, split each occupied root into distinct angles in an arc of half-width ε_N/N, where $\varepsilon_N > 0$ tends to zero. These arcs can be chosen disjoint. At least $N - M_N$ spacings are then at most $2\varepsilon_N/N$, or ε_N/π after unfolding by $N/(2\pi)$. The extra norm change is at most ε_N/N and preserves Egorov. For any fixed $\eta > 0$ the limiting mass in the closed interval $[0, \eta]$ is at least $1 - 1/L$ by the portmanteau inequality; let $\eta \downarrow 0$. □

This conclusion applies to the stated weak quantization axioms regardless of the classical map. It is not a counterexample to a formulation imposing a fixed smooth microlocal symbol class or a specified canonical quantization: the spectral rounding is not shown to belong to such a class. In particular, neither this construction nor the clock pair can replace the missing Heisenberg-scale joint-trace estimate for the maintained kicked propagator. Their purpose is to reject proof attempts that use only first-order Egorov, classical covariance positivity, control Lie rank, or coarse determinant data.

Microscopic rounding with simultaneous low traces. The microscopic-rounding refinement selects the lattice rounding by an unbiased second-moment argument. For any specified $K_N = o(N^{1/3})$, its simple-spectrum perturbation preserves the whole vector of unnormalized traces through K_N with ℓ^2 error $o(1)$. Operator-norm continuity also preserves the coarse position-chain gap, entry flatness, reversed-entry decay and fixed-observable first-order Egorov bounds. Nevertheless a fixed stable-band pair statistic has $\liminf > 1/2$. The canonical proof and exact scope are in `mathematics/microscopic_rounding_control.md`. This extends the preceding project rounding proposition and changes the quantum operator; it retains no claim of exact quantization or subprincipal-symbol preservation. The actual deterministic BGS target remains open.

Chapter 12 — Fixed-map frontier

70 Fixed-map spectral statistics: exact reductions and the remaining estimate

Collective-diffusion continuation claim: BGS-FM-019.

Claim IDs: BGS-FM-005, BGS-FM-006, BGS-FM-007, BGS-FM-008, BGS-FM-009, BGS-FM-010, BGS-FM-011, BGS-FM-012, BGS-RES-005, BGS-RES-006, BGS-RES-007, BGS-LOOP-001, BGS-TRACE-005, BGS-TRACE-007, BGS-TRACE-008, BGS-CUE-001, BGS-FLOW-001, BGS-FLOW-002, BGS-FLOW-003, BGS-FLOW-004, BGS-FLOW-005, BGS-FLOW-006, BGS-FLOW-007, BGS-FLOW-008, BGS-FLOW-009, BGS-FLOW-010, BGS-FLOW-011, BGS-FLOW-012, BGS-FLOW-013, BGS-FLOW-014, BGS-FLOW-015, BGS-FLOW-016, BGS-FLOW-017, BGS-FLOW-018, BGS-FLOW-019, BGS-FLOW-020, BGS-FLOW-021, BGS-FM-018, BGS-FM-017, BGS-FM-013, BGS-FM-016, BGS-TRACE-013, BGS-TRACE-010, BGS-TRACE-011, BGS-BGS-001

Status, 6 October 2026. The fixed-map BGS target is open. This chapter consolidates the recovered research, states the deterministic all-order criterion and proves its auxiliary original-chain version. The supplement contains full derivations, programs, finite certificates and dated audits. External priority is not claimed. Formalization covers only explicitly stated algebraic kernels.

70.1 The unchanged target

For odd $N \geq 3$, use raw modular Weyl operators

$$(T_N(a,b)\psi)(x) = e_N\big(b(x+2^{-1}a)\big)\psi(x+a), \qquad e_N(t) = \exp(2\pi i t/N),$$

with modular inverses. Let $S = \left(\begin{smallmatrix} 3 & 4 \\ 2 & 3 \end{smallmatrix}\right)$ and

$$(M_N)_{xy} = N^{-1/2} e_N\big(-8^{-1}(3x^2 - 2xy + 3y^2)\big), \qquad U_N = \exp(i\tau N A_N) M_N, \quad \tau = 10^{-80}.$$

Here A_N quantizes $f - 1/10$ using $e^{2\pi i(kq+lp)} \mapsto T_N(l,k)$, where

$$\begin{aligned} r(q) &= (1-\cos 14\pi q)(1+\cos 2\pi q), \\ f(q,p) &= \cos 14\pi q + \frac{r(q)}{10} + \frac{r(q)\sin 2\pi q}{100} + \frac{r(q)\cos(2\pi p - \alpha_0)}{100}, \\ e^{i\alpha_0} &= (-3+4i)/5. \end{aligned}$$

Centering changes only a common spectral rotation. Neither an auxiliary coupling average nor independent phases replace this fixed family.

Writing $p_m = \operatorname{Tr} U_N^m$, the pair target can be expressed as

$$L_N(s) := N^{-2} \sum_{m \geq 1} e^{-2sm/N} |p_m|^2 \longrightarrow \frac{1 - e^{-2s}}{4s^2}, \qquad s > 0.$$

The determinant-ratio reduction proves equivalence with all compact weighted pair tests, with an explicit positive-definite tail bound. No estimate establishing this limit for the displayed U_N has been obtained. All-order universality needs more than this pair target.

70.2 An exact adjoint-resolvent form of the pair target

Claim ID: `BGS-RES-006`. See Appendix A.

Theorem 70.1 (Exact adjoint-resolvent form of the pair target)**.** *This identity is finite-dimensional and holds for every unitary $N \times N$ matrix, not just the frozen family. On the Hilbert–Schmidt space $M_N(\mathbb{C})$ define*

$$\mathcal{U}_N(X) = U_N X U_N^*, \qquad \mathcal{D}_N = \frac{N^2}{4}\,(2I - \mathcal{U}_N - \mathcal{U}_N^*)\,, \qquad T_N(s) = \frac{1}{N}\,\mathrm{Tr}_{HS}(\mathcal{D}_N + s^2 I)^{-1}.$$

Summing the Poisson series with $q = e^{-2s/N}$ gives the exact identity

$$1 + 2L_N(s) = N \sinh(s/N) \cosh(s/N)\, T_N(N \sinh(s/N))\,.$$

For every unitary spectrum and every $s > 0$ one also has the uniform comparison

$$0 \le \frac{1 + 2L_N(s)}{s} - T_N(s) \le \frac{2}{3N} \cosh(2s/N). \tag{R1}$$

The map $\mathcal{U}_N$ is unitary on Hilbert–Schmidt space, so $\mathcal{D}_N \ge 0$. If $e^{i\theta_j}$ are the eigenvalues of U_N, the matrix units in its eigenbasis diagonalize $\mathcal{D}_N$, with eigenvalues $N^2 \sin^2((\theta_j - \theta_k)/2)$.

To see the sign and the error, set $a = s/N$ and $y_{jk} = N^2 \sin^2((\theta_j - \theta_k)/2)$. The ratio of the (j,k) summand in $(1 + 2L_N(s))/s$ to its summand in $T_N(s)$ is

$$\frac{\sinh(2a)}{2a}\,\frac{y_{jk} + s^2}{y_{jk} + N^2 \sinh^2 a}.$$

It lies between 1 and $\sinh(2a)/(2a)$. Also $T_N(s) \le N/s^2$, and $\sinh x/x - 1 \le x^2 \cosh(x)/6$ for $x \ge 0$, which proves (R1). Consequently, for each fixed $s > 0$,

$$L_N(s) \longrightarrow \frac{1 - e^{-2s}}{4s^2} \quad \Longleftrightarrow \quad T_N(s) \longrightarrow \frac{1}{s} + \frac{1 - e^{-2s}}{2s^3}.$$

This is a reformulation of the same open pair limit. It gives no estimate for $T_N(s)$ for the prescribed U_N and no higher-order joint-trace comparison. The detailed audit and the companion exact spectral-flow and control identities also rule out collapsing the actual iterated kicks into an unordered single exponential; they do not estimate the ordered product at $m \asymp N$.

70.3 A universal finite-dimensional Abel floor

Claim ID: `BGS-TRACE-007`. See Appendix A.

Theorem 70.2 (Universal finite-dimensional Abel floor)**.** *Every N-point unitary spectrum satisfies*

$$L_N(s) \ge \frac{1}{e^{2s} - 1}, \qquad s > 0, \tag{R2}$$

and the bound is attained by a rotated regular N-gon.

Here is the precise energy reduction. For $N = 1$ the bound is equality directly. For $N \ge 2$, put $z_j = e^{i\theta_j}$ and

$$f_q(r) = \frac{1 - q^2}{(1 - q)^2 + qr}, \qquad 0 < q < 1.$$

This is strictly completely monotone on $[0,4]$. The regular N-gon is a sharp configuration on S^1: it is a circular $(N-1)$-design and has $\lfloor N/2 \rfloor$ distinct inner products between different vertices. Cohn–Kumar, Theorem 1.2 in *Universally optimal distribution of points on spheres*, therefore minimizes $\sum_{j\neq k} f_q(|z_j - z_k|^2)$. The diagonal terms of the full ordered sum are configuration-independent. If eigenangles coincide, perturb them to distinct angles and pass to the limit; f_q is continuous at zero. Since

$$\frac{1-q^2}{|1-qe^{it}|^2} = \sum_{m\in\mathbb{Z}} q^{|m|} e^{imt},$$

the minimized full sum is $N^2 + 2\sum_{m\geq 1} q^m |p_m|^2$. For the regular polygon $p_m = N$ exactly when $N \mid m$, and is zero otherwise. Taking $q = e^{-2s/N}$ yields (R2).

The CUE benchmark is strictly above this floor:

$$\frac{1-e^{-2s}}{4s^2} - \frac{1}{e^{2s}-1} = \frac{e^{-s}(\sinh^2 s - s^2)}{2s^2 \sinh s} > 0.$$

Thus (R2) is a sharp universal lower bound but does not identify the frozen model's limit or give a uniform excess over the clock value.

70.4 Smooth resolvent transfer for the seven-cut *full* propagator

Claim ID: `BGS-RES-007`. See Appendix A.

Theorem 70.3 (Smooth resolvent transfer for the seven-cut full propagator)**.** *Let U_N and V_N be the original and seven-cut **full** propagators of `BGS-FM-007`. For an arbitrary unitary W write $L_{W,N}$ and $T_{W,N}$ for the pair Abel and adjoint-resolvent functions above. For every compact $K \subset (0,\infty)$ and every integer $r \geq 0$, there are constants $C_{K,r}, N_{K,r}$ such that, for odd $N \geq N_{K,r}$,*

$$\sup_{s\in K} |\partial_s^r (L_{U,N}(s) - L_{V,N}(s))| \leq C_{K,r} \frac{\tau}{\log N}, \tag{R3}$$

$$\sup_{s\in K} |\partial_s^r (T_{U,N}(s) - T_{V,N}(s))| \leq C_{K,r} \left(\frac{\tau}{\log N} + \frac{1}{N} \right). \tag{R4}$$

Thus $T_{U,N} - T_{V,N} \to 0$ in $C^\infty_{\mathrm{loc}}(0,\infty)$.

This is equivalence of any possible scalar-resolvent limits, not convergence of either resolvent to the CUE value.

Proof. Formula (13) of `BGS-FM-007` expresses $L_{W,N}$ as $\frac{1}{2} C^W_{2,N}(h_s)$ plus terms independent of W, where $h_s(x) = s/(s^2 + \pi^2 x^2)$. All Abel series and their fixed-order s derivatives converge absolutely on K at each N. For every r, the derivatives $\partial_s^r h_s$ and their x derivatives have uniformly summable unit-interval suprema as s runs over K; the x derivative is $O_{K,r}(|x|^{-3})$ at infinity. Apply the phase-coupling estimate (7) there to these differentiated tests. Its coupled total displacement is $O(\tau)$, and the seven-cut occupancy is $O(N/\log N)$. The spectrum-independent terms cancel, giving (R3) uniformly on K.

It remains to pass from the Abel functions to the resolvents **with derivatives**, since (R1) alone bounds only their values. Set $y_{jk} = N^2 \sin^2((\theta_j - \theta_k)/2)$, $c_N(s) = \sinh(2s/N)/(2s/N)$, and $b_N(s) = N^2 \sinh^2(s/N)$. The exact identity preceding (R1) gives

$$R_{W,N}(s) := \frac{1 + 2L_{W,N}(s)}{s} - T_{W,N}(s) = \frac{1}{N} \sum_{j,k=1}^{N} \left[\frac{c_N(s)}{y_{jk} + b_N(s)} - \frac{1}{y_{jk} + s^2} \right]. \tag{R5}$$

On a fixed small complex neighborhood of K in $\Re s > 0$ and for all sufficiently large N, the expansions $c_N(s) = 1 + O_K(N^{-2})$ and $b_N(s) = s^2 + O_K(N^{-2})$ hold uniformly. Both denominators have modulus at least $c_K(1 + y_{jk})$ there. Hence each bracket in (R5) is bounded by $C_K N^{-2}/(1 + y_{jk})$. The N^2 terms and the exterior $1/N$ give $\sup |R_{W,N}| \le C_K/N$ uniformly over **every** unitary W. Cauchy's derivative estimates on a slightly smaller neighborhood yield $\sup_{s\in K} |\partial_s^r R_{W,N}(s)| \le C_{K,r}/N$. Subtract (R5) for U_N, V_N and use (R3), together with the bounded derivatives of $1/s$ on K, to obtain (R4). □

The proof exploits the already established full-propagator phase coupling; it neither decomposes V_N into independent spectra nor evaluates its scalar resolvent at Heisenberg times. The all-order mixed-resolvent obligation `BGS-TRACE-005` therefore remains open. The separate `BGS-TRACE-008` two-clock example also proves that the necessary Gram moment $4/3$, simple phases, macroscopic Haar convergence, uniformly bounded unfolded unit occupancy and nonvanishing traces cannot replace that obligation.

70.5 What the recovered work establishes

For the actual frozen kick, the exact Jacobi-chain decomposition has $\gcd(N, 7)$ irreducible blocks for every odd dimension. With the cited Hermite–Lindemann input, it implies $\operatorname{rank}(U_N - zM_N) \ge N - \gcd(N, 7)$ for every complex z and every nonzero algebraic coupling. Its approximate-rank bound applies only before eigenphase wrapping, $|\tau N|(79/50) \le \pi$; it supplies no fixed-strength asymptotic control. Separately, the full-lag classical gap proves a uniform translated Riesz bound for the exact physical symbol under the *linear* cat map, including small weighted-Fourier perturbations. It does not treat transport by the nonlinear kicked map. The nonlinear bracket pressure theorem shows that this same λ^{-r} normalization cannot have a uniform upper bound for the **actual** nonlinear map: its individual bracket norms grow at a strictly larger exponential rate. This result is classical and uses thermodynamic-formalism inputs; none of these results bounds the Heisenberg-scale joint traces needed for `BGS-TRACE-005`.

An exact seven-cut reduction now gives $\|U_N - e^{i\tau N A_N^{\mathrm{cut}}} M_N\|_1 \le 28\pi^2\tau/(25N)$ for the same fixed coupling. The two **full** propagators have equivalent fixed-order sine-kernel targets. Their pair and full Abel statistics differ by $O_s(\tau/\log N)$ without assuming a CUE limit. The seven paths belong only to the generator: every proper union of paths exchanges Hilbert–Schmidt mass $d(N - d)/N$ with its complement under the full propagator. Thus `BGS-TRACE-005` can be attacked for either full propagator, with no claim of independent seven-block spectra. The all-minor and simple-spectrum theorem removes exact level multiplicity when $7 \nmid N$, while the integer commutator-cube certificate excludes a common antiunitary reversal of (A_N, M_N) except at $N = 7$. These results are qualitative and do not estimate near-degenerate spacings. The arithmetic trace identity and rigidity theorem also proves exact nonvanishing of every $\operatorname{Tr} U_N^m$ on the odd sequence and classifies scalar powers. The effective gap floor makes spectral simplicity quantitative for $7 \nmid N$, but its bound is far below the mean spacing N^{-1}. Neither result supplies an upper bound for Heisenberg-time traces or the open Abel limit.

Exact algebraic arguments exclude scalar-algebraic returns for the stated frozen family except its explicitly treated raw $N = 7$ exception. They do not give microscopic eigenvalue separation. The new qualitative simplicity theorem covers all odd $7 \nmid N$, whereas the older *quantitative* finite-size certificates cover only specified dimensions. Structured ergodic comparison controls unfolded RMS motion of subprincipal perturbations under its stated Egorov and ergodicity hypotheses; transferring factorial correlations also needs local occupation moments.

The block-memory analysis distinguishes closed quantum propagation from reset evolution. Mixing of the positive reset chain does not imply decay of coherent quantum memory. Actual weighted principal-minor computations show cancellation within and across frequency windows. They do not justify independent-phase substitution. A rotated deleted clock has exactly the CUE characteristic-polynomial covariance and nevertheless regular spacings; covariance alone

therefore does not determine pair statistics.

Full statements, assumptions and proofs are in the topic-indexed supplement. The shadow-tube analysis, recovered from the remote branch, likewise keeps its geometric equidistribution assumption separate from its proved symbolic filling bound. Both local and remote research lines are retained.

70.6 A conditional route to every compact correlation order

For eigenangles θ_j in radians and a uniform common origin $\alpha \in [0, 2\pi)$, put

$$s_N(w) = \frac{1}{N} \sum_j \cot\left(\frac{\theta_j - \alpha}{2} - \frac{w}{N}\right).$$

This Herglotz function has unit-mass particles at $N(\theta_j - \alpha)/2 + \pi N\ell$, and intensity $1/\pi$. Define, using holomorphic extension at coincident arguments,

$$\mathcal{E}_{p,N}(\mathbf{w}) = \mathbb{E}[(s_N(w_1)^2 + 1) \prod_{j=2}^{p} s_N(w_j)] + \sum_{j=2}^{p} \mathbb{E}\left[\partial_{w_j} \frac{s_N(w_1) - s_N(w_j)}{w_1 - w_j} \prod_{\ell=2, \ell\neq j}^{p} s_N(w_\ell)\right].$$

Theorem 1.9 of Bourgade–Huang, *Loop Equations Characterize Random Matrix Statistics*, arXiv:2607.07617v2, provides the external convergence theorem. Our sufficient model hypothesis is

$$\sup_{\mathbf{w}\in K^p} \frac{|\mathcal{E}_{p,N}(\mathbf{w})|}{1 + \mathbb{E}\max_j |s_N(w_j)|^{p+1}} \longrightarrow 0$$

for every fixed p and every compact $K \subset \mathbb{C} \setminus \mathbb{R}$. The event in that theorem is the whole origin probability space. Its moment bootstrap then bounds all local particle-count moments, since $\mu_N([-B, B]) \le (1 + B^2)|s_N(i)|$. Uniform integrability upgrades weak point-process convergence to all compact factorial correlation tests. Dividing positions by π and integrating the origin gives exactly the usual deterministic anchored statistic, with normalization $1/N$. The detailed bridge checks these steps. This is a conditional implication; the model hypothesis is unproved. It does not assert pointwise correlation densities.

The single-anchor and dense-test reduction shows that all fixed anchor orders and pointwise tests on countably dense mixed tuples suffice for this whole-space compact-uniform hypothesis. FM-007 further permits normalized first-mixed short-leg deletion with error $O(N^{-1/40} \log N)$ on fixed positive compact height intervals, keeping the original full-resolvent denominator and leaving an exact long-long tensor-resolvent target; the actual anchors, signed target and full hierarchy remain open. FM-007 is needed only for this deletion, not for the external characterization or the abstract testing reduction.

70.7 The first two mixed equations

For arguments in the same half-plane every loop residual vanishes for every unitary spectrum. Indeed, the nonconstant Fourier frequencies in the uniform origin all have the same strict sign. Such tests cannot verify universality.

Fix $s > 0$, $w = is/2$, $v = -is/2$, $q = e^{-2s/N}$, and define

$$C_N = N^{-3} \sum_{a,b\geq 1} q^{a+b} p_{-a} p_{-b} p_{a+b}.$$

Absolute Fourier convergence gives the exact mixed identity

$$\mathcal{E}_{2,N}(w,v) = 8i \left(L_N + C_N - \frac{1}{4s^2} \right).$$

The cubic quantity C_N need not be real. Finite circular CUE satisfies $\mathbb{E}(L_N + C_N) = 1/(4N^2 \sinh^2(s/N))$, so the limiting straight-line equation has a finite-size correction even in the reference ensemble.

Put $Z = s_N(w)$, $M_r^{\mathrm{mom}} = \mathbb{E}|Z|^r$, $e_2 = L_N + \Re C_N - 1/(4s^2)$, and $e_3 = \mathcal{E}_{3,N}(w,v,v)$. Then

$$e_3 = M_4^{\mathrm{mom}} - 1 - \frac{2}{s^2}(M_2^{\mathrm{mom}} + 1) \in \mathbb{R}.$$

To prove this, write $T = s_N(v) = \bar{Z}$. Same-half-plane identities give $\mathbb{E}T^2 = -1$ and $\mathbb{E}(T'T) = 0$. Each differentiated quotient, multiplied by T, has expectation $-(M_2^{\mathrm{mom}} + 1)/s^2$. There are two such terms; the leading expectation is $M_4^{\mathrm{mom}} - 1$.

For $P = (Z/i - 1)/2$ and $Q = \mathbb{E}|P|^4$, Fourier orthogonality yields $M_2^{\mathrm{mom}} = 1 + 4L_N$ and $M_4^{\mathrm{mom}} = 1 + 16L_N + 32\Re C_N + 16Q$. Eliminating C_N gives

$$Q = \left(1 + \frac{1}{2s^2}\right) L_N - \frac{1}{4s^2} + \frac{e_3}{16} - 2e_2.$$

Consequently $Q \geq L_N^2$ implies

$$L_N^2 - \left(1 + \frac{1}{2s^2}\right) L_N + \frac{1}{4s^2} \leq \frac{e_3}{16} - 2e_2.$$

These conclusions require no probabilistic independence. If $|e_3| \leq \delta_N(1 + M_4^{\mathrm{mom}})$ with $\delta_N \to 0$, Jensen gives, for $x = M_2^{\mathrm{mom}}$ and $t = s^{-2}$,

$$(1 - \delta_N)x^2 - 2tx - (1 + \delta_N + 2t) \leq 0.$$

Thus $M_2^{\mathrm{mom}}, M_4^{\mathrm{mom}}$ are bounded and $e_3 \to 0$. A normalized vanishing $p = 2$ residual now also vanishes without normalization. Every subsequential limit ℓ of $L_N(s)$ lies in

$$\frac{1}{2} + \frac{1}{4s^2} - \frac{1}{2}\sqrt{1 + \frac{1}{4s^4}} \leq \ell \leq \frac{1}{2s^2}.$$

The interval has positive width. These low-order bounds do not identify the CUE value or prove convergence; no realizable non-CUE process satisfying the two vanishing residuals is asserted either. Establishing those residual hypotheses for the actual map remains a dynamical problem.

70.8 Why a direct noise replacement does not close the estimate

There is also an exact identity for depolarizing noise. Let $\mathcal{P}_N(X) = \mathrm{Tr}(X)I/N$ and $\Phi_{\eta,N} = e^{-\eta}\mathcal{U}_N + (1 - e^{-\eta})\mathcal{P}_N$ for $\eta \geq 0$. The scalar matrices have eigenvalue 1 under $\Phi_{\eta,N}$; on traceless matrices it acts as $e^{-\eta}\mathcal{U}_N$. Therefore

$$\mathrm{Tr}_{HS}\, \Phi_{\eta,N}^m = 1 + e^{-m\eta}(|p_m|^2 - 1).$$

For $L_{\eta,N}(s) = N^{-2} \sum_{m\geq 1} e^{-2sm/N} \mathrm{Tr}_{HS}\, \Phi_{\eta,N}^m$, this gives

$$L_{\eta,N}(s) = L_N(s + N\eta/2) + \rho_N(s,\eta), \qquad \rho_N(s,\eta) = \frac{1}{N^2}\sum_{m\geq 1} e^{-2sm/N}(1 - e^{-m\eta}),$$

with $0 \leq \rho_N(s,\eta) \leq (2Ns)^{-1}$. Noise shifts the unknown parameter from s to $s + N\eta/2$; this identity alone does not recover the fixed-s pair limit or control Heisenberg-time traces.

For the unitary Brownian generator $G = (2N)^{-1}\sum_a D_{H_a}^2$, with $\mathrm{Tr}(H_a H_b) = \delta_{ab}$, matrix completeness gives

$$G|p_m|^2 = m^2 - m|p_m|^2 - \frac{m}{N}\Re\left(\bar{p}_m \sum_{a=1}^{m-1} p_a p_{m-a}\right).$$

Hence exactly

$$\frac{1}{N}GL_N(s) = \frac{1}{2}\partial_s\left[L_N(s) + \Re C_N(s) - \frac{q}{N^2(1-q)^2}\right].$$

The unknown mixed defect reappears in the attempt to remove noise. For an exact clock it is nonzero. Moreover, transport directions $V^j A V^{-j}$ have identical diagonal entries in a simple eigenbasis of V; Gaussian linear combinations therefore give rank at most one for instantaneous eigenphase covariance. Nonlinear bracket effects are not excluded.

A separate elementary obstruction concerns global entropy. On the unfolded circle of length N, let d_N be RMS matching distance to a deterministic configuration, minimized over permutations and common rotation. Labeled CUE density is at most $1/N!$ by Hadamard's inequality. A rotation mesh and a union of Euclidean balls therefore give

$$Q_N(d_N \leq \delta) \leq \min\left\{1, \left\lceil\frac{N}{\delta}\right\rceil (\sqrt{8\pi e}\,\delta)^N\right\}.$$

Binary relative-entropy contraction shows that $d_N \to 0$ in P_N probability forces $D(P_N\|Q_N)/N \to \infty$. Thus global entropy $o(N)$ and microscopic RMS preservation cannot both justify the proposed comparison. This does not exclude local entropy methods or universality. The full proof includes the torus-volume and labeling factors.

70.9 The singular reference boundary and a coboundary control

The Haar-CUE reference proof (BGS-CUE-001) evaluates an unregularized object beyond the fixed-positive-s pair Abel statistic: if $F_{N,2} = \sum_{j\neq k} |e^{i\theta_j} - e^{i\theta_k}|^{-2}$, then $\mathbb{E}F_{N,2} = N(N^2-1)/6$ while its variance is infinite. Centered $N^{-8/3}F_{N,2}$ has a spectrally positive 3/2-stable law, derived from the published Haar-CUE small-gap Poisson limit and a project determinantal tail bound. The same proof covers inverse chord powers $3/2 < p < 3$. Moving one eigenphase by $O(N^{-1})$ can preserve weak local process limits while making the normalized unregularized inverse-gap energy diverge. Therefore neither a weak local limit nor the exact positive-s adjoint-resolvent identity permits taking the singular boundary without an additional gap/uniform-integrability estimate. None has been established for the maintained fixed propagator.

The finite coboundary spectral-flow theorem (BGS-FLOW-001) controls RMS unfolded phase motion by an adjoint Birkhoff average of $H = iN[G, UGU^*]$, with explicit higher-order commutator errors and a sharp abstract clock example. Its needed logarithmic-time average and nested-commutator bounds remain unproved for the frozen family. The related infinitesimal calculation (BGS-FLOW-002) shows that Gaussian coboundary kicks produce a sine drift and no radial Dyson diffusion. The second-order three-exponential normal form itself was previously proved in the Pell coboundary note. These observations leave the mixed Heisenberg-time trace obligation unchanged.

70.10 Audited periodic-point control from the 27 September upload

The all-time annihilator theorem (`BGS-FM-012`) writes $C^m = \left(\begin{smallmatrix} x_m & 2y_m \\ y_m & x_m \end{smallmatrix}\right)$ for $C = \left(\begin{smallmatrix} 1 & 2 \\ 1 & 1 \end{smallmatrix}\right)$, $C^2 = S$, and proves

$$(S^m - I)^T \mathbb{Z}^2 = \begin{cases} 2x_m \mathbb{Z}^2, & m \text{ odd}, \\ 2y_m \mathbb{Z} \times 4y_m \mathbb{Z}, & m \text{ even}. \end{cases}$$

Thus the physical ordinary symbol averages exactly $-1/10, -1/20, 0$ over the linear-cat fixed-point sets at times $1, 2, \geq 3$. This unweighted identity does not estimate an oscillatory Birkhoff sum or quantum trace. The submission audit rejects the proposed upgrade from pair-only TG-mean estimates to all-order CUE. It also corrects the submitted ordinary-symbol translation, bispectrum Fourier sign, distinct-index convention and Abel cutoff accuracy. These controls do not change the Heisenberg-scale joint-trace obligation.

70.11 Quadratic noise and its deterministic-transfer boundary

The quadratic-noise theorem (`BGS-FLOW-003`) goes beyond the second-order radial obstruction. For normalized GUE G, put $X = i[G, UGU^*]/\sqrt{2} - \mathbb{E}(i[G, UGU^*]/\sqrt{2})$ and $m_2 = \operatorname{Tr} U^2/N$. Its exact covariance obeys

$$\frac{1 - |m_2|}{N} \operatorname{Tr} T^2 \leq \mathbb{E}(\operatorname{Tr} XT)^2 \leq \frac{2 + |m_2|}{N} \operatorname{Tr} T^2$$

for every traceless Hermitian T. It is nonsingular precisely when the spectrum is not contained in two antipodal points. Inverting the square root of its covariance yields the infinitesimal traceless Dyson generator. The raw noise is different: for $N \geq 4$, $\int \mathcal{L}_X |\operatorname{Tr} U^2|^2 \, d\mathrm{Haar} = 4 + 2/N^2$.

For a compactly supported smooth local pair statistic, microscopic-time generators differ by at most $C_{R,\phi}|m_2|M^2(1 + \log N) + 2N^{-2} \sup |x^2 \phi''|$ if every mean-spacing arc contains at most M phases. That occupancy bound and its propagation are not proved for the frozen map. A quadratic noise built from d Gaussian directions has covariance rank at most $d(d+1)/2$; finitely many smooth directions and logarithmically many frozen-frame time translates do not yield full instantaneous radial rank. None of these results supplies deterministic noise removal or the required all-order Heisenberg trace comparison.

70.12 Quantitative quadratic-noise dynamics: a scoped continuation

The coupling theorem (`BGS-FLOW-004`) compares the raw diffusion U_t with traceless unitary Brownian motion V_t from the same matrix. For $0 \leq t \leq 1$,

$$N\mathbb{E}\|U_t - V_t\|_{\mathrm{HS}}^2 \leq 10e^3 t(1 + |\operatorname{Tr} U_0|^2 + |\operatorname{Tr} U_0^2|^2).$$

Symmetry of the covariance square root cancels the relative Itô commutator. Every-even-moment estimates strengthen this to $N\|U_{t_N} - V_{t_N}\|_{\mathrm{op}} \to 0$ in probability when the initial first two traces are bounded and $t_N \log N \to 0$. A separate integrated translation estimate gives phase-averaged local Laplace and finite-count comparison for every $t_N \to 0$ with bounded initial low traces, without an occupancy assumption. Those earlier weak-law estimates alone do not give unbounded moments; FLOW-016 supplies a separate high-moment transport below.

The reversible correction (`BGS-FLOW-005`) is the commuting traceless Hermitian drift

$$A = -\operatorname{Im}[(m_2 + m_1^2/2)U^{*2} + (m_2 \bar{m}_1/2)U^*], \qquad m_j = N^{-1} \operatorname{Tr} U^j.$$

Adding D_A puts the generator in Haar divergence form. Moreover $A = -(4N)^{-1}\operatorname{grad}(|p_2|^2 + N^{-1}\operatorname{Re}(p_2\bar{p}_1^2))$. For $N \geq 4K$, initial $|p_1|, |p_2| \leq K/2$, and $t \leq 1/12$, Girsanov and an explicit exit estimate give

$$d_{\mathrm{TV}}(P_X^{[0,t]}, P_R^{[0,t]}) \leq \min(1, 2K\sqrt{t} + 16e^{-K^2/(768t)}).$$

This compares full path laws with the same covariance, not path laws with Dyson covariance. Haar reversibility does not imply ergodicity; scalar matrices are absorbing.

The fixed-trace theorem (`BGS-FLOW-006`) assumes every fixed initial trace is uniformly bounded and the selected initial traces converge. For fixed K, T, it proves joint path convergence to independent circular complex OU processes with variance coefficient $b_k^2 = k^2(1 + \mathbf{1}_{k=2}/2)$ and rates

$$a_X(k) = k/2 - \mathbf{1}_{k=2}/2, \qquad a_R(k) = k/2 + \mathbf{1}_{k=2}/2.$$

Uniform maximal trace moments justify the limit. From zero initial traces the raw second-trace variance is $6(1-e^{-t})$, whereas Dyson has $2(1-e^{-2t})$. Their difference prohibits microscopic operator coupling at fixed positive time. The reversible limit has variance $2(1-e^{-3t})$, with Haar-compatible stationary value two; no finite-N stationary convergence or growing-order theorem is implied.

For the odd clock start, the local-limit theorem (`BGS-FLOW-007`) uses the established circular Dyson clock kernel of Forrester, with a rederived odd-N normalization and explicit alias bound. Both quadratic diffusions converge after phase averaging to Sine_2 at $t_N \to 0$, $Nt_N \to \infty$, with Laplace error $O_f(\sqrt{t} + (Nt)^{-1} + N^{-2})$. The initial clock has exactly one point per unit interval while Sine_2 has a positive hole probability. Thus low traces and bounded occupancy cannot justify generic noise removal.

The direct fixed-time model input (`BGS-FM-013`) is now **proved at project/author-review level**: every fixed nonzero power trace is O_j(1), along all sufficiently large odd N. Its exact Gaussian frame and sampled packet residual are separate from Faure's even-dimensional torus formula. The historical source route, and its sharper action-phase amplitudes remain conditional. The direct Gaussian overlap bound now gives limsup values below 4 and 9 for the first two traces, closing K=20 for sufficiently large odd N. No effective threshold, uniform Heisenberg estimate or all-N initial trace convergence follows. These results leave `BGS-TRACE-005` and `BGS-BGS-001` open. The current audit records the strengthened input and its proof boundary.

70.13 From finite coboundary kicks to a spectral diffusion

The fourth-order calculation (`BGS-FLOW-008`) distinguishes the actual step $U \mapsto e^{i\varepsilon(G-UGU^*)}U$ from the raw quadratic diffusion. In its conjugated logarithm, the expected second-degree Hermitian field is $J = -\operatorname{Im}(m_1U^*)$; a further fourth-degree drift remains even after frozen subtraction of J. Thus covariance alone cannot identify the limit. Both logarithmic mean terms vanish at $m_1 = 0$.

The Möbius reduction (`BGS-FLOW-009`) uses $\Pi(U) = (U - wI)(I - \bar{w}U)^{-1}$, with the small w chosen to set $\operatorname{Tr}\Pi(U) = 0$. The faster field J lies on common Möbius orbits. Phase-averaged local Laplace functionals change by $O_f(|m_1|)$ under centering, without an occupancy assumption. At fixed dimension, killed trace moments control the fast mode and identify the stopped spectral quotient limit at step $\varepsilon^4/2$ with generator $\overline{L}f = L_X(f \circ \Pi)$ on the centered slice. The projected diffusion has an explicit additional Itô drift of operator norm at most $80/N^2$ for $|m_2| \leq 1/2$; it admits a dimension-uniform short-time comparison with Dyson motion.

Consequently, for independent normalized GUE kicks and odd clock start, if $t_N \to 0$ and $Nt_N \to \infty$,

$$\lim_{N\to\infty,\ N\ \text{odd}} \limsup_{\varepsilon\downarrow 0} \left|\mathbb{E}\Phi_{N,f}(U^{\varepsilon}_{\lfloor 2t_N/\varepsilon^4\rfloor}) - \Phi_{\mathrm{Sine}}(f)\right| = 0.$$

The chain contains no centering drift or covariance whitening. This proof first fixes dimension; the following extension supplies an explicit joint regime. No convergence of the original matrix paths is asserted. The initial clock remains a counterexample to generic noise removal. This auxiliary theorem does not establish the deterministic map's Heisenberg joint-trace estimate. The continuation audit records the proof obligations, exact algebra and finite projection checks.

70.14 A dimension-quantitative joint regime for the random chain

The uniform-step theorem (`BGS-FLOW-010`) strengthens the fast-coordinate estimate to $\mathbb{E}[|m_1|; j < \sigma] \le C(\varepsilon/N + \varepsilon^4)$. Its full quartic mean and absolute sixth-order remainder are retained. Smooth localization gives backward derivative growth e^{CNt}; a discrete second-trace Lyapunov estimate controls exits directly. For smooth nonnegative compactly supported tests, the original chain obeys

$$\left|\mathbb{E}\Phi_{N,f}(U^{\varepsilon}_{\lfloor 2t/\varepsilon^4\rfloor}) - \Phi_{\mathrm{Sine}}(f)\right| \le C_f[\sqrt{t} + (Nt)^{-1} + N^{-2} + N^{20}e^{CNt}\varepsilon], \qquad 1/N \le t \le 1/12.$$

Hence $\varepsilon_N = e^{-N}$, $t_N \to 0$ and $Nt_N \to \infty$ give a simultaneous local weak limit. At $t_N = N^{-2/3}$ the smooth-test rate is $O_f(N^{-1/3})$. Nonnegative C^1_c tests follow by approximation, without this rate. The displayed exponential step count is not practical. Its bound also gives a slower polynomial regime at $t = L_N/N$, $L_N \to \infty$, $L_N = o(\log N)$, $\varepsilon = N^{-a}$, $a > 20$; the earlier denial of any polynomial regime is corrected by ERR-026. No deterministic noise removal follows. The author audit separates the analytic proof from exact algebra and finite Frechet-gradient checks.

70.15 Polynomial original-coboundary steps with the previous local rate

The polynomial continuation (`BGS-FLOW-011`) proves for the same original chain, odd clock start and nonnegative $f \in C^\infty_c(\mathbb{R})$ that

$$\left|\mathbb{E}\Phi_{N,f}(U^{\varepsilon}_{\lfloor 2t/\varepsilon^4\rfloor}) - \Phi_{\mathrm{Sine}_2}(f)\right| \le C_f[\sqrt{t}+(Nt)^{-1}+N^{-2}+N^4\varepsilon+tN^9\varepsilon+tN^{10}\varepsilon^2], \quad 1/N \le t \le 1/12.$$

Here the original kicks retain independent normalized GUE and do not insert centering or covariance whitening. The proof improves two dimension estimates. On the localized region $|m_2| \le 3/5$, writing $B = N\mathcal{C}_U$ on the fixed traceless Hermitian Hilbert space, positive-order derivatives satisfy $\|D^r B\|_{\mathrm{op}} \le C_r$ and $\|D^r B\|_{\mathrm{HS}} \le C_r\sqrt{N}$. In the differentiated square-root resolvent, placing one derivative in the latter norm cancels the exterior $N^{-1/2}$, giving $\|D^r\mathcal{C}_U^{1/2}\|_{\mathrm{HS}} \le C_r$. The rank-two Möbius projection has the same normalized structure. Its contracted noise drift follows from the exact identities

$$Q_K = I - \tfrac{1}{2}(m_2\bar{m}_1U^* + \bar{m}_2m_1U), \qquad Q_L = Q_K + 2i(T^*U^* - TU) + 2vI - \zeta U^{*2} - \bar{\zeta}U^2,$$

where $T = \sum_a \bar{w}_aK_a$, $v = \sum_a |w_a|^2$, $\zeta = \sum_a w_a^2$ are the FLOW-009 coefficients. The derivative norms of drift and noise are dimension-uniform, including the normalized-trace cutoffs. Stochastic-flow variation estimates yield $\|S_sF\|_6 \le Ce^{Cs}\|F\|_6$. Retaining the fast coordinate $\varepsilon/N + \varepsilon^4$ in the weak sum gives $tN^4\|F\|_6(\varepsilon/N + \varepsilon^2)$.

For $A = \operatorname{Tr} f_{N,\alpha}(U) \ge 0$, positivity bounds its first derivative by $C_fN\sqrt{A}$. For orders $j \ge 2$, two direction factors in Schatten-2 norm bound differentiated Fourier traces by $C_{f,j}N^j$. Partition

differentiation of e^{-A} absorbs all remaining powers of A, proving $\|\Phi_{N,f}\|_6 \le C_f N^6$, even at collisions. This completes the stated estimate.

Taking $t = N^{-2/3}$ and $\varepsilon = N^{-26/3}$ gives exactly $2N^{34}$ steps and error $O_f(N^{-1/3})$. Nonnegative C_c^1 tests have the weak limit by approximation. The review records the independent internal check of the new estimates and finite controls. For a fixed bounded interval I, the local counting generating function converges to $\det(I-(1-z)K_I)$ for $0 \le z \le 1$; the hole probability converges to $\det(I-K_I)$, with K_I the restricted sine kernel. Smooth L^1 approximation uses unit intensity; no interval-test rate or higher-factorial-moment uniform integrability is asserted. The polynomial exponent is not optimized. FLOW-014 supplies actual-start relaxation separately at a logarithmic time. Deterministic noise removal and all-order transfer at that polynomial step schedule remain separate obligations. FLOW-017 below closes all orders at a different superpolynomial schedule; deterministic BGS-TRACE-005 and BGS-BGS-001 remain open.

70.16 Retaining the actual initial spectrum in the stochastic comparison

The general-start proof (`BGS-FLOW-012`) starts the original uncentered Gaussian coboundary chain at any odd unitary A_N with $|p_1| + |p_2| \le K$. Its normalized traceless Dyson reference starts at exactly $\Pi(A_N)$; the algorithm inserts no centering. For nonnegative compact smooth local Laplace tests, writing $n = \lfloor 2t/\varepsilon^4 \rfloor$ and $1/N \le t \le 1/12$,

$$|\mathbb{E}\Phi_{N,f}(U_n^\varepsilon) - \mathbb{E}\Phi_{N,f}(D_t[\Pi A_N])| \le C_{f,K}[\sqrt{t} + N^{-1} + N^4\varepsilon + N^9\varepsilon^2 + tN^9\varepsilon + tN^{10}\varepsilon^2].$$

The fast coordinate has a decaying initial part as well as the stationary scale $\rho = \varepsilon/N + \varepsilon^4$. Summation retains $h\sum_{j<n}\mathbb{E}[|m_1(U_j)|; j < \sigma] \le C\varepsilon^2|m_1(A_N)| + Ct\rho$. The frozen stopped sixth moment bounds exits without an inverse-step loss from the initial term. The centered initial second trace is $O_K(1)$, giving an N^{-2} initial exit cost. Backward telescoping then adds $N^9\varepsilon^2$ to the previous error. At $t = N^{-2/3}$, $\varepsilon = N^{-26/3}$, exactly $2N^{34}$ steps have $O_f(N^{-1/3})$ comparison error. FM-013 supplies $K = 20$ for the actual frozen map eventually; the Dyson limit at that faster time is still unknown.

The prescribed-time obstruction (`BGS-FLOW-013`) constructs, for every selected vanishing t_N, a schedule-dependent globally uniform simple spectrum with bounded fixed traces and every positive power trace nonzero. Its Dyson local limit is empty at that time, both before and after centering. The proof combines $\mathbb{E}N^{-1}\|B_tA - A\|_{\mathrm{HS}}^2 \le t$ with spectral matching and Q narrow balanced clusters. With width δ and $I \subset [-R,R]$,

$$\mathbb{P}(X_{N,t}(I) > 0) \le \frac{Q}{\pi}(b + 2\pi R/N) + |I|\frac{2\delta^2 + (\pi^2/2)t}{b^2}.$$

Taking $Q \le \min(N^{1/10}, t_N^{-1/6})$ and $b = Q^{-2}$ proves the empty limit. Existing EXT-POPESCU-2023 supplies all-power nonvanishing after one algebraic-exponent perturbation. Under $Nt_N \to \infty$, FLOW-012 also transfers the empty limit to the chain at $\varepsilon = N^{-10}$. The finite laws retain unit intensity without uniform integrability. No actual-map counterexample or obstruction to every family-dependent slower schedule is asserted.

70.17 Logarithmic initial regularity and local heat-phase stationary points

The corollary (`BGS-FM-017`) uses the existing FM-007 bound $|p_j(U_N)| \le CN^{39/40}$ for $1 \le j \le J_N = \lfloor c\log N \rfloor$. For $H_A(z) = N^{-1}\operatorname{Tr}[(A+z)(A-z)^{-1}]$ and $d_N = CN^{-1/40}$, the absolutely convergent trace series gives

$$\sup_{|z|\le e^{-g}} |H_{U_N}(z) - 1| \le \frac{2[d_N + e^{-(J_N+1)g}]}{1 - e^{-g}}.$$

At $g_N = 4\log J_N/J_N$ this tends to zero. Centering contributes only $C/[N(1-e^{-g_N})^2]$, so the exact reference initial spectrum has uniform Poisson-smoothed density down to that logarithmic scale. Every fixed angular derivative also converges at the larger scale $J_N^{-1/2}$. This is not a microscopic local law.

At $t_N = \sqrt{g_N}$, FLOW-012 gives vanishing comparison error $O_f(g_N^{1/4}) + o(1)$ for $\varepsilon = N^{-10}$. For the centered eigenangles α_j, the local phase

$$\mathcal{S}_N(\zeta;\theta) = \frac{(\zeta-\theta)^2}{2t_N} + \frac{1}{N}\sum_j \log\left[2\sin\frac{\zeta-\alpha_j}{2}\right]$$

has unique stationary points in the radius-$t_N/8$ disks about $\theta \pm it_N/2$, with shifts $o(t_N)$ and $t_N\mathcal{S}_N'' = 1 + o(1)$ uniformly in θ. A contraction proof uses $\mathcal{S}_N' = (\zeta-\theta)/t_N - iH_A(e^{i\zeta})/2$ and Cauchy bounds in the two off-real strips. The route note credits the prior arbitrary-start determinantal formula and checks its normalization. The real logarithmic potential additionally bounds the unweighted vertical phase tails by $C\sqrt{t_N/N}\, e^{Na_N(\theta,t_N/2)-cNt_N}$ outside the two stationary neighborhoods. This uses positivity of the initial Herglotz real part near the real axis and the trace-series bound away from it, with no lower-gap assumption. The full kernel's amplitudes are not bounded by this estimate alone; the separate FLOW-014 argument below handles them. Even the whole logarithmic trace window cannot abstractly select $t = N^{-2/3}$: the FLOW-013 family at that time satisfies it and still has an empty limit.

The review states the independent internal proof-checking boundary. Deterministic noise removal and the original signed Heisenberg hierarchy remain open.

70.18 Actual-start circular relaxation and the original-chain Sine law

The full proof (`BGS-FLOW-014`) uses the prior arbitrary-start circular Brownian biorthogonal kernel of Kieburg–Li–Zhang–Forrester, with source time $t/(2N)$ and angular measure $d\theta/(2\pi)$. Its sine-product interpolation and backward Gaussian heat representation give an exact double contour plus a crossing term. The crossing-pole residue is negative before rearrangement; the resulting kernel receives the full positive residue, not a half-residue. All circle windings are retained.

Let E_N be the initial Herglotz error on the strips $|\Im z| \ge t/8$, and v_N the real-potential error beyond $g \le t/8$, as explicitly defined in the full proof. When $E_N \to 0$, $v_N/t \to 0$ and $Nt \to \infty$, Cauchy bounds give global horizontal convexity and vertical concavity of the phase, not merely nonzero local Hessians. The near-axis central barrier comes from FM-017. With $w = \sqrt{t/N}$, the remaining pole amplitude is controlled by

$$\int_{\mathbb{R}^2} \frac{e^{-|z-a|^2/(4w^2)}}{|z-c|}\, d^2z \le Cw,$$

uniformly even when the Gaussian center and pole coincide. Period-cell summation and the N/t prefactor then give a remainder $C_R\sqrt{N/t}$ in each fixed microscopic window. There are no separately estimated initial-gap denominators. Define the unfolded kernel by

$$\widehat{K}_{N,t}^{\theta_0}(x,y) = N^{-1}K_{N,t}(\theta_0 + 2\pi x/N, \theta_0 + 2\pi y/N).$$

The explicit crossing term yields, uniformly in θ_0 and $|x|, |y| \le R$,

$$\left|\widehat{K}_{N,t}^{\theta_0}(x,y) - \frac{\sin\pi(x-y)}{\pi(x-y)}\right| \le C_R[E_N + (Nt)^{-1/2}].$$

Hadamard bounds and determinant telescoping are summable after division by $k!$, so the entire Laplace/Fredholm series converges with that error. Gap-independent estimates pass to

repeated initial levels by continuity of the bounded Laplace test. Full and traceless Brownian motion have exactly the same uniform-origin observable.

For $A_N = \Pi U_N$, the actual logarithmic trace input gives the required errors at $g_N = 4\log J_N/J_N$, $t_N = 8g_N$, and $E_N + (Nt_N)^{-1/2} = o(\sqrt{g_N})$. Balanced comparison parameters are

$$\varepsilon_N = N^{-9}/\sqrt{t_N}, \qquad n_N = \lfloor 2t_N^3 N^{36} \rfloor.$$

FLOW-012's six terms reduce to $\sqrt{t}$, N^{-1}, $N^{-5}/\sqrt{t}$, N^{-9}/t, $\sqrt{t}$, N^{-8}, all $O(\sqrt{t})$ for $t \geq 1/N$. Thus the original uncentered chain from the unchanged U_N has

$$\mathbb{E}\Phi_{N,f}(U_{n_N}^{\varepsilon_N}) = \mathcal{L}_{\mathrm{Sine}_2}(f) + O_f(\sqrt{g_N}), \qquad 0 \leq f \in C_c^\infty.$$

This closes the actual-start stochastic relaxation step at the stated logarithmic time, with an ineffective startup threshold. It does not compare the noisy spectrum with its deterministic initial spectrum, prove relaxation at $N^{-2/3}$, or transfer unbounded correlations of the original chain. Deterministic BGS remains open. The new analytic implications received an independent internal audit; the prior finite kernel and inherited project proofs retain their separate trust boundaries. No external referee report, priority or prize-level significance is asserted.

70.19 All local moments and the original-chain signed criterion

Claim IDs: `BGS-FLOW-016`, `BGS-FLOW-017`. See Appendix A.

Theorem 70.4 (Local moment transport and projected diffusion comparison)**.** *The transport proof matches two period-N fields with averaged unfolded displacement cost $c_{p,N} = N^{-1}\sum_j |d_j|^p$. Let $M(u) = \sup_{r\geq 1} B_N([u-r, u+r])/(2r)$ be the reference maximal density. Its fixed moments are bounded by the periodic Hardy–Littlewood estimate and the reference's fixed unit-window moments. If $n = A_N([u-R, u+R]) > 16(R+1)M$, matching and packing give*

$$\sum_{a_j+\ell N\in[u-R,u+R]} |d_j|^p \geq \frac{3n^{p+1}}{4\,16^p M^p}.$$

Hölder and origin integration bound all target count moments $1 < s < p+1$. If $\mathbb{E}c_{p,N} \to 0$, positive Laplace tests transfer the weak law; the moment bound supplies uniform integrability for every factorial order $k < p+1$. All fixed vanishing transport moments therefore transfer every fixed factorial order.

Along a unitary logarithm path, Jensen for eigenangle speeds and Schatten triangle give the matching inequality

$$c_{p,N}(V,W) \leq \frac{N^{p-1}}{4^p}\operatorname{Tr}|V-W|^p.$$

For the projected diffusion Y and its coupled Dyson reference V, stop the comparison at $|p_2(Y)| = K_{N,r}$ with $K_{N,r}^2 = C_{K,r}(1+t\log N)$. The inherited covariance and drift estimates bound both matrix brackets by $(4K_{N,r}^2+44)N^{-2}I\,ds$. The Hermitian-dilation moment inequality and the exponential trace exit tail yield

$$\mathbb{E}c_{2r,N}(Y_t,V_t) \leq C_{K,r}(t+t^2\log N)^r + C_{K,r}N^{-2r-4} \to 0.$$

This holds when $t \to 0$, $t^2\log N \to 0$, including the actual $t = 8g$ with $g = 4\log J/J$, $J = \lfloor c\log N\rfloor$.

Thus the projected diffusion has all fixed local factorial limits.

Collapsing K clock points at one location has cost $N^{-1}\sum_{j<K} j^p$ and kth factorial mean $(K)_k/N$. At $N \sim K^k$ this refutes extending a single pth cost to $k > p+1$. No critical-endpoint or growing-order theorem is asserted.

It proves no largest-displacement estimate at this time. FLOW-014's gap-independent kernel bounds supply reference count moments, including repeated starts by fixed-N continuity of bounded smooth factorial majorants.

Theorem 70.5 (All fixed factorial orders and one-chain signed CUE comparison)**.** *The original-chain proof strengthens localization to $\mathbb{P}(\sigma \le n) \le C_{L,K}(N^{-L} + N^4\varepsilon)$ for every fixed L. It uses the frozen stopped fast moments and $|p_2|^{2r}$ Lyapunov tests; no inverse-power initial-layer loss occurs. The exact partition/Fourier formula gives polynomial C^6 derivative norms for each fixed factorial test without spectral-gap denominators. The dimension-uniform backward semigroup and fast terminal centering then transfer every fixed test for one sequence satisfying $N^b\varepsilon_N \to 0$ for all fixed b. The algorithm remains*

$$U_{j+1} = \exp[i\varepsilon_N(G_j - U_jG_jU_j^*)]U_j, \qquad U_0 = U_N,$$

with fresh normalized GUE. At the actual logarithmic time one may take $\varepsilon_N = N^{-\log\log N}$ and $n_N = \lfloor 2t_N N^{4\log\log N}\rfloor$.

Spatial ergodicity of Sine$_2$, from Soshnikov's prior Theorem 7, turns the barycenter limit and moment bounds into one-chain origin-average convergence in L^2. For a fixed block average $\mathcal{A}_S\varphi$, periodicity and Jensen give

$$\mathbb{E}_\omega|F_N(\omega) - c|^2 \le \int |\mathcal{A}_S\varphi - c|^2 dQ_N.$$

Take $N \to \infty$ at fixed S using uniform integrability, then $S \to \infty$ using the mean ergodic theorem. Schwartz tails also remove tuples using two copies of the same circle label, whose diameter is at least N. Hence, writing $W_N(m;A) = \sum_\pi \mu(\pi)\prod_{B\in\pi} p_{\sum_{i\in B} m_i}(A)$, for every fixed $\widehat{\psi} \in C_c^\infty(\mathbb{R}^k)$,

$$\mathbb{E}_\omega\left|\frac{1}{N^k}\sum_{\sum_i m_i=0}\widehat{\psi}(m/N)\,[W_N(m;U_{n_N}) - \mathbb{E}_{\mathrm{CUE}}W_N(m;V_N)]\right|^2 \to 0.$$

Every partition counterterm and $p_0 = N$ remains present; the modulus stays outside the whole signed sum. This is a stochastic theorem. For the original one-step Markov operator T_ε, the exact discrete defect is

$$\mathfrak{D}_{N,\psi}^{\mathrm{chain}} = \sum_{j<n_N}\mathbb{E}[(T_{\varepsilon_N} - I)F_{N,\psi}(U_j)] = \mathbb{E}F_{N,\psi}(U_{n_N}) - F_{N,\psi}(U_N).$$

The deterministic complete signed CUE target for each fixed test is now equivalent to $\mathfrak{D}_{N,\psi}^{\mathrm{chain}} \to 0$, or to $\mathbb{E}|F_{N,\psi}(U_{n_N}) - F_{N,\psi}(U_N)|^2 \to 0$.

This schedule is superpolynomial; the earlier polynomial bounded-Laplace result remains valid with its original scope.

Its actual vanishing remains open. Independent internal review checked the new implications; inherited inputs keep their stated evidence levels. The finite packet and source review do not establish deterministic noise removal, world priority, external refereeing, an all-order polynomial schedule, or full BGS.

70.20 Critical-noise compensation and the entire fixed-degree projection

Claim IDs: `BGS-FLOW-018`, `BGS-FLOW-019`. See Appendix A.

These results sharpen the transfer route and retain its open deterministic complement. The clock proof begins with the prior odd-dimensional Dyson kernel, and the projection proof uses stable Haar trace moments in exactly the range $2Q \le N$.

Theorem 70.6 (Exact clock covariance and critical noise law)**.** *For the clock start and $\tau = Nt$, the exact covariance spectral measure is*

$$\mathcal{S}_{N,t} = \frac{1}{N^2} \sum_{m \neq 0} \mathbb{E}|\operatorname{Tr}(B_t C_N)^m|^2 \delta_{m/N}.$$

Define $b_{q,\tau}(\xi) = (1 - |\xi - q|)_+ e^{-\tau q \xi}$ and

$$a_{q,N}(\tau) = e^{-\tau q^2/2} \frac{\sinh(\tau q/2)}{N \sinh(\tau q/(2N))}.$$

Exact angular Fourier selection gives $\mathbb{E}|p_m|^2 = N[1 - \sum_q b_{q,\tau}(m/N)] + \mathbf{1}_{N|m} N^2 a^2_{m/N,N}$, where the $q = 0$ summand is $(1 - |\xi|)_+$. The negative $q \neq 0$ mesh mass equals $a^2_{q,N}$ exactly. Consequently

$$\mathcal{S}_{N,t} - \mathcal{S}_N^{\mathrm{CUE}} = \sum_{q \neq 0} a^2_{q,N} (\delta_q - \nu_{q,N,\tau}), \qquad \|\mathcal{S}_{N,t} - \mathcal{S}_N^{\mathrm{CUE}}\|_{\mathrm{TV}} \le 8/\tau^2$$

for $\tau \ge 1$, with ν a probability mesh measure. This signed pair rate improves the absolute kernel bound by one factor of Nt. It is sharp: if $\tau \to \infty$ and $t \to 0$, the rescaled difference converges against compact weights to $\delta_1 + \delta_{-1} - 2\delta_0$. The positive-frequency Abel difference is strictly negative for every finite positive time and has leading coefficient $e^{-2s} - 1$ in this joint regime.

At $Nt \to \tau \in (0, \infty)$ the critical local law is the known lattice-relaxation phase mixture, with all fixed factorial moments. Its covariance density on $0 < |\xi| < 1$ is $|\xi|(1 - e^{-\tau|\xi|})$, and it retains nonzero Bragg atoms. The ordered-lift Ito estimate

$$\mathbb{E} \frac{1}{N} \sum_j |N(\theta_j(t) - 2\pi j/N)/(2\pi)|^{2r} \le (2r-1)!![Nt/(4\pi^2)]^r$$

and synchronous particle-noise contractivity transfer this trichotomy to starts with vanishing unfolded matching moments of every fixed order. Clock preservation requires $Nt \to 0$, while its Sine relaxation requires $Nt \to \infty$. The original uncentered chain has the critical all-order corollary in the stated $t \ge 1/N$ superpolynomial step regime.

Its lower floor remains TRACE-007; no monotonicity in time is asserted.

No such clock matching is established for the actual U_N.

Theorem 70.7 (Fixed-degree neutral projections and projected CUE rate)**.** *The actual-map estimate uses the neutral trace span $\mathcal{H}_{Q,N}$ of positive and negative weighted degree at most a fixed Q. Its Haar Gram matrix is dimension independent and positive at $N \ge 2Q$. D5 preserves it with generator coefficients $a + b/N$; Haar self-adjointness makes its projection $\mathsf{P}_{Q,N}$ commute with heat evolution. The bounded fixed traces of FM-013 therefore imply*

$$|(P_t - I)\mathsf{P}_{Q,N} \Phi_{N,f}(U_N)| \le C_{Q,f} t.$$

A finite CUE cross-window determinant bound additionally proves $\operatorname{Var}_{\mathrm{CUE}} F_N \le C_F/N$ for compact factorial or Laplace tests. Thus, uniformly at every heat time $t \ge 0$,

$$|P_t \mathsf{P}_{Q,N} F_N(U_N) - \mathbb{E}_{\mathrm{CUE}} F_N| \le C_{Q,F} N^{-1/2}.$$

Schwartz factorial tests have qualitative $o(1)$ only. These statements concern the projected observable, not the full local statistic. The complementary evaluation remains open. For a narrow positive Laplace test, its complement has Haar norm $O(N^{-1/2})$ yet has a nonzero limiting value on the clock. A small Haar norm cannot replace an actual complementary pointwise estimate. Neither claim closes TRACE-005 or deterministic BGS. Prior kernels and standard mechanisms are credited; world priority is unverified.

70.21 Growing neutral degree and the particular missed local test

Claim ID: `BGS-FLOW-020`. See Appendix A.

Theorem 70.8 (Exact reproducing-kernel bound)**.** *The full proof replaces the fixed-degree Gram bound by its exact reproducing kernel. For $2Q \le N$, normalized complex Wick polynomials give an orthonormal basis $E_{\boldsymbol{a},\boldsymbol{b}}$ of the neutral span. With $\mathcal{K}_Q(A) = \sum |E_{\boldsymbol{a},\boldsymbol{b}}(A)|^2$, centered point evaluation has norm $\sqrt{\mathcal{K}_Q(A) - 1}$. The prior complex Mehler identity gives, for every $s > 0$,*

$$\log \mathcal{K}_Q(A) \le 2sQ + \frac{\pi^2}{12s} + 2 \sum_{m \le Q} \frac{e^{-sm}}{1 + e^{-sm}} \frac{|p_m(A)|^2}{m}.$$

This environment is part (1) of the BGS-FLOW-020 claim, whose full subsection arc has three parts: (1) the kernel bound above (this environment); (2) the logarithmic projection rate $N^{-1/4}$ (the prose block following, beginning "For the unchanged actual odd-N operator"); (3) the missed-test obstruction $J_{N,w}$ (the prose block following that, beginning "The universal cost obeys"). Parts (2) and (3) stay as prose and are not separate environments; nothing in the arc is dropped or reordered.

For the unchanged actual odd-N operator, a fixed-order interaction packet expansion proves $|p_j(U_N)| \le C_0 e^{\alpha|j|}$ uniformly in dimension. An order-ℓ word has at most ℓ insertions and degree at most 3ℓ after removing accumulated quadratic evolution. Fixed Schwartz seminorms, weighted sampling and torus lattice overlap therefore grow exponentially through a logarithmic window. The exact frame and uniform return-volume bound give the trace estimate there; unitarity extends the coarse exponential bound outside that window. It supplies no cancellation at $j \asymp N$.

Set $s = 2\alpha + 1$, $\gamma = 1/(4s)$ and $Q_N = \lfloor \gamma \log N \rfloor$. The evaluation norm is at most $CN^{1/4}$, whereas compact local CUE deviation has Haar norm $C_F N^{-1/2}$. Consequently, uniformly at all heat times,

$$|P_t \mathsf{P}_{Q_N} F_N(U_N) - \mathbb{E}_{\mathrm{CUE}} F_N| \le C'_F N^{-1/4}.$$

This is a growing-degree projection theorem. A separate stronger bound $|p_m(A_N)| \le K$ through $m \le Q_N$ gives convergence for $Q_N \le c(\log N)^2$, $c < 3/(2\pi^2)$, but that input is not proved for the actual operator. No quantitative Schwartz rate follows.

The universal cost obeys $\mathcal{K}_Q(A) \ge (33/32)^L$ with $L = \lfloor (\sqrt{1 + 4Q} - 1)/2 \rfloor$, for every unitary in the stable range. Thus a whole-filtration norm estimate becomes superpolynomial when $Q/(\log N)^2 \to \infty$. More concretely, choose fixed nonzero $0 \le w \in C_c^\infty((a,b))$, $0 < a < b < 1/2$, and put

$$J_{N,w} = N^{-2} \sum_m w(m/N)(|p_m|^2 - m).$$

It is the fixed real even Schwartz pair comparison with $\widehat{\varphi}(\xi) = w(|\xi|)/2$. Exactly,

$$\mathsf{P}_Q J_{N,w} = 0 \quad (Q < aN), \qquad \|J_{N,w}\|_2^2 = N^{-4} \sum_m m^2 |w(m/N)|^2.$$

Hence every sublinear degree misses this same microscopic observable. Its Haar norm is $O(N^{-1/2})$, but its clock value tends to $-\int_a^b \xi w(\xi) d\xi < 0$.

Permutation cut-and-join, transported through the Wick isometry, gives the attained decay interval $q \pm q(q-1)/N$ on $\mathcal{H}_{q,N} \ominus \mathcal{H}_{q-1,N}$. Each participating pair term has degree m, so

$$\|P_t J_{N,w}\|_2 \le e^{-a(1-b)Nt} \|J_{N,w}\|_2.$$

A left inverse recovering this particular local test therefore has Haar-L^2 norm at least $e^{a(1-b)Nt}$. This is a norm obstruction, not actual point evaluation. The specific signed vector $J_{N,w}(U_N)$ remains unestimated, as do the full higher-order counterterms. The review checks the new analytic implications; the finite packet does not certify their all-degree limits, priority or BGS. The complete deterministic target remains OPEN.

70.22 Exact Heisenberg pair heat compensation and the actual smoothed band

Claim ID: `BGS-FLOW-021`. See Appendix A.

Theorem 70.9 (Exact Heisenberg pair heat compensation)**.** *The full proof diagonalizes the particular stable pair vector $Q_m = |p_m|^2 - m$, $2m \le N$, directly from determinant-ratio hook modes. Define*

$$H_m(A; y) = \frac{[x^m]}{1-y} \frac{\det(I - yxA)}{\det(I - xA)}, \qquad H_m(A; 1) = p_m(A),$$

$$K_m(A; y) = |H_m(A; y)|^2 - m y^{m-1} - (1-y)^2 \sum_{k=1}^{m-1} (m-k) y^{m-k-1} |H_k(A; y)|^2.$$

The complete pointwise identity, including every Wick contraction, is

$$P_t Q_m(A) = e^{-t[m+m(m-1)/N]} K_m(A; e^{mt/N}).$$

It holds at repeated eigenvalues and does not require a minimum gap. The m^2 orthonormal modes have rates $m + m(m-1-r-s)/N$, $0 \le r, s < m$, with their repeated-sum multiplicities retained. In particular

$$\|P_t Q_m\|_2^2 = e^{-2mt} \left[\frac{\sinh(m^2 t/N)}{\sinh(mt/N)} \right]^2.$$

The clock value is separately $P_t Q_m(C_N) = -m e^{-mt}$.

This is a Haar-L^2 identity, not point evaluation. The local band has an exact triangular limiting heat spectral measure at time s/N.

The same determinant formula also gives a quantitative actual-start estimate. Suppose $|p_j(A)| \le Nd$ through $j \le J$ and put $\vartheta = a(1-b)$ for fixed $0 < a < b < 1/2$. If $0 < t \le 1$ and

$$d(1 + \log J) + \frac{8e^{-aJt/4}}{aJt} \le \vartheta t/16,$$

then throughout $aN \le m \le bN$,

$$|P_tQ_m(A)| \le \left[e^b(at)^{-2} + N^2/2 + N\right] e^{-\vartheta Nt/4}.$$

The proof reflects the numerator outside the unit disk and takes the coefficient circle at $e^{-3(m/N)t/4}$. The factor e^b retains the finite-dimension exponent correction. FLOW-020's actual exponential trace bound gives $J_N = \lfloor \log N/[2(\alpha+1)] \rfloor$ and $d_N = C_0 N^{-1/2}$. At $t_N = \max(32, 8/a) \log J_N/J_N$, the entire actual smoothed stable band agrees with the CUE form factor to $o(N^{-H})$ for every fixed $H > 0$. The time constant depends on the band endpoint a; no effective starting dimension or unperturbed universality is inferred.

For every sublinear trace window there are two simple globally uniform spectra with identical trace data and common exponential bounds, but one unperturbed pair vector has a negative clock limit while the other grows as $L \int w > 0$. Both smoothed vectors vanish at this schedule. Thus the full signed noise defects remain distinguishable even when all the specified short-trace data match. The proof and programme preserve this input limitation without presenting a chaotic counterexample.

For the original U_N, the determinant formula makes the full signed defect explicit, but does not estimate it. The actual $N^{-2}\sum_m w(m/N)(|p_m(U_N)|^2 - m)$, other Fourier bands and all higher factorial counterterms remain open. Exact determinant self-inversiveness supplies no missing uniform quantum remainder. The complete deterministic BGS target remains OPEN.

70.23 Actual Weyl coercivity, regularized transfer and common-noise invariants

Claim ID: `BGS-FM-018`. See Appendix A.

Theorem 70.10 (Actual Weyl coercivity)**.** *The full proof keeps the actual $U_N = e^{i\tau N A_N} M_N$, $\tau = 10^{-80}$, and its ordinary Weyl dictionary. Use normalized Hilbert–Schmidt norm on the traceless space, $\mathcal{P}_N = \operatorname{Ad} U_N$, and let Π_R^0 project onto $0 < |\nu| \le R$ in the centered integer-label Weyl basis. Set*

$$\mathscr{L}_N = \frac{N^2}{2}\sum_{j=1}^{2}(2I - \operatorname{Ad} T_N(e_j) - \operatorname{Ad} T_N(e_j)^*), \quad \mathcal{T}_{u,N} = e^{-u\mathscr{L}_N}\mathcal{P}_N.$$

There are fixed $c, \gamma > 0$ and an eventual odd-dimensional threshold such that, at $T_N = \lfloor c \log N \rfloor$, $R_N = \lfloor N^\gamma \rfloor < N/4$,

$$\sup_{|\zeta|=1} \left\| \frac{1}{T_N} \sum_{j=0}^{T_N-1} \zeta^{-j} \mathcal{P}_N^j \Pi_{R_N}^0 \right\|^2 \le \frac{1}{16}.$$

The proof controls every linear combination in the growing band, using the exact sin commutator, uniform Egorov and product-Sobolev-norm classical correlations. It introduces no independent-eigenphase input. For every complete traceless adjoint spectral arc of angular width $2/T_N$, its low-frequency mass is at most one quarter. Consequently its diffusion energy is at least $6R_N^2$ times its squared norm, including every traceless commuting operator. Arbitrary degeneracies are allowed; the positive exponent and startup are noneffective.

For every $u > 0$, the elementary dissipation argument gives

$$\inf_{|\zeta|=1} \sigma_{\min}(\mathcal{T}_{u,N}|_{\mathcal{H}_N^0} - \zeta I) \ge d_{u,N} := \min\{(128T_N^2)^{-1}, (1 - e^{-16uR_N^2})/8\}.$$

No normality is assumed. Two ordered Gaussian Hamiltonian kicks built from the real and imaginary parts of $T_N(0,1)$ and $T_N(1,0)$ realize $\mathbb{E}\operatorname{Ad} V_{u,\xi} = \mathcal{T}_{u,N}$ exactly. For $u_N = N^{-\alpha}$, $0 <$

$\alpha < 1+2\gamma$, one has $N d_{u_N,N}/\log N \to \infty$. Thus the traceless quadratic block $I - e^{-s/N+i\omega/N}\mathcal{T}_{u_N,N}$ has minimum singular value at least $d_{u_N,N}/2$ eventually, for fixed $s \geq 0$ and real ω. This is phase-space diffusion, distinct from FLOW-021 matrix heat.

The full-band classical generator comparison has an actual linear-in-N discrepancy. The channel trace at Heisenberg time tends to 1 in the stated noise regime, rather than the spectral ramp, because fresh-step noise and fixed-matrix spectral statistics are different. O25 gives an exact finite discriminator: uniform Weyl and Weyl-Fourier ensembles have the same averaged adjoint, while their fourth-power trace moments are respectively 1 and N^2. Their rank-one nonlinear determinant factors also differ. These controls are not perturbations of the actual map. A gapped Hessian alone does not determine the nonlinear integral.

Retaining the same Gaussian parameters in k copies instead yields the ordered common-noise channel

$$\mathbb{E}\,\mathrm{Ad}(V_{u,\xi}^{\otimes k}) = e^{-uN^2 Q_{q,k}/2} e^{-uN^2 Q_{p,k}/2}\,\mathrm{Ad}(U_N^{\otimes k}),$$

where $Q_{q,k}, Q_{p,k}$ are sums of squares of collective commutator generators. For $k = 2$ the pair factors do not commute. Full Gaussian support and explicit Weyl Lie generation prove, for every finite odd $N \geq 3$, $u > 0$, that its whole peripheral space is exactly $\mathrm{span}\{I, \mathsf{S}_N\}$, with eigenvalue 1. This also covers composite N. On the orthogonal complement this argument gives a positive finite-N singular gap. The uniform shrinking-noise rate left open at this checkpoint is subsequently proved for the original two-kick law in FM-020. Independent-copy tensorization would erase these common-ensemble moments.

The source review credits prior noisy-relaxation methods and checks the exact classical mixing leaf. The internal audit reconstructed O1–O29 relative to the inherited Anosov/Gibbs inputs. Exact finite controls do not prove the analytic asymptotics or priority. The microscopic reduced rate for the original two-kick law was open at this checkpoint and is subsequently closed in FM-020. Higher-copy quantitative rates, the full nonlinear functional and the original signed Heisenberg noise defect remain OPEN. No deterministic BGS conclusion follows from the regularized gap or finite uniqueness.

70.24 Uniform collective diffusion and a refined common-noise microscopic mass

Claim ID: `BGS-FM-019`. See Appendix A.

Theorem 70.11 (Uniform collective diffusion gap)**.** *The full proof keeps both physical copies under the same noise. For the collective sums of the real/imaginary parts of $Z = T_N(0,1)$ and $X = T_N(1,0)$, put*

$$A_N^{\mathrm{diff}} = \frac{N^2}{2}\sum_{\ell=1}^{2}(\mathrm{ad}\,H_\ell^{[2]})^2, \quad B_N^{\mathrm{diff}} = \frac{N^2}{2}\sum_{\ell=3}^{4}(\mathrm{ad}\,H_\ell^{[2]})^2, \quad L_N = A_N^{\mathrm{diff}} + B_N^{\mathrm{diff}}.$$

These are diffusion superoperators, distinct from the physical kick Hamiltonian. Let Q_N project onto the whole normalized-HS orthogonal complement of $\mathrm{span}\{I, \mathsf{S}_N\}$*. There is a universal $c_0 > 0$ such that, for every odd $N \geq 3$,*

$$\ker L_N = \mathrm{span}\{I, \mathsf{S}_N\}, \qquad L_N \geq c_0 Q_N.$$

This follows analytically from N^2 exact total-Weyl blocks of dimension N^2, including composite N. Translation and a sign gauge give weighted integral or half-integral lattices with cross antiperiodic holonomy. A dyadic annular Poincare inequality has no logarithmic loss. Nonzero

even total labels have an additional positive form whose compression on the two otherwise invariant directions gives a uniform bound. The sole zero block represents identity/swap.

The original ordered two-pair law is still not the exponential of the sum. Exact pair-kernel angle $N^{-1/2}$ gives the explicit bound

$$C_{N,u,1} = e^{-uA_N^{\mathrm{diff}}} e^{-uB_N^{\mathrm{diff}}} \operatorname{Ad}(U^{\otimes 2}),$$
$$\inf_{|\zeta|=1} \sigma_{\min}[(C_{N,u,1} - \zeta I)|_{\operatorname{ran} Q_N}] \geq \frac{(1 - N^{-1/2})(1 - e^{-uN^2 q_N})}{10},$$
$$q_N = 16 \sin^2 \frac{\pi}{2N} \sin^2 \frac{\pi}{N}.$$

Its small-u lower bound has order u/N^2. The subsequent FM-020 result below closes a stronger microscopic m=1 rate. FM-019 also closes a quantitative obligation for an explicitly **refined** ensemble: sample m independent Gaussian four-vectors, keep their complete sequence common to the copies, and multiply their sub-kicks before the single U. The higher-copy law changes, while its one-copy channel remains exactly $D_{u,N} \operatorname{Ad} U$ at every m. Its two-copy channel is

$$C_{N,u,m} = (e^{-(u/m)A_N^{\mathrm{diff}}} e^{-(u/m)B_N^{\mathrm{diff}}})^m \operatorname{Ad}(U^{\otimes 2}), \qquad \|C_{N,u,m} Q_N\| \leq e^{-c_0 u} + 64\pi u^2 N^3/m.$$

The error uses a direct collective-Weyl commutator bound and contraction Duhamel, not a formal interchange of N and the product limit. At $u_N = N^{-\alpha}$, $0 < \alpha < 1$, $m_N = N^3$, the error is $o(u_N)$. For fixed $s \geq 0$, real ω, $z_N = e^{-s/N + i\omega/N}$,

$$\sigma_{\min}[(I - z_N C_{N,u_N,m_N})|_{\operatorname{ran} Q_N}] \geq c_0 u_N/4,$$

eventually, so its mass dominates N^{-1}. This holds for any U, and in particular for the unchanged actual fixed-coupling U_N. No normality or simple-spectrum assumption is used.

For each fixed smooth symbol a the refined actual-map ensemble satisfies

$$\mathbb{E}\| \operatorname{Ad}(\widetilde{V}_{N,u,m}) \operatorname{Op}_N(a) - \operatorname{Op}_N(a \circ G)\|_{HS}^2 \leq C_a(u + N^{-4}),$$

uniformly in m. This precise retention of the classical map follows from exact one-copy averaging and inherited one-step Egorov. It is normalized-HS mean square on fixed smooth observables, without a long-product Egorov or microscopic operator-norm statement.

The internal review reconstructed D1–D25; the source check credits prior moment-semigroup/design and shear-mixing methods without importing an external gap theorem. Exact finite controls and float64 diagnostics are separate. The full nonlinear quenched functional, higher-copy quantitative rates and actual original signed noise defect remain open. The universal regulator gap does not estimate the original $J_{N,w}(U_N)$; TRACE-005 and BGS-BGS-001 retain their OPEN status.

70.25 Original ordered two-pair microscopic mass

Claim: `BGS-FM-020`. See Appendix A.

The canonical proof keeps the original ordered common-noise law with m=1.

Theorem 70.12 (Original ordered two-pair microscopic mass)**.** *Put*

$$\ell_N = 1 + \log N, \qquad \lambda_N = (1024\ell_N)^{-2}.$$

For every odd $N \geq 3$, $u \geq 0$ and unitary U,

$$\|e^{-uA_N^{\mathrm{diff}}} e^{-uB_N^{\mathrm{diff}}} Q_N\| \leq 1 - \frac{1 - e^{-2u\lambda_N}}{50}.$$

The quantity $(1 - e^{-2u\lambda_N})/50$ is a lower bound for the unit-circle singular gap after right multiplication by $\mathrm{Ad}(U^{\otimes 2})$, and for $I - zC_{N,u,1}$ when $|z| \leq 1$.

At $u = N^{-\alpha}$, $0 < \alpha < 1$, this displayed lower bound is asymptotic to $u/(25 \cdot 1024^2 \ell_N^2)$ and dominates N^{-1}. This is not an asymptotic formula for the actual gap.

In every total-label block, the number of positive eigenvalues of either pair generator below λ is at most $4N\sqrt{\lambda}(1 + \log N)$. The shifted integral/half-integral lattice count treats nonzero axes separately. The explicit eigenbases $\delta_a \otimes \mathrm{Fourier}_j$ and $\mathrm{Fourier}_i \otimes \delta_b$ have every overlap of modulus $1/N$. Their low-positive ranks and the existing exact kernel angle therefore give a low-spectral projection angle below $4/5$ on Q. The two-contraction loss estimate yields the stated ordered-product bound without a Trotter approximation.

The independent internal review reconstructs all parity sectors, projection reduction and loss constants. Exact rational counting checks and float64 spectral diagnostics have distinct scopes. Neither the summed-generator gap nor the refined noise law is used. The bound holds for every U and supplies no model-specific signed spectral estimate. Higher-copy quantitative control, the nonlinear quenched functional, original noise removal and deterministic BGS remain open.

70.26 The exact deterministic noise-removal obligation

Claim ID: `BGS-FLOW-015`. See Appendix A.

Theorem 70.13 (Exact deterministic noise-removal obligation)**.** *The full proof retains the unchanged U_N and the positive uniform-origin local Laplace test $\Phi_{N,f}$. For normalized unitary Brownian motion, let $P_t F(A) = \mathbb{E}F(B_t A)$ and*

$$G_N = \frac{1}{2N} \sum_j \partial_j^2 + \frac{1}{2N} \sum_j \sum_{k \neq j} \cot \frac{\theta_j - \theta_k}{2} \partial_j.$$

The generator applied to the smooth Laplace function extends through collisions by cancellation of its paired derivative differences. At each fixed dimension, the exact signed identity is

$$\mathfrak{D}_{N,f}(A, t) = \int_0^t P_s(G_N \Phi_{N,f})(A) ds = P_t \Phi_{N,f}(A) - \Phi_{N,f}(A).$$

FLOW-014 applies directly to the uncentered U_N using FM-007's logarithmic trace estimate. Consequently its deterministic weak local Sine$_2$ law is equivalent to $\mathfrak{D}_{N,f}(U_N, t_N) \to 0$ for every nonnegative $f \in C_c^\infty$, at the proved logarithmic time $t_N = 8g_N$. The modulus stays outside the full signed integral.

This equivalence does not establish its vanishing or the stronger unbounded factorial-moment target TRACE-005.

The full mixed-trace generator splits and merges factor positions, including the factor $p_0 = N$ on opposite-index merging. It preserves the finite spaces of charge-zero products with positive degree at most q. Those spanning symbols need not be independent when $q \geq N$; finite closure does not give uniformly bounded complexity or a fixed-correlation-order closure. The exact inclusion-exclusion/Fourier formula still involves signed mixed traces with indices of order N. FM-007's e^{Cm} error control does not reach that range.

Four precise boundaries follow.

1. For a test supported in width $w < 1$, a spectrum with minimum unfolded gap greater than w has $\Phi_{N,f} = 1 - a_f$ on an open neighborhood, where $a_f = \int(1 - e^{-f})$. Hence every initial generator iterate and right heat derivative is zero. Nevertheless the simple family $A_N = \operatorname{diag}(e^{iN^{-4}}, e^{2\pi i/N}, \dots, e^{2\pi i(N-1)/N})$ has all positive traces nonzero and the whole logarithmic trace window, but whenever $t_N \to 0$, $Nt_N \to \infty$,

$$\mathfrak{D}_{N,f}(A_N, t_N) \to \delta_f \geq (1 - a_f/3)e_2 > 0,$$

$$e_2 = \frac{1}{2}\iint h(x)h(y)\left[1 - \left(\frac{\sin \pi(x-y)}{\pi(x-y)}\right)^2\right] dxdy, \qquad h = 1 - e^{-f}.$$

   The proof uses the prior clock theorem and a uniform-origin $O_f(N^{-3})$ comparison. It refutes the heat-jet transfer shortcut. Its strict-gap hypothesis is not established for the actual map.
2. At a deterministic matrix put $X = |p_1|^2$, $W = X^2$ and $Z = \Re(p_2\bar{p}_1^2)$. The exact equations $G_N X = 1 - X$ and $G_N W = -2W - 2Z/N + 4X$ give

$$\max\{|G_N X|, |G_N W|\} \geq (\sqrt{17} - 1)/4 - 2K^3/N \quad \text{if } |p_1| + |p_2| \leq K.$$

   FM-013 makes this an eventual actual-map bound with K=20. Unscaled full-class stationarity fails; the normalized microscopic mixed loop hierarchy remains a possible route.
3. The normalized character of highest weight $(r, 0, \dots, 0, -r)$ has sup norm one and heat multiplier $e^{-tr(r+N-1)/N}$. At $r = N$, any sup-norm left inverse containing this mode has norm at least $e^{(2N-1)t}$, exceeding every fixed polynomial at the chosen logarithmic time. This standard character calculation concerns a global growing-degree function space, not instability for one fixed local Laplace test.
4. The stationary periodic point field of every finite deterministic spectrum has infinite relative entropy to Poisson on every interval of length at least two, and infinite specific entropy. An adjacent gap $d \leq 1$ produces an event of a fixed pair distance with positive probability $(2 - d)/N$, whereas Poisson gives that event probability zero. The published Sine-beta free-energy minimizer characterization does not supply finite-entropy near-minimality of those raw fields. A limit-level or justified regularized variational argument remains open.

The source audit credits the character heat input and entropy definition. A bounded packet checks 348 independent exact radial/trace identities, 630 charge-closure cases, literal low-trace rows, finite character and rational gap controls, and separate quadrature diagnostics. Two internal proof passes checked the new implications. These controls do not establish the outstanding model-specific defect estimate, external referee validation or priority.

70.27 Classical action completion and the ideal Fourier alphabet

The Heisenberg-note audit retains three scoped results. For the unchanged physical symbol, actual q-circle separation at least 1/7 gives

$$\max_{1\leq j\leq 14} |D_q^j \Delta_\delta g| \geq \kappa_0^{\mathrm{thr}}, \qquad \max_{2\leq j\leq 15} |D_q^j(\Delta_\delta g + \ell q)| \geq \kappa_0^{\mathrm{thr}}, \quad \kappa_0^{\mathrm{thr}} = \frac{4}{3}\frac{(\sqrt{101}\sin(\pi/7) - 1)/100}{6409325/2286144}.$$

The finite-type proof (`BGS-FM-014`) includes uniform scalar cancellation of order 1/14 for the plain phase and 1/15 with arbitrary linear additions. It also limits the submitted half-shift Hessian argument to its local Morse–Bott neighborhoods.

The classical action theorem (`BGS-FM-015`) constructs a local mixed-coordinate generating function for $\Phi_{g_\delta}^t \circ \Phi_g^{-t}$ with

$$s_{t,\delta} = t\Delta_\delta g + E_{t,\delta}, \qquad \|E_{t,\delta}\|_{C^{15}} \le 10^{61} t^2, \qquad \max_{1\le j\le 15} \|D_q^j E_{t,\delta}\|_\infty \le 10^{49} t^2.$$

The analytic flow construction holds for $|t| \le 10^{-8}$; finite type is preserved for $0 < |t| \le 10^{-52}$, including the maintained $t = \tau/(2\pi)$. This is a classical chart statement. A uniform quantum torus FIO with controlled amplitudes and an operator norm gap is still missing. Different strip labels do not imply the separation hypothesis.

The ideal-alphabet theorem (`BGS-FUP-001`) uses Cohen's pinned prior Fourier criterion and a corrected limiting-set geometry proof. For $\mathcal{A}_M = \{(r,s) : r \ne s\}$ and $N = M^k$:

Base	Ideal two-dimensional Fourier compression
Odd $M \ge 3$	Some $CN^{-\beta}$ bound with $\beta > 0$
$M = 2$	Exact norm $\sqrt{2/N}$ for $k \ge 1$
Even $M \ge 4$	Exact norm 1 at every scale, by an explicit Fourier-supported unit vector

In base 7 the limiting set contains a skew line of direction $(1, -6)$; the submitted exclusion of such lines is false. Absence of perpendicular pairs, with closed-drawing boundary issues handled, is the relevant prior criterion. The 2026 anisotropic Jin–Zhang–Zhang theorem assumes unequal bases and cannot be specialized directly to this equal-base alphabet. The source audit pins the exact theorems and limits their source-to-model use.

The supporting lemmas give the complete finite factorial CUE cycle benchmark, corrected raw/ordinary Weyl dictionary, an explicit conditional Fejer error budget, and near-unitary transfer for positive and negative times. None supplies the remaining actual-map quantum normal form, universal renewal weights, CUE plateau or joint traces at every order. BGS-TRACE-005 and BGS-BGS-001 remain open.

70.28 Prime-resonant blocks and the actual sector obligation

The prime-resonant audit (`BGS-TRACE-009`) retains the exact left-to-right block

$$C_{p,d_p} = \prod_{j=0}^{d_p-1} e^{i\tau p M_p^j A_p M_p^{-j}}, \qquad U_p^{r d_p} = \zeta_p^r C_{p,d_p}^r.$$

The known order relation $d_p \mid p-(2/p)$ and the integer determinant give $d_p > 2\log p/\log(3+2\sqrt{2})$. For a compact window $K \subset [a,b] \subset (0,\infty)$, let $S_p(K)$ be the normalized absolute resonant comparison and $E_p(K)$ the normalized actual trace-square mass. Then

$$|S_p(K) - E_p(K)| \le (b-a)/d_p + 1/p.$$

Thus the whole-sector comparison vanishes exactly when its actual mass does. A single deterministic trace need not equal its CUE ensemble expectation. The correct reference block is a Haar power, whose smaller independent component law is Rains' prior theorem; it is not Haar CUE of dimension p.

The exact Fejer pullback is an average of microscopic pair kernels centered at all rational shifts h/d_p. A baseline $1 + o(1)$ for that comb would dispose of the sector; this quantitative hypothesis remains unproved for the frozen map. Conversely, a trace of size at least cp at positive bounded Heisenberg times on an infinite prime subsequence would already obstruct the pair law through a positive time-mass. Finite coherence at $|\tau|p^2 \ll 1$ is not such a subsequence.

The selected invariant -98 and candidate prime list were reproduced exactly. At 29 the selected orbit sum is 903/1000; at 97 the norm-colliding candidate points lie in another orbit and the sum

is 9/10. Neither prime needs exclusion from the scalar test. The stronger scalar-power exclusion is prior FM-010 and gives no upper trace bound. Nonresonant powers, composite dimensions and higher joint correlations remain outside this reduction. TRACE-005 and BGS-BGS-001 remain open.

70.29 A universal boundary on the norm used for the deterministic target

TRACE-010 proves that the unaveraged all-order absolute CUE trace criterion is impossible. If $N^{-2}\sum_{j\in[N/8,N/2]}||p_j|^2-j|\to 0$, then

$$\liminf N^{-1/2}N^{-3}\sum_{m,n\in[N/8,N/4]}|D_N(m,n,-m-n)-2(N-m-n)|\geq 1/1024.$$

The same obstruction grows as $N^{k/2-1}$ at every fixed $k\geq 3$ on the stated positive cones. It does not refute BGS: the exact microscopic bridge uses the weighted **signed** sum, with a modulus outside the whole sum. TRACE-003's absolute sufficient implication remains correct, including averages taken before the modulus; its simultaneous unaveraged all-order premise is excluded.

The continuation calibrates the signed factorial cubic with an exact Haar-CUE complex variance for arbitrary low-band weights. Its reference root mean square scale is $O(N^{-1/2})$, with variance $O(1/(Nd)+1/N^2)$ on the three resonant strips. These are reference laws, not estimates for this map. A formal quadratic-phase trace table exhibits rapid signed cancellation but fails moment positivity already at $N=3$; it is not a unitary construction.

The next direct model-specific attack is the full signed factorial cubic, separating $d_p\mid m$, $d_p\mid n$, and $d_p\mid m+n$ with the actual ordered block and retaining the nonzero CUE counterterms above the ramp. The pair-sector mass alone loses a factor of order $\sqrt{N/d}$ in the available absolute Cauchy–Schwarz estimate. No such signed actual-map bound is proved. TRACE-005 and BGS-BGS-001 remain open.

70.30 Resonant cubic deletion from local spectral density

Claim ID: `BGS-TRACE-011`. The full proof closes a conditional sector question and identifies a sharp tail obstruction. It gives no unconditional Heisenberg estimate for the frozen map.

For the occupancies n_h of the N intervals of length $1/N$, put $M_s=N^{-1}\sum_h n_h^s$. Fix $0<a<b$, $2b<1$, positive times in $[aN,bN]$, and let $E_N=N^{-2}\sum_{j\in[aN,2bN],d|j}|p_j|^2$. For every compact smooth complex weight W, the signed union of the three resonant strips satisfies

$$\left|N^{-3}\sum_{\substack{m,n\in[aN,bN]\\ d|m\text{ or }d|n\text{ or }d|(m+n)}}W(m/N,n/N)p_mp_n\overline{p_{m+n}}\right|\leq C_W\sqrt{E_NM_4}.$$

The modulus is outside the entire sum. More generally, for $3<s\leq 4$, the bound is $C_{W,s}E_N^{(s-3)/(s-2)}M_s^{1/(s-2)}$. It has no cardinality loss. With $Q_N=N^{-2}\sum_{j\in[aN,2bN]}|p_j|^2$ and $\ell=|[aN,bN]\cap\mathbb{Z}|/N$, the factorial counterterms cost at most

$$\|W\|_\infty\left[3\ell E_N+6(\ell/d+1/N)Q_N+12b\ell(\ell/d+1/N)\right].$$

Therefore d tending to infinity, E tending to zero and bounded M_s for some $s>3$ delete the full resonant factorial cubic. Critical cubic uniform integrability is an alternative to the higher moment. Bounded M_3 alone fails.

The proof retains phases. A positive Gaussian density at scale 1/N has Fourier coefficients $e^{-(m/N)^2}p_m/N$, and its L^s norm to the s power is comparable to M_s. Smooth frequency filters

have uniformly integrable convolution kernels. The projection onto d-divisible modes is the positive average over shifts h/d. Parseval supplies its L^2 bound by the square root of E; Holder supplies L^4 bounds for the other two factors. General W follows by Fourier inversion with translation-independent kernel norms. The union indicator equals the three strip indicators minus twice the common intersection. Direct incidence counting gives the displayed counterterm budget, including the $m+n$ strip.

The critical obstruction is realizable. Take $N = dq^3$, q copies of a d-clock and an $(N-dq)$-clock background. Its active traces are $dq\,\mathbf{1}_{d|m}$, its E is $O(1/q)$, M_3 is bounded and M_4 has order q. Yet its resonant factorial cubic tends to the integral of W. Tiny phase perturbations make the spectrum simple and preserve these limits. The interpolation exponents have constant-order budget on this family.

There is a stronger variant: for odd $q = 2^l + 1$, $d = q$ and $N = q^4$, select an $(N-q^2)$-dimensional CUE background instead. Deterministic simple realizations exist which satisfy every compact smooth signed CUE pair comparison and have vanishing resonant mass and bounded M_3, while both the resonant and full-box factorial cubic tend to the integral of W for $0 < a < b < 1/4$. All-band pair variance follows from a finite projection cumulant bound $|\kappa_4| \le 26(N-q^2)$. The second/fourth/sixth mixed cubic moments use Diaconis–Evans only when $4bN \le N - q^2$. Microscopic count covariances are controlled by positive block determinants, not by independent-bin assumptions. Uniform negative-Sobolev bounds and summability justify all smooth cubic weights. These are abstract spectra, not the frozen map, prime-period theorems or weak-process counterexamples.

For the actual primes, the oriented block from TRACE-009 gives exactly the E above. The needed block-mass bound and spectral density tails remain open. The quantitative alternative $E = O(1/d_p)$ with $M_4 = o(d_p)$ also suffices. When d_p is comparable to p, bounded Q and E tending to zero already close the sector by the earlier finite-cardinality budget. The new mechanism addresses growing sectors without assuming they exhaust the prime orders. Nonresonant signed cubic cancellation, other cones, composites and every higher factorial order remain separate obligations. A raw k-factor version has budget $C_{W,k}\sqrt{E_N M_{2(k-1)}}$, but does not automatically treat proper subset resonances and CUE counterterms. TRACE-005 and BGS-BGS-001 remain open; no change to the coupling, unfolding or symmetry class is made.

70.31 Folded-square refinement and block crowding

Claim ID: `BGS-TRACE-012`. See Appendix A.

Theorem 70.14 (Folded-square cubic control)**.** *The proof sharpens the preceding sufficient density input. With the same positive Gaussian density and arithmetic projection, put* $G_d = \|P_d(\rho_N^2)\|_2^2$. *For every smooth compact complex weight, the signed three-strip cubic satisfies*

$$|Z_N(W)| \le C_W\sqrt{E_N G_d}, \qquad 1 \le \|\rho_N\|_2^4 \le G_d \le \|\rho_N\|_4^4.$$

The modulus stays outside the full sum. The factorial counterterm budget from TRACE-011 is unchanged. Since $Q_N \le e^{8b^2}\sqrt{G_d}$, $E_N G_d \to 0$ *and* $(1/d + 1/N)\sqrt{G_d} \to 0$ *suffice for deletion. For actual prime orders* $d \le N+1$, $G_d = o(d^2)$ *suffices for the second condition.*

These actual-map estimates are not proved.

The improvement comes from moving the self-adjoint projection onto the product of the other two filtered factors. Positivity of the multiplier-kernel majorants gives $\|P_d(F_u F_v)\|_2 \le C_{u,v}\|P_d(\rho_N^2)\|_2$. Projected factors and the common intersection obey the same estimate. Fourier inversion with modulation-independent kernel norms treats W. An explicit simple-spectrum separation uses $N = L^4$, $d = L^3$, $K = L^2$: an $(N-K)$-clock plus a tiny split cluster gives $E_N M_4 \gtrsim L$, but $E_N G_d = O(L^{-2})$ and a vanishing resonant factorial cubic. This abstract spectrum is not claimed to satisfy CUE pairs or be the map.

Write $C = U^d$ and $R = N/d$. For $C = \zeta^{-1}U^d$, rotate the original phases by a d-th-root phase of ζ^{-1}; E, Q and G_d are unchanged. The normalized block density is $f_{C,R}(u) = d^{-1}\sum_{j,v}\eta(R(u+dx_j+v))$. For $g_h(u) = \rho_N((u+h)/d)$ define $B_d = d\sum_h g_h^2/(\sum_h g_h)^2$. Then exactly

$$P_d\rho_N(x) = f_{C,R}(dx), \qquad G_d = \int B_d(u)^2 f_{C,R}(u)^4 du, \qquad 1 \le B_d \le d.$$

The block spectrum does not determine root allocation B_d. Concentrated and balanced lifts of the same $C = I$ give a d^2 energy ratio when N/d tends to infinity. Keeping only the block density therefore loses relevant information. Equivalent positive formulas are

$$G_d = \frac{1}{d}\sum_{h=0}^{d-1}\int \rho_N(x)^2\rho_N(x+h/d)^2 dx = \sum_{r\in\mathbb{Z}} |\widehat{\rho_N^2}(rd)|^2.$$

The proof expands the latter as an absolutely convergent fourth joint-trace energy, providing an explicit dynamical target.

A separate obstruction needs no root-allocation assumption. If K block phases lie in an arc of turn length $\varepsilon d/N < 1$, the positive anchored factorial r-point measure satisfies

$$\mu_{N,r}([-\varepsilon,\varepsilon]^{r-1}) \ge \frac{(d-s)(t)_r + s(t+1)_r}{N} \ge \frac{(K-(r-1)d)_+^r}{Nd^{r-1}}, \quad K = dt+s.$$

This follows by lifting the arc to d microscopic intervals and balancing their counts, using discrete convexity of the falling factorial. When $d = o(N)$, $\varepsilon \to 0$ and $K \ge c(Nd^{r-1})^{1/r}$, this excludes Sine factorial comparison. The argument uses a fixed smooth positive neighborhood before shrinking it; no weak-process conclusion is inferred from moment mass alone.

The Sine cubic cube mass, and the finite CUE expectation of that mass, are at most $(3\pi^6/5)\varepsilon^8$. A probability-frequency Gram identity, Newton divided differences and Hadamard prove this uniform nonoptimal bound. The actual $p = 239$, $d = 7$ operator gives a finite application: $\|A_p + I/10\| \le 26/25$ and telescoping put all $K = p$ block phases in the required arc with $\varepsilon < 3\cdot 10^{-76}$. Its factorial cubic mass is at least $254694/239$, while the finite CUE expectation is below 10^{-600}. This is analytically certified at the actual coupling; it is finite coherence, not an infinite-prime BGS obstruction. The balanced count is a minimum, not a computation of the actual root multiplicities.

The available degree-L ordered Duhamel remainder is at most $H^{L+1}/(L+1)!$, with $H = (26/25)|\tau|Nm$. At Heisenberg times $m \asymp N$, H grows as $|\tau|N^2$; a fixed-degree certificate does not decay. Fixed orbit moments alone do not justify the required growing word expansion. Nonperturbative E/G estimates, nonresonant signed cancellation and the complete TRACE-005/BGS-BGS-001 goals remain open. The review separates proof, finite controls and prior work.

70.32 Fixed-time bounded traces and quantizer overlap

Claim IDs: `BGS-FM-013`, `BGS-FM-016`. See Appendix A.

Theorem 70.15 (Bounded fixed-time traces)**.** *The packet proof retains the exact frozen odd-N family and establishes* $|\operatorname{Tr} U_N^j| = O_j(1)$ *for every fixed nonzero integer j.*

The method is standard finite-time coherent propagation, reconstructed with the exact ordinary Weyl symbol and boundary character.

For z=(q,p), the frame is

$$\psi_{z,N}(x) = \left(\frac{2}{N}\right)^{1/4}\sum_{k\in\mathbb{Z}} e^{-\pi N(x/N-q-k)^2} e^{2\pi i N p(x/N-q-k)}, \qquad N\int |\psi_{z,N}\rangle\langle\psi_{z,N}| dz = I.$$

The p integral sets x=y,k=l; the remaining Gaussian integral is exactly 1/N. Odd midpoint Weyl sampling, the invariant lattice character under S0 mod2, and clock/shift irreducibility identify the actual matrix M_N. A degree-five triangular packet expansion has residual O(hˆ4); sampling that weighted-Schwartz residual before finite-dimensional Duhamel gives error O(hˆ3). Its overlap with the original packet is bounded by $C_{j,M}(1+\operatorname{dist}(F^j z,z)/\sqrt{h})^{-M}+C_j h^3$. Anosov nondegeneracy of fixed-time periodic points gives sublevel area O_j(rˆ2), and the exact frame trace gives O_j(Nh)=O_j(1). One-step Egorov or a first-order Gaussian approximation alone would not justify this bound. Keeping the leading Gaussian gives the explicit fixed-time limsup upper bound

$$\sqrt{2}\sum_{z\in\operatorname{Fix}(F^j)}\frac{\det(I+DF^j(z)DF^j(z)^{\mathsf{T}})^{1/4}}{|\det(DF^j(z)-I)|}.$$

Localized higher packets contribute O(sqrt(h)) after the trace integral; the norm remainder contributes O(Nhˆ3). The classical derivative bound and the 4/32 fixed-point counts yield eventual $|\operatorname{Tr}U_N|<4$, $|\operatorname{Tr}U_N^2|<9$ and sum below 13. Thus K=20 is available. No effective dimension threshold or estimate with j of order N is supplied.

FM-007 had already proved normalized fixed-time traces tend to zero in an actual logarithmic time window. The new bounded unnormalized input improves the starting FLOW-003 covariance rate to |m2|=O(1/N), and closes the fixed-trace boundedness assumptions, with subsequences for convergence of the initial traces. It does not remove the external noise.

The Egorov overlap lemma states that

Theorem 70.16 (Egorov overlap rigidity)**.** *if two fixed quantized classical maps F,G coincide on area zero, then*

$$N^{-1}\operatorname{Tr}(K_N^* L_N)\to 0,\qquad \inf_{|\zeta|=1}N^{-1/2}\|K_N-\zeta L_N\|_{HS}\to\sqrt{2}.$$

If the maps differ at one point, the operator-norm liminf is at least sqrt(2), by localized coherent states. The four actual fixed-point Hessians show every fixed nonzero power remains non-affine under C1 conjugacy. Fixed smooth conjugators therefore give the operator separation; fixed analytic ones also give the HS separation.

N-dependent spectral basis changes are outside these assertions.

70.33 Coherent words despite auxiliary expansion

Claim ID: `BGS-TRACE-013`. See Appendix A.

Theorem 70.17 (Arithmetic coherent-words counterexample)**.** *The complete counterexample uses three fixed integral hyperbolic matrices*

$$R=\begin{pmatrix}4&9\\19&43\end{pmatrix},\quad S=\begin{pmatrix}3&2\\1&1\end{pmatrix},\quad V=\begin{pmatrix}-5&19\\-24&91\end{pmatrix},\quad VS=R.$$

R has no integer/topological root and no topological reversor.

The rational fourth root (R+8I)/21 gives $d_p=\operatorname{ord}_p R\mid(p-1)/4$ for all primes p=1 mod20. Thus the even Weil sector of odd dimension N=(p+1)/2 has a clock of unfolded pitch N/d>2, failing Sine2 and its weak local law.

The proofs exhaust determinant signs and use incompatible modulo-three/seven Lucas periods, respectively.

The non-elementary fixed generators S,V have a uniform Cayley gap by Bourgain–Gamburd. Representation transfer and even-sector irreducibility therefore give a uniform one-copy quantum

mixing gap. Yet its two-copy invariant rank is D2=(p+7)/4, not the Haar rank two. More generally

$$D_{k,p} = \frac{2((p+1)/2)^{2k} + (p-3)p(p+1)/2 + (p^2-1)2^{1-2k}\sum_{l=0}^{k}\binom{2k}{2l}p^l}{p(p^2-1)}.$$

The character/frame-potential method is prior Gross–Audenaert–Eisert mathematics. The complete class calculation gives extensive invariant memory and, for k>=3, D_k~Nˆ(2k-3)/4.

On the full p-space the symplectic two-copy invariant rank is 2p+1. Adding a fixed Weyl translation only to the auxiliary channel leaves I and swap, and compresses every other orbit by -1/(pˆ2-1). A two-case Dirichlet argument gives a uniform two-copy gap gamma/18. This channel mixes parity; the added translation never enters the selected word VS. It therefore excludes a full-space auxiliary-gap sufficiency criterion, without asserting a sector-preserving two-copy counterexample.

Label an algebraic eigenbasis of rho(R) by j=1,...,p and perturb by $D_p = \exp(iA_p/p!)$, $A_p = \sum_j (j/p)|\psi_j\rangle\langle\psi_j|$. Hermite–Lindemann proves simple spectra, nonzero traces of every positive power and absence of scalar powers. The perturbation is analytic flat on each fixed strip; the enlarged two-copy gap remains gamma/36. Nevertheless, for the even sector,

$$N^{-2}\sum_{\substack{N\le m\le 2N\\ d_p\mid m}}\Big||\operatorname{Tr}((D_p\rho(R))^m)|^2 - N\Big| \ge 2 - o(1).$$

The full p-space lower bound is 4-o(1). At every fixed nonzero time, the regular Weil character and flat perturbation give bounded traces 1+o(1). Thus all these exact exclusions, fixed-time bounds and auxiliary gaps coexist with coherent Heisenberg returns. Fixed nonlinear shear conjugation preserves the joint construction, but retains smooth arithmetic conjugacy. Earlier project nonlinear arithmetic examples are credited. This is a different map, not an obstruction established for the frozen operator.

The current research audit records exact rational/modular controls, finite character and channel tests, odd coherent frames, checked source leaves and prior-scope comparison. The actual frozen block mass, shifted microscopic energy and nonresonant signed hierarchy remain unproved.

70.34 Verification and remaining work

The recovered supplement includes exact arithmetic replay, an independent C++ verifier, independent matrix assemblies and numerical identity checks. The new quartic relation was checked in 27 finite cases and by a separate internal algebra audit. Its source code and proof are retained. Auxiliary kick strengths in these tests do not resolve $\tau = 10^{-80}$.

The Lean module `CueLoopKernels` states the cubic positive-part identity and a cleared-denominator elimination kernel over integers. These are a small algebraic layer; they do not formalize the analytic loop criterion. Current compilation results are recorded in the publication review.

The remaining task is a model-specific Heisenberg-scale cancellation estimate for the actual mixed resolvent products, or an equally strong direct trace comparison. No noise approximation, source theorem or finite computation in this chapter supplies that missing estimate.

71 Fixed-map continuation: symmetry, crossover, and resonance coercivity

This section records the final continuation after the recovered fixed-map frontier. The detailed proofs and status boundaries are maintained in the corresponding Markdown notes; none of the results below proves the fixed-τ BGS conjecture.

71.1 The physical Weyl symbol and classical symmetry exclusion

For the raw modular convention of the frozen operator, the ordinary midpoint Weyl matrices satisfy $W_N(a,b) = (-1)^{ab}T_N(a,b)$. Consequently the same matrix A_N is the ordinary Weyl quantization of

$$g(q,p) = \cos 14\pi q + \frac{r(q)}{10} + \frac{r(q)\sin 2\pi q}{100} + \frac{r(q+\frac{1}{2})\cos(2\pi p-\alpha)}{100} - \frac{1}{10},$$

where $r(q) = (1-\cos 14\pi q)(1+\cos 2\pi q)$, $\cos\alpha = -3/5$, and $\sin\alpha = 4/5$. For $G_\kappa = S\circ\Phi_g^\kappa$ and $0<|\kappa|\le 10^{-12}$, the project proof shows that G_κ is analytic Anosov, its complete C^1 centralizer is $\{G_\kappa^j : j\in\mathbb{Z}\}$, it has no C^1 reversor, and it is not C^1-conjugate to an affine toral map with the required homological linear part. The actual classical map is

$$F_\tau = \Phi_g^{-\tau/(2\pi)}\circ S^{-1} = G_{\tau/(2\pi)}^{-1}.$$

Thus inherited smooth classical symmetries are excluded for the frozen model, but N-dependent quantum commutants and Heisenberg-time cancellation are not.

71.2 Odd-Pell full-symbol crossover

For odd K, let $N_K + F_K\sqrt{2} = (1+\sqrt{2})^K$ and $\overline{A}_K = K^{-1}\sum_{j=0}^{K-1} M_N^j A_N M_N^{-j}$. The empirical spectral measure of $\sqrt{K}\,\overline{A}_K$ has the Gaussian limit with exact variance

$$\sigma^2 = \frac{1650797}{4000000}.$$

Here η_K is the scalar defined by $M_{N_K}^K = \eta_K I$ on the Pell sequence, so $|\eta_K| = 1$ (canonical source: `mathematics/fixed_model_pell_crossover.md`). If $\epsilon_K\sqrt{K}\to t$ for $U_{\epsilon,N} = \exp(i\epsilon A_N)M_N$, then, for each fixed m,

$$\eta_K^{-m}N_K^{-1}\operatorname{Tr}U_{\epsilon_K,N_K}^{mK} \longrightarrow e^{-m^2t^2\sigma^2/2}.$$

Since $|\eta_K^{-m}| = 1$, the squared moduli satisfy $N_K^{-2}|\operatorname{Tr}U_{\epsilon_K,N_K}^{mK}|^2 \to e^{-m^2t^2\sigma^2}$; with $K/N_K\to 0$ the Abel factors e^{-2smK/N_K} in $L_{N_K}(s)$ tend to 1 at each fixed m. Consequently

$$\liminf L_{N_K}(s) \ge \sum_{m\ge 1} e^{-m^2t^2\sigma^2},$$

so CUE pair behaviour along these Pell dimensions requires $|\epsilon_K|\sqrt{K}\to\infty$. The frozen choice $\epsilon_K = \tau N_K$ eventually satisfies this necessary condition and is therefore outside the proved crossover obstruction.

71.3 A stronger conditional mixed-loop barrier

Let $w = is/2$, $v = -is/2$, and impose normalized vanishing of the three mixed residuals $\mathcal{E}_{2,N}(w,v)$, $\mathcal{E}_{3,N}(w,v,v)$, and $\mathcal{E}_{3,N}(w,w,v)$ at every fixed $s>0$. The additional orientation gives a derivative identity for $R = E(P^3\bar{P})$. Passing positivity of the limiting real moment matrix and using a scalar differential barrier yields

$$\liminf_{N\to\infty} L_N(s) \ge \frac{\sqrt{1+s^{-2}}-1}{2} + \frac{s^3}{64(1+s^2)^{7/2}}.$$

At $s=1$ this raises the earlier lower bound from $0.2071067812\ldots$ to $0.2084878491\ldots$, still below the CUE value $0.2161661792\ldots$. Repeated-clock spectra show that arbitrarily many one-upper residual tests at one selected height do not suffice.

71.4 Sharp Fourier-line resonance coercivity

Let $A \in SL(2,\mathbb{Z})$, $|\operatorname{tr} A| > 2$, and $g(x) = G(v \cdot x)$ with real nonconstant $G \in H^{1+\beta}(\mathbb{T})$. For $h = \{g, C_A g\}/(4\pi)$, the selected orbit sums J_m satisfy

$$\pi|D|\kappa_\beta E_\beta \le \sum_{m\ge 1} m^{2\beta}|J_m| \le \pi|D|(1+\rho_\beta)E_\beta,$$

provided

$$\rho_\beta = 2\sum_{j\ge 0}(a_j a_{j+1})^{-\beta} < 1, \qquad \kappa_\beta = 1-\rho_\beta.$$

The lower constant is sharp over real trigonometric polynomials. In particular

$$\kappa_1 = \sqrt{\frac{T-2}{T+2}},$$

and for the project matrix $\left(\begin{smallmatrix} 3 & 2 \\ 4 & 3 \end{smallmatrix}\right)$, $v = (1,0)$, one has $D = 4$ and $\kappa_1 = 1/\sqrt{2}$. If $|\operatorname{tr} A| \ge 6$, every nonconstant Fourier-line $H^{3/2}$ observable already has positive dynamical variance.

This solves the all-members resonance question on a complete infinite-dimensional Fourier-line class, not for arbitrary analytic two-dimensional symbols. An explicit local polynomial coboundary shows that fixed-point Taylor jets alone cannot close that remaining global cohomological problem.

71.5 Remaining obligation

The fixed-model pair target and all-order microscopic BGS statistics still require a model-specific Heisenberg-scale cancellation estimate for the actual frozen propagator. Classical symmetry exclusion, positive Fourier-line dynamical variance, weak-kick crossover obstructions, and finite exact regressions are logically separate from that quantum estimate.

72 Position-chain mixing and two scoped path contributions

Keep the unchanged odd-dimensional quantization $U_N = \exp(i\tau N A_N)M_N$ at the fixed coupling $\tau = 10^{-80}$. All limits in this section are through odd N. These are subsidiary project results with independent internal review, not a deterministic BGS claim, a priority claim, or an external referee certificate.

72.1 Position-chain mixing and the cyclic contribution

Put $P_N(x,y) = |U_N(x,y)|^2$ and $J_N(x,y) = 1/N$. The position-chain estimate is

$$\limsup_{N\to\infty} \|P_N - J_N\| \le 10^{-64}.$$

Consequently, for all sufficiently large odd N, simultaneously for every integer $L \ge 0$,

$$\|P_N^L - J_N\| \le 2^{-L}.$$

Indeed, P_N is doubly stochastic, so $P_N J_N = J_N P_N = J_N$ and hence $(P_N - J_N)^L = P_N^L - J_N$; the displayed estimate $\limsup \|P_N - J_N\| \le 10^{-64} < 1/2$ then gives $\|P_N^L - J_N\| = \|(P_N - J_N)^L\| \le \|P_N - J_N\|^L \le 2^{-L}$. The threshold is not effective. This is a singular-norm estimate and does not assume that P_N is normal.

Here is the proof mechanism, with complete details in the canonical source. An exact four-term quadratic Gauss identity, followed by two finite-difference transport equations and unitary Duhamel, gives a four-branch kernel with entry error $O(N^{-3/2})$. Its four branch-diagonal

squared amplitudes yield $N^{-1}d(x,y)$ with $|d-1|<10^{-64}$. For every distinct branch pair, the mixed derivative of the difference phase has a Fourier coefficient whose linear term is

$$-\frac{\pi\tau}{128}e^{-i\alpha}\bigl(e^{\pi ir/2}-e^{\pi is/2}\bigr),$$

and whose remainder is bounded by $10^{50}\tau^2$. It is nonzero at the specified coupling. The mixed derivatives also have absolute value less than $1/4$. A local discrete TT^* summation-by-parts argument therefore excludes all lattice aliases and gives squared norm $O(\log N/N)$ away from the analytic zero set. A small-area cutoff is controlled in unnormalized Hilbert–Schmidt norm. Hence the twelve off-diagonal branch kernels are $o(1)$ in operator norm. Double stochasticity gives the stated simultaneous power estimate.

For a cyclic coordinate word x of length m, let $a(x)$ be its path amplitude and $d(x)$ its least cyclic period. The exact cyclic contribution is $D_m=\sum_x d(x)|a(x)|^2$. Writing $q_N=\max_{x,y}|U_N(x,y)|^2=O(N^{-1})$,

$$|\operatorname{Tr}P_N^m-1|\le N2^{-m},\qquad 0\le m\operatorname{Tr}P_N^m-D_m\le m^2Nq_N^{m/2}.$$

Thus $D_m/m\to1$ uniformly on each fixed band $aN\le m\le bN$, $0<a<b<\infty$.

72.2 A precisely marked three-link contraction

With cyclic indices modulo three, define

$$E_{\mathrm{dist}}(L)=\sum_{\substack{a_1,a_2,a_3\ \text{distinct}\\ b_1,b_2,b_3}}\prod_{i=1}^{3}(P_N^{L_i})_{a_ib_i}\prod_{i=1}^{3}U_{b_ia_{i+1}}\overline{U_{b_ia_{i-1}}},\qquad L_i\ge0.$$

Replacing any one link by J_N gives zero exactly, by distinct-column orthogonality. Replacing the longest link by $P_N^{L_i}-J_N$ then gives

$$|E_{\mathrm{dist}}(L)|\le C^3N\,2^{-\max_i L_i},\qquad q_N\le C/N.$$

This includes zero-length links. For $m\ge3$, summing all $L_1+L_2+L_3=m-3$ and multiplying by m^2 costs at most

$$C^3m^2\binom{m-1}{2}N\,2^{-(m-3)/3}.$$

Its normalized smooth compact positive Heisenberg-band sums vanish. This is a marked expression only: finite primitive path pairs have different representation multiplicities, and canonical masks can destroy the free endpoint contraction. No once-counted unmarked family is estimated here.

72.3 The reversed-coordinate-path family

Let $Q_N(x,y)=U_N(x,y)\overline{U_N(y,x)}$. The separate reversal theorem states $\|Q_N\|\to0$. Its sixteen phases use $S_r(x,y)-S_s(y,x)$, whose mixed coefficient has magnitude $\pi\tau/128$ with remainder at most $10^{50}\tau^2$. The same shared discrete norm lemma applies. Let R_m^{rev} sum ordered pairs related by reverse cyclic rotation, excluding all cyclic pairs. Exact periodic-word and reflection counts give

$$|R_m^{\mathrm{rev}}-m\operatorname{Tr}Q_N^m|\le2m^2Nq_N^{\lfloor m/2\rfloor}.$$

For the even edge-axis reflection there are $m/2-1$ paired edges and two endpoint loops; both loop factors are retained. It follows that $R_m^{\mathrm{rev}}/m\to0$ uniformly on every fixed positive Heisenberg band.

72.4 The signed boundary remains open

For every $w \in C_c^\infty((0,\infty))$,

$$N^{-2}\sum_m w(m/N)D_m \longrightarrow \int_0^\infty \xi w(\xi)\,d\xi.$$

Consequently the still-unestimated other-path remainder $R_m^{\mathrm{other}} = |\operatorname{Tr} U_N^m|^2 - D_m - R_m^{\mathrm{rev}}$ must satisfy

$$N^{-2}\sum_m w(m/N)R_m^{\mathrm{other}} \longrightarrow \int_0^\infty (\min(\xi,1)-\xi)w(\xi)\,d\xi$$

for the pair CUE target. The correction is negative above $\xi = 1$. Neither the marked estimate nor position-chain mixing proves this signed limit. The higher factorial hierarchy, TRACE-005 and deterministic BGS remain open. Exact finite verification supports only its stated algebraic coverage; no analytic theorem here has been formally certified.

73 Once-counted distinct-port three-block cancellation

For the unchanged frozen propagator, put $P_{xy} = |U_{xy}|^2$ and $J_{xy} = N^{-1}$. The position-chain theorem supplies, for all sufficiently large odd N, $\max P_{xy} \le C/N$ and $\|P^L - J\| \le 2^{-L}$ for every integer $L \ge 0$. Let T_m be the set of rooted ordered coordinate-word pairs obtained by cutting the first cyclic word into three nonempty blocks A, B, C with pairwise distinct initial vertex values, then taking a cyclic rotation of ACB for the second word. This is a set: each ordered pair is counted once, even if several cuts produce it. Write $\mathcal{A}(x) = \prod_{j \bmod m} U_{x_j x_{j+1}}$.

Theorem (BGS-TRACE-015). For every smooth w compactly supported in $(0,\infty)$,

$$\frac{1}{N^2}\sum_m w(m/N) \sum_{(x,y)\in T_m} \mathcal{A}(x)\overline{\mathcal{A}(y)} \longrightarrow 0.$$

For $K = \lceil \log N \rceil$ and support in $[a,b] \subset (0,\infty)$, the error is bounded by

$$O_w(K^5/N) + O_w\Big(N^8(C/N)^{K/2}\Big) + O_w\Big(N^5 2^{-(aN-3)/3}\Big).$$

The startup threshold inherited from the position-chain theorem is not made effective.

Proof mechanism. In the marked three-link contraction, replacing any link by J gives zero by column orthogonality and distinct ports. Using the longest link therefore suppresses the full marked sum exponentially, including zero internal link lengths. Passing to a once-counted family is a separate combinatorial step. Words with repeated length-K substrings impose at least $K/2$ equality constraints and have superpolynomially small exceptional absolute weight. For the remaining words, two blocks of length at least $4K$ force a unique cut triple, using uniquely located common substring anchors. Ambiguity is consequently confined to two short blocks. Competing cuts lie in an $O(K)$ neighborhood, giving $O(K^3)$ alternatives and $O(K^2)$ original length choices. A rank-one relative positional permutation would be a transposition; conjugating predecessor cycles then forces equal-valued positions to be two cut heads, contradicting distinct ports. Thus every competing representation imposes at least two equalities. Exposing the three starts first pays each equality through a transition bounded by C/N. After the two trace roots and band normalization, the exceptional mass is $O(K^5/N)$. Every nonexceptional pair has exactly three marked representations, corresponding to cyclic block labels. The canonical proof gives all equality, cyclic-root and overlap counts explicitly.

Boundary. This is a deterministic, once-counted, fixed-complexity family estimate. The combinatorial theorem uses only unitarity, flat entries and the stated position-chain mixing bound; actual-model input enters through BGS-FM-021. It is compatible with spectrally nonuniversal families satisfying those matrix estimates. It does not cover repeated ports, arbitrary reconnections, growing encounter complexity or higher joint traces. The complete signed remainder, including restoration of the super-Heisenberg plateau, remains open; no BGS or local-statistics theorem follows from this family alone.

74 Rank-two four-cut classification and its bounded certificate

For two distinct reduced four-block decompositions with pairwise distinct port values, a specified rank-two relative position permutation between cyclically aligned equal outputs is a product of two transpositions on four consecutive alternating source vertices. Its cut sets are exactly $\{s, s+1, s+4, z\}$ and $\{s, s+3, s+4, z\}$, where z lies outside the cluster and its endpoint. A three-cycle cannot occur. The canonical mathematical proof in `mathematics/four_cut_rank_two_classification.md` retains the support, its immediate successors, and the first cut set, reducing to at most twelve or ten vertices and preserving genuine cuts. This contraction and the endpoint-preserving lift were independently reviewed.

The compiled `Bgs.FourCutKernel` module certifies only the executable finite predicates: 3858 pair configurations with 560 directed local matches for sizes five through twelve, and 650 oriented triple configurations with no matches for sizes five through ten. These use `native_decide` and its declared native-evaluation trust. The size-four exclusions use ordinary `decide`. The abstract compression, cyclic-word correspondence, and analytic cancellation are not Lean theorems here. This bounded certificate alone asserts neither complete four-block unmarking nor BGS universality.

75 Once-counted reduced four-block cancellation

Let U_N be unitary, $P_{xy} = |U_{xy}|^2$, and $J_{xy} = N^{-1}$. Assume, eventually, $\max P_{xy} \le C/N$ and $\|P^L - J\| \le 2^{-L}$ for every integer $L \ge 0$. BGS-FM-021 supplies these bounds for the unchanged frozen map. Let $T_m^{(4)}$ be the set of rooted ordered coordinate-word pairs obtained by cutting the first cyclic word into four nonempty blocks A, B, C, D, whose initial vertex values are pairwise distinct, and taking a cyclic rotation of $ADCB$ for the second word. Each pair is counted once, regardless of its witnesses. This is the unique reduced four-block ordering: all four original successor boundaries change. Put $\mathcal{A}(x) = \prod_{j \bmod m} U_{x_j x_{j+1}}$.

Theorem (BGS-TRACE-016). For every smooth w compactly supported in $(0, \infty)$,

$$\frac{1}{N^2} \sum_m w(m/N) \sum_{(x,y) \in T_m^{(4)}} \mathcal{A}(x)\overline{\mathcal{A}(y)} \longrightarrow 0.$$

For $K = \lceil \log N \rceil$ and support in $[a, b] \subset (0, \infty)$, the absolute value of this complete signed band sum is bounded by

$$O_w(K^{10}/N) + O_w\Big(N^{10}(C/N)^{K/2}\Big) + O_w\Big(N^5 2^{-(aN-4)/4}\Big).$$

The startup inherited from the mixing theorem is not effective.

Marked cancellation and exact correction. The marked four-link sum M_m has two independent trace-root factors. Replacing any one link by J gives zero by column orthogonality

and distinct ports, so its longest link makes M_m exponentially small. The local overlap O_m restricts the first two blocks to $A = (a)$ and $B = (b, a, b)$, with arbitrary nonempty C, D. It retains a free long endpoint: uniform C gives $\delta_{db}/N = 0$, and uniform D gives $\delta_{ac}/N = 0$. Its normalized sum is likewise exponentially small. The once-counted correction is $M_m/4 - O_m$, up to the stated exceptional error; a uniform multiplicity division is not imposed on all pairs.

Combinatorial control. A word with distinct length-K cyclic substrings localizes every competing cut within $O(K)$ of an original cut. Indeed a common refinement interval of length at least $(r + 1)K$ fixes its translation offset. Each mismatch component begins at a common genuine cut, since otherwise two cut heads would have equal vertex values. A longest common interval fixes cyclic alignment, without an extra factor m. Three fixed anchors in different original four-block pieces force cut uniqueness by prefix cancellation. The corrected argument handles all three locations of the unanchored block, including empty prefixes and equal lengths. Thus ambiguity requires two short original blocks. There are $O(K^2 m)$ original length choices and $O(K^4)$ competing cut sets. Exposing all four starts first pays every equality-rank constraint by a transition bounded by C/N, giving $O(K^6/N)$ for rank at least three.

The separately proved computer-assisted classification identifies every rank-two relation with the local flip $(s\ s + 2)(s + 1\ s + 3)$ on an alternating four-vertex cluster. Its finite kernel is certified separately by `Bgs/FourCutKernel.lean`; the universal compression and the analytic argument here are not formalized by that bounded certificate. Two distinct local flips compose to rank four, except neighboring flips, whose product is the forbidden three-cycle case. The representation graph is therefore triangle-free. A pair without a higher-rank relation has one or two cut sets; subtracting its single local edge counts it once. Higher multiplicities cost at most another $O(K^4)$, yielding $O(K^{10}/N)$. Repeated length-K substrings, including periodic words, have superpolynomially small exceptional marked absolute mass.

Boundary. This is a fixed-family theorem under generic unitarity, flatness and position-chain mixing hypotheses. It is compatible with nonuniversal spectra satisfying those bounds. Repeated ports, non-permutation pairs, overlaps among families, higher or growing reconnection complexity, plateau restoration and the full signed hierarchy remain open. The rounding obstruction precludes inferring the exact spectral target from these coarse controls alone. No deterministic BGS conclusion follows.

76 Finite-order WKB reduction and its local analytic boundary

The general near-unitary integer-power and complete partition perturbation estimate is already proved in the maintained Heisenberg trace-support source, Section 3. The present contribution instantiates it through the actual arbitrary-fixed-order WKB construction and explicit signed-table conventions. For each fixed factorial order $k \geq 2$ and fixed compact smooth Fourier weight, choose a fixed transport order $J \geq k$. The direct finite-shift transport construction gives $\|U_N - V_{N,J}\| \leq C_J N^{-J-1/2}$. For either the genuine-inverse Laurent table or the signed-adjoint table, the complete partition–Möbius Fourier transform changes by $O_{k,W,J}(N^{k-J-1/2})$. The estimate retains every partition and zero block. The proof uses the exact rising-factorial identity $\sum_\pi |\mu(\pi)| N^{|\pi|} = N(N + 1) \cdots (N + k - 1)$ and finite-time power telescoping. The signed comparison to the same ensemble reference is therefore equivalent at this fixed order. It remains open.

The two tables must not be conflated: negative Laurent powers are inverses, whereas the adjoint table uses conjugated positive traces after forming each block sum. Neither table is assigned a positive circle spectral process. The every-bandwidth positive-measure extension belongs to the actual unitary spectra only. No fixed finite J covers all orders by this estimate.

Separately, exact finite Gaussian reciprocity represents every finite $\operatorname{Tr} V_{N,J}^m$, including all branches and winding classes, without coprimality restrictions. It uses rational matrix inverses and center-based Abel regularization on a periodic Wiener class. Its unitary L^2 extension is not a point-evaluation or cancellation bound.

The massive local cyclic covariance yields formal connected coefficients $|\ell_{q,m}| \le mC^q q!$. For the explicitly defined conditioned rotated Gaussian box, $h = (2\pi N)^{-1}$ and $mh \le b$, the actual local theorem is $Z_{m,h} = \exp(h\ell_{1,m})(1 + O_b(\sqrt{h}))$. The unconditioned box integral additionally carries its Gaussian box probability. The separate global rotated-box theorem now controls the original-trace contour and its summed tail in a specified finite band. It does not justify absolute summation of local first-loop errors over exponentially many classes. Signed evaluation, unrestricted bands, TRACE-005 and deterministic BGS remain open.

Sources and verification boundary. The maintained proofs are `factorial_wkb_reduction.md`, `exact_gaussian_reciprocity.md`, `uniform_cluster_lemma.md`, and `rotated_local_first_loop.md` in `mathematics/`. The sole external local inequality is Kolesnikov–Milman [33], arXiv:1310.2526v7, Theorem 3.1(1), p. 13 and (3.2), p. 14, with convex exhaustion. Independent internal review and exact finite controls are separate from external referee review and analytic formalization.

The unitary polar factor of the principal WKB matrix. Let $h = N^{-1}$ and retain the actual fixed coupling and ordinary Weyl convention. For the explicitly glued principal four-branch WKB matrix $W_{0,N}$, the additional amplitude transport and direct discrete Weyl composition give

$$\|W_{0,N}^* W_{0,N} - I\|_{\mathrm{op}} = O(h^2), \qquad V_N^{\mathrm{pol}} := W_{0,N}(W_{0,N}^* W_{0,N})^{-1/2} = \exp\bigl(ih\,\mathrm{Op}_N(b)\bigr)U_N + O_{\mathrm{op}}(h^2),$$

for a fixed smooth real symbol $b = -\operatorname{Im}(a_1/a_0)$ expressed on the final canonical graph. In particular $\|V_N^{\mathrm{pol}} - W_{0,N}\|_{\mathrm{op}} = O(h^2)$. The polar factor V_N^{pol} is a different object from the order-J WKB approximation $V_{N,J}$ used above. The smooth symbol is obtained from the exact branch gluing, rather than inferred from an operator-norm estimate. The preexisting ergodic spectral-transport theorem, with the actual logarithmic Egorov and mixing inputs, supplies a bounded unfolded matching of V_N^{pol} and U_N with mean-square displacement $O(1/\log N)$ after a common rotation. Consequently, if either unitary endpoint has all compact factorial correlation limits, the other has the same limits; the all-order reference bounds supply the needed local occupation moments. No such limit is evaluated here. The principal matrix $W_{0,N}$ may be nonnormal, and its small polar correction is not an eigenvalue-matching theorem for that matrix or a replacement of the order-J trace-table reduction by $J = 0$.

77 Global rotated-box representation at finite Heisenberg bandwidth

Keep the prescribed odd-dimensional quantization and $\tau = 10^{-80}$. For each fixed transport order J, let $V_{N,J}$ be the explicit finite four-branch kernel. Write $h = (2\pi N)^{-1}$; its amplitude polynomial is $\sum_{\ell=0}^{J} N^{-\ell} a_{r,\ell} = \sum_{\ell=0}^{J} (2\pi h)^{\ell} a_{r,\ell}$. The two expansion parameters must not be conflated.

Put $B_m = 6I - P_m - P_m^{\mathsf{T}}$, where P_m is the cyclic shift, and $D_m = \det B_m$. In particular $B_1 = 4$, while B_2 has diagonal entries 6 and off-diagonal entries -2. For each branch vector $r \in \{0,1,2,3\}^m$ and winding representative $\nu \in \mathbb{Z}^m / B_m\mathbb{Z}^m$, let $x_{r,\nu}^*$ be the unique real stationary point of the exact lifted phase $\Psi_{r,\nu}(x) = \sum_j S_{r_j}(\tau, x_j, x_{j+1}) + \nu \cdot x$. Let $A_{r,J}$ be the product of the finite one-step amplitudes and set

$$\mathcal{B}_{N,m,J} = \frac{e(3mN/8)N^{m/2}}{8^m} \sum_{r} \sum_{\nu \bmod B_m} \int_{x_{r,\nu}^* + (1-i)[-1,1]^m} e(N\Psi_{r,\nu}(x))\, A_{r,J}(x)\, dx.$$

The integrals retain the full nonlinear phase and their product orientation. The class list is not asserted to consist of distinct physical orbits.

Conservative finite-band theorem. For every fixed $J \geq 1$, uniformly for $1 \leq m \leq 2N$,

$$\operatorname{Tr} U_N^m = \mathcal{B}_{N,m,J} + O_J(N^{3/2-J}) + O_J\Big(\sqrt{N}e^{-4N/5}\Big).$$

The first error is the separate operator-telescoping error; the second is an absolutely summed contour-tail error. Negative traces use conjugation. More generally, the latter error is $O_{J,\beta}(\sqrt{N}\exp[-(8\pi/5-\beta\log 7)N])$ for $m \leq \beta N$ and $\beta < 8\pi/(5\log 7)$, with a strict endpoint. No coprimality condition is imposed on N and D_m.

Contour mechanism and normalization. Exact Gaussian reciprocity first gives global, center-based Abel/Fresnel integrals. The literal symbol bounds certify a holomorphic endpoint strip of width 2, with cyclic nonlinear Hessian row bound below $1/10$. After centering at $x^*_{r,\nu}$, deform to $z = u - i\,\mathrm{clip}(u,-1,1)$. Since $K_0 = B_m/4 = I + L_{\mathrm{cycle}}/4$, monotonicity of clipping gives

$$u^{\mathsf{T}} K_0\,\mathrm{clip}(u) \geq \sum_j \min(u_j^2, |u_j|).$$

The nonlinear correction leaves imaginary-phase damping at least $(4/5)\sum_j \min(u_j^2,|u_j|)$. Cauchy deformation is performed with positive Gaussian regularization; dominated convergence removes it on the saturated contour. The proof retains the full $4^m D_m$ class count, $D_m \leq 6^m$, amplitude bound 2^m, Jacobian bound $2^{m/2}$, and the factor $N^{m/2}/8^m$. Their scalar integral estimate produces the displayed summed error.

Conditional tail without an artificial volume factor. For one integral normalize by its actual leading amplitude times $(2\pi h)^{m/2}/\sqrt{\det K}$, where $K = -\Psi''(x^*)$. Partition the saturated contour by its actual large-coordinate set S. The retained core uses the exact principal Gaussian K_{TT}^{-1}, $T = S^c$, on its box. Incident interactions cost only $C^{|S|}$; convex-box moment and Poincaré estimates bound the core by e^{Cmh}. The Schur determinant ratio also costs only $C^{|S|}$. Consequently the normalized outside-box integral is at most

$$e^{Cmh}\left[\Big(1 + C\sqrt{h}\,e^{-2/(5h)}\Big)^m - 1\right].$$

For each fixed b_h with $mh \leq b_h$, this is $O_{b_h}(m\sqrt{h}\,e^{-2/(5h)})$. The conditional Gaussian normalization is retained: a local box integral divided by the full Gaussian factor is $\mathbb{P}_G(B_h)Z_{m,h}$. Completing the core Gaussian on all of Euclidean space before bounding its cross terms is not permitted.

Separate alternative corollary. With $\eta = 10^{-4}$, $\gamma = 10^{-7}$ and

$$\kappa = \log(1+\sqrt{2}) + \gamma - \tfrac{1}{2}\log(1-4\eta),$$

the conditional bound gives total tail $O_{J,\beta}(\sqrt{N}\exp[-(4\pi/5-\beta\kappa)N])$ for $\beta < (4\pi/5)/\kappa$. Indeed $a_* \leq e^{\gamma m}$ and $\det K \geq (1-4\eta)^m D_m/4^m$; summing the leading Gaussian weights therefore costs at most $e^{\kappa m}$. This is a separate corollary, not a replacement for the stronger conservative rate at $\beta = 2$.

Scope and verification. The canonical proof is `mathematics/global_rotated_box_trace.md`. The analytic chain has fresh independent internal review and portable exact controls; it is not analytically formalized or externally refereed. This finite-band contour step is closed. Signed evaluation of the exact box sum, optimal local remainders compatible with exponential class mass, unrestricted bands, TRACE-005 and full BGS remain open. In particular, the local $O(N^{-1/2})$ first-loop errors are not summed absolutely here.

78 Exact paired characters and phase-only winding localization

For the physical odd-dimensional quantization, let $G_m = \mathrm{Fix}(S^m : \mathbb{T}^2 \to \mathbb{T}^2)$ and $D_m = |G_m|$, with $S = \left(\begin{smallmatrix} 3 & 4 \\ 2 & 3 \end{smallmatrix}\right)$. Integer Hamilton–Jacobi gluing quotients the original branch/winding list by exactly 4^m identical copies per physical label. This is an exact quotient, not cancellation and not a repair of the earlier normalization. The physical group cannot generally be replaced by the winding group alone. Let Q_m be the cat quadratic action and B_m its perfect polarization. For the actual full weight α from the exact global representation,

$$\begin{aligned} \operatorname{Tr} V_{N,J}^m &= \zeta_{N,m} D_m^{-1/2} \sum_{z \in G_m} e(NQ_m(z))\alpha(z), \\ |\operatorname{Tr} V_{N,J}^m|^2 &= \sum_{d \in G_m} e(NQ_m(d))\widehat{C}_d(-Nd). \end{aligned}$$

where $C_d(z) = \alpha(z+d)\overline{\alpha(z)}$ and $\widehat{C}_d(\xi) = D_m^{-1} \sum_z C_d(z) e(-B_m(z,\xi))$. For arbitrary weights the same identity concerns the normalized quadratic sum; physical trace notation requires the actual weight and odd N. At zero coupling global Fresnel weights equal one, retaining the exact arithmetic torsion contribution. The separate first-response identity uses the derivative divided by iN; no torsion suppression follows from that jet.

Actual action defect. For the unit-winding displacement $d_m = (S^m - I)^{-1}(0,1)$, the exact nonlinear stationary-action defect has a periodic infinite-chain limit $\mathcal{D}_\infty$. Put $\lambda = 3 + 2\sqrt{2}$, $\beta = 29/5$ and $\eta = 10^{-70} + 10^{-110}$. The reviewed refinement gives

$$\begin{aligned} \|\mathcal{D}_\infty\|_\infty &\le 2\eta, \\ |\mathcal{D}_m - \mathcal{D}_\infty| &\le 4 \cdot 10^6 \eta \beta^{-m}, \\ [\mathcal{D}_\infty]_{C^{\log\beta/\log\lambda}} &\le 2 \cdot 10^7 \eta. \end{aligned}$$

The phase-only selected coefficient satisfies $K_{m,N} = \widehat{\mu}_\tau(-N) + O(N\beta^{-m/2})$, where $\mu_\tau = (q + \mathcal{D}_\infty(q,p) \bmod 1)_*(dq\,dp)$. Exact finite-grid aliases are retained. The Pell/grid input is prior project work, not a new identity supplied by this checkpoint.

Monotonicity and averaged squares. The actual phase map is strictly increasing along each q fibre and has inverse bound $H_\tau(q+\delta,p) - H_\tau(q,p) \ge \delta^{p_*}/28$, $p_* = \log 7/\log\lambda$. Thus, for circle arcs I and $\gamma = \log\lambda/\log 7$,

$$\mu_\tau(I) \le (28|I|)^\gamma, \qquad \sum_{n=1}^{M} |\widehat{\mu}_\tau(n)|^2 \le 112 M^{1-\gamma}.$$

The measure is non-atomic; a density and pointwise Fourier decay are not proved. The average runs over Fourier index/dimension, not over time at a fixed dimension. Only Sections 1–3 of the source transversality note are promoted. Complete determinant, transport and quantum-box weights, other displacements and torsion sectors, the full signed hierarchy, TRACE-005 and BGS remain open. These are reviewed analytic results with bounded exact controls, not analytic formalization.

An exact stationary potential and homoclinic defect. At the literal coupling $\tau = 10^{-80}$, the actual real periodic action correction is a sum of one stationary potential V_τ along the cat orbit. Its exact unit-winding limit is the uniformly convergent homoclinic cocycle

$$D_\infty(z) = \sum_{j\in\mathbb{Z}} [V_\tau(S^j(z+d)) - V_\tau(S^j z)], \qquad d = (1/\sqrt{2}, -1/2).$$

With $\alpha = \log(29/5)/\log(3+2\sqrt{2})$, the full bound $\|V_\tau - \tau g/(2\pi)\|_{C^\alpha} \le 10^{60}\tau^2$ is explicit at that coupling. The actual isolated Fourier orbit has sum $|A_{(1,0)}| \ge \tau/1000$. Exact Fourier factorization therefore excludes an additive L^2 translation transfer and shows that the distributional q derivative of D_∞ is not represented by an L^1 function. The positive derivative measure of $q + D_\infty$ has a nonzero singular part. These use the project's prior Fourier orbit-sum and constant-tail methods. They give neither a multiplicative-gauge obstruction nor a decay statement for the distinct inverse-fibre pushforward measure. Complete quantum weights, signed trace comparison and BGS remain open.

79 A selected full-weight homoclinic family

Fix the literal coupling and any finite WKB order $J \ge 0$. The exact trace representation of $V_{N,J}$ has coefficients

$$A_{N,m,J}(d) = D_m^{-1} \sum_{z\in G_m} \beta(z+d)\overline{\beta(z)}, \qquad \beta(z) = e(NQ_m(z))\alpha_{N,m,J}(z).$$

For $d_m = (S^m - I)^{-1}(0,1)$ and every fixed positive band $0 < a_0 \le m/N \le b_0 \le 2$, the independently reviewed interface theorem gives $A_{N,m,J}(d_m) = O_{J,a_0,b_0}(N^{-1/2})$ through odd N. This is a coefficient of the fixed-J WKB representation, not an intrinsic orbit summand of the exact unitary.

The physical leading coefficient retains the geometric-mean stability weight. Its limiting observable is $\sqrt{J_d}\, e(NA)$ in physical area, with $\partial_q A \ge 1/2$ and a half-Hölder transformed density. Norm-uniform stability-weighted periodic equidistribution applies to its $O(1+N)$ Hölder norm. The exact paired quantum factor approaches one; the individual conditioned factor need not. Positive periodic mass and Cauchy–Schwarz prevent an exponential class-count loss. Summed absolute contour tails transfer the selected box coefficient to the global Fresnel coefficient only in the certified band $m \le 2N$.

Let $\mathcal{O}_m = \{\pm S^j d_m : 0 \le j < m\}$ count each displacement once. Exact cyclic symmetry and negative autocorrelation symmetry give $|\mathcal{O}_m| \le 2m$, including short-orbit overlaps. For fixed smooth w compactly supported in $(0,2)$,

$$N^{-2} \sum_m w(m/N) \sum_{d\in\mathcal{O}_m} A_{N,m,J}(d) = O_{J,w}(N^{-1/2}).$$

The inherited aggregate transfer to U_N costs $O_{J,w}(N^{3/2-J})$; fixed $J \ge 2$ therefore preserves this error order. No individual U_N coefficient is assigned. The unchanged CUE benchmark belongs to the complete pair comparison. The exponentially larger complement, its torsion and plateau mechanism, higher factorial orders, TRACE-005 and BGS remain open.

The classical source bundle uses Parry–Pollicott's complex-potential remainder with its proof and boundary transfer, Ruelle's entropy inequality, and Mañé's 1981 entropy formula together with the mandatory 1983 errata [35, 36]. The original Pesin theorem-number locator remains unverified. Each W, Q, EQ and combined-interface review retains its own hypotheses and scope.

A separately reviewed supporting decomposition removes the stationary-label cyclic ordered pairs exactly once. Their positive contribution is $D_m^{-1}\sum_z p_m(z)|\alpha(z)|^2$, with the actual least

period $p_m(z)$; proper periods are retained. An order-two character proves exact disjointness from the selected homoclinic family. On a compact central window, the remaining signed ordered-pair sum equals the original pair discrepancy up to $O(N^{-1/2})$ for fixed $J \geq 2$. Only unremoved portions of torsion sectors remain. The prior position-path ramp is FM-021; no identification of its pair family with these stationary labels is asserted. Above the plateau threshold, the signed complement must still provide $N - m$; no such estimate is proved.

80 The centered collective occupation hierarchy

Work in the full ordered $\mathrm{End}((\mathbb{C}^N)^{\otimes m})$ with ordinary Hilbert–Schmidt metric and the two-sided slot cycle. For traceless diagonal coefficients of degree q, full Haar removal has Gram matrix $G_{q,N} = I - H_{q,N}$, where $H_{q,N} = \mathbb{E}_{V\ \mathrm{Haar}} \mathrm{Sym}^q(|V|^2 - J)$. The exact collective trace is

$$S_{m,N}(U) = \sum_{q=1}^{m} \mathrm{Tr}\Big[G_{q,N}^{-1}(\mathrm{Sym}^q(T_U) - H_{q,N})\Big], \qquad T_U = (|U|^2 - J)|_{\mathbf{1}^\perp}.$$

The inverse is legitimate at every degree, by the following Gram bound (canonical source: `mathematics/collective_occupation_hierarchy.md`, (C4)–(C5)). The exact Haar fourth moments $\mathbb{E}|V_{ia}|^4 = 2/[N(N+1)]$ and $\mathbb{E}|V_{ia}|^2|V_{ib}|^2 = 1/[N(N+1)]$ $(a \neq b)$ give $\mathbb{E}_V(T_V^* T_V) = \mathbb{E}_V(T_V T_V^*) = I/(N+1)$ on the traceless diagonal space, where $T_V = (|V|^2 - J)|_{\mathbf{1}^\perp}$. For $q \geq 2$, factor two distinct tensor legs of $\langle v, \mathrm{Sym}^q(T_V)v\rangle$, bound the remaining legs by $\|T_V\| \leq 1$, and apply Cauchy–Schwarz in V; this gives $|\langle v, H_{q,N}v\rangle| \leq \|v\|^2/(N+1)$. Since $H_{q,N}$ is self-adjoint, its spectrum lies in $[-(N+1)^{-1}, (N+1)^{-1}]$, so $G_{q,N} = I - H_{q,N}$ is invertible with $\|G_{q,N}^{-1}\| \leq (N+1)/N$ for every $q \geq 2$; at $q = 1$, $H_{1,N} = \mathbb{E}_V T_V = 0$ and $G_{1,N} = I$. There is no conditioning singularity at $q = N$. An all-degree Gram bound and a finite $q \leq bN$ Haar-tail argument, together with the explicitly inherited actual FM-021 gap and flatness estimates, give $S_{m,N}(U_N) = O_b(1)$ through fixed positive Heisenberg bands. In outline (source (C9)–(C14)): the Haar tail $\Pr(\|T_V\| > \sqrt{159/160}) \leq 4N^2 e^{-N/160}$ and the rank bound $\mathrm{rank}\, H_{q,N} \leq p(q) \leq e^{3\sqrt{q}}$ give $\sum_{q=2}^{m} \mathrm{Tr}[H_{q,N}(I - H_{q,N})^{-1}] = O_{b,\varepsilon}(N^{-1+\varepsilon})$ for $m \leq bN$; the inherited FM-021 inputs $\|T_{U_N}\| \leq 1/2$ eventually and $P_{U_N}(a,b) \leq C/N$ make $\log\det(I - zT_{U_N})^{-1}$ uniformly bounded on $|z| = 3/2$, so Cauchy summation of the symmetric powers yields $S_{m,N}(U_N) = \det(I - T_{U_N})^{-1} - 1 + O_{b,\varepsilon}(N^{-1+\varepsilon} + (2/3)^m) = O_b(1)$. The original N^{-2} compact-time aggregate is therefore $O_w(N^{-1})$. The $N = 2$ unbounded-degree failure control prevents replacing this finite band theorem by an infinite-degree determinant expectation.

With P_H the full invariant projector and P_{coll} the centered collective projector, the exact remaining space is

$$Q = I - P_H - P_{\mathrm{coll}} = (\Pi - P_H - P_{\mathrm{coll}}) + (I - \Pi).$$

The full pair trace is $\min(m, N) + S_{m,N}(U_N) + \mathrm{Tr}(R_m Q C_U)$. Some vectors in Q are cycle-fixed. This is not a decomposition into only nontrivial permutation sectors, nor a bound on the whole balanced trace. The separately reviewed quadratic and slow-source examples retain their single-sector or single-vector scopes. No identification with WKB stationary masks, complete signed residual estimate, plateau mechanism or BGS follows.

A distinct ordered-diagonal companion. Removing full Haar invariants from the entire ordered diagonal algebra gives an orthogonal projector P_{ord} of rank $N^m - 1$. Only full traceless slot support survives its cyclic trace. An exact matrix-product Neumann identity and the independently reconstructed Haar tail show, for each fixed $a > 0$, uniformly over integers $m \geq aN$ and sufficiently large odd N,

$$|\,\mathrm{Tr}(R_m P_{\mathrm{ord}} C_U)| \leq 12N^3 e^{-c_a N}, \qquad c_a = \min\{a\log 2, -a\log\sqrt{159/160}, 1/160\} > 0.$$

This bound is exponential in N, not in m at fixed N; the exact $N = 2$ even-degree control has a $-1/m$ tail. Different cycle cancellations mean this theorem does not imply the collective bound above. The exact residual $Q_{\mathrm{ord}} = I - P_H - P_{\mathrm{ord}}$ still contains remaining balanced and off-charge ranges. No bound on that full signed trace or BGS conclusion follows. Only the inherited actual FM-021 singular gap is used.

Every fixed factorial order under entrywise flatness. For canonical positive and negative consecutive cycle presentations, the Haar-removed ordered-diagonal sector admits a true Gram-inverse compression even when the mixed two-sided permutation does not preserve its range. Cutting at every interval endpoint leaves only open strings; zero strings are exact identity deltas and all repeated ports remain. For a unitary with $N \max_{a,b} |U_{ab}|^2 \le C$, this sector trace is bounded by $C_{h,C}(1 + \log N)^{2h}$ independently of all cycle lengths. The proof runs as follows (canonical source: `mathematics/factorial_ordered_diagonal.md`, G1–G5). The centered Gram operator H_c obeys the same exact Haar moment bound as in the collective hierarchy, $\|H_c\| \le 1/(N+1)$, so the centered Neumann series for $(I - H_c)^{-1}$ converges absolutely in operator norm. Writing the sector trace through this series, each summand is a difference $F_Q(U) - \mathbb{E}_V F_Q(V)$ over independent Haar products Q; the common scalar $F_J(U) = N^{\mathrm{cycles}(\sigma\pi^{-1})-M}$ is independent of U and cancels inside each summand. Exact Haar second moments give $\mathbb{E}\|Q_j - J\|^2$ decay $(N+1)^{-j}$ up to the dimension factor, so a fixed cutoff $L = 4h + 4$ makes the tail $O_h(N^{-3/2})$; the finitely many initial terms are controlled by maximum-entry Haar moments $(\mathbb{E}Y_N^p)^{1/p} \le 4 \log N + 2(p!)^{1/p}$, $Y_N = N \max_{a,b} |V_{ab}|^2$, which produce the displayed $(1 + \log N)^{2h}$ bound. It uses no position-chain singular gap.

The exact partition–Möbius formula, including every zero-power stratum, therefore gives an $O_{k,\widehat{\phi},C}((1 + \log N)^{2k}/N)$ contribution for each fixed factorial order $k \ge 2$ and bounded compact Fourier weight. All-zero block powers remain exact constants with no positive replica degree. The actual-map application inherits only the accepted entrywise flatness. A separate singular-gap companion gives exponential-in-N estimates for fixed-many matched long cycles and the two-positive/one-negative long–long pattern, using shared Haar draws across cycles. Neither estimate controls the full signed projector complement or proves TRACE-005/BGS.

81 A disjoint Hamming and low-switch pair residual

For the actual odd-dimensional map at fixed $\tau = 10^{-80}$, work in $E = \mathrm{End}((\mathbb{C}^N)^{\otimes m})$ with ordinary Hilbert–Schmidt structure. Let R be the two-sided full slot cycle, $\mathcal{C}_U = \mathrm{Ad}(U^{\otimes m})$, P_H the full Haar invariant projector, Π the occupation-balanced projector, and F_r the raw coordinate-Hamming-at-most-r mask. Define the genuine centered range projectors and the raw off-charge mask by

$$P_r = \mathrm{Proj}((I - P_H)\,\mathrm{ran}(\Pi F_r)), \qquad \widetilde{P}_r = \mathrm{Proj}((I - P_H)\,\mathrm{ran}\,F_r), \qquad O_r = (I - \Pi)F_r.$$

Exactly $\widetilde{P}_r = P_r + O_r$. The centered balanced projector is not a raw path mask. The earlier sparse-marker theorem already covers $r \le \kappa N/\log N$; the independently reviewed position-block refinement reaches a fixed positive fraction of the cycle length.

Put $D = N^2 - 1$, $n = m - 2r$, $M = \sum_{s=0}^{r} \binom{m}{s}$, $t = \|P_U - J\|$, and $a_n = \mathbb{E}_{V\,\mathrm{Haar}}\|P_V - J\|^n$. For $n \ge 1$ and $Ma_n < 1$, both centered projectors satisfy

$$|\,\mathrm{Tr}(RPC_U)| \le DM\frac{t^n + a_n}{1 - Ma_n}.$$

The separate denominator-free estimate is $|\,\mathrm{Tr}(RO_rC_U)| \le 2DMt^n$ for $n \ge 0$, with $0^0 = 1$. These bounds retain the exact Gram inverse, repeated labels, whole offdiagonal local spaces, and shifted cyclic seam.

Fix $0 < a < b < \infty$, let $h(\delta)$ be binary entropy, and put $\lambda = \frac{1}{2}\log(160/159)$. The balanced radius $r_{\mathrm{bal}} = \lfloor \delta m \rfloor$ requires both

$$h(\delta) < (1-2\delta)\lambda, \qquad bh(\delta) < 1/160.$$

The inherited actual singular bound and explicit Haar tail then give exponential decay on $aN \le m \le bN$, with strict exponent slack. Separately $r_{\mathrm{off}} = \lfloor 0.49m \rfloor$ is valid for O_r, using the eventual bound $t_N \le 10^{-63}$. The inherited statement is $\limsup t_N \le 10^{-64}$; no eventual 10^{-64} bound is inferred. The larger radius is not asserted for the whole balanced or centered ball.

Actual coordinate translations and low-switch fibers. Define

$$B_c(j,k) = U(j,k)\overline{U(j+c,k+c)}.$$

A separately reviewed proof gives one existential finite L with $\|B_c^L\| \le 3/4$ for every nonzero c and all sufficiently large odd N. Growing offsets have one-step norm $o(1)$; fixed offsets use a genuine compact norm limit, fixed-observable Egorov, and mixing for the actual invariant normalized area. These are inherited FM-018 inputs with their original source-review boundaries, not consequences of topological conjugacy alone. No quantitative startup threshold is given.

For rooted words put $c_j = y_j - x_j$ modulo N, with cyclic switch count $s(c)$ and checksum $t(c) = \sum_j c_j$ modulo N. The exact block contraction for each difference word, together with FM-021 for zero-charge runs, bounds the entire family $s(c) \le R_N = \lfloor \kappa N/\log N \rfloor$, $t(c) \ne 0$ by $O(e^{-\eta N})$. Its original N^{-2} time aggregate is $O_w(N^{-1}e^{-\eta N})$. The checksum selects whole fibers and implies off-charge membership; checksum zero does not imply occupation balance. Safe constant translations are included and must not be added again.

One exact residual, still unestimated. Let L_{high} select the low-switch, nonzero-checksum fibers with the additional condition $\#\{j : c_j \ne 0\} > r_{\mathrm{off}}$. This cuts no fiber internally and is disjoint from $O_{r_{\mathrm{off}}}$. The four removed projectors are pairwise orthogonal and commute with R, so

$$Q = I - P_H - P_{r_{\mathrm{bal}}} - O_{r_{\mathrm{off}}} - L_{\mathrm{high}}$$

is an orthogonal projector. No commutation with $\mathcal{C}_U$ is asserted. Using the exact Haar-cycle trace gives uniformly on the fixed positive band

$$|\operatorname{Tr} U_N^m|^2 - \min(m,N) = \operatorname{Tr}(RQ\mathcal{C}_{U_N}) + O(e^{-cN}).$$

TRACE-021 is this pair residual reduction, for this earlier generation Q; the current generation is the distinct operator Q_{cur} of Section 82, and the name TRACE-021 refers to Q_{cur}. It leaves the signed Q trace, plateau compensation and higher factorial comparison open; it does not strengthen TRACE-020 into a full fixed-order theorem or prove BGS.

Auxiliary half-radius limitation. Separate Fourier and symplectic witnesses have an order-N off-charge trace at radius $m/2$ despite exact flatness and $\|P_U - J\| = 0$. The symplectic witness also has reverse-kernel norm $N^{-1/2}$. These change the operator and credit the prior broad rounding obstruction; they are not counterexamples for the actual quantizer. They show that the listed coarse data alone do not justify a uniform half-radius extension.

82 Reviewed phase and signed-trace checkpoint

This section records the 7 October 2026 reviewed subsidiary frontier. The authoritative statement and exact source/review locators are `mathematics/phase_trace_checkpoint.md`; source copies retain their reviewed bytes and historical preparation headings. The deterministic BGS claim and TRACE-005 remain open.

Charge and exact pair residual. FM-027 gives an eventual, map-dependent uniform contraction of every nonzero ordinary cyclic charge block for the unchanged odd-dimensional operator at the literal fixed coupling. Its scoped actual phase/parametrix A2–A5 inputs do not include classical mixing or Egorov. Occupation-uniform row twists remove every low-switch offcharge partial fiber, including checksum zero. The current TRACE-021 residual is

$$Q_{\mathrm{cur}} = I - P_H - P_{\mathrm{bal}} - O_{\mathrm{low}} - L_{\mathrm{all}}.$$

This Q_{cur} is not the operator Q of Section 81 (the earlier generation $Q = I - P_H - P_{r_{\mathrm{bal}}} - O_{r_{\mathrm{off}}} - L_{\mathrm{high}}$): the occupation-uniform O_{low} and the all-fiber L_{all} replace $O_{r_{\mathrm{off}}}$ and L_{high}. The name TRACE-021 refers to the current residual Q_{cur}; the exact trace identity displayed for the earlier Q in that section is a statement about the earlier generation only. Its balanced component is $\Pi - P_H - P_{\mathrm{bal}}$; its remaining offcharge component has the literal high-Hamming, high-switch occupation-mismatch mask. The full switch-damped aggregate is negligible when $\kappa a N(1 - z_N) - 2\log N \to +\infty$. The undamped signed marked-switch boundary integral, full residual trace and plateau compensation are unestimated. This strengthens the older checksum-restricted generation without changing its historical proof or counting translations twice.

Completed phase source and balanced reductions. TRACE-022 proves the exact positive Schur-energy and expected three-radius reduction. Positivity is after phase expectation; the reverse compact-to-Abel direction retains its near-zero majorant. TRACE-023 proves the complete finite-size color-one pair source identity using four compact coefficient functionals, the universal complete-pair bound $F_N(s) \le \coth(s/N)/(2N)$, and dimension-uniform complex microscopic source regularity with uniform integrability for each fixed flavor count. The latter is a theorem about the original phase ensemble, not an explicit higher-flavor compact CFT. Complete signed growing-degree hook covariance and microscopic identification remain open. Sectorwise absolute bounds cannot replace the completed signed source.

The accepted finite-N phase Ward identity reduces the balanced Abel limit to its stated signed overlap estimate. Its v2 wording retains the infinite smooth Poisson expansion and does not infer CUE eigenvalues from Haar eigenvectors alone. The actual signed overlap estimate remains open. The independently reviewed Fourier/tensor example has infinite unregularized one-channel boundary-delay variance despite exact flatness, perfect position mixing and reverse decay. It excludes a finite-L^2 delay premise, not the balanced CUE law.

Radial boundary and fixed words. FM-028 proves collective compactness of the whole actual charge pinching and the radial shell/resolvent theorem for $q = 1, \theta = 0$: at every $\delta_N \to 0$, the centered radius is $1 - \delta_N + o(\delta_N)$ and the worst unit-circle singular-resolvent norm is $\delta_N^{-1}(1 + o(1))$. This theorem really uses inherited Egorov and classical mixing. It supplies no occupation-uniform, power-norm, Schatten or signed trace conclusion. FM-029 establishes literal kick/cat normalized fixed-word moment freeness and its relative-word chord and rank laws. The independently reviewed quantitative continuation covers only the precise triple-logarithmic word window. A fixed commutator polynomial has $\liminf \lambda_{\max} \ge 1 - 10^{-148}$ versus free radius $\sqrt{3}/2$, refuting strong freeness. These laws do not estimate Heisenberg-scale words or the microscopic signed hierarchy.

Ordinary bosonic body and source qualification. FM-030 proves scalar and nonnormal block ordinary-body coercivity $S \ge H/(2K^2)$, where $K = 1 + \sqrt{C-1}/(1-\rho)$ under the stated flatness, singular-gap and strict centered-block assumptions. The actual family has eventual $c = 1/50$. The sharp real-absorption determinant inequality yields $S_\alpha \ge cH - ks/2$ at $\alpha = e^{-s/N}$. The checked external leaf is Bhatia's generalized exponential metric increase theorem. These results do not construct a block gauge, super-Berezinian, full superintegral positivity or normalized asymptotic tail estimate.

The audited ordinary bulk-only color-one disk/plane formula misses even its zero-source normalization. Its prescribed boundary completion and the completed compact pair identity resolve that finite-size source issue. This narrow correction does not invalidate all invariant prescriptions, graph universality or FM-018's direct finite-matrix Gaussian/Hessian and determinant results. Earlier graph-template source language is qualified by the exact-source record.

A separately accepted changed growing-period Fourier/cubic family has flatness, reverse decay and small nonzero cyclic-charge norms but divergent genuine high-Hamming/high-switch offcharge boundary contribution. It is a changed-family obstruction to abstract closure, not a counterexample to the literal BGS operator. All higher factorial comparison, deterministic noise removal and BGS/CUE identification remain open.

Actual one-pinch disappearance. TRACE-024 combines the independently accepted actual fixed-lag displacement-frame overlap estimate and annealed one-pinch deletion in one scoped theorem. For every smooth complex weight on a fixed positive Heisenberg band,

$$N^{-2}\sum_m w(m/N)\left[E_{N,m}(0)-\int E_{N,m}(\theta)\,d\theta\right]\longrightarrow 0.$$

The untwisted positive aggregate separately vanishes for bounded compact-band weights. Fixed-lag constants give no aggregate rate or effective startup. The annealed leaf extends to one consecutive block of pinches; the actual positive block hierarchy is not asserted. Arbitrary separated insertions, occupation/Hamming/high-switch masks and boundary deconvolution remain open. FM-028 C4 and TRACE-023 Abel theorem (A) enter only the annealed leaf.

Fixed-size ordinary centering and qualified endpoint barriers. FM-031 proves ordinary centering almost everywhere on labelled stable strata for odd $N \geq 3$, retaining original normalized circle/disk measures, the coarea factor and gauge determinant $\Delta \geq 2N-1$. Its completed signed source and fixed holomorphic derivatives satisfy fixed-N cutoff bounds. It gives no positive full signed-source measure, super-Berezinian or normalized dimension-uniform microscopic tail. TRACE-025 gives changed-quantizer rounding and simple-spectrum split constructions. A successful endpoint proof requires a property or estimate excluding these witnesses and an identified use of it. This qualification supersedes the stronger strategy sentence in the immutable rounding source; no particular raw-Weyl/all-order proof method is required. Conditional Schatten preservation does not establish an actual Schatten theorem. The fixed-grid rounded family has a positive lower bound, while the separate growing-period split family has the stated divergent bound; these scopes are not interchanged.

Corrected finite hook and signed-current reduction. TRACE-026 extends exterior characters by $e_j = 0$ for $j > N$ before the full marked cycle selection. Its exact all-N, m identities cancel disconnected slot cycles before phase expectation or absolute values and retain repeated coordinates, edges and periodic words in the Eulerian directed-tree formula. Once-counted cyclic/reverse/overlap subtraction defines the complete signed current residual $H_m(U_N)$. Under the stated eventual flatness and strict P/Q singular-gap assumptions, the removed Abel contribution is $1/(4s^2)$, and balanced identification is equivalent to

$$N^{-2}\sum_m e^{-2sm/N}H_m(U_N)\longrightarrow -\frac{e^{-2s}}{4s^2},\qquad s>0.$$

The required negative plateau correction is not a small-error estimate and remains unproved. Finite signed sums are completed before summation in degree. Original v1 and its audit remain history; corrected v2 and its acceptance are separately preserved. The bounded phase-source literature audit adds no load-bearing graph-universality leaf or priority claim.

Weighted coherence and upper-half slow-window Ward law. TRACE-027 proves finite-unitary spatial covariance with its nontrivial classical Green–Kubo leading term $H = R + R^T - I$, where $R = (I - P + J)^{-1}$. The exact signed Hessian-plus-cotangent Ward action on $|p_m|^2$ is controlled for $m \geq 3$. For the unchanged odd-dimensional actual operator, convergence holds uniformly only on $\lceil T_N/2 \rceil \leq m \leq T_N$ for the explicitly constrained slowly growing T_N. This does not assert convergence at every fixed m, on the full prefix, or at Heisenberg scale. The actual unconjugated transpose-product kernel $B_{ab} = U_{ab}U_{ba}$ separately has norm $O_\tau(\sqrt{\log(2N)/N})$, removing only the degree-two exception for the specified mixed-moment family. It does not identify all fixed-degree CUE moments. The exact exported proof and fresh independent controls are sealed; the unavailable original ZIP verifier and receipts were not replayed. The absorbed infinite Poisson/Ward obligation (13), complete balanced microscopic law, literal masked T16, TRACE-005, higher factorial comparison and deterministic BGS remain open. No formal certification is asserted.

83 Complete unmasked insertion closure and the deconvolution boundary

This reviewed continuation supersedes earlier open wording for complete unmasked separated insertion sums at their stated normalization. It does not estimate a masked coefficient, source derivative, or the undamped zero-insertion spectral difference. The exact sources and reviewed scope are `mathematics/right_frame_unmasked_pinch_hierarchy.md` and `mathematics/convex_pinch_weak_noise_and_deconvolution.md`.

Right frame and separated averages. TRACE-028 proves, for every fixed $r \geq 1$, the actual right-multiplication compression bound $\|C^* R_U^r C\| = O_r(\sqrt{\log(2N)/N})$. The exact transpose identity and its analytic input are independently checked; left multiplication is not silently used for the arbitrary-B step. The interval frame bound is uniform over arbitrary dimension-dependent unitary B, provided B is fixed during the coherent-time sum. It does not permit an uncontrolled sequence B_n. Every complete fixed-j weak-composition average $A_{j,N,m}$ has vanishing normalized positive-band contribution, both actual and annealed. Zero gaps, repeated occupations and periodic slot patterns remain included. The exact rooted-slot sum is $\binom{m}{j} A_{j,N,m}$, retaining every marking multiplicity. The N^{-2} normalization applies to the average; raw composition and rooted-slot sums instead use $N^{-(j+1)}$ and $N^{-(j+2)}$.

Three pinches close the complete convex hierarchy. For a gap vector $\mathbf{n} = (n_1, \ldots, n_j)$ with $n_i \geq 0$, define

$$G_{j,N}(\mathbf{n}; \theta) = \mathrm{Tr}_H[D_\theta T_\theta^{n_1} \cdots D_\theta T_\theta^{n_j}],$$

where $T_\theta = \mathrm{Ad}(Z_\theta U)$ and D_θ is the corresponding centered pinch superoperator, as in the canonical source `mathematics/right_frame_unmasked_pinch_hierarchy.md` and its exported proof (equations (18)–(23) of `evidence/wave3_checked_20261007/raw/right_overlap/authorized_exports/right_overlap_and_unmasked_insertions.md`). On the ordinary HS traceless space, collective compactness gives, for each $\varepsilon > 0$, a fixed R with $s_{R+1}(D_0) \leq \varepsilon$ eventually, where s_i are the singular values of D_0. Splitting the singular values and using $\|D_0\| \leq 1$ with the flatness estimate $\|D_0\|_{S_2}^2 \leq C_{\mathrm{flat}} N$ gives

$$\|D_0\|_{S_3}^3 = \sum_i s_i^3 \leq R + \varepsilon \|D_0\|_{S_2}^2 \leq R + \varepsilon C_{\mathrm{flat}} N, \qquad \text{hence} \qquad \eta_N := N^{-1} \|D_0\|_{S_3}^3 \longrightarrow 0.$$

Thus $\|D_0\|_{S_3}^3 = o(N)$, with no quantitative rate for η_N. Since V_θ is unitary, D_θ has the same singular values as D_0 on the traceless space. Schatten–Hölder with three S_3 factors and bounded

remaining factors therefore gives, for every $j \geq 3$,

$$|G_{j,N}(\mathbf{n};\theta) - 1| \leq \|D_0\|_{S_3}^3 = N\eta_N,$$

uniformly in occupation phases, all gaps and insertion count. The identity contribution 1 is retained exactly. No pointwise $o(1)$ bound on $G_{j,N} - 1$ is asserted or needed: the closure below is at the normalized aggregate level. Averaging over the $\binom{m-1}{j-1}$ gap compositions gives $0 \leq A_{j,N,m}(\theta) \leq 1 + N\eta_N$ for $j \geq 3$. In the expansion of $\mathrm{Tr}[zT_\theta + (1-z)D_\theta]^m$ the order-j coefficients are the binomial weights $\binom{m}{j} z^{m-j}(1-z)^j$, whose sum over $j \geq 1$ is $1 - z^m \leq 1$. Hence, with $O(N)$ band terms at the N^{-2} normalization, the $j \geq 3$ tail contributes at most $N^{-2} \cdot O(N) \cdot (1 + N\eta_N) = O(N^{-1}) + O(\eta_N) = o(1)$. This does not bound the first two orders pointwise by compactness alone. Their independently controlled complete averages (the preceding paragraph), combined with the three-pinch tail, close every complete unmasked convex insertion mixture with mass at most one, uniformly for every $z_N \in [0,1]$, at the stated N^{-2} band normalization. There is no quantitative decay rate, growing-r stationary phase estimate or effective startup.

Exact convex bounds and bounded-noise signed stability. TRACE-029 derives exact finite-N resummation and deconvolution estimates by first-pinch factorization. The tail length is fixed while coherent time is summed; its unitary-word distribution is independent of that summed time. Every binary word, adjacent pinch, all-D word and short run is counted. With the actual right-frame input and the separately reviewed compactness/Abel annealed companion, complete signed weak-noise stability holds for bounded $u_N = N(1 - z_N)$ and arbitrary bounded compact-band weights. This is an unmasked conclusion, without source differentiation or boundary extraction.

Sharp unsigned logarithmic obstruction. The actual whole-centered contraction, with $\kappa = 1 - \rho > 0$, gives $F_m(z) = 1 + o(1)$ uniformly on $aN \leq m \leq bN$ when $1 - z = A_0 \log N/N$ and $A_0 > 2/(\kappa a)$. The unweighted complete positive interval sum satisfies

$$N^{-2} \sum_{m=\lceil aN \rceil}^{\lfloor bN \rfloor} z^{-m} F_m(z) = (1 + o(1)) \frac{N^{bA_0 - 1}}{A_0 \log N} \longrightarrow \infty.$$

This is an actual unsigned obstruction. At finite size the pinch correction's common scalar part is $1 - z^m$; it cancels exactly after signed occupation subtraction and is excluded by the relevant projection. The divergence does not refute T16. A signed proof must retain this cancellation before absolute bounds. Hamming/high-switch extraction, source derivatives and rates, signed centered deconvolution, undamped zero-insertion comparison, microscopic Ward (13), balanced identification, TRACE-005 and BGS remain open. The unavailable author's 222-control suite was not replayed; fresh independent bounded controls and exact exported proof bytes support the stated review.

84 Quantitative exclusion of arithmetic translation algebras

Keep the unchanged odd-dimensional operator $U_N = \exp(i\tau N A_N) M_N$, with $\tau = 10^{-80}$, and the physical midpoint symbol, including its shifted momentum coefficient. Put $\kappa = \tau/(2\pi)$ and equip matrices with $\langle X, Y \rangle_N = N^{-1} \mathrm{Tr}(X^* Y)$. For an odd divisor $d \geq 3$ of N, let

$$E_{d,N}^0 = \mathrm{span}\{T_N((N/d)v) : v \in (\mathbb{Z}/d\mathbb{Z})^2 \setminus \{0\}\}, \qquad \Pi_{d,N}^0 : \mathcal{H}_N \longrightarrow E_{d,N}^0.$$

The $d^2 - 1$ displayed raw Weyl modes are exactly orthonormal. No coprimality of d and N/d or CRT factorization is assumed.

Theorem 84.1 (Arithmetic translation-algebra exclusion)**.** *There is C_τ, independent of N, d and the grid labels, such that*

$$\|\Pi^0_{d,N}\,\mathrm{Ad}(U_N)\Pi^0_{d,N}\| \le \min\left\{1, C_\tau\left(\frac{d^{262}}{N}\right)^{1/129}\right\}.$$

For every fixed $\delta < 1/262$, uniformly when $d \mid N$ and $d \le N^\delta$, the compression is

$$O_\tau\big(N^{-(1-262\delta)/129}\big).$$

In particular, for $d \le N^{1/524}$,

$$\sup_{\substack{X\in E^0_{d,N}\\ \|X\|_N=1}} \Big|\|U_N X U_N^* - X\|_N^2 - 2\Big| \le 2C_\tau N^{-1/258}.$$

The fixed-d statement follows immediately. Since the cat propagator permutes these labels, the conclusion applies to normalized arithmetic cat orbit sums. If a projection P_N lies in $\mathbb{C}I \oplus E^0_{d,N}$ and $0 < \alpha_N = \mathrm{rank}(P_N)/N < 1$, its nonzero centered part $X_N = (P_N - \alpha_N I)/\sqrt{\alpha_N(1-\alpha_N)}$ satisfies the same unit-vector estimate. The projections $0, I$ are excluded from this normalization. Arbitrary spectral projections are not covered.

Uniform finite type at fixed coupling. Write the kick flow as $\Psi = \Phi_g^{-\kappa} = \mathrm{Id} + u$. Complex-strip bounds give $u = -\kappa X_g + R$. Crucially, for a translation ξ and $r = \|\xi\|_\infty$ in its centered representative,

$$\|D^\alpha(R(\cdot+\xi) - R)\|_\infty \le C\kappa^2 r \qquad (|\alpha| \le 16),$$

using one extra derivative. Degree-16 differential filters isolate the primitive $(-1,0)$ and $(0,-1)$ Fourier modes of the actual symbol. Their coefficient norms are $6409325/1143072 < 6$, and their surviving multipliers have moduli $4/3$ and 2. Explicit Cauchy estimates bound the filtered remainder by $10^{55}\kappa^2 r$, smaller than the leading $|\kappa|r$ signal at the literal coupling. A selected real component of $\xi - \eta + u(x+\xi) - u(x)$ has finite type at most 16, uniformly in the translations. Directional slicing and the one-dimensional sublevel lemma therefore give coincidence-sublevel area $O_\tau((d\varepsilon)^{1/16})$ for a nonzero d-grid translation.

Quantum overlap and dimension loss. Exact Weyl multiplication gives the actual kick Egorov error $O_\tau(N^{-2}\|a\|_{H^5})$. It remains uniform after inserting a translation, since translations preserve Sobolev norms. In the trace intertwining argument, division by squared map separation costs ε^{-4} in C^3. Hilbert–Schmidt Cauchy–Schwarz on the bad set retains the *square root* of its area, while the aliased Fourier tail costs $N^{-2}\varepsilon^{-3}$. Thus every matrix overlap is bounded by

$$C_\tau\left((d\varepsilon)^{1/32} + N^{-1}\varepsilon^{-4} + N^{-2}\varepsilon^{-3}\right).$$

Choosing $\varepsilon = N^{-32/129}d^{-1/129}$ gives $C_\tau(d^4/N)^{1/129}$ per entry. Passing to the entire (d^2-1)-dimensional span costs a further factor d^2, yielding the stated $262/129$ exponent. The residual estimate follows from unitarity and $\|\mathrm{Ad}(U_N)X - X\|_N^2 = 2 - 2\,\mathrm{Re}\langle X, \mathrm{Ad}(U_N)X\rangle_N$.

Scope. This is a project proof with independent internal review and exact finite controls, not an external referee certificate or analytic formalization. The exponents are not claimed optimal; the endpoint $\delta = 1/262$ is not a decay statement. Constants depend on the fixed coupling, with no useful numerical startup or uniform shrinking-coupling estimate supplied. Faster-growing divisors, general quantum commutants, coherent returns and the signed Heisenberg joint-trace comparison remain open. The theorem does not establish microscopic BGS universality.

84.1 The actual higher-seven-primary commutant

For every odd N satisfying $49 \mid N$ and $5 \nmid N$, the prescribed pair A_N, M_N has common commutant $\mathbb{C}I$ (FM-024). This is an actual-model conclusion. Reducing $1000A_N$ at the Gaussian prime with $i \mapsto 2$ kills the diagonal and lower bonds, while every internal upper bond and the raw metaplectic Fourier kernel remain units. Both matrices are conjugated by the same chain-normalizing gauge.

The exact last and penultimate block-row equations force the leading Toeplitz coefficient to be scalar. Their sufficient conditions are tied by polynomial identities to six numerators and a degree-828 norm. Complete factor products and Rabin tests exclude factors whose degrees are divisible by $42 = \mathrm{ord}_{49}(5)$, uniformly excluding zeros at all roots of order divisible by 49. Minimal-tail elimination handles $N \geq 147$; an exact actual-residue cyclic-rank certificate at $N = 49$ handles the short-chain base. A nonzero commutator-rank minor then lifts to characteristic zero.

For an algebraic-entry matrix X, writing the algebraic Hermitian spectral resolution $A_N = \sum_j a_j P_j$ in $[X, e^{itA_N} M_N] = 0$ gives $\sum_j e^{ita_j}(XP_jM_N - P_jM_NX) = 0$. For nonzero algebraic real t, Lindemann–Weierstrass coefficient separation [34] yields each coefficient zero. Summing shows $[X, M_N] = 0$, then $[X, P_j] = 0$, hence $[X, A_N] = 0$. Thus X is scalar, including at $t = 10^{-80}N$. This is only an algebraic-entry commuter restriction.

The higher-seven component excludes $5 \mid N$ and $7 \parallel N$; the separate first-seven extension follows below. No spectral simplicity, quantitative gap, signed Heisenberg comparison, or BGS conclusion is asserted. The canonical proof, completed independent review, and portable exact certificate replays are separate from the bounded Lean predicate and its native-evaluation trust boundary.

Finite formal certificate. The seven-primary Lean source has compiled proofs of 22 executable array predicates, with independent semantic review and coefficient binding. These use `native_decide`; generic polynomial correctness, Rabin soundness, finite-field bridges and the commutant proof remain unformalized.

Extension to the first unramified seven-primary level. A separate independently reviewed exact certificate closes the remaining unramified case: for odd $N > 7$ with $7 \mid N$, $49 \nmid N$ and $5 \nmid N$, the actual pair (A_N, M_N) has scalar complex joint commutant. Four corner-block equations eliminate the long endpoint gauge and give a 63×48 polynomial system over $\mathbb{F}_{5^6}$. In each of six residue classes, three minors have degree at most 936. Their 937-node identities, plus ten extra nodes each, are independently rebuilt in 17,046 exact determinant evaluations. Gcd, Bezout and finite-field odd-order exclusions then establish the rank for every relevant root; this is not dimensional sampling. Invertibility of the corner block removes all Toeplitz tails, and a nonzero commutator minor lifts to characteristic zero. Together with the higher-seven result this extends FM-024 to all odd $N > 7$ with $7 \mid N$ and $5 \nmid N$. Combining with the separate prior FM-008 result covers odd $N \geq 3$ with $35 \nmid N$, except $N = 7$. At that stage the remaining ramified family was the odd multiples of 35; the chain-trace extension below closes its first-seven part. This does not make the full complex commutant of U_N scalar, prove seven-divisible spectral simplicity or establish BGS. FORMAL-003 remains restricted to its earlier executable-array predicates.

First-seven closure with arbitrary five-adic ramification. A separate independently reviewed argument removes the condition $5 \nmid N$ at the first seven-primary level: the actual pair (A_N, M_N) has scalar complex joint commutant for every odd $N > 7$ with $7 \mid N$ and $49 \nmid N$. A valuation gauge preserves a rank-seven endpoint block even when the full reduced Fourier matrix is singular. Before reduction, an integral block-trace consequence of the actual commutator supplies an additional diagonal constraint. Its two normalized harmonics cannot vanish together, by an exact polynomial Bezout identity. All nine collision partitions have enough singleton groups; nonzero Fourier minors of orders at most three force every block scalar. A nonzero minor

of the augmented integral system then lifts to characteristic zero. The finite controls support this uniform argument, not dimensional extrapolation. Together with the separate higher-seven and FM-008 results, the then-remaining uniform arithmetic family was the odd multiples of 245, closed by the separate corrected lattice argument below, besides the genuine exception $N = 7$. No single-U complex scalar-commutant, seven-divisible simplicity, quantitative gap or BGS conclusion follows.

Closing the higher-seven ramified common-commutant branch. The separately reviewed corrected v2 argument proves scalar joint commutant of the actual pair for every odd N divisible by 245. Write $N = 5^e n$, $5 \nmid n$, $49 \mid n$. A normalized integral lattice of actual commuters, rather than the naive reduced pair, has leading coefficients controlled by squared Gaussian binomials. Uniform graded valuation bounds remove all lower/diagonal corrections. The elementary q-Lucas identity then identifies a mixed-digit upper-Toeplitz algebra. An invertible highest-to-lowest high-digit Fourier block reduces its leading coefficient to a shifted prime-to-five base commutant; induction removes all remaining high-digit coefficients. Saturation of the actual integral commuter lattice lifts scalarity to characteristic zero.

The shifted-base step uses the already reviewed polynomial identity with independently variable pole parameter. Its new finite checks retain six literal effective-gauge witnesses and six distinct unit-upper-gauge witnesses at dimension 49. The original finite-witness naming discrepancy is explicitly corrected by geometric-gauge transfer; the old rank-42 endpoint attempt is not counted as a successful certificate. Finite bases and dictionaries alone do not replace the uniform lattice proof.

Combining the retained seven-primary components with the separately attributed FM-008 result yields scalar complex joint commutant of (A_N, M_N) for every odd $N \geq 3$ except the genuine exception $N = 7$. Each component retains its own review and source boundary. The literal-U_N corollary is restricted to algebraic-entry commuters via the existing coefficient-separation argument. No scalar full complex commutant of a single U_N, seven-divisible spectral simplicity, quantitative spectral gap, signed trace comparison or BGS follows.

At the exceptional dimension seven, a separate exact characteristic-zero check gives $T^2 = -7$ and $G^3 = -7TI = T^3 I$, where $T = \operatorname{tr} G$ and $G_{xy} = \zeta_7^{-(3x^2-2xy+3y^2)}$. Thus $H = G/T$ has $H^3 = I$ and trace one; its cube-root multiplicities are $(3, 2, 2)$. Since $A_7 = (9/10)I$, the complex pair commutant has dimension $3^2 + 2^2 + 2^2 = 17$. This is a characteristic-zero identity, with no residue nullity lifting.

Three distinct spectral extensions. A separately reviewed coefficientwise cyclic-arc induction proves simplicity of the actual Hermitian generator $A_{7\cdot 5^e}$ for every $e \geq 1$. Its finite residue and derivative inputs are independently certified; the uniform implication uses the repaired two-index analytic argument. It does not establish simplicity of the propagator throughout that family.

For the four dimensions $N \in \{21, 35, 49, 63\}$, independent exact outward interval certificates give simple A_N and all decreasing-eigenvalue leading spectral minors of M_N nonzero. The prior large-imaginary-time argument and registered Lindemann–Weierstrass input then give simple $e^{itA_N} M_N$ for nonzero real algebraic t. This is a finite-dimensional result.

A separate exact Laurent-polynomial certificate proves

$$\operatorname{Tr}(M_N) \operatorname{Tr}(A_N^2 M_N) - \operatorname{Tr}(A_N M_N)^2 \neq 0 \qquad (N \geq 3 \text{ odd},\ N \neq 7).$$

The degree/root-order reduction makes its cyclotomic screen exhaustive. The unique extreme active spectral difference and the same registered transcendence input imply that $|\operatorname{Tr}(e^{itA_N} M_N)|^2$ is transcendental at every nonzero real algebraic t. At seven it equals one. This extends the prior prime-to-seven result only for one-step traces; it requires no simple-spectrum assumption. These

statements give no uniform quantitative gap, higher-power seven-divisible trace theorem, signed Heisenberg comparison or BGS conclusion.

Literal-time multiplicity bounds on the seven–five family. For $N = 7 \cdot 5^e$, $e \geq 1$, every eigenvalue of the actual $e^{itA_N} M_N$ has multiplicity at most seven at every nonzero real algebraic time, including $t = 10^{-80}N$. At least $612N/625 + 7$ eigenvalues are simple for even $e \geq 4$, and at least $612N/625 + 175$ for even $e \geq 6$; the odd counts are $24N/25$ for $e \geq 3$, $24N/25 + 35$ for $e \geq 5$, and $24N/25 + 875$ for $e \geq 7$. The exact special-dimension lower bounds are 4333 at $N = 4375$ and 107443 at $N = 109375$. The counts are assembled additively over disjoint complete cat eigenspaces (canonical source: `mathematics/literal_time_multiplicity_bounds.md`). For even $e \geq 4$: the newest sector contributes $24N/25 = 600N/625$ simple branches; the first-old quotient of dimension $24N/625$ contributes half of its blocks, $12N/625$, via the norm classes $c = 2, 4 \bmod 5$ with simple first-order slopes; the disjoint deepest even core contributes 7 at second order. The total is $612N/625 + 7$. For even $e \geq 6$ the residual-25 theorem splits all 24 size-seven blocks of the intermediate layer ($24 \cdot 7 = 168$) together with the 7 deepest dimensions, adding a disjoint 175 in place of the core 7 alone, giving $612N/625 + 175$. At $N = 4375$ exactly 18 first-old blocks split at first order, giving $4200 + 18 \cdot 7 + 7 = 4333$; at $N = 109375$ exactly 324 first-old blocks split and the residual-25 layer adds 175, giving $105000 + 324 \cdot 7 + 175 = 107443$. For odd e the newest sector gives $24N/25$ ($e \geq 3$), the deepest odd core adds 35 ($e \geq 5$), and the residual-125 theorem splits 120 further size-seven groups ($120 \cdot 7 = 840$), giving $24N/25 + 875$ with the 35 counted once ($e \geq 7$). The separately reviewed cat, complete oldform, deepest-core and newest-sector arguments supply the local splitting premises. Polynomial minors in $U(t)$ are algebraic exponential polynomials, so the registered Lindemann–Weierstrass input transfers witnessed ranks to each nonzero algebraic real time. The rank of $\chi'(U)$ counts simple eigenvalues; higher derivative stacks recover the dimensions of multiplicity-bounded eigenspaces.

A finite first-order companion proves simple U at $N = 35, 175$; the latter is the additional bounded case. Zero oldform first-order compressions defeat a criterion, not simplicity of the actual propagator. The repeated-eigenspace count does not assert persistence of a fixed oldform invariant subspace. Full uniform simplicity, quantitative gaps and BGS remain open.

The underlying inert-torus multiplicity-free mechanism has prior provenance in Kelmer [37]; the actual cyclic/CRT and jet arguments are independently derived here.

Intermediate old layers and the strengthened partial count. For the unchanged family $N = 7 \cdot 5^e$ and every nonzero real algebraic t, the preceding multiplicity bound remains at most seven. The exact first-old-layer classifier strengthens the simple-eigenvalue lower bound to $612N/625 + 7$ for even $e \geq 4$. Its regular norm classes contribute half of the first-old quotient; the singular classes are decided by an exact cyclotomic coefficient-balancing test. Nonempty intersection alone does not ensure a nonzero compression: at residual dimension 625, the norm label $c = 13$ has five intersections but six zero blocks and 24 simple blocks. The general return operator is a character-normalized matrix inverse with initial factors and the resonant projector term retained. Its corrected coherent-lattice residue is an affine orbit sum, and integral circlewise six-cycle bases prove second-order splitting of all 24 intermediate seven-dimensional blocks at residual dimension 25. Including the prior $3, 2, 2$ deepest blocks gives $612N/625 + 175$ simple eigenvalues for even $e \geq 6$. Exact first-old counts give 4333 at $N = 4375$ and 107443 at $N = 109375$. The counts add over disjoint complete unperturbed cat eigenspaces and pass to algebraic time through the existing exponential-rank theorem and its registered Lindemann–Weierstrass input. They do not identify fixed-time eigenvectors with old invariant subspaces. Larger intermediate residuals and their wild-character denominators remain open; no full simplicity, gap estimate, or BGS conclusion follows.

The separately audited odd newest sector contributes $24N/25$ first-order branches at odd depths $e \geq 3$. The deepest odd core adds 35 for $e \geq 5$. At odd $e \geq 7$, the residual-125 bridge

supplies 840 further second-order branches; nesting places these and the deepest core in W_1, disjoint from the newest complete cat groups. Thus the addend is $875 = 840 + 35$, counted once. The inherited algebraic-time rank transfer applies; no fixed-time oldform invariance, full simplicity or quantitative gap is asserted.

Chapter 11 — Temporal resonance and spectral modules (continued)

85 All-order moment-channel proximity with a clock spectrum

The following negative control concerns the same ensemble on both sides: its balanced moment channels approach Haar in induced Hilbert–Schmidt operator norm, while every realization has a rigid clock spectrum. It does not impose the Egorov or frozen-map hypotheses of the deterministic problem.

Theorem 85.1 (Clock moment-channel obstruction)**.** *Let $N \geq 3$, $D_N = \operatorname{diag}(1, \zeta, \dots, \zeta^{N-1})$, $\zeta = e^{2\pi i/N}$, and $V_N = e^{i\Theta} Q D_N Q^*$, with Q Haar on $U(N)$ and Θ independent uniform. Let $\mathcal{M}_{N,k}$ be $\mathbb{E} \operatorname{Ad}(V_N^{\otimes k})$ and $\mathcal{P}_{N,k}$ the Haar moment projector. Then, for every integer $k \geq 1$,*

$$\|\mathcal{M}_{N,k} - \mathcal{P}_{N,k}\|_{2\to 2} \leq N^{-1}.$$

At $k = 2$ the error is exactly $(N^2-1)^{-1}$. The channel eigenvalues and their multiplicities are

$$\begin{array}{c|c} 1 & 2 \\ -1/(N^2-1) & 4(N^2-1) \\ 0 & N^2(N^2-3)/2 \\ 4/((N^2-1)(N^2-4)) & (N^2-1)(N^2-4)/2. \end{array}$$

Every unfolded consecutive spacing equals one. In particular this ensemble does not have the Sine_2 local spectral limit.

Proof. Vectorize the channel using $R_k(U) = U^{\otimes k} \otimes \overline{U}^{\otimes k}$. On each irreducible isotypic summand, Haar conjugacy averaging is the scalar $\chi_\lambda(D_N)/d_\lambda$, with identity on the multiplicity factor. The Haar moment projector retains exactly the trivial summands. The Weyl alternant formula at all Nth roots gives $\chi_\lambda(D_N) \in \{0, 1, -1\}$: repeated numerator exponent residues annihilate the determinant; otherwise its rows permute the denominator rows. Every summand is centrally balanced. A nontrivial summand cannot therefore be a determinant character. Its differential injects the simple Lie algebra $\mathfrak{su}(N)$ into $\mathfrak{u}(d_\lambda)$, implying $d_\lambda^2 \geq N^2 - 1$ and hence $d_\lambda \geq N$. This proves the uniform all-order bound.

For the exact two-copy calculation split $\mathbb{C}^N \otimes \mathbb{C}^N$ into $S = \operatorname{Sym}^2 \mathbb{C}^N$ and $A = \Lambda^2 \mathbb{C}^N$. Each endomorphism block contains one trivial and one adjoint summand; each cross block contains one adjoint summand. The remaining irreducible summands have highest weights $(2, 0, \dots, 0, -2)$, $(1, 1, 0, \dots, 0, -1, -1)$, and $(2, 0, \dots, 0, -1, -1)$ and its dual. Their dimensions are respectively

$$\frac{N^2(N-1)(N+3)}{4}, \qquad \frac{N^2(N+1)(N-3)}{4}, \qquad \frac{(N^2-1)(N^2-4)}{4}.$$

The second residual is absent at $N = 3$. These constituents exhaust the blocks by dimension. Explicit highest-weight vectors and adjoint embeddings are supplied in the canonical proof. Since $\operatorname{Tr} D_N = \operatorname{Tr} D_N^2 = 0$, the characters of S, A vanish, the adjoint character is -1, the two endomorphism residual characters vanish, and the cross residual characters equal 1. Dividing by

dimensions yields the displayed spectrum. The invariant space is spanned by identity and swap. On its complement the maximum absolute eigenvalue is $(N^2-1)^{-1}$, including $N=3$.

Finally conjugation preserves the regular N-gon spectrum and the scalar phase only rotates it. Its local unfolded limit is a uniformly translated integer lattice; an integer-length half-open interval has deterministic point count. The Sine_2 count variance on a nonempty finite interval is strictly positive, so the two local laws differ. □

This is an induced $2 \to 2$ estimate, not diamond-norm control or an exact design. A growing-order spectral test requires its own quantitative extraction bound. For example $N^{-2}|\operatorname{Tr} V_N^N|^2 = 1$, whereas the corresponding Haar expectation is N^{-1}. No deterministic BGS or noise-removal conclusion follows. Coxeter character rigidity and isospectral twirling are prior; the accompanying source note limits attribution and makes no global novelty claim.

Chapter 14 — Late trace results

86 Phase memory, real-flow sources, and sharp matrix jets

This section records three project proofs reconstructed and independently reviewed internally on 8 October 2026; the statements and proofs are given in full in this section. The canonical sources are `max_entry_phase_channel.md`, `phase_channel_real_flow.md`, and `sharp_inner_jet.md` in the mathematical corpus. No external priority claim or deterministic BGS conclusion is made.

86.1 Max-entry phase memory and averaged pinching (TRACE-053)

Let $S \in U(m)$, $P_{ij} = |S_{ij}|^2$, $\delta = \max_{ij} P_{ij}$, and let D have independent uniform coordinate phases. The same D is reused in every power. On the full complex Frobenius space write $A = \Pi \operatorname{Ad}(S)$, where Π is diagonal pinching, and $\mathcal{T}_n = \mathbb{E}_D \operatorname{Ad}((DS)^n)$.

Theorem 86.1. *For $n \geq 2$,*

$$\|\mathcal{T}_n - A^n\|_{2\to 2} \leq (n!-1)\sqrt{\delta} + n!n(n-1)\delta.$$

Degree one is exact. At degree two the error has exact norm $\max_{i\neq j} |S_{ij}S_{ji}| \leq \delta$.

Proof. Put $Z = SXS^*$ and pair with an arbitrary complex mark Y. For $a = (a_1, \ldots, a_n)$ put $u(a) = \prod_{r<n} S_{a_{r+1},a_r}$. Phase integration gives

$$\langle Y, \mathcal{T}_n X\rangle = \sum_{\mathrm{occ}(a)=\mathrm{occ}(b)} \overline{Y_{a_n b_n}} Z_{a_1 b_1} u(a)\overline{u(b)}.$$

For each time permutation σ, first sum without collision restrictions with $b_r = a_{\sigma(r)}$. Cauchy–Schwarz assigns $|Y|^2$ to the positive P chain and $|Z|^2$ to the reordered negative P chain. Distinct-time marginals of a bistochastic counting-measure path are entries of positive powers of P, all bounded by δ. A moved endpoint therefore costs $\sqrt{\delta}$, and two moved endpoints cost δ, times $\|Y\|_F\|Z\|_F$. If both endpoints are fixed, their unrestricted sums peel exactly, replacing the marks by $S^*\Pi(Y)S$ and $S\Pi(Z)S^*$, both contractions. Any nonidentity permutation eventually has a moved endpoint. The identity sum is A^n.

A prescribed collision at distinct formal times has probability at most δ conditionally on an initial endpoint, since

$$\sum_c (P^{i-1})_{ce}(P^{j-i})_{cc} \leq \delta \quad (i<j).$$

The reversed bistochastic chain handles a terminal endpoint. The Cauchy estimate therefore costs δ for a collision even with unrestricted complex marks. Partition repeated paths by their first collision. The true occupation-matched repeated part has at most $n!$ canonical matching classes and costs $n!\binom{n}{2}\delta$. Separately, the repeated part of the unrestricted permutation sum costs the same amount. Both corrections are needed. The remaining $n!-1$ pairings give the theorem. At degree two direct summation gives off-diagonal entries $S_{ij}\overline{S_{ji}}(SXS^*)_{ji}$ and zero diagonal, proving the exact norm formula. □

Let $\mathcal{E}_V$ denote full eigenspace pinching, including degeneracies. For $L \geq 2$ the positive Fejer operator

$$\mathcal{F}_L(V) = L^{-1} \sum_{|n|<L} (1 - |n|/L) \operatorname{Ad}(V)^n$$

has eigenmatrix multipliers $L^{-2}|\sum_{r=0}^{L-1} e^{ir\theta}|^2$, so $0 \leq \mathcal{E}_V \leq \mathcal{F}_L(V)$. If $\|P - J_m\| \leq \rho < 1$, then on traceless matrices $\|A^n\| \leq \rho^{n-1}$ and hence

$$\beta_m := \|\mathbb{E}_D \mathcal{E}_{DS}\| \leq \frac{1 + 2/(1-\rho)}{L} + \frac{2}{L} \sum_{n=2}^{L-1} n!\{\sqrt{\delta} + n(n-1)\delta\}.$$

For $X = \log(1/\delta)$ take $L = \lfloor X/(8 \log X) \rfloor$. The last term is at most $2e^{-3X/8} + 2L^2 e^{-7X/8}$, so $\beta_m = O_\rho(\log X/X)$. This is the norm of the average. The samplewise pinching norm need not vanish.

Spectral extraction boundary. For the symmetric Fourier matrix, $P = J_m$ and $\delta = 1/m$, whereas

$$\mathbb{E}|\operatorname{Tr}(DS)^2|^2 = 2 - 1/m, \qquad \mathbb{E}|\operatorname{Tr}(DS)^3|^2 = 6 - 9/m + 4/m^2.$$

At degree three the occupation multiplicities contribute $36\binom{m}{3} + 9m(m-1) + m$ divided by m^3. The vanishing degree-two channel norm coexists with a nonvanishing superoperator-trace defect. This witness is not the fixed deterministic map and supplies no microscopic limiting law.

86.2 Direct real-flow transport (TRACE-056)

Let $S = S_m \in U(m)$ be a deterministic unitary family with $\delta = \delta_m := \max_{ij} |S_{ij}|^2 \to 0$ and uniform intensity singular gap $\||S|^2 - J_m\|_{\mathrm{op}} \leq \rho < 1$, where J_m has all entries $1/m$. Let $H = H_m = H_m^*$ be deterministic and independent of the random coordinate phases, with $\operatorname{Tr} H = m$ and $\operatorname{Tr} H^2 \leq C_H m$. Positivity of H is unnecessary for this result. Put

$$W_x = DSe^{ixH/m}, \quad V_x = e^{ix/m}DS, \quad Z_m(s;V) = m^{-1}\operatorname{Tr}[(I - e^{-s/m}V)^{-1} - I].$$

For bounded real $|x| \leq R$, the preceding pinching bound holds uniformly with $\beta_{m,R} = O_{\rho,R,C_H}(\log X/X)$. Indeed $\delta \geq 1/m$, the new maximum entry squared is at most $(2 + 2R^2 C_H)\delta$, and the intensity matrix changes in norm by at most $2R\sqrt{C_H/m}$.

Theorem 86.2. *For* $\operatorname{Re} s > 0$,

$$\mathbb{E}|Z_m(s;W_x) - Z_m(s;V_x)| \leq \frac{|x|\sqrt{(C_H-1)\beta_{m,R}}}{2m \sinh(\operatorname{Re} s/m)} \leq \frac{|x|\sqrt{(C_H-1)\beta_{m,R}}}{2 \operatorname{Re} s}.$$

The difference tends to zero locally uniformly in absorbed L^1*, including each fixed source derivative.*

Proof. Set $B = H - I$. Absolutely continuous eigenangle branches of W_x have almost-everywhere velocities given by eigenvalues of the compressed generator H/m, including at multiple eigenvalues. Thus

$$\sum_j |\theta_j'(x) - 1/m|^2 = m^{-2}\|\mathcal{E}_{W_x}B\|_F^2, \qquad \mathbb{E}\|\mathcal{E}_{W_x}B\|_F^2 \leq \beta_{m,R}(C_H - 1)m.$$

Cauchy–Schwarz and integration give a branch matching with

$$\mathbb{E}\sum_j |\theta_j(x)-\theta_j(0)-x/m| \le |x|\sqrt{(C_H-1)\beta_{m,R}}.$$

The coupled endpoint law has a uniform common phase. For $r = e^{-\operatorname{Re} s/m}$ and $k_r(\phi) = re^{i\phi}/(1-re^{i\phi})$,

$$\|k_r'\|_{L^1(d\phi/(2\pi))} = \frac{r}{1-r^2} = \frac{1}{2\sinh(\operatorname{Re} s/m)}.$$

Apply the translated-kernel L^1 bound to each matched branch, average over the common phase, and divide by m. Cauchy contour estimates on a fixed absorbed neighborhood give convergence of every fixed derivative. □

This is an unweighted scalar trace with normal scalar absorption at both unitary endpoints. It does not compare the anisotropic physical source with $e^{-sH/m}$, identify the scalar phase law, control mixed products, remove absorption or set $D = I$.

86.3 Optimal matrix inner second jets (TRACE-054)

Let $F(z) = zG(z)$, where G is square rational inner and analytic on the closed disk after removal of removable singularities. On $z = e^{i\theta}$ write $d\mu = d\theta/(2\pi)$, $T = F^*zF' \ge I$, and $\dot T = dT/d\theta$.

Theorem 86.3.

$$\int \operatorname{Tr}(\dot T T^{-1}\dot T)\,d\mu \le \frac{1}{3}\int \operatorname{Tr}(T^3-T)\,d\mu.$$

Consequently

$$\int \|F''T^{-1/2}\|_F^2 d\mu \le \frac{4}{3}\int \operatorname{Tr} T^3 d\mu - 2\int \operatorname{Tr} T^2 d\mu + \frac{2}{3}\int \operatorname{Tr} T\,d\mu.$$

The leading constant $4/3$ *is optimal in every matrix dimension. For nonnegative periodic* C^2 *weights* ω*, the weighted versions have the additional right-hand term* $(2/3)\int \omega''\operatorname{Tr} T\,d\mu$.

Proof. Finite factorization can be obtained directly. If $\det G$ has a disk zero a, choose a unit vector $v \in \ker G(a)$, $P = vv^*$ and $B = I + (b_a - 1)P$, with $b_a = (z-a)/(1-\bar a z)$. Right division GB^{-1} is analytic because $G(z)v$ vanishes at a. It is rational inner and its determinant degree drops by one. Repeat. A zero-free rational inner determinant is constant unimodular; the maximum principle applied to both residual matrix and its inverse forces unitary values throughout the disk, hence a constant matrix. Thus F starts at zU and appends finitely many such right factors.

For $A > 0$, a projection P and $p > 0$, arbitrary matrices X, Y with $PY = Y$ obey

$$\|(A+pP)^{-1/2}(X+Y)\|_F^2 \le \|A^{-1/2}X\|_F^2 + p^{-1}\|Y\|_F^2.$$

The block row $(A+pP)^{-1/2}[A^{1/2}, \sqrt{p}P]$ is a coisometry; apply its contraction bound to $[A^{-1/2}X; p^{-1/2}Y]$. Singular pP causes no inversion problem.

Append $B = I + (b_a - 1)P$ to a previous product. Its Poisson kernel is $p = \bar b_a z b_a' > 0$. Set $A = B^*T_{\text{old}}B$ and $C = B^*\dot T_{\text{old}}B$. Then

$$T = A + pP, \qquad \dot T = C + ip(AP - PA) + p'P.$$

Use the asymmetric split $X = C + ipAP$, $Y = p'P - ipPA$; it has $PY = Y$. Expansion under the trace, retaining noncommutativity, gives

$$\operatorname{Tr}(\dot T T^{-1}\dot T) \le \operatorname{Tr}(CA^{-1}C) + p^2\operatorname{Tr}(PA) + p\operatorname{Tr}(PA^2) + (p'^2/p)\operatorname{Tr} P,$$
$$\operatorname{Tr}(T^3 - A^3) = 3p\operatorname{Tr}(PA^2) + 3p^2\operatorname{Tr}(PA) + p^3\operatorname{Tr} P.$$

All cross traces in the first expansion vanish by cyclicity, not by matrix commutation. The exact scalar identity

$$3p'^2/p = p^3 - p + 2p''$$

shows that the integrated cubic-minus-linear derivative functional cannot increase at a factor. With ω, integrate by parts twice, giving $\int \omega p'' = \int \omega'' p$. Accordingly the functional

$$\Phi_\omega(T) = \int \omega \operatorname{Tr}(\dot{T}T^{-1}\dot{T})\, d\mu - \frac{1}{3}\int \omega \operatorname{Tr}(T^3 - T)\, d\mu - \frac{2}{3}\int \omega'' \operatorname{Tr} T\, d\mu$$

cannot increase. Its base value at $T = I$ is zero. Induction proves both derivative inequalities.

The exact complex-derivative identity is $F^* z^2 F'' = T^2 - T - i\dot{T}$. Since T^{-1} commutes with $T^2 - T$, the cross trace vanishes and

$$\|F''T^{-1/2}\|_F^2 = \operatorname{Tr}(T^3 - 2T^2 + T) + \operatorname{Tr}(\dot{T}T^{-1}\dot{T}).$$

This proves the asserted right-weighted jet bounds.

For sharpness take $F_r = zb_r$ with $r \uparrow 1$ and $T = 1 + p_r$. Put

$$M_2 = \frac{1 + r^2}{1 - r^2}, \qquad M_3 = \frac{1 + 4r^2 + r^4}{(1 - r^2)^2}.$$

The derivative energy loss from replacing p_r by $1 + p_r$ is between zero and $\int (p_r'/p_r)^2 = M_2 - 1$. Thus the jet energy is between $(4/3)M_3 + 2/3$ and $(4/3)M_3 + M_2 - 1/3$, while $\int T^3 = M_3 + 3M_2 + 4$. The ratio tends to $4/3$. Embedding this scalar factor beside $m - 1$ copies of z gives sharpness in each fixed matrix dimension. □

Localization and physical scope. The derivative penalty cannot be dropped for an arbitrary angular localization or stopping event. For $a = (25 + 5i)/26$ and $F = zb_a/b_a(1)$ one has $T(0) = 2$ and $F''(1) = 2 - 10i$, so the pointwise energy is 52, larger even than $(13/3)T^3 = 104/3$. The integrated result strengthens the supplied manuscript Section 89.2; it supplies no cubic-delay moment bound for the actual cavity. The submitted deep-source hierarchy (TRACE-055) remains verification-pending: its joint delay barriers, killed moment, ordinary-prefix covariance and physical-source continuation have not been reconstructed here. The scalar cavity law, TRACE-005 and BGS remain open.

11 October continuation. Covariance, physical-source continuation and a conditional joint-delay barrier are now reconstructed in Section 87. The full TRACE-055 hierarchy, all-time flatness and scalar microscopic law remain unresolved.

87 Maximal microscopic paths and compatible circular phase laws

This checkpoint records the checked 8–11 October continuation. Complete proofs and exact boundaries are in the mathematical corpus and the 11 October integration record. Internal review and finite controls do not establish external priority or deterministic BGS universality.

87.1 A deterministic maximal path bound (TRACE-072)

Let U_N be unitary, $B_N = B_N^*$, $\operatorname{Tr} B_N = 0$ and $\|B_N\|_{\mathrm{op}} \le C$. Put $U_N(x) = U_N e^{ixB_N/N}$ and

$$A_{N,T} = T^{-1} \sum_{k=0}^{T-1} U_N^{-k} B_N U_N^k, \qquad v_{N,T} = \|A_{N,T}\|_F/\sqrt{N}.$$

For the periodically unfolded point process with one shared uniform circle origin, use the bounded vague-topology Laplace metric d_L specified in `deterministic_path_transport_20261011.md`. Then

$$\mathbb{E}_\phi \sup_{|x|\le R} d_L(\Xi_N(x;\phi), \Xi_N(0;\phi)) \le \frac{R v_{N,T}}{\pi} + \frac{C^2 R^2 (T-1)}{2\pi N}.$$

At multiplicities, the eigenangle velocities are the eigenvalues of the compression of B_N/N. Full eigenspace pinching fixes the time average; unitary-power telescoping gives

$$\sum_j |\theta_j'(x)| \le v_{N,T} + C^2 |x| (T-1)/N.$$

Common-origin averaging of a periodized test derivative contributes $\|f'\|_1/(2\pi)$; integrating on $[-R, R]$ proves the bound. No level-spacing denominator or simple-spectrum assumption occurs.

For a fixed smooth exactly centered observable of the maintained map, FM-018 and the polar-WKB trace pairing give $v_{N,T} = O((\log N)^{-1/2})$ for sufficiently small $T = c \log N$. Thus every $R_N = o(\sqrt{\log N})$ is permitted. The time-average endpoint transfer is already in the maintained ergodic-transfer supplement. The added conclusion is the maximal path and growing interval, not identification of the initial local spectral law.

The estimate also survives the prior microscopic rounding construction. For the same observable and $\epsilon = \|V_N - U_N\|_{\mathrm{op}}$, power telescoping gives

$$|v_{N,T}(V_N, B_N) - v_{N,T}(U_N, B_N)| \le C(T-1)\epsilon.$$

Thus $O(N^{-1})$ rounding preserves the logarithmic maximal-path conclusions for every fixed smooth observable. The prior construction can preserve its coarse controls and prescribed low-trace window while violating a fixed signed CUE pair target. These combined abstract controls still leave that target undetermined; the rounded operator changes the exact quantization.

87.2 Phase paths and physical sources (TRACE-057–060,071)

For $W_x = DS_m e^{ixH_m/m}$, $\operatorname{Tr} H_m = m$ and $\operatorname{Tr} H_m^2 \le C_H m$, the averaged full pinching norm $\beta_{m,R}$ on traceless matrices gives the centered bound

$$\mathbb{E}_D \sup_{|x|\le R} d_L(\Xi_m(x), \Xi_m(0)) \le (R/\pi)\sqrt{(C_H - 1)\beta_{m,R}}.$$

Endpoint flatness $\delta_m \to 0$ and a fixed intensity singular gap imply vanishing on intervals with $R_m^2 \log\log(1/\delta_m)/\log(1/\delta_m) \to 0$. Every fixed normally absorbed source product and derivative transfers by the all-order source bounds and interpolation. The physical anisotropic source theorem is only asserted in L^2.

For a strict contraction F, $K = (D^* - F)^{-1}$ and arbitrary complex B, the exact covariance satisfies

$$\mathbb{E}|\operatorname{Tr} BK|^2 \le \min\{\operatorname{Tr}[B(I - F^*F)^{-1}B^*], \operatorname{Tr}[B^*(I - FF^*)^{-1}B]\}.$$

The preceding joint Schur delay barrier uses stopped scalar Freedman [43] conditional on all-time entry flatness. Its killed second and third moments are upper bounds. The later equality-law extensions are stated separately below; their all-time flatness premise remains unproved.

87.3 Frozen circular backgrounds (TRACE-064,067–070)

For a nonzero balanced Laurent Jack signature, $\alpha > 0$ and $K = \min(\lambda_1, -\lambda_N) \geq 1$, the constant-coefficient ratio obeys

$$\frac{c_0}{P_\lambda(1^N)} \leq \left(1 + \frac{\alpha E_\lambda}{2K}\right)^{-1} \leq \frac{1}{N + \alpha K} \leq \frac{1}{N + \alpha}.$$

The highest-root signature is the unique final equality mode. Spherical identification [38, 39] gives, for spectral $f \in L^2$ and $\beta = 1, 2, 4$, $\alpha = 2/\beta$,

$$\mathrm{Var}_Q \, \mathbb{E}_D f(DQ) \leq \mathrm{Var}_{\mathrm{circ},\beta}(f)/(N + \alpha).$$

CSE uses N paired phases and half traces. Circular local convergence [40] then yields the full conditional weak Sine-β law in probability over the once-sampled background Q. This is not a theorem for a prescribed deterministic scattering matrix. The all-degree one-sided Jack norm bound uses [41] at its stated scope.

The exact CUE variance, CSE second/fourth cumulants and limiting curves are prior formulas of [42]. The written all-order critical calculation proves $p_n/\sqrt{\mathbb{E}|p_n|^2}$ converges to a standard complex Gaussian whenever $n/N \to 1$, $d = |n - N| = o(N)$ and $\log[N/(d+1)] \to \infty$, with compatible frozen-phase convergence. The complete CUE/CSE form-factor curves converge in mean $L^2(0, T)$ using $n = \lceil N\tau \rceil$. This does not assert a temporal supremum estimate; the logarithmic CSE critical peak is integrable in L^2.

87.4 Finite-field coherence and exact boundaries (TRACE-061–066)

The signed finite-field Fourier calculus gives a leading growing ramp for $n = o(N^{1/8})$ and the symmetric first correction on $n = o(N^{1/9})$. A proposed $N^{1/7}$ window is unconfirmed. Birthday-rank Poisson control uses [44] with its TV convention converted explicitly. Flat clock completions can preserve intensity mixing and exact frame designs while changing clock–eigenbasis coupling. These are qualified obstructions in other constructed families, not counterexamples to the literal map.

The complete balanced phase law, the scalar cavity microscopic law, signed Heisenberg TRACE-005 and deterministic BGS remain open. None of the preceding path comparisons evaluates the initial microscopic process.

87.5 All fixed killed clock moments and equality laws (TRACE-060)

Consider the fresh-phase Schur process with uniformly selected remaining coordinate, $T_N = Q_N = I$, and $\ell = \log(2N)$. Fix $K, \kappa > 0$ and set $m = \lceil \ell^{1+\kappa} \rceil < N$. Write

$$\mathcal{A}_m = \{\max_{ij} |(S_u)_{ij}|^2 \leq K\ell/u \text{ for every } N \geq u \geq m\}.$$

The independently reconstructed extension of TRACE-060 supplies joint incoming/outgoing first-delay concentration, with error $\epsilon = \ell^{-\kappa/16}$, in intersection probability at most $C_{K,\kappa}\ell^{-\kappa/16}$. Every fixed killed normalized moment

$$M_k(u) = \mathbf{1}_{\{\text{alive at } u\}} \frac{u^{k-1} \operatorname{Tr} T_u^k}{N^k}$$

has a dimension-independent expected upper bound. Order k is fixed; no growing-order bound or unrestricted exceptional-event moment estimate is claimed.

The rank-two trace expansion and positive spectral-moment interpolation close the killed upper recurrence. Convex raw trace powers give a frozen submartingale, including the complete

crossing state. The near-flatness cutoff follows from its scale-dependent cap and a dyadic scalar maximal Freedman argument. All-time flatness probability is a separate obligation.

For the original un-killed $Y_k(u) = u^{k-1} \operatorname{Tr} T_u^k / N^k$, a separately reconstructed extension proves

$$\Pr\Big(\mathcal{A}_m \cap \Big\{ \max_{N \ge u \ge m,\ k=2,3} |Y_k(u) - f_k(u/N)| > C_{K,\kappa} \ell^{-\kappa/32} \Big\}\Big) \le C_{K,\kappa} \ell^{-\kappa/16},$$

where $f_2(x) = 2 - x$ and $f_3(x) = 6 - 6x + x^2$. Squared cyclic words give

$$\mathbb{E}[(\Delta Y_k)^2 \mid \mathcal{F}_u] \le C_k (1 + Y_{2k-1}(u))/u^2.$$

The already bounded killed moments therefore control the expected stopped bracket by C_k/m. Scalar Doob concentration and the exact contractive second/third drifts yield the uniform comparison. The normalized crossing value is frozen without subsequent renormalization. If $\Pr(\mathcal{A}_m^c) \to 0$ is separately proved, this also gives convergence in probability and convergence of *killed* second/third expectations. The subsequent fourth-order and repaired all-fixed-order extensions below are independently reconstructed under the same flatness premise.

87.6 Diagonal squares and the fourth delay law (TRACE-060)

Under the same $\mathcal{A}_m$ premise, a separate reconstruction proves uniform incoming/outgoing diagonal-square concentration about $f_2(u/N)$ and the fourth normalized trace law

$$Y_4(u) = f_4(u/N) + o(1), \qquad f_4(x) = 22 - 32x + 12x^2 - x^3,$$

where the probability of a uniform failure, intersected with $\mathcal{A}_m$, tends to zero. The proof neither bounds $\Pr(\mathcal{A}_m^c)$ nor controls unrestricted moments on its complement.

The exact mixed clock $\Gamma = ST$ satisfies, on surviving labels,

$$\Gamma'_{jk} = \Gamma_{jk} + q(S_{je}\Gamma_{ek} + S_{ek}\Gamma_{je}) + q^2 S_{je} S_{ek} \Gamma_{ee}.$$

Both phase monomials have zero mean, including at nonzero $a = S_{ee}$. A live-pair scalar martingale estimate controls the coherent term in the diagonal-square variance. Its deterministic envelope violation is explicitly included in the joint stop; no future good event is used for conditioning. For a retained j, write $d = 1 - |a|^2$, $t = T_{ee}$, $\gamma = \Gamma_{ee}$. The exact phase average is

$$\mathbb{E}_w (T'^2)_{jj} = (T^2)_{jj} + \frac{|S_{ej}|^2 (T^2)_{ee} + 2t\Re(\overline{S_{ej}}\Gamma_{ej})}{d} + \frac{2|S_{ej}|^2 t \Re(\bar{a}\gamma)}{d^2}.$$

The reference value freezes when its label is removed. Its conditional increment is $(u-1)/(uN)$; the ideal residual is $-2/u^2 + 2/u^3$.

For $t_r = (T^r)_{ee}$, $s_r = (Q^r)_{ee}$, $s = s_1$ and $\beta = (ST^2)_{ee}$, the full nonzero-a fourth identity is

$$\begin{aligned} \mathbb{E}_w \operatorname{Tr} T'^4 = {} & \operatorname{Tr} T^4 + \frac{4}{d}(ts_3 + t_2 s_2 + s t_3) \\ & + \frac{8}{d^2} \Re[\bar{a}\{\gamma(st_2 + ts_2) + ts\beta\}] \\ & + \frac{1 + |a|^2}{d^3}(2t^2 s^2 + 4ts|\gamma|^2) + \frac{8ts}{d^3}\Re(\bar{a}^2\gamma^2). \end{aligned}$$

The diagonal-square laws and vanishing mixed clock identify its normalized forcing as $8f_3 + 4f_2^2 + 2$. This gives $-xf_4' = -3f_4 + 8f_3 + 4f_2^2 + 2$, with $f_4(1) = 1$. The expected stopped trace bracket is bounded using the killed seventh moment, and complete crossing values are retained. If separately $\Pr(\mathcal{A}_m^c) \to 0$, actual probability convergence and convergence of killed fourth expectations follow. The known proper-delay value 22 is credited to Cunden, Mezzadri, Simm and Vivo [45]; finite rational and Wolfram executions do not certify the asymptotic theorem. The proof source is `mathematics/fourth_delay_law_20261011.md`.

87.7 Repaired all-fixed-order clock law and the rank atom (TRACE-060)

Put $H_u = uT_u/N$, $V_u = uQ_u/N$ and $\Gamma_{r,u} = S_u H_u^r$. For every fixed L, the probability, intersected with $\mathcal{A}_m$, of any uniform failure of the following entry limits tends to zero:

$$\begin{aligned}(H_u^r)_{jj}, (V_u^r)_{jj} &= f_r(u/N) + o(1),\\ (H_u^r)_{jk}, (V_u^r)_{jk} &= o(1) \quad (j \neq k),\\ (\Gamma_{r,u})_{jk} &= o(1), \quad r \le L,\ u \ge m.\end{aligned}$$

The references solve the triangular equation, with $\tau = -\log x$,

$$\partial_\tau f_k = -(k-1)f_k + \mathcal{P}_k, \quad \mathcal{P}_k = k[z^k]\{-\log(1-M^2)\}, \quad M = \sum_{r\ge 1} f_r z^r, \quad f_k(1) = 1.$$

This is a fixed-order conditional theorem, with no probability bound for $\mathcal{A}_m^c$. The original general-word proof required two substantive repairs before this promotion.

With $v = S^* e$ and $\alpha_k = (1-1/u)^k$, exact open words come from

$$H'^k = \alpha_k K^* H (R_w H)^{k-1} K, \qquad \Gamma'_k = \alpha_k J^* S H (R_w H)^{k-1} K, \qquad R_w = I + qev^* + \bar{q}ve^*.$$

Their interior gaps are positive integers summing to k. Alternating orientations supply diagonal clock factors; identical adjacent orientations supply a lower mixed factor. For the sum of both orientations of an even alternating word with $2q$ interior insertions and no boundary insertion, excluding a retained endpoint has a leading diagonal contribution. After replacement of lower moments, the drift coefficient, with the common factor α_k/u outside, is

$$\left(\prod_{i=1}^{2q-1} f_{n_i}\right)\left(2f_{n_0+n_{2q}} - f_{n_0} f_{n_{2q}}\right).$$

The alternating two-boundary term compensates the negative product. This is compensation in the ideal drift polynomial, not a pathwise word identity. If $C_{2q}(k)$ denotes the positive $2q$-gap product sum, merged endpoint gaps contribute $2(k-2q)C_{2q}(k)/(2q)$ and the two one-boundary orientations contribute $2C_{2q}(k)$. Their total is $(k/q)C_{2q}(k)$, including the empty merged sum at $k = 2q$. The surviving lower-matrix polynomial uses only powers at most $k-1$. The sole unknown top diagonal has coefficient $\alpha_k \overline{S_{ei}} S_{ej}/[u(1-|S_{ee}|^2)]$; its absolute transport row sum is at most $(1+O(\delta_u))/u$. The prospective mixed top transport has phase q^2 and exact zero mean. The own scaling thus gives absolute live-row contraction at least $(k-1)/(2u)$ for the input/output errors. Complex transport is controlled by absolute sums, without a positivity assumption.

The second repair separates interior words with both boundary insertions. They need not contain a small lower clock endpoint. Their scattering endpoint products instead satisfy, for example,

$$\frac{1}{u}\sum_e |S_{ei}S_{ej}|^2 \le \frac{\delta_u}{u}, \qquad \frac{1}{u}\sum_e |S_{ie}S_{ej}|^2 \le \frac{\delta_u}{u}.$$

Stopped top diagonals bound their internal scalar factors. The zero-interior two-boundary top term and its outgoing/mixed analogues obey the same raw second-moment estimate by the top-diagonal cap and $|q| \le 2$. Other interior words use a lower-entry envelope. With $R_p = 1 + (uB_u)^{1/p}$, the resulting predictable bounds for each real component are

$$\begin{aligned}\mathrm{Var}(\Delta) &\le C_{k,p}\left[\frac{\ell R_p^{2k}}{u^2} + \frac{E_{k-1}^2 R_p^{2k}}{u} + \frac{1}{u^2}\right],\\ |\Delta| &\le C_{k,p}(\sqrt{\delta_u} + E_{k-1})R_p^k + C_{k,p}/u.\end{aligned}$$

The lower drift error is at most $C_k(E_{k-1}+\sqrt{\delta_u})R_p^{2k}/u + C_k/u^2$. Nonzero-$a$ phase remainders are included; pure phase powers average to zero exactly. Reference freezing contributes $(1-1/u)[f_k((u-1)/N) - f_k(u/N)] = -xf_k'(x)/u + O(u^{-2})$.

Take $D_k = 2k(k+3)$, $\zeta = 1/4$, $B = \ell^b$ and fixed p sufficiently large that $D_L(1+\zeta)/p < 1/8$ and $2D_L(1+\kappa+b)/p < \kappa/4$. A closing local envelope is

$$E_k(u) = C_{k,p}R_p^{D_k}\left[\sqrt{\ell/u}\log(2u)^{k+3} + \ell/u\right].$$

Dyadic scalar Freedman controls only the pairs alive at the block start. Pathwise absolute-row stability propagates their raw noise without future-dependent martingale transforms. Spare powers R_p^{-2k-4} in the drift and R_p^{-3k-4} in the lower noise close the induction. Joint first-exit stops retain complete crossing states and all current envelopes. Deleted labels keep their previous-size reference; surviving labels receive the new-size reference before the complete value and reference are frozen at a global exit. The killed p-cap costs $O(C_p\ell^{-b})$; $E_L(m) \to 0$. The maintained proof is `mathematics/conditional_clock_universality_20261011.md`.

The formal characteristics give

$$z = \frac{M(1-M)}{(1+M)(1-xM)}, \qquad (1-xz)M^2 - [1-(1-x)z]M + z = 0.$$

Put $a_x = 3-x-2\sqrt{2(1-x)}$ and $b_x = 3-x+2\sqrt{2(1-x)}$. Cauchy inversion gives, for $0 \le x < 1$,

$$\mu_x = \omega_x\delta_x + \frac{\sqrt{(b_x-h)(h-a_x)}}{2\pi h(h-x)}\mathbf{1}_{[a_x,b_x]}(h)\,dh, \qquad \omega_x = \begin{cases} 0 & x \le 1/2, \\ 2-1/x & x > 1/2. \end{cases}$$

At $x = 1$ set $\mu_1 = \delta_1$. Its Cauchy transform is

$$G_x(w) = \frac{3w-1-x-\sqrt{w^2-(6-2x)w+(1+x)^2}}{2w(w-x)}, \qquad G_x(w) \sim w^{-1}.$$

The square root is asymptotic to w and negative below a_x. The apparent pole at zero is removable; $G_x(0) = -2/(1+x)$. At $x = 1/2$ the lower branch endpoint gives an integrable inverse square-root density and no atom. The atom test, rather than an ordinary residue at that endpoint, handles the merging singularity. All fixed moments and compact moment determinacy identify this weak clock law on the same intersection with $\mathcal{A}_m$. Its endpoint moments are the prior Schröder sequence [45].

An independent pathwise observation explains the atom. The product of feedback embeddings has unchanged surviving-coordinate rows, hence $T_u = L_u^*L_u = I + X_u^*X_u$ with X_u having $N-u$ rows. Thus

$$\operatorname{rank}(T_u - I) \le N-u, \qquad \dim\ker(T_u - I) \ge (2u-N)_+.$$

The empirical uT_u/N measure has at least $(2-1/x)_+$ mass at x; finite-rank equality for arbitrary trajectories is not asserted. For comparison with the prior free-Jacobi family [46], set $p = x/(1+x)$, $q = 2x/(1+x)$ and view $Y = PQP$ in the normalized P corner for free reference projections of traces p, q. Subtract $(1-p)\delta_0$ from the full-trace law and divide by p. Then x/Y has precisely μ_x; this dictionary makes no claim of freeness of the actual matrices. Its $x = 0$ limit is established directly by the Cauchy formula, since the corner with $p = 0$ is undefined. The separate proposition is `mathematics/full_depth_clock_measure_20261011.md`. All-time flatness probability, growing-order/edge laws, the physical microscopic cavity law, signed TRACE-005 and global BGS remain open.

87.8 Classical singular-spectrum censoring and a quantum gap (TRACE-073)

Let P be doubly stochastic, A a retained set of size a, and D its complement. Suppose $I - P_{DD}$ is invertible. Its stochastic complement is

$$P^c = P_{AA} + P_{AD}(I - P_{DD})^{-1}P_{DA}.$$

Every ordered centered singular value contracts:

$$\sigma_j(P^c - J_a) \le \sigma_j(P - J_N), \qquad 1 \le j < a.$$

This requires no reversibility. For $x \perp \mathbf{1}_A$ put $t = (I - P_{DD})^{-1}P_{DA}x$, $z = (x, t)$ and $z_0 = z - N^{-1}(\mathbf{1}^*z)\mathbf{1}$. The exact lift identity is

$$\|(P - J_N)z_0\|^2 - s^2\|z_0\|^2 = \|(P^c - J_a)x\|^2 - s^2\|x\|^2 + (1 - s^2)(\|t\|^2 - |\mathbf{1}^*t|^2/N).$$

For $0 \le s \le 1$, its last term is nonnegative. The lift is injective, and positive inertia proves the entire ordered singular-value inequality.

In a quantum fresh-phase elimination, the classical complement is the conditional mean of the new intensity matrix, rather than a realization bound. The centered coherent increment has operator norm at most $8\delta_u$ when $\delta_u = \max_{ij} |(S_u)_{ij}|^2 \le 1/4$. For even $p \ge 2$, the self-contained Schatten potential $F_r(X) = (\|X\|_p - r)_+^2$ satisfies

$$F_r(X + H) \le F_r(X) + DF_r(X)[H] + (p - 1)\|H\|_p^2.$$

Pad the *centered* matrices with zeros to the original dimension. If $\|P_N - J_N\|_{\mathrm{op}} \le \rho_0$, $m \ge 4K\ell$, and $g > N^{1/p}\rho_0$, the stopped convex potential yields

$$\Pr\left(\mathcal{A}_m \cap \left\{\max_{N \ge u \ge m} \|P_u - J_u\|_{\mathrm{op}} \ge g\right\}\right) \le \frac{64N^{2/p}(p - 1)K^2\ell^2}{m(g - N^{1/p}\rho_0)^2}.$$

With $p = 2\lceil \ell \rceil$, $m = \lceil \ell^{3+\kappa} \rceil$ and a fixed $g > \sqrt{e}\,\rho_0$, the bound is $O_{K,\kappa,g,\rho_0}(\ell^{-\kappa})$. FM-021 supplies a sufficiently small initial gap for the maintained map; it does not supply $\Pr(\mathcal{A}_m^c) \to 0$.

The exact Walsh four-port example has $P = J_4$ but one feedback realization is a permutation on three ports, with centered gap one. It separates conditional-mean contraction from sample contraction, without constituting an asymptotic counterexample to the maintained map. Complete proofs and review inputs are in the dated Schur continuation. Finite Python, C++ and Wolfram controls check algebra only. These results leave the scalar cavity law, signed TRACE-005 and deterministic BGS open.

87.9 General contractions and a reversal gap (TRACE-073)

A separate independent reconstruction extends the harmonic lift to every complex contraction C, with no positivity, normality or Perron mode. If $I - C_{DD}$ is invertible, its algebraic Schur complement obeys $\sigma_j(C^c) \le \sigma_j(C)$ for $1 \le j \le |A|$. For arbitrary x, set $t = (I - C_{DD})^{-1}C_{DA}x$, $z = (x, t)$. The exact identity

$$\|Cz\|^2 - s^2\|z\|^2 = \|C^cx\|^2 - s^2\|x\|^2 + (1 - s^2)\|t\|^2$$

first proves contraction of C^c at $s = 1$, then proves the entire ordered singular-value bound by positive inertia and the injective lift.

For unitary S, the Hermitian reversal matrix $R(S)_{ij} = S_{ij}\overline{S_{ji}}$ is a contraction by Cauchy–Schwarz on absolute row and column sums. Write $S = \left[\begin{smallmatrix} A & c \\ b & a \end{smallmatrix}\right]$, $q = w/(1 - aw)$, $d = 1 - |a|^2$, $C_0 = \mathrm{diag}(c)A^*\,\mathrm{diag}(b)$ and $v_i = c_i\overline{b_i}$. Exact fresh-phase expansion gives

$$R(S') = R_{AA} + qC_0 + \bar{q}C_0^* + |q|^2vv^*, \qquad \mathbb{E}_wR(S') = R_{AA} + vv^*/d = R^c.$$

The centered noise has norm at most $8\delta_u$. Applying the same stopped Schatten potential directly to R_u, with zero padding and *no* J subtraction, gives

$$\Pr\Big(\mathcal{A}_m \cap \Big\{\max_{N\ge u\ge m}\|R_u\|_{\mathrm{op}} \ge g\Big\}\Big) \le \frac{64N^{2/p}(p-1)K^2\ell^2}{m(g-N^{1/p}\rho_R)^2}, \qquad g > N^{1/p}\rho_R,$$

when $\|R_N\|_{\mathrm{op}} \le \rho_R$. FM-022 proves precisely this initial matrix norm tends to zero for the maintained odd-dimensional family. Thus both intensity and reversal gaps are controlled, in intersection probability, down to $\lceil \ell^{3+\kappa}\rceil$. This is neither a complete symmetry classification nor a microscopic local-law theorem; all-time flatness probability and deterministic BGS remain open.

87.10 A two-step obstruction inside unitary intensities (TRACE-074)

For every sufficiently large n there is $S_n \in U(n)$, with $P_{ij} = |(S_n)_{ij}|^2$ and $J_{ij} = 1/n$, such that

$$\delta_n := \max P_{ij} \le \frac{8\log(2n)}{n}, \qquad \rho_n := \|P-J\|_{\mathrm{op}} \le n^{-1/8} + 16\sqrt{\frac{\log(2n)}{n}} \longrightarrow 0,$$

but, for two fixed distinct labels i, j,

$$(P^2)_{ij} = \frac{1}{b_n}, \qquad b_n = \left\lfloor \frac{n}{\log(2n)} \right\rfloor, \qquad n(P^2)_{ij} \sim \log(2n).$$

The norm is a singular operator norm. The construction is genuinely unistochastic and refutes a universal estimate $|((P-J)^2)_{ij}| \le C\rho_n\delta_n$.

For unit vectors c, d use the rank-one unitary correction

$$H(c,d) = I - \frac{(c-d)(c-d)^*}{1-d^*c} \quad (c \ne d), \qquad H(c,c) = I.$$

Indeed $v = c-d$ satisfies $v^*c = 1-d^*c$ and $\|v\|^2 = (1-d^*c)+(1-c^*d)$, proving $Hc = d$ and $H^*H = I$. Choose $a = \beta = b_n^{-1/2}\mathbf{1}_B$, with B disjoint from i, j, and a Haar unitary Z. Put

$$L = H(Ze_j, \beta), \quad W = LZ, \quad r = W^*e_i, \quad R = H(a,r), \quad S = WR.$$

Then $r \perp e_j$, $Re_j = e_j$, $Se_j = \beta$ and $e_i^*S = a^*$. Thus $(P^2)_{ij} = 1/b_n$ exactly, while $\mathrm{rank}(S-Z) \le 2$ and $\|S-Z\|_{\mathrm{HS}} \le 2\sqrt{2}$.

The remaining completion is Haar from $\{e_j, a\}^\perp$ to $\{\beta, e_i\}^\perp$. To justify this dependent construction, let $\mathcal{G}$ fix both β, e_i. Under $Z \mapsto GZ$, $L \mapsto GLG^*$, $W \mapsto GW$, r, R stay fixed, and $S \mapsto GS$. The joint law of (r, S) is invariant. The group acts freely and transitively on the completions, so the conditional completion law is the unique Haar law. The residual dimension is $n-2$; each bulk entry is a Haar entry multiplied by projection norms at most one. Consequently, with $t_n = 8\log(2n)/n$,

$$\Pr(\max|S_{kl}|^2 > t_n) \le n^2 e^{-(n-3)t_n}, \qquad \Pr(\max|Z_{kl}|^2 > t_n) \le n^2 e^{-(n-1)t_n}.$$

The necessary Haar singular bound is derived internally. For $p_l = |Z_{1l}|^2$ and $s_q = \sum_l p_l^q$, Dirichlet moments give

$$\mathbb{E}s_2 = \frac{2}{n+1}, \quad \mathbb{E}s_3 = \frac{6}{(n+1)(n+2)}, \quad \mathbb{E}s_2^2 = \frac{4(n+5)}{(n+1)(n+2)(n+3)}.$$

Conditional on the first row vector $\alpha = Z^*e_1$, the second is uniform on the complex sphere with projector $\Pi = I - \alpha\alpha^*$. For $D = \mathrm{diag}(p)$, its quadratic-form second moment is $[(\mathrm{Tr}\, D\Pi)^2 +$

$\mathrm{Tr}(D\Pi D\Pi)]/[n(n-1)]$, namely $(1-s_2-2s_3+2s_2^2)/[n(n-1)]$. Summing equal and distinct rows and removing the Perron mode gives

$$\mathbb{E}\big\||Z|^2-J\big\|_{S_4}^4=\frac{2(n-1)(n+4)}{(n+1)(n+2)(n+3)}\le 2/n.$$

Markov gives $\Pr(\||Z|^2-J\|_{\mathrm{op}}>n^{-1/8})\le 2n^{-1/2}$. On the two flatness events, $\|P-|Z|^2\|_{\mathrm{op}}\le 2\sqrt{t_n}\|S-Z\|_{\mathrm{HS}}\le 4\sqrt{2t_n}$. A union bound, without independence assumptions, proves simultaneous witnesses for every sufficiently large n.

The positive higher-step statement, valid for every doubly stochastic P, is

$$|(P^h)_{ij}-1/n|\le\rho^{h-2}\sqrt{\left(\sum_k P_{ik}^2-\frac{1}{n}\right)\left(\sum_k P_{kj}^2-\frac{1}{n}\right)}\le\rho^{h-2}(\delta-1/n),\qquad h\ge 2.$$

It follows from $E^h=P^h-J$ and $(E^h)_{ij}=(e_i^*E)E^{h-2}(Ee_j)$. At $h=2$, the factor is one, including $\rho=0$; the prescribed row and column attain the first bound exactly. Our witnesses satisfy $\max_{ij} n|(P^3)_{ij}-1/n|\to 0$. Nevertheless their frozen-entry one-step Schur bracket is

$$\mathbb{E}|\Delta S_{ij}|^2=\frac{1}{n}\sum_{e\notin\{i,j\}}\frac{P_{ie}P_{ej}}{1-P_{ee}}\sim\frac{\log(2n)}{n^2}.$$

Here $\Delta S_{ij}=0$ if an endpoint is chosen. Thus three-step mixing does not replace the two-step kernel in the exact bracket.

The Haar unistochastic ensemble is prior work [47]. Its numerical singular statistics are not imported as a theorem. The canonical proof is `mathematics/unitary_two_step_obstruction_20261012.md`. No maintained trajectory, chaotic quantizer, BGS counterexample, analytic Lean or world-priority claim is supplied.

87.11 Fresh-phase restoring drift of a two-step peak (TRACE-075)

Let $S\in U(u)$, $P=|S|^2$, $K=P^2$, $\delta=\max P<1$, and $i\neq j$. Pick e uniformly among all u labels, independently of a fresh uniform phase w. Freeze K_{ij} if $e=i$ or j; otherwise use $S'_{kl}=S_{kl}+qS_{ke}S_{el}$, $q=w/(1-S_{ee}w)$, on survivors. With $a=S_{ee}$, $\eta_e=1-|a|^2$, define

$$A=S_{ie}S_{ek}\overline{S_{ik}},\quad B=P_{ie}P_{ek},\qquad C=S_{ke}S_{ej}\overline{S_{kj}},\quad D=P_{ke}P_{ej}.$$

The circle moments are $\mathbb{E}q=\mathbb{E}q^2=0$, $\mathbb{E}|q|^2=1/\eta_e$, $\mathbb{E}(q|q|^2)=\bar a/\eta_e^2$ and $\mathbb{E}|q|^4=(1+|a|^2)/\eta_e^3$. Hence

$$F_{e,k}:=\mathbb{E}_w(P'_{ik}P'_{kj})=P_{ik}P_{kj}+\frac{P_{ik}D+P_{kj}B+2\,\mathrm{Re}(A\bar C)}{\eta_e}+\frac{2\,\mathrm{Re}\{\bar a(AD+CB)\}}{\eta_e^2}+\frac{(1+|a|^2)BD}{\eta_e^3}.$$

The frozen-label average is exactly

$$u\,\mathbb{E}(\Delta K_{ij}\mid S)=\sum_{e\notin\{i,j\}}\left[\sum_{k\neq e}F_{e,k}-K_{ij}\right].$$

The subtraction lies inside the sum; survival conditioning would instead have denominator $u-2$.

The coherent term is a swap pairing of three-step path amplitudes:

$$T_{e,k}=S_{ie}S_{ek}S_{kj},\qquad A\bar C=T_{e,k}\overline{T_{k,e}}.$$

On $\mathcal{D} = \{(e,k) : e \notin \{i,j\},\ k \neq e\}$, both square sums are bounded by $(P^3)_{ij}$, even though $\mathcal{D}$ is not swap invariant. Weighted Cauchy–Schwarz therefore gives

$$\left|\sum_{\mathcal{D}} \frac{A\bar{C}}{\eta_e}\right| \leq \frac{(P^3)_{ij}}{1-\delta}.$$

The base loss is $-K_{ij}+P_{ij}(P_{ii}+P_{jj})$; the two positive quadratic terms total at most $2(P^3)_{ij}/(1-\delta)$. For the nonzero-diagonal cubic terms, AM–GM yields

$$|aAD| \leq \frac{\delta}{2}P_{ej}(P_{ie}P_{ek} + P_{ik}P_{ke}), \qquad |aCB| \leq \frac{\delta}{2}P_{ie}(P_{ke}P_{ej} + P_{kj}P_{ek}).$$

Each sum is at most $\delta[K_{ij} + (P^3)_{ij}]/2$. The quartic term totals at most $\delta(1+\delta)K_{ij}/(1-\delta)^3$ since $(P^2)_{ee} \leq \delta$. Thus

$$u\,\mathbb{E}(\Delta K_{ij} \mid S) \leq -(1-c(\delta))K_{ij} + 2\delta P_{ij} + d(\delta)(P^3)_{ij},$$

where

$$c(\delta) = \frac{2\delta}{(1-\delta)^2} + \frac{\delta(1+\delta)}{(1-\delta)^3}, \qquad d(\delta) = \frac{4}{1-\delta} + \frac{2\delta}{(1-\delta)^2}.$$

The increasing functions satisfy $c(1/16) = 752/3375 < 1/2$, $d(1/16) = 992/225 < 5$. Therefore

$$\mathbb{E}(\Delta K_{ij} \mid S) \leq \frac{-K_{ij}/2 + 5(P^3)_{ij} + 2\delta P_{ij}}{u} \qquad (\delta \leq 1/16).$$

The preceding higher-step theorem gives $(P^3)_{ij} \leq 1/u + \rho(\delta - 1/u)$. At the TRACE-074 witnesses, $P_{ij} = 0$, $K_{ij} \sim \log(2u)/u$ and $(P^3)_{ij} = (1+o(1))/u$. All nonbase terms have absolute sum $o(K_{ij})$, and hence

$$\mathbb{E}(\Delta K_{ij} \mid S) = -(1+o(1))K_{ij}/u.$$

This proves a restoring mean mechanism for each constructed input. It does not establish high-probability or sustained decay, preservation of its hypotheses, higher moments, or control of other peaks. The canonical proof is `mathematics/two_step_restoring_drift_20261012.md`. Independent exact residue and 64-dimensional unitary controls are finite checks. All-time flatness probability, scalar microscopic identification, full TRACE-055, signed TRACE-005 and deterministic BGS remain open.

87.12 Exact Haar feedback and entropy-stable clocks (TRACE-076)

Take an initial Haar unitary U_N independent of the fresh feedback phases and uniform coordinate order, with $F_N(z) = zU_N$ and initial incoming and outgoing delays equal to the identity. Retain a predetermined set of u coordinates and put $q = N - u$. Absorb any fixed feedback phases in the eliminated rows: $\widehat{U} = \mathrm{diag}(I_u, W_E)U_N$. Conditional on the independent phases and order, this matrix is still Haar. Define

$$X(z) = (I_q - z\widehat{U}_{EE})^{-1}z\widehat{U}_{ER}, \qquad F_u(z) = z\widehat{U}_{RR} + z\widehat{U}_{RE}X(z).$$

Sequential and block feedback solve the same linear system in the unit disk and agree at regular boundary points. All required inverses exist almost surely at $z = 1$: a fixed vector of the eliminated principal block would, by unitarity and the norm identity, extend to a fixed vector of $\widehat{U}$, an event of Haar measure zero.

Set $S = F_u(1)$, $T = S^*F_u'(1)$ and $Q = STS^*$. The exact joint law is

$$\begin{aligned} &S \sim \mathsf{H}_u, \qquad S \text{ independent of } T, \qquad T = I_u + X^*X, \\ &d\mathbb{P}(X) \propto \det(I_u + X^*X)^{-(N+u)}\,dX, \qquad X \in \mathbb{C}^{q\times u}. \end{aligned} \tag{HF.1}$$

The X law is invariant under right unitaries, so Q has the same law as T and is separately independent of S. The pair (T,Q) need not be independent of S, and no independence across sizes is asserted. At $q = 0$, $T = Q = I_u$.

Derivative and complete block factorization. On $z = e^{i\theta}$, the unitary colligation identity is

$$z\widehat{U}\begin{pmatrix} I_u \\ X(z) \end{pmatrix} = \begin{pmatrix} F_u(z) \\ X(z) \end{pmatrix}.$$

Differentiate in θ and left multiply by $(F_u(z)^*, X(z)^*)$. The terms $X(z)^*\dot{X}(z)$ cancel, giving $-iF_u(z)^*\dot{F}_u(z) = I_u + X(z)^*X(z)$. At $\theta = 0$ this is the delay identity in (HF.1).

For the distribution, use the Hermitian Cayley variable

$$C = i(I + \widehat{U})(I - \widehat{U})^{-1} = \begin{pmatrix} A & B^* \\ B & D \end{pmatrix}, \qquad \widehat{U} = (C - iI)(C + iI)^{-1}.$$

The differential $-i\widehat{U}^* d\widehat{U} = 2(C - iI)^{-1} dC (C + iI)^{-1}$ gives Haar density proportional to $\det(I + C^2)^{-N} dC$. Here Hermitian congruence in dimension n by L has real Jacobian $|\det L|^{2n}$: singular values s_j scale the real diagonal coordinates by s_j^2 and each complex off-diagonal coordinate by $s_j s_k$, whose real two-dimensional contribution is $s_j^2 s_k^2$.

The block colligation gives $S = (A - iI)(A + iI)^{-1}$ and $X = B(A + iI)^{-1}$. With $G = I_q + XX^*$ make the change

$$B = X(A + iI), \qquad D = XAX^* + G^{1/2} Y G^{1/2}, \qquad Y = Y^*.$$

Its real Jacobian is $\det(I_u + A^2)^q \det(G)^q$. The full derivative is block triangular in the order (A, X, Y); the derivatives of the square root contribute no extra factor. Since $B(I + iA)^{-1}B^* = X(I - iA)X^*$, the Schur determinant gives

$$\det(I + iC) = \det(I + iA) \det G \det(I + iY).$$

Consequently the transformed density is proportional to

$$\det(I_u + A^2)^{-u} \det(G)^{-(N+u)} \det(I_q + Y^2)^{-N} \, dA \, dX \, dY. \tag{HF.2}$$

The original mass is finite and positive; Tonelli therefore makes each independent factor finite and positive. The A factor is the Cayley image of Haar in dimension u, proving (HF.1). Haar-compatible Schur projections and Cauchy ensembles are prior objects; see [48]. Rectangular matrix-beta integrals and their singular-value reductions are prior as well [49]. The displayed factorization and its delay coupling are derived here; no world-priority claim is made.

Inverse-Jacobi spectrum, atom and moments. Let G_0 and Y_0 be independent standard complex Gaussian matrices of sizes $q \times u$ and $2u \times u$, respectively. Put $W = Y_0^* Y_0$. Then

$$X \overset{d}{=} G_0 W^{-1/2}.$$

Indeed, the Wishart density of W is proportional to $e^{-\operatorname{Tr} W} \det(W)^u$, and $G_0 = XW^{1/2}$ has real Jacobian $\det(W)^q$. Integrating W gives exponent $-(N + u)$. For $B_0 = G_0^* G_0$, the eigenvalues of T^{-1} are those of

$$J = (W + B_0)^{-1/2} W (W + B_0)^{-1/2}. \tag{HF.3}$$

This is a spectral identity, not an equality of differently ordered noncommuting matrices.

Writing $r = \min(q, u)$, the squared-singular-value volume of a complex $q \times u$ matrix is $\prod_{\ell=1}^{r} t_\ell^{|q-u|} \prod_{\ell<j} (t_\ell - t_j)^2 \, dt$ up to angular volume. The angular tangent between each two singular directions contributes $(\sigma_\ell^2 - \sigma_j^2)^2$; the unmatched directions contribute $\sigma_\ell^{2|q-u|}$, and the diagonal polar coordinate contributes $\sigma_\ell \, d\sigma_\ell$. The substitution $y_\ell = (1 + t_\ell)^{-1}$ in (HF.1) gives

$$d\mathbb{P}(y) \propto \prod_{\ell=1}^{r} y_\ell^u (1 - y_\ell)^{|q-u|} \prod_{\ell<j} (y_\ell - y_j)^2 \, dy, \quad 0 < y_\ell < 1. \tag{HF.4}$$

The power of y_ℓ is $N+u-|q-u|-2r=u$. If $q<u$, the remaining $u-q$ eigenvalues of T^{-1} equal one. Thus $H=uT/N$ has exactly the atom $\max(0,2-N/u)$ at u/N. For $q>0$ and $k>0$,

$$\mathbb{E}\,\mathrm{Tr}\,T^k<\infty \quad\Longleftrightarrow\quad k<u+1.$$

The necessity and sufficiency follow from the density near $y=0$; restricting the other eigenvalues to separated interior intervals establishes necessity. Finite-size spectra remain unbounded.

Projecting one Gaussian column of Y_0 orthogonally to the other $u-1$ columns leaves $u+1$ complex Gaussian directions. The Schur complement is gamma of shape $u+1$, with inverse mean $1/u$. Unitary invariance gives $\mathbb{E}W^{-1}=I_u/u$; independence and $\mathbb{E}G_0^*G_0=qI_u$ then yield the exact normalization

$$\mathbb{E}T=(N/u)I_u, \qquad \mathbb{E}H=I_u. \tag{HF.5}$$

Whole-trajectory Haar probability and unconditional clocks. Let $\ell=\log(2N)$, $m=\lceil\ell^{1+\kappa}\rceil$, $\kappa>0$, and let $A_m(K)$ require max entry intensity at most $K\ell/u$ simultaneously for every size $u\ge m$. Every scattering marginal is Haar. The Haar entry tail $(1-t)^{u-1}$ follows from a normalized complex Gaussian column, and a union bound gives

$$\mathbb{P}_{\mathsf{H}_N}(A_m(K)^c)\le\sum_{u=m}^{N}u^2e^{-K\ell(u-1)/u}\le N^3(2N)^{-K(1-1/m)}=:b_N(K). \tag{HF.6}$$

The entry failure probability is zero if the threshold is at least one. For each fixed $K>3$, $b_N(K)=2^{-K}N^{3-K}(1+o(1))$, because $\ell/m\to0$. Independent depths were not used.

Write $\mathcal{E}_{L,N}$ for the maximum error in all conditional TRACE-060 clock conclusions through a fixed power L, at all $u\ge m$. The theorem for every deterministic initial sequence supplies a common $\varepsilon_N(L,K,\kappa)\to0$ with

$$\sup_{U\in U(N)}\mathbb{P}_U\big(A_m(K)\cap\{\mathcal{E}_{L,N}>\varepsilon_N\}\big)\longrightarrow0. \tag{HF.7}$$

If uniformity failed, a violating initial matrix could be chosen at infinitely many dimensions, contradicting that deterministic-sequence quantifier. Integrate (HF.7) over the initial Haar law and add (HF.6). Incoming/outgoing diagonal power laws, off-diagonal decay, mixed-entry decay and all fixed empirical moments are now unconditional in probability, uniformly over the full retained depth.

The empirical probability measure of H converges uniformly to the maintained family $\mu_{u/N}$ in the bounded-Lipschitz metric. For clarity, the finite-moment criterion is uniform over $x\in[0,1]$ and probability measures on $[0,\infty)$. If no finite number of moments forced a given positive weak-metric tolerance, choose counterexamples with the number of controlled moments tending to infinity and tolerances to zero. Pass to $x\to x_0$. First moments give tightness; each next higher fixed moment gives uniform integrability. Every weak limit has all moments of μ_{x_0}. Common compact reference support and moment determinacy identify that limit, contradicting the chosen tolerance. This applies a finite fixed L to (HF.7), rather than inserting a growing order into TRACE-060.

Relative entropy and its microscopic limitation. Let an independent initial law ν_N have finite $D_N=D(\nu_N\|\mathsf{H}_N)$. The joint entropy including fresh phase/order randomness is still D_N. If p_N and q_N are the respective failure probabilities of $A_m(K)$, binary Jensen gives

$$D_N\ge p_N\log(p_N/q_N)+(1-p_N)\log((1-p_N)/(1-q_N))\ge p_N\log(1/q_N)-\log2.$$

For $q_N=0$, absolute continuity gives $p_N=0$. Otherwise (HF.6) yields, for all large N,

$$p_N\le\frac{D_N+\log2}{\log(1/b_N(K))}. \tag{HF.8}$$

Hence $D_N = o(\log N)$ proves high-probability flatness for each fixed $K > 3$. More generally, if $C = \limsup_N D_N/\log N < \infty$, integrating (HF.7) gives for every fixed error tolerance $\eta > 0$

$$\limsup_N \mathbb{P}_{\nu_N}(\mathcal{E}_{L,N} > \eta) \le C/(K-3).$$

Let K tend to infinity *after* this limsup. Every fixed clock conclusion and the full-depth empirical weak law therefore hold unconditionally when $D_N = O(\log N)$. This argument gives no high-probability flatness statement for one prescribed finite K throughout that larger entropy class.

The entropy condition alone cannot identify a prescribed-origin microscopic spectral law. For $0 < c < 1$, a fixed arc of angular length $2\pi c/N$ has Haar expected count c, so its void event E_N has probability at least $1-c$. Conditioning Haar on E_N costs $-\log \mathbb{P}(E_N) \le -\log(1-c)$ entropy and preserves the clock conclusions above. Its initial rescaled point process has a forbidden interval of length c. A nonzero nonnegative continuous test function compactly supported inside that interval integrates to zero for every N. The intensity-one Sine_2 process gives a positive expected integral and hence a positive probability of a nonzero integral. Local point-process convergence to Sine_2 at that origin is impossible. This is not an origin-averaged counterexample. A deterministic initial matrix has infinite relative entropy to Haar. Classical dynamical entropy is a different quantity. The maintained deterministic trajectory probability, full physical TRACE-055, signed TRACE-005 and global BGS remain open.

The canonical source is `mathematics/haar_feedback_clock_law_20261012.md`. Independent reconstruction and maintained-transfer review are separate from five exact colligation controls, symbolic Jacobians, twelve one-port moment integrals and two actually executed Wolfram identities. None of those finite controls proves the probability limits, entropy transfer, mathematical priority or deterministic BGS.

88 Green compression and shrinking stopped intensity gaps

The project proof BGS-TRACE-077 concerns an unconditional stopped prefix, not the probability of surviving to that prefix. Write $J_n = \mathbf{1}\mathbf{1}^*/n$, $Q_n = I - J_n$ and

$$G(P) = (I - P + J_n)^{-1} - J_n, \qquad M(P) = G(P) - Q_n.$$

Here P is nonnegative doubly stochastic and $I - P$ is invertible on $\mathbf{1}^\perp$; normality is not required. For a retained set A and deleted set D, assume $I - P_{DD}$ is invertible and form the stochastic complement $P^c = P_{AA} + P_{AD}(I - P_{DD})^{-1}P_{DA}$. Then

$$G(P^c) = Q_u G(P)_{AA} Q_u, \qquad M(P^c) = Q_u M(P)_{AA} Q_u. \tag{GC.1}$$

Indeed, solve $(I - P)x = (b, 0)$ with $\sum b = 0$, eliminate x_D, and center x_A. A kernel vector of $I - P^c$ lifts to a kernel vector of $I - P$, hence is constant; the reduced centered inverse therefore exists. Block stochastic sums and the nonnegative geometric inverse of $I - P_{DD}$ give double stochasticity of P^c. On the original coordinate space the projections Q_A onto supported zero-sum vectors are nested: $Q_A Q_B = Q_A$ when $A \subset B$.

Let S_N be deterministic unitary, $P_N = |S_N|^2$, $\max P_N \le C_0/N$, $C_0 \ge 1$, and $\rho_N = \|P_N - J_N\|_{\mathrm{op}} \le 1/5$. Geometric expansion and the correct higher-step row–column estimate give

$$\|M_N\| \le 1/4, \quad \max|(M_N)_{ij}| \le B/N, \quad B = C_0 + \frac{C_0 - 1}{1 - \rho_N}, \quad \|Q_A M_N Q_A\| \le \min(1/4, Bu/N). \tag{GC.2}$$

In particular the original contribution decays with retained depth.

At each step choose the port using the past, then generate a fresh independent uniform circle phase. The exact one-step intensity is $P' = P^c + \Xi$, with conditional mean zero, $\Xi\mathbf{1} = \mathbf{1}^*\Xi = 0$

and $\|\Xi\| \le 8\delta_v$ when $\delta_v = \max P_v \le 1/4$. Put $\ell = \log(2N)$, choose $m < N$ with $m \ge 32K\ell$ and $C_0 \le K\ell$, and stop at the first $\delta_u > K\ell/u$ or $\|M_u\| > 1/2$, retaining the full crossing state. The old admissible state gives $\|G^c\| \le 3/2$ and $\|G^c\Xi\| \le 3/8$. Thus the new inverse exists even at the first crossing, with $\|G'\| \le 3$, and the ordered resolvent expansion is

$$G' = G^c + L_v + D_v, \quad L_v = G^c\Xi G^c, \quad D_v = G^c\Xi G^c\Xi G', \quad \|L_v\| \le 18\delta_v, \quad \|D_v\| \le 432\delta_v^2. \tag{GC.3}$$

The L_v are adapted martingale differences with predictable coefficients and stop mask. The D_v are bounded adapted remainders, with no centering assumption. Embed both in the original N coordinates, set them to zero after stopping, and put $Z_u = \sum_{v>u} L_v$, $E_u = \sum_{v>u} D_v$. Nested compression gives, pathwise,

$$M_u = Q_u M_N Q_u + Q_u Z_u Q_u + Q_u E_u Q_u. \tag{GC.4}$$

Future projections occur only outside the martingale sum. This is why adaptive port choices preserve centering without conditioning on future flatness.

For even $p \ge 2$, the internally reconstructed Schatten smoothness bound of TRACE-073 gives $\mathbb{E}\|Z_m\|_p^2 \le 324N^{2/p}(p-1)K^2\ell^2/m$. Conditional Jensen and scalar Doob L^2 then give

$$\mathbb{P}\{\max_u \|Z_u\| > \eta\} \le \frac{1296N^{2/p}(p-1)K^2\ell^2}{m\eta^2}, \qquad \|E_u\| \le \beta := 432K^2\ell^2/m. \tag{GC.5}$$

If $\eta + \beta < 1/4$, (GC.4) excludes a first Green exit before or simultaneously with the entry exit. Consequently, outside (GC.5), every state through the first entry exit satisfies $\|M_u\| \le \min(1/4, Bu/N) + \eta + \beta$. For $h_u = Bu/N + \eta + \beta < 1$, the centered identity $P_u - J_u = M_u(I + M_u)^{-1}$ yields $\rho_u \le h_u/(1 - h_u)$.

Choose $p = 2\lceil \ell \rceil$, $m = \lceil \ell^{7+\kappa} \rceil$, $\eta = \ell^{-2}$, $\kappa > 0$. On all reached entry-admissible states $m \le u \le \lfloor N/\ell^2 \rfloor$ simultaneously,

$$\rho_u = O_{C_0,K}(\ell^{-2}), \qquad (P_u^3)_{ij} \le 1/u + \rho_u(\delta_u - 1/u) \le [1 + O_{C_0,K}(\ell^{-1})]/u, \tag{GC.6}$$

except with probability $O_K(\ell^{-\kappa})$. The second inequality is not asserted at an entry-crossing state whose entry bound fails. The interval can be empty because of early entry exit. Its survival probability and sustained two-step concentration remain open. FM-021 supplies only the named eventual initial inputs for the actual odd-N fixed-coupling family; no effective startup is obtained. TRACE-005, physical microscopic identification and global BGS remain open.

The canonical source is `mathematics/green_compression_gap_20261013.md`. Group inverses and stochastic complementation are prior structures; no world-priority claim is made. Independent reconstruction and its maintained-source transfer check are distinct from four exact rational nonnormal compression controls, two rejected algebraic mutations and four complex-double adaptive feedback steps. No new Lean certificate or C++ verification of the analytic theorem is asserted.

89 Second moment of the complete two-step feedback increment

BGS-TRACE-078 complements the one-step drift with a fluctuation bound. For any unitary S of size $u \ge 3$, put $P = |S|^2$, $K = P^2$ and $\delta = \max P \le 1/4$. Fix distinct labels i, j. Choose a port uniformly among all u labels, freeze $K'_{ij} = K_{ij}$ if either endpoint is chosen, and otherwise apply exact Schur feedback. For arbitrary circle phases, even depending on the selected port,

$$\mathbb{E}[(K'_{ij} - K_{ij})^2 \mid S] \le \frac{2600\delta^2}{u}\{K_{ij} + (P^3)_{ij}\}. \tag{SV.1}$$

No phase independence is needed for this raw second moment. Uniform port selection is needed for its factor $1/u$.

For $e \notin \{i,j\}$ and $k \neq e$, write $x = P_{ik}$, $y = P_{kj}$, $\alpha = S_{ie}S_{ek}\overline{S_{ik}}$, $\gamma = S_{ke}S_{ej}\overline{S_{kj}}$, $b = P_{ie}P_{ek}$, $d = P_{ke}P_{ej}$ and $r_e = P_{ie}P_{ej}$. With all following sums over $k \neq e$, define

$$\begin{aligned} A_e &= \sum(y\alpha + x\gamma), \quad & B_e &= \sum(xd + yb + 2\,\mathrm{Re}(\alpha\bar{\gamma})), \\ C_e &= \sum \alpha\gamma, & D_e &= \sum(\alpha d + \gamma b), \qquad & F_e &= \sum bd. \end{aligned}$$

The exact real increment, $q_e = w_e/(1 - S_{ee}w_e)$, is

$$\Delta_e = -r_e + 2\,\mathrm{Re}(q_e A_e) + |q_e|^2 B_e + 2\,\mathrm{Re}(q_e^2 C_e) + 2\,\mathrm{Re}(q_e|q_e|^2 D_e) + |q_e|^4 F_e. \tag{SV.2}$$

The subtraction $-r_e$ is the removed old $k = e$ term. Since $|q_e| \leq 2$, six-group Cauchy bounds its square. The key complete-coordinate bounds are

$$\begin{aligned} \sum r_e^2 &\leq \delta^2 K_{ij}, & \sum |A_e|^2 &\leq 10\delta^2 K_{ij}, \\ \sum B_e^2 &\leq 6\delta^2 (P^3)_{ij} + 12\delta^2 K_{ij}, & \sum |C_e|^2 &\leq \delta^2 K_{ij}, \\ \sum |D_e|^2 &\leq 4\delta^3 (P^3)_{ij}, & \sum F_e^2 &\leq \delta^4 K_{ij}. \end{aligned} \tag{SV.3}$$

For A_e, apply Parseval to the full unitary row and column sums first; each component's squared sum is at most $\delta^2 K_{ij}$. The omitted term is $2S_{ee}r_e$, separately bounded by $4\delta^3 K_{ij}$. For the coherent parts of B_e, C_e, Cauchy gives $r_e \delta K_{ij}$ per port. For D_e, weighted Cauchy gives $\delta P_{ie}P_{ej}^2 K_{ie}$ and $\delta P_{ej}P_{ie}^2 K_{ej}$ for the two components; coordinate sums are bounded using P^3. These yield (SV.3) without decorrelation. Consequently, for any simultaneous choice of phases,

$$\sum_{e \notin \{i,j\}} \Delta_e^2 \leq \delta^2 \{2598 K_{ij} + 2112 (P^3)_{ij}\}. \tag{SV.4}$$

Uniform coordinate averaging proves (SV.1). This also bounds conditional variance, but does not prove its accumulated concentration. Combining it with the restoring mean and stopped three-step forcing requires a separate dynamic argument respecting endpoint deletion and full first exits. Early entry survival, deterministic microscopic identification and BGS remain open. The canonical proof is `mathematics/two_step_increment_variance_20261013.md`. Independent reconstruction is separate from exact Walsh expansion checks and finite complex-double port-sum diagnostics. No priority or new formal certificate is asserted.

90 Simultaneous stopped two-step concentration

BGS-TRACE-079 upgrades the restoring mean and raw second moment to a simultaneous stopped estimate. Start with any deterministic $S_b \in U(b)$, put $P = |S|^2$, and assume $\max P_b \leq D/b$, $\max P_b^3 \leq C/b$. Choose uniform retained ports and fresh independent uniform phases. For integers $b > m \geq 3$, $m \geq 16D$, stop after the full first new state violating either bound $\max P_u \leq D/u$, $\max P_u^3 \leq C/u$. For every ordered pair, including $i = j$, freeze the old unscaled $K_{ij} = (P^2)_{ij}$ on its endpoint-removing transition, then deactivate that scalar pair. This is a scalar convention, never an entry of an absent later matrix.

The distinct-label one-step bounds extend to the diagonal: the mean base loss is $-K_{ii} + P_{ii}^2$, with the same positive majorants, and the omitted linear coefficient remains $2S_{ee}P_{ie}P_{ei}$. Thus

$$\mathbb{E}\Delta K_{ij} \leq \frac{-K_{ij}/2 + 5(P^3)_{ij} + 2\delta P_{ij}}{v}, \qquad \mathbb{E}(\Delta K_{ij})^2 \leq \frac{2600\delta^2\{K_{ij} + (P^3)_{ij}\}}{v}. \tag{SF.1}$$

The complete six-group expansion also gives $|\Delta K_{ij}| \leq 32\delta^{3/2}$. These averages divide by all v ports, including the zero frozen increments.

Put $a = 4/3$, $F = 5C + 2D^2/m$, $\beta = F/a$, $V = 2600D^2(D + C)$ and $R = 64D^{3/2}$. For $Y_v = vK_{ij}$,

$$\mathbb{E}\Delta Y \le -aY_v/v + F/v, \quad \mathrm{Var}(\Delta Y \mid \mathrm{past}) \le V/v^2, \quad |\Delta Y - \mathbb{E}\Delta Y| \le R/\sqrt{v}. \tag{SF.2}$$

After either scalar deactivation or the global stop continue only the bookkeeping recurrence $Y_{v-1} = (1 - a/v)Y_v + F/v$. Its centered noise is zero. For each fixed final depth the positive deterministic integrating factors telescope with a nonnegative drift slack:

$$Y_u \le \beta + r_{b,u}(Y_b - \beta) + \sum_{v=u+1}^{b} w_{v,u}\xi_v, \quad r_{b,u} = \prod_{v=u+1}^{b}(1 - a/v), \quad 0 < w_{v,u} \le 1. \tag{SF.3}$$

The weights depend only on sizes. Endpoint survival is not conditioned upon, and the entire first crossing is retained.

The conditional Taylor series for a centered $|X| \le c$ yields $\mathbb{E}(e^{\lambda X} \mid \mathcal{F}) \le \exp\{\lambda^2\mathbb{E}(X^2 \mid \mathcal{F})/[2(1 - \lambda c/3)]\}$. A variance-compensated supermartingale and Markov give the usual scalar Bernstein tail. Here the fixed-depth variance is at most V/u and the range at most $R/\sqrt{u}$. Union over all original ordered pairs and all depths proves

$$\mathbb{P}\{\exists u, i, j \text{ retained}: \ uK_{ij} > \beta + Dr_{b,u} + \eta\} \le b^3 \exp\left[-\frac{\eta^2}{2\{V/m + R\eta/(3\sqrt{m})\}}\right]. \tag{SF.4}$$

The event is restricted to the stopped prefix, through its full crossing. Also $r_{b,u} \le [(u + 1)/(b + 1)]^{4/3}$.

Combined with TRACE-077, take $\ell = \log(2N)$, $m = \lceil \ell^{7+\kappa} \rceil$, $b = \lfloor N/\ell^2 \rfloor$, $D = K_0\ell$, $C = 2$ and $\eta = 1$. At a revealed startup satisfying both barriers, the estimate is uniform in that starting matrix. The scalar martingale is proved before unioning its failure with the future Green failure; it is not conditioned on that future good event. On every reached depth $m \le u \le \lfloor b/\ell \rfloor$ through the entry crossing, simultaneously for all retained pairs, $(P_u^2)_{ij} \le 9/u$ except with probability $O(\ell^{-\kappa})$. This corollary alone does not estimate early entry survival. Its canonical source is `mathematics/two_step_stopped_concentration_20261013.md`.

91 Unconditional feedback flatness from deterministic starts

BGS-TRACE-080 closes the trajectory probability for a specified deterministic startup class. Fix $C_0 \ge 1$, $\kappa > 0$ and any deterministic $S_N \in U(N)$ with $\max|S_{N,ij}|^2 \le C_0/N$ and $\|P_N - J_N\| \le 1/100$. Ports are uniform and phases fresh independent uniform. Put

$$\begin{aligned} \ell &= \log(2N), & L &= C_0 + 7, & g &= 1/10, \\ C &= 1 + g(L - 1), & T &= C_0 + 4C + 3, & K_0 &= 32T, \\ D &= K_0\ell, & m &= \lceil \ell^{4+\kappa} \rceil, & p &= 2\lceil \ell \rceil. \end{aligned} \tag{UF.1}$$

For sufficiently large N, simultaneously at every $N \ge u \ge m$,

$$\begin{gathered} \max P_u \le D/u, \qquad \|P_u - J_u\| \le g, \qquad \max(P_u^2)_{ij} \le T/u, \\ \max_i \sum_j P_{u,ij}^2 \le L/u, \qquad \max_j \sum_i P_{u,ij}^2 \le L/u, \end{gathered} \tag{UF.2}$$

with failure $O_{C_0,\kappa}(\ell^{-1-\kappa})$. All ordered pairs include the diagonal. No random initial ensemble or deterministic microscopic reference law is assumed.

The additional mechanism controls fourth-power energies of entire rows and columns before estimating individual entries. Write $E_i = \sum_j P_{ij}^2$ and $A_i = \sum_j P_{ij}(P^2)_{ij}$. For a possibly nonnormal intensity matrix,

$$A_i \le 1/v + \rho(E_i - 1/v), \qquad \rho = \|P - J_v\|. \tag{UF.3}$$

This is the centered row inner product against its image under $P - J_v$. Exact feedback, with the removing row frozen, gives

$$v\mathbb{E}\Delta E_i \le -(1 - c(\delta))E_i + d(\delta)A_i + P_{ii}^2. \tag{UF.4}$$

The functions c, d are those of the exact two-step drift. At $\delta \le 1/16$, $c + d/10 \le 448/675 < 2/3$. Hence when $\rho \le g$, $\mathbb{E}\Delta E_i \le -E_i/(3v) + 5/v^2 + \delta^2/v$. Unitary row Parseval before truncation, weighted Cauchy and the complete feedback expansion give

$$\mathbb{E}(\Delta E_i)^2 \le 3500\delta^2(E_i + A_i)/v, \qquad |\Delta E_i| \le 8\delta\sqrt{E_i} + 34\delta^2. \tag{UF.5}$$

Transpose gives the same column estimates. The positive forcing in (UF.3) depends on row energy rather than the logarithmic entry maximum.

Use one joint first-exit rule for the entry cap, the Green cap $\|M_u\| \le 1/11$, the two energy caps, and the two-step cap. All estimates use the old admissible state and keep the complete new crossing, including simultaneous crossings. On the zero-sum space, $\|G^c\| \le 12/11$, $\|\Xi\| \le 1/2$, so $\|G^c\Xi\| \le 6/11$ and $\|G'\| \le 12/5 < 3$ even at a crossing. The same Green martingale constants $18,432$ remain valid. The initial $\|M_N\| \le 1/99$ and $\eta_G = 1/40$ provide a fixed positive margin.

For the scaled row energy $Y_v = vE_i$ the drift may be weakened to $-Y_v/v + (5 + D^2/m)/v$. Its initial value is at most C_0, and its old value is at most L under the joint stop. Define

$$\begin{aligned} V_E &= 3500D^2[(1+g)L + 1], \quad & c_E(u) &= 16D\sqrt{L}/\sqrt{u} + 68D^2/u, \\ V_K &= 2600D^2(T + C), & c_K(u) &= 24D\sqrt{T}/\sqrt{u} + 36D^2/u. \end{aligned} \tag{UF.6}$$

The variance per active row step is at most V_E/v^2 and the centered range at most $c_E(v)$. The scalar integrating factors give a row bound $uE_i \le \max(C_0, 5 + D^2/m) + 1 < L$ outside its exception. The old energy and Green caps imply $(P^3)_{ij} \le [1 + g(L-1)]/v = C/v$. For the all-pair two-step scalar, (SF.1)–(SF.3) and the sharper range $|\Delta K_{ij}| \le 12\delta\sqrt{K_{ij}} + 18\delta^2$ give $uK_{ij} \le \max(C_0, 3(5C + 2D^2/m)/4) + 1 < T$ outside its exception. There is no burn-in, since the initial scaled energies and two-step entries are already bounded by C_0.

For each original amplitude, freeze on endpoint removal and continue only $W_{v-1} = \sqrt{(v-1)/v}\, W_v$ after deactivation, where $W_v = \sqrt{v}\, S_{ij}$. For fixed target u, its centered martingale increments are $\sqrt{u}\, qS_{ie}S_{ej}$ with the predictable old stop mask. Uniform coordinate choice and the two-step cap give total real/imaginary variation at most $\sigma^2 = 16T/15$ and range at most $c_A(u) = 2D/\sqrt{u}$. The initial term has modulus at most $\sqrt{C_0}$. Put $t = \sqrt{D} - \sqrt{C_0}$. The four signed real/imaginary tails, plus ordered pairs and depths, exclude the entry crossing.

The explicit exceptional probabilities are

$$\begin{aligned} q_G &= \frac{1296N^{2/p}(p-1)D^2}{m\eta_G^2}, \\ q_E &= 2N^2 \exp[-1/(2\{V_E/m + c_E(m)/3\})], \\ q_K &= N^3 \exp[-1/(2\{V_K/m + c_K(m)/3\})], \\ q_A &= 4N^3 \exp[-t^2/(4\{\sigma^2 + c_A(m)t/(3\sqrt{2})\})]. \end{aligned} \tag{UF.7}$$

Their ordinary union excludes every possible first joint exit. No proof is conditioned on any future good event. Here $q_G = O(\ell^{-1-\kappa})$, q_E, q_K are superpolynomially small in N, and $q_A = N^{-9/2+o(1)}$. This proves (UF.2). The row/column energy mechanism removes the previous early-entry-survival premise for this cutoff and startup class.

For $F_N(z) = zS_N$ and $T_N = I$, revealing an original unused independent phase at each uniformly selected port is the same process. TRACE-060 with $K = K_0$, $\kappa_{\mathrm{clock}} = 3 + \kappa$ has precisely this cutoff m. Its intersection estimate plus the flatness probability gives unconditional convergence in probability for every fixed clock order and the empirical weak law with reference

$\mu_{u/N}$, uniformly for $u \geq m$. The reference atom and pathwise rank lower bound retain their source scope; no equality of actual finite-N atom weight is asserted. The entire upstream clock induction is a named internal input, not newly certified here.

Separately, if the initial Hermitian reversal channel $B_{N,ij} = S_{N,ij}\overline{S_{N,ji}}$ has $\|B_N\| \to 0$, TRACE-073 (C9) and the same flatness probability imply $\max_{u\geq m} \|B_u\| \to 0$ in probability. This extra assumption is confined to the reversal corollary. FM-021 and FM-022 supply only the named eventual initial hypotheses for the actual odd-N fixed-coupling family, without an effective startup. Smaller logarithmic cutoffs, growing orders, edge laws, physical microscopic identification, signed TRACE-005 and BGS remain open. The canonical source is `mathematics/unconditional_feedback_flatness_20261013.md`. Independent reconstruction and exact/floating finite controls have distinct roles; scalar Bernstein/Freedman is prior, and no new formal certificate or world-priority claim is made.

92 Perfect intensity mixing need not determine a microscopic law

The abstract result BGS-TRACE-081 distinguishes intensity mixing from the coherent information in an initial spectrum. It does not concern the unchanged odd-dimensional smooth quantizer. Let p_k be any sequence of odd primes tending to infinity, $k \geq 2$, and put

$$n_k = 4^k, \qquad N_k = n_k p_k, \qquad \epsilon_k = N_k^{-4}.$$

There are deterministic $U_k \in U(N_k)$ with simple spectrum such that

$$P(U_k) = J_{N_k}, \qquad \|P(U_k) - J_{N_k}\| = 0, \qquad \|B(U_k)\| = 2^{-k}, \quad B(U)_{ij} = U_{ij}\overline{U_{ji}}.$$

Their macroscopic eigenphase measure converges to uniform circle measure. However, the empirical unfolded spacing measure converges weakly to δ_0, and their microscopic point process about a uniform random origin converges vaguely in probability to the empty measure. For times

$$t_k = 6p_k\lfloor 4^k/6\rfloor = N_k - 4p_k, \qquad t_k/N_k \longrightarrow 1,$$

the normalized form factor satisfies

$$N_k^{-1}|\operatorname{Tr} U_k^{t_k}|^2 \geq N_k(1 - t_k\epsilon_k)^2 \longrightarrow \infty.$$

The vanishing reversal norm is a coordinate statement, not a full antiunitary symmetry classification.

Exact construction. Set

$$H = \frac{1}{2}\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & -1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \end{pmatrix}.$$

Row orthogonality gives $H^*H = I$; H^2 permutes the last three coordinates cyclically, so $H^6 = I$. The sign matrix $4B(H)$ has Gram matrix $4I$, whence $\|B(H)\| = 1/2$. For $\omega = e^{2\pi i/p}$, define

$$(F_p)_{ab} = p^{-1/2}\omega^{ab}, \qquad D_p = \operatorname{diag}(\omega^{b^2}), \qquad V_p = F_p^* D_p F_p.$$

Then $V_p^p = I$, and

$$(V_p)_{ij} = p^{-1}\sum_b \omega^{b^2+(j-i)b}.$$

For each $d \in \mathbb{F}_p$, the bijection $(b, c) \mapsto (b - c, b + c + d)$ proves

$$\left| \sum_b \omega^{b^2+db} \right|^2 = \sum_{a,h} \omega^{ah} = p.$$

Thus every entry of V_p has squared modulus $1/p$. Replacing b by $-b$ proves $V_p^T = V_p$, so $B(V_p) = J_p$. The matrix $A_k = H^{\otimes k} \otimes V_{p_k}$ consequently has exactly flat intensities, reversal norm 2^{-k}, and $A_k^{6p_k} = I$. Its phases lie in at most $6p_k$ root-of-unity centers. For fixed $q > 0$, eventually $p_k \nmid q$, and the same Gauss argument gives

$$\frac{|\operatorname{Tr} A_k^q|}{N_k} \le \frac{|\operatorname{Tr} V_{p_k}^q|}{p_k} = p_k^{-1/2}.$$

Removing every exact degeneracy. For every invertible complex matrix A, the polynomial $f(\lambda, z) = \det(\lambda I - A \operatorname{diag} z)$ is monic in λ and multi-affine in z. If an irreducible factor repeated, its degree in every z_j would be zero. It would then divide $f(\lambda, 0) = \lambda^N$, but λ cannot divide f since its constant coefficient is $(-1)^N \det A \prod_j z_j \neq 0$. Monicity and factorization over the fraction field show that the discriminant in λ is a nonzero polynomial. Its zero set has measure zero on every product of circle arcs, by the univariate identity principle and Fubini. Thus arbitrarily small generic column phases split the spectrum of every invertible A.

Choose $E_k = \operatorname{diag}(e^{i\alpha_j})$, with $|\alpha_j| < \epsilon_k$, giving simple spectrum, and put $U_k = A_k E_k$. This choice can be deterministic. Then

$$\|U_k - A_k\| \le \epsilon_k, \quad P(U_k) = P(A_k), \quad B(U_k) = E_k^* B(A_k) E_k, \quad \|U_k^q - A_k^q\| \le q\epsilon_k.$$

For every fixed positive q, $|\operatorname{Tr} U_k^q|/N_k \le p_k^{-1/2} + q\epsilon_k$. Conjugation gives the negative Fourier coefficients, and trigonometric approximation proves the uniform macroscopic limit.

Local limits and rare counts. Normality of A_k implies that each eigenvalue of U_k lies within chord distance ϵ_k of an old center: apply $(\lambda I - A_k)x = (U_k - A_k)x$ to a unit eigenvector. All eigenphases therefore lie in at most $6p_k$ disjoint arcs of radius $2\epsilon_k$, for large k. At least $N_k - 6p_k$ cyclic consecutive gaps are internal to these arcs and at most $4\epsilon_k$. Their unfolded values are at most $2N_k\epsilon_k/\pi \to 0$, and the exceptional fraction is at most $6/4^k$, proving the weak spacing limit.

For an independent uniform origin Θ, let

$$\Xi_k = \sum_j \delta_{N_k \operatorname{Arg}(e^{i(\theta_j - \Theta)})/(2\pi)}.$$

For each fixed $R > 0$, a union over old centers gives

$$\mathbb{P}\{\Xi_k([-R, R]) > 0\} \le \frac{12 p_k \epsilon_k}{\pi} + \frac{12R}{4^k} \longrightarrow 0.$$

Every compactly supported continuous test is consequently zero with probability tending to one. This is vague convergence in probability to the empty measure. Nonetheless the exact origin average is $\mathbb{E}\Xi_k([-R, R]) = 2R$ for $N_k > 2R$; uniform integrability fails. At t_k, the norm bound and $A_k^{t_k} = I$ give $|\operatorname{Tr} U_k^{t_k} - N_k| \le N_k t_k \epsilon_k$, which proves the announced form-factor lower bound.

The construction uses classical Fourier, tensor and discriminant mechanisms; no priority is claimed. Full phase-averaged formulations [50, 51] are different objects. The example meets the algebraic startup bounds of TRACE-080, so its random-feedback conclusions cannot alone identify this initial local law. No fixed smooth chaotic quantization or BGS counterexample is asserted.

93 Exact coherence memory and the dimension-time norm barrier

BGS-TRACE-082 concerns every unitary U of dimension $N \geq 2$. On Hilbert–Schmidt matrix space let $\mathcal{T}(X) = UXU^*$, let $\mathcal{D}$ be the orthogonal diagonal projection, and let $\mathcal{Q} = I - \mathcal{D}$. With diagonal/off-diagonal domains, write

$$P = \mathcal{D}\mathcal{T}\mathcal{D}, \quad \mathcal{L} = \mathcal{D}\mathcal{T}\mathcal{Q}, \quad \mathcal{C} = \mathcal{Q}\mathcal{T}\mathcal{D}, \quad \mathcal{E} = \mathcal{Q}\mathcal{T}\mathcal{Q}.$$

Here $P_{ij} = |U_{ij}|^2$. Block elimination, valid for $|z| < 1$, gives the exact identity on the diagonal space

$$\mathcal{D}(I - z\mathcal{T})^{-1}\mathcal{D} = \left[I - zP - z^2\mathcal{L}(I - z\mathcal{E})^{-1}\mathcal{C}\right]^{-1}.$$

For diagonal X_0, putting $a_n = \mathcal{D}\mathcal{T}^n X_0$ and $c_n = \mathcal{Q}\mathcal{T}^n X_0$, one has $c_0 = 0$ and

$$a_{n+1} = Pa_n + \sum_{s=0}^{n-1} \mathcal{L}\mathcal{E}^{n-1-s}\mathcal{C}a_s.$$

The terms retain coherent return memory. Repeated dephasing gives P^n; unmeasured evolution gives $\mathcal{D}\mathcal{T}^n\mathcal{D}$. They need not agree, but do agree for special matrices, including diagonal and monomial unitaries.

Universal norm plateau. For every integer $0 \leq r < N$,

$$\|\mathcal{E}^r\|_{\mathrm{HS}\to\mathrm{HS}} = 1.$$

Indeed work in the complex trace-zero space of dimension $N^2 - 1$. Each constraint $\mathcal{D}\mathcal{T}^j X = 0$ has codimension at most $N - 1$, since trace is preserved. The constraints for $j = 0, \ldots, r$ leave dimension at least

$$N^2 - 1 - (r+1)(N-1) = (N-1)(N-r) > 0.$$

A nonzero witness stays off diagonal at all these times, so induction gives $\mathcal{E}^j X = \mathcal{T}^j X$ for $j \leq r$. Hilbert–Schmidt unitarity yields norm preservation; contractivity gives the opposite bound. The witness may depend on U and r. This is a singular-norm statement, not a spectral-radius or every-source nondecay statement. Exact finite controls include an $N = 2$ compression with norm one but square zero.

An observed Heisenberg return. For the family in the preceding section, $\|U_k^{t_k} - I\| \leq t_k\epsilon_k$ implies $\|\mathcal{T}^{t_k} - I\|_{\mathrm{HS}\to\mathrm{HS}} \leq 2t_k\epsilon_k$. Diagonal compression and $\|I - J\| = 1$ therefore give

$$\left\||U_k^{t_k}|^{\circ 2} - I\right\| \leq 2t_k\epsilon_k, \qquad \left\||U_k^{t_k}|^{\circ 2} - P(U_k)^{t_k}\right\| \geq 1 - 2t_k\epsilon_k.$$

This proves order-one complete observed return deviation. It is not a lower bound for one individually selected $\mathcal{L}\mathcal{E}^r\mathcal{C}$ kernel. Neither the universal plateau nor this abstract example estimates the actual fixed-map signed source.

The independent mathematical audits are distinct from exact finite Python/C++ controls and from the active Lean arithmetic kernel. That kernel proves only positivity of $(N-1)(N-r)$ for natural $N \geq 2, r < N$; it does not formalize codimension, the operator theorem, the generic discriminant lemma or a limiting point process.

94 Observed first-return pulses and cyclic coherence

This section records BGS-TRACE-083. Its authoritative proof is `mathematics/coherent_first_return_pulse_20261014.md`. It uses the abstract family of TRACE-081 and the exact block recurrence of TRACE-082. The unchanged fixed-map TRACE-005 and BGS limit remain open. The first-return mechanism itself is prior mathematics: Bourgain, Grünbaum, Velázquez and Wilkening [52], equation (1) on printed page 4 and the renewal equations on printed page 6. Here the Hilbert space is a matrix Hilbert–Schmidt space; amplitude energies are not identified with literal measurements of density matrices.

Let $n = 4^k$, $N = np$, $k \geq 2$, p an odd prime, $\epsilon = N^{-4}$, and use TRACE-081's $A = H^{\otimes k} \otimes V_p$ and simple-spectrum $U = AE$, with $\|U - A\| \leq \epsilon$. Put $S = (H^2)^{\otimes k}$, so $S^3 = I$, and $\Pi_n = I - J_n$, $\Pi_p = I - J_p$. For diagonal projection $\mathcal{D}$, $\mathcal{Q} = I - \mathcal{D}$ and $\mathcal{T}_U(X) = UXU^*$, define

$$F_t(U) = \mathcal{D}\mathcal{T}_U(\mathcal{Q}\mathcal{T}_U)^{t-1}\mathcal{D}, \qquad t \geq 1.$$

For $t \geq 2$, this is the individual memory coefficient $\mathcal{L}_U\mathcal{E}_U^{t-2}\mathcal{C}_U$, not the full unmeasured return $\mathcal{D}\mathcal{T}_U^t\mathcal{D}$.

Exact finite support. The complete first-return amplitudes for A are

$$F_1(A) = J_n \otimes J_p, \quad F_2(A) = (S - J_n) \otimes J_p,$$
$$F_p(A) = J_n \otimes \Pi_p, \quad F_{2p}(A) = (S^p - J_n) \otimes \Pi_p,$$

with every other positive-time coefficient zero. Indeed $R_m = \mathcal{D}\mathcal{T}_A^m\mathcal{D}$ equals $S^{m/2}$ in the first factor for even m, and J_n for odd m; the second factor is I_p if $p \mid m$, and J_p otherwise. This follows from the core periodicity and the same nonzero-coefficient Gauss identity as TRACE-081. The exact recurrence

$$R_m = \sum_{t=1}^{m} F_t R_{m-t}, \qquad R_0 = I,$$

uniquely determines F_t. In the four orthogonal diagonal sectors given by constants or their complements in each tensor factor, the first nonzero return times are respectively $1, 2, p, 2p$. Later returns are powers of the respective isometry. Substitution proves both the four displayed coefficients and all the zero ones, including times beyond a period.

Large observed pulse and small tail. Let D_0 be centered diagonal space, of dimension $N - 1$, and $W = \mathrm{Ran}(\Pi_n \otimes \Pi_p)$, of dimension $d_W = (n-1)(p-1)$. Write $\delta = 4p\epsilon$. The estimate $\|\mathcal{T}_U - \mathcal{T}_A\|_{\mathrm{HS}\to\mathrm{HS}} \leq 2\epsilon$ and contraction telescoping give

$$\|F_t(U) - F_t(A)\|_{2\to 2} \leq 2t\epsilon.$$

Since $F_{2p}(A)|_W = S^p \otimes I$ is an isometry,

$$\|F_{2p}(U)x\|_2 \geq (1-\delta)\|x\|_2 \quad (x \in W), \qquad 1 \geq \frac{\|F_{2p}(U)|_{D_0}\|_{\mathrm{HS}}^2}{N-1} \geq \frac{d_W}{N-1}(1-\delta)^2 \longrightarrow 1.$$

The finite-dimensional min-max principle also gives at least d_W singular values at least $1 - \delta$.

This large pulse coexists with the all-source bound

$$\sum_{t>2p} \|F_t(U)x\|_2^2 \leq \delta^2\|x\|_2^2, \qquad \frac{1}{N-1}\sum_{t>2p} \|F_t(U)|_{D_0}\|_{\mathrm{HS}}^2 \leq \delta^2 \leq 16N^{-6}.$$

For completeness set $Y_s(U) = (\mathcal{Q}\mathcal{T}_U)^s\mathcal{D}$. Unitarity and orthogonal projection give

$$\|Y_{t-1}(U)x\|_{\mathrm{HS}}^2 - \|Y_t(U)x\|_{\mathrm{HS}}^2 = \|F_t(U)x\|_2^2.$$

The four $F_t(A)^*F_t(A)$ are the four sector projections summing to I, so $Y_{2p}(A) = 0$. Telescoping yields $\|Y_{2p}(U)\| \leq \delta$. Finite tail sums are bounded by that survival energy, and the increasing limit proves the infinite bound. No weighted-time bound or completeness theorem is imported.

Cyclic compositions and the time scale. For $p \neq 3$, let $M_A = F_{2p}(A)|_{D_0}$ and $M_U = F_{2p}(U)|_{D_0}$. The three-cycle permutation H^2 and its square each fix precisely one coordinate, giving $\operatorname{Tr} S = \operatorname{Tr} S^2 = 1$. Thus for every positive integer q,

$$M_A^q = (S^{pq} - J_n) \otimes \Pi_p, \qquad \left| \frac{\operatorname{Tr}_{D_0} M_U^q}{N-1} - \frac{d_W}{N-1} \mathbf{1}_{3|q} \right| \leq q\delta.$$

For fixed q, normalized traces converge to $\mathbf{1}_{3|q}$: the first two vanish while the cubic one tends to one. These are unconjugated powers; $\operatorname{Tr}(M_U M_U^*)/(N-1) \to 1$, rather than zero. Nor is M_U^q a later first-return coefficient: $F_{6p}(A) = 0$, while $R_{6p}(A)|_W = M_A^3|_W = I_W$. Completed short excursions can therefore recycle coherence despite a negligible monitored tail.

Taking any odd primes $p_k \geq 4^{k^2}$, with $t_k = 2p_k$, gives

$$\frac{t_k}{N_k} = \frac{2}{4^k} = 2\|B(U_k)\|^2 \longrightarrow 0, \qquad \frac{t_k}{N_k^\alpha} = \frac{2p_k^{1-\alpha}}{4^{\alpha k}} \longrightarrow \infty \quad (0 < \alpha < 1 \text{ fixed}).$$

Unboundedness of primes suffices. Exact flat intensities, vanishing coordinate reversal, simple spectrum and macroscopic uniformity cannot therefore supply a uniform norm-decay shortcut for these observed coefficients. The result concerns the abstract family, not the fixed smooth chaotic map, its specific physical cavity source, a conjugated covariance theorem, a phase-averaged ensemble or BGS universality. The proof has independent internal reconstruction; finite exact replays and unconfirmed compiler/build executions have separate scope.

95 Full spectral functionals and the 15 October research record

Status and provenance. This section summarizes the canonical proofs integrated on 16 October 2026. The preserved record contains 103 numbered sections, 16 original notes and 90 historical claim entries. That inventory is not a new certification of 90 theorems. The integration consists of the author's own check and fresh finite checks, without an external referee or a new analytic Lean certificate. The unchanged model is $U_N = \exp(i10^{-80} N A_N) M_N$, for odd N. The model-specific signed hierarchy TRACE-005 and microscopic BGS remain open.

95.1 From the complete signed hierarchy to a local process

For a nonnegative band-limited Schwartz function b with $\sup b < 1$, let $M_{k,N}(b)$ be the origin-averaged elementary factorial coefficient of $b_N(t) = \sum_{\ell \in \mathbb{Z}} b(N(t+\ell))$, and put $\Delta_{k,N} = M_{k,N}^U - M_{k,N}^{\mathrm{CUE}}$. TRACE-086 proves the implication

$$\Delta_{k,N}(b) \longrightarrow 0 \quad \text{for every fixed } k \text{ and every such } b \quad \Longrightarrow \quad \Xi_N \Rightarrow \mathrm{Sine}_2.$$

It also supplies uniform integrability and convergence of every fixed local count moment. No separate UI premise is needed under this complete all-order hypothesis. For $h = 1 - e^{-f}$, $f \geq 0$ smooth and compactly supported, positive band-limited approximants b give the finite certificate

$$|L_N^U(f) - L_N^{\mathrm{CUE}}(f)| \leq 2\|h - b\|_1 + \sum_{k=2}^{J} |\Delta_{k,N}(b)| + \frac{(\int b)^J}{J!}, \qquad J \text{ odd}.$$

First choose the approximation and finite J, then take N to infinity. The actual signed coefficients are not estimated by this implication.

95.2 Parameter bandwidth and the missing averaged input

For $U_N(t) = e^{itNA_N} M_N$, $\|A_N\| \le C$, and a fixed real $w \in C_c^\infty((0,B))$, define

$$J_{N,w}(t) = \frac{1}{N^2} \sum_{m \ge 1} w(m/N)(|\operatorname{Tr} U_N(t)^m|^2 - \min(m,N)).$$

TRACE-087 gives exact angular parameter bandwidth at most $2CBN^2$ and global bound $O_w(N)$. A reproducing-kernel argument yields, for $K \Subset I$, $q > 1$ and every fixed $A > 0$,

$$\sup_K |J_{N,w}| \le C_{I,K,w,q} N^{2/q} \|J_{N,w}\|_{L^q(I)} + O_{I,K,w,A}(N^{1-2A}).$$

Thus a physical sixth moment $\int_I |J_{N,w}|^6 = O(N^{-3})$ would imply $O(N^{-1/6})$ fixed-coupling decay. It is unproved. The corresponding Haar-CUE moment is only a reference statement. The exact CUE covariance (CUE-002), valid for all positive m, n, is

$$\operatorname{Cov}(|p_m|^2, |p_n|^2) = \delta_{mn} \min(m,N)^2 + 2(N - \max(m,n))_+ - (N - m - n)_+ - (N - |m-n|)_+.$$

Gaussian trace moment identities are used only in their weighted-degree stable range.

95.3 Equal-edge multiplicities and their limited cancellation

For a rooted closed word let ν_{ab} be its directed edge counts, $r_a = \sum_b \nu_{ab}$, and $t(\nu)$ the directed tree cofactor on the active support. TRACE-088 proves the rooted multiplicity

$$c(\nu) = \frac{m\, t(\nu) \prod_{a: r_a > 0} (r_a - 1)!}{\prod_{a,b} \nu_{ab}!}.$$

For $P_{ab} = |U_{ab}|^2$, set $\mathcal{B}_{N,m} = \sum_\nu c(\nu)^2 \prod_{a,b} P_{ab}^{\nu_{ab}}$. Assume $P_{ab} \le C/N$ and $\|P - \mathbf{J}\|_{2\to 2} \le \rho < 1$, with fixed C, ρ and $\mathbf{J}_{ab} = 1/N$. At $m/N \to \xi > 0$,

$$\liminf \frac{1}{N} \log \mathcal{B}_{N,m} \ge \log 2\, [\xi - 2 + (\xi + 2) e^{-\xi}] > 0.$$

Unitarity forces the complementary contribution to equal $-\mathcal{B}_{N,m} + O(N^2)$, which does not identify a CUE limit. At the different scale $m \to \infty$, $m/\sqrt{N} \to c < \infty$, TRACE-089 gives

$$\frac{\mathcal{B}_{N,m}}{m} \longrightarrow \frac{\sinh(c^2/2)}{c^2/2},$$

with value one at $c = 0$. Its proof requires factorially weighted control of triple visits and the classical Harer–Zagier chord identity. Substituting $c = \sqrt{N}$ is outside the theorem. The later reported Gaussian comparison at $m = o(N^{3/4})$ lacks an available source proof and is not promoted here.

95.4 Contractive ratios and exact positive Hardy remainders

Let U be unitary of size N, $p(z) = \det(I - zU)$, $0 < s < t$, $a = e^{-s/N}$, $b = e^{-t/N}$, $x = e^{-s}$, $y = e^{-t}$. TRACE-090 uses the full bounded observable

$$\mathcal{F}_{N;s,t}(\alpha) = e^{-(t-s)} \frac{|p(ae^{-2\pi i\alpha})|^2}{|p(be^{-2\pi i\alpha})|^2} \in (0,1].$$

All mixed shifted moments characterize the weak local process, but do not alone supply its unbounded correlation moments. The CUE mean is exactly

$$e^{-(t-s)} \left[1 + \frac{(a-b)^2}{1-b^2} \frac{1-a^{2N}}{1-a^2} \right].$$

The sampled Toeplitz contraction argument requires an angular grid of at least N points; it is not valid under arbitrary undersampling.

Write $p(z) = \sum c_k z^k$, $q(z) = p(bz)$, $\mathscr{B}(z) = z^N \overline{q(1/\bar{z})}/q(z)$, and

$$u = \frac{1-xy}{1-y^2}, \qquad v = \frac{x-y}{1-y^2}, \qquad E(z) = \frac{p(az)}{p(bz)} - u - c_N v \mathscr{B}(z).$$

TRACE-091 proves the exact orthogonal decomposition

$$Z_U(s,t) = e^{-(t-s)} \left[1 + \frac{(x-y)^2}{1-y^2} + \|E\|_{H^2}^2 \right].$$

The minimum is attained exactly by a rotated regular N-clock. All roots of p are required to lie on the unit circle; self-inversiveness alone does not suffice. For $r = e^{-s/N}$, $q_r = p(rz)$ and $\mathscr{B}_r = q_r^*/q_r$, put

$$Y(z) = \frac{z}{N} \frac{q_r'(z)}{q_r(z)} + \frac{e^{-2s}}{1-e^{-2s}} - \frac{c_N e^{-s}}{1-e^{-2s}} \mathscr{B}_r(z).$$

Then the complete Abel energy satisfies

$$L_N(s) = \frac{1}{N^2} \sum_{m \geq 1} e^{-2sm/N} |\operatorname{Tr} U^m|^2 = \frac{1}{e^{2s}-1} + \|Y\|_{H^2}^2.$$

The required CUE excess is positive:

$$\|Y\|_{H^2}^2 \longrightarrow \frac{1-e^{-2s}}{4s^2} - \frac{1}{e^{2s}-1}.$$

This limit is unproved for the frozen family. Deleted-clock spectra have strictly positive finite excess of order N^{-1} and bounded fixed traces, so neither observation determines the desired full Hardy norm.

95.5 Cyclic cohomology, root labels, and the unchanged gap

FM-032 evaluates the additive defect for primes $p > 99$: $\inf_{B=B^*} \|A_p - (B - M_p B M_p^{-1})\|_p = \|PA_p\|_p$, where $P = d^{-1} \sum_{j=0}^{d-1} \mathrm{Ad}(M_p)^j$. For $\Sigma_p = d\|PA_p\|_p^2$, the principal harmonic contributes $81/200$ at odd d and $81/100$ at even d, with an explicit bounded nonnegative remainder. This is initial spectral response, not a nonperturbative physical trace estimate. FM-033's stronger all-label Weyl overlap is separately conditional on its named quantitative analytic inputs.

TRACE-084 is a sufficient mesoscopic number-variance route for deleting a chosen resonant sector. TRACE-085 proves it is not necessary: the root lift with phases $(a + \varphi_j)/d$, $0 \leq a < d$, has exactly the same uniformly centered unfolded process as its base spectrum. Its full positive trace energy can lie at times divisible by d. Full signed discrepancies can therefore cancel between sectors. A powered block loses root labels; $U = ZR_d(U^d)$ requires $Z^d = I$ and $[Z, U^d] = 0$ as joint data. No new estimate of those physical data, the signed Heisenberg hierarchy, or a microscopic BGS law is asserted.

96 Abel rigidity and the flat equal-edge Gaussian comparison

Scope. Claims `BGS-TRACE-092` and `BGS-TRACE-093`. The canonical full proofs are `mathematics/abel_clock_rigidity_20261016.md` and `mathematics/gaussian_equal_edge_20261016.md`. This section records their statements and proof mechanisms. They do not establish the signed Heisenberg hierarchy of the maintained fixed propagator. The Gaussian benchmark and its double-scaling function are classical [53, 54].

96.1 A positive decomposition with a geometric remainder

For eigenvalues $e^{2\pi i\theta_j}$ of $U \in U(N)$ put $x_j = N\theta_j$, $T_m = \operatorname{Tr} U^m$ and

$$L_N(s) = N^{-2} \sum_{m\ge 1} e^{-2sm/N}|T_m|^2, \qquad \delta_N(s) = L_N(s) - (e^{2s}-1)^{-1}.$$

For $s > 0$ define

$$f_s(x) = \frac{s}{s^2+\pi^2x^2}, \qquad q_s(x) = \frac{s^3}{\sinh^2 s}\frac{(\sin(\pi x)/(\pi x))^2}{s^2+\pi^2x^2}, \qquad H_s = f_s - q_s.$$

Continuous values are used at zero. With Fourier convention $\widehat{g}(t) = \int g(x)e^{-2\pi itx}\,dx$,

$$\widehat{H}_s(t) = \begin{cases} [\sinh(2s(1-|t|)) - 2s(1-|t|)]/(2\sinh^2 s), & |t| \le 1, \\ 0, & |t| > 1. \end{cases}$$

Theorem 96.1 (Exact positive Abel decomposition)**.** *For every $N \ge 1$, including repeated eigenvalues,*

$$\delta_N(s) = N^{-2}\sum_{m=1}^{N-1} \widehat{H}_s(m/N)|T_m|^2 + \frac{1}{2N}\sum_{j\ne k}\sum_{\ell\in\mathbb{Z}} q_s(x_j - x_k + \ell N). \tag{96.1}$$

Both terms are nonnegative. Equality holds precisely for a rotated clock.

Proof. The transform of q_s is $(s^2/\sinh^2 s)$ times the convolution of $(1-|t|)_+$ and $e^{-2s|t|}$. Elementary integration gives the displayed transform of H_s. Periodize all kernels at the integer period N. The Fourier series are absolutely convergent, and the H_s series has finite support. The diagonal of the q_s sum contributes exactly $q_s(0)/2$, since $q_s(\ell N) = 0$ for nonzero integers ℓ. Finally $(\widehat{H}_s(0) + q_s(0) - 1)/2 = (e^{2s}-1)^{-1}$. □

96.2 An optimal square-root stability theorem

Let d_R be the bounded local semimetric obtained by taking the supremum of $|\int \phi\, d(\mu-\nu)|$ over functions supported on $[-R,R]$ with supremum norm and Lipschitz constant at most one, and then truncating at one.

Theorem 96.2 (Stationary clock rigidity)**.** *Let Ξ be a stationary counting process on $\mathbb{R}$ with intensity one, allowing multiplicities and an empty component. Suppose its ordered q_s pair-energy density is $e < \infty$. For every fixed R,*

$$W_{d_R}(\mathcal{L}(\Xi), \mathcal{L}(\mathbb{Z}+V)) \le \min\{1, C_{s,R}\sqrt{e}\}, \qquad V \sim \mathrm{Unif}[0,1).$$

For the uniform-origin unfolded spectrum, $e \le 2\delta_N(s)$, so $W_{d_R} \le \min\{1, C'_{s,R}\sqrt{\delta_N(s)}\}$ uniformly in N. The exponent $1/2$ is optimal already at $R = 2$ over these spectra.

Proof. Let ν be the intensity-normalized outgoing-gap law, retaining zero gaps between coincident particles. Coverage of the line by the positive gaps gives $\int g\,d\nu = 1 - p_0$, where p_0 is the empty-component probability. Consequently

$$A := \int (1-g)_+\,d\nu = \Pr\{\Xi([0,1)) = 0\}, \qquad \int |g-1|\,d\nu \le 2A.$$

The chord bound $q_s(g) \ge c_s(1-g)^2$ on $[0,1]$, where $c_s = 4s^3/[\pi^2\sinh^2 s(s^2+\pi^2)]$, yields $A \le \sqrt{e/c_s}$. On a fixed enlarged window, the expected sum of incoming and outgoing gap deviations is $O_R(A)$. Unless that sum exceeds $1/4$ or the anchor unit interval is empty, consecutive-index matching

with the anchor clock has total local displacement $O_R(A)$ in expectation. The boundary gaps are included in the marks. Translating the matched configurations by an independent uniform variable preserves the stationary process law and makes the clock phase exactly uniform. This proves the estimate without a separate count-integrability premise.

For optimality take even N and $x_j = j + \epsilon(-1)^j$. Writing $z = e^{-s}$,

$$\delta_N(s) = \frac{z(1-z)(1-\cos(2\pi\epsilon))}{2(1+z)(1+2z\cos(2\pi\epsilon)+z^2)} \asymp_s \epsilon^2.$$

Every translate of this configuration is at d_2 distance at least ϵ from every clock: two consecutive points in $[-1,1)$ have fractional separation 2ϵ, detected by the nonnegative admissible test $\min\{\mathrm{dist}(x,\mathbb{Z}+t),(2-|x|)_+,1\}$. □

The CUE excess is positive. The limiting mean low-mode part of (96.1) is

$$I_s = \frac{\sinh(2s) - 2s - 4s^3/3}{8s^2\sinh^2 s},$$

and the geometric part tends to $\frac{1}{2}\int q_s(x)[1-(\sin(\pi x)/(\pi x))^2],dx$. Neither limit is established here for the physical fixed map. Vanishing excess would instead force its clock law.

96.3 All sublinear powers and the Heisenberg exponential rate

Let $c(\nu)$ count rooted cyclic words with directed-edge table ν, and put

$$B_{N,m} = N^{-m}\sum_{\nu} c(\nu)^2, \qquad G_{N,m} = N^{-m}\sum_{\nu} c(\nu)^2 \prod_{a,b} \nu_{ab}!.$$

For an exactly flat unitary, $B_{N,m}$ is its positive equal-edge sector. The classical Gaussian benchmark is

$$G_{N,m} = \frac{\prod_{j=0}^{m}(N+j) - \prod_{j=0}^{m}(N-j)}{(m+1)N^m},$$

with falling product zero for $m \geq N$ [53].

Theorem 96.3 (Flat equal-edge comparison and collision mean)**.** *For $N \geq 3$ and $3 \leq m \leq N$,*

$$0 \leq 1 - B_{N,m}/G_{N,m} \leq \frac{m(m-1)}{N(N+1)} + \frac{15m}{N^2}.$$

At $m = 1$ the deficit is zero and at $m = 2$ it is $1/(2N)$. Thus $B_{N,m}/G_{N,m} \to 1$ for every $m = o(N)$. For $m/N \to \alpha > 0$,

$$\liminf B_{N,m}/G_{N,m} \geq 2^{-2\alpha^2}, \qquad \frac{\log B_{N,m}}{N} \longrightarrow (1+\alpha)\log(1+\alpha) - \alpha.$$

Under the exact Wick table law proportional to $c(\nu)^2\prod \nu_{ab}!$,

$$\mathbb{E}_W \sum_{a,b} \binom{\nu_{ab}}{2} \longrightarrow 2\alpha^2, \qquad \Pr_W\{\max_{a,b}\nu_{ab} \geq 3\} \longrightarrow 0.$$

For $m \geq N+2$ the finite mean is exactly

$$F(N,m) = \frac{2m\{(N+5)m(m+1) - (N+2)(N+3)\}}{(N+1)(N+2)(N+3)(m+2)}.$$

On every fixed positive compact range of m/N, the error from this rational expression is exponentially small in N. With $\alpha_N = m/N$ the mean is $2\alpha_N^2 - 2(\alpha_N^2+\alpha_N+1)/N + O(N^{-2})$.

Proof. For unscaled Ginibre Y, put $p_m = \operatorname{Tr} Y^m$ and $h_m = \mathbb{E}|p_m|^2$. Gaussian Fock adjointness makes the repeated-edge mean a self-coordinate Hessian norm divided by $2h_m$; the total Hessian norm is $m(m-1)h_m$. Two sphere fourth moments control off-diagonal coordinates. The 24 diagonal eighth-moment pairings reduce to four explicit contractions. The essential contracted matrix satisfies

$$\|S\|_{L^2}^2 = \langle p_m, \operatorname{Tr}(Y^2 D^2) p_m \rangle \le m(N+m-1)h_m.$$

Here $D_{ij} = \partial/\partial Y_{ji}$. The bound follows from hook Schur orthogonality and the finite identity $\sum_{a=1}^{m-1} p_a p_{m-a} = \sum_j (-1)^j (m-1-2j) s_{(m-j,1^j)}$. The exact partial upper-index swap of the Hessian gives the collision mean limit. Sixth sphere moments make triple edges negligible. Keeping the exact 24 and six pairing sums yields the finite hook formula; stars-and-bars sums of the untruncated hook weights give $F(N,m)$. The discarded hook tail is exponentially small at m/N in a positive compact interval. Jensen then bounds B/G away from zero, transferring the classical Gaussian exponential rate. □

The following continuation, `BGS-TRACE-094`, closes the limiting collision distribution and exact flat ratio. Its high-derivative argument is an additional input beyond the first-moment calculation above. For a flat unitary, unitarity forces the remaining signed sector to equal $-B_{N,m} + O(N^2)$. Obtaining its $O(N)$ CUE remainder, extending the sharp flat comparison to the actual non-flat profile, and proving the full every-order signed hierarchy remain distinct unresolved obligations.

96.4 Complete Wick collision law at Heisenberg powers

The canonical proof is `mathematics/wick_collision_poisson_20261016.md`. Let $\mathbb{W}(\nu) = c(\nu)^2 \prod_e \nu_e!/h_m$ and put $C_N = \sum_e \binom{\nu_e}{2}$ and $D_N = \#\{e : \nu_e \ge 2\}$. Directed loops are included.

Theorem 96.4 (Complete flat collision comparison)**.** *For every fixed $0 < a < b < \infty$ and positive integer r, uniformly for $aN \le m \le bN$,*

$$\mathbb{E}_{\mathbb{W}}(D_N)_r = [2(m/N)^2]^r + O_{a,b,r}(N^{-1}), \qquad \Pr_{\mathbb{W}}\{\max_e \nu_e \ge 3\} = O_{a,b}(N^{-1}).$$

Consequently, if $m/N \to \alpha > 0$,

$$C_N \implies \text{Poisson}(2\alpha^2), \qquad \frac{B_{N,m}}{G_{N,m}} \longrightarrow e^{-\alpha^2}.$$

This is a collision law of an auxiliary positive measure, not an energy-level spacing law.

Proof. For the kth derivative tensor $H^{(k)} = (\partial_{a_1 b_1} \cdots \partial_{a_k b_k} p_m)$, every partial matching P of upper and lower indices obeys

$$\|C_P H^{(k)}\|_2^2 \le C_k m(N+m)^{k-1} h_m \le C_{a,b,k} (m)_k h_m. \tag{96.2}$$

Here k is fixed and N is sufficiently large. To prove the first bound, expand the marked trace into $(k-1)!$ cyclic orders and compositions $r_1 + \cdots + r_k = m - k$. Move all k derivatives in one factor of the squared norm to creation variables by Fock adjointness. The resulting index network consists of k unit creation edges and k segments of lengths r_i. Partial matches and the remaining cross-copy links form directed cycles. Each zero-length cycle consumes a distinct zero r_i. If exactly q gaps are zero there are $\binom{k}{q}\binom{m-k-1}{k-q-1}$ compositions and at most q factors N from zero traces. A product of $\ell \le 2k$ positive traces has overlap with p_m at most $2^\ell h_m$: its hook coefficients are coefficients of $\prod_v (1-t^v)/(1-t)$ and hence have modulus at most 2^ℓ. Summing over $q \le k-1$ proves (96.2).

Conjugation invariance now permits the classical finite Haar formula [55]. At degree $2k$ its Gram matrix is $G_{\sigma\tau} = N^{\#\,\text{cycles}(\sigma^{-1}\tau)}$. For fixed k, $G^{-1} = N^{-2k}(I + O_k(N^{-1}))$ entrywise. Every

input pairing is the inner product of two partial contractions in (96.2). Therefore, uniformly over all coordinate choices, including coincidences,

$$\frac{\|\partial_{e_1}\cdots\partial_{e_k}p_m\|_2^2}{h_m}\le C_{a,b,k}N^{-k}.$$

If the prescribed upper and lower label sets are disjoint, the leading pairings are fully cross. Upper multiplicities A_u and lower multiplicities B_v give $\prod_u A_u!\prod_v B_v!$ such pairings. For $k\ge 2$, mixed Fock homogeneity transfers the Hessian swap identity exactly:

$$K_k=(m-2)_{k-2}K_2,\qquad K_k/\|H^{(k)}\|_2^2=1-2\bar{j}/(m-1),\qquad \bar{j}\le (N-1)/m.$$

A bounded number of upper transpositions makes each leading overlap $1+O_{a,b,k}(N^{-1})$ times the total energy. Thus the coordinate norm equals $(m)_kN^{-2k}[\prod A_u!\prod B_v!+O_{a,b,k}(N^{-1})]h_m$.

The exact coefficient identity identifies these derivative norms with $\mathbb{E}_{\mathbb{W}}\prod_e(\nu_e)_{k_e}$. For r distinct edges each used twice, with all $2r$ endpoints distinct, the multiplicity is 4^r and the number of tuples is $(N)_{2r}$. Dividing by 2^r gives the asserted leading factorial moment. All other tuples number $O_r(N^{2r-1})$ and the uniform coordinate bound makes their total $O(N^{-1})$. Replacing binomial counts by indicators costs a sum of derivative order $2r+1$, again $O(N^{-1})$. Order three bounds the probability of any triple edge by $O(N^{-1})$. Finite Bonferroni bounds, with truncation removed only after the limit in N, give $\mathbb{E}(1-q)^{D_N}\to e^{-2\alpha^2q}$ for $0\le q\le 1$. Finally $C_N=D_N$ and $\prod\nu_e!=2^{C_N}$ outside the triple-edge event. These bounded transforms prove both conclusions. □

The same argument gives independent Poisson counts in arbitrary disjoint deterministic edge classes of limiting densities β_j, with means $2\alpha^2\beta_j$. For fixed distinct vertices, the outgoing counts $n_a=\sum_b\nu_{ab}$ instead have independent geometric limits of mean α: their mixed factorial moments tend to $\prod_i k_i!\alpha^{\sum_i k_i}$. Sum the coordinate derivative formula over the lower labels; all distinct lower labels outside the fixed upper set give this main term, and the remaining $O(N^{K-1})$ choices are controlled uniformly. Exchangeability also gives convergence in probability of the empirical vertex-occupation distribution. Incoming and outgoing counts of one vertex remain equal exactly.

As a separate corollary, let $Z_{ab}=\xi_{ab}/\sqrt{N}$ be iid circular complex entries with $\mathbb{E}|\xi|^2=1$ and $\mathbb{E}|\xi|^{2j}\le j!$ for every j. With $\kappa=\mathbb{E}|\xi|^4$,

$$\frac{\mathbb{E}|\operatorname{Tr}Z^m|^2}{G_{N,m}}\longrightarrow e^{(\kappa-2)\alpha^2}.$$

Indeed the exact ratio is $\mathbb{E}_{\mathbb{W}}\prod_e q_{\nu_e}$, $q_j=\mathbb{E}|\xi|^{2j}/j!\in[0,1]$. Away from triple edges this is $(\kappa/2)^{C_N}$. The bounded transform limit applies. This proof does not cover $\kappa>2$ or the non-flat deterministic physical profile.

97 Circular-phase trace schemes and the remaining signed boundary

The 16–19 October continuation concerns auxiliary phase ensembles, not the fixed-coupling BGS conclusion. The maintained sources and exact proof revisions are indexed in `mathematics/phase_ensemble_continuation_20261019.md` and `archive/record/chat_integration_20261019.md`. The new statements have author proofs, checked by the author, with exact finite computations where specified; independent review and priority are not established.

For an odd prime p, a nondegenerate alternating form Ω and $N=p^{2d}$, put

$$S_{xy}=N^{-1/2}\exp(2\pi i\Omega(x,y)/p),\qquad U=DS,\qquad V_m=\mathbb{E}|\operatorname{Tr}U^m|^2,$$

where D has independent uniform phases. Then $S = S^*$ and $S^2 = I$. For every fixed number of trace powers the joint law is

$$(\operatorname{Tr} U, \operatorname{Tr} U^2, \dots, \operatorname{Tr} U^M) \implies (Z, Z^2, \sqrt{3}Z_3, \dots, \sqrt{M}Z_M).$$

The Gaussian variables displayed separately are independent. The exact identity $\operatorname{Tr} U^2 = (\operatorname{Tr} U)^2$ prevents identifying the first two global traces with independent CUE traces. This does not by itself decide the local microscopic law. The polynomial trace-window results likewise do not reach microscopic absorption.

97.1 Statements of TRACE-095–097 and TRACE-099

The following statements (canonical source: `mathematics/phase_ensemble_continuation_20261019.md`) are recorded here because this section is their registered manuscript source.

TRACE-095 (conditional). Under the pinned feedback trajectory, covariance and moment inputs and the reversal hypothesis of the source, the Gaussian limit of every fixed selected block extends to all $u_N \to \infty$ with $u_N/N \to 0$, with error $O((u_N/N)^{1/4} + u_N^{-1/8})$ for the stated smooth bounded tests of disjoint entries. The substitution $u = 2$ is not permitted.

TRACE-096. For a deterministic unitary S with independent row phases, spectral averaging identifies the law of the open-port vector X_i with the eigenvector-ratio law weighted by $|\psi_l(i)|^2$. Closing one port of a two-port cavity with transmission T by a fresh uniform phase gives $J = T/|1 - \sqrt{1-T}\, e^{i\phi}|^2$ and $Y = J/(1+J)$; the resulting probability kernel is weakly continuous and injective, so the transmission laws converge to Uniform$[0,1]$ if and only if the corresponding Y laws converge to Beta$(1,2)$. This is an equivalence criterion; it does not prove either convergence.

TRACE-097 (conditional). Conditional on the fixed-level TV compactness input TC of the source (the full G or rank condition R stated there), the complete normalized positive path-clock current is controlled at arbitrary cutoffs $K_N \to \infty$. The statement evaluates neither the finite-channel transmission law nor the complete signed trace sum, and TC is not discharged in this section.

TRACE-099. For the symplectic seeds the signed mixed-moment comparison reaches trace powers $o(N^{1/5})$; for generic fixed bilinear seeds of dimension b it reaches $o(N^{1/(6b)})$, with the actual Gauss phases retained. The associated disk laws concern fixed compact subsets of $|z| < 1$; they do not reach $z = e^{-s/N}$ or microscopic absorption.

97.2 Exact schemes and a joint quartic limit

The signed phase expansion assigns order $j = c + \frac{1}{2}\operatorname{rank}(H_\Pi^{\mathsf{T}} K H_\Pi)$ and defect $\delta = 2j - e$ to an open connected cluster with e common-edge ports. Connected clusters satisfy $j \le \delta + 1$. Exact neutral peeling leaves primitive order at most 2δ and size at most 16δ. This controls each fixed defect without summing all connected types.

The saturated-support classification through rank six bounds the connected four-port core by size sixteen. Exact enumeration gives a factor $1 - z$ in each of its twenty-four open channels. The physical quotient at $z = 1$ is $-191/12$. Together with the earlier primitives this completes the total-defect-two sector for $0 < \tau < 1$, where $m/N \to \tau$. For fixed $h \ge 3$, the stable coefficient has degree at most $2h - 3$ when $m > 8h$; its vanishing is not asserted.

There is also a different joint family: all neutral additions and arbitrary numbers of connected clusters of individual defects one or two. Its correction $\mathcal{R}_{\le 2}$ is normalized to V_m/m. The gluing uses Stanley [56], Lemmas 2.1–2.2 and Theorem 3.1; the project supplies the port weights, cumulants and uniform majorant. Define

$$v(x) = \frac{7}{12} + \frac{5}{3}x, \quad q_3(x) = 2 + \frac{3}{2}x + \frac{1}{2}x^2, \quad q_4(x) = 6 + \frac{25}{2}x + \frac{19}{2}x^2 + 2x^3.$$

For each $x \in [0,1]$, let w_x be the unique positive solution of

$$\tau = w_x + w_x^2 + \frac{q_3(x)}{2} w_x^3 + \frac{q_4(x)}{6} w_x^4, \qquad D_x = 1 + 2w_x + \frac{3}{2} q_3(x) w_x^2 + \frac{2}{3} q_4(x) w_x^3.$$

For $0 < \tau < 1/9$ the exact five-index sum has limit

$$L_{\le 2}(\tau) = \int_0^1 \left[\exp\{w_x + v(x) w_x^2\} \frac{\tau}{w_x D_x} - 1 \right] dx.$$

One positive majorant controls all five indices, with tail $11(9\tau_0)^{H+1}$ on $m/N \le \tau_0 < 1/9$ and a uniform $O_{\tau_0}(N^{-1})$ finite-size error for $9 \le m \le \tau_0 N$. The result includes the earlier overlapping cluster families and must not be added to them again. Its expansion begins

$$L_{\le 2}(\tau) = -3\tau^3 + \frac{20893}{864} \tau^4 + O(\tau^5).$$

The omitted individual defects remain essential. The connected $(j, \delta, e) = (3, 3, 3)$ source is known only through a finite size-nine prefix in the latest check. Its full value is not assigned by fitting the desired CUE answer.

97.3 What exact Cauchy averaging does not identify

For any Hermitian involution the local iid-phase problem admits the quantitative Cayley comparison with $S - \mathrm{diag}(\mathrm{Cauchy})$ stated in the maintained source. Universality of this particular dense deterministic hopping model is still required.

An exact finite-dimensional consequence of bounded holomorphic averaging is useful for separating that requirement. If $B = B^*$, E is real, T is iid standard Cauchy diagonal and $A \ge 0$ is nonzero, then

$$Q_A = \mathrm{Tr}(A(B - T - EI)^{-1})$$

is Cauchy with complex parameter $\mathrm{Tr}(A(B - EI + iI)^{-1})$ in the lower half-plane. For $B = S = S^{-1}$ and $E = 0$, its centre and scale are $\mathrm{Tr}(AS)/2$ and $\mathrm{Tr}(A)/2$. The contour is applied to a bounded transform of Q_A, not to its undefined first moment. The general Cauchy/Herglotz principle is classical; see Aizenman–Warzel, arXiv:1312.7769v2.

Even all these positive projections do not determine a matrix law. For the two-dimensional exchange involution, compare

$$X = (S - \mathrm{diag}(t_1, t_2))^{-1}, \qquad Y = S/2 + \mathrm{diag}(c_1, c_2)/2,$$

with independent standard Cauchy variables. Every positive projection $\mathrm{Tr}(AX)$ and $\mathrm{Tr}(AY)$ agrees in law. For $A = \mathrm{diag}(1, -1)$ the signed tail constants are instead $4/\pi^2$ and $2/\pi$. More generally, for every non-diagonal Hermitian involution, including the exact Fourier seed, H^{-1} and $H/2$ have identical positive scalar projections but matrix laws at total variation distance one. A rank-two signed source separates a deterministic value from a probability-zero polynomial level set. This obstructs the proposed positive-source closure, not BGS. Finite-dimensional total-variation separation does not exclude coincident limits for particular spectral observables.

All higher connected types, the complete signed mixed hierarchy and the comparison with the unchanged frozen propagator remain open. The conditional feedback and TV-compactness routes retain their named inputs.

98 Uniform physical-kick decay and controlled polynomial replacement

This section records BGS-TRACE-106 and BGS-TRACE-107, with analytic author proofs and an internal check by the author on 21 October 2026. The complete maintained proofs are

`mathematics/physical_kick_trace_decay.md` and `mathematics/physical_trace_polynomial_approximation.md`. The imported research snapshot and the limited review coverage are identified in `record/spectral_research/spectral_master_audit.md`.

For the unchanged modular Weyl quantization A_N and odd N, put $E_N(t) = \exp(itNA_N)$ and $\mathrm{tr}_N = N^{-1}\,\mathrm{Tr}$. There is a finite constant C, independent of N, t, such that

$$|\,\mathrm{tr}_N E_N(t)| \le C(N|t|)^{-1/9}, \qquad 0 < |t| \le 10^{-20}, \quad N|t| \ge 1.$$

The prefactor has not been evaluated numerically. The original $t = 10^{-80}$ is included. This strengthens the imported exponent $1/25$. For every bounded interval $|t| \le T$ there is also a positive exponent c_T and a finite prefactor C_T.

The ordinary-half Weyl convention is $\mathsf{W}_N(k,l) = (-1)^{kl}T_N(l,k)$. The exact symbol in this convention is

$$a(q,p) = \cos 14\pi q + \frac{r(q)}{10} + \frac{r(q)\sin 2\pi q}{100} - \frac{1}{10} + \frac{r(q+1/2)\cos(2\pi p - \alpha_0)}{100}.$$

In particular, replacing the last coefficient by $r(q)$ would change the classical symbol and invalidate this proof for the original matrices.

Here is the quantitative localization mechanism. For the flow κ_t of $(a_p, -a_q)/(2\pi)$, set $b_1 = e^{2\pi iq}$, $b_2 = e^{2\pi ip}$, $d_j = (b_j - b_j \circ \kappa_t)/t$ and $v_t = \sum_j |d_j|^2$, extended smoothly at zero. Direct derivative estimates give, uniformly on the displayed small-time interval,

$$\mathrm{vol}\{v_t \le \delta\} \le \frac{7}{34\pi}\sqrt{\delta}.$$

Egorov for the two fixed characters has error $O_T(|t|N^{-2})$. Cyclicity then gives

$$|\,\mathrm{tr}_N(Q_N(ud_j)E_N(t))| \le C_T(N|t|)^{-1}|u|_1.$$

Choose a real smooth cutoff $\eta = \rho(v_t/\delta)$ and use its square. The symmetric Weyl product cancels its first-order term exactly:

$$\|Q_N(\eta)^2 - Q_N(\eta^2)\| \le CN^{-2}|\eta|_2^2, \qquad \mathrm{tr}_N\, Q_N(\eta)^2 \le \|\eta\|_{L^2}^2 + CN^{-2}|\eta|_2^2.$$

Since $Q_N(\eta)^2$ is positive, its pairing with a unitary is bounded by its trace. Fourier Cauchy–Schwarz in dimension two requires only C^3 for $|u|_1$ and C^4 for $|\eta|_2$. Writing $R = N|t|$, the resulting estimate is

$$|\,\mathrm{tr}_N E_N(t)| \le C_T\big(\delta^{1/2} + R^{-1}\delta^{-4} + R^{-2}\delta^{-8}\big).$$

The choice $\delta = R^{-2/9}$ proves the exponent $1/9$. For arbitrary finite T, the stationary circle $q = 0$ has nonzero nilpotent flow linearization. Thus no nonzero time has identity flow; analytic finite type on the compact time interval supplies a positive sublevel exponent and the stated bounded-time result.

Consequently, if $0 < |t_N| \le T$ and $N|t_N| \to \infty$, the empirical eigenvalue measure of $E_N(t_N)$ converges to circle Haar measure, and

$$\inf_{|z|=1} N^{-1/2}\|E_N(t_N) - zI\|_{\mathrm{HS}} \to \sqrt{2}, \qquad \inf_{|z|=1}\|E_N(t_N) - zI\|_{\mathrm{op}} \to 2.$$

In particular $\inf_{|z|=1}\|U_N - zM_N\|_{\mathrm{op}} \to 2$. This is a matrix comparison, not a comparison of unordered spectra after arbitrary conjugation.

There is also a controlled polynomial replacement at the actual argument $N\tau$. Since $\|A_N\| \le a_* = 59/50$, a Chebyshev–Bessel polynomial of degree $2a_*\tau N + O_K(\log N)$ approximates $E_N(\tau)$ with norm error N^{-K}. Telescoping trace products and the partition–Möbius formula give an error $O_{k,b}(N^{k-K})$ for each fixed factorial order k and fixed compact Fourier bandwidth.

Negative trace indices of the nonunitary approximant are defined by adjoint powers. A single degree $2a_*\tau N + O((\log N)^2)$ works for all fixed orders by choosing $K = \log N$. No positivity or point-process interpretation of the approximating trace polynomial is asserted.

On the Pell dimensions $N = xy$, $(1+\sqrt{2})^r = x + y\sqrt{2}$ with odd r, the first nonzero scalar two-copy alias has length exactly $x \asymp \sqrt{N}$. Hence the fixed-word two-copy approximation does not extend uniformly to the degrees needed by the physical exponential. The complete remaining signed trace sum is not evaluated. BGS-TRACE-005 and BGS-BGS-001 remain open.

99 The final spectral research corpus: one-step returns and weighted Pell curvature

This section records the final fourteen-round source edition integrated on 21 October 2026. Canonical proofs are the four maintained Markdown notes for TRACE-108–111; the complete individual author sources, internal reviews, reference programs and execution receipts are organized by mathematical topic under `mathematics/spectral_research/`. Source preservation is complete for the supplied master; mathematical adoption has the bounded scope below. No statement here proves the original microscopic hierarchy or BGS.

99.1 Original one-step coordinate return

Retain exactly the modular Weyl family and fixed $\tau = 10^{-80}$ of the preceding section. The ordinary-half symbol has the translated coefficient $r(q+1/2)$ in its momentum-dependent term. Four lifted Gauss sheets and their transported amplitudes give

$$\lim_{N\to\infty,\ N \text{ odd}} \sum_x |U_N(x,x)|^2 = 1 - \frac{25823\pi^4}{4800000000000}\tau^4 + E, \qquad |E| \le 10^{56}\tau^5.$$

This is TRACE-108, an analytic author proof, checked by the author. The first three Taylor coefficients vanish under the four-sheet filter. The fourth follows from an exact Fourier convolution; Cauchy bounds on the holomorphic Hamilton–Jacobi phase control the remainder at fixed coupling. Cross-sheet discrete aliases are included. This one-step result does not evaluate the Heisenberg-scale sum; its individual contribution there is divided by N^2.

99.2 Exact projections and subsequent trace closure

For any unitary U and Hermitian A, put $Q = \sum_\lambda P_\lambda A P_\lambda$, with full spectral projectors. Orthogonal projection on matrix Hilbert space gives

$$\inf_B \|A - (B - UBU^*)\|_{\mathrm{HS}} = \|Q\|_{\mathrm{HS}}, \qquad \|Q\|_{\mathrm{HS}} \ge |\operatorname{Tr}(AU)|/\sqrt{N}.$$

On two copies the projected generator is ad_Q and

$$\|\operatorname{ad}_Q\|_{\mathrm{HS}}^2 = 2N\|Q\|_{\mathrm{HS}}^2 - 2(\operatorname{Tr} Q)^2.$$

The 21 October integration retained two analytic input qualifications: the original inserted-trace asymptotic for TRACE-109 and the uniform power-trace bound for TRACE-110. Both are closed in Section 100. The first now has a finite-grid proof with $O(N^{-1})$ error; no uncontrolled remainder is differentiated. The second has one dimension threshold for all positive iterates. Consequently the global spectral distance allowing arbitrary conjugation is $O(1/\log N)$, with matching lower order on Pell dimensions. This does not determine microscopic spacings.

99.3 The full weighted sum at the initial parameter

Let $\phi(\xi) = e^{-1/(1-\xi^2)}\mathbf{1}_{|\xi|<1}$, $I = \int \phi$, and $w = (\phi * \phi)^2/(4I^4)$. Define

$$S_N(t) = N^{-2} \sum_{1\le m<2N} w(m/N)|\operatorname{Tr}(e^{itNA_N} M_N)^m|^2.$$

On odd Pell dimensions $N = (\lambda^r - \lambda^{-r})/(4\sqrt{2})$, $\lambda = 3 + 2\sqrt{2}$, $d = 2r$, TRACE-111 proves

$$S_N(0) = \frac{N}{d} I_w + O(1), \qquad S_N'(0) = O_L(N^{-L}), \qquad S_N''(0) \le -\frac{289 J_2}{80000}\frac{N^5}{d^2},$$

where $I_w = \int_0^2 w$ and $J_2 = \int_0^2 x^2 w(x)\,dx$. The proof sums all weighted times and every off-clock Hessian term in a fixed eigenblock basis, without differentiating eigenvalue branches. Positivity and a uniform third-derivative bound then force positive curvature at some $t_N = O(d^3/N^2)$. Thus initial concavity cannot persist to a fixed positive parameter for all sufficiently large Pell dimensions. This does not place a positive first derivative before that parameter.

The continuation

$$\mathcal{S}_N(z) = N^{-2} \sum_m w(m/N) \operatorname{Tr} U_N(z)^m \operatorname{Tr} U_N(z)^{-m}$$

is entire. The inverse power, rather than an adjoint at complex z, is essential. If $(d/N)\mathcal{S}_N$ is bounded by a fixed M_0 on a disk of radius R_N, Cauchy's inequality forces $R_N \le \sqrt{2M_0/c}\sqrt{d}/N^2$, where $c = 289J_2/80000$. This bounds a uniformly controlled window, not the radius of convergence.

99.4 The still missing plateau correction

The exact rotational class of cyclic histories supplies the linear ramp, under the stated intensity flatness and singular-gap inputs. Hence its complement must contribute

$$-\delta_*, \qquad \delta_* = \int_1^2 (t-1)w(t)\,dt > 0,$$

to the normalized weighted pair sum. An estimate making that entire complement vanish would miss the plateau. Arbitrarily accurate one-step replacement does not evaluate this coherent remainder. Neither the original pair limit at $t = \tau$ nor its higher factorial analogues is proved here.

The edition preserves 535 source units and exact reconstruction of the uploaded bytes. Fresh execution covers 56 bounded research checks and an additional independent finite Hessian comparison. These computations, formula parsing and the manuscript build have separate evidence scopes; none is a formal or external referee certification of the asymptotic claims.

100 Closure of the inserted trace and global spectral comparison

This continuation closes the two analytic input qualifications recorded on 21 October for TRACE-109 and TRACE-110. The canonical proofs are the finite-grid, displacement-density and uniform-trace notes in `mathematics/`, with the exact dependency review in `record/spectral_research/trace_closure.md`. These are project proofs with adversarial checking by the author, not external refereeing or formal certification. The original modular matrices and fixed $\tau = 10^{-80}$ are unchanged. The microscopic signed hierarchy remains open.

100.1 A finite-grid inserted trace with the required remainder

Put $h=(2\pi N)^{-1}$ and $t_*=\tau/(2\pi)$. The original cat matrix has four exact Gauss sheets with initial phase

$$F_0(Q,s)=-\frac{3Q^2-2Qs+3s^2}{8},\qquad M_N(x,y)=\frac{e^{3\pi iN/4}}{2\sqrt{N}}\sum_{j=0}^{3}e^{iF_0(x/N,y/N+j)/h}.$$

For the ordinary-half symbol a, solve $F_t=a(Q,F_Q)$. On a fixed analytic strip and a fixed short time interval the characteristic equation is a contraction. Its derivative bounds are uniform over the real plane. The exact conjugated finite Weyl generator has the expansion $L_h=L_0+hL_1+O(h^2)$, where

$$L_0b=a_p\partial_Qb+\tfrac12(a_{Qp}+a_{pp}F_{QQ})b.$$

Solve $\partial_tB_0=L_0B_0$, $B_0(0)=1$, and $\partial_tB_1=L_0B_1+L_1B_0$, $B_1(0)=0$. Then $B_0=2\sqrt{F_{Qs}}$ and B_0+hB_1 has residual $O(h^2)$. The exact deck identities

$$F(Q,s+4)=F(Q,s)+Q-3s-6,\qquad F(Q+1,s)=F(Q,s-3)-2s+3$$

pass to both amplitudes. Their four-sheet sum therefore has the exact wrapped grid action of the original finite matrices. Its entry residual is $O(h^2/\sqrt{N})$, so unitary Duhamel gives

$$\|U_N-V_N\|_{\mathrm{HS}}=O(N^{-3/2}),\qquad|\operatorname{Tr}(U_N-V_N)|+|\operatorname{Tr}(A_N(U_N-V_N))|=O(N^{-1}).$$

Keeping only B_0 would not provide this trace error.

Poisson summation with smooth lifted circle cutoffs includes every alias. The diagonal phases have second derivative $-1+O(\tau)$; their four stationary classes are the protected fixed points $(q,p)\in\{0,1/2\}^2$. The kick is stationary there, with values $9/10,9/10,-11/10,-11/10$. The one-dimensional stationary remainder and the hB_1 term each contribute $O(N^{-1})$ after summation. Deck permutations make the cutoff weights sum to one without duplicate stationary contributions. Applying the finite Weyl sum before taking the trace changes the principal amplitude to $a(Q,F_Q)B_0$; no asymptotic remainder is differentiated. This proves both original and inserted formulas of TRACE-109. In its notation,

$$\liminf_{N\to\infty,\ N\ \mathrm{odd}}|\operatorname{Tr}(A_NU_N)|\ge\tfrac{11}{10}\mathsf{B}-\tfrac{9}{10}|\mathsf{A}|>\tfrac{11}{10}.$$

The commutant projection and two-copy obstruction consequently retain their stated constants for arbitrary spectral multiplicities.

100.2 Uniform classical displacement and an independent upper route

Let $\Psi=\phi_{-a}^{t_*}\circ L$ with $L=\left(\begin{smallmatrix}3&-4\\-2&3\end{smallmatrix}\right)$. For uniform area on the torus, TRACE-112 proves that $(x\mapsto\Psi^mx-x)_*dx$ has a continuous density q_m and

$$\|q_m-1\|_\infty\le\epsilon_Rm\vartheta^{m-1}\le8\cdot10^{-53},\qquad\epsilon_R<10^{-56},\quad\vartheta=1-1/8000,\quad m\ge1.$$

In Fourier coordinates use the stable/unstable weights $w_k=\exp(b(|s(k)|-|u(k)|))$, $b=.001$. The resulting weighted Koopman operator R is trace class and $\|R-R_0\|_1=\epsilon_R$. Both nonconstant blocks have operator norm at most ϑ. The linear block has zero diagonal in every positive power. Hence

$$\|R_\perp^m-R_{0,\perp}^m\|_1\le\epsilon_Rm\vartheta^{m-1}.$$

Finite Fourier compressions telescope the weights exactly; their trace-norm and unweighted strong limits identify each diagonal with a displacement Fourier coefficient. This avoids assuming

a bounded similarity for the unbounded weights. Absolute summability of those coefficients proves the density assertion, also for inverse iterates.

For $b_1 = e^{2\pi i q}$, $b_2 = e^{2\pi i p}$ and $G = \Psi^{-1}$ put $v_m = \sum_j |b_j - b_j \circ G^m|^2$. The density bound gives $\mathrm{vol}\{v_m \le \delta\} \le C\delta$ uniformly in m. Fixed-character Egorov has error $Ce^{C_0 m} N^{-2}$. Using a positive-square cutoff near $v_m = 0$, the Weyl product estimates give

$$|\operatorname{tr}_N U_N^m| \le C\big[\delta + e^{Km} N^{-1} \delta^{-4} + e^{Km} N^{-2} \delta^{-8}\big].$$

Choosing $\delta = N^{-1/5}$ yields $Ce^{Km} N^{-1/5}$, in particular a uniform $CN^{-1/10}$ bound for $m \le \log N/(10K)$. Squared Fejér smoothing gives arc discrepancy $O(1/\log N)$ for both U_N and the bare cat matrix. Sorting eigenphases and matching orthonormal eigenbases proves the upper global spectral comparison below, independently of the all-time quantum trace formula and pressure.

100.3 One power-trace threshold for every iterate

TRACE-113 reconstructs the exact odd comb scalar, amplitude and sector identification, and the centered trace-class Gaussian cocycle. Its splitting is

$$K_h(x) = \gamma_h(x) v_h(\Psi x) \ell_h(x) + \mathcal{B}_h(x), \quad \ell_h v_h = 1, \quad \mathcal{B}_h v_h = 0, \quad \ell_h(\Psi x)\mathcal{B}_h(x) = 0.$$

The residual operator and trace norms are less than .174 and .209. Exact vanishing of mixed products and the fixed-point count give

$$\operatorname{Tr} U_N^m = \zeta_N^m \sum_{x \in \operatorname{Fix} \Psi^m} \omega_{m,N}(x) \lambda^{-m/2} \Gamma_h(x, m) + E_{m,N}, \qquad |E_{m,N}| \le 2(.421)^m,$$

where $|\omega_{m,N}| = 1$, $\lambda = 3 + 2\sqrt{2}$ and $\Gamma_h = \prod_{j=0}^{m-1} \gamma_h(\Psi^j x)$. The centered saturated contour has integrable linear tails and the trace-class convergence is uniform on the torus. Graph contraction therefore gives $\gamma_h \to \gamma_0 > 0$ uniformly. Exact Gaussian repetitions give $\lambda^{-m/2}\Gamma_0 = \Lambda_m^{-1/2}$, where Λ_m is the unstable periodic multiplier. Thus

$$\lambda^{-m/2} |\Gamma_h| \le e^{m\varepsilon(h)} \Lambda_m^{-1/2}, \qquad \varepsilon(h) \longrightarrow 0.$$

Only continuity is required; the older $O(h^2)$ modulus expansion is not.

The actual area-preserving map is conjugate to L and has entropy $H_0 = \log \lambda$. Bowen–Ruelle equilibrium and volume theorems [57, Theorem 3.3, Lemmas 4.1–4.2, Proposition 4.4(b), Remark 5.2], checked in the primary source for the constant-roof suspension, apply without a mixing-flow premise. For $\varphi = \log J_\Psi^u$ one has $P(0) = H_0$ and $P(-\varphi) = 0$. Equality at the midpoint would identify smooth area with the maximal-entropy equilibrium measure. At $(0, 0)$,

$$\operatorname{Tr} D\Psi = 6 + 39212\pi\tau/125 > 6,$$

so its unstable multiplier exceeds λ; the Gibbs lower and volume upper bounds contradict that identification. Therefore $\Delta = H_0/2 - P(-\varphi/2) > 0$. Separated periodic points give an upper pressure bound for their weighted sum. Choosing once and for all $\varepsilon(h) \le \Delta/4$ proves

$$\boxed{|\operatorname{Tr} U_N^m| \le C a_*^m, \qquad 1 < a_* < \sqrt{\lambda}, \quad m \ge 1, \quad N \ge N_0 \text{ odd.}}$$

The threshold N_0 is independent of m. Neither it nor Δ is numerically evaluated. Exact source locators and the hypothesis dictionary are in `literature/bowen_ruelle_trace_closure_2026 1021.md`.

100.4 Global distance and the remaining microscopic gap

For $d_N = \inf_{W,|z|=1} \|U_N - zWM_NW^*\|_{\mathrm{op}}$, TRACE-110 now proves

$$d_N = O(1/\log N), \qquad \liminf_{r\to\infty,\ r\ \mathrm{odd}} (\log N_r) d_{N_r} \geq \frac{\pi \log \lambda}{4}, \qquad N_r = \frac{\lambda^r - \lambda^{-r}}{4\sqrt{2}}.$$

The lower bound uses $M_{N_r}^{2r}$ scalar and $|\operatorname{Tr} U_{N_r}^{2r}|/N_r \to 0$ from $a_*^2 < \lambda$. This establishes the matching order $\Theta(1/\log N_r)$, not an equality for its leading constant. The upper comparison allows arbitrary conjugation and does not contradict the distance without that conjugation. At $m \asymp N$ the exponential trace bound exceeds the trivial bound N; the global distance is much larger than $1/N$. Neither route evaluates the negative signed plateau correction or any higher microscopic factorial defect. TRACE-005 and BGS remain open.

Chapter 15 — Conclusion and outlook

101 Conclusion and outlook

Draft note. This chapter is a draft conclusion. It summarises only what the preceding chapters and the claim registry already record, and it marks every open obligation as open. It is not a claim of completion.

101.1 What this volume establishes

The chapters fall into distinct regimes, and their results do not combine into a single universality claim. The finite-field and Cayley chapters give exact combinatorial reductions (pair keys, Cayley energy, fixed-degree row polynomials, scalar supports) together with range-restricted or conditional arithmetic estimates. The return-law chapters give asymptotic laws on explicitly declared arithmetic subsequences, regularity scales, and observables. The resonance and cohomology chapters give sharp coercivity and rigidity results, counterexamples in distinct smooth, analytic, linear-transport, and nonlinear regimes, and an explicit noncommuting return-word exception. The fixed-map chapters give exact trace and resolvent identities, selected interference sectors, and conditional bridges toward the microscopic target. The tomography and minimax chapters are a separate branch, with their own dynamics and source and measurement assumptions, and are not results about the unchanged deterministic fixed-strength spectrum.

Two reductions deserve explicit mention because later chapters build on them. The fixed-map position-chain singular mixing, cyclic ramp, and marked three-block results are registered as proved, and the phase-model transfer built on them is proved in the repo-model reading of the transfer while remaining conditional in the literal reading of the external transfer text; the written dictionary between the two readings has since landed as mathematics/fixed-map/r1_dictionary.md (90c3b18), leaving only the R-L6 residual at $R \leq 3$. No chapter in this volume silently upgrades a conditional bridge into a proved theorem.

101.2 What is conditional, pending, or unresolved

As registered, the volume contains 199 claims: 173 proved, 12 verification-pending, 9 conditional, and 5 open. Conditional claims are labelled as such at their statements, and verification-pending claims are not presented as proved. Among the latter, the radius-eight resonance nondegeneracy result stays verification-pending; its promotion is a separate gated decision, not a consequence of this chapter. External results used in the finite-field and spectral chapters retain their named

external hypotheses, and the bibliography collects uncited entries separately under Further reading rather than attaching them to points of use.

The reported proofs have not received a blanket independent external referee or novelty certification. That is a statement about the present status of the volume, not a disclaimer about any individual proof: individual results carry their own evidence, finite certificates, or replay records, as recorded in Appendix B.

101.3 The open target

The terminal open obligation is the signed Heisenberg-scale desymmetrized joint-trace estimate for the fixed map (BGS-TRACE-005), and with it the fixed-strength microscopic universality claim (BGS-BGS-001). Both are registered open, and this chapter does not close them. For the fixed deterministic family the comparison is a signed weighted sum: the necessary pair-scale plateau correction is negative, classical intensity mixing cannot supply the cancellation by itself, and a global Gaussian observable law, a finite polynomial correction, an open-dense nondegeneracy statement, or an exceptional all-orders conjugacy does not supply the missing comparison. The absolute sufficient implication remains correct, including averages taken before the modulus; the simultaneous unaveraged all-order absolute strengthening is excluded.

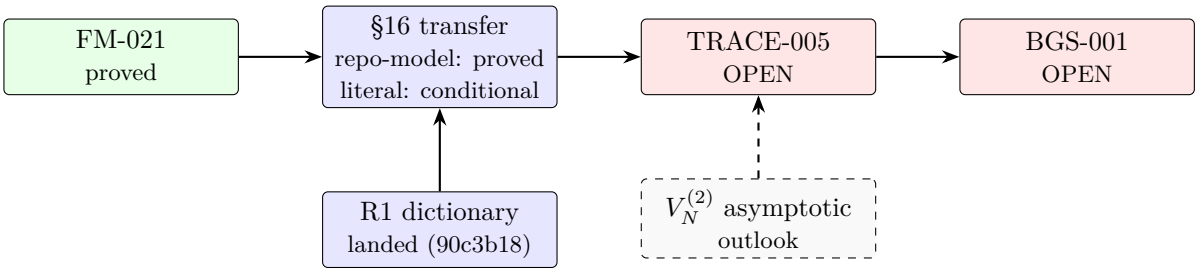


Figure 2: Flagship dependency chain (schematic; registry statuses). FM-021 proved; the §16 transfer is proved in the repo-model reading and conditional in the literal external-text reading; TRACE-005 and BGS-001 open. The R1 dictionary box is landed (90c3b18); the $V_N^{(2)}$ box is outlook, not a result.

The manuscript-scope decision inside the fixed-map frontier chapter has been taken: the lead kernel-bound block of the growing-degree projection claim is Theorem 70.8, and the projection-rate and missed-test blocks stay as prose parts of the same claim. Its registry state is proved. That decision does not affect any other claim in this volume.

101.4 Outlook

The next obligations, in order, are: a uniform-in-word-length asymptotic for the second model, aimed at the pair-scale plateau correction; and then the higher factorial orders. In parallel, the remaining verification programme is the re-basing of the replay notes that could not be pinned during the restructure, the continuation of the formalization (whose current modules cover the finite and algebraic layer, with the open target formalized as a statement only), and the splitting of this compendium into focused articles, each with its own claim manifest and publication-review requirements. Until those articles pass external review, the correct description of this volume is the one its abstract already gives: a continuing research monograph of exact results and an open frontier, not a proof of universality.

Appendix A: Claim index

This index is generated from `record/claims.json` at HEAD 0ed548d (199 claims: 173 proved, 12 verification-pending, 9 conditional, 5 open). It maps each registry claim ID to its mentions in the manuscript body, with file, line, and chapter from the Pass 0 chapter map. Mentions were rescanned after the Pass-2 chapter assembly and claim conversions (base HEAD 4b3a577); body mentions remain 122 hits / 68 distinct IDs, only fixed-map-frontier line numbers moved. A claim with no manuscript mention is registry-only: its statement and evidence live in the registry and its source files, not in this compendium's body text. Claim IDs remain in the body text in Pass 2 by decision: the 13 converted fixed-map-frontier paragraphs carry pointer sentences to this index, and 3 paragraphs stay as prose (chapter index list, FLOW-020 uncertain scope, TRACE-011 diffuse assertions); moving IDs fully into this index remains an open chapter-definition-of-done decision. Repository paths likewise remain in the body text. Repository paths in the body text, by pattern (Pass 0 definitions, re-verified in Pass 1): `.../...` commands: 41; `.../...` commands: 24; bare repo-rooted paths (`mathematics/`, `computations/`, `record/`, `literature/`, `formalization/`, `evidence/` outside / spans): 103; total pattern hits 168. Family codes (`(XX.N)` pattern, broad definition): 240 hits across families ND, RW, CB, FN, LC, PL, FA, FO, MQ, HF, GC, SV, UF, SF, PB, TW.

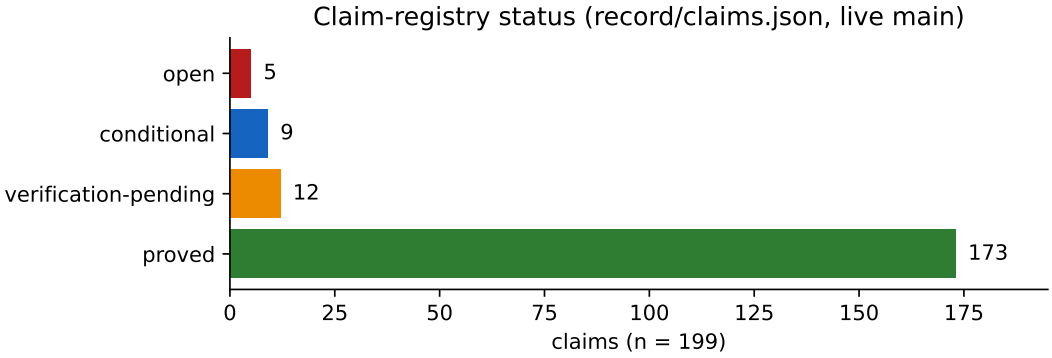


Figure 3: Claim-registry status counts from `record/claims.json` at live main HEAD d9a729a (blob 6fbc6b7c9713187fb7fc25b6b16e8a8bbbc9ee90), counted 2026-10-25: 199 claims — 173 proved, 12 verification-pending, 9 conditional, 5 open.

Claim ID	State	Manuscript mentions (file:line, chapter)
BGS-BGS-001	open	unified_sections/fixed_map_frontier.tex:8 (Ch. 12); unified_sections/fixed_map_frontier.tex:689 (Ch. 12); unified_sections/fixed_map_frontier.tex:828 (Ch. 12); unified_sections/fixed_map_frontier.tex:1616 (Ch. 12); unified_sections/fixed_map_frontier.tex:1829 (Ch. 12); unified_sections/fixed_map_frontier.tex:1868 (Ch. 12); unified_sections/fixed_map_frontier.tex:1900 (Ch. 12); unified_sections/fixed_map_frontier.tex:1971 (Ch. 12); unified_sections/fixed_map_frontier.tex:2064 (Ch. 12); unified_sections/physical_kick_decay.tex:97 (Ch. ?)
BGS-CAYLEY-001	proved	— (registry only)
BGS-CAYLEY-002	proved	— (registry only)
BGS-CAYLEY-003	verification-pending	— (registry only)
BGS-CAYLEY-004	proved	— (registry only)
BGS-CAYLEY-005	open	— (registry only)
BGS-FM-001	proved	— (registry only)
BGS-FM-002	proved	— (registry only)
BGS-FM-003	proved	— (registry only)
BGS-FM-004	conditional	— (registry only)
BGS-FM-005	proved	unified_sections/fixed_map_frontier.tex:8 (Ch. 12)
BGS-FM-006	proved	unified_sections/fixed_map_frontier.tex:8 (Ch. 12)
BGS-FM-007	proved	unified_sections/fixed_map_frontier.tex:8 (Ch. 12); unified_sections/fixed_map_frontier.tex:189 (Ch. 12); unified_sections/fixed_map_frontier.tex:212 (Ch. 12)
BGS-FM-008	proved	unified_sections/fixed_map_frontier.tex:8 (Ch. 12)
BGS-FM-009	proved	unified_sections/fixed_map_frontier.tex:8 (Ch. 12)
BGS-FM-010	proved	unified_sections/fixed_map_frontier.tex:8 (Ch. 12)
BGS-FM-011	proved	unified_sections/fixed_map_frontier.tex:8 (Ch. 12)
BGS-FORMAL-001	proved	— (registry only)
BGS-INFO-001	open	— (registry only)
BGS-LOOP-001	conditional	unified_sections/fixed_map_frontier.tex:8 (Ch. 12)
BGS-RES-001	proved	— (registry only)
BGS-RES-002	proved	— (registry only)
BGS-RES-003	open	— (registry only)
BGS-RES-004	proved	unified_sections/finite_orbit_resonance.tex:5 (Ch. 10)

Claim ID	State	Manuscript mentions (file:line, chapter)
BGS-RES-005	proved	unified_sections/fixed_map_frontier.tex:8 (Ch. 12)
BGS-RES-006	proved	unified_sections/fixed_map_frontier.tex:8 (Ch. 12); unified_sections/fixed_map_frontier.tex:66 (Ch. 12)
BGS-RES-007	proved	unified_sections/fixed_map_frontier.tex:8 (Ch. 12); unified_sections/fixed_map_frontier.tex:184 (Ch. 12)
BGS-RET-001	proved	— (registry only)
BGS-RET-002	proved	— (registry only)
BGS-TRACE-001	proved	— (registry only)
BGS-TRACE-002	proved	— (registry only)
BGS-TRACE-003	proved	— (registry only)
BGS-TRACE-004	proved	— (registry only)
BGS-TRACE-005	open	unified_sections/fixed_map_frontier.tex:8 (Ch. 12); unified_sections/fixed_map_frontier.tex:251 (Ch. 12); unified_sections/fixed_map_frontier.tex:275 (Ch. 12); unified_sections/fixed_map_frontier.tex:284 (Ch. 12); unified_sections/fixed_map_frontier.tex:689 (Ch. 12); unified_sections/fixed_map_frontier.tex:828 (Ch. 12); unified_sections/fixed_map_frontier.tex:1829 (Ch. 12); unified_sections/physical_kick_decay.tex:97 (Ch. ?)
BGS-TRACE-006	proved	— (registry only)
BGS-TRACE-007	proved	unified_sections/fixed_map_frontier.tex:8 (Ch. 12); unified_sections/fixed_map_frontier.tex:130 (Ch. 12)
BGS-TRACE-008	proved	unified_sections/fixed_map_frontier.tex:8 (Ch. 12); unified_sections/fixed_map_frontier.tex:252 (Ch. 12)
BGS-CUE-001	proved	unified_sections/fixed_map_frontier.tex:8 (Ch. 12); unified_sections/fixed_map_frontier.tex:521 (Ch. 12)
BGS-FLOW-001	proved	unified_sections/fixed_map_frontier.tex:8 (Ch. 12); unified_sections/fixed_map_frontier.tex:540 (Ch. 12)
BGS-FLOW-002	proved	unified_sections/fixed_map_frontier.tex:8 (Ch. 12); unified_sections/fixed_map_frontier.tex:546 (Ch. 12)
BGS-FM-012	proved	unified_sections/fixed_map_frontier.tex:8 (Ch. 12); unified_sections/fixed_map_frontier.tex:558 (Ch. 12)

Claim ID	State	Manuscript mentions (file:line, chapter)
BGS-FLOW-003	proved	unified_sections/fixed_map_frontier.tex:8 (Ch. 12); unified_sections/fixed_map_frontier.tex:583 (Ch. 12)
BGS-FLOW-004	proved	unified_sections/fixed_map_frontier.tex:8 (Ch. 12); unified_sections/fixed_map_frontier.tex:613 (Ch. 12)
BGS-FLOW-005	proved	unified_sections/fixed_map_frontier.tex:8 (Ch. 12); unified_sections/fixed_map_frontier.tex:632 (Ch. 12)
BGS-FLOW-006	proved	unified_sections/fixed_map_frontier.tex:8 (Ch. 12); unified_sections/fixed_map_frontier.tex:654 (Ch. 12)
BGS-FLOW-007	proved	unified_sections/fixed_map_frontier.tex:8 (Ch. 12); unified_sections/fixed_map_frontier.tex:672 (Ch. 12)
BGS-FM-013	proved	unified_sections/fixed_map_frontier.tex:8 (Ch. 12); unified_sections/fixed_map_frontier.tex:681 (Ch. 12); unified_sections/fixed_map_frontier.tex:2071 (Ch. 12)
BGS-FLOW-008	proved	unified_sections/fixed_map_frontier.tex:8 (Ch. 12); unified_sections/fixed_map_frontier.tex:697 (Ch. 12)
BGS-FLOW-009	proved	unified_sections/fixed_map_frontier.tex:8 (Ch. 12); unified_sections/fixed_map_frontier.tex:705 (Ch. 12)
BGS-FLOW-010	proved	unified_sections/fixed_map_frontier.tex:8 (Ch. 12); unified_sections/fixed_map_frontier.tex:738 (Ch. 12)
BGS-FLOW-011	proved	unified_sections/fixed_map_frontier.tex:8 (Ch. 12); unified_sections/fixed_map_frontier.tex:769 (Ch. 12)
BGS-FM-014	proved	unified_sections/fixed_map_frontier.tex:1774 (Ch. 12)
BGS-FM-015	proved	unified_sections/fixed_map_frontier.tex:1779 (Ch. 12)
BGS-FUP-001	proved	unified_sections/fixed_map_frontier.tex:1795 (Ch. 12)
BGS-TRACE-009	proved	unified_sections/fixed_map_frontier.tex:1834 (Ch. 12)
BGS-TRACE-010	proved	unified_sections/fixed_map_frontier.tex:8 (Ch. 12)
BGS-TRACE-011	proved	unified_sections/fixed_map_frontier.tex:8 (Ch. 12); unified_sections/fixed_map_frontier.tex:1905 (Ch. 12)

Claim ID	State	Manuscript mentions (file:line, chapter)
BGS-TRACE-012	proved	unified_sections/fixed_map_frontier.tex:1977 (Ch. 12)
BGS-FM-016	proved	unified_sections/fixed_map_frontier.tex:8 (Ch. 12); unified_sections/fixed_map_frontier.tex:2071 (Ch. 12)
BGS-TRACE-013	proved	unified_sections/fixed_map_frontier.tex:8 (Ch. 12); unified_sections/fixed_map_frontier.tex:2144 (Ch. 12)
BGS-FLOW-012	proved	unified_sections/fixed_map_frontier.tex:8 (Ch. 12); unified_sections/fixed_map_frontier.tex:833 (Ch. 12)
BGS-FLOW-013	proved	unified_sections/fixed_map_frontier.tex:8 (Ch. 12); unified_sections/fixed_map_frontier.tex:859 (Ch. 12)
BGS-FM-017	proved	unified_sections/fixed_map_frontier.tex:8 (Ch. 12); unified_sections/fixed_map_frontier.tex:883 (Ch. 12)
BGS-FLOW-014	proved	unified_sections/fixed_map_frontier.tex:8 (Ch. 12); unified_sections/fixed_map_frontier.tex:932 (Ch. 12)
BGS-FLOW-015	proved	unified_sections/fixed_map_frontier.tex:8 (Ch. 12); unified_sections/fixed_map_frontier.tex:1649 (Ch. 12)
BGS-FLOW-016	proved	unified_sections/fixed_map_frontier.tex:8 (Ch. 12); unified_sections/fixed_map_frontier.tex:1009 (Ch. 12)
BGS-FLOW-017	proved	unified_sections/fixed_map_frontier.tex:8 (Ch. 12); unified_sections/fixed_map_frontier.tex:1009 (Ch. 12)
BGS-FLOW-018	proved	unified_sections/fixed_map_frontier.tex:8 (Ch. 12); unified_sections/fixed_map_frontier.tex:1143 (Ch. 12)
BGS-FLOW-019	proved	unified_sections/fixed_map_frontier.tex:8 (Ch. 12); unified_sections/fixed_map_frontier.tex:1143 (Ch. 12)
BGS-FLOW-020	proved	unified_sections/fixed_map_frontier.tex:8 (Ch. 12); unified_sections/fixed_map_frontier.tex:1247 (Ch. 12)
BGS-FLOW-021	proved	unified_sections/fixed_map_frontier.tex:8 (Ch. 12); unified_sections/fixed_map_frontier.tex:1332 (Ch. 12)

Claim ID	State	Manuscript mentions (file:line, chapter)
BGS-FM-018	proved	unified_sections/fixed_map_frontier.tex:8 (Ch. 12); unified_sections/fixed_map_frontier.tex:1418 (Ch. 12)
BGS-FM-019	proved	unified_sections/fixed_map_frontier.tex:6 (Ch. 12); unified_sections/fixed_map_frontier.tex:1516 (Ch. 12)
BGS-FM-020	proved	unified_sections/fixed_map_frontier.tex:1621 (Ch. 12)
BGS-TRACE-014	proved	— (registry only)
BGS-FM-021	proved	unified_sections/four_block_unmarking.tex:6 (Ch. 12); unified_sections/three_block_unmarking.tex:54 (Ch. 12)
BGS-FM-022	proved	— (registry only)
BGS-TRACE-015	proved	unified_sections/three_block_unmarking.tex:14 (Ch. 12)
BGS-FM-023	proved	— (registry only)
BGS-TRACE-016	proved	unified_sections/four_block_unmarking.tex:15 (Ch. 12)
BGS-FORMAL-002	proved	— (registry only)
BGS-TRACE-017	proved	— (registry only)
BGS-FM-024	proved	— (registry only)
BGS-FORMAL-003	proved	— (registry only)
BGS-TRACE-018	proved	— (registry only)
BGS-FM-025	proved	— (registry only)
BGS-TRACE-019	proved	— (registry only)
BGS-TRACE-020	proved	— (registry only)
BGS-FM-026	proved	— (registry only)
BGS-TRACE-021	proved	— (registry only)
BGS-FM-027	proved	— (registry only)
BGS-TRACE-022	proved	— (registry only)
BGS-TRACE-023	proved	— (registry only)
BGS-FM-028	proved	— (registry only)
BGS-FM-029	proved	— (registry only)
BGS-FM-030	proved	— (registry only)
BGS-TRACE-024	proved	— (registry only)
BGS-FM-031	proved	— (registry only)
BGS-TRACE-025	proved	— (registry only)
BGS-TRACE-026	proved	— (registry only)
BGS-TRACE-027	proved	— (registry only)
BGS-TRACE-028	proved	— (registry only)
BGS-TRACE-029	proved	— (registry only)
BGS-TRACE-053	proved	— (registry only)
BGS-TRACE-054	proved	— (registry only)
BGS-TRACE-055	verification-pending	— (registry only)
BGS-TRACE-056	proved	— (registry only)
BGS-TRACE-057	proved	— (registry only)
BGS-TRACE-058	proved	— (registry only)
BGS-TRACE-059	proved	— (registry only)
BGS-TRACE-060	conditional	— (registry only)
BGS-TRACE-061	proved	— (registry only)
BGS-TRACE-062	proved	— (registry only)
BGS-TRACE-063	proved	— (registry only)
BGS-TRACE-064	proved	— (registry only)

Claim ID	State	Manuscript mentions (file:line, chapter)
BGS-TRACE-065	proved	— (registry only)
BGS-TRACE-066	proved	— (registry only)
BGS-TRACE-067	proved	— (registry only)
BGS-TRACE-068	proved	— (registry only)
BGS-TRACE-069	proved	— (registry only)
BGS-TRACE-070	proved	— (registry only)
BGS-TRACE-071	proved	— (registry only)
BGS-TRACE-072	proved	— (registry only)
BGS-TRACE-073	conditional	— (registry only)
BGS-TRACE-074	proved	— (registry only)
BGS-TRACE-075	proved	— (registry only)
BGS-TRACE-076	proved	— (registry only)
BGS-TRACE-077	proved	unified_sections/microscopic_paths_and_phase_laws.tex:887 (Ch. ?)
BGS-TRACE-078	proved	unified_sections/microscopic_paths_and_phase_laws.tex:992 (Ch. ?)
BGS-TRACE-079	proved	unified_sections/microscopic_paths_and_phase_laws.tex:1063 (Ch. ?)
BGS-TRACE-080	proved	unified_sections/microscopic_paths_and_phase_laws.tex:1137 (Ch. ?)
BGS-TRACE-081	proved	unified_sections/microscopic_paths_and_phase_laws.tex:1271 (Ch. ?)
BGS-TRACE-082	proved	unified_sections/microscopic_paths_and_phase_laws.tex:1404 (Ch. ?)
BGS-TRACE-083	proved	unified_sections/microscopic_paths_and_phase_laws.tex:1479 (Ch. ?)
BGS-FM-032	proved	— (registry only)
BGS-TRACE-084	proved	— (registry only)
BGS-TRACE-085	proved	— (registry only)
BGS-TRACE-086	proved	— (registry only)
BGS-FM-033	conditional	— (registry only)
BGS-CUE-002	proved	— (registry only)
BGS-TRACE-087	proved	— (registry only)
BGS-TRACE-088	proved	— (registry only)
BGS-TRACE-089	proved	— (registry only)
BGS-TRACE-090	proved	— (registry only)
BGS-TRACE-091	proved	— (registry only)
BGS-TRACE-092	proved	unified_sections/abel_gaussian_rigidity.tex:3 (Ch. ?)
BGS-TRACE-093	proved	unified_sections/abel_gaussian_rigidity.tex:4 (Ch. ?)
BGS-TRACE-094	proved	unified_sections/abel_gaussian_rigidity.tex:178 (Ch. ?)
BGS-TRACE-095	conditional	— (registry only)
BGS-TRACE-096	verification-pending	— (registry only)
BGS-TRACE-097	conditional	— (registry only)
BGS-TRACE-098	verification-pending	— (registry only)
BGS-TRACE-099	verification-pending	— (registry only)
BGS-TRACE-100	verification-pending	— (registry only)
BGS-TRACE-101	verification-pending	— (registry only)
BGS-TRACE-102	verification-pending	— (registry only)

Claim ID	State	Manuscript mentions (file:line, chapter)
BGS-TRACE-103	verification-pending	— (registry only)
BGS-TRACE-104	verification-pending	— (registry only)
BGS-TRACE-105	verification-pending	— (registry only)
BGS-CAYLEY-006	proved	— (registry only)
BGS-CAYLEY-007	proved	— (registry only)
BGS-CAYLEY-008	proved	— (registry only)
BGS-CAYLEY-009	proved	— (registry only)
BGS-CAYLEY-010	proved	— (registry only)
BGS-FLOW-022	proved	— (registry only)
BGS-FLOW-023	proved	— (registry only)
BGS-FLOW-024	proved	— (registry only)
BGS-FLOW-025	proved	— (registry only)
BGS-FM-034	proved	— (registry only)
BGS-FM-035	proved	— (registry only)
BGS-FM-036	proved	— (registry only)
BGS-RES-008	proved	— (registry only)
BGS-RES-009	proved	— (registry only)
BGS-RES-010	proved	— (registry only)
BGS-RES-011	proved	— (registry only)
BGS-RES-012	proved	— (registry only)
BGS-RES-013	proved	— (registry only)
BGS-RES-014	conditional	— (registry only)
BGS-RES-015	proved	— (registry only)
BGS-RES-016	proved	— (registry only)
BGS-RES-017	proved	— (registry only)
BGS-RES-018	proved	— (registry only)
BGS-RES-019	proved	— (registry only)
BGS-RES-020	proved	— (registry only)
BGS-RES-021	proved	— (registry only)
BGS-RES-022	verification-pending	— (registry only)
BGS-RET-003	conditional	— (registry only)
BGS-RET-004	proved	— (registry only)
BGS-RES-023	proved	— (registry only)
BGS-TRACE-106	proved	unified_sections/physical_kick_decay.tex:4 (Ch. ?)
BGS-TRACE-107	proved	unified_sections/physical_kick_decay.tex:4 (Ch. ?)
BGS-TRACE-108	proved	— (registry only)
BGS-TRACE-109	proved	— (registry only)
BGS-TRACE-110	proved	— (registry only)
BGS-TRACE-111	proved	— (registry only)
BGS-TRACE-112	proved	— (registry only)
BGS-TRACE-113	proved	— (registry only)
BGS-CAYLEY-011	proved	— (registry only)
BGS-FM-037	proved	— (registry only)

Supplement: verbatim listings

The five verbatim blocks below are the verifier invocations that stood in the return chapters. They are reproduced here verbatim (content preserved, not cut); each original location now

carries a pointer sentence to its listing.

Listing S1: `fixed_harmonic_return_gas.py`

From the return-gas chapter (fourth-coefficient verification discussion) (source file `unified_sections/return_gas.tex`).

```
OPENBLAS_NUM_THREADS=1 python computations/verification/fixed_harmonic_return_gas.py \
  --output build/fixed-harmonic-return-gas-precheck.json
```

Listing S2: `general_cat_return_laws.py`

From the return-general chapter, "Verification and the remaining boundary" (source file `unified_sections/return_general.tex`).

```
OPENBLAS_NUM_THREADS=1 python computations/verification/general_cat_return_laws.py \
  --output build/general-cat-return-laws-precheck.json
```

Listing S3: `periodic_point_trace_comparison.py`

From the return-periodic chapter (periodic-point comparison diagnostics) (source file `unified_sections/return_periodic.tex`).

```
OPENBLAS_NUM_THREADS=1 python
↪  computations/verification/periodic_point_trace_comparison.py \
  --output build/periodic-point-trace-comparison-precheck.json
```

Listing S4: `phase_space_return_gas.py`

From the return-phase chapter, "Verification and scope" (source file `unified_sections/return_phase.tex`).

```
OPENBLAS_NUM_THREADS=1 python computations/verification/phase_space_return_gas.py \
  --output build/phase-space-return-gas-precheck.json
```

Listing S5: `analytic_kick_return_gas.py`

From the return-position chapter (winding-term verification discussion) (source file `unified_sections/return_position.tex`).

```
OPENBLAS_NUM_THREADS=1 python computations/verification/analytic_kick_return_gas.py \
  --output build/analytic-kick-return-gas-precheck.json
```

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